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Crazy Merged 4

Sections 13–17

One merged book: the condensed books and the Paper 1–4 answers combined, section by section, with each topic appearing once. Read each chapter top to bottom. Where a topic was answered in the papers, the full paper answer sits in its place (marked "Paper answer", with the papers it was asked in), followed by a dashed box of anything extra the book had. Paper answers not tied to a chapter are at the end of each section, then the section's Rapid Revision Sheet.

Section	Title
13	Kidney & UTUC
14	Adrenal & Retroperitoneum
15	Renal, Nephrology & Transplant
16	BPH
17	Prostate Cancer

SECTION 13

Kidney & UTUC

136 Benign Renal Tumours

1 · The benign small renal mass

Many resected "cancers" are benign.

- Up to **~25%** of surgically treated small renal masses (SRMs) are benign.
- Only the **simple cyst** and **fat-containing angiomyolipoma (AML)** are imaging diagnoses; size, sex and age are unreliable.
- Management follows the risk of renal cell carcinoma (RCC); the SRM algorithm is in Ch 137.

2 · Renal cysts and the Bosniak classification

- A **simple cyst** (any size, homogeneous, **≤20 Hounsfield units (HU)**) needs no further imaging. Otherwise use Bosniak.

Class	Features (v2019)	Malignancy	Management (Canadian Urological Association, CUA 2023)
I	Thin (**≤2 mm**) smooth wall (may enhance); simple fluid; no septa or calcification	Benign	No follow-up
II	Thin wall plus (a) **≤3** thin enhancing septa, any calcification; or (b) homogeneous marked T2 hyperintensity; or (c) marked T1 hyperintensity	Highly likely benign	No follow-up
IIF	Minimally thickened (**3 mm**) enhancing wall or septum, or **≥4** thin septa; or heterogeneous T1 hyperintensity on fat-saturated MRI	Likely benign	Active surveillance up to **5 yr**
III	Enhancing thick (**≥4 mm**) or irregular (≤3 mm obtuse-margined protrusion) wall or septum	**~50%**	≤2 cm: surveillance preferred; 2–4 cm: surveillance or excision (shared decision); >4 cm: excision, partial (PN) over radical nephrectomy (RN) when feasible
IV	Enhancing nodule: ≥4 mm obtuse-margined, or any-size acute-margined protrusion	**~90%**	

KEY POINT

Ladder: wall ≤2 mm, 3 mm, ≥4 mm, irregular, then nodule. Traps: any calcification fits II, a class I wall may enhance, and growth of I/II is not malignancy.

KEY POINT

Why surveil small III/IV? Cystic RCC is indolent and mostly fluid, so size predicts poorly (proposed **pT1c**) and the number needed to treat is very high.

- European Association of Urology (EAU): treat Bosniak IV as a solid mass, and biopsy a solid component only if it changes management.

Diffuse cystic disease

SAQ ADPKD (Adult polycystic kidney disease)

Paper answer · P2

Type: SAQ (10 marks)
Source: 2017-Winter Paper 2, Q3d

Genetics

- ADPKD (autosomal dominant polycystic kidney disease): commonest inherited cystic kidney disease (about 1 in 400–1000); 7–15% of patients on haemodialysis.

Gene	Chromosome	Share	ESRD (end-stage renal disease)
PKD1 (polycystin-1)	16	~85%	50s
PKD2 (polycystin-2)	4	~15%	70s

- Polycystins (primary cilium) restrain proliferation.

Clinical features (30–50 years)

- **Pain** (commonest): mass, cyst haemorrhage, infection, stones. **Haematuria** about 50%; **stones** 20–30%.
- **Hypertension:** about 50% at 20–35 years with normal function; stretched vessels release renin.
- **Renal failure:** tracks kidney volume; worse with PKD1, male sex.
- **Extrarenal:** liver cysts (commonest; up to 90% by 50); **intracranial aneurysm 10–30%**; seminal vesicle (40%), pancreatic, arachnoid cysts; mitral valve prolapse; diverticulosis.
- RCC (renal cell carcinoma) is not commoner, but is more often bilateral and sarcomatoid.

Diagnosis (ultrasound in at-risk relatives)

Age	Ravine (PKD1 families)	Pei unified (2009)
15–29	≥2 cysts, one or both kidneys	≥3 cysts in total (15–39)
30–59	≥2 cysts in each kidney	≥2 in each kidney (40–59)
≥60	≥4 cysts in each kidney	≥4 in each kidney

- CT or MRI shows TKV (total kidney volume), which predicts progression. Genetic testing if equivocal or for a living donor.

Management

- **BP <130/80 mmHg** with ACE (angiotensin-converting enzyme) inhibitor or ARB (angiotensin receptor blocker); HALT-PKD: tighter control slowed TKV growth.
- **Tolvaptan** (V2-receptor antagonist; lowers cyclic adenosine monophosphate): **TEMPO 3:4 (2012)** slowed TKV growth and function loss; **REPRISE (2017)** slowed eGFR (estimated glomerular filtration rate) decline in later disease. Causes polyuria and liver toxicity.
- **Pain:** exclude infection, stone, tumour; avoid NSAIDs (non-steroidal anti-inflammatory drugs); aspiration, then laparoscopic unroofing.

- **Infected cyst:** lipid-soluble antibiotics (trimethoprim–sulfamethoxazole, fluoroquinolones); drain if no response.
- **Haemorrhage:** usually settles with rest. **Aneurysm screening** (MR angiography) if family history.
- **ESRD:** dialysis or transplant (no recurrence). **Native nephrectomy** before or at transplant for recurrent infection, bleeding, pain, suspected tumour or no room for the graft.

Diagram to draw: large cystic kidneys, liver cysts and berry aneurysm.

> **Key points to remember:**
> - PKD1: chromosome 16, 85%, earlier ESRD.
> - Ravine: <30 years ≥2 cysts; 30–59 ≥2 each kidney; ≥60 ≥4 each.
> - Tolvaptan (TEMPO 3:4, REPRISE); ACE inhibitor or ARB.

SAQ Acquired renal cystic disease

Paper answer · P2

Type: SAQ (10 marks)
Source: 2016-Summer Paper 2, Q3b

Definition

- **ARCD (acquired renal cystic disease)**: bilateral cysts in kidneys of patients with **ESRD (end-stage renal disease)** not due to inherited cystic disease. Diagnosis: **≥3 cysts in each kidney**.
- Described in 1977; occurs with haemodialysis and peritoneal dialysis.

Epidemiology

Duration of renal failure	Prevalence
Before dialysis	About 10%
3 years of dialysis	44%
5 years	60%
≥10 years	>90%
- Male:female 3:1; independent of the cause of renal failure.	

Aetiology

- **Uraemia, not dialysis**, is the cause; dialysis only gives cysts time to form.
- Tubular obstruction (fibrosis, oxalate crystals); uraemic toxins; growth factors (EGF (epidermal growth factor)); compensatory hyperplasia.
- Cysts may regress after transplant (data conflict); cancer risk persists.

Clinical features

- Mostly asymptomatic.
- **Bleeding into cysts (up to 50%)**: loin pain, haematuria, retroperitoneal haematoma; worsened by uraemic coagulopathy and heparin.
- Cyst infection, stones, polycythaemia (erythropoietin).
- **Neoplasia**: adenomas in 20–25%. **RCC (renal cell carcinoma)** risk about 3 times that without ARCD. **Acquired cystic disease-associated RCC** is a distinct subtype (oxalate

crystals, good prognosis); papillary RCC is also common.

Evaluation

- **Ultrasound**: small echogenic kidneys, many small cysts.
- **CT (computed tomography) with and without contrast**: best for tumours. MRI (magnetic resonance imaging) for small lesions; gadolinium risks NSF (nephrogenic systemic fibrosis).

Feature	ARCD	ADPKD (autosomal dominant polycystic kidney disease)
Kidney size	Small (<300 g)	Large (>800 g)
Cysts	Small	Large
Extrarenal	None	Liver cysts, aneurysms

Management

- **Bleeding**: conservative; persistent → **embolisation or nephrectomy**; peritoneal dialysis avoids heparin.
- Painful cyst: aspiration; infected cyst: drainage, nephrectomy if it fails.
- **Tumour >3 cm: nephrectomy**. <3 cm: nephrectomy in fit patients, or yearly CT and excise if growing.
- **Screening** (debated): younger fit patients after 3 years of dialysis, and before transplant.
- After transplant, native-kidney RCC is more aggressive; >70% of transplant RCCs arise there.

Diagram to draw: small kidney with multiple cysts and a small solid tumour.

> **Key points to remember:**
> - ≥3 cysts per kidney in ESRD; >90% after 10 years.
> - Uraemia causes it; transplant may regress cysts, not risk.
> - RCC risk 3 times; image dialysis patients with haematuria.

Also, from the condensed book (not in the paper answer):

- Acquired cystic kidney disease (ACKD) affects **80%** by 10 yr of dialysis, with RCC risk **≥10×** (multifocal, bilateral, often papillary).
- PKD1 relatives aged 15–29 need **≥3 cysts** for diagnosis; a normal scan excludes disease only from age 40.

3 · Oncocytoma

Biopsy cannot reliably separate it from chromophobe RCC.

- It is the **commonest benign enhancing renal mass**, up to half of benign SRMs <4 cm.
- **Birt–Hogg–Dubé** (folliculin, chr 17) gives fibrofolliculomas, lung cysts with pneumothorax, oncocytoma and chromophobe RCC (chRCC).
- A **stellate scar** is non-specific, whereas **segmental enhancement inversion** favours oncocytoma.

	Oncocytoma	Chromophobe RCC
Cytokeratin 7 (CK7)	Rare, single-cell; KIT positive	Diffuse positive

	Oncocytoma	Chromophobe RCC
Cytogenetics	Normal, −1, −Y, 11q13 rearrangement	Extensive abnormalities

KEY POINT

Why can a core not clear oncocytoma? chRCC has oncocytic areas, so some biopsy "oncocytomas" are RCC. EAU: no definitive core diagnosis.

4 · Angiomyolipoma

AML can bleed catastrophically, so treatment follows bleeding risk.

SAQ Renal angiomyolipoma and its management

Paper answer · P2 · P3 · merged from 2

Asked as: P2 SAQ "Renal aangiomyolipoma" (Practice Set 7 Paper 2 Q3f) · P3 SAQ "Management of AML." (2021-Summer Paper 3 Q3d)

Assumed meaning: AML = renal angiomyolipoma (not acute myeloid leukaemia).

Introduction and pathogenesis

- Benign **PEComa (perivascular epithelioid cell tumour)** of dysmorphic **vessels, smooth muscle and fat**. Female predominance; fourth–fifth decades.
- **Sporadic** (solitary, small) or with **TSC (tuberous sclerosis complex**; AML in 55–90%: younger, bilateral, multifocal, larger) or LAM (lymphangioleiomyomatosis).
- **Mechanism:** loss of **TSC2** (tuberin, 16p13) or TSC1 (hamartin, 9q34) leads to unchecked **mTOR (mammalian target of rapamycin)** signalling, which drives growth and angiogenesis.
- **Epithelioid AML**: little fat; aggressive (venous extension, metastasis); treat as RCC (renal cell carcinoma).

Clinical features

- Mostly incidental. Large tumours bleed: **Wunderlich syndrome** (spontaneous retroperitoneal haemorrhage; Lenk triad: flank pain, mass, shock).
- CKD (chronic kidney disease) in TSC.

Diagnosis

- **CT (computed tomography): macroscopic fat (below about −15 to −20 HU [Hounsfield units]) on non-contrast images is diagnostic.**
- US (ultrasound): hyperechoic (so are some RCCs).
- MRI (magnetic resonance imaging): signal loss on fat suppression, India-ink artefact.
- Fat-poor lesion: MRI ± biopsy (HMB-45 [human melanoma black-45], Melan-A, smooth muscle actin positive).
- **Aneurysm ≥5 mm** on CT angiography predicts bleeding.

Management algorithm (Restrepo 2022)

Situation	Management
Acute haemorrhage	Resuscitate, then **SAE (selective arterial embolisation)**; nephrectomy if SAE fails or is unavailable

Situation	Management
Symptomatic (pain, haematuria) or poor access to care	SAE, NSS (nephron-sparing surgery), ablation in selected cases
Asymptomatic sporadic with **>6 cm, aneurysm >5 mm, growth >0.25 cm/year, childbearing age or pregnancy**	Active treatment
None of these	**Surveillance**: imaging (US or CT) at 6 and 12 months, then yearly
TSC, asymptomatic, **>3 cm**	**Everolimus**; MRI every 1–3 years

- The old **4 cm** cut-off overtreats and should not be a general trigger; EAU (European Association of Urology) 2023 favours surveillance for most.

Treatments

- **SAE:** first line; **post-embolisation syndrome** (fever, pain, leucocytosis) is common; abscess, infarction.
- **Partial nephrectomy:** preferred surgery; essential in TSC (multifocal, recurrent).
- **Thermal ablation** (radiofrequency, cryo): limited data.
- **Everolimus (EXIST-2):** ≥50% shrinkage in **42% vs 0%** (placebo); side effects stomatitis, acne, hyperlipidaemia.

Diagram to draw

- Management algorithm by bleeding, symptoms, size and TSC.

> **Key points to remember:**
> - Macroscopic fat on non-contrast CT is diagnostic.
> - Treat if >6 cm, aneurysm >5 mm, growth, pregnancy or symptoms.
> - SAE first for bleeding; partial nephrectomy preferred; everolimus for TSC >3 cm.

Also, from the condensed book (not in the paper answer):

- An observed AML ≥4 cm needs CT angiography to look for an aneurysm.
- EAU (sporadic AML): use a mammalian target of rapamycin (mTOR) inhibitor only if unfit for embolisation or surgery.

ALGORITHM

1. **Spontaneous retroperitoneal haemorrhage** (Wunderlich): resuscitate, then selective arterial embolisation (**SAE**). Nephrectomy only if SAE is unavailable or unsuitable.
2. **Symptomatic** (pain, haematuria, barriers to care): SAE, then nephron-sparing surgery (NSS) or RN; ablation in selected cases.
3. **Asymptomatic sporadic**: if >**6 cm**, aneurysm **≥5 mm**, growth >**0.25 cm/yr**, or childbearing age/pregnancy, then treat; otherwise surveil.
4. **Tuberous sclerosis (TSC)**: if >**3 cm** or growing, then everolimus first line; SAE or kidney-sparing resection second.

> **KEY POINT**
>
> **Why abandon 4 cm?** It was poorly specific and caused overtreatment. "Barriers to care" counts because surveillance needs access to imaging and emergency care.

5 · Other benign epithelial and mesenchymal tumours

- **Papillary adenoma** is **≤15 mm** and cannot metastasise, so it is acceptable in a donor kidney.
- **Metanephric adenoma** (women in their 50s, polycythaemia) is **WT1 positive, CK7 negative**; excise it by NSS.
- **Mixed epithelial and stromal tumour and adult cystic nephroma** form one family and are managed by Bosniak.
- **Leiomyoma** is often really a lipid-poor AML; **Melan-A** is positive only in AML.

Tumour	Key trait
Haemangioma	Sturge–Weber, Klippel–Trénaunay; usually excised
Haemangioblastoma	Von Hippel–Lindau (VHL); mimics RCC; cured by excision
Lymphangioma	Perinephric cysts; chyluria
Reninoma (juxtaglomerular cell tumour)	Young women; refractory hypertension, hypokalaemia, high renin; excision cures
Solitary fibrous tumour	~10% aggressive: age ≥55, ≥15 cm, mitoses, necrosis

6 · Landmark trial

Trial	Design	Result		Take-home
EXIST-2	Phase III randomised trial, everolimus vs placebo, AML in TSC or lymphangioleiomyomatosis	≥50% volume reduction in **42% vs 0%**; no bleeding needing embolisation		Approved; mTOR inhibition first line in TSC

137 Malignant Renal Tumours

1 · Overview, epidemiology & aetiology

Most kidney cancers are incidental, so overtreatment is the modern risk.

- Renal cell carcinoma (RCC) makes up 2–3% of adult cancers and is the most lethal common urological cancer. Male to female ratio is **2:1**, peak age 55–75 y, and **4–6% are familial**.
- Over 60% are incidental and ~70% are organ-confined.
- Risk factors: **tobacco** (best accepted; relative risk ~1.4, dose-dependent); **obesity** (the "obesity paradox" of higher BMI but lower stage); **hypertension** (the disease, not its drugs); chronic kidney disease (CKD) and end-stage renal disease (ESRD); family history (2–4×).
- In children and young adults the commonest RCC is **Xp11/TFE3 translocation RCC**. These patients do well with surgery even when node-positive, so do a formal lymph node dissection (LND) at radical nephrectomy (RN).

2 · Classification & pathology

WHO 2022 classification (key groups)

Group	Entities
Renal cell tumours	Clear cell; multilocular cystic neoplasm of low malignant potential; papillary; chromophobe; other oncocytic; RCC not otherwise specified (NOS)
Molecularly defined (new)	TFE3/TFEB-rearranged; ELOC-mutated; fumarate hydratase (FH)-deficient; succinate dehydrogenase (SDH)-deficient; ALK-rearranged; SMARCB1-deficient medullary
Other	Collecting duct; mucinous tubular & spindle cell; tubulocystic; acquired cystic disease-associated; eosinophilic solid & cystic; clear cell papillary tumour
Non-RCC malignant	Urothelial/squamous; sarcomas; nephroblastic; neuroendocrine; lymphoma; **epithelioid angiomyolipoma**; metastases

> **KEY POINT**
>
> Trap: epithelioid angiomyolipoma (AML) is malignant, whereas classic AML, papillary adenoma and oncocytoma are benign. The papillary type 1/2 split was **abandoned** because "type 2" was several diseases (MET-altered, MiT translocation, FH-deficient), so WHO moved to molecular definitions.

Gross features

- RCC has a **pseudocapsule** of compressed parenchyma and is not infiltrative, except collecting duct and sarcomatoid tumours. Cystic degeneration occurs in 10–25% (better prognosis).
- Venous involvement occurs in ~10% and can reach the right atrium.
- Sporadic RCC is bilateral in 2–4% and multicentric in 10–20%; occult satellites are the main cause of ipsilateral recurrence after partial nephrectomy (PN).

Subtype (share)	Genetics	Hallmarks	Behaviour
Clear cell (70–80%)	3p loss/VHL (>90% 3p altered); +5q; PBRM1, SETD2, BAP1	Hypervascular; clear cytoplasm; carbonic anhydrase IX (CA-IX) positive; sarcomatoid 5–8%	Most venous invasion; worse when localised, better when metastatic
Papillary (10–15%)	**Trisomy 7, 17**, loss of Y; MET altered 81%	Hypovascular; CK7, AMACR; multicentric up to 40%	Good (5-y cancer-specific survival (CSS) 86–92%); common in acquired cystic disease
Chromophobe (3–5%)	Losses 1, 2, 6, 10, 13, 17; TP53, PTEN; FLCN in Birt–Hogg–Dubé (BHD)	"Plant cells", perinuclear halo, raisinoid nuclei; diffuse CK7	From intercalated cells; >90% cancer-free at 5 y unless sarcomatoid
Clear cell papillary tumour (<5%)	—	Linear papillae; CK7; **cup-like CA-IX**	Seen in ESRD and VHL; now benign
Collecting duct (Bellini)	Multiple losses	Central, infiltrative, desmoplastic; PAX8 positive; diagnosis of exclusion	Poor; urothelial-like, so may respond to cisplatin/gemcitabine
SMARCB1-deficient medullary	**Sickle cell trait**	Young Black patients; infiltrative	Dismal, median 8–13 mo
FH-deficient	FH (1q42–43)	Macronucleolus with perinucleolar halo; FH loss	Metastasises early
MiT translocation	Xp11.2 TFE3, TFEB	Clear cells with papillae; FISH confirms	40% of paediatric RCC; late nodal recurrence
SDH-deficient	SDH genes	Vacuolated eosinophilic cells; SDHB loss	Young; usually favourable; often germline
ALK-rearranged	ALK (2p23)	Test ALK before calling a tumour "unclassified"	Responds to ALK inhibitors in some
Unclassified (1–5%)	—	Shrinking catch-all	Poor

- **Indolent types:** multilocular cystic, tubulocystic, mucinous tubular & spindle cell, eosinophilic solid & cystic, ELOC and clear cell papillary. **Lethal types:** medullary, FH-deficient, collecting duct, unclassified and sarcomatoid.
- Because chromophobe and oncocytoma are hard to separate, WHO 2022 added "oncocytic renal neoplasm of low malignant potential".
- **Sarcomatoid** change (1–5%) is dedifferentiation of any subtype, with median survival <1 y. Sarcomatoid or rhabdoid features mean automatic grade 4 and must be reported in every RCC.

> **KEY POINT**
>
> Localised clear cell RCC does worse than papillary or chromophobe because it is intrinsically aggressive. Yet metastatic clear cell does better, because VHL-driven angiogenesis and immunogenicity make it respond to tyrosine kinase inhibitors (TKIs) and immune checkpoint inhibitors (ICIs).

WHO/ISUP grading: replaced Fuhrman; validated for clear cell & papillary

Grade	Criterion
1	Nucleoli absent/inconspicuous, basophilic at ×400
2	Nucleoli conspicuous, eosinophilic at ×400; not prominent at ×100
3	Nucleoli conspicuous, eosinophilic at **×100**
4	Extreme pleomorphism, giant cells, **rhabdoid/sarcomatoid**

Nuclear grade is an independent prognostic factor.

3 · Molecular biology & hereditary syndromes

The VHL pathway explains clear cell vascularity and drug targets.

SAQ VHL gene and von Hippel–Lindau disease

Paper answer · P2

Asked as:

- VHL gene. — *Practice Set 2 Paper II Q3f (SAQ)*
- Von Hippel Lindau disease — *Practice Set 8 Paper II Q3c (SAQ)*

Type: SAQ (10 marks)

(Practice Set 8 Paper II Q3c: VHL gene also answered in Practice Set 2 Paper 2, Q3f)

Introduction

- **VHL (von Hippel–Lindau)** is a **tumour suppressor gene on chromosome 3p25**, identified in 1993; three exons, 213-amino-acid protein.
- Germline mutation causes **VHL disease**: an **autosomal dominant** familial cancer syndrome (about 1 in 36,000), penetrance >95% by 65 years.
- It is the familial form of **clear cell RCC** (renal cell carcinoma). Somatic VHL loss occurs in most sporadic clear cell RCC; chromosome 3p is altered in over 90%.

Function / genetics and pathogenesis: the oxygen-sensing pathway

- VHL protein binds elongin B, elongin C, cullin-2 and RBX1 to form (VHL protein forms) an **E3 ubiquitin ligase** that degrades **HIF (hypoxia-inducible factor)** in normoxia.
- **Normoxia:** hydroxylated **HIF -1α and -2α** are tagged for proteasomal destruction.
- **VHL loss (or hypoxia):** loss of VHL stabilises HIF, which (chiefly HIF-2α) accumulates, joins ARNT (aryl hydrocarbon receptor nuclear translocator) and switches on **VEGF (vascular endothelial growth factor)**, PDGF (platelet-derived growth factor), TGF-α (transforming growth factor-α), GLUT1, erythropoietin and CA-IX (carbonic anhydrase IX).
- Hence the **hypervascularity** (**highly vascular** tumours) of clear cell RCC.
- **Knudson two-hit:** one germline plus one somatic hit, so VHL tumours are **early, bilateral and multifocal**.

VHL disease manifestations

Lesion	Frequency
Craniospinal haemangioblastoma (cerebellum commonest)	60–80%
Retinal angioma (earliest)	25–60%

Lesion	Frequency
Clear cell RCC (mean age 39; commonest cause of death)	25–75%
Renal cysts (commonest renal lesion)	Up to 76%
Phaeochromocytoma	10–25%
Pancreatic cysts / neuroendocrine tumours	45–75% / 11–17%
Epididymal cystadenoma	25–60% of men
- Also endolymphatic sac tumour.	
- **RCC is now the commonest cause of death.**	

Genotype–phenotype

Type	Mutation	Phaeochromocytoma	RCC
1	Deletions, truncations	Low	High
2A	Missense	High	Low
2B	Missense	High	High
2C	Missense	Phaeochromocytoma only	Absent

Diagnosis

- Family history plus one lesion; otherwise ≥2 haemangioblastomas, or 1 plus a visceral lesion. Genetic testing of relatives.

Surveillance (VHL Alliance 2020)

- Eye check (examination) yearly from infancy; metanephrines yearly from 5 years; CNS (brain and spine) MRI every 2 years from 11; **abdominal MRI every 2 years from 15**.

Clinical implications / urological management

- **3 cm rule:** observe solid renal tumours <3 cm; intervene when largest reaches 3 cm; then nephron-sparing surgery (enucleation) or ablation, after excluding phaeochromocytoma.
- **Nephron-sparing surgery** (enucleation) is standard; not curative: local recurrence 75–85%, yet 10-year disease-specific survival 81–94%. Ablation for repeated small tumours.
- Exclude phaeochromocytoma before any surgery (normetanephrine raised; α-blockade). **Partial (cortical-sparing) adrenalectomy.**
- **Targeted therapy:** VEGF TKIs (tyrosine kinase inhibitors), bevacizumab, mTOR (mammalian target of rapamycin) inhibitors.
- **Belzutifan** (HIF-2α inhibitor): FDA (US Food and Drug Administration) approved 2021 for VHL disease; renal response 30–50% (another source: responses in about half of renal tumours); anaemia.
- **Also know:** PBRM1, BAP1, SETD2 on 3p.

Diagram to draw: VHL–HIF pathway in normoxia versus (vs) VHL loss, with drug targets.

> **Key points to remember:**
>
> - AD; 3p25 tumour suppressor; VHL protein degrades HIF.
> - VHL loss: HIF-2α drives (→) VEGF, giving vascular clear cell RCC.
> - Type 2 = phaeochromocytoma; 3 cm rule for nephron sparing (nephron-sparing surgery).
> - Belzutifan (HIF-2α inhibitor) approved 2021.

*See the paper answer **Familial / hereditary urological malignancies** (Section 2, Ch 25).*

Also, from the condensed book (not in the paper answer):

- BAP1 mutation (up to 14% of sporadic clear cell) gives high grade and rapid growth, so these tumours are unsuitable for active surveillance (AS).
- Loss of **CDKN2A** (9p) carries a poor prognosis in all subtypes.

ALGORITHM

1. Normoxia: the VHL protein joins elongin B/C, cullin-2 and RBX1 to form an E3 ligase, which degrades hypoxia-inducible factor (HIF)-1α/2α.
2. If VHL is lost (or in hypoxia), **HIF-2α** accumulates and pairs with ARNT. This raises VEGF, PDGF, TGF-α, GLUT1, erythropoietin and CA-IX → hypervascular clear cell RCC.

Drug targets: VEGF ligand (bevacizumab); VEGF-receptor TKIs (sunitinib, pazopanib, axitinib, cabozantinib); mTOR (temsirolimus, everolimus); HIF-2α (**belzutifan**, FDA-approved in 2021 for VHL, ~50% response); CTLA-4 (ipilimumab); PD-1 (nivolumab, pembrolizumab); PD-L1 (atezolizumab, avelumab).

Syndrome (beyond topics)	Gene (chr)	Renal tumour	Other / management
Hereditary leiomyomatosis and RCC (HLRCC)	FH (1q42–43)	Aggressive, often **solitary**	Uterine leiomyosarcoma; **prompt excision at any size, wide margins**; PN only if a wide margin is feasible; FDG-PET finds small metastases
SDH-RCC	SDHA/B/C/D, SDHAF2	Onset ≤40 y	Paraganglioma/phaeochromocytoma, GIST; excision, PN preferred
Cowden	PTEN (10q23)	Mostly papillary (lifetime 34%)	Breast, thyroid; 3-cm rule
Tuberous sclerosis	TSC1/TSC2	AMLs; clear cell 1–3%	AML: surveil <3 cm, everolimus 3–5 cm, embolise or excise ≥5 cm
BAP1	BAP1 (3p21)	High-grade clear cell	Uveal/cutaneous melanoma, mesothelioma; nephron-sparing excision
MiTF-associated	MiTF p.E318K	RCC >5×	Melanoma; management undefined

4 · Clinical presentation

- The **classic triad** (flank pain, gross haematuria, mass) is rare and means advanced disease. Other clues are weight loss, bone pain, a **non-reducing or right-sided varicocele**, and leg oedema from inferior vena cava (IVC) obstruction.
- Over 50% of spontaneous perirenal haematomas are due to tumour, so repeat CT later.

Paraneoplastic syndromes (10–20%)

Syndrome	Frequency / mechanism	Key point
Raised erythrocyte sedimentation rate (ESR)	Commonest	Marker of advanced disease

Syndrome	Frequency / mechanism	Key point
Hypercalcaemia	Up to 13%; parathyroid hormone-related protein (PTHrP) or osteolytic metastases	**The only one managed medically**: hydration, then furosemide; zoledronic acid 4 mg IV 4-weekly (withhold in renal insufficiency); denosumab if refractory; steroids, calcitonin. Definitive: nephrectomy with or without metastasectomy
Hypertension	Renin, artery compression, arteriovenous fistula	—
Polycythaemia	Erythropoietin	—
Stauffer syndrome	3–20%; interleukin-6	Non-metastatic liver dysfunction (ALP raised almost always); exclude liver metastases; normalises after nephrectomy in 60–70%; persistence means viable tumour and poor prognosis
Others	Rare	Cushing, hyperglycaemia, galactorrhoea, neuromyopathy, clotting disorders

- Paraneoplastic syndromes resolve only with excision or systemic therapy, so persistence after nephrectomy means residual disease.

5 · Diagnosis

Imaging sorts a mass into cyst, fat or enhancing tumour.

LAQ Radiology for solid renal masses.

Paper answer · P4

Type: LAQ (25 marks)
Source: 2018-Summer Paper 4, Q1

Introduction

- Most renal masses are now found incidentally. Imaging is the **strongest predictor of malignancy**; size and **enhancement** are the two key variables.
- About 20% of solid enhancing cT1 masses are benign (oncocytoma, fat-poor AML (angiomyolipoma)).
- **Goals:** solid or cystic? enhancing? fat? subtype? stage and surgical anatomy.

A. CT (computed tomography): the standard test

Renal mass protocol (dedicated, thin slices):

Phase	Timing	Use
Non-contrast	–	Baseline HU (Hounsfield units), fat, calcification, haemorrhage
Corticomedullary	30–70 s	Hypervascular masses; arteries; clear cell vs other subtypes
Nephrographic	90–120 s	**Most sensitive** for detection (tumour enhances less than parenchyma); renal vein and IVC (inferior vena cava) thrombus
Excretory	7–10 min	Collecting system relation

Enhancement thresholds (rise from unenhanced value):

Rise	Meaning
≥20 HU	Enhancing: solid tumour
10–20 HU	Equivocal (MRI, magnetic resonance imaging, or CEUS)
<10 HU	Non-enhancing: cyst

- **Pitfall:** pseudoenhancement of small (<10–15 mm) cysts; dual-energy CT iodine maps help.
- **Macroscopic fat = AML:** fat ≤ −10 to −20 HU on unenhanced CT is diagnostic. Fat **with calcification** suggests RCC (renal cell carcinoma).
- **Fat-poor AML:** hyperdense on unenhanced CT, homogeneous, avid early enhancement and washout.
- **Subtype clues:** clear cell RCC enhances early, avidly and heterogeneously (necrosis); papillary RCC is homogeneous and **hypoenhancing**; chromophobe and oncocytoma intermediate, with a central scar. Overlap is large.

B. MRI

- **Indications:** contrast allergy, CKD (chronic kidney disease), equivocal CT enhancement, young patients, thrombus. Enhancement = signal rise ≥15–20%. Macrocyclic gadolinium carries very low NSF (nephrogenic systemic fibrosis) risk.

Feature	Clear cell	Papillary	Chromophobe / oncocytoma	AML
T2 signal	High, heterogeneous	**Low**, homogeneous (haemosiderin)	Central T2-bright scar	Fat signal
Enhancement	Avid (>200%)	Weak	Intermediate / avid	Avid
Diffusion (ADC, apparent diffusion coefficient)	High	Lowest	Intermediate	–
Chemical shift	**Signal drop** (intracellular lipid, ~59%)	No drop	–	**India-ink** artefact; loss on fat saturation

- **ccLS (clear cell likelihood score, 1–5):** structured MRI algorithm for solid cT1a masses using T2 signal, corticomedullary enhancement and microscopic fat, plus diffusion and **segmental enhancement inversion** (favours oncocytoma). Score 4–5 suggests clear cell RCC; 1–2 a non-clear-cell tumour. It guides biopsy versus surveillance.

C. Ultrasound and CEUS (contrast-enhanced ultrasound)

- Grey-scale: cyst versus solid; poor below 2 cm. AML is hyperechoic, but so are some small RCCs.
- **CEUS** (microbubbles, not nephrotoxic): shows enhancement; used in CKD or allergy, indeterminate lesions, ablation and surveillance.

D. Cystic masses: Bosniak 2019 (CT and MRI)

Class	Key feature	Malignancy / action
I	Simple, thin wall ≤2 mm, no septa	Benign; none
II	1–3 thin septa; homogeneous hyperdense (≥70 HU unenhanced)	Benign; none

Class	Key feature	Malignancy / action
IIF	Wall or septa 3 mm, or ≥4 thin septa	<10%; follow 6, 12 months, then yearly for 5 years
III	Wall or septa ≥4 mm, or irregular	~50%; surgery or surveillance
IV	**Enhancing nodule**	~90%; treat

E. Functional and molecular imaging

- **99mTc-sestamibi SPECT/CT** (single-photon emission CT): mitochondria-rich **oncocytoma** and hybrid oncocytic or chromophobe tumours take it up; clear cell RCC does not.
- **Girentuximab PET (positron emission tomography)** targets CA-IX (carbonic anhydrase IX) on clear cell RCC. **ZIRCON** (^{89}Zr-girentuximab, indeterminate cT1 masses): about 85% sensitivity and 87% specificity for clear cell RCC.
- **FDG-PET** (fluorodeoxyglucose PET): not recommended (low sensitivity). PSMA (prostate-specific membrane antigen) PET: experimental.

F. Renal mass biopsy

- **Indications:** result may change management; suspected lymphoma, metastasis or infection; before ablation or surveillance; metastatic disease. Not needed if surgery is planned regardless.
- Cores: PPV (positive predictive value) 98.8%; subtype accuracy ≥90%, grade 60–80%; non-diagnostic ~14%. Seeding almost nil. Oncocytoma cannot be confirmed on cores.

G. Staging and surgical planning

- **Venous thrombus:** CT sensitivity 78% (renal vein), 96% (IVC). MRI or multiplanar CT shows the cephalad tip; enhancing thrombus is tumour, non-enhancing is bland.

Mayo level	Extent	T stage
0	Renal vein	T3a
I	IVC ≤2 cm above renal vein	T3b
II	IVC below hepatic veins	T3b
III	Intrahepatic IVC, below diaphragm	T3b
IV	Above diaphragm or atrium	T3c

- Nodes ≥2 cm usually malignant. Chest CT; bone scan only if symptomatic.
- **Nephrometry:** R.E.N.A.L. (Radius, Exophytic/endophytic, Nearness to collecting system, Anterior/posterior, Location), PADUA, C-index: predict complexity and complications; guide partial versus radical nephrectomy.
- **After ablation:** persistent or new enhancement = residual or recurrence.
- **Recent:** radiomics and AI (artificial intelligence).

Diagram to draw: algorithm: incidental mass → renal-protocol CT/MRI → fat (AML) / cyst (Bosniak) / enhancing solid → biopsy or treat.

Key points to remember:

- Enhancement ≥20 HU = solid tumour; macroscopic fat = AML.
- MRI: clear cell T2-bright with chemical-shift drop; papillary T2-dark and hypoenhancing; ccLS 1–5.
- Sestamibi for oncocytoma; ^{89}Zr-girentuximab PET (ZIRCON) for clear cell.
- Biopsy PPV ~99%, non-diagnostic ~14%.

Also, from the condensed book (not in the paper answer):

- The strongest predictors of malignancy are **male sex (~3×)** and **size**: 20–30% of masses ≤4 cm and ~40% of ≤2 cm are benign, while few ≥7 cm are benign.
- A cT1 mass is ~20% benign, ~60% indolent and ~20% potentially aggressive.
- Renal mass biopsy (RMB): **multiple cores over fine-needle aspiration**; haematoma ~5%, specificity 94%. A non-malignant result does not prove the mass is benign.

- **Multiphase CT** is the single most important test: enhancement **>15–20 Hounsfield units (HU)** suggests RCC, and fat **<−20 HU** means AML. MRI is best for IVC thrombus and organ invasion.

ALGORITHM

1. If simple cyst, then observe.
2. If complex or solid: eGFR >45 → CT or MRI with/without contrast; eGFR 30–45 → hydrate, then the same; eGFR <30 → non-contrast CT or gadolinium MRI; contrast allergy → ultrasound (USG), contrast USG or MRI.
3. Then: fat → AML; complex cyst → Bosniak; solid enhancing → management options, with or without RMB.

- **Bosniak** (see Ch 136): malignancy is 3–10% in IIF, ~50% in III and 75–90% in IV. The key discriminator is unequivocal **measurable enhancement**. Avoid ablation and biopsy of cystic masses; AS is acceptable for III/IV.
- Do RMB when the result may change management, or when lymphoma, metastasis or infection is suspected. Do not biopsy patients who will be managed the same way regardless. Biopsy before ablation.
- **Screening:** there is none for the population or for autosomal dominant polycystic kidney disease. In ESRD, screen with USG or CT from the 3rd year of dialysis if fit, and continue after transplant.
- Offer genetic counselling for RCC at **≤46 y**, multifocal or bilateral tumours, or a suggestive history or pathology.

6 · Staging & prognosis

Pathological stage is the strongest predictor of outcome.

TNM (AJCC 8th)	Definition	5-y survival
T1a / T1b	≤4 / >4–7 cm, confined	95–100 / 85–95%

TNM (AJCC 8th)	Definition	5-y survival
T2a / T2b	>7–10 / >10 cm	65–85 / 55–75%
T3a	Renal vein/segmental branches, **pelvicalyceal system**, perirenal/sinus fat, within Gerota's	45–75%
T3b	IVC below diaphragm	40–55%
T3c	IVC above diaphragm or **caval wall invasion**	20–45%
T4	Beyond Gerota's, incl. **contiguous ipsilateral adrenal**	0–35%
N1 / M1	Regional node / distant (non-contiguous adrenal = M1)	0–30%

Stage groups: I is T1N0; II is T2N0; III is T1–2N1 or T3 any N; IV is T4 or M1. The 8th edition moved pelvicalyceal invasion to T3a.

> **KEY POINT**
>
> TNM limitation: stage is set by the single worst feature, so co-existing adverse features (e.g. adrenal invasion plus a supradiaphragmatic thrombus) are lost. Record all of them and use integrated nomograms.

- **Work-up:** abdominal CT and chest X-ray (chest CT if symptoms or high risk). Bone scan only with raised ALP, bone pain or poor performance status (PS); no PET. Treat an abnormal adrenal as involved.
- **Pathological stage** is the most important factor, then grade, size, subtype, necrosis and sarcomatoid change. Node-positive 5-y survival is 5–30%, and with a thrombus but otherwise confined disease it is 45–69%.
- Venous wall invasion is as important as thrombus level, and collecting-system invasion is adverse.
- Nomograms: **UISS** uses TNM, grade and ECOG PS; **SSIGN** (Mayo) uses Stage, Size, Grade and Necrosis; **Leibovich** predicts recurrence in clear cell RCC.

7 · Management of localised RCC

Every option trades cancer control against kidney function.

LAQ Nephron-sparing surgery for renal masses: indications, approaches and recent advances

Paper answer · P1 · P2 · P3 · P4 · merged from 4

Asked as: P1 LAQ "Embryology and anatomy of kidney; nephron-sparing surgery in small renal mass and renal malignancy" · P2 LAQ "Nephron-sparing surgery / partial nephrectomy in RCC: role, indications and approaches" · P3 LAQ "Nephron-sparing surgery for renal masses: methods, recent advances and neoadjuvant therapy" · P4 SAQ "Nephron Saving Surgery (NSS)"

"Nephron sparing nephrectomy" and "nephron saving surgery" are read as nephron-sparing surgery (NSS): partial nephrectomy and ablation. "Nepron" is read as nephron.

Embryology of the kidney

- Three successive kidneys form from **intermediate mesoderm**, in cranial-to-caudal order:

Kidney	Time and site	Fate
Pronephros	Week 3–4, cervical	Non-functional; regresses
Mesonephros	Week 4, thoracolumbar	Functions briefly; tubules become efferent ductules; **mesonephric (Wolffian) duct** persists
Metanephros	Week 5, sacral	**Definitive kidney**

- **Reciprocal induction:** the **ureteric bud (UB)** grows from the caudal Wolffian duct (about week 5) into the **metanephric blastema**.
- UB branches into ureter, pelvis, calyces and collecting ducts.
- Blastema, induced by UB tips, forms nephrons (glomerulus to distal tubule).
- **Molecular control:** GDNF (glial cell line-derived neurotrophic factor) from the blastema acts on RET on the UB; SLIT2/ROBO2 restricts GDNF to one bud. Also WT1, PAX2.
- Urine production is established by 14 weeks; nephrogenesis ends before birth.
- **Ascent and rotation (weeks 6–9):** the kidney rises from the pelvis to the lumbar region and rotates 90° so the hilum faces medially. It takes successively higher arteries; persistent lower ones become accessory arteries.
- **Clinical correlation:** no UB gives agenesis; abnormal induction gives MCDK (multicystic dysplastic kidney); failed ascent gives ectopia; fused lower poles trapped by the IMA (inferior mesenteric artery) give horseshoe kidney; also malrotation, duplication.

Surgical anatomy of the kidney

- **Size and position:** about 11 cm, 125–170 g; T12–L3, lies on the cone-shaped psoas, so the upper pole is more medial and posterior, and the kidney is angled 30–50° behind the coronal plane. The right kidney is 1–2 cm lower (liver).
- **Relations:**

Aspect	Right	Left
Superior	Liver, adrenal, 12th rib	Spleen, stomach, adrenal, 11th–12th ribs
Medial/anterior	Duodenum (2nd part), hepatic flexure	Pancreatic tail with splenic vessels, splenic flexure, jejunum
Posterior (both)	Diaphragm, psoas, quadratus lumborum, subcostal nerve	Same

- **Coverings (inside out):** true capsule, perirenal fat, **Gerota fascia**, pararenal fat. Gerota fascia is closed above and laterally but open below. So it contains tumour and abscess (collections), yet fluid tracks into the pelvis.
- **Hilum (front to back):** renal vein, artery, pelvis, then the posterior segmental artery.
- **Vascular anatomy (the basis of partial nephrectomy)**
- **Arteries:** renal artery from the aorta at L1–L2; the right passes behind the IVC (inferior vena cava).
- **Five segmental end arteries**: posterior (first branch, behind the pelvis) plus four from the anterior division (apical, upper, middle, lower). With no anastomoses, occlusion causes segmental infarction, but each segment can be clamped or resected separately. Then interlobar (columns of Bertin), arcuate, interlobular arteries.
- **Accessory arteries** in 25–28%, end arteries too.

- **Brödel line:** an avascular plane between anterior (two-thirds) and posterior (one-third) territories, just behind the lateral border. Incision here bleeds least (anatrophic nephrolithotomy).
- **Veins anastomose freely**, so a segmental vein can be ligated safely. The long left renal vein receives adrenal, gonadal and lumbar veins.
- **Parenchyma / internal structure:** cortex and 8–18 pyramids; 0.4–1.2 million nephrons per kidney.
- **Collecting system:** papilla, minor calyces (7–9), major calyces, pelvis. Puncture through the **calyceal fornix**, not the infundibulum (Sampaio).
- **Also know:** lymph drains to para-aortic (left) or paracaval/interaortocaval nodes (right); sympathetic T8–L1.

Small renal mass (SRM)

- **Definition:** an enhancing renal mass **≤4 cm (cT1a)**, mostly incidental.
- **Biology:** 20–30% of masses ≤4 cm and about 40% ≤2 cm are benign (oncocytoma, fat-poor angiomyolipoma) (P2: about **20%** of enhancing cT1 masses are benign); most cancers are indolent (<2% of the cancers metastasise). About 20% benign, 60% indolent, 20% potentially aggressive.

Definition and rationale of NSS

- NSS (nephron-sparing surgery) is complete local removal of the tumour with **maximal preservation of functioning parenchyma**. PN (partial nephrectomy) is the standard form; thermal ablation is the non-excisional form.
- **Why spare nephrons:** RN (radical nephrectomy) causes CKD (chronic kidney disease), which raises cardiovascular events and death. PN gives equal cancer control for small masses (cT1a) with better function and keeps about **90%** of global function (RN 65–70%).
- **CKD from surgery alone** is stable, with near-normal survival if the new GFR (glomerular filtration rate) stays >45; **CKD from medical disease** (diabetes, hypertension) progresses. So NSS matters most when medical renal disease coexists.
- **Concept:** function after PN depends mainly on the **parenchyma preserved** (preserved parenchymal volume), less on warm ischaemia time.

Indications for nephron-sparing surgery

Category	Situations
Imperative (absolute)	Solitary (functioning) kidney; bilateral tumours (bilateral RCC, renal cell carcinoma); RN would mean dialysis
Relative	Contralateral kidney threatened: stones, renal artery stenosis, hydronephrosis, diabetes, (severe) hypertension, proteinuria; hereditary RCC
Elective	Localised tumour with a normal contralateral kidney; **cT1a (≤4 cm) with normal contralateral kidney: standard of care**; cT1b–T2 when feasible
Benign	Symptomatic AML (angiomyolipoma); focal atrophy in duplex kidney; infected calyceal diverticulum

- Now extended (increasingly used) for selected cT1b–T2, complex or hilar tumours when function matters (better function).
- **Relative contraindications: <20% of nephron mass retained**; pedicle encasement; collecting-system invasion; **venous tumour thrombus**; cT4; cN1 (nodes).

Role of NSS by clinical setting

Setting	Role of NSS
cT1a (≤4 cm)	PN is the **standard of care** when intervention is needed; RN is overtreatment
cT1b–T2	PN when feasible and function matters; otherwise debatable
Hereditary RCC	Observe until the largest tumour is **3 cm**, then enucleate (except HLRCC [hereditary leiomyomatosis and RCC]: wide excision)
Bilateral tumours, solitary kidney	Imperative PN, staged if bilateral; avoids dialysis
Frail, comorbid, cT1a <3 cm	Thermal ablation or active surveillance
Locally advanced, thrombus, cN1, metastatic	Generally RN or cytoreductive surgery

Choosing the option for a small renal mass (AUA, American Urological Association, 2021)

Option	Role
Active surveillance	<2 cm or predominantly cystic; elderly, comorbid (frail patients), life expectancy <5 years. Imaging every 3–6 months. Intervene if >3 cm or growth >5 mm a year (DISSRM registry data: almost no metastases)
Thermal ablation (cryoablation, radiofrequency)	cT1a **<3 cm**, percutaneous, biopsy first. Higher local recurrence than PN, salvaged by repeat ablation
Partial nephrectomy	**Prioritise PN for cT1a** needing treatment. Imperative: solitary kidney, bilateral tumours, CKD or proteinuria; also familial RCC, young patients
Radical nephrectomy	Preferred only if **all three** apply: high complexity; no CKD or proteinuria; normal contralateral kidney with new (post-RN) eGFR likely **>45**. Otherwise PN

- **PN principles:** negative margin (width does not matter); maximise preserved parenchyma; avoid long warm ischaemia; enucleation for familial or multifocal tumours.

Preoperative evaluation and planning

- Multiphase CT (computed tomography) or MRI (magnetic resonance imaging) with 3D (three-dimensional) reconstruction; chest imaging; eGFR (estimated glomerular filtration rate) and proteinuria (CKD stage).
- **Renal mass biopsy (RMB)** when it may change management, and before (preferred) or at ablation (non-diagnostic about 14%). Genetic counselling if age ≤46, multifocal or bilateral.
- **Nephrometry scores** predict complexity, complications, ischaemia time and conversion:

R.E.N.A.L. component	Scored as
Radius	≤4, 4–7, ≥7 cm
Exophytic/endophytic	≥50% exophytic, <50%, entirely endophytic
Nearness to collecting system	≥7, 4–7, ≤4 mm
Anterior/posterior	Descriptor

R.E.N.A.L. component	Scored as
Location to polar lines	Polar, crossing, central
- Total: **low 4–6, moderate 7–9, high 10–12** complexity. **PADUA** (Preoperative Aspects and Dimensions Used for an Anatomical) and C-index are alternatives. 3D models help (shorten ischaemia in a randomised trial).	

Methods of NSS

Group	Methods
Surgical PN	Open, laparoscopic, robot-assisted (RAPN)
Excision technique	Standard (wedge) resection with a rim; enucleation; polar resection or heminephrectomy (≥30% of parenchyma; large polar tumours)
Thermal ablation	Cryoablation, RFA (radiofrequency ablation), MWA (microwave ablation)
Developmental	SBRT (stereotactic body radiotherapy), HIFU (high-intensity focused ultrasound), irreversible electroporation

Surgical approaches for PN

Approach	Features
Open, extraperitoneal flank (supracostal 11th/12th rib, 8 cm mini-flank)	Posterior exposure; solitary kidney, complex or central tumours, cold ischaemia
Open transperitoneal (subcostal, chevron, midline)	Bilateral or large tumours, vascular reconstruction
Laparoscopic transperitoneal	Large space (more space), easier suturing; anterior tumours
Laparoscopic retroperitoneal	No bowel mobilisation, direct hilum; posterior tumours
Robot-assisted (RAPN)	Easier suturing, shorter warm ischaemia, fewer conversions; transperitoneal for anterior and retroperitoneal for posterior tumours; equal oncology

- Open PN suits complex tumours in a solitary kidney.
- First laparoscopic PN 1993 (Winfield).
- **Robotic vs laparoscopic:** shorter warm ischaemia, fewer conversions. **Robotic vs open:** less blood loss, shorter stay, fewer complications; same margins and eGFR.
- Haemostasis: sliding-clip renorrhaphy, barbed sutures, sealants.

Operative steps of PN

1. Flank extraperitoneal or transperitoneal (open), laparoscopic or robotic approach.
2. **Early hilar control** of artery, vein and accessory vessels, even if off-clamp is planned.
3. Expose and defat the kidney; **intraoperative ultrasound** maps the tumour borders and depth.
4. Vascular control: clamp the artery (± vein); selective or off-clamp for small tumours. Use **ice slush** (cold ischaemia) if ischaemia is expected to be **≥25–30 minutes** (solitary kidney, CKD).
5. Excise with a thin rim of normal parenchyma, or **enucleate** along the pseudocapsule (hereditary, familial, multifocal). **A negative margin is the priority; its width does not**

matter.

6. **Two-layer renorrhaphy** (sliding clip): inner layer closes vessels and collecting system; outer capsular layer over a haemostatic bolster.
7. Unclamp; check haemostasis; close Gerota fascia; drain.

Enucleation vs standard resection

	Standard (wedge) resection	Enucleation
Plane	Through normal parenchyma, thin rim	Along the tumour pseudocapsule
Advantage	Fewer positive margins	Less bleeding, maximal parenchyma, eases zero ischaemia
Use	Default	Hereditary RCC (e.g. VHL [von Hippel–Lindau]), familial, multifocal, solitary kidney, severe CKD

Vascular control and ischaemia

Strategy	Verdict
En bloc or artery-only clamp	Equivalent; traditional warm ischaemia <25–30 minutes; ice slush if longer is expected
Early unclamping (after inner layer)	May reduce pseudoaneurysm
Selective or super-selective segmental clamping	Avoids global ischaemia; no functional gain in a randomised trial
Off-clamp (zero ischaemia)	Small exophytic tumours; more blood loss; no long-term eGFR benefit in randomised trials
Parenchymal compression	Peripheral tumours
Hypothermia, mannitol	No benefit with a normal contralateral kidney

- Function depends mainly on the **quantity of parenchyma preserved**; ischaemia is secondary within accepted limits.
- **Trifecta:** negative margin + no major complication + warm ischaemia ≤25 minutes. RAPN: shorter ischaemia than laparoscopy, less blood loss than open.

Thermal ablation (ablative NSS)

- **Cryoablation** (double freeze–thaw), **RFA**, **MWA**; mostly **percutaneous** (CT or ultrasound guided).
- AUA 2021: alternative for **cT1a <3 cm**, especially elderly, comorbid, solitary kidney or VHL; biopsy before (preferred) or at ablation.
- **Cryoablation:** cell death at ≤−40 °C; iceball 5–10 mm beyond the tumour. **RFA/MWA:** heat causes coagulative necrosis.
- Success = loss of enhancement on CT/MRI. Local recurrence is higher than PN but treatable by repeat ablation (equal after repeat ablation).
- **Also know:** SBRT for non-surgical candidates; active surveillance for frail patients.

Special situations

- **Hereditary RCC (VHL):** observe until the largest tumour is **3 cm**, then enucleate (except HLRCC: wide excision).
- **Bilateral tumours:** bilateral PN, usually staged; if one side needs RN, do the PN first.

- **Solitary kidney:** temporary dialysis about 3.5%; local recurrence 3–5%.
- **Also know:** bilateral Wilms tumour; kidney-sparing upper-tract urothelial cancer surgery.

Role of neoadjuvant therapy

- **Aim:** downsize a tumour so that PN becomes possible, mainly an **imperative indication** (solitary kidney, bilateral tumours) where radical nephrectomy means dialysis.
- **TKIs (tyrosine kinase inhibitors):** pazopanib, sunitinib, axitinib. Tumour volume falls **10–30%**; resection becomes possible in **40–45%**, including PN candidates. Best in clear cell RCC.
- **Toxicity:** serious adverse events about 20%, mainly **wound healing**. Stop TKIs several half-lives before and after surgery.
- **Checkpoint inhibitors:** nivolumab is safe but shrinks little; trials ongoing.
- **Status:** not standard (trials only); **upfront surgery whenever feasible**. Adjuvant pembrolizumab (KEYNOTE-564) suits high-risk resected disease.

Complications of PN

- **Urine leak** (drain, then stent); **haemorrhage, pseudoaneurysm or AV (arteriovenous) fistula** (**delayed haematuria**: angioembolisation); AKI (acute kidney injury), renal insufficiency; temporary dialysis in a solitary kidney.
- **Positive margin 0–7%:** surveillance, not reoperation.
- **Hyperfiltration injury** when >50% of a solitary kidney is lost: proteinuria, FSGS (focal segmental glomerulosclerosis); give an ACE (angiotensin-converting enzyme) inhibitor.

Follow-up

- AUA risk-based imaging; eGFR and urine protein at every visit.

Outcomes and evidence

- Elective PN: **local recurrence 1–2%**, cancer-free survival >90%; imperative 3–5%. Margin width does not matter if negative.
- **EORTC (European Organisation for Research and Treatment of Cancer) 30904 (2011):** RN vs PN for ≤5 cm tumours: PN better function, similar oncology (cancer control); overall survival slightly favoured RN (10-year overall survival 81% RN vs 76% PN), but the trial is flawed.
- **DISSRM (Delayed Intervention and Surveillance for Small Renal Masses) registry:** surveillance almost never led to metastasis; about 24% had delayed intervention.

Recent advances

Area	Advance and evidence
Planning	R.E.N.A.L. and PADUA scores; **3D models**: randomised trial showed shorter operation and ischaemia, less blood loss
Platform	**RAPN:** shorter warm ischaemia than laparoscopic; less blood loss than open. **Single-port robot:** shorter stay, less opioid
Vascular control	Off-clamp, segmental, selective or super-selective clamping (guided by **NIRF (near-infrared fluorescence)** with ICG, indocyanine green), early unclamping. **Randomised trials: no long-term eGFR gain**
Excision	**Enucleation** along the pseudocapsule: less blood loss; hereditary, multifocal, solitary kidney

Area	Advance and evidence
Reconstruction	Sliding-clip and **barbed suture** renorrhaphy; single-layer gives better function, fewer pseudoaneurysms; sutureless with sealants in selected
Neoadjuvant	TKI downsizing in imperative cases (see above)
Outcome reporting	**Trifecta:** negative margin + no major complication + warm ischaemia ≤25 minutes

Diagrams to draw: segmental arteries with the Brödel line on a transverse kidney section; clamped renal artery, tumour excised with a rim, two-layer renorrhaphy; RAPN with selective segmental clamp and renorrhaphy.

Key points to remember:

- Ureteric bud gives collecting system; blastema gives nephrons (GDNF–RET).
- Five segmental end arteries; veins anastomose; Brödel line avascular; puncture at the fornix.
- SRM ≤4 cm: 20–30% benign; PN is standard for cT1a, ablation <3 cm (after biopsy), surveillance <2 cm or frail.
- Imperative, relative, elective; elective PN is standard for cT1a.
- AUA: RN only if complex + no CKD + new GFR likely >45.
- R.E.N.A.L. 4–6 / 7–9 / 10–12.
- Retroperitoneal for posterior, transperitoneal for anterior tumours.
- Preserved parenchyma, not clamp time, governs function; trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
- Enucleation for hereditary/multifocal.
- Off-clamp and super-selective clamping: no long-term eGFR gain in randomised trials; 3D models: randomised benefit.
- EORTC 30904: PN better function, similar cancer control.
- Neoadjuvant TKI shrinks tumours 10–30%; for imperative PN only.

Also, from the condensed book (not in the paper answer):

- AUA 2021: refer to nephrology if GFR <45, confirmed proteinuria, diabetic CKD, or expected post-op GFR <30. Do LND for clinically positive nodes and adrenalectomy only if suspicious.
- CKD follows RN in 65% vs PN in 20%; surgical CKD progresses ~0.7%/y vs medical CKD ~4.7%/y. **Predicted GFR after RN ≈ global GFR × contralateral split function × 1.25**.
- **Triggers to intervene on AS**: >3 cm, growth >5 mm/y, stage progression, adverse biopsy, symptoms.

- On AS, small masses grow 0.1–0.3 cm/y and metastasise in <2%; image every 3–6 mo.
- PN function depends first on parenchymal quality, then on the **amount of vascularised parenchyma preserved (the main modifiable factor)**, and only secondarily on ischaemia.
- Leak and bleeding are commoner after PN than RN (steps: Ch 142).

- In a solitary kidney at least 20–30% of the kidney is needed. If bilateral tumours make PN impossible, options are RN with dialysis or neoadjuvant TKI.
- Enucleation is controversial in sporadic RCC because 40–50% infiltrate the pseudocapsule.

Radical nephrectomy

- Robson's principles (early vascular ligation, removal outside Gerota's, adrenalectomy, extended LND) are now modified: perifascial dissection is still essential (~25% of cT1b/T2 invade fat), but there is **no routine adrenalectomy** (unless enlarged on imaging or extensive tumour) and **no routine extended LND**.
- Laparoscopic RN is standard without gross invasion, major venous involvement or bulky nodes; open surgery uses a subcostal or thoracoabdominal approach (see Ch 141).

Thermal ablation (see Ch 143)

- Thermal ablation (TA) works best <2.5–3 cm and gives no pathological staging. Cryoablation is lethal at ≤−20 °C with a double freeze–thaw; neither cryoablation nor radiofrequency is superior.
- Recurrence is mostly salvaged by repeat TA, and nodular enhancement means recurrence.

Familial RCC: 3-cm rule (National Cancer Institute)

- VHL tumours <3 cm had no metastases at ~5 y, while ~25% metastasise when larger. **HLRCC is the exception**.
- Recurrence is near-universal, so surveillance is lifelong. **Exclude phaeochromocytoma** and central nervous system haemangioblastoma before surgery.

8 · Locally advanced RCC

Even with a caval thrombus, complete surgery can cure.

IVC tumour thrombus (venous involvement 4–10%; cure 45–70%)

Level	Extent	T	Vascular control
Renal vein	Vein/branches	T3a	Standard RN
I	At renal vein ostium	T3b	Satinsky clamp/stapler; keep ≥50% lumen; patch if needed
II	Below hepatic veins	T3b	Clamp infrarenal IVC, then contralateral renal vein, then suprarenal IVC; occlude lumbars
III	Intrahepatic, below diaphragm	T3b	Mobilise liver and milk thrombus below hepatic veins; or suprahepatic clamp with Pringle; venovenous bypass often
IV	Above diaphragm	T3c	Cardiopulmonary bypass with or without hypothermic arrest; avoid opening the heart if thrombus can be brought below the atrium
Any, wall invasion	Caval wall	T3c	Resect to negative margins; graft/reconstruct

- Image close to surgery, because the thrombus grows. **No routine renal artery embolisation** (more deaths). Do coronary angiography before bypass (levels III–IV) and use intraoperative transoesophageal echocardiography.
- Level and wall invasion predict survival, and level IV without metastasis is curable. Mortality is 5–10% for supradiaphragmatic thrombus.

Locally invasive (T4)

- Only ~40% of suspected invasion is confirmed. En bloc resection is the only cure, but prognosis is poor.

Lymph node dissection

- Only 30–43% of enlarged nodes are malignant. With ≥2 risk factors (≥10 cm, cT3/T4, grade 3–4, sarcomatoid change, necrosis), >40% are node-positive.
- LND gave no survival benefit (EORTC 30881), so it is done for suspicious nodes, mainly for staging.

9 · Recurrence, adjuvant therapy & follow-up

A third of recurrences come late; one adjuvant drug works.

Local recurrence	After RN	After PN	After TA
Rate	2–4%	1.4–10%	Cryo 3–10%, radiofrequency ablation 5–20%
Site	Fossa, adrenal, vein stump, nodes	Remnant, usually away from bed	Ablation zone
Management	Only 20–40% isolated, so do a full work-up; en bloc resection gives 30–50% long-term cancer-free	Repeat PN, completion nephrectomy, TA, AS	Repeat TA, AS, salvage surgery

- A positive PN margin is usually surveilled; completion nephrectomy is only for gross involvement.
- About 20–35% metastasise after "curative" surgery, but cytokines, vaccines and most TKIs failed as adjuvants. **Adjuvant pembrolizumab** is FDA-approved for high risk.

Follow-up (AUA risk groups)

Risk	Definition	Chest imaging
Low	pT1 grade 1–2	X-ray
Intermediate	pT1 grade 3–4 or pT2	X-ray
High	pT3	CT
Very high	pT4, pN1, sarcomatoid/rhabdoid, gross positive margin	CT

- Imaging starts at 3–6 mo and intensifies with risk; very high risk is followed for up to 10 y. A microscopic positive margin moves the patient up one level. **30% of recurrences occur after 5 y**.
- Benign pathology needs no imaging. After TA, do contrast imaging within 6 mo, then follow the intermediate schedule.

10 · Other renal malignancies

Lymphoma and metastases need biopsy and drugs, not nephrectomy.

Tumour	Clues	Management
Sarcoma (1–2%)	**Leiomyosarcoma commonest** (50–60%); capsule/sinus origin, huge, no nodes, hypovascular	Wide en bloc excision, negative margins; the first resection is the best chance; margin and grade give prognosis
Lymphoma	Non-Hodgkin; multiple masses (45%); massive adenopathy, splenomegaly; prior lymphoma means recurrence until proven otherwise	**Biopsy, avoid nephrectomy**; R-CHOP with or without radiotherapy
Metastases	Commonest renal malignancy (12% at autopsy); lung primary commonest; multifocal, 5–30 HU	Biopsy; systemic therapy; nephrectomy only for refractory bleeding
Carcinoid	**Horseshoe kidney (82×)**	Excision; good prognosis
Small cell	Usually advanced	Surgery and platinum
Primitive neuroectodermal tumour	Ewing family; CD99; t(11;22)	Multimodal; poor
Adult Wilms	3% of Wilms; triphasic	Paediatric protocols; worse than in children

- In a suspected sarcoma, any RCC feature excludes it (the differential is sarcomatoid RCC).

> **KEY POINT**
>
> Non-RCC malignancies are almost all **hypovascular**. The key decision is surgical (sarcoma, carcinoid) vs medical (lymphoma, metastasis: biopsy, don't resect). RCC nodes are interaortocaval (right) or para-aortic (left), so nodes elsewhere suggest lymphoma.

11 · Landmark trials

Trial	Design	Result	Take-home
EORTC 30904	Randomised trial (RCT), RN vs elective PN, ≤5 cm, normal contralateral kidney	PN better function; 10-y overall survival (OS) better with RN (81 vs 76%)	Flawed; PN's functional gain may be smaller than believed
EORTC 30881	RCT, RN with or without LND	No survival benefit	Underpowered, low-risk (4% node-positive); LND for staging
S-TRAC	Sunitinib vs placebo, high-risk clear cell	Disease-free survival (DFS) 6.8 vs 5.6 y; no OS gain	Only positive adjuvant TKI trial
KEYNOTE-564	Pembrolizumab vs placebo, high-risk	DFS hazard ratio 0.63	FDA 2021; only positive adjuvant ICI trial
DISSRM	Registry, AS vs intervention, small masses	1 death/metastasis on AS; 24% delayed intervention	AS safe for small masses

138 Urothelial Tumours of the Upper Urinary Tract and Ureter

1 · Epidemiology & risk factors

UTUC is often invasive at diagnosis, and its causes act as a field effect.

LAQ Upper tract urothelial carcinoma

Paper answer · P2 · P3 · merged from 2

Asked as: P3 LAQ "Upper Tract Urothelial Carcinoma (including TCC of Ureter): Diagnosis, Staging, Minimally Invasive, Endoscopic and Radical Management" · P2 LAQ "Etiopathogenesis and management of Upper tract TCC"

TCC (transitional cell carcinoma), including TCC of the ureter, is taken as UTUC (upper tract urothelial carcinoma).

Introduction and epidemiology

- **UTUC** is 5–10% of urothelial cancers. Pelvicalyceal tumours are about twice as common as ureteric ones. Incidence is rising, **driven by ureteric tumours**.
- Peak age 70–90; male:female 2:1.
- About 60% are invasive at diagnosis, because the upper tract has a thin muscle wall. Concurrent bladder cancer in 17%.
- Bladder recurrence follows in 22–47%, mostly from downstream seeding.
- **Site (ureter): distal ureter 70%, mid 25%, proximal 5%**, explained by downstream implantation.
- Standard is **RNU (radical nephroureterectomy) with bladder cuff**; kidney-sparing versus radical surgery is chosen by **risk**.
- URS (ureteroscopy) biopsy is small and shallow, so it **cannot stage**. Stage is inferred from grade, endoscopic appearance, imaging and cytology together.

Aetiology (etiopathogenesis)

Factor	Key facts and mechanism
Tobacco	Commonest and most important modifiable factor; risk 2.5–7-fold. Aromatic amines are activated to mutagenic N-hydroxylamines
Aristolochic acid	Balkan endemic nephropathy (UTUC 40% of renal tumours there) and Chinese herb nephropathy. Aristolactam–DNA adducts cause A:T to T:A transversions in TP53. Familial but **not inherited**; often bilateral
Lynch syndrome (HNPCC, hereditary nonpolyposis colorectal cancer)	Germline MMR (mismatch repair) mutation: MLH1, MSH2, MSH6, PMS2. Risk ~14-fold, highest with MSH2; found in 5–9% of screened UTUC
Occupation	"Amino tumours": aniline dyes, benzidine, β-naphthylamine (dyes, rubber, petrochemicals); latency up to 20 years
Analgesic abuse	Phenacetin: papillary necrosis; capillarosclerosis is pathognomonic; history in 11% of ureteric tumours
Arsenic	Well water in south-west Taiwan (blackfoot disease)

- **Also know:** cyclophosphamide (acrolein); chronic infection and stones (squamous carcinoma); alcohol.
- **Lynch screening:** suspect if young (age <60), or a personal or family history of Lynch-spectrum cancer (colon, endometrium). Test the tumour for MSI (microsatellite instability) or MMR immunohistochemistry, then germline testing.

Pathogenesis

- Two theories both operate. **Intraluminal seeding** (one clone spreading downstream) explains distal predominance and frequent bladder recurrence (22–47%). **Field effect / field change** (carcinogen in urine, independent clones) explains multifocality.
- Sequence: hyperplasia, dysplasia, CIS (carcinoma in situ). Variant histology in ~25%. Molecular subtypes include FGFR3-mutant (low grade), TP53/MDM2-mutant and hypermutated (Lynch).

Pathology and spread

- 85% papillary, 15% sessile; WHO (World Health Organization) 2004/2016 low or high grade. Ureteric tumours are low grade and low stage in 55–75%.
- **Why ureteric tumours do worse:** the renal parenchyma is a barrier for pelvic tumours, but periureteric fat holds a rich lymphovascular plexus. At equal stage, ureteric tumours have more local and distant failure.
- Spread: direct, lymphatic (**mid and distal ureter to pelvic nodes**; upper ureter to hilar and retroperitoneal), blood (lung, liver, bone).

Clinical features

- **Haematuria** (56–98%), **flank pain** (20–32%, dull from hydronephrosis/obstruction or colic from clots); lumbar mass about 10%; incidental about 15%.
- Weight loss, bone pain or a mass suggest advanced disease.

Diagnosis (investigations)

1. Cystoscopy: always, to exclude coexisting bladder cancer (concurrent in 17%). Positive cytology with a normal bladder and urethra means the upper tracts must be examined.

2. Imaging

Test	Role and key points
CTU (computed tomography urography)	First line; best test; highest accuracy (sensitivity ~92%, specificity ~95%). Radiolucent **filling defect**, hydronephrosis, ureteric wall thickening, non-visualisation. Tumour ~46 HU (Hounsfield units) versus uric acid stone ~100 HU. Misses flat lesions (CIS)
MRU (magnetic resonance urography)	If contrast or radiation is contraindicated; sensitivity ~75% for tumours <2 cm
Retrograde pyelography	Poorly filled calyces; accuracy ~75%; done with ureteroscopy. **Goblet (champagne-glass) sign**, the ureter dilated just below the tumour; **Bergman's sign**, the catheter coiling below it
FDG-PET/CT (fluorodeoxyglucose positron emission tomography)	Nodes and metastases when contrast is contraindicated
Renogram	Split function; contralateral kidney must be assessed

Test	Role and key points
- **Hydronephrosis** predicts invasive disease (invasive ureteric tumour in up to 80%).	

3. Urine cytology

- Sensitivity ~65%, grade-dependent (grade 1 ~20%, grade 3 ~75%). **Selective** ureteric washings are more accurate than voided urine. Take it before contrast. FISH (fluorescence in situ hybridisation) is optional for surveillance.

4. Flexible URS with biopsy: the key test

- Adds accuracy (75–85% to ~90%). Biopsy gives **grade correctly in ~90%**, but **stage cannot be assessed** (shallow samples; a Ta biopsy is often T3 at surgery).
- Indications: diagnostic doubt, planned kidney-sparing treatment, solitary kidney.
- Diagnostic URS before RNU raises bladder recurrence, without changing survival.

Staging (TNM [tumour, node, metastasis] 8th edition, 2017)

T	Renal pelvis	Ureter
Ta / Tis	Non-invasive papillary / carcinoma in situ	Papillary non-invasive / carcinoma in situ
T1	Subepithelial connective tissue	Subepithelial connective tissue
T2	Muscularis	Muscularis
T3	**Peripelvic fat or renal parenchyma**	Periureteric fat
T4	Adjacent organs, or **through the kidney into perinephric fat**	Adjacent organs
- **N1** single node ≤2 cm; **N2** node >2 cm or multiple. **M1** distant metastasis.		
- **AJCC (American Joint Committee on Cancer) grouping:** stage I = T1, II = T2, III = T3; **any N+, T4 or M1 = stage IV**.		
- **Grade:** WHO 2004/2016: PUNLMP (papillary urothelial neoplasm of low malignant potential), low grade, high grade.		
- Regional nodes: pelvis and upper ureter to hilar and retroperitoneal; lower ureter to pelvic.		

Risk stratification

EAU (European Association of Urology)

Low risk (all needed)	High risk (any one)
Unifocal	Multifocal
Under 2 cm (<2 cm)	2 cm or more (P2: >2 cm)
Low-grade cytology and biopsy	High-grade cytology or biopsy

Low risk (all needed)	High risk (any one)
No invasion on CTU	Hydronephrosis; invasion on imaging
—	Previous radical cystectomy; variant histology

- **AUA/SUO (American Urological Association/Society of Urologic Oncology) 2023** splits each group into favourable and unfavourable (obstruction, multifocality, lower-tract involvement). Ablation is **preferred** in low-risk favourable, **offered** in low-risk unfavourable, **rare** in high-risk favourable, **palliative** in high-risk unfavourable. Endoscopic therapy suits tumours **under 1.5 cm**.
- **Prognostic factors:** stage (strongest), grade, LVI (lymphovascular invasion), nodes, sessile architecture, necrosis >10%, ureteric site, surgical delay >12 weeks.

Management A: kidney-sparing (minimally invasive and endoscopic) treatment

Indications

- **Elective**: low-risk disease, whatever the other kidney.
- **Imperative**: solitary kidney, bilateral tumours, CKD (chronic kidney disease), unfit for RNU, after shared decision.
- The patient must accept frequent recurrence and **lifelong endoscopic surveillance**.

Ureteroscopic (retrograde) management

- Closed system, lowest morbidity; for small ureteric and renal tumours.
- **Steps**: cystoscopy, then selective cytology and saline barbotage, then no-touch semirigid then flexible URS, map all lesions, biopsy (3 Fr forceps or basket) into fixative, ablate the base with laser (holmium or thulium, after biopsy), stent optional.

Laser	Penetration	Note
Ho:YAG (holmium:yttrium-aluminium-garnet)	Under 0.5 mm	Safe in thin ureter
Nd:YAG (neodymium:YAG)	Up to 5–6 mm	Coagulative necrosis; transmural injury risk
Thulium	Under 0.2 mm	Strong haemostasis

- Avoid circumferential fulguration or ablation (stricture); stage large tumours.
- Results: upper-tract recurrence 23–90% (P2: about half), bladder recurrence 15–45%, progression 0–20%. **Grade is the strongest predictor.**

Percutaneous (antegrade) management

- Percutaneous resection for large pelvic tumours, lower-pole or inaccessible calyces and urinary diversions.
- **Steps**: retrograde catheter, puncture a calyx away from the tumour (upper or middle calyx for pelvic tumours), dilate to 30 Fr, cold-cup, bipolar resection or laser with the base sent separately, nephrostomy left, **second-look nephroscopy**.
- Removes more tumour and can **stage**; the tract allows topical therapy. But it is an **open system**: tract seeding about 0.3%, bleeding.

Segmental ureterectomy

- **Distal ureterectomy with bladder cuff and reimplantation** (psoas hitch or Boari flap) for distal tumours, including high-grade ones when the kidney must be saved. It gives a

staging specimen; add pelvic lymphadenectomy if high grade.

- First examine the whole tract above for multifocality. Lower-ureteric tumours with unfavourable features need resection, not endoscopy.

Site	Operation
Distal ureter	Distal ureterectomy with bladder cuff and reimplantation; psoas hitch (~5 cm extra) or Boari flap (10–15 cm)
Mid or proximal	Excision with 1–2 cm margins (frozen section) and ureteroureterostomy (rare)
Whole ureter	Ileal ureter or autotransplantation
- Refluxing reimplant eases surveillance; non-refluxing limits seeding (debated).	
- Laparoscopic or robotic routes are feasible; outcomes equal RNU in selected patients.	
- Adjuvant upper-tract BCG (bacillus Calmette–Guérin) or MMC (mitomycin C); UGN-101 mitomycin gel for low grade (see topical therapy). Lifelong endoscopic surveillance.	

Management B: RNU with bladder cuff (standard for high risk)

- **Principles**: kidney outside Gerota's fascia; clip the ureter early; never open the collecting system; remove the **whole intramural ureter and cuff** (stump recurrence 30–75% if left); bag extraction intact; watertight bladder closure; template lymphadenectomy for high risk.
- **Laparoscopic and robotic RNU steps**: modified flank, 3–4 ports; reflect colon; clip ureter; artery then vein; kidney outside Gerota's fascia (adrenal spared); ureter traced under the superior vesical artery; cuff excised (extravesical or open); two-layer closure.
- Abandoned (higher recurrence): transurethral (endoscopic) "pluck", stripping, laparoscopic stapling.
- Open, laparoscopic and robotic routes are oncologically equivalent: laparoscopic **equivalent to open** (AUA/SUO), with less blood loss and shorter stay; less favoured in bulky T3/T4. Robotic yields more nodes.
- **Template lymphadenectomy** for high risk: hilar plus paracaval (right) or para-aortic (left); pelvic nodes for lower ureteric tumours.

Management C: adjuvant topical therapy

- **Single postoperative intravesical chemotherapy** (AUA/SUO strong): **ODMIT-C (2011)** MMC 40 mg cut bladder recurrence 28% to 16% at 1 year; **Ito (2013)** pirarubicin, 42% to 17% at 2 years.
- **Upper-tract instillation** via nephrostomy (low pressure, 25 cm H2O), ureteric catheter or reflux up a double-J. **BCG** (or MMC): 6-week course for upper-tract CIS, and after complete ablation of high-risk favourable Ta/T1. Risks: sepsis, stricture.
- **UGN-101 (Jelmyto)**: mitomycin reverse-thermal gel, **FDA (Food and Drug Administration) 2020** for low-grade UTUC not manageable endoscopically. **OLYMPUS**: complete response 59%, 56% disease-free at 12 months; ureteric stricture about 44%.

Management D: perioperative (systemic) chemotherapy

- **POUT (2020)**: adjuvant platinum after RNU for pT2–T4 or N+: 3-year DFS (disease-free survival) **71% vs 46%**. Gemcitabine–cisplatin; carboplatin if creatinine clearance 30–49.
- **Neoadjuvant** platinum is preferred when renal function is borderline, i.e. when post-RNU renal function would rule out cisplatin (RNU removes cisplatin eligibility).
- **Also know:** adjuvant nivolumab (CheckMate 274) if platinum-ineligible, but no clear benefit in the UTUC subgroup.

Management E: metastatic disease

- First line: **enfortumab vedotin + pembrolizumab** (EV-302, 2023): overall survival ~31 vs ~16 months versus platinum chemotherapy. Alternative: gemcitabine–platinum, then avelumab maintenance.
- Later: pembrolizumab (KEYNOTE-045), erdafitinib for FGFR3 alterations. Node-positive: chemotherapy first, then consolidative surgery.

Prognosis

- 5-year disease-specific survival: localised ~89%, regional ~63%, distant ~17%. Stage is the strongest factor.

Surveillance and follow-up (AUA/SUO)

- **Kidney-sparing**: cystoscopy and URS within 3 months, URS again at 6 and 12 months; imaging 3–6-monthly (high risk) or 6–9-monthly (low risk), then yearly to 5 years; URS and CTU at closer intervals, for life.
- **After RNU**: cystoscopy and cytology from 3 months; CTU yearly (more often if pT2 or more).

Recent advances

- Lynch syndrome screening; thulium fibre laser; single-use digital ureteroscopes.

Diagrams to draw: ureteroscopic laser ablation of a pelvic tumour beside RNU with bladder cuff; RNU specimen with bladder cuff beside lymph node templates; TNM levels of a renal pelvic tumour (T3 parenchyma versus T4 perinephric fat), beside the diagnostic flowchart (CTU, cystoscopy, URS biopsy, risk group); distal ureterectomy with psoas-hitch reimplantation, beside the goblet sign.

Key points to remember:

- Tobacco, aristolochic acid (A:T to T:A in TP53) and Lynch (MSH2) are the key causes.
- CTU is the best test; URS biopsy gives grade, not stage.
- Ureteroscopy is closed but cannot stage; percutaneous stages but can seed the tract.
- Pelvic tumour into parenchyma is T3; through kidney to perinephric fat is T4; any N+ is stage IV.
- EAU low risk needs all favourable features (unifocal, under 2 cm, low grade, no invasion); any single adverse feature makes it high risk.
- Ureteric tumours: 70% distal; hydronephrosis predicts invasion (up to 80%).
- Ureteric T3 does worse than pelvic T3 (no parenchymal barrier).
- Distal ureterectomy for selected distal tumours.
- RNU with complete bladder cuff + single postoperative intravesical MMC: bladder recurrence 28% to 16% (ODMIT-C).
- POUT: adjuvant platinum, 3-year DFS 71% vs 46%; EV-302 for metastatic disease.

Also, from the condensed book (not in the paper answer):

- Upper tract urothelial carcinoma (UTUC) is **5–10%** of urothelial cancers, but **40%** of renal neoplasms in the Balkans. Concurrent bladder cancer occurs in **17%**, ~60% are invasive at diagnosis, and multifocality is **10–20%**.
- After UTUC, the bladder recurs in **22–47%**.
- **EAU: screen for Lynch syndrome (LS) at the first interview** if age <60, a personal LS-spectrum cancer, a first-degree relative (FDR) <50 with LS cancer, or two FDRs with LS cancer.

KEY POINT

Why UTUC presents invasive: the muscle layer is thin, so tumours reach muscle, parenchyma or periureteric fat earlier than bladder tumours.

- **Smoking** carries a relative risk (RR) of **2.5–7**. Chronic infection or stones (RR 1.5–2) favour squamous cell carcinoma (SCC). Cyclophosphamide, long-term anthranoid laxatives and pelvic radiotherapy are iatrogenic causes.

KEY POINT

Aristolochic acid, arsenic, phenacetin and Lynch all behave as a **field disease** (multiple, bilateral recurrences). This favours kidney-sparing treatment, but Balkan endemic nephropathy patients often have renal insufficiency.

- **Lynch**: UTUC risk is ≈**14×**, and ≈**75×** in MSH2 carriers. If mismatch repair proteins are lost on immunohistochemistry, do germline testing.

2 · Pathology & molecular biology

- Over 90% are urothelial carcinoma. Variant histology (≈**25%**: squamous, glandular, sarcomatoid, micropapillary, neuroendocrine) is aggressive, and **micropapillary** is independently adverse.
- Ureteric inverted papilloma harbours malignancy in ≈**18%**, so follow it for ≥2 yr.

- SCC (0.7–7%) is commoner in the pelvis and invasive at presentation; adenocarcinoma is rare and linked to stones.
- Microsatellite instability distinguishes UTUC from bladder cancer, is prognostic and flags LS; the **RAS-mutated** subtype does worst.

3 · Diagnosis

Biopsy gives grade but not stage.

- Haematuria is the commonest presentation, then flank pain. **Always do cystoscopy**; positive cytology with normal cystoscopy (including urethra) means the upper tracts must be evaluated.
- **CT urography (CTU)** is first line (EAU prefers it to MR urography): sensitivity ≈**92%**, specificity ≈95%, but flat lesions are missed. Hydronephrosis means an invasive ureteric tumour in up to 80%.
- For MR urography, avoid gadolinium if creatinine clearance <30 mL/min. Do a split-function scan before radical surgery.
- Cytology detects ≈65% overall (G1 20%, G3 75%); selective washing beats voided urine, and it should be **collected before contrast**.
- Flexible ureteroscopy (URS) with biopsy raises accuracy to ≈90%; biopsy grade is ≈90% accurate, but visual grading only ≈70%.

> **KEY POINT**
>
> **Why URS biopsy cannot stage:** samples are small and shallow, so a Ta biopsy is often T3 at resection. Stage is **inferred** from grade, appearance, hydronephrosis and cytology, which is why kidney-sparing patients need intensive follow-up.

- **Diagnostic URS indications:** uncertainty, planned kidney-sparing treatment, solitary kidney. **Pitfall:** URS before radical nephroureterectomy (RNU) increases intravesical recurrence without changing survival, so avoid it when CTU and cytology already justify RNU.

4 · Staging & grading

Pelvic T3 vs T4 is a classic trap.

TNM 8th	Definition
Ta / Tis	Non-invasive papillary / CIS
T1	Subepithelial connective tissue
T2	Muscularis
T3	Pelvis: peripelvic fat or **renal parenchyma**; ureter: periureteric fat. pT3a = microscopic parenchymal; pT3b = macroscopic or peripelvic fat (worse)
T4	Adjacent organs, or through kidney into perinephric fat
N1 / N2	Single node ≤2 cm / single >2 cm or multiple

- AJCC stages: I = T1; II = T2; III = T3; **IV = T4, any N+ or M1**.
- Pelvic and upper ureteric tumours drain to hilar and retroperitoneal nodes, and lower ureteric tumours to pelvic nodes. WHO grades are PUNLMP (managed as low grade), low and

high grade.

5 · Prognosis & risk stratification

The risk group decides between kidney-sparing treatment and RNU.

- **Stage matters most**, then grade. SEER 5-yr cancer-specific survival after RNU: ≤T1 86%, pT2 77%, pT3 63%, pT4/N+ 39%.
- **Ureteric and multifocal tumours do worse** than pelvic ones at the same stage, because there is no parenchymal barrier and the periureteric lymphovascular plexus is rich.
- Age alone is never a reason to deny RNU. Do **RNU within 12 wk** for invasive disease. Size **2 cm** is the best cut-off for ≥pT2.
- Post-op adverse factors include nodal disease, lymphovascular invasion (never seen on biopsy, so report it on every specimen), positive margins, necrosis, sessile growth, CIS and variant histology.

Low risk — ALL	High risk — ANY ONE (EAU 2020)
Unifocal	Multifocal
<2 cm	>2 cm
Low-grade cytology	High-grade cytology
Low-grade URS biopsy	High-grade URS biopsy
No invasive aspect on CTU	Hydronephrosis
—	Previous radical cystectomy; variant histology

> **KEY POINT**
>
> Every pre-op factor is a stand-in for invasion, so one adverse feature means high risk. **Trap:** a small, low-grade, unifocal tumour **with hydronephrosis is high risk**.

6 · Recurrence patterns

- Bladder recurrence after UTUC is 15–75% at 5 yr, usually within 2 yr.
- Upper tract tumours after bladder cancer occur in 2–4%, late (median >3 yr). About **30%** of bacillus Calmette–Guérin (BCG)-treated bladder CIS (or cystectomy for BCG-refractory disease) develop UTUC.

> **KEY POINT**
>
> **Asymmetric recurrence:** antegrade seeding makes bladder recurrences common, early and monoclonal, while the field effect gives multifocal tumours of different clones. Both operate.

- Intravesical recurrence after RNU is linked to male sex, prior bladder cancer, ureteric or multifocal tumours, laparoscopic or extravesical cuff technique, positive margins and diagnostic URS.
- **Pan-urothelial disease** (bladder plus ≥2 extravesical sites) has no evidence-based algorithm; note that cystectomy before RNU may lose cisplatin eligibility.

7 · Management overview & follow-up

ALGORITHM

1. Work-up: cystoscopy and CTU, with or without URS biopsy and cytology; assess contralateral function; screen for LS; then assign the risk group.
2. If **low risk**, then kidney-sparing treatment (endoscopic or segmental) with intensive surveillance.
3. If **high risk**, then RNU with bladder cuff, with or without LND, within 12 wk.
4. If **cN+**, then systemic chemotherapy first, and surgery only if there is a response.
5. After RNU, give a single intravesical chemotherapy dose; all patients have cystoscopic follow-up.

- Pre-op nomograms (≥pT2 or non-organ-confined) select patients for neoadjuvant chemotherapy, given before RNU because nephrectomy often removes cisplatin eligibility.
- Lymph node dissection (LND) gives the best staging, but its curative role is debated (technique and POUT: Ch 139).
- Follow-up: cystoscopy for all. After kidney-sparing treatment, add URS, in-situ cytology and CTU. After high-risk bladder cancer (BCG-treated CIS, BCG-refractory), do yearly upper-tract CT. Surveillance is prolonged because the risk is constant.

139 Surgical Management of Upper Urinary Tract Urothelial Tumours

Epidemiology, risk factors, pathology, staging and imaging: Ch 138

1 · Principles & patient selection

Tumour risk and renal reserve together decide between removing and saving the kidney.

*See the paper answer **Upper tract urothelial carcinoma** (Section 13, Ch 138).*

Also, from the condensed book (not in the paper answer):

- Endoscopy carries ~3× more reinterventions than RNU; ipsilateral recurrence after segmental resection is 33–55%; no routine adrenalectomy.
- **Standard** treatment of upper tract urothelial carcinoma (UTUC) is radical nephroureterectomy (RNU) with bladder cuff, if the other kidney is normal and the patient fit.
- UTUC is **unforgiving of spillage**, so every approach keeps the collecting system closed.
- Work-up is CT or MR urogram, cystoscopy, then ureteroscopy (URS) with biopsy and selective cytology; URS raises accuracy from ≈75% to 85–90%.
- Keep diagnostic endoscopy brief and low-pressure to avoid seeding.
- Low-grade (LG) tumours are mostly Ta/T1; high-grade (HG) tumours are mostly invasive.

*AUA/SUO risk group	Features	Ablation	Systemic therapy
Low – favourable	LG, negative cytology, unifocal, papillary, no obstruction/invasion	**Preferred**	No
Low – unfavourable	LG with obstruction, multifocality or prior lower-tract tumour	May be offered	No
High – favourable	HG, unifocal, papillary, no invasion/obstruction	Rare, select cases	Neoadjuvant/adjuvant
High – unfavourable	HG cytology, invasion on imaging, obstruction, suspicious nodes, multifocal, sessile/flat	Palliation only	Neoadjuvant/adjuvant

*American Urological Association / Society of Urologic Oncology.

KEY POINT

Low grade does not mean automatic ablation: obstruction or multifocality makes an LG tumour unfavourable, where RNU keeps a role.

KEY POINT

Two axes: tumour risk and renal reserve. High risk with adequate reserve means extirpation; low risk means preservation. Imperative indications are where they conflict.

ALGORITHM

1. Imaging suspicion: CT or MR urogram.
2. Endoscopy with biopsy and cytology; assign risk group; shared decision.
3. If high risk with adequate glomerular filtration rate (GFR), then RNU with or without neoadjuvant chemotherapy.
4. If low risk or poor GFR, then kidney-sparing management (KSM).

- Imperative KSM must not significantly compromise cancer control; dialysis risk is weighed against oncological risk.
- Unfavourable lower ureteric tumours need distal ureterectomy, not endoscopy.
- **Expectant management** (AUA/SUO option) is only for major comorbidity or high risk of end-stage renal disease (ESRD); untreated UTUC progresses.

2 · Endoscopic (kidney-sparing) management

- **Grade** is the key selector and the strongest predictor of recurrence. AUA/SUO: tumours **<1.5 cm**; caution at ≥1.5 cm.
- For a bulky or suspected HG tumour, use a **staged approach**: biopsy and partial ablation only, avoiding perforation, and complete it once pathology confirms suitability.
- Versus RNU, survival is similar but there are **~3× more reinterventions** and lifelong surveillance.

	Retrograde ureteroscopic	**Antegrade percutaneous**
System	**Closed**	**Open** – tract seeding ≈0.3%, all HG
Best for	Low-volume ureteric, small renal	Large pelvic/upper ureteric, lower-pole calyx, diversion
Staging	Not possible	Possible – base sent separately
Morbidity	Low	Higher (≈25%), admission
Recurrence	Higher bladder and upper-tract	Lower
Second look / topical therapy	Repeat URS; catheter or stent reflux	Via retained nephrostomy

- **URS:** saline barbotage improves cytology, and a tight ureter is pre-stented rather than forcefully dilated. The usual method is a **3-Fr biopsy, then laser**; resection never goes beyond the lamina propria.
- A tumour protruding from the orifice needs ablation or resection of the whole intramural ureter.
- **Never fulgurate circumferentially**, because a stricture follows.
- Lasers: holmium:YAG <0.5 mm, ideal for the ureter; neodymium:YAG 5–6 mm, with transmural risk; thulium <0.2 mm, with strong coagulation.
- **Percutaneous:** puncture the calyx distal to a peripheral tumour, never through it, and dilate to 30 Fr. Resect with cold-cup forceps or a **bipolar** resectoscope, since monopolar risks hypo-osmotic absorption.
- A **nephrostomy is always left** for second look and topical therapy. HG or invasive histology means RNU.
- In a diversion, use flexible or antegrade endoscopy.

KEY POINT

Does URS spread tumour? Prior ureteroscopy increases bladder recurrence (seeding), but pre-RNU URS has not increased metastasis. A single post-op intravesical instillation mitigates this.

3 · Segmental ureterectomy

- It saves nephrons **and** gives a staging specimen; in selected patients it equals RNU.
- **URS of the entire tract above is mandatory**. For HG, add ipsilateral pelvic lymph node dissection (LND).
- Ipsilateral recurrence is **33–55%**, mostly distal and lifelong.
- Segmental resection with uretero-ureterostomy is rare and needs 1–2 cm margins with frozen section.
- Distal ureterectomy with reimplant suits LG low-stage tumours, or selected HG distal ones when the kidney must be kept.
- **Boari flap:** contraindicated by a small or irradiated bladder. The base is ≥2 cm wider than the apex, and the width ≥3× the ureteric diameter.
- **Ileal ureter:** 20–25 cm of ileum ≥15 cm from the ileocaecal valve, isoperistaltic.

Manoeuvre	Length gained	Condition
Psoas hitch	≈5 cm	Mobile bladder; avoid nerve entrapment
Boari flap with hitch	Further 10–15 cm	Adequate capacity
Renal descensus with nephropexy	Up to 8–10 cm	Better on left (longer renal vein)
Ileal ureter / autotransplant	Whole ureter	Autotransplant in skilled hands

4 · Positive cytology & carcinoma in situ (CIS)

Trace a positive cytology before removing any kidney.

ALGORITHM

1. Positive voided cytology: repeat it.
2. CT urogram and bladder biopsies. If the bladder is positive, then treat the bladder.
3. If still positive with a negative bladder (or after BCG "6+1"), then selective upper-tract cytology, retrograde pyelography (RPG), prostatic urethral biopsy (men) and URS.

- Upper-tract CIS is a diagnosis of exclusion: persistently positive selective cytology with normal URS and imaging.
- **No RNU for cytology alone**, because of false positives and later contralateral disease.
- AUA/SUO: **6-week BCG induction**, preferably via nephrostomy; RNU if more than surface disease is proven.
- CIS with HG papillary UTUC needs radical or multimodal treatment.

5 · Topical therapy in the upper tract

- Indications are solitary kidney, bilateral UTUC and ESRD risk. Routes are nephrostomy (≤25 cm H_2O), ureteric catheter or stent reflux.
- Always use **low pressure and sterile urine**, as sepsis deaths are reported.

- Adjuvant instillation after papillary ablation has **no proven benefit**. BCG is acceptable after complete ablation of HG favourable Ta/T1; its commonest complication is renal granulomas.
- **UGN-101 (Jelmyto)** is mitomycin C (MMC) in a reverse-thermal gel, FDA-approved in 2020 for LG UTUC **not endoscopically manageable**. Stricture occurs in ≈44%.

6 · Radical nephroureterectomy

Remove the whole ureter and cuff without spillage.

- **Indications:** non-metastatic high-risk pelvic or proximal ureteric tumour (HG, obstruction, parenchymal invasion), with cuff and LND; or LG with unfavourable features or failed or infeasible endoscopy.
- **Kidney first, cuff last. Clip the ureter early**, control the hilum before manipulation and stay outside Gerota's fascia. **Adrenalectomy is not routine.**
- Remove the **entire intramural ureter and orifice**: a retained stump recurs in **30–75%** and cannot be surveyed. After bowel diversion, excise the ureteroenteric anastomosis.
- Extravesical or transvesical excision is acceptable; lasso, pluck, stripping and stapling are abandoned because of spillage and positive margins. Close in two layers.
- Laparoscopic or robotic: wide perihilar margin and extract **intact in a bag**.
- Ureteric blood supply is anteromedial in the upper third, medial in the middle and **lateral** in the lower third.

	Open	Laparoscopic	Robotic
Oncology	Reference	Equivalent	Comparable
Morbidity	Highest	Less blood loss, shorter stay	Lowest; fewer positive margins
Caveat	—	Possibly more bladder recurrence; less favoured for T3/T4	Longest; more LND done

> **KEY POINT**
>
> Principle beats approach: port-site seeding and carcinomatosis follow spillage, not the approach.

Lymphadenectomy

*See the paper answer **Muscle-invasive bladder cancer: management, neoadjuvant therapy and lymphadenectomy** (Section 12, Ch 131).*

Also, from the condensed book (not in the paper answer):

- Outcome ranks pN0 better than pNx, better than pN+; the prognostic cliff is T2 to T3.
- LND has value in **T2–T4**, especially for pelvis and proximal ureter, and the surgeon, not the approach, sets the yield.
- Caudal limits: renal pelvis to the inferior mesenteric artery; upper two-thirds of ureter to the aortic bifurcation; lower third to the pelvis.

7 · Perioperative adjuncts

- Bladder tumours follow in 30–50% after RNU or endoscopy, mostly within 2 yr. **AUA/SUO strongly recommend** a single dose after RNU (optional after endoscopy).

- Post-op (day 3–7), do a cystogram, then give MMC or gemcitabine with a 1–2 h dwell. An **intraoperative** instillation, clamped until the ureter is clipped, needs no cystogram.
- **Renal function is the hinge**, so give platinum **before** nephrectomy. Neoadjuvant is favoured if post-op creatinine clearance (CrCl) is expected **<u><50</u>** mL/min.
- In EA8141, patients with CrCl >50 received accelerated MVAC ×4.
- Adjuvant (POUT): gemcitabine–cisplatin (CrCl >50) or gemcitabine–carboplatin (30–49). cN+ disease receives chemotherapy, then consolidative surgery.

8 · Complications

Procedure	Complication	Management / prevention
URS	Perforation; stricture	Stent; pre-stent tight ureter; no circumferential fulguration
Percutaneous	Bleeding, irrigant absorption, perforation, secondary ureteropelvic junction obstruction	Bipolar resection; avoid major vessels; nephrostomy
Topical therapy	Sepsis, absorption, stricture	Sterile urine, low pressure
Cuff closure	Urine leak	Two-layer closure; catheter; cystogram
RNU	Renal insufficiency	Nephrology referral

9 · Follow-up

- Every visit includes **cystoscopy**, cytology and a metabolic panel; after KSM add ipsilateral URS.
- Use CT or MR urography (RPG if no contrast) and chest imaging.

Group	Cystoscopy	Upper-tract endoscopy	Imaging
KSM low risk	3 mo, then 6–9-monthly ×2 yr, then yearly	3 mo, 6 mo, 1 yr	6–9-monthly ×2 yr, yearly to 5 yr
KSM high risk	3 mo, then 3–6-monthly ×3 yr, then yearly	3 mo, 6 mo, 1 yr	3–6-monthly ×3 yr, yearly to 5 yr
RNU <pT2N0	With cytology at 3 mo, then by grade	—	CT/MRI by 6 mo, yearly ≥5 yr
RNU ≥pT2	3 mo, 3–6-monthly ×3 yr, yearly	—	Abdomen/pelvis 3–6-monthly ×2 yr; chest CT 6–12-monthly ×5 yr

10 · Landmark trials

Trial	Design	Result	Take-home
ODMIT-C	Randomised controlled trial (RCT), MMC 40 mg single dose at catheter removal after RNU	Bladder recurrence 28% vs 16% at 1 yr	Single-dose instillation standard
Pirarubicin trial (Ito)	RCT, single pirarubicin dose within 5 days of RNU	Bladder recurrence 42% vs 17% at 2 yr	Confirms single post-RNU instillation
Laparoscopic vs open RNU	Only RCT; small	Equal 5-yr survival; less blood loss	Minimally invasive acceptable

Trial	Design	Result	Take-home
UGN-101 phase III	Single-arm, LG UTUC	Complete response 59%; 56% disease-free at 12 mo	Gel chemoablation option
POUT	Phase III adjuvant platinum vs surveillance, pT2–4/N+	Stopped early; disease-free survival (DFS) hazard ratio **0.45**; 3-yr DFS 71% vs 46%	Adjuvant platinum standard after RNU for ≥pT2/N+

141 Contemporary Open Surgery of the Kidney

1 · Role & preoperative work-up

- Open surgery offers **exposure and early great-vessel control**. It suits renal cell carcinoma (RCC) with inferior vena cava (IVC) thrombus, cytoreduction, complex tumours and a hostile abdomen (minimally invasive surgery: Ch 142).
- **Renal function** decides partial (PN) versus radical nephrectomy (RN). Estimate glomerular filtration rate (GFR) with the **CKD-EPI 2021 race-free** equation, which beats MDRD.
- If GFR is **<60** or there is significant proteinuria, ask for a perioperative nephrology consult.
- In advanced RCC, check blood count and metabolic panel (MSKCC and IMDC risk models), liver tests (**Stauffer syndrome**) and cardiopulmonary fitness.
- Contrast is generally safe if GFR is **>30**. If thrombus is suspected, MR or CT venography must be **≤1 month old**, because thrombus propagates rapidly.
- Give cefazolin ≤1 h pre-incision, pneumatic compression and subcutaneous heparin 5000 IU; no routine bowel preparation.
- **Renal artery embolisation** lets the vein be tied first when the hilum is hidden, but it is no longer routine because of postinfarction syndrome and harder surgery if delayed >48 h.

2 · Surgical anatomy & approaches

Most disasters come from misread anatomy.

- At risk on the **right** are liver, duodenum and colon; on the **left**, spleen, pancreatic tail, colon and superior mesenteric artery (SMA).
- The **right renal artery** lies behind the renal vein and IVC and can be reached in the **interaortocaval space** by retracting the left renal vein (LRV) upward.
- The **SMA** leaves the front of the aorta **1–2 cm above LRV** and veers left, so it can be mistaken for the left renal artery.
- The right gonadal and adrenal veins drain into the IVC; the left ones drain into the LRV.
- Anterior exposure reflects the colon along the **white line of Toldt**. On the right, add a **Kocher** manoeuvre without cautery, because duodenal injury is devastating.
- **Cattell-Braasch** (right colon and midgut mobilisation) and partial **Mattox** (left medial visceral rotation) expose the great vessels.

Incision	Best for	Limitations
Midline	Trauma; advanced tumour or IVC thrombus (early aortic access); bilateral	Bowel mobilisation needed
Anterior subcostal	Unilateral transperitoneal	Muscle division
Chevron	Renovascular; RN with thrombectomy; both pedicles	Extensive muscle division
Subcostal flank (below 12th rib)	Lower pole, pelviureteric junction, proximal ureter	Poor hilar access; not for PN
Supracostal (11th/12th)	RN or PN; 8-cm mini-flank above the 11th	Pleura
Transcostal (11th, rib resected)	Retroperitoneum at an old rib tip	Rib division

Incision	Best for	Limitations
Thoracoabdominal, dorsal lumbotomy	Largely historical; thoracoabdominal is still an option for level III–IV thrombus	

- The **flank** position lowers venous return and lung function, so it is poor in cardiopulmonary disease; pad against rhabdomyolysis.
- In the **supracostal** approach, push the pleura up and divide the intercostals close to the **upper border of the rib** to spare the bundle. Spare the **subcostal nerve**.

3 · Simple nephrectomy

- Stay **within Gerota's fascia**, on the capsule, for a benign nonfunctioning kidney (obstruction, infection, stones, reflux), and spare the adrenal.
- Benign PN suits focal atrophy, a duplex pole or a symptomatic angiomyolipoma.

4 · Radical nephrectomy

- Robson's operation included adrenal and extended nodes; now the **normal adrenal is spared** and lymph node dissection (LND) is selective.
- **Indications:** tumours not amenable to PN, such as a nonfunctioning kidney, tumour replacing the kidney, nodes, cytoreduction or venous thrombus.
- Staging is abdominal CT or MRI and **chest CT**; a bone scan only for symptoms or raised alkaline phosphatase.
- Biopsy only if it changes the plan, such as an infiltrative mass or before systemic therapy.
- T4 invasion means altered enhancement of the organ, not just contact.

ALGORITHM

1. **Right:** survey the abdomen; open Toldt's line down to the iliac vessels; take down the hepatic flexure; Kocher; expose the IVC.
2. Loop and palpate the renal vein; find the artery behind it; tie **artery, then vein**; remove the kidney outside Gerota's.
3. **Difficult hilum:** ligate the artery in the interaortocaval space, so the kidney decompresses; then double-ligate the flaccid vein.
4. Alternative: start at the lower pole and lift the kidney from posterior to anterior to show the artery.
5. **Left:** mobilise the splenic flexure down to the iliacs; divide the renocolic ligament (pad the pancreas).
6. Follow the aorta or gonadal vein to the LRV. Tie the artery above and behind it, confirm occlusion, then the vein.

KEY POINT

Why artery first? Tying the vein first leaves the kidney engorged, so collaterals bleed. Arterial ligation shrinks the kidney and softens the vein, which can then be checked for thrombus. A persisting pulse means an accessory artery, incomplete control, or the SMA has been tied.

- **Lumbar veins:** use silk ties and **no clips on the IVC side**, because clips block emergency caval clamping.

- **Lost pedicle control:** tell the anaesthetist, compress, use two suckers and **never clamp blindly**.
- **Adrenalectomy** only for an upper-pole tumour with a lost plane, an adrenal mass or diffuse involvement. Metastasis occurs in **<5%**, and contralateral metastasis is as common as ipsilateral.
- **LND:** nodes are positive in ≈5%, and the EORTC RCT showed no overall survival benefit, though it included many low-stage tumours.
- **Do LND if nodes are enlarged**: a short axis **≥1.5 cm ≈ 50% positive**. Consider it in high risk (>10 cm, grade ≥3, sarcomatoid, thrombus, T4).
- The template runs from crus to bifurcation. **Clip lymphatics above the renal vein** to prevent chylous ascites.
- **Closure:** fill with saline and give a big breath; bubbles mean a pleural tear, which is sutured over a 14-Fr catheter. Take a **chest X-ray in all**.

5 · IVC tumour thrombectomy

Surgery is the only cure; thrombus level decides the plan.

- RCC is the commonest cause (**5–10%** of RCC). Untreated, most die within 1 yr, so surgery (RN and LND, with or without systemic therapy) is the backbone.
- **Tumour thrombus** differs from **bland thrombus** (clot below the obstruction). **MR venography** best shows extent, wall invasion and bland thrombus.
- Anticoagulate for venous thromboembolism, bland thrombus or IVC occlusion. Use **no suprarenal filters**, because thrombus grows onto the device.
- Use **transoesophageal echo (TEE)** for levels II–IV, and involve a **cardiothoracic surgeon for III–IV**.

Mayo level	Upper extent	Management
0	Renal vein	RN (commonest)
I	IVC **≤2 cm** above ostium	Milk back, Satinsky partial clamp, ostial cavotomy
II	>2 cm, below hepatic veins	Full IVC control; infrahepatic cavotomy
III	At or above major hepatic veins, below diaphragm	Liver mobilisation with or without Pringle; occlusive thrombus: suprahepatic clamp; nonocclusive: venovenous bypass (VVB)
IV	Above diaphragm	Sternotomy, cardiopulmonary bypass (CPB) with deep hypothermic circulatory arrest (DHCA), atriotomy

Bland group	Location	Action
A	None	None
B	Iliac veins	Infrarenal filter
C	Infrarenal IVC, separate	Interrupt (clip or staple); do not extract
D	Mixed with tumour	Resect infrarenal IVC; keep lumbars as collaterals

ALGORITHM

1. **Level I (right):** control the IVC before handling the tumour; ligate the right artery interaortocavally.
2. Loop the LRV and the suprarenal and infrarenal IVC with **Rummel tourniquets**.
3. Milk the thrombus toward the ostium; place a Satinsky round the ostium with the thrombus inside the jaws.
4. Do an ostial cavotomy keeping ≥half the IVC width, and extract.
5. **Level I–II:** ligate the renal artery early. Tighten tourniquets in order: **infrarenal IVC, contralateral renal vein, suprarenal IVC**.
6. Open the cava, extract, flush and inspect the wall. Close with 4-0 Prolene.
7. Release the **caudal clamp first**, flush on a held breath, then release the cranial clamp.

- A **left tumour (level II)** is harder: tie the left artery first, then Kocher, **split-and-roll** the IVC and excise the LRV ostium en bloc. Do not clamp the right renal artery.
- **Level III:** keep bypass on standby and mobilise the liver as in transplantation. The **major hepatic veins are never divided**, and **Budd–Chiari syndrome is a contraindication**.
- The **Pringle** manoeuvre is needed if clamping above the hepatic veins. Limit it to ≤60 min, ideally **≤20 min**.
- **Level III sequence:** clamp infrarenal IVC, contralateral renal vein, porta hepatis, then suprahepatic IVC; open and extract. Unscrapable wall invasion needs IVC resection.
- Do a 2–5 min **clamp trial**. If cardiac output falls by >50% or mean arterial pressure by >30%, use bypass. Occlusive thrombus tolerates clamping because of azygos collaterals; free-floating thrombus does not.
- Incomplete tumour removal is universally fatal, so never compromise IVC control to save time.

	VVB	**CPB with DHCA**
Access	Femoral to jugular, percutaneous	Sternotomy; cool to 16–18 °C and drain 95% of blood
Heparin	Partial	Full; coagulopathy
Drawback	Lumbar and intercostal back-bleeding	Cardiac and neurological events
Use	Least invasive; not with atrial or iliac bland thrombus, Budd–Chiari	Level IV; bloodless field; ≤60 min (<40 better)

- In **VVB**, place cannula tips far from the thrombus, because dislodgement causes massive pulmonary embolism (PE).

LAQ Vascular grafts and vascular surgery in urology

Paper answer · P1 · P2 · P3 · merged from 3

Asked as: P3 LAQ "Vascular grafts and vascular surgery in urology" (asked as "Vascular Grafts in Urology", Summer 2021 Paper III Q2 LAQ; "Vascular Surgery in Urology", Winter 2024 Paper III Q3a SAQ) · P1 SAQ "Vascular grafts in urology" (Practice Set 5 Paper 1 Q3e) · P2 SAQ "Role of vascular grafts in urology" (2015-Summer Paper 2 Q3e)

Assumed meaning: blood-vessel grafts and patches (conduits) used in urological surgery: transplantation, renal and IVC (inferior vena cava) tumour surgery, RPLND (retroperitoneal

lymph node dissection), sarcoma, renovascular repair and autotransplantation, dialysis access and penile surgery. Tissue grafts (buccal mucosa) are not included.

Assumed meaning (vascular surgery): vascular procedures performed or met by the urologist (control and repair of vessels, caval surgery, transplant and renovascular surgery, penile vascular surgery and access), not vascular surgery as a specialty.

Introduction

- A vascular graft **replaces, bypasses, extends or widens** a vessel when primary repair would fail or leave tension or narrowing.
- **Patch:** widens a partly resected wall. **Interposition graft:** replaces a segment. **Bypass:** routes flow around a lesion.

Classification of graft materials

Type	Examples	Advantages	Disadvantages
Autologous vein	Great saphenous, gonadal, superficial femoral, donor external iliac vein	Resists infection; suits small vessels and contaminated fields	Limited length; valves; harvest wound
Autologous artery	Internal iliac (hypogastric), **Carrel aortic patch**	Best for renal artery repair	Limited supply
Allograft (homograft)	Cadaveric vein or artery	Off the shelf; avoids xenograft	Availability; degeneration
Xenograft (biological)	**Bovine or porcine pericardium**	Pliable, strong patch, low infection	Patch only; cost
Synthetic	**PTFE (polytetrafluoroethylene, Gore-Tex)**, ringed PTFE, Dacron (polyethylene terephthalate), collagen-impregnated Dacron	Off-the-shelf, large calibre, any length	Infection needs removal; venous grafts thrombose

Principles

- **Control first:** proximal and distal control with vessel loops, Rummel tourniquets and atraumatic clamps (bulldog, Satinsky, DeBakey). Artery before vein.
- **Choice of graft: autologous** vein or artery for small vessels and contaminated fields; **synthetic** (PTFE or Dacron) for large, high-flow vessels with no bowel contamination.
- **Venous (low-pressure) grafts thrombose** more than arterial ones, so use the **widest graft** possible.
- Patch: **smooth side luminal**, rough thrombogenic side outward; edges must not invert.
- Systemic heparin before clamping; flush air and debris before tying; tension-free, everting, spatulated anastomosis with fine monofilament polypropylene (Prolene).
- **Sutures:** polypropylene (Prolene), double-armed. IVC or aorta 3-0/4-0; renal vessels 4-0/5-0; transplant anastomoses 5-0/6-0.
- Antiplatelet or anticoagulant afterwards as indicated.
- Wrap synthetic grafts in **omentum** or retroperitoneal fat, away from bowel and urine.

Application 1: renal transplantation

- Renal vein and artery end-to-side to external iliac vessels; Carrel patch.

- **Short right renal vein:** extend with donor IVC, free donor external iliac vein, or **spiral gonadal vein graft**.
- **Multiple arteries (18–43%):** Carrel patch; on the back table, conjoined side-to-side or end-to-side (bench) reconstruction. **Preserve the lower-pole artery** (it supplies the ureter).
- **Atherosclerotic recipient iliac artery:** mount the graft artery on a **Gore-Tex or bovine pericardial patch**.
- **Children under 15 kg:** aorta and IVC anastomosis, transperitoneal.
- **Transplant renal artery stenosis (1–23%):** angioplasty or stent first; surgical interposition graft or bypass if it fails, which risks graft loss.

Application 2: RCC (renal cell carcinoma) with IVC tumour thrombus

- **Mayo levels 0–IV:** level I milking and partial caval clamp; II full caval control; III liver mobilisation ± venovenous bypass; IV cardiopulmonary bypass ± deep hypothermic arrest.

Situation	Repair
Residual lumen ≥50%	Primary closure
Residual lumen <50%	**Patch cavoplasty**: bovine or autologous pericardium, PTFE, Dacron or saphenous vein
Circumferential wall invasion	**Caval replacement: PTFE 16–20 mm**; contralateral renal vein reimplanted into the graft
Infrarenal bland thrombus	Filter or caval interruption, sparing lumbar collaterals

- **Leibovich–Bower patch width** = normal caval diameter × % diameter resected × 3.
- Left tumour with suprarenal caval resection: the right kidney has no collaterals, so give it new drainage (autotransplantation, or **saphenous vein graft** to the splenic, portal or inferior mesenteric vein).
- After caval replacement: low-dose heparin, then **lifelong aspirin 81 mg**.

Application 3: PC-RPLND (post-chemotherapy RPLND)

- Vascular procedures in **9–15%**; aorta most often.
- **IVC:** narrowing <25%: primary repair; >25%: patch or interposition graft. **Ligation is tolerated** (slow occlusion builds collaterals).
- **Aorta:** never ligated; must be repaired or grafted. Small injury: repair or pericardial patch; resection: **PTFE or Dacron** (about 1% need replacement). Graft also if the duodenum is injured, to prevent aortoduodenal fistula.

Application 4: retroperitoneal sarcoma

- Caval resection ± reconstruction (Dacron, PTFE, homograft); grafts **thrombose in 10–40%**; anticoagulate at least 3 months.

Application 5: renovascular repair and autotransplantation

- Aortorenal bypass (saphenous vein, internal iliac artery, PTFE); splenorenal or hepatorenal bypass.
- Ex vivo bench repair of branch aneurysm or stenosis (or ureteric loss) with internal iliac artery or saphenous vein, then **autotransplantation** (see Q3e). Angioplasty or stent first for fibromuscular dysplasia.

- Traumatic renal artery injury: endovascular stent for the main renal artery; attempt open repair only for solitary kidney or bilateral injury (warm ischaemia <120 minutes).

Application 6: dialysis access

- Arteriovenous fistula first; **AV (arteriovenous) graft (PTFE)** when no native fistula is possible (poor veins): usable in 2–3 weeks; more stenosis, thrombosis and infection than AVF (arteriovenous fistula).

Application 7: iatrogenic or traumatic vascular injury

- Interposition graft.
- SMA (superior mesenteric artery): repair or graft bypass. SMV (superior mesenteric vein): autologous vein graft; Gore-Tex only if no vein.

Application 8: penile surgery

- Priapism: **Grayhack** saphenous vein–cavernosal shunt.
- Vein grafts for Peyronie tunical defects (survive by endothelial perfusion); synthetic grafts not recommended.
- **Arterial revascularisation** (inferior epigastric artery to dorsal artery) for young men with post-traumatic pudendal stenosis; venous ligation investigational.

Application 9: endovascular

- Angioembolisation for trauma, AML (angiomyolipoma) bleeding, pseudoaneurysm and AVF after partial nephrectomy.

Complications

- **Thrombosis** (caval grafts 10–40%), anastomotic stenosis (intimal hyperplasia), bleeding, pseudoaneurysm.
- **Graft infection** (synthetic grafts usually need removal); graft–enteric or urinary fistula.
- **Ureteric:** transient hydronephrosis after aorto-iliac grafts (12–20%), symptomatic stricture 1–2%, **ureteroarterial fistula** (treat with endovascular covered stent).

Recent advances

- Endovascular covered stent grafts; bioengineered acellular vessels (under study); robotic autotransplantation.

Diagrams to draw

- IVC patch cavoplasty and PTFE caval replacement with left renal vein reimplanted.
- Mayo IVC thrombus levels with the matching vascular control.
- Bench arterial reconstruction.

> **Key points to remember:**
>
> - Proximal and distal control, heparin, flush, everting Prolene anastomosis.
> - Autologous for small or contaminated fields; PTFE or Dacron for large vessels.
> - Mayo levels decide caval control and bypass.
> - IVC: patch if lumen <50%; PTFE 16–20 mm replacement if wall invaded.
> - PC-RPLND: IVC may be ligated; aorta must be grafted.
> - Transplant: Carrel patch, bench reconstruction; preserve the lower-pole artery.
> - Bench surgery plus autotransplantation for complex renovascular lesions.

Also, from the condensed book (not in the paper answer):

- Suprarenal IVC ligation with a right tumour is acceptable if the LRV is tied caval to its tributaries, which act as collaterals.

KEY POINT

"Shortcuts" that become "longcuts": opening the IVC before all inflow is controlled; side-biting a level I; clips on the IVC side; incomplete arterial control; forceful milking (embolus, tear); clamping before volume loading or skipping the test clamp.

6 · Partial nephrectomy

*See the paper answer **Nephron-sparing surgery for renal masses: indications, approaches and recent advances** (Section 13, Ch 137).*

Also, from the condensed book (not in the paper answer):

- Hyperfiltration injury (focal segmental glomerulosclerosis, proteinuria) is rare unless nephron mass falls by >80%.
- Multifocality is ≈2% in clear cell and ≈10% in papillary RCC, so inspect the whole kidney with intraoperative ultrasound.
- PN is size-independent if it is safe; collecting-system invasion is a relative contraindication.

Question	Evidence
Warm ischaemia	Aim **<30 min**; within limits, preserved parenchyma dominates
Cold vs none	Randomised controlled trial (RCT): no difference
Mannitol	RCT: no benefit
On- vs off-clamp	2 RCTs: no difference
3D models	RCT: less ischaemia, blood loss and stay

KEY POINT

PN trades nephrons lost to the margin against nephrons injured by ischaemia. Within accepted limits, the **volume of parenchyma preserved** matters more than clamp minutes. Hypothermia and mannitol are optional.

- Always **control the hilum first**, even if not clamping, so a clamp can go on at once.
- **Enucleation** follows the pseudocapsule plane; **resection** takes a 5–10 mm rim and is usually clamped.
- **Renorrhaphy:** the inner layer closes vessels and collecting system with absorbable sutures (permanent sutures become a stone nidus); the outer layer is capsular over a bolster.

7 · Complications & management

Injury	Management
SMA	Injured when mistaken for the left renal artery or by Cattell-Braasch traction. Potentially lethal, so call a vascular surgeon for primary repair or graft

Injury	Management
Inferior mesenteric artery or vein (IMA/IMV)	Ligation safe if the marginal artery of Drummond is patent
Superior mesenteric vein (SMV)	Ligation causes bowel ischaemia, splanchnic hypertension and hypotension. **Never ligate**; repair with 6-0 Prolene venorrhaphy. If transected, reanastomose or use autologous vein; leave the abdomen open
Duodenum	Haematoma narrowing the lumen: incise serosa and muscularis, drain. Tear: debride, 2-layer closure, omentum, drain, nasogastric tube
Pancreas	Glue and drain (check amylase); ductal injury: general surgeon; late collection: percutaneous drain, nil by mouth, octreotide
Pleura	Purse-string over a 14-Fr catheter, evacuate on a positive-pressure breath; large tear: chest tube

- In thrombectomy, air embolism is prevented by releasing the caudal clamp first. Use **no cell saver**, because it spreads tumour.
- A **urine leak** mostly appears within 72 h. If drain output is >50 mL/24 h, a drain creatinine above serum means urine; a persisting urinoma needs a double-J stent and Foley, and most resolve in 4–6 wk.
- A significant **pseudoaneurysm** (flank pain, haematuria or shock) needs **angiography** with **selective embolisation**, because CT angiography can miss it and cannot treat it.
- An **arteriovenous fistula** is observed, then coil-embolised, with surgery if complex.
- **Renal insufficiency** is usually mild and transient; exclude obstruction, drug toxicity and thrombosis.

142 Laparoscopic and Robotic Surgery of the Kidney

1 · Principles & patient selection

Fitness here means fitness for the gas.

- First laparoscopic nephrectomy: **Clayman, 1991** MCQ; first laparoscopic partial nephrectomy (PN): Winfield, 1993.
- Minimally invasive surgery (MIS) gives less pain, blood loss and convalescence than open, with **equivalent long-term outcomes** and the same indications.
- **General anaesthesia is mandatory.** The CO_2 pneumoperitoneum impairs ventilation, venous return and cardiac output, so risk is high in chronic obstructive pulmonary disease (COPD) and cardiac disease.
- In severe COPD with hypercarbia, lower the pressure, then use helium, then convert.
- With the robot, **docking and undocking delay emergency access**, so the team must communicate closely.
- In obesity, **visceral adipose tissue**, not body mass index (BMI), predicts difficulty. Prior surgery is **not a contraindication**.

> **KEY POINT**
>
> The patient must tolerate the **pneumoperitoneum**, not the renal operation: cardiopulmonary reserve, not age or BMI, decides fitness.

2 · Access & port placement

Routes are transperitoneal (TP, most used), retroperitoneal (RP), hand-assisted (HAL) and single-site (LESS).

	Transperitoneal	Retroperitoneal
Space / landmarks	Large; familiar	Limited, poor triangulation; less familiar
Bowel	Colon reflection; adhesiolysis	Peritoneum not entered
Hilum	Reached after colon reflection	Direct, especially the posterior aspect
Preferred for	Most surgery; suturing; anterior tumours	Posterior tumours, pyeloplasty, cysts, biopsy; dense adhesions, peritoneal dialysis, pregnancy, morbid obesity

- **TP** uses a 30–45° modified flank position with 3–5 trocars.
- **Robotic** arms are spaced **8 cm (Si) / 6 cm (Xi)**, forming a **broad-based triangle** with the camera at the apex.
- **RP** uses the full flank position; the space is made by finger or balloon below the 12th rib in the posterior axillary line.

> **KEY POINT**
>
> RP pitfalls: a balloon inflated within muscle causes a flank hernia; too anterior injures the peritoneum or colon; too posterior makes the quadratus or psoas bleed. Bail-out: open the peritoneum and continue transperitoneally.

- **HAL** gives tactile feedback and helps novices, donors and scarred kidneys; a hand through the port can compress a bleeding vessel.
- **LESS and the single-port robot** improve cosmesis but lose triangulation, with unchanged outcomes.

3 · Nephrectomy — operative steps

ALGORITHM

1. Reflect colon; find ureter and gonadal vessels on the psoas.
2. Lift kidney to expose the hilum; artery, then vein.
3. Upper pole (adrenal spared or taken); ureter and lateral attachments; bag; extract.

- Keep the **lateral attachments** until late, or the kidney rolls over the hilum. On the right, dissect cold near the duodenum and do a **Kocher**.
- The **ureter** lies just behind the gonadal vessels and is a useful handle.
- **Hilum:** divide the **artery first, then the vein**, with an endo-GIA stapler. **Never leave clips in the staple line.**
- **En bloc hilar stapling** is safe (no arteriovenous fistula, AVF) and saves ~45 min.
- **Extraction:** intact in a bag, or morcellated in an impermeable **LapSac**, which keeps type and grade but loses size and stage.

4 · Simple vs radical nephrectomy

	Simple	Radical
Gerota's fascia and fat	May be left	Removed intact with specimen
Pathology	Benign, often inflammatory	Renal cell carcinoma (RCC)
Difficulty / conversion	Can exceed radical; higher	Planes preserved; lower

- **Simple** nephrectomy is for benign nonfunctioning kidneys, such as chronic pyelonephritis, xanthogranulomatous pyelonephritis, obstruction, reflux or tuberculosis; after pyonephrosis or stones, use hand assistance or a subcapsular approach.
- **Radical nephrectomy (RN):** MIS equals open oncologically at 5 and 10 yr. In RCTs of RP versus TP, the only consistent RP advantage is **shorter operative time**.
- **Lymph node dissection (LND)** has staging value only, with no overall survival benefit; the AUA advises removing clinically positive nodes.
- Tumours >10 cm are feasible, though landmarks are lost; conversion is higher laparoscopic than robotic.
- Selected **inferior vena cava (IVC) thrombus** cases can be done by experts, with early arterial ligation that may retract the thrombus (Ch 141). Laparoscopic **cytoreductive** nephrectomy allows earlier systemic therapy.

KEY POINT

Access changes morbidity and recovery, never oncology: early arterial control, en bloc removal and intact bagged extraction carry over from open surgery.

5 · Partial nephrectomy

*See the paper answer **Nephron-sparing surgery for renal masses: indications, approaches and recent advances** (Section 13, Ch 137).*

SAQ Role of renal nephrometry in management of RCC.

Paper answer · P1

Type: SAQ (10 marks)
Source: 2020-Summer Paper 1, Q3f

Definition

- **Nephrometry scores** grade the anatomical complexity of a renal mass on CT (computed tomography) or MRI (magnetic resonance imaging), to standardise RCC (renal cell carcinoma) reporting and plan treatment.

R.E.N.A.L. score (Kutikov and Uzzo, 2009)

Component	1 point	2 points	3 points
Radius (maximum diameter)	≤4 cm	>4 to <7 cm	≥7 cm
Exophytic/endophytic	≥50% exophytic	<50% exophytic	Entirely endophytic
Nearness to collecting system or sinus	≥7 mm	>4 to <7 mm	≤4 mm
Anterior/posterior	Suffix a, p or x (no points)		
Location to polar lines	Entirely above or below	Crosses a polar line	>50% across, crosses midline, or entirely between
- Suffix "h": touches main renal vessels. **Total 4–12:** low 4–6, moderate 7–9, high 10–12.			

PADUA (Preoperative Aspects and Dimensions Used for an Anatomical classification) score (Ficarra, 2009)

- Polar location, exophytic rate, renal rim (lateral or medial), renal sinus, collecting system and size, plus anterior/posterior face. **Total 6–14:** 6–7 low, 8–9 intermediate, ≥10 high risk.

C-index (centrality index)

- Distance from tumour centre to kidney centre ÷ tumour radius. **Lower value = more central, more complex.**

Role in management

Decision	Use
Partial vs radical nephrectomy	Feasibility and risk of PN (partial nephrectomy) vs RN (radical nephrectomy)
Approach	Open vs minimally invasive; hilar clamping
Predict outcomes	Warm ischaemia, blood loss, complications, conversion, collecting-system entry; urine leak (R and N components)
Function	Parenchymal loss, eGFR (estimated glomerular filtration rate) decline

Decision	Use
Ablation	Cryoablation complications rise with complexity
Pathology	Higher scores: malignancy, higher grade

- **AUA (American Urological Association) 2021:** with high oncological potential, prefer RN only if PN would be challenging even for experts, there is no CKD (chronic kidney disease) or proteinuria, and new baseline eGFR will likely exceed 45.

Limitations

- Interobserver variation; ignores patient factors and surgeon experience.

Also know: ABC (arterial based complexity), DAP (diameter–axial–polar) scores; **MAP (Mayo Adhesive Probability) score** for adherent perinephric fat.

Diagram to draw: kidney with polar lines and the five R.E.N.A.L. components.

> **Key points to remember:**
> - RENAL 4–12: low 4–6, moderate 7–9, high 10–12.
> - PADUA 6–14; C-index lower = more central.
> - Predicts PN feasibility, ischaemia time and complications.
> - MAP score: adherent perinephric fat.

Also, from the condensed book (not in the paper answer):

- In cT1b/T2 tumours, PN gives better estimated GFR with equal oncology, but **tumours >5 cm upstaged to pT3a recur more after PN**.
- The collecting system is entered in about 60%, so always drain.

ALGORITHM

1. Intraoperative ultrasound (size, depth); open Gerota's away from the lesion; score the capsule.
2. Clamp the hilum (Satinsky en bloc or bulldogs); cold-scissor excision.
3. Close vessels and collecting system; renorrhaphy; unclamp; drain.

- **Trifecta:** negative margin, no major complication within 30 days, and **WIT ≤25 min**.
- **Enucleation** gives less blood and function loss but more positive margins, with the same survival. It suits familial, multifocal, solitary-kidney and severe chronic kidney disease (CKD) cases.
- **Renorrhaphy:** sliding-clip and barbed sutures shorten warm ischaemia time (WIT), and **single-layer** closure gives better function and fewer pseudoaneurysms.
- The **collecting system** is entered in **~60%**; repair it with absorbable suture and **always drain**.
- **Ischaemia:** the historical warm limit is **30 min**, but **preserved parenchymal volume predicts function better than WIT**. Use cold ischaemia for a solitary kidney, CKD or a long expected clamp time.

Hilar strategy	Best for	Verdict
Artery-only vs artery and vein	Any	Equivalent

Hilar strategy	Best for	Verdict
Early unclamping	Any	Possibly fewer pseudoaneurysms
Selective / super-selective (near-infrared fluorescence, duplex)	Mapped segmental supply	Randomised: no 6-month benefit
Off-clamp (zero ischaemia)	Small, exophytic	Better early eGFR; no 2-yr difference; more blood loss
Parenchymal compression	Peripheral	Feasible, safe
Hypothermia	Solitary kidney, long WIT	Benefit unproven

> **KEY POINT**
>
> No clamping strategy has beaten standard clamping on long-term function: what is kept matters more than how long it is clamped.

- Robotic PN has fewer conversions and shorter WIT than laparoscopic, and less blood loss than open.
- **Positive margins (0–7%)** matter only in high-risk tumours, so the answer is **surveillance, not reoperation**.
- **Port-site recurrence is ~0.1%** and follows spillage, not the gas.
- A **hilar** tumour touches the renal artery or vein; consider enucleation with superficial renorrhaphy bites.
- Salvage surgery after PN or ablation often ends in RN, because the planes are lost.

6 · Other renal procedures

Procedure	Indication	Key points
Cyst decortication	Pain, infection, obstruction, suspected malignancy	Percutaneous aspiration first proves the cyst causes the pain; excise the wall, biopsy, fulgurate the base
Laparoscopic renal biopsy	Failed percutaneous biopsy, bleeding diathesis, solitary kidney, morbid obesity	RP, two ports, lower-pole cortex
Nephropexy	Descent >5 cm or 2 vertebrae from supine to erect, with pain, reduced flow or obstruction	Usually right side, slim young women; 20% asymptomatic. Fix the capsule to psoas or quadratus fascia
Calyceal diverticulectomy	Symptomatic; endoscopy failed	Suture the neck, obliterate the cavity with fat
Nephrolysis	Chyluria, loin pain–haematuria	Lateralise by cystoscopy; hilar lymphatic ligation with or without omental wrap
Ablation	≤3 cm, comorbid, solitary kidney, hereditary RCC	Percutaneous preferred. Cryo: two freeze–thaw cycles, iceball 0.5–1 cm beyond tumour. Radiofrequency ablation (RFA): 60–100°C (>105°C vaporises), 1 cm margin, no real-time imaging; more retreatment

7 · Complications & management

Dangerous complications happen out of view, so seek them.

*See the paper answer **Pneumoperitoneum physiology and complications of laparoscopy and retroperitoneoscopy** (Section 1, Ch 18).*

Also, from the condensed book (not in the paper answer):

- The overall complication rate is **~15%** (PN > RN > simple), with a learning curve of about 20 nephrectomies.
- **In left RN, the superior mesenteric artery (SMA) can be mistaken for the renal artery**, leading to bowel necrosis.

ALGORITHM

1. Major bleed: locate; temporary control (grasper or pressure); inform the anaesthetist (blood, open set).
2. Clear the field: raise pneumoperitoneum or add a trocar; then definitive clip or suture.
3. If that fails, then convert and call a vascular surgeon.

- Vascular injury happens at **access** or at the hilum. **Never clip or suture blindly.**
- At the end of the case, **lower the insufflation pressure** to unmask venous bleeding.
- **Haematuria at ~day 12 after PN** suggests a pseudoaneurysm or AVF and is treated by selective embolisation.
- A **urine leak** after PN (0.8–10%) is drained percutaneously, with a stent if it persists.
- **Bowel injury** (<1%, mortality ~3%) is thermal in about half. It presents with **pain at a single trocar site**, distension, diarrhoea and a **normal or low white count**.
- Pancreatic injury is insidious, so inspect at the end. A diaphragm injury shows as a sudden rise in airway pressure and is closed with a purse-string, followed by a chest X-ray.
- Bowel injury is repaired laparoscopically if stable, or by laparotomy if unstable or with gross spillage.
- **Neuromuscular** injury (2.7%) includes flank rhabdomyolysis, treated by hydration.

KEY POINT

MIS complications are the open ones in altered form and timing: the lethal ones (access vessel injury, thermal bowel injury) happen out of view and present late or atypically.

8 · Laparoscopic live donor nephrectomy †

P4 P1 *Laparoscopic donor nephrectomy — left preferred (longer vein), donor keeps better kidney; gonadal–ureter packet; artery then vein; no Hem-o-lok on donor artery; mortality 3/10 000; lifetime ESRD risk*

- † Donors need lifelong annual blood pressure and renal function checks.

143 Focal Therapy for Renal Tumours

1 · Rationale & indications

Ablation suits patients whom surgery harms more than the tumour.

- Small renal cell carcinoma (RCC) rarely kills; chronic kidney disease (CKD) and comorbidity do (Ch 137).
- **Partial nephrectomy (PN) is the gold standard** for cT1a. Thermal ablation (TA) is the alternative: cryoablation (CA), radiofrequency ablation (RFA) or microwave ablation (MWA).
- **Indications:** high surgical risk, solitary kidney, bilateral tumours, or hereditary syndromes (von Hippel–Lindau, VHL) needing repeat treatment. Recovery is faster and renal function equal or better than PN.

> **KEY POINT**
>
> **Trap:** TA is never the default for a fit patient.

Guideline	Position on TA
American Urological Association (AUA) 2021	Consider for cT1a **≤3 cm**, especially comorbid; PN prioritised; percutaneous preferred
European Association of Urology (EAU) 2022	For frail/comorbid; TA not routinely **>3 cm**; CA not **>4 cm**
National Comprehensive Cancer Network (NCCN) 2022	**Only one allowing cT1b** (non-surgical candidates); may need multiple treatments
American Society of Clinical Oncology (ASCO) 2017	Option if complete ablation achievable

2 · Selection & work-up

The tumour stays in place, so biopsy gives the only histology.

ALGORITHM

1. Counsel: age, comorbidity, CKD risk, size and location.
2. Biopsy, then re-counsel with histology.
3. Choose PN, surveillance or percutaneous ablation.

- **The AUA advises biopsy of all**, ideally at a **separate earlier session**. Post-ablation, biopsy only a new enhancing lesion.
- Treat urinary infection first; colonised urine (ileal conduit) risks abscess.
- Complication predictors are the RENAL score (low 4–6, moderate 7–9, high 10–12), American Society of Anesthesiologists (ASA) score >3 (about 3×) and $(MC)^2$ (CA only). Day-case suits ≤3 cm, low complexity and no anticoagulation.

3 · Mechanisms

*See the paper answer **Cryotherapy for CA prostate** (Section 17, Ch 163).*

Also, from the condensed book (not in the paper answer):

- Cryo target is **≤−40 °C**; the iceball extends **5–10 mm beyond tumour**, pushing the −40 °C isotherm past the tumour edge.
- Heat: **≥60 °C** causes coagulative necrosis; ≥100 °C causes vaporisation and charring. Lethal: 45 °C for 60 min, 55 °C for 5 min, **70 °C for 1 min**.
- **Heat sink:** large vessels carry heat or cold away, so perivascular tumour survives. Counter with colder or longer freezes, more probes, or MWA.

4 · Choice of energy

- RFA electrodes vary: the cool-tip gives the largest zone, the umbrella the most spherical, and a single monopolar tip suits only ≤2 cm.

	Cryoablation	**RFA**	**MWA**
Physics	Argon cools (**Joule–Thomson**); helium warms	450–1200 kHz current causes ionic friction	915 MHz/2.45 GHz causes **dielectric hysteresis**
Target	≤−40 °C	≥70 °C; set 90–105 °C	Often >100 °C
Cycles	**Double freeze–thaw** (10 + 6–8 min)	Two, 30-s cool-down	Single dose (e.g. 65 W × 5 min)
Margin	Iceball 5–10 mm	≥5 mm	≥5 mm
Charring limits	No	**Yes**	No
Grounding pad	No	Yes	No
Heat sink	Yes	Yes	**Least**
Zone visible	**Iceball**	**No**; inferred from impedance and temperature	Gas bubbles on ultrasound (US)
Pros	Spares collecting system	Longest follow-up	Fast; fewest complications
Cons	Bleeding, tumour fracture	Heat sink, pad burns	Risk to adjacent organs

> **KEY POINT**
>
> **Why RFA is self-limiting:** charring raises impedance and halts current (cap 105 °C, electrodes cooled). MWA radiates energy, so it works through desiccated tissue and blood, heats actively to 2 cm (RFA: millimetres) and outruns the heat sink.

- In CA the second freeze enlarges necrosis. MWA for cT1 uses 45–65 W for 5–15 min.

5 · Technique

- **Percutaneous is standard.** Laparoscopic is kept for prior scarring or failed hydrodisplacement. General anaesthesia with jet ventilation reduces renal motion.

ALGORITHM

1. Give intravenous (IV) antibiotics.
2. Position prone or flank on the computed tomography (CT) gantry.
3. Place probes under image guidance; confirm on non-contrast CT.
4. Ablate, then contrast CT for completeness and haematoma.

- **Hydrodisplacement** pushes organs away (colon >1 cm) with saline or 5% dextrose (**D5W for RFA**, as saline conducts current). Use caution in cardiac failure.
- **Pyeloperfusion** (central tumours) irrigates a ureteral catheter at the **opposite temperature**: cold for RFA/MWA, warm for CA.
- **CA:** cryoprobes sit **1–2 cm apart** and come out only after full thaw. Bleeding rises with size and probe number (>4 cm). **RFA:** ≥70 °C in each quadrant during cool-down excludes heat sink; overlap ablations for larger lesions.
- **MWA:** antennas go tangentially, never into the collecting system: 1 for ≤2.5 cm, 2 for 2.5–3.5, 3 for 3.5–4.5 cm.

6 · Follow-up, success & salvage

With no specimen, imaging alone shows success.

- **Success is complete loss of contrast enhancement** on CT/MRI.
- **Incomplete ablation** enhances on the first scan and is re-ablated. **Local recurrence** later enlarges or enhances and needs biopsy. **Enlargement after any modality is recurrence until proved otherwise.**
- CA zones shrink >50% in year 1; RFA zones barely shrink (**fibrotic halo**).
- **AUA:** CT/MRI with and without contrast plus chest imaging at **6, 12, 24, 36, 48, 60 months**, then shared decision. US is not routine unless a contrast-enhanced (CEUS) protocol exists.
- Salvage is re-ablation, PN, radical nephrectomy (RN), or stereotactic body radiotherapy (SBRT) if comorbid.

7 · Oncological outcomes vs PN

- There is no randomised trial. Local recurrence is commoner after TA but **equal once re-ablation is counted**.
- **Metastasis-free and cancer-specific survival are equal**; overall survival is lower in these older, comorbid patients.
- **Size predicts failure:** RFA does worse at ≥3 cm, and T1b worse than T1a. No energy is superior.

KEY POINT

Why comparisons are biased: benign and under-graded tumours flatter TA; occult pT3a penalises it; non-surgical candidates die of other causes first. Few RCC deaths occur, so trials need huge cohorts.

8 · Complications & management

- Major complications (Clavien ≥3) are **≤5%**, fewer than PN; percutaneous beats laparoscopic, and MWA is lowest.

Complication	Prevention	Management
Haemorrhage – commonest; transfusion 2–3%	Full thaw before CA probe removal; avoid intercostal arteries	Ablate (RFA/MWA coagulate); otherwise selective embolisation
Urinoma, fistula, pelviureteric junction (PUJ) or ureteric stricture	Pyeloperfusion; CA for central tumours	Conservative or stent; drain a large urinoma
Bowel or other organ injury (tumour <1 cm away)	Hydrodisplacement, positioning	Fistula: stent, low-residue diet; diversion if persistent
Pneumothorax (access above 12th rib)	Recognise during procedure	Aspirate; chest tube rarely
Nerve injury / RFA pad burn	Pads level on posterior thighs	Neuralgia is self-limiting
Infection <1%, may be lethal; abscess at 1–6 months	Urine culture, prophylaxis	†Drain and antibiotics

9 · Newer modalities (investigational)

- **High-intensity focused ultrasound (HIFU)** ablates incompletely; **histotripsy** (non-thermal) has no clinical data yet.
- **SBRT** needs no anaesthesia but leaves viable tumour on biopsy; it may have a salvage role.
- **Irreversible electroporation** spares vessels but controls cancer poorly.

144 Treatment of Advanced Renal Cell Carcinoma

1 · Overview

In metastatic RCC, drug choice rests on risk group and mechanism.

LAQ Metastatic RCC: assessment and management

Paper answer · P2 · P3 · merged from 2

Asked as: P2 LAQ "Metastatic RCC (mRCC): assessment and management, including recent concepts" · P3 SAQ "Discuss the management of Metastatic Renal Cell Carcinoma."

Introduction

- About one-third of RCC (renal cell carcinoma) patients have metastases at diagnosis; 20–40% of localised tumours metastasise later.
- Five-year survival is about 12%, but median survival now approaches 3–5 years.
- Treatment moved from cytokines to VEGF (vascular endothelial growth factor)-targeted TKIs (tyrosine kinase inhibitors), and now to ICI (immune checkpoint inhibitor) combinations. Surgery has become selective.

Assessment (before treatment)

- **Clinical:** symptoms of metastases (cough, bone pain, neurological deficit), weight loss; paraneoplastic syndromes (hypercalcaemia, polycythaemia, hypertension, Stauffer syndrome); performance status (Karnofsky or ECOG [Eastern Cooperative Oncology Group]); comorbidity.
- **Laboratory:** haemoglobin, corrected calcium, neutrophils, platelets, LDH (lactate dehydrogenase), ALP (alkaline phosphatase), creatinine, liver tests.
- **Imaging:** contrast CT (computed tomography) of chest, abdomen and pelvis (or MRI, also for IVC [inferior vena cava] thrombus). Bone scan only with bone pain, raised ALP or poor performance status; brain MRI (magnetic resonance imaging) with neurological symptoms; that is, bone scan or brain MRI only when symptoms or tests suggest spread; PET not routine.
- **Histology:** nephrectomy specimen or **percutaneous biopsy** before systemic therapy. It separates clear cell from non-clear cell RCC and shows sarcomatoid change, which alters drug choice.

Prognostic risk groups (decide first-line therapy)

Adverse factor	MSKCC (Memorial Sloan Kettering), cytokine era	IMDC (International Metastatic RCC Database Consortium), TKI era
Karnofsky <80%	Yes	Yes
Haemoglobin below lower limit of normal	Yes	Yes
Corrected calcium raised	>10 mg/dL	> upper limit of normal (ULN)
LDH >1.5 × ULN	Yes	No
Time factor	No prior nephrectomy	<1 year from diagnosis to systemic therapy

Adverse factor	MSKCC (Memorial Sloan Kettering), cytokine era	IMDC (International Metastatic RCC Database Consortium), TKI era
Neutrophils, platelets > ULN	No	Yes (two factors)

- IMDC 6 factors; MSKCC model: 5 factors including LDH.
- Favourable 0 factors, intermediate 1–2, poor ≥3. IMDC median survival: 43.2, 22.5 and 7.8 months.

Management: principles

- Multidisciplinary care: systemic therapy is the backbone; surgery and radiotherapy are added selectively.

1. First-line systemic therapy, clear cell RCC (NCCN [National Comprehensive Cancer Network] and EAU [European Association of Urology])

IMDC risk	Preferred	Alternatives (if IO, immuno-oncology, not possible)
Favourable	Pembrolizumab + axitinib; nivolumab + cabozantinib; pembrolizumab + lenvatinib	Sunitinib or pazopanib; active surveillance for slow, low-volume (indolent) disease
Intermediate/poor	The same three, **or nivolumab + ipilimumab**	Cabozantinib; sunitinib; pazopanib

- **Rationale:** VHL (von Hippel–Lindau) loss lets HIF (hypoxia-inducible factor) accumulate and drive VEGF (the TKI target). ICIs block PD-1 (programmed death-1) or CTLA-4 (cytotoxic T-lymphocyte antigen-4), releasing T-cell brakes and giving durable responses.
- **Trials (all versus sunitinib):** CheckMate 214 (nivolumab + ipilimumab: OS [overall survival] HR [hazard ratio] 0.63; overall survival benefit in intermediate/poor risk); KEYNOTE-426 (pembrolizumab + axitinib); CheckMate 9ER (nivolumab + cabozantinib); CLEAR (pembrolizumab + lenvatinib: PFS [progression-free survival] 23.9 vs 9.2 months). Benefit is independent of PD-L1 (programmed death ligand-1) status, so PD-L1 does not select patients.

2. Later lines: switch mechanism

- After an IO combination: a VEGF TKI not used before (cabozantinib, axitinib, tivozanib) or **belzutifan** (HIF-2α inhibitor). IO rechallenge is not advised (CONTACT-03 negative).
- After both IO and TKI: **belzutifan**; LITESPARK-005 showed better PFS than everolimus.
- After TKI alone: nivolumab (CheckMate 025) or cabozantinib (METEOR), both with OS gain.

3. Cytoreductive nephrectomy (CN): from routine to selective

- Cytokine era: SWOG 8949 and EORTC (European Organisation for Research and Treatment of Cancer) trials showed longer survival with CN + interferon-α.
- **CARMENA (2018):** sunitinib alone was non-inferior to CN + sunitinib in MSKCC intermediate/poor risk (median OS 18.4 vs 13.9 months).
- **SURTIME:** deferred CN after sunitinib was safe; OS favoured deferral.
- **Now:** immediate (upfront) CN only for fit patients with low metastatic burden who do not need immediate systemic therapy; **deferred CN** in responders. Avoid in poor risk, poor performance status or brain metastases.

- Palliative nephrectomy or angioembolisation for haematuria (bleeding) or pain.

4. Metastasectomy and local therapy (radiotherapy)

- Complete resection of solitary or few metastases gives 5-year survival of 35–50%. Better with complete resection, solitary lung lesion, disease-free interval >1 year, metachronous disease.
- **SBRT (stereotactic body radiotherapy)** or ablation for oligometastases; radiosurgery for brain metastases; palliative radiotherapy or surgery for cord compression and weight-bearing bones; palliative radiotherapy for bone pain.

5. Adjuvant therapy after complete resection

- **KEYNOTE-564:** pembrolizumab for 1 year after surgery in high-risk clear cell RCC, including **M1 NED** (no evidence of disease after complete metastasectomy): disease-free survival HR 0.63; FDA (Food and Drug Administration) approval 2021; a later report showed OS benefit.

6. Supportive care and special groups (non-clear cell RCC)

- Zoledronic acid or denosumab for bone metastases. Immune-related adverse events (colitis, hepatitis, hypophysitis, thyroiditis) need steroids and drug interruption.
- Non-clear cell RCC: clinical trials preferred; cabozantinib or sunitinib; gemcitabine + carboplatin (platinum) for collecting duct carcinoma.
- **Also know:** high-dose IL-2 (interleukin-2) gives durable complete response in 7–9% (another source: in a few percent); temsirolimus for poor risk; chemotherapy and hormones are ineffective.

Follow-up

- CT every 8–12 weeks on therapy (response by RECIST [Response Evaluation Criteria in Solid Tumours]); monitor thyroid, liver, blood pressure and cortisol.

Recent advances

- Belzutifan, also approved for VHL-associated tumours; neoadjuvant ICI trials; biomarkers to choose IO/IO or IO/TKI.

Algorithm

```
Metastatic RCC -> biopsy + IMDC risk -> IO/TKI (all risks) or nivo + ipi (intermediate/poor) -> deferred CN / metastasectomy / SBRT in responders -> progression: switch class (TKI, belzutifan)
```

Diagram to draw: the flowchart above (metastatic RCC, biopsy and IMDC risk, IO combination ± deferred CN in responders, metastasectomy or SBRT, switch drug class on progression).

> **Key points to remember:**
>
> - IMDC (6 factors): Karnofsky, haemoglobin, calcium, neutrophils, platelets, <1 year to therapy; poor = ≥3.
> - IO/TKI combinations for all risk groups; nivolumab + ipilimumab for intermediate/poor only.
> - CARMENA and SURTIME: no routine upfront CN.
> - Adjuvant pembrolizumab (KEYNOTE-564), including M1 NED.
> - Belzutifan after IO and TKI.

Also, from the condensed book (not in the paper answer):

- About **1/3** present with metastases; a further **20–40%** of localised tumours metastasise later. mRCC 5-yr survival is about **12%**; 10-yr <5%.

Standard care is now **immune checkpoint (immuno-oncology, IO) combinations**.

2 · Prognostic models

- **MSKCC (Motzer)** came from the cytokine era; **IMDC (Heng)** came from first-line VEGF-targeted therapy.

Factor	MSKCC (5)	IMDC (6)
Performance status	Karnofsky <80%	Karnofsky <80%
Haemoglobin	< lower limit of normal (LLN)	< LLN
Corrected calcium	**>10 mg/dL**	> upper limit of normal (ULN)
Lactate dehydrogenase (LDH)	**>1.5 × ULN**	—
Time factor	No prior nephrectomy	**<1 yr** from diagnosis to systemic therapy
Neutrophils; platelets	—	> ULN (each a factor)
Favourable (0) – median overall survival (OS)	20 mo	43.2 mo
Intermediate (1–2)	10 mo	22.5 mo
Poor (≥3)	4 mo	7.8 mo

> **KEY POINT**
>
> Three factors are shared. IMDC swaps LDH and prior nephrectomy for neutrophilia, thrombocytosis and short interval. The IMDC risk group **decides the first-line regimen**.

3 · Biology and drug targets

- The **VHL** protein (3p) tags **hypoxia-inducible factor (HIF)-α** for degradation; when VHL is lost, HIF drives VEGF, PDGF, TGF-α, GLUT-1 and erythropoietin. Drugs block the ligand (bevacizumab), receptors (tyrosine kinase inhibitors, TKIs), HIF translation (mTOR inhibitors) or HIF-2α (belzutifan). MET/AXL/FGFR escape explains broader later TKIs.
- **CTLA-4** and **PD-1/PD-L1** switch off T cells. Blocking them gives autoimmune toxicity, and benefit shows as OS and durability rather than progression-free survival (PFS).
- VEGF inhibitors are immunomodulatory, the rationale for IO plus TKI. Clear cell RCC (ccRCC) is chemoresistant.

Agent	Target / route	Signature toxicity / note
Bevacizumab	Antibody vs VEGF-A; IV	Bleeding, proteinuria, hypertension (HTN)
Sorafenib	VEGFR2, PDGFR-β, raf-1	Rash, hand–foot syndrome (HFS); rarely used
Sunitinib	VEGFR2, PDGFR-β, c-KIT, FLT3; **50 mg, 4 wk on / 2 off**	HFS, fatigue, HTN, thrombocytopenia, **hypothyroidism**

Agent	Target / route	Signature toxicity / note
Pazopanib	VEGFR, PDGFR, KIT	**Hepatotoxicity (raised ALT)**; less fatigue/HFS
Axitinib	VEGFR1–3 only – most selective	Diarrhoea, fatigue, HTN
Tivozanib	VEGFR1–3, c-KIT, PDGFR-β	Third line
Cabozantinib	VEGFR, **MET, AXL**	Fatigue, diarrhoea, HFS; frequent dose reduction
Lenvatinib	VEGFR1–3, **FGFR1–4**, PDGFR-α, RET, KIT	Partner of everolimus or pembrolizumab
Temsirolimus	mTOR; **IV weekly**	Mucositis, hyperglycaemia, hyperlipidaemia, pneumonitis
Everolimus	mTOR; oral daily	Asthenia, anaemia
Belzutifan	**HIF-2α inhibitor**; oral	After IO and TKI; also VHL-associated RCC
Nivolumab, pembrolizumab	Anti-PD-1; IV	Immune-related: colitis, hypophysitis, vitiligo
Avelumab	**Anti-PD-L1**; IV	
Ipilimumab	Anti-CTLA-4; IV	

4 · Surgery in metastatic disease

Surgery now serves systemic therapy; the question is who and when.

- **Cytoreductive nephrectomy (CN) rationale:** spontaneous regression of metastases is <1%, and a bulky primary suppresses T cells. CN helps only as a platform for systemic therapy.
- Fewer than 1/4 reach systemic therapy with ECOG 2, multiple sites, or brain or liver metastases.
- **CARMENA caveats:** about 45% were poor risk, about 25% of the CN arm were off-protocol, and it stopped early underpowered. It does **not** show harm in fit, low-burden, favourable patients.

ALGORITHM

1. Assess mRCC: histology, IMDC risk, PS and metastatic burden.
2. If fit, with low burden, favourable features and a resectable primary, then consider upfront CN followed by systemic therapy.
3. If intermediate/poor risk or high burden, then give systemic therapy first and offer deferred CN to responders.

KEY POINT

CN helps only patients fit enough, with disease slow enough, to reach systemic therapy; operating first on poor-risk patients delays it.

- **Metastasectomy** does best with complete resection, a solitary lesion, age <60, disease-free interval >1 yr, lung site, size <4 cm and metachronous disease, all markers of indolent biology.

- **Palliative** nephrectomy treats pain and paraneoplastic syndromes; try **angioembolisation for haematuria** first. Brain and spinal metastases get surgery and radiation.

5 · Systemic therapy for ccRCC

First line follows IMDC risk; later lines switch mechanism.

	IFN-α	High-dose IL-2
Response	10–15%	15–20%
Durable complete response (CR)	<2%	**7–9%**; >60% of CRs stay disease-free
Regimen	Schedule/route immaterial	**600,000–720,000 IU/kg** IV bolus 8-hourly, max 15 doses
Toxicity	Constitutional	**Vascular leak**: hypotension, multiorgan failure; early mortality 2–5%

- IL-2 is now kept for carefully selected fit patients with clear cell histology and prior nephrectomy; IFN-α is not used first line.

Risk	Preferred / standard	Alternatives (incl. if IO not possible)	Certain circumstances
Favourable	Pembrolizumab + axitinib; nivolumab + cabozantinib; pembrolizumab + lenvatinib	Avelumab + axitinib, cabozantinib, ipilimumab + nivolumab, pazopanib, sunitinib	**Active surveillance**, axitinib, high-dose IL-2
Intermediate / poor	Same three **+ ipilimumab + nivolumab**	Cabozantinib, avelumab + axitinib, pazopanib, sunitinib	Axitinib, high-dose IL-2, **temsirolimus**

KEY POINT

Choosing: IO/TKI gives higher response and PFS and is better tolerated; IO/IO gives deeper durable responses (CR 9%) with autoimmune toxicity; never compared directly. Ipilimumab + nivolumab is restricted to intermediate/poor risk (favourable did better on sunitinib in CheckMate 214). PD-L1 does not select. Use a TKI or temsirolimus if IO is contraindicated.

Prior therapy	EAU standard	NCCN recommended (no preferred)
TKI only (IO-naive)	**Nivolumab or cabozantinib** (alternative axitinib)	IO combinations, nivolumab, cabozantinib, lenvatinib + everolimus
Prior IO	Any VEGF-targeted agent not previously used	Axitinib, **belzutifan**, cabozantinib, lenvatinib + everolimus, **tivozanib**

- Tivozanib is used after ≥2 prior therapies. TKI + mTOR and nivolumab + sunitinib/pazopanib combinations failed.
- **Ipilimumab + nivolumab toxicity:** grade 3–4 events **46%**, high-dose steroids 35%, discontinuation 22%. Dose: nivolumab 3 mg/kg + ipilimumab 1 mg/kg 3-weekly × 4.
- † For immune-related events, hold IO and give steroids for grade ≥2 (prednisolone 1–2 mg/kg for grade 3–4), tapering over ≥4 weeks. Grade 4 means permanent stopping.
- TKI toxicity is managed by dose reduction or interruption; mTOR inhibitors need glucose, lipid and phosphate monitoring.

6 · Non-clear cell RCC

- Non-clear cell RCC (nccRCC) is 15–25%. No agent is proven, so offer a **trial to all**.
- Papillary tumours respond to sunitinib in only 5–10%; temsirolimus beat IFN in the nccRCC subgroup.
- **Hereditary papillary RCC (HPRC)** carries a germline activating **MET** mutation (chromosome 7) and gives type 1 papillary RCC. Sporadic papillary has **trisomy 7 in >2/3**. MET inhibitors (savolitinib, crizotinib) are active here.
- **Hereditary leiomyomatosis and RCC (HLRCC)**: fumarate hydratase loss gives type 2 papillary RCC. The cells depend on glycolysis, so **bevacizumab + erlotinib** restricts glucose (partial response about 2/3).
- **Collecting duct** carcinoma is treated with **gemcitabine + carboplatin**; **sarcomatoid** tumours with gemcitabine-based therapy.

7 · Landmark trials

Trial	Design	Result	Take-home
SWOG 8949 / EORTC	CN + IFN vs IFN	OS 11.1 vs 8.1; 17 vs 7 mo; response same	CN in cytokine era
CARMENA	CN + sunitinib vs sunitinib; noninferiority; n=450	OS 13.9 vs **18.4** mo, hazard ratio (HR) 0.89	No upfront CN in intermediate/poor
SURTIME	Immediate vs deferred CN	OS 15.0 vs **32.4** mo, HR 0.57	Systemic therapy first is safe
Sunitinib vs IFN	Phase III, first line	PFS 11 vs 5 mo; response 31 vs 6%	Later comparator
COMPARZ	Pazopanib vs sunitinib; noninferiority	PFS HR 1.05; OS similar	Equivalent; patients preferred pazopanib (PISCES)
CALGB 90206 / AVOREN	Bevacizumab + IFN vs IFN	PFS better, OS not	Little used
Temsirolimus phase III	vs IFN vs combination; poor risk, all histologies	OS 10.9 vs 7.3 mo	Only mTOR inhibitor with OS gain
TARGET	Sorafenib vs placebo after cytokines	PFS 5.5 vs 2.8 mo; OS same	Historical
Everolimus phase III	vs placebo after sunitinib/sorafenib	PFS 4.0 vs 1.9 mo; OS same	Supplanted as monotherapy
Lenvatinib ± everolimus	Phase II, after VEGF therapy	Combination PFS longer	Lenvatinib + everolimus
AXIS	Axitinib vs sorafenib, second line	PFS 6.7 vs 4.7; OS same	Second-line axitinib
METEOR	Cabozantinib vs everolimus after VEGF	PFS HR 0.58; **OS HR 0.70**	Only TKI with OS gain over active comparator
CABOSUN	Phase II; cabozantinib vs sunitinib; intermediate/poor	PFS 8.2 vs 5.6, HR 0.66	First-line cabozantinib
TIVO-3	Tivozanib vs sorafenib, after 2–3 lines	PFS HR 0.73; OS same	Third-line tivozanib

Trial	Design	Result	Take-home
LITESPARK-005	Belzutifan vs everolimus after IO + TKI	PFS HR 0.75 (curves separate late)	HIF-2α inhibition approved
CheckMate 025	Nivolumab vs everolimus after TKI; primary OS	OS 25.0 vs 19.6, **HR 0.73**; PFS same	IO benefit shows as OS
CheckMate 214	Ipilimumab + nivolumab vs sunitinib	Intermediate/poor OS HR 0.63; CR 9 vs 1%	Only IO/IO; not for favourable
KEYNOTE-426	Pembrolizumab + axitinib vs sunitinib	PFS and OS better; response 60 vs 40%	All risk groups
JAVELIN Renal 101	Avelumab + axitinib vs sunitinib	PFS better; OS not significant	Not NCCN-preferred
CheckMate 9ER	Nivolumab + cabozantinib vs sunitinib	PFS 16.6 vs 8.3, HR 0.51; OS HR 0.70	All risk groups
CLEAR	Pembrolizumab + lenvatinib vs sunitinib	PFS 23.9 vs 9.2, **HR 0.39**; response 71%	Largest PFS/response effect

★ Putting it together

A · Small renal mass

	Active surveillance (AS)	Ablation (Ch 143)	Partial nephrectomy (PN)	Radical nephrectomy (RN)
Best for	Frail, life expectancy <5 y; <2 cm or cystic	cT1a **≤3 cm**, comorbid; biopsy first	cT1a standard; imperative if solitary, bilateral, hereditary or CKD	Only if high complexity, no CKD/proteinuria, and new eGFR **>45**
Numbers	Growth 0.1–0.3 cm/y; treat if >3 cm or >5 mm/y	Cryo: 2 freeze–thaw cycles; RFA/MWA >60–70 °C	Trifecta: R0, no complication, warm ischaemia ≤25 min	New GFR ≈ GFR × contralateral split × 1.25
Weakness	Imaging adherence	More local recurrence; no pathology	Leak, bleeding	CKD 65% vs 20% after PN

B · Renal lesions

Lesion	Imaging	Genetics / IHC	Action
Simple cyst	≤20 HU, no enhancement	—	Nothing
Complex cyst	Bosniak v2019	—	≤2 cm surveil; >4 cm excise
Angiomyolipoma (AML)	Fat ≤−20 HU	TSC2/TSC1; HMB-45, Melan-A+	Embolise if >6 cm, aneurysm >5 mm or bleeding; everolimus in TSC
Oncocytoma	Central scar	KIT+, CK7 single cells; Birt–Hogg–Dubé	Surveil or PN (chromophobe mimic)
Clear cell RCC	Hypervascular	3p/VHL loss activates HIF-2α; CA-IX	PN/RN/ablation/AS
Papillary RCC	Hypoenhancing	Trisomy 7, 17; loss of Y; MET	As RCC
Chromophobe RCC	Moderate enhancement	Multiple losses; diffuse CK7	As RCC; good prognosis
Lymphoma/metastasis	Infiltrative	—	**Biopsy; don't operate**
UTUC	Filling defect, ~46 HU, cytology+	MSI suggests Lynch	Risk-stratify, then KSM or RNU

C · UTUC: kidney-sparing (KSM) vs radical nephroureterectomy (RNU)

	KSM (endoscopic or segmental)	RNU + bladder cuff
Who	Low risk: ALL of unifocal, <2 cm, low grade, no invasion on CT urogram; or imperative (solitary, CKD, bilateral)	ANY of multifocal, >2 cm, high grade, hydronephrosis, prior cystectomy, variant

	KSM (endoscopic or segmental)	RNU + bladder cuff
Extras	Surveillance ureteroscopy; UGN-101 mitomycin gel	≤12 wk; template node dissection; single post-op intravesical chemo
Systemic	—	Neoadjuvant platinum if post-op CrCl <50; adjuvant (POUT) ≥pT2/N+

D · Open vs minimally invasive

- **Open (Ch 141)** is kept for IVC thrombus needing bypass, complex anatomy, prior surgery and trauma.
- **Laparoscopic/robotic (Ch 142)** is standard for organ-confined RN and most PN, with faster recovery and the same oncology.

E · Metastatic RCC

ALGORITHM

1. **Score IMDC (6 factors, Ch 144)**: 0 is favourable, 1–2 intermediate, ≥3 poor.
2. **First line**: IO + TKI (pembro + axitinib, nivo + cabozantinib, pembro + lenvatinib) at any risk; ipi + nivo only if intermediate/poor.
3. **Cytoreductive nephrectomy**: upfront only if fit and low burden; otherwise deferred (CARMENA, SURTIME).
4. **Later lines**: switch mechanism; belzutifan after IO + TKI.

Other paper questions in this section

Asked in the papers but not tied to a specific place in the book above.

LAQ Discuss differential diagnosis of small renal tumours. Describe techniques to achieve hemostasis during laparoscopic partial nephrectomy.

Paper answer · P3

Type: LAQ (25 marks)
Source: 2019-Summer Paper 3, Q1

Definition and biology

- **SRM (small renal mass):** an enhancing solid or complex cystic renal mass **≤4 cm (cT1a)**, usually found incidentally.
- **Benign share falls with size:** 20–30% of masses ≤4 cm and about 40% of masses ≤2 cm are benign. At ≥7 cm few are benign and about half are high grade.
- Male sex triples the risk of malignancy. Under 2% of small cancers metastasise.

Differential diagnosis

Group	Entities	Pointers
RCC (renal cell carcinoma)	**Clear cell** (commonest), papillary (10–15%), chromophobe (3–5%), cystic RCC, rare molecular subtypes	Clear cell enhances strongly; papillary enhances weakly
Other malignant	Urothelial carcinoma, lymphoma, metastasis, sarcoma, epithelioid AML (angiomyolipoma)	Infiltrative growth
Benign solid	Oncocytoma, fat-poor AML, papillary adenoma, metanephric adenoma	See below
Cystic	Haemorrhagic or complex cyst, cystic nephroma	Bosniak category
Inflammatory, vascular	Abscess, focal pyelonephritis, tuberculosis; aneurysm, AVM (arteriovenous malformation), infarct, pseudotumour	Clinical picture

- **Oncocytoma:** commonest benign enhancing SRM (up to half of benign SRMs). Central stellate scar and segmental enhancement inversion suggest it but are not specific. A core cannot separate it from chromophobe RCC, which has oncocytic areas. Sestamibi SPECT (single-photon emission computed tomography) is under evaluation.
- **Fat-poor AML:** no visible fat (fat below −20 HU [Hounsfield units] diagnoses classic AML); homogeneous, T2-dark, mimics RCC; biopsy HMB-45 and Melan-A positive.
- **Also know:** metanephric adenoma (hypovascular, polycythaemia, WT1 positive); papillary adenoma (≤15 mm).

Narrowing the differential

- **Multiphase CT (computed tomography):** enhancement over 15–20 HU means solid tissue. MRI (magnetic resonance imaging) helps below 2 cm.
- **RMB (renal mass biopsy)** if it will change management (AUA [American Urological Association]); seeding almost unreported; non-diagnostic about 14%.
- Then surveillance, ablation or **PN (partial nephrectomy)**, preferred for cT1a.

Haemostasis in LPN (laparoscopic partial nephrectomy)

Principle: a clear field for excision, then secure closure of the bed. Plan with R.E.N.A.L. score and 3D (three-dimensional) CT; intraoperative ultrasound.

1. Vascular control

Technique	Method	Comment
Main artery ± vein clamp	Bulldog clamps, or laparoscopic Satinsky en bloc	Artery-only equals artery–vein. Aim warm ischaemia ≤25 min (trifecta)
Selective or super-selective	Clamp the tumour's segmental branch; 3D CT, NIRF (near-infrared fluorescence)	Avoids global ischaemia; no functional gain in an RCT (randomised controlled trial)
Off-clamp ("zero ischaemia")	No clamping; small exophytic tumours	More blood loss; same long-term eGFR (estimated glomerular filtration rate)
Early unclamping	Release after the inner suture layer	Shorter ischaemia; possibly fewer pseudoaneurysms
Cold ischaemia	Ice slush, cold saline via ureteric catheter, cold arterial perfusion	Solitary kidney, CKD (chronic kidney disease), long repair; benefit unproven

- Function depends mainly on parenchyma preserved. Mannitol: no benefit in an RCT. **Also know:** parenchymal clamp for peripheral tumours.

2. Excision: score the capsule, cut with cold scissors, clip visible vessels. Enucleation along the pseudocapsule bleeds less.

3. Renorrhaphy (most reliable step)

- **Inner layer:** running 3-0 or 2-0 absorbable suture closes cut vessels and collecting system (opened in about 60%).
- **Outer layer, sliding-clip technique:** mattress sutures anchored by a locking clip; clips slid down to the capsule give tension without knots. Pledgets prevent capsular tears; oxidised cellulose bolster fills the defect.
- **Barbed suture:** holds tension without knots, so ischaemia is shorter.
- Single layer gave better function and fewer pseudoaneurysms than two layers.

4. Energy: argon beam or bipolar coagulation of the bed. Argon can destroy collecting-system sutures and adds gas, so vent the ports.

5. Haemostatic agents (adjuncts, no proven benefit in RCTs): gelatin–thrombin matrix; fibrin sealant; oxidised cellulose; PEG (polyethylene glycol) hydrogel; cyanoacrylate. Sutureless closure only for selected small tumours.

6. Final check: unclamp, inspect, **lower pneumoperitoneum** to unmask venous bleeding; drain.

Postoperative bleeding

- Early (hypotension, falling haemoglobin): resuscitate, CT, angioembolisation.
- **Delayed haematuria (about day 12):** pseudoaneurysm or arteriovenous fistula; selective angioembolisation preserves function. Endophytic and posterior tumours carry higher risk.

Diagram to draw: sliding-clip renorrhaphy over a bolster, with bulldog clamp on the artery.

Key points to remember:

- SRM ≤4 cm: 20–30% benign; ≤2 cm about 40%.
- Oncocytoma is the commonest benign mimic; biopsy cannot exclude chromophobe RCC.
- Trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
- Delayed haematuria after PN means pseudoaneurysm: angioembolise.

SAQ Immunotherapy / immunomodulation in RCC

Paper answer · P2 · P4 · merged from 2

Asked as: P2 SAQ "Immunotherapy in (metastatic) RCC" (2024-Winter Paper II Q3e; Practice Set 4 Paper II Q3f) · P4 SAQ "Immuno modulation in Renal, cell carcinoma" (2025-Winter Paper 4 Q3a)

Assumed meaning: immunomodulation (immunotherapy) in renal cell carcinoma. Overall management of metastatic RCC: see 2024-Winter Paper 3, Q3b.

Rationale

- RCC (renal cell carcinoma) is immunogenic and chemoresistant: rare (<1%) spontaneous regression after nephrectomy, and tumour-infiltrating T cells.
- **Checkpoints:** PD-1 (programmed death-1) on CD8 T cells binds tumour PD-L1 and switches the T cell off; CTLA-4 (cytotoxic T-lymphocyte antigen-4) on regulatory T cells damps priming. Blockade **releases the brake** on existing anti-tumour immunity.
- VEGF (vascular endothelial growth factor) is immunosuppressive, so TKIs (tyrosine kinase inhibitors) synergise with checkpoint inhibitors.

Cytokine era (non-specific stimulation)

	IFN-α (interferon-alpha)	High-dose IL-2 (interleukin-2)
Response	10–15%	15–20%
Durable complete response	<2%	**7–9%**
Toxicity	Flu-like	**Capillary leak**; needs intensive care
Role	Obsolete	Selected fit patients only

- High-dose IL-2 dose: 600,000–720,000 IU/kg intravenously every 8 hours, maximum 15 doses; over 60% of complete responders stay disease-free.

Checkpoint inhibitors: trials (all clear cell)

Trial	Regimen vs comparator	Result
CheckMate 025	Nivolumab vs everolimus, second line after TKI	OS (overall survival) 25.0 vs 19.6 months
CheckMate 214	Nivolumab + ipilimumab vs sunitinib	Intermediate/poor risk: OS HR (hazard ratio) 0.63; complete response 9%
KEYNOTE-426	Pembrolizumab + axitinib vs sunitinib	Better PFS (progression-free survival) and OS
CheckMate 9ER	Nivolumab + cabozantinib vs sunitinib	Better PFS and OS; PFS 16.6 vs 8.3 months
CLEAR	Pembrolizumab + lenvatinib vs sunitinib	PFS 23.9 vs 9.2 months; response 71%

Trial	Regimen vs comparator	Result
JAVELIN	Avelumab + axitinib	No OS benefit

Treatment sequencing

- **First line:** an IO/TKI (immuno-oncology plus TKI) combination for all IMDC (International Metastatic RCC Database Consortium) risk groups; nivolumab + ipilimumab for intermediate/poor risk only.
- **After IO:** switch to a VEGF-targeted TKI. IO rechallenge is not recommended: it failed in CONTACT-03 and TiNivo-2. Belzutifan after IO and TKI.

Adjuvant

- **KEYNOTE-564:** pembrolizumab for 1 year after nephrectomy in high-risk clear cell RCC, including resected metastases (M1 NED, no evidence of disease). DFS (disease-free survival) HR 0.68 (P4 and another source: HR 0.63); later OS benefit (HR 0.62).
- Atezolizumab and nivolumab-based adjuvant trials were negative.

Toxicity: irAEs (immune-related adverse events)

- Colitis, hepatitis, pneumonitis, thyroiditis, hypophysitis, nephritis, myocarditis, rash.

Grade	Action
1	Continue, monitor
2	Hold; prednisolone 0.5–1 mg/kg
3–4	Stop; high-dose steroids (methylprednisolone 1–2 mg/kg); infliximab or mycophenolate if refractory

- Endocrinopathies: hormone replacement.

Biomarkers and also know

- **PD-L1 does not select patients.** Sarcomatoid tumours respond well. Gene signatures are investigational; antibiotics may reduce benefit.
- Non-clear cell data are scant.
- Vaccines and allogeneic transplant failed.

Diagram to draw

- T cell–tumour synapse with PD-1/PD-L1 and CTLA-4/B7 blocked.

> **Key points to remember:**
> - IO/TKI for all risks; nivolumab + ipilimumab for intermediate/poor only.
> - High-dose IL-2: durable complete response 7–9%.
> - CheckMate 214: nivolumab + ipilimumab, OS HR 0.63, complete response 9%.
> - KEYNOTE-564: adjuvant pembrolizumab, DFS and OS benefit.
> - irAE grade ≥2: hold drug and give steroids.

SAQ Biomarkers for patients with advanced RCC

Paper answer · P2

Type: SAQ (10 marks)
Source: 2023-Winter Paper 2, Q3d

Introduction

- A **prognostic** biomarker predicts outcome regardless of treatment; a **predictive** biomarker predicts benefit from a specific drug.
- In advanced RCC (renal cell carcinoma), only clinical risk models are used routinely. No molecular biomarker is yet validated to select treatment (EAU, European Association of Urology; NCCN, National Comprehensive Cancer Network).

Clinical and laboratory: risk models

Factor	MSKCC (Memorial Sloan-Kettering Cancer Center)	IMDC (International Metastatic RCC Database Consortium)
Karnofsky performance status <80%	Yes	Yes
Haemoglobin < lower limit of normal	Yes	Yes
Corrected calcium high	>10 mg/dL	> upper limit of normal
LDH (lactate dehydrogenase) >1.5 × normal	Yes	No
Time factor	No prior nephrectomy	<1 year from diagnosis to treatment
Neutrophils, platelets > normal	No	Yes

- IMDC: favourable 0 factors (median survival 43 months), intermediate 1–2 (22.5 months), poor ≥3 (7.8 months).
- IMDC group guides first-line choice (nivolumab–ipilimumab not for favourable risk).
- **Also know:** neutrophil-to-lymphocyte ratio, CRP (C-reactive protein), liver or bone metastases (poor prognosis).

Tissue biomarkers

- **PD-L1 (programmed death-ligand 1):** benefit of checkpoint combinations is **independent of PD-L1**, so it is not used to select patients.
- **Sarcomatoid features:** respond well to checkpoint inhibitors.
- **Gene expression signatures:** IMmotion150 showed high T-effector signature tumours do better with atezolizumab–bevacizumab; high angiogenic signature tumours do well with sunitinib. Investigational.
- **Genomic:** BAP1 loss and CDKN2A loss indicate poor prognosis; PBRM1 data conflicting. MET alterations in papillary RCC (cabozantinib); FH (fumarate hydratase) deficiency. TMB (tumour mutational burden) is low and not predictive.

Circulating and other biomarkers

- **KIM-1 (kidney injury molecule-1):** high serum levels predict recurrence and benefit from adjuvant immunotherapy (investigational).
- **ctDNA (circulating tumour DNA):** RCC sheds little DNA; methylation assays are promising.
- **Also know:** antibiotics during checkpoint therapy reduced survival (microbiome); sunitinib-induced hypertension is linked to better outcome.

Diagram to draw: IMDC risk groups with survival curves.

Key points to remember:

- IMDC: 6 factors; poor risk ≥3; LDH only in MSKCC.
- PD-L1 does not select patients in RCC.
- Sarcomatoid tumours respond well to immunotherapy.

SAQ Tivozanib

Paper answer · P2

Type: SAQ (10 marks)
Source: 2024-Winter Paper 2, Q3b

Introduction

- Oral **VEGFR TKI (vascular endothelial growth factor receptor tyrosine kinase inhibitor)** used in advanced RCC (renal cell carcinoma).
- Targets **VEGFR-1, -2 and -3** (also c-KIT and PDGFR-β, platelet-derived growth factor receptor-β); highly potent and selective.

Mechanism and rationale

- In clear cell RCC, **VHL (von Hippel–Lindau) loss** lets HIF (hypoxia-inducible factor) accumulate, driving VEGF and angiogenesis.
- Tivozanib blocks VEGFR signalling, starving the tumour of new vessels.
- High selectivity means fewer off-target effects (less hand-foot skin reaction and diarrhoea than sorafenib).

Dose and pharmacology

- **1.34 mg orally once daily, 21 days of a 28-day cycle.**
- Long half-life (about 4–5 days); dose reduction for toxicity or hepatic impairment.

Key trials

Trial	Setting	Result
TIVO-1 (phase III)	0–1 prior therapy, versus sorafenib	PFS (progression-free survival) 11.9 vs 9.1 months; response 33% vs 23%; OS (overall survival) not better (28.8 vs 29.3 months)
TIVO-3 (phase III)	After 2–3 prior systemic therapies including a VEGFR TKI, versus sorafenib	PFS 5.6 vs 3.9 months (HR, hazard ratio, 0.73); response 18% vs 8%; OS not different (HR 0.97); benefit also after prior checkpoint inhibitor
TiNivo-2 (phase III)	After checkpoint inhibitor; tivozanib ± nivolumab	**Negative:** adding nivolumab did not improve PFS (no immunotherapy re-challenge)

Approval and place

- FDA (Food and Drug Administration) declined in 2013 because of the OS concern in TIVO-1; EMA (European Medicines Agency) approved first-line use in 2017.
- **FDA March 2021:** relapsed or refractory advanced RCC **after ≥2 prior systemic therapies**.
- NCCN (National Comprehensive Cancer Network): later-line option, mainly after immunotherapy combinations.

Adverse effects

- **Hypertension** (commonest; control before starting), dysphonia, fatigue, diarrhoea, proteinuria, hypothyroidism.
- Bleeding, arterial thrombosis, cardiac failure, **impaired wound healing** (stop before surgery), RPLS (reversible posterior leukoencephalopathy syndrome).
- **Monitoring:** blood pressure, thyroid tests, urine protein, liver tests.

Diagram to draw: VHL loss → HIF → VEGF → VEGFR, blocked by tivozanib.

> **Key points to remember:**
> - Selective VEGFR-1–3 TKI; 1.34 mg daily, 3 weeks on, 1 off.
> - TIVO-3: PFS 5.6 vs 3.9 months; OS not different.
> - FDA 2021: after ≥2 prior systemic therapies.
> - TiNivo-2 negative; main toxicity hypertension.

SAQ Zero-ischaemia partial nephrectomy (laparoscopic and robotic)

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 3 Paper 4, Q3e (clamping strategies and laparoscopic partial nephrectomy haemostasis also in 2019-Summer Paper 3, Q1)

Definition and rationale

- **Zero-ischaemia PN (partial nephrectomy):** excision and renorrhaphy **without clamping the main renal artery**, so the kidney stays perfused (Gill and colleagues, 2011).
- **Why:** warm ischaemia injures nephrons (historical limit 30 minutes; trifecta ≤25). Most relevant in a **solitary kidney or CKD (chronic kidney disease)**.

Techniques

Technique	Method
Off-clamp	Hilum looped for rescue but not clamped; cold-scissor excision with strong suction
Anatomical (super-selective)	Microdissect the higher-order arterial branches entering the tumour; occlude only these with small bulldogs
Selective segmental clamping	Only the tumour's segment is ischaemic ("minimal ischaemia")
Parenchymal compression	Parenchymal clamp across a polar tumour
Enucleation	Pseudocapsule plane bleeds less, so suits off-clamp work

Aids

- **3D (three-dimensional) CT (computed tomography) reconstruction** maps segmental arteries; intraoperative Doppler ultrasound.
- **NIRF (near-infrared fluorescence) with ICG (indocyanine green)** on robotic consoles: after selective clamping, perfused kidney glows and the ischaemic tumour zone stays dark.
- **Robot:** wristed instruments and fast suturing ease off-clamp work.

- **Haemostasis:** clip or suture bed vessels as met; sliding-clip or barbed renorrhaphy; clamp if bleeding hides the margin.

Selection

- **Ideal:** small (cT1a), exophytic, peripheral tumour; experienced surgeon.
- **Caution:** endophytic, hilar or central tumours.

Evidence

- Off- versus on-clamp: **better early eGFR (estimated glomerular filtration rate), more blood loss**; **no difference beyond the early period**.
- **Robotic RCTs (randomised controlled trials, e.g. CLOCK):** no difference in eGFR or split function up to 2 years.
- RCT of super-selective clamping with NIRF: no 6-month gain; stopped for futility.
- **Concept:** parenchymal volume preserved, not ischaemia time within accepted limits, governs function.
- **Also know:** early unclamping; hypothermia for long ischaemia; mannitol no benefit.

Diagram to draw: hilum with a micro-bulldog on a tertiary branch; tumour zone dark on NIRF.

> **Key points to remember:**
> - Zero ischaemia: no main-artery clamp (Gill, 2011); off-clamp or super-selective.
> - NIRF with ICG confirms the right branch.
> - RCTs: no long-term eGFR gain; more blood loss.
> - Parenchyma preserved matters more than clamp time.

Rapid Revision Sheet · Section 13

ADPKD (Adult polycystic kidney disease) (SAQ)

- PKD1: chromosome 16, 85%, earlier ESRD.
- Ravine: <30 years ≥2 cysts; 30–59 ≥2 each kidney; ≥60 ≥4 each.
- Tolvaptan (TEMPO 3:4, REPRISE); ACE inhibitor or ARB.

Acquired renal cystic disease (SAQ)

- ≥3 cysts per kidney in ESRD; >90% after 10 years.
- Uraemia causes it; transplant may regress cysts, not risk.
- RCC risk 3 times; image dialysis patients with haematuria.

Renal angiomyolipoma and its management (SAQ)

- Macroscopic fat on non-contrast CT is diagnostic.
- Treat if >6 cm, aneurysm >5 mm, growth, pregnancy or symptoms.
- SAE first for bleeding; partial nephrectomy preferred; everolimus for TSC >3 cm.

VHL gene and von Hippel–Lindau disease (SAQ)

- AD; 3p25 tumour suppressor; VHL protein degrades HIF.
- VHL loss: HIF-2α drives (→) VEGF, giving vascular clear cell RCC.
- Type 2 = phaeochromocytoma; 3 cm rule for nephron sparing (nephron-sparing surgery).
- Belzutifan (HIF-2α inhibitor) approved 2021.

Radiology for solid renal masses. (LAQ)

- Enhancement ≥20 HU = solid tumour; macroscopic fat = AML.
- MRI: clear cell T2-bright with chemical-shift drop; papillary T2-dark and hypoenhancing; ccLS 1–5.
- Sestamibi for oncocytoma; ^{89}Zr-girentuximab PET (ZIRCON) for clear cell.
- Biopsy PPV ~99%, non-diagnostic ~14%.

Nephron-sparing surgery for renal masses: indications, approaches and recent advances (LAQ)

- Ureteric bud gives collecting system; blastema gives nephrons (GDNF–RET).
- Five segmental end arteries; veins anastomose; Brödel line avascular; puncture at the fornix.
- SRM ≤4 cm: 20–30% benign; PN is standard for cT1a, ablation <3 cm, surveillance <2 cm.
- Imperative, relative, elective; elective PN is standard for cT1a; ablation or surveillance for frail patients with tumours <3 cm.
- R.E.N.A.L. 4–6 / 7–9 / 10–12; trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
- AUA: RN only if complex + no CKD + new eGFR/GFR likely >45.
- Retroperitoneal for posterior, transperitoneal for anterior tumours.
- Preserved parenchyma, not clamp time, governs function.
- Enucleation for hereditary/multifocal; ablation for cT1a <3 cm after biopsy.
- Off-clamp and super-selective clamping: no long-term eGFR gain in randomised trials.
- 3D models: randomised benefit.

- EORTC 30904: PN better function, similar cancer control.
- Neoadjuvant TKI shrinks tumours 10–30%; for imperative PN only.

Upper tract urothelial carcinoma (LAQ)

- Tobacco, aristolochic acid (A:T → T:A in TP53) and Lynch (MSH2) are the key causes.
- CTU is the best test; URS biopsy gives grade, not stage.
- Ureteroscopy is closed but cannot stage; percutaneous stages but can seed the tract.
- Pelvic tumour into parenchyma is T3; through kidney to perinephric fat is T4; any N+ is stage IV.
- EAU low risk needs all favourable features (unifocal, under 2 cm, low grade, no invasion); any single adverse feature makes it high risk.
- Ureteric tumours: 70% distal; hydronephrosis predicts invasion (up to 80%).
- Ureteric T3 does worse than pelvic T3 (no parenchymal barrier).
- Distal ureterectomy for selected distal tumours.
- RNU with complete bladder cuff + single postoperative intravesical MMC: bladder recurrence 28% to 16% (ODMIT-C).
- POUT: adjuvant platinum, 3-year DFS 71% vs 46%; EV-302 for metastatic disease.

Vascular grafts and vascular surgery in urology (LAQ)

- Proximal and distal control, heparin, flush, everting Prolene anastomosis.
- Autologous for small or contaminated fields; PTFE or Dacron for large vessels.
- Mayo levels decide caval control and bypass.
- IVC: patch if lumen <50%; PTFE 16–20 mm replacement if wall invaded.
- PC-RPLND: IVC may be ligated; aorta must be grafted.
- Transplant: Carrel patch, bench reconstruction; preserve the lower-pole artery.
- Bench surgery plus autotransplantation for complex renovascular lesions.

Role of renal nephrometry in management of RCC. (SAQ)

- RENAL 4–12: low 4–6, moderate 7–9, high 10–12.
- PADUA 6–14; C-index lower = more central.
- Predicts PN feasibility, ischaemia time and complications.
- MAP score: adherent perinephric fat. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Metastatic RCC: assessment and management (LAQ)

- IMDC (6 factors): Karnofsky, haemoglobin, calcium, neutrophils, platelets, <1 year to therapy; poor = ≥3.
- IO/TKI (IO combinations) for all risk groups; nivolumab + ipilimumab for intermediate/poor only.
- CARMENA and SURTIME: no routine upfront CN.
- Adjuvant pembrolizumab (KEYNOTE-564), including M1 NED.
- Belzutifan after IO and TKI.

Discuss differential diagnosis of small renal tumours. Describe techniques to achieve hemostasis during laparoscopic partial nephrectomy. (LAQ)

- SRM ≤4 cm: 20–30% benign; ≤2 cm about 40%.
- Oncocytoma is the commonest benign mimic; biopsy cannot exclude chromophobe RCC.

- Trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
- Delayed haematuria after PN means pseudoaneurysm: angioembolise.

Immunotherapy / immunomodulation in RCC (SAQ)

- IO/TKI for all risks; nivolumab + ipilimumab for intermediate/poor only.
- High-dose IL-2: durable complete response 7–9%.
- CheckMate 214: nivolumab + ipilimumab, OS HR 0.63, complete response 9%.
- KEYNOTE-564: adjuvant pembrolizumab, DFS and OS benefit.
- irAE grade ≥2: hold drug and give steroids.

Biomarkers for patients with advanced RCC (SAQ)

- IMDC: 6 factors; poor risk ≥3; LDH only in MSKCC.
- PD-L1 does not select patients in RCC.
- Sarcomatoid tumours respond well to immunotherapy.

Tivozanib (SAQ)

- Selective VEGFR-1–3 TKI; 1.34 mg daily, 3 weeks on, 1 off.
- TIVO-3: PFS 5.6 vs 3.9 months; OS not different.
- FDA 2021: after ≥2 prior systemic therapies.
- TiNivo-2 negative; main toxicity hypertension.

Zero-ischaemia partial nephrectomy (laparoscopic and robotic) (SAQ)

- Zero ischaemia: no main-artery clamp (Gill, 2011); off-clamp or super-selective.
- NIRF with ICG confirms the right branch.
- RCTs: no long-term eGFR gain; more blood loss.
- Parenchyma preserved matters more than clamp time. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

SECTION 14

Adrenal & Retroperitoneum

145 Surgical and Radiographic Anatomy of the Adrenals

LAQ Phaeochromocytoma: adrenal anatomy, hormones and perioperative management

Paper answer · P1 · P2 · merged from 2

Asked as: P1 LAQ "Describe anatomy of arenal glands AND steps in production of adrenal hormones & perioperative management of a patient of pheochromocytoma" (Practice Set 6 Paper 1 Q1) · P2 SAQ "Phaeochromocytoma: overview, pre-operative and intra-operative management" (asked as "Pre operative management of pheochromocytoma", Summer 2020 Paper II Q3c; "Intra op mgt of pheo", Practice Set 7 Paper II Q3c; "Pheochromocytoma", Practice Set 1 Paper II Q3d)

Assumed meaning: "arenal" = adrenal glands.

Adrenal anatomy

Gross

- Paired retroperitoneal glands on the upper renal poles (11th–12th ribs), inside Gerota fascia, separated from the kidney by thin connective tissue. Yellow-orange, 4–6 × 2–3 cm, about 5 g each.

Feature	Right	Left
Shape, level	Triangular; higher	Crescentic; lower (to L1)
Relations	Liver anteriorly, duodenum anteromedially, IVC (inferior vena cava) medially, diaphragmatic crus behind	Pancreas body and splenic vessels anteriorly, aorta medially, crus behind
Vein	**Short (<1 cm), into posterior IVC**: tears easily	**Longer, into left renal vein**, often joined by inferior phrenic vein

Blood supply

- **"Many arteries in, one vein out."** Superior adrenal (from inferior phrenic), middle (aorta) and inferior (renal artery) arteries break into 10–50 small branches.
- Cortical sinusoids drain into the medulla, so cortisol-rich blood reaches the chromaffin cells. **Cortisol induces PNMT (phenylethanolamine-N-methyltransferase)**, which is why the medulla makes adrenaline.

Nerves: preganglionic sympathetic fibres (T11–L2, via splanchnic nerves, no synapse in the coeliac ganglion) end directly on chromaffin cells. The medulla is a modified sympathetic ganglion.

Embryology: cortex from coelomic mesoderm (adrenogonadal ridge, weeks 3–4); medulla from neural crest. Hence adrenals stay orthotopic in renal agenesis, and adrenal rests occur along the path of gonadal descent.

Histology (zones from outside in: glomerulosa, fasciculata, reticularis = "salt, sugar, sex")

Zone	Share	Product	Main control
Glomerulosa	~15% of cortex	Aldosterone	Angiotensin II, potassium
Fasciculata	~80%	Cortisol (up to 20 mg/day)	ACTH (adrenocorticotropic hormone)
Reticularis	5–7%	DHEA (dehydroepiandrosterone), DHEA-S, androstenedione	ACTH
Medulla	~10% of gland	Adrenaline 80%, noradrenaline 19%, dopamine 1%	Sympathetic nerves

Steps in adrenal hormone production

1. Steroids (cortex)

- Cholesterol (mainly from LDL [low-density lipoprotein]) is carried into mitochondria by **StAR (steroidogenic acute regulatory protein)**: the rate-limiting step, driven by ACTH.
- **Side-chain cleavage enzyme (CYP11A1)** converts it to **pregnenolone**. The enzymes present in each zone then decide the product: `Pregnenolone -3βHSD-> Progesterone -21OH-> 11-deoxycorticosterone -11βOH-> Corticosterone -aldosterone synthase (CYP11B2)-> ALDOSTERONE (glomerulosa only) Pregnenolone -17αOH-> 17-OH pregnenolone/progesterone -21OH-> 11-deoxycortisol -11βOH-> CORTISOL (fasciculata) 17-OH pregnenolone -17,20-lyase-> DHEA -> androstenedione (reticularis)`
- 3βHSD (3β-hydroxysteroid dehydrogenase). Steroids are not stored; they diffuse out and act on intracellular receptors.
- **Regulation:** CRH (corticotropin-releasing hormone) drives ACTH, which drives cortisol, with negative feedback and a morning peak; RAAS (renin–angiotensin–aldosterone system) and potassium control aldosterone.
- **Clinical link:** 21-hydroxylase deficiency is the commonest cause of CAH (congenital adrenal hyperplasia).

2. Catecholamines (medulla) `Tyrosine -tyrosine hydroxylase (rate-limiting)-> L-DOPA -decarboxylase-> Dopamine -dopamine β-hydroxylase-> Noradrenaline -PNMT (cortisol-induced)-> Adrenaline`

- Stored in vesicles, released by exocytosis. **COMT (catechol-O-methyltransferase)** inside chromaffin cells forms **metanephrine and normetanephrine** continuously, which is why metanephrines are the best test. MAO (monoamine oxidase) and the liver form VMA (vanillylmandelic acid).

Phaeochromocytoma: definition and epidemiology

- Catecholamine-secreting tumour of adrenal medullary chromaffin cells; **paraganglioma** is the extra-adrenal form (organ of Zuckerkandl, bladder).
- 1–2 per 100,000 a year; about 5% of adrenal incidentalomas.
- **"Rule of 10" is outdated:** extra-adrenal up to 25%, **hereditary about one-third**, bilateral commoner in hereditary disease.

Genetics

- **MEN2 (multiple endocrine neoplasia type 2)**, RET: about 50% risk, adrenal, low malignancy.
- **VHL (von Hippel–Lindau)**: normetanephrine only.
- **NF1 (neurofibromatosis type 1)**.
- **SDHB (succinate dehydrogenase subunit B):** extra-adrenal, **30–50% metastatic**. (SDHB/SDHD = succinate dehydrogenase subunits.)
- Offer genetic testing to all: about one-third hereditary (MEN2, VHL, NF1, SDHB/SDHD).

Clinical features

- Triad: **headache, sweating, palpitations**. Sustained hypertension is commoner than paroxysmal; pallor, hyperglycaemia; >20% asymptomatic. Epinephrine-dominant tumours can cause hypotensive episodes.

Why preparation is needed

- Handling the tumour releases catecholamines: hypertensive crisis, arrhythmia, MI (myocardial infarction), stroke. After vein ligation, sudden withdrawal causes hypotension and hypoglycaemia.
- Phaeochromocytoma surgery is a **two-phase haemodynamic problem**: catecholamine **surge** while the tumour is handled, then catecholamine **withdrawal** once the vein is tied.
- Surges occur at intubation, positioning, pneumoperitoneum and tumour manipulation.
- Mortality fell from up to 50% to <3% with blockade, now advised for **all** proven tumours, even normotensive.

Pre-operative step 1: confirm, localise, test genes

- **Biochemistry first:** plasma free or 24-hour urinary fractionated **metanephrines**. Tumours convert catecholamines continuously, so metanephrines are more sensitive than episodic catecholamines. Chromogranin A; clonidine suppression for borderline results.
- **Imaging:** CT (computed tomography): >10 HU (Hounsfield units) unenhanced, no rapid washout; MRI (magnetic resonance imaging).
- **Functional imaging** if extra-adrenal or hereditary: ^{68}Ga-DOTATATE PET (positron emission tomography) or MIBG (metaiodobenzylguanidine); FDG (fluorodeoxyglucose) PET (best for SDHB); ^{123}I-MIBG (best in MEN2).
- **Never biopsy** (hypertensive crisis). Iodinated contrast is safe.
- Genetic testing (about one-third hereditary).

Pre-operative step 2: cardiac assessment

- ECG (electrocardiogram) and echocardiography for catecholamine cardiomyopathy; joint plan with the anaesthetist.

Pre-operative step 3: drug blockade (7–14 days)

Drug	Action	Points
Phenoxybenzamine (first line)	Irreversible non-selective α-blocker	10 mg twice daily, increased every 2–3 days (titrated); about 1 mg/kg total
Doxazosin, prazosin	Selective α1-blockers	Fewer side effects

Drug	Action	Points
Calcium channel blocker	Smooth muscle relaxation	Alternative strategy or add-on
β1-blocker (atenolol, metoprolol)	Controls tachycardia	**Only after adequate α-blockade**
Metyrosine	Blocks tyrosine hydroxylase	Add-on; sedation, extrapyramidal effects

- **α-blockade for all**, 7–14 days.
- **Never β before α:** β-blockade removes β2 vasodilatation, leaving α-vasoconstriction unopposed, which causes crisis.
- **Target:** seated BP (blood pressure) 120–130/80 mmHg; a mild postural drop (systolic >90 mmHg) shows adequate blockade.

Pre-operative step 4: volume expansion

- High-salt diet and fluids once blockade starts; 1–2 L IV (intravenous) fluid the night before. This prevents post-excision hypotension.

Pre-operative step 5: day of surgery

- Last phenoxybenzamine dose the night before surgery; omit the morning dose.

Intra-operative management

Prerequisites

- α-blockade 7–14 days (phenoxybenzamine or doxazosin), β-blocker only after α, volume expansion; ECG and echocardiography; joint plan with the anaesthetist.

Monitoring

- **Arterial line before induction**, central venous line, ECG, urine output, frequent glucose; cardiac output or transoesophageal echo if cardiomyopathy.

Anaesthesia

- Deep, smooth induction (propofol, opioid such as fentanyl or remifentanil); attenuate the intubation response.
- **Avoid triggers:** histamine-releasing drugs (morphine, atracurium), ketamine, ephedrine, metoclopramide, droperidol, desflurane.
- Sevoflurane or isoflurane; vecuronium or rocuronium; magnesium sulphate as an adjunct (reduces catecholamine release and arrhythmia).

Haemodynamic control

Event	Action
Hypertensive surge	**Pause manipulation**; nitroprusside, nicardipine or clevidipine, nitroglycerin, phentolamine
Tachyarrhythmia	**Esmolol**, lidocaine; amiodarone if needed
Hypotension after vein ligation	**Aggressive volume**, then α-agonist (phenylephrine, noradrenaline); vasopressin if refractory
Hypoglycaemia (insulin rebound)	Dextrose infusion, glucose checks

Surgical principles

- **Laparoscopic adrenalectomy** (transperitoneal or posterior retroperitoneoscopic) is standard; open for large or invasive tumours.
- **Minimal handling**; "no-touch" dissection away from the tumour.
- **Early adrenal vein ligation** where possible, after warning the anaesthetist.
- Remove intact in a bag (avoid capsule rupture and seeding).
- **Cortical-sparing** (partial) adrenalectomy in MEN2, VHL and bilateral tumours.
- Steroid cover if bilateral adrenalectomy.

Post-operative management

- HDU (high dependency unit) overnight: blood pressure, heart rate, glucose.
- Fluids ± vasopressors for **hypotension**.
- Glucose monitoring for **rebound hypoglycaemia** (insulin no longer suppressed).
- Steroid cover after bilateral adrenalectomy.

Follow-up and metastatic disease

- Metanephrines after surgery, then **yearly for life** (recurrence up to 16%).
- Malignancy is defined only by **metastases**. Treat with metastasectomy, ^{131}I-MIBG, CVD (cyclophosphamide, vincristine, dacarbazine).

Diagrams to draw

- Posterior view of both adrenals with the three arteries and the two veins.
- Flowchart: metanephrines, imaging, α-blockade, β-blockade, salt and fluids, surgery.
- Intra-operative BP curve: surges during handling, fall after vein ligation, with drugs marked.
- Catecholamine pathway to metanephrines (COMT).

> **Key points to remember:**
> - Right vein short into IVC; left vein into left renal vein.
> - StAR is rate-limiting for steroids; tyrosine hydroxylase for catecholamines.
> - Cortisol induces PNMT; COMT forms metanephrines inside the tumour.
> - Metanephrines first; never biopsy.
> - α-blockade 7–14 days for all; never β before α (α before β), plus volume expansion.
> - Volume expansion prevents hypotension; watch glucose.
> - Arterial line before induction; avoid histamine releasers and ephedrine.
> - Surge: pause handling, nitroprusside or phentolamine; esmolol for tachycardia.
> - After vein ligation: fluids, vasopressors, glucose.
> - One-third hereditary; SDHB most often metastatic.

Also, from the condensed book (not in the paper answer):

- Neonatal gland weight and adrenal rests (§1).

1 · Embryology

- The cortex and gonad share the adrenogonadal ridge, which explains adrenal rests; neural crest forms the medulla.
- At birth the gland is **2×** adult weight. The **fetal zone involutes in year 1**.
- In **renal agenesis** the adrenal is **discoid**.

- **Adrenal rests** occur in ≈1% of adults and in 7.5–15% of neonates along gonadal descent; in congenital adrenal hyperplasia (CAH) they form **testicular masses**.

2 · Zones and hormones

Cortex is ≈90% and medulla ≈10%. †Outside inwards: **GFR = salt, sugar, sex**.

- Glomerulosa ~15% makes aldosterone; fasciculata **~80%** makes cortisol (lipid-laden "clear cells"); reticularis 5–7% makes DHEAS; the medulla (~10%) holds chromaffin cells.
- The medulla is a **modified sympathetic ganglion** of axonless postganglionic neurones.

3 · Relations and blood supply

- Each gland weighs ≈5 g. The right is **retrocaval**, directly above the upper pole; the left lies medial to the upper pole.
- **Exposure**: on the right, liver above, inferior vena cava (IVC) medial, duodenum and kidney below. On the left, pancreas and spleen above, splenic flexure medial.
- Three arteries enter via many twigs, but **one vein drains**. Cortical blood drains into the medulla, so cortisol drives **epinephrine synthesis**.
- The **right vein** is short and enters the **posterior IVC**, so it can avulse. The **left** enters the left renal vein, joined by the inferior phrenic.

> **KEY POINT**
>
> The right side is riskier: high, retrocaval, with a short vein whose avulsion tears the cava.

5 · Imaging appearance

- **CT** shows an inverted **V/Y** with limbs of 3–6 mm. Unenhanced **≤10 HU** means lipid-rich.
- **MRI** has the best contrast; on **T2 the gland resembles fat**. Ultrasound separates solid from cystic.
- Adrenal venous sampling (AVS) is the only **functional** study; cavography shows intraluminal tumour.

146 Pathophysiology, Evaluation and Medical Management of Adrenal Disorders

1 · Applied anatomy and physiology

*See the paper answer **Phaeochromocytoma: adrenal anatomy, hormones and perioperative management** (Section 14, Ch 145).*

Also, from the condensed book (not in the paper answer):

- Anomalous venous drainage occurs in up to **9%**, which matters for adrenal vein sampling (AVS).
- **Pituitary failure or exogenous steroids spare aldosterone**, because the glomerulosa obeys angiotensin II and potassium; the inner zones atrophy without ACTH.

2 · Evaluation of an adrenal mass – principles

KEY POINT

Two questions: **(1) Is it malignant?** (imaging). **(2) Is it hormonally active?** (biochemistry). Yes to either means treat, usually by resection. Imaging cannot identify functional adenomas, so biochemistry is never omitted.

Category	Indications for adrenalectomy (Box 146.1)
Functional	Autonomous cortisol secretion; phaeochromocytoma; lateralised aldosterone excess
Oncological	>4 cm (except myelolipoma); suspicious imaging (lipid-poor, heterogeneous, irregular, infiltrating); growth >1 cm on follow-up; isolated metastasis (multidisciplinary)
Myelolipoma	Very large and/or symptomatic
At nephrectomy for renal cell carcinoma (RCC)	Adrenal abnormal or not visualised; thrombus to adrenal vein level
Bilateral	Failed pituitary surgery for Cushing disease; selected ectopic ACTH; AIMAH; PPNAD (§7)

3 · Imaging

LAQ Adrenal mass evaluation, incidentaloma and adrenocortical carcinoma

Paper answer · P2 · P4 · merged from 2

Asked as: P4 LAQ "Adrenal mass evaluation and adrenocortical carcinoma: diagnosis and management" (asked as "Diagnosis and management for Adrenal Carcinoma", Summer 2018 Paper IV Q2 LAQ; "Approach to evaluation of adrenal mass/s", Summer 2022 Paper IV Q3f SAQ) · P2 SAQ "Algorithm for management of an incidentaloma" (Practice Set 3 Paper 2 Q3d)

Assumed meaning (P2): adrenal incidentaloma.

Definition and epidemiology

- **Adrenal incidentaloma:** mass **≥1 cm** found on imaging done for another reason.

- Two questions: **Is it malignant? (imaging)** and **Is it functioning? (biochemistry).**
- ACC (adrenocortical carcinoma): rare (0.7–2 per million per year), aggressive cortical tumour. **Bimodal** (children <10 years; adults 40–50 years); slight female excess.
- **ACC syndromes:** Li–Fraumeni (TP53), Beckwith–Wiedemann (IGF-2, insulin-like growth factor 2), Lynch, MEN-1 (multiple endocrine neoplasia 1), Carney complex, FAP (familial adenomatous polyposis). Surgery is the only cure.

Differential diagnosis of an adrenal mass

- Non-functioning adenoma (commonest); cortisol-secreting adenoma 5.3%; phaeochromocytoma 5.1%; aldosteronoma 1%; ACC 4.7%; metastasis 2.5%; myelolipoma, cyst, haemorrhage.
- **Bilateral:** metastases, lymphoma, **tuberculosis** and histoplasmosis, haemorrhage, CAH (congenital adrenal hyperplasia), macronodular hyperplasia, bilateral phaeochromocytoma (MEN-2 [multiple endocrine neoplasia type 2], VHL [von Hippel–Lindau]).

Clinical features of ACC

- **Functional in 50–79% of adults** (90% of children):
- Cortisol (commonest): rapidly progressive Cushing syndrome, often with virilisation.
- Androgens: virilisation in women. **Oestrogen in a man (feminisation) suggests carcinoma.**
- Aldosterone: rare.
- Non-functional: pain, fullness, weight loss, or incidental.

Incidentaloma algorithm

```
Adrenal mass ≥1 cm ├─ IMAGING: unenhanced CT (computed tomography) │ ≤10 HU (Hounsfield units), homogeneous → benign (lipid-rich) adenoma │ (or MRI chemical-shift signal drop-out → lipid-rich adenoma) │ >10 HU → washout CT (15 min) or chemical-shift MRI │ APW >60% or RPW >40% → adenoma │ indeterminate → MDT (multidisciplinary team): repeat imaging 6-12 months, FDG-PET or surgery │ Macroscopic fat → myelolipoma ├─ HORMONES (all patients) │ 1-mg overnight DST (dexamethasone suppression test) │ plasma or urine metanephrines │ ARR (aldosterone-renin ratio) if hypertension or low potassium │ sex steroids only if ACC suspected └─ DECISION functioning, suspicious, >4 cm, or growing → adrenalectomy benign, non-functioning → discharge
```

Hormonal work-up (all lesions; ESE–ENSAT, European Society of Endocrinology–European Network for the Study of Adrenal Tumours)

Axis / hormone	Tests
Glucocorticoid (cortisol)	**1-mg overnight DST** (or 24-h urine or late-night salivary cortisol): ≤1.8 µg/dL (50 nmol/L) excludes; 1.8–5.0 µg/dL = possible autonomous secretion (**MACS**, mild autonomous cortisol secretion); >5.0 µg/dL autonomous; ACTH (adrenocorticotropic hormone)
Sex steroids and precursors	DHEA-S (dehydroepiandrosterone sulphate), androstenedione, testosterone, 17-OH-progesterone, 11-deoxycortisol; oestradiol in men and postmenopausal women (only if ACC suspected)
Mineralocorticoid (aldosterone)	**ARR** (aldosterone-to-renin ratio) if hypertension or hypokalaemia

Axis / hormone	Tests
Catecholamines	**Plasma free or 24-h urinary fractionated metanephrines**: must exclude phaeochromocytoma
Bilateral	17-hydroxyprogesterone (CAH); exclude adrenal insufficiency

- Why: confirms cortical origin, plans steroid cover, gives **tumour markers** for follow-up.
- **Never biopsy or operate before excluding phaeochromocytoma.** Biopsy only for suspected metastasis in known cancer.

Imaging

- Abbreviations: CT, HU, MRI (chemical-shift magnetic resonance imaging), MDT, FDG-PET (fluorodeoxyglucose positron emission tomography).
- **APW** (absolute percentage washout) = (enhanced − delayed) ÷ (enhanced − unenhanced) × 100; **RPW** (relative) = (enhanced − delayed) ÷ enhanced × 100.
- Malignant features: heterogeneous, irregular, invasion, >4 cm. Macroscopic fat = myelolipoma.
- **Size and malignancy:** <4 cm 0.1%; 4–6 cm 2.4%; >6 cm 19.5% (another source: carcinoma in ~2% <4 cm, ~6% at 4–6 cm, ~25% >6 cm). Benign imaging overrides size.
- **ACC size:** mean 10–12 cm; >90% exceed 5 cm.
- **CT in ACC:** irregular, heterogeneous, necrosis, calcification; **unenhanced >10 HU** (mean ~39 HU vs ~8 HU in adenoma); **slow washout** (absolute <60%, relative <40% at 15 min). Growth ~1.9 cm/year.
- **MRI in ACC:** no chemical-shift signal loss; best for IVC (inferior vena cava) thrombus and invasion.
- **FDG-PET/CT:** ACC is avid, but PET cannot separate ACC from metastasis.
- **Staging:** CT chest, abdomen, pelvis; bone or brain only if symptomatic.
- **No biopsy** if ACC is suspected (seeding, poor accuracy); only for unresectable or metastatic disease, after excluding phaeochromocytoma.

Management of incidentaloma (ESE–ENSAT 2023)

- **Surgery:** cortisol excess, phaeochromocytoma (after alpha-blockade), aldosteronoma (after AVS, adrenal vein sampling), malignant features or suspicious imaging, >4 cm in fit patients, significant growth (≥5 mm and >20%). Laparoscopic for benign; open for suspected ACC.
- **MACS:** screen for hypertension, diabetes and osteoporosis; surgery individualised.
- **Follow-up:** homogeneous **≤10 HU, non-functioning: benign, no further imaging whatever the size**; metanephrines may be omitted. Indeterminate non-resected: re-image at 6–12 months; growth >20% and ≥5 mm is concerning.
- **Bilateral:** metastases, lymphoma, tuberculosis, haemorrhage, CAH (17-hydroxyprogesterone); exclude adrenal insufficiency.

ACC pathology: Weiss score (≥3 of 9 = malignant)

Domain	Criteria
Nuclear / cytology	High nuclear grade; >5 mitoses per 50 high-power fields; atypical mitoses
Architecture	≤25% clear cells; diffuse architecture; necrosis
Invasion	Venous; sinusoidal; capsular

- SF-1 (steroidogenic factor 1) confirms cortical origin. **Ki-67** and mitotic rate grade the tumour (low vs high grade).

ENSAT staging

Stage	Definition	5-year disease-specific survival (ENSAT 2009)
I	T1 (≤5 cm), N0 M0	~80%
II	T2 (>5 cm), N0 M0	~60%
III	T3 (periadrenal fat) or T4 (adjacent organ, venous thrombus) or N1	~50%
IV	Distant metastasis only	~13%

- ENSAT differs from older UICC (Union for International Cancer Control) staging by moving T4 and N1 without metastasis into stage III.

Management of ACC

1. Surgery (stages I–III)

- **Open transperitoneal adrenalectomy** for suspected ACC, especially >6 cm or invasive. Laparoscopy risks capsule rupture and peritoneal carcinomatosis; only for small organ-confined tumours by experts.
- **R0, intact capsule:** rupture gives dismal results. **En bloc** removal of involved kidney, liver, spleen, pancreas or IVC; IVC thrombectomy (bypass if needed).
- **Locoregional lymphadenectomy** (periadrenal fat, renal hilum, para-aortic or paracaval) improves staging and may reduce recurrence.
- High-volume surgeon (ESE–ENSAT: >6, preferably ≥20 adrenalectomies a year). Hydrocortisone cover if cortisol-secreting.

2. Adjuvant therapy

- **Mitotane** (adrenolytic DDT [dichlorodiphenyltrichloroethane] derivative) for **high risk**: stage III, R1/Rx, or Ki-67 >10% (ESE–ENSAT 2018), for at least 2 years.
- Target blood level **14–20 mg/L**; needs **glucocorticoid replacement** (it destroys the cortex and raises cortisol-binding globulin).
- Toxicity: gastrointestinal, neurological, hepatic, hypothyroidism, gynaecomastia.
- **ADIUVO** (RCT, randomised controlled trial; R0, stage I–III, Ki-67 ≤10%): no recurrence-free survival benefit, so observe low-risk patients.
- **Adjuvant radiotherapy** (tumour bed): R1 or Rx, or stage III. It cuts local recurrence but not overall survival.

3. Advanced or metastatic (stage IV)

- **EDP-M** (etoposide, doxorubicin, cisplatin + mitotane) is first line. **FIRM-ACT** (2012): versus streptozotocin–mitotane, response 23% vs 9% and PFS (progression-free survival) 5.0 vs 2.1 months; overall survival not significantly better.
- Mitotane alone for low-volume, slow disease.
- **Surgery:** resect oligometastases or late recurrence; debulk only if >90% removable.
- **Hormone control:** metyrapone, ketoconazole, mifepristone, etomidate. Radiotherapy for bone and brain metastases.

Follow-up of ACC

- CT chest, abdomen, pelvis and baseline-raised hormones **every 3 months for 2 years**, then 6-monthly to 5 years.

Prognosis of ACC

- Overall 5-year survival **20–47%**. Bad factors: stage, R1, high Ki-67 or mitoses, cortisol excess, age, size >12 cm.

Recent advances

- Molecular classes (CIMP, CpG island methylator phenotype) predict prognosis.
- Checkpoint inhibitors give modest responses; IGF-1R (IGF-1 receptor) inhibitor linsitinib failed (GALACCTIC).
- ADIUVO-2 tests mitotane ± platinum–etoposide in high risk.

Diagrams to draw

- The incidentaloma imaging and biochemistry algorithm above.
- ACC algorithm: 1. Adrenal mass: hormones + CT (HU, washout). 2. Suspected ACC: open R0 en bloc resection + LND (lymph node dissection). 3. Risk-based adjuvant mitotane or radiotherapy.

Key points to remember:

- ≤10 HU = lipid-rich adenoma; APW >60%, RPW >40%.
- 1-mg DST: >1.8 µg/dL = MACS.
- Exclude phaeochromocytoma before biopsy or surgery.
- Surgery for incidentaloma: functioning, suspicious, >4 cm or growing.
- Feminisation in a man, >6 cm, >10 HU, poor washout: think ACC. Never biopsy before resection.
- Weiss ≥3/9; ENSAT stage IV = distant metastasis only.
- Open, R0, intact capsule, en bloc + lymphadenectomy.
- Mitotane level 14–20 mg/L; ADIUVO (low risk: no benefit); FIRM-ACT: EDP-M first line.

Also, from the condensed book (not in the paper answer):

- Up to **7%** of metastases are <10 HU, so every lesion still needs biochemistry; RCC and hepatoma metastases can also wash out.
- **Unenhanced CT**: **≤10 Hounsfield units (HU)** means a lipid-rich adenoma, which covers >70% of adenomas. Chemical-shift MRI drop-out is about as accurate.
- **CT washout** is the gold standard for lipid-poor lesions, with phases at 0, 60 s and 15 min. It is an adenoma if the absolute percentage washout (APW) is **>60%** or the relative washout (RPW) **>40%**.

ALGORITHM

1. Unenhanced CT: if ≤10 HU or MRI drop-out, then adenoma.
2. If >10 HU, then do a 15-min washout CT.
3. If APW >60% / RPW >40%, then it is an adenoma; if below, manage as indeterminate.

Size	Malignancy (large unselected series)	Surgical series
<4 cm	0.1%	≈2–5% adrenocortical carcinoma (ACC)
4–6 cm	2.4%	≈6–10%
>6 cm	19.5%	≈25%

- **4 cm** is the consensus threshold (sensitive, poorly specific); 6 cm is more specific.
- **Benign imaging overrides size**: no <10 HU lesion was malignant in the largest series.
- Heterogeneity, irregular margins and invasion are suspicious. **Macroscopic fat** (<−30 HU) means myelolipoma. **Calcification is not reassuring**, as it occurs in up to 30% of ACC.
- Classic follow-up re-images at 6 and 12 months; growth of ≥1 cm means consider resection.

Functional imaging

SAQ MIBG scan

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 4 Paper 2, Q3c

Principle

- MIBG (meta-iodobenzylguanidine) is a guanethidine analogue that resembles **noradrenaline**.
- It enters chromaffin and sympathetic cells through the **noradrenaline transporter** and is stored in **adrenergic secretory vesicles**. So it maps catecholamine-handling tissue: function, not just anatomy.

Isotopes

	^{123}I-MIBG	^{131}I-MIBG
Use	**Diagnostic** (preferred)	Older diagnostic; mainly **therapeutic**
Image quality	Better; allows SPECT-CT (single-photon emission computed tomography)	Poorer
Emission	Gamma	Beta (therapy) plus gamma

Technique

- **Thyroid blockade** with potassium iodide before and after injection (free iodine).
- **Stop interfering drugs:** labetalol, tricyclic antidepressants, sympathomimetics, reserpine, some calcium channel blockers.
- Intravenous injection; planar and SPECT-CT images at about 24 hours.
- **Normal uptake:** salivary glands, liver, heart, bladder, thyroid if unblocked; normal adrenals faint.

Uses

1. Phaeochromocytoma and paraganglioma

- **Extra-adrenal, multiple, metastatic or recurrent disease**; biochemistry positive but imaging negative.
- **Isolated normetanephrine excess** may indicate an extra-adrenal tumour; MIBG finds it.

- Tumour >5 cm: MIBG or FDG (fluorodeoxyglucose) PET (positron emission tomography) to exclude metastases before surgery.
- **Best in MEN-2 (multiple endocrine neoplasia type 2).** Raised metanephrines after surgery: MIBG can unmask metastases.

2. Neuroblastoma

- ^{123}I-MIBG is **required by COG (Children's Oncology Group) for staging**, replacing skeletal survey and bone scan. More sensitive for relapse. Non-avid tumours: PET.

3. Therapy (^{131}I-MIBG)

- Confirm uptake first. Metastatic phaeochromocytoma: symptom response in up to two-thirds; complete response <5%. High-specific-activity form FDA (Food and Drug Administration)-approved 2018.
- Relapsed or refractory neuroblastoma: response about 28–39%.

Limitations

- **Cold scans in SDHB (succinate dehydrogenase subunit B) and VHL (von Hippel–Lindau)**; aggressive metastases lose uptake.
- **FDG PET is now gold standard** for staging, and **^{68}Ga-DOTATATE PET/CT** is replacing it at some centres.
- **Also know:** carcinoid and medullary thyroid cancer imaging; cardiac sympathetic imaging.

Diagram to draw: whole-body MIBG showing an adrenal hot spot with normal liver, heart and bladder uptake.

> **Key points to remember:**
> - Noradrenaline analogue; block thyroid; stop labetalol.
> - ^{123}I diagnostic, ^{131}I therapeutic.
> - Best for extra-adrenal, metastatic and MEN-2 tumours; cold in SDHB and VHL.

Also, from the condensed book (not in the paper answer):

- For phaeochromocytoma >5 cm, do MIBG or FDG PET (best in SDHB/VHL) pre-op to exclude metastases. Iodinated contrast does **not** precipitate crisis.

4 · Hormonal work-up (all lesions >1 cm)

> **KEY POINT**
>
> Every incidentaloma gets a **1-mg dexamethasone suppression test (DST) and metanephrines**. Add an **aldosterone-to-renin ratio (ARR)** if there is hypertension or unexplained hypokalaemia. **Never biopsy, sample or excise before excluding phaeochromocytoma**.

Incidentalomas comprise cortisol adenoma 5%, phaeochromocytoma 5%, aldosteronoma 1%, ACC 5% and metastasis 2.5%, so **≈18%** are potentially surgical.

Cortisol

SAQ Cushing syndrome and the dexamethasone suppression test

Paper answer · P2

Asked as:

- Cushing Syndrome — *Practice Set 2 Paper II Q3c (SAQ)*
- Dexamethasone suppression test — *Practice Set 5 Paper II Q3c (SAQ)*

Type: SAQ (10 marks)

Definitions

- **Cushing syndrome:** symptomatic hypercortisolism. **Cushing disease:** due to a pituitary ACTH (adrenocorticotropic hormone)-secreting adenoma.
- **MACS (mild autonomous cortisol secretion):** autonomous cortisol without clinical signs (replaces "subclinical Cushing"); raises cardiovascular, bone and diabetes risk.

Causes

Group	Cause	Share
Exogenous	Glucocorticoids (oral, inhaled, topical)	Commonest overall
ACTH-dependent (80–85% of endogenous)	Pituitary adenoma (Cushing disease)	About 70%
	Ectopic ACTH (bronchial carcinoid, small cell lung cancer, phaeochromocytoma)	10–15%
ACTH-independent	Adrenal adenoma, ACC (adrenocortical carcinoma), AIMAH (macronodular hyperplasia), PPNAD (pigmented nodular disease; Carney complex)	About 10%

Clinical features

- Central obesity, moon face, plethora (90–100%); purple striae, hypertension, hirsutism, ED (70–90%).
- **Discriminating:** proximal weakness, bruising.
- **Urological:** nephrolithiasis in up to 50%; hypogonadotropic hypogonadism (low libido, ED).

Diagnosis

Step 1: exclude exogenous steroids (including creams and sprays) first. Step 2: confirm cortisol excess (two tests)

Test	Method	Abnormal
Overnight 1-mg DST (dexamethasone suppression test)	1 mg at 11 pm; 8 am cortisol	>1.8 µg/dL (50 nmol/L)
Late-night salivary cortisol	11 pm, twice	Raised
24-h UFC (urinary free cortisol)	Two collections	Raised; used in pregnancy

Step 3: locate the source

- **Low ACTH:** adrenal; adrenal CT or MRI.
- **High ACTH:** pituitary MRI; **IPSS (inferior petrosal sinus sampling) with CRH (corticotropin-releasing hormone)**: central gradient = pituitary; none = ectopic (chest CT).
- Pitfalls: pseudo-Cushing (depression, alcohol), oestrogens, enzyme inducers.

Dexamethasone suppression test

Principle

- **Dexamethasone** is a potent synthetic glucocorticoid that is **not detected by the cortisol assay**. In a normal axis it suppresses pituitary ACTH (adrenocorticotropic hormone) by negative feedback, so morning cortisol falls.
- **Autonomous** cortisol (adrenal adenoma) or excess ACTH drive (pituitary, ectopic) fails to suppress.
- "Low-dose" 1 mg is still 3–4 times physiological glucocorticoid.

Indications

- Suspected Cushing syndrome; **every adrenal incidentaloma** (5–8% secrete cortisol); suspected ACC (adrenocortical carcinoma).

Types

Test	Method	Abnormal
Overnight 1-mg LD-DST (low-dose DST; preferred screen)	Dexamethasone 1 mg at 11 pm–midnight; serum cortisol 8–9 am	See below
48-hour 2-mg LD-DST	0.5 mg 6-hourly for 48 hours; cortisol at 48 hours	>1.8 µg/dL; improves accuracy
High-dose DST	8 mg overnight or 2 mg 6-hourly for 48 hours	Over 50% suppression suggests pituitary source; **no longer routine** (some ectopic tumours also suppress)

Interpreting the 1-mg test

Morning cortisol	Meaning
≤1.8 µg/dL (50 nmol/L)	Normal; >90% sensitive, rules out
1.8–5.0 µg/dL	Possible autonomous cortisol secretion
>5.0 µg/dL (140 nmol/L)	Autonomous secretion; >95% specific
- ESE (European Society of Endocrinology) 2023 incidentaloma guideline: cortisol >1.8 µg/dL without overt signs = **MACS (mild autonomous cortisol secretion)**.	
- If abnormal: repeat testing and measure **morning ACTH** (low = adrenal; high = pituitary or ectopic, IPSS [inferior petrosal sinus sampling]).	

Pitfalls

- **False positive:** oral contraceptives (raise cortisol-binding globulin; up to 50%); CYP3A4 inducers (phenytoin, carbamazepine, rifampicin); missed dose; pseudo-Cushing states (depression, alcohol, obesity, stress).
- **False negative:** CYP3A4 inhibitors (itraconazole, ritonavir, diltiazem).
- **Choice of test:** pregnancy → UFC (urinary free cortisol); epilepsy → LNSC (late-night salivary cortisol); renal failure → DST (not affected by GFR [glomerular filtration rate]).

Management

Cause	Treatment
Cushing disease	Trans-sphenoidal surgery (cure 60–80%); radiotherapy
Adrenal adenoma	**Laparoscopic adrenalectomy**
ACC	Open radical adrenalectomy, mitotane
Ectopic	Resect source; otherwise bilateral adrenalectomy
AIMAH, PPNAD	Bilateral adrenalectomy

- **Bilateral adrenalectomy:** for failed pituitary surgery or life-threatening hypercortisolism. Cost: lifelong replacement, **Nelson syndrome** (8–29%).
- **Medical (bridge):** metyrapone, ketoconazole, etomidate, mitotane, mifepristone.

Perioperative care (urological relevance)

- **Contralateral adrenal is suppressed:** autonomous cortisol suppresses the opposite adrenal; adrenal insufficiency in about 99.7% after surgery for overt Cushing.
- Give **perioperative steroid cover** and taper after adrenalectomy: **Hydrocortisone 100 mg IV** on the day of surgery, then slow oral taper; recovery takes a median 11 months.

Diagram to draw: HPA (hypothalamic–pituitary–adrenal) axis with dexamethasone feedback and with ACTH-dependent and ACTH-independent sites.

> **Key points to remember:**
> - Exclude exogenous steroids; 1-mg DST cortisol >1.8 µg/dL abnormal.
> - 1 mg at 11 pm; 8 am cortisol ≤1.8 µg/dL excludes; >5.0 µg/dL confirms autonomy.
> - Oral contraceptives and enzyme inducers cause false positives.
> - ACTH low = adrenal; high = pituitary/ectopic (IPSS); ACTH separates adrenal from pituitary/ectopic.
> - High-dose DST no longer routine.
> - Steroid cover after adrenalectomy; Nelson syndrome after bilateral.

Also, from the condensed book (not in the paper answer):

- 1-mg DST: **≤1.8 µg/dL (50 nmol/L)** excludes autonomy; 1.8–5.0 means possible autonomy; **>5.0 µg/dL (140 nmol/L)** is >95% specific but misses up to 18% of Cushing syndrome.
- Urinary free cortisol (UFC) is insensitive for mild excess and unreliable if GFR is <60 mL/min.

Catecholamines

	Plasma free metanephrines	24-h urinary fractionated metanephrines
Performance	Sensitivity 97–99% – best rule-out	>97% by tandem mass spectrometry
Conditions	**Supine after 20 min rest**, fasting	Normal renal function
Highly indicative	Normetanephrine >2.2 nmol/L and/or metanephrine >1.2 nmol/L (≈4× upper limit of normal)	—

- **False positives** come from paracetamol, mesalazine, sulfasalazine (stop ≥5 days), tricyclics, **phenoxybenzamine**, severe illness and seated sampling.
- If equivocal, repeat supine off β-blockers, check chromogranin A, or do a **clonidine suppression** test.

5 · Biopsy and incidentaloma management

- Biopsy answers only "is this a metastasis?", because cytology cannot separate adenoma from carcinoma. ESE–ENSAT advises **no biopsy without a history of malignancy**; complications occur in 3–12%, with seeding in ACC.

ALGORITHM

1. If imaging characterises the lesion, then do not biopsy.
2. If it is functional, then do not biopsy.
3. If ACC is suspected, then do not biopsy.
4. If the result will change management (e.g., known primary) and justify risk, then biopsy, **after excluding phaeochromocytoma**.

- Incidentalomas appear on ≈5% of scans; ≈93% are inactive and need no treatment.

6 · Primary aldosteronism (PA)

LAQ How will you evaluate and manage 38 yr male with uncontrolled hypertension

Paper answer · P2

Type: LAQ (25 marks)
Source: Practice Set 7 Paper 2, Q1 (renovascular hypertension also answered in Practice Set 2 Paper 2, Q1; phaeochromocytoma in Practice Set 6 Paper 1, Q1)

Assumed focus: secondary (urological, renal and adrenal) causes of hypertension in a young adult.

Introduction

- **Resistant hypertension:** BP (blood pressure) above goal despite 3 drugs of different classes, including a diuretic, at optimal doses; or needing 4 or more drugs.
- Poor control below 40 years is a classic trigger to search for a **secondary cause**; many are curable.

Causes

Group	Causes	Clues
Renovascular	FMD (fibromuscular dysplasia), atherosclerotic RAS (renal artery stenosis), Takayasu arteritis, reninoma	Bruit; creatinine rise >30% on ACE (angiotensin-converting enzyme) inhibitor; flash pulmonary oedema; kidneys differing >1.5 cm

Group	Causes	Clues
Renal parenchymal	CKD (chronic kidney disease), ADPKD (autosomal dominant polycystic kidney disease), reflux nephropathy, glomerulonephritis, Page kidney	Raised creatinine, proteinuria, haematuria
Obstructive uropathy	Bilateral obstruction (about 77% hypertensive), unilateral (about 20%), chronic retention	LUTS (lower urinary tract symptoms), palpable bladder, hydronephrosis
Primary aldosteronism (PA)	Adenoma (Conn), bilateral hyperplasia	Hypokalaemia (present in <40% only)
Phaeochromocytoma, paraganglioma	Sporadic or hereditary	Paroxysmal headache, sweating, palpitations
Cushing syndrome	Exogenous steroid, adrenal adenoma, pituitary	Central obesity, striae, diabetes
- **Also know:** sleep apnoea; drugs (NSAIDs [non-steroidal anti-inflammatory drugs], steroids, decongestants, liquorice); alcohol; thyroid and parathyroid disease; coarctation.		

Evaluation

Step 1: Exclude pseudo-resistance

- Correct cuff technique; home or ambulatory monitoring to exclude white-coat effect; check adherence, doses, interfering drugs, salt and alcohol.

Step 2: History and examination

- Paroxysms, weakness or polyuria (hypokalaemia), stones, UTI (urinary tract infection) or childhood reflux, snoring, family history.
- BP both arms and radio-femoral delay; abdominal bruit; palpable kidneys or bladder; cushingoid habitus; café-au-lait spots (NF1 [neurofibromatosis type 1]); fundus.

Step 3: Baseline tests and target-organ damage

- Creatinine, eGFR (estimated glomerular filtration rate), electrolytes, urinalysis with albumin–creatinine ratio, glucose, lipids, ECG (electrocardiogram), echocardiography.
- **Ultrasound KUB (kidneys, ureters, bladder):** renal size and asymmetry, scars, cysts, masses, hydronephrosis, residual urine.

Step 4: Targeted screening (method and interpretation)

Cause	Screen	Confirm or localise
PA	Morning **ARR (aldosterone-to-renin ratio)** >20–30 with aldosterone ≥15–20 ng/dL; stop spironolactone first; correct potassium	**Intravenous saline 2 L over 4 h** (aldosterone stays >5 ng/dL) or oral salt loading → adrenal CT (computed tomography) → **AVS (adrenal vein sampling)**; skip AVS if age <40 with a clear unilateral adenoma >1 cm

Cause	Screen	Confirm or localise
Phaeochromocytoma	**Plasma free or 24-h urinary fractionated metanephrines**	CT or MRI (magnetic resonance imaging); ^{68}Ga-DOTATATE PET (positron emission tomography) or MIBG (metaiodobenzylguanidine); genetic testing
Cushing	**1-mg overnight dexamethasone suppression** (cortisol ≤1.8 µg/dL excludes); late-night salivary cortisol; 24-h urinary free cortisol	ACTH (adrenocorticotropic hormone): low = adrenal, high = pituitary or ectopic
Renovascular	**Duplex Doppler**: peak systolic velocity >180–200 cm/s, renal/aortic ratio >3.5	CT or MR angiography; catheter angiography with pressure gradient if intervention planned
Parenchymal, obstructive	Ultrasound, urine tests	Renogram (DTPA [diethylenetriaminepentaacetic acid] or MAG3 [mercaptoacetyltriglycine]) for split function and drainage; CT urography

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Uncontrolled BP, age 38 → exclude pseudo-resistance → baseline tests + ultrasound KUB → ARR, metanephrines, dexamethasone test, renal Doppler ├─ positive → confirm → localise → treat cause └─ all negative → essential: optimise drugs
```

Management

1. General (all patients)

- Lifestyle: salt restriction, weight loss, exercise, limit alcohol.
- Optimise **ACE inhibitor or ARB (angiotensin receptor blocker) + calcium channel blocker + thiazide-like diuretic**. Fourth drug: **spironolactone** (PATHWAY-2, 2015: best add-on in resistant hypertension).

2. Cause-specific

Cause	Treatment
PA, lateralised	**Laparoscopic adrenalectomy**: biochemical cure about 94%, but only 37–52% need no antihypertensives
PA, bilateral or unfit	Spironolactone 25–50 mg/day (up to 400 mg) or eplerenone
Phaeochromocytoma	**α-blockade 7–14 days**, β-blocker only after α, volume expansion → laparoscopic adrenalectomy; metanephrines yearly for life
Adrenal Cushing	Adrenalectomy with steroid cover; pituitary disease → trans-sphenoidal surgery
FMD	**Angioplasty alone** (cure common in the young)
Atherosclerotic RAS	Medical therapy (**ASTRAL 2009, CORAL 2014**: stenting no benefit); stent for flash pulmonary oedema or falling function
Branch disease, aneurysm	Surgical bypass or ex vivo repair with autotransplantation
Reninoma	Partial nephrectomy (cures most)
Small non-functioning kidney (reflux, segmental hypoplasia)	**Nephrectomy** if renal vein renin lateralises (ratio >1.5)
Page kidney	Drain or decapsulate

Cause	Treatment
Obstruction	**Relieve it** (catheter, nephrostomy, pyeloplasty, prostate surgery); watch post-obstructive diuresis
CKD	ACE inhibitor or ARB for proteinuria; scheduled diuretics

3. Follow-up: home BP, creatinine, potassium; duplex after angioplasty.

Recent advances

- **Catheter renal denervation** (radiofrequency or ultrasound), approved in the USA in 2023 for selected resistant hypertension; SPARTACUS trial questioned routine AVS.

Diagram to draw: stepwise algorithm from pseudo-resistance to cause-specific treatment.

> **Key points to remember:**
> - Hypertension below 40 years: screen for secondary causes.
> - ARR → saline suppression → CT → AVS (skip if <40 with adenoma >1 cm).
> - Metanephrines for phaeochromocytoma; α before β.
> - FMD: angioplasty; atherosclerotic RAS: medical (ASTRAL, CORAL).

Also, from the condensed book (not in the paper answer):

- PA causes 3–13% of hypertension; most patients (63–91%) are **normokalaemic**.

Subtype	%	Surgically curable?
Bilateral (idiopathic) hyperplasia	60	No – mineralocorticoid receptor antagonist (MRA)
Aldosterone-producing adenoma	35	Yes
Unilateral hyperplasia	2	Yes
Carcinoma, ectopic (ovarian), familial hyperaldosteronism (FH)	<1 each	Carcinoma/ectopic yes; FH no

- **FH-I** (glucocorticoid-remediable) results from a chimeric gene that puts aldosterone under ACTH control; it causes early severe hypertension and is treated with glucocorticoid.
- **Screen (Endocrine Society 2016)**: resistant hypertension (≥3 drugs including a diuretic) or control on ≥4 drugs; BP ≥150/100 on three days; hypertension and hypokalaemia; **incidentaloma with hypertension**; sleep apnoea; onset <20 y or stroke <50 y; family history of PA; disproportionate target-organ damage.
- **ARR** is measured in the morning, seated, as plasma aldosterone concentration (PAC, ng/dL) divided by plasma renin activity (PRA, ng/mL/h). It is positive if **ARR >20–30 with PAC >15–20 ng/dL**. Floor PRA at 0.2 and correct K^+ first.

Drug	Effect on ARR	Action
Spironolactone, eplerenone, amiloride	Profound	Stop ≥4–6 weeks; use α1-blocker or calcium channel blocker (CCB)
β-blockers	Lower renin, so **false positive**	May continue (or stop)
ACE inhibitors / angiotensin receptor blockers	Raise renin, so false negative	Continue; undetectable renin is suspicious

Drug	Effect on ARR	Action
Calcium channel blockers	Neutral	Continue

Confirmatory test	Protocol	Positive
Oral sodium loading	3 days high salt; 24-h urine Na >200 mmol	Urine aldosterone >12 µg/day
IV saline infusion	2 L 0.9% saline over 4 h, recumbent	PAC >5 ng/dL (>10 suggests adenoma)
Captopril challenge	25–50 mg, when salt load is contraindicated	PAC stays >15 ng/dL

- **Skip confirmation** if there is spontaneous hypokalaemia, undetectable renin and PAC >20 ng/dL.
- CT all surgical candidates to exclude ACC, but ≈20% of aldosteronomas are <1 cm, so CT alone misdirects ≈25%.
- **AVS** (sensitivity 95%, specificity 100% if done well) is done in the morning with K^+ corrected, sampling both adrenal veins and the IVC for aldosterone and cortisol.
- **Selectivity** proves the catheter is in the vein (adrenal-to-IVC cortisol >1.1–5:1). **Lateralisation**: a cortisol-corrected aldosterone ratio **>2–4:1** indicates ipsilateral adrenalectomy.
- **Omit AVS** if age <40 with a clear unilateral adenoma >1 cm and a normal contralateral gland, or if ACC is suspected.

ALGORITHM

1. If there is a screening indication, measure the ARR.
2. If the ARR is positive, confirm with a suppression test. If there is a family history or onset <20 y, test for FH-I.
3. If not a surgical candidate, then MRA.
4. If a surgical candidate, do CT. If age <40 with a unilateral >1 cm nodule, then adrenalectomy.
5. Otherwise do AVS: if lateralised, laparoscopic adrenalectomy; if not lateralised, MRA.

- After surgery, check PAC and ARR; † stop K^+ and MRA, and watch for **hyperkalaemia** for weeks because the contralateral glomerulosa is suppressed. BP settles over 6 months; residual hypertension is likelier in older, obese men with long-standing disease.
- MRA: spironolactone 25–50 mg, up to a maximum of 400 mg (gynaecomastia, erectile dysfunction); eplerenone 25 to 100 mg (selective).

7 · Cushing syndrome and cortisol excess

*See the paper answer **Cushing syndrome and the dexamethasone suppression test** (Section 14, Ch 146).*

Also, from the condensed book (not in the paper answer):

- Post-op insufficiency follows ≈65% of MACS resections; the HPA axis recovers in a median of 6.5 (MACS) to 11 (Cushing) months.
- **Urological**: hypogonadotropic hypogonadism and **urolithiasis in up to 50%**, so test cushingoid stone formers.

ALGORITHM

1. Exclude exogenous steroid.
2. Screen with 1-mg DST, late-night salivary cortisol (LNSC) ×2 or UFC ×2.
3. If abnormal, endocrinologist confirms; measure **morning ACTH**.
4. If ACTH low: adrenal, do CT.
5. If normal/high: pituitary MRI and **inferior petrosal sinus sampling after CRH**. A central-to-peripheral gradient means Cushing disease; no gradient means ectopic.

- **Bilateral adrenalectomy** is for failed pituitary treatment, life-threatening hypercortisolism, or unresectable or occult ectopic ACTH. The cost is lifelong replacement, crisis risk and **Nelson syndrome** (chiasmal compression, oculomotor palsies).

	ACTH-independent macronodular adrenal hyperplasia (AIMAH)	**Primary pigmented nodular adrenocortical disease (PPNAD)**
Glands	Massive, nodules up to 4 cm, >60 g each	Normal size; small black/brown nodules in atrophic cortex
Association	McCune–Albright	Carney complex (≈half)
Clue	Looks ACTH-driven – check ACTH	Paradoxical rise in UFC after dexamethasone
Treatment	Bilateral adrenalectomy; unilateral removal of the larger gland has worked in AIMAH	

- **Mild autonomous cortisol secretion (MACS)**: individualise surgery for hypertension, obesity, dysglycaemia or osteopenia; progression to overt Cushing is very unlikely.
- **Post-op adrenal insufficiency is expected** after overt Cushing (≈99.7%), because the contralateral gland is atrophic.
- Give hydrocortisone 100 mg IV on the day of surgery, then an oral taper (25–50 mg/day). Alternatively give dexamethasone 0.5 mg 6-hourly, where a morning cortisol <50 nmol/L confirms cure. Maintenance is 10–12 mg/m²/day in 2–3 doses.

8 · Phaeochromocytoma and paraganglioma (PPGL)

These can kill on the table, so block before surgery.

*See the paper answer **Phaeochromocytoma: adrenal anatomy, hormones and perioperative management** (Section 14, Ch 145).*

Also, from the condensed book (not in the paper answer):

- ≈24% of apparently sporadic cases carry germline mutations.
- 75–90% are adrenal; paraganglioma is extra-adrenal. Headache occurs in 60–90%.

KEY POINT

The rule of 10s is outdated – every patient with PPGL should be offered genetic counselling/testing (Endocrine Society 2014).

Syndrome	Gene	Features	PPGL risk / malignancy
MEN-2A	RET 10q11.2	Medullary thyroid cancer (MTC), hyperparathyroidism	50% / ≈3%; adrenal, bilateral
MEN-2B	RET	MTC, mucosal neuromas, marfanoid	50% / ≈3%
VHL type 2	VHL 3p25	RCC, haemangioblastoma, epididymal cystadenoma	10–20% / ≈5%; normetanephrine only
Neurofibromatosis 1 (NF1)	NF1 17q11	Neurofibromas, café-au-lait	1% / ≈11%
PGL-4	**SDHB** 1p36	Abdominal/thoracic paragangliomas	≈20% / **30–50%**
PGL-1	SDHD 11q23 (**maternal imprinting** – paternally transmitted)	Head and neck, multifocal	≈20% / <3%

All are autosomal dominant. VHL type 1 has no phaeochromocytoma, whereas type 2 does.

- On CT the mean is ≈35 HU; the T2 "light bulb" sign is unreliable.
- **Block all with abnormal biochemistry, even if normotensive.** Phenoxybenzamine causes hypotension after resection; α1-selective drugs do so less. CCBs can be sole therapy in mild cases.
- **Metyrosine** 250 mg 2–3 times a day, increased to 1.5–2 g, is added to α-blockade for large or metastatic tumours; it causes sedation and **extrapyramidal** effects.

KEY POINT

Why never β before α? β-blockade removes β2-mediated arteriolar dilatation. Catecholamines then act unopposed on α1 receptors and cause a hypertensive crisis. Hence use selective β1 agents.

- **Targets**: seated BP 120–130/80; standing systolic >80–90 mmHg; † heart rate 60–70/min. **Volume repletion** is key.
- In **MEN-2 and VHL**, do **cortical-sparing partial adrenalectomy**.
- Malignancy is defined **only by metastases**; risk factors are SDHB, extra-adrenal site, >5 cm and dopamine secretion.
- Check metanephrines at ≈6 months, then annually for life, because recurrence occurs even >15 years later.

9 · Non-functional benign lesions

Lesion	Key features	Management
Myelolipoma	Fat and haematopoietic tissue; macroscopic fat diagnostic; may exceed 10 cm	Exception to 4-cm rule; still do hormonal work-up. Excise if symptomatic or large; embolise a bleed

Lesion	Key features	Management
Cysts	Endothelial 45%, pseudocyst 39% (harbours malignancy ≈7%); >30 HU means haemorrhage	Endocrine work-up. **Excise if >5 cm, thick-walled, heterogeneous or symptomatic**
Ganglioneuroma	Young; encases vessels without narrowing them	Benign; diagnosed at resection
Oncocytoma	Rare; large; ≈10% functional	Resect; >30% malignant (unlike renal)

10 · Adrenocortical carcinoma (ACC)

*See the paper answer **Adrenal mass evaluation, incidentaloma and adrenocortical carcinoma** (Section 14, Ch 146).*

Also, from the condensed book (not in the paper answer):

- Overall 5-year survival is 20–47% (stage IV <5%), and 60–80% recur after resection.

ENSAT stage	Definition	At diagnosis
I	T1 (≤5 cm) N0 M0	3–4%
II	T2 (>5 cm) N0 M0	29–46%
III	T3 (periadrenal fat) or T4 (adjacent organ) or N1 or venous thrombus	11–19%
IV	Distant metastasis only	39–49%

- Resect recurrences if low grade with a long disease-free interval.
- † Mitotane often needs higher glucocorticoid doses, because it raises binding globulins and speeds clearance.
- **Paediatric ACC** is >90% functional (virilisation), warrants TP53 testing and has a better prognosis.

11 · Adrenal metastases

- With known malignancy, **>50%** of new adrenal lesions are metastases. RCC and hepatoma deposits may wash out, so growth and biopsy decide. **Still exclude phaeochromocytoma**.
- Resect an isolated metastasis in selected patients: non-small cell lung cancer (level 1 evidence) and melanoma, and RCC as part of cytoreduction.

12 · Adrenal insufficiency

	Primary (Addison)	Secondary (pituitary/hypothalamic)
Causes	Autoimmune (anti-21-hydroxylase), **tuberculosis**, bilateral haemorrhage, infiltration, **bilateral adrenalectomy**, drugs (ketoconazole, abiraterone)	**Exogenous glucocorticoid suppression** (commonest for surgeons), pituitary tumour/apoplexy, Sheehan, critical illness
Aldosterone	Lost	Preserved
Skin	Hyperpigmentation (high ACTH)	Pallor
Electrolytes	Low Na^+, **high K^+**; salt craving	Low Na^+, normal K^+
Replacement	Hydrocortisone + fludrocortisone	Hydrocortisone only

- A **crisis** mimics an acute abdomen. Those at risk have a cortisol-secreting tumour, bilateral adrenalectomy or a **previous contralateral nephrectomy/adrenalectomy**, so review old notes.
- Diagnose with morning cortisol and ACTH, then ACTH stimulation. Give ½–⅔ of the hydrocortisone dose on waking.

> **KEY POINT**
>
> † **Adrenal crisis**: do not wait for results. Draw cortisol and ACTH, give **hydrocortisone 100 mg IV bolus**, then 200 mg/24 h (infusion or 50 mg 6-hourly). Give 0.9% saline (1 L in the first hour) with dextrose, treat the precipitant, and taper over 1–3 days. Fludrocortisone is unnecessary while hydrocortisone is >50 mg/day. Teach sick-day rules and issue a steroid card.

Surgical stress (chronic steroids)	Example	Cover
Minor	Inguinal hernia, <1 h	Hydrocortisone 25 mg
Moderate	Joint replacement	Usual dose + hydrocortisone 50–75 mg intra-op
Major	Oesophagectomy, bypass	Usual dose + hydrocortisone 50 mg IV 8-hourly for 48–72 h

Glucocorticoid	Potency	Replacement (mg/day)	Half-life (h)
Hydrocortisone	1	20	8–12
Prednisone / prednisolone	4	5	18–36
Dexamethasone	25–50	0.5	36–54

13 · Congenital adrenal hyperplasia (CAH)

- CAH is autosomal recessive, and **21-hydroxylase deficiency is >95%**; low cortisol raises ACTH, causing hyperplasia and androgen excess (see Ch 59).
- Non-classic CAH may present as an adult mass, and testicular rests mimic tumour, so consider them before orchidectomy.

14 · Landmark trials

Trial	Design	Result	Take-home
SPARTACUS	RCT: CT- vs AVS-guided PA treatment	Same antihypertensive intensity at 1 y; CT and AVS discordant in half; persistent aldosteronism 20% (CT) vs 10% (AVS)	AVS remains standard for biochemical cure
FIRM-ACT	RCT: mitotane + etoposide, doxorubicin, cisplatin (EDP) vs mitotane + streptozotocin, advanced ACC	Better response and progression-free survival; overall survival similar	M-EDP first-line
ADJUVO	RCT: adjuvant mitotane vs observation, low/intermediate-risk R0 stage I–III ACC	No difference in recurrence	No routine adjuvant mitotane in low risk

15 · MCQ one-liners

MCQ ONE-LINERS

1. Commonest ectopic ACTH source: bronchial carcinoid.

147 Surgery of the Adrenal Glands

SAQ Adrenal Sx.

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 6 Paper 2, Q3d (preoperative phaeochromocytoma management also in 2020-Summer Paper 2, Q3c)

Assumed meaning: surgery of the adrenal gland (adrenalectomy).

Surgical anatomy

- Right: pyramidal; **short vein into posterior IVC (inferior vena cava)**, so it is the riskier side. Left: crescentic; long vein into left renal vein. Three arteries (inferior phrenic, aorta, renal), one vein.

Indications (AAES [American Association of Endocrine Surgeons] 2022)

- **Functional, any size:** aldosteronoma (Conn; after AVS [adrenal venous sampling]), cortisol-secreting adenoma, phaeochromocytoma, sex-hormone tumour, failed pituitary surgery for Cushing disease.
- **Malignant potential:** ≥4 cm; growth; >10 HU (Hounsfield units) unenhanced; washout <50%; suspected ACC (adrenocortical carcinoma); solitary metastasis.
- Myelolipoma >7 cm or symptomatic; cysts.
- **Contraindications (correct first):** uncontrolled hypertension, coagulopathy, peritonitis (transperitoneal); pregnancy relative.

Preoperative preparation

- Biochemistry: metanephrines in **every** case before surgery.
- **Phaeochromocytoma:** α-blockade 2 weeks (phenoxybenzamine or doxazosin), β-blocker only after α, salt and fluids.
- Cushing: hydrocortisone 100 mg IV on the day of surgery. Conn: correct potassium.

Approaches

Approach	Use
Laparoscopic transperitoneal (gold standard since 1992)	Familiar view, larger tumours
Retroperitoneoscopic (lateral or prone)	Prior abdominal surgery, small tumours
Robotic	Either route
Open (subcostal, chevron, thoracoabdominal, flank)	**ACC**, large or invasive tumours

Steps (laparoscopic transperitoneal)

- **Right:** elevate liver → Kocher manoeuvre → follow IVC → **control adrenal vein early** (warn anaesthetist) → free gland to psoas → bag intact.
- **Left:** reflect colon and spleen → follow left renal vein to adrenal vein → clip → free from pancreas and splenic vessels.
- Kidney attachment last (handle). ACC: open en bloc, no rupture.
- **Partial adrenalectomy:** bilateral or hereditary phaeochromocytoma, solitary gland.

Complications

- Bleeding (adrenal vein avulsion, IVC); spleen, pancreatic tail, liver, bowel injury; pneumothorax.
- **Phaeochromocytoma:** surges during handling, **hypotension after vein ligation**, rebound hypoglycaemia.
- **Adrenal insufficiency** after Cushing surgery (~99.7%): steroid taper.
- Conn: **hyperkalaemia** for weeks (high-salt diet).

Diagram to draw: right and left adrenal veins and relations.

> **Key points to remember:**
> - Function (any size) or malignancy risk (≥4 cm, >10 HU) indicates surgery.
> - Control the vein early; right vein short into IVC.
> - Open en bloc surgery for ACC.

Also, from the condensed book (not in the paper answer):

- Size and imaging criteria for incidentaloma (§2).

1 · Surgical anatomy (essentials)

- The **right** gland lies behind the bare area of the liver. Its vein is **<1 cm** long and enters the **posterior IVC at 45°**. The **left** gland is more mobile.

2 · Indications

Two gates: any functional tumour, or malignant potential.

Category	Indication
Functional (any size)	Aldosteronoma; cortisol adenoma; Cushing disease after failed trans-sphenoidal surgery; phaeochromocytoma (PCC); virilising/feminising tumour
Non-functional	Adrenocortical carcinoma (ACC); symptomatic cyst/myelolipoma; solitary metastasis
Size	**≥4 cm** ; >2 cm growing ≥1 cm in 12 months
Imaging	Non-contrast CT **>10 HU**; washout <50% with delayed >35 HU; chemical-shift MRI not adenoma; FDG-PET uptake equal to or above liver

> **KEY POINT**
>
> Trap: bilateral hyperplasia causing aldosteronism is **not** surgical, so adrenal venous sampling (AVS) comes before adrenalectomy for aldosteronism.

Incidentaloma work-up

- ACC is found in ~2% and autonomous secretion in up to 20%. Measure sex steroids if ≥4 cm or virilised.
- All get a 1-mg dexamethasone suppression test (never spot cortisol) and metanephrines (>2× upper limit is significant).
- If hypertensive, check the aldosterone-to-renin ratio (ARR >20 with aldosterone >20 ng/dL), then salt loading, then AVS; up to 40% are normokalaemic.
- Inert, ≤4 cm, ≤10 HU lesions get CT at 12 months. Operate on myelolipoma (≤ −20 HU) only if **>7 cm**, symptomatic or atypical.

3 · Contraindications

- Correct physiology first. Low-dose aspirin may continue; necessary surgery in pregnancy can be done in any trimester. Confirm that a prior nephrectomy has not removed the ipsilateral adrenal before contralateral adrenalectomy.

4 · Pre-operative preparation

Pathology	Preparation	Why
PCC	α-blockade for **2 weeks** (phenoxybenzamine; doxazosin, calcium channel blocker, metyrosine alternatives); β **only after α**; volume expansion	Contracted patients collapse once vein tied
Cushing	†Hydrocortisone 100 mg at induction, tapered over months; thromboprophylaxis	Contralateral gland is suppressed
Aldosteronoma	†Correct K^+; control BP	†Hypokalaemia causes arrhythmia

5 · Approaches

- **Minimally invasive surgery is standard** (since 1992).
- The **retroperitoneoscopic** route (lateral or **prone**) avoids bowel, so it suits prior abdominal surgery; bilateral small tumours can be done prone without repositioning.
- **Open** surgery is for suspected **ACC**, large tumours, or invasion of the IVC or organs.

6 · Operative steps

Control the vein, then free the gland.

ALGORITHM

1. **Right**:
2. Elevate the liver; Kocher the duodenum; expose the IVC.
3. Dissect up from the renal vein to the short adrenal vein.
4. **Warn anaesthesia**, then clip early.
5. Dissect superiorly and laterally down to psoas; bag the gland intact.

ALGORITHM

1. **Left**:
2. Reflect the colon at Toldt, outside Gerota; lift spleen and pancreas.
3. Follow the left renal vein to the adrenal vein; clip after warning.
4. Protect pancreas and splenic artery; divide superomedial tissue; bag.

- **Retroperitoneoscopic**: enter below the 12th rib tip with a balloon. Along psoas the **renal artery is met first**; the right vein enters the posterolateral IVC above the hilum.

> **KEY POINT**
>
> Ligate the vein early in PCC to limit the catecholamine surge; hypotension follows, so tell anaesthesia first.

7 · Special pathologies

- **PCC intra-op**: treat a surge with nitroprusside, nitroglycerin or phentolamine, and arrhythmia with esmolol.
- **ACC** needs a high-volume surgeon (>6, ideally ≥20/year) and an **open, intact, en bloc** resection.
- **Metastasis** (two-thirds from lung or kidney) is resected when the adrenal is the only disease; renal origin and <5 cm do best.
- Avoid ablation of PCC or ACC outside trials.

8 · Partial (cortical-sparing) adrenalectomy

- **Indications**: bilateral or hereditary PCC, familial syndromes, prior contralateral adrenalectomy, children, Conn adenoma and cysts.
- Arteries enter superiorly, medially and inferiorly, so **lateral dissection is safe**; the main vein may be divided. †Retain about one-third of a gland, guided by intra-op ultrasound.

9 · Complications

- For bleeding, apply pressure, raise the pneumoperitoneum, side-bite the IVC, and convert if uncontrolled.
- †Check glucose hourly for 24–48 h after PCC. Treat insufficiency with IV hydrocortisone 100 mg and saline. For hyperkalaemia after aldosteronoma, stop K^+ and spironolactone, occasionally adding fludrocortisone.

10 · Outcomes

- PCC recurs in up to **16%** over life, so check BP and metanephrines annually.
- †Aldosteronoma surgery corrects hypokalaemia in nearly all and cures hypertension in about half. Advanced-stage ACC has a 5-year survival **<15%**.

140 Retroperitoneal Tumours

1 · Anatomy of the retroperitoneum

See the paper answer ***Describe in brief the anatomy of the retroperitoneum. Explain the principles of laparoscopic retroperitoneal surgery while mentioning the salient steps of retroperitoneal laparoscopic nephrectomy.*** *(Section 10, Ch 78).*

Also, from the condensed book (not in the paper answer):

- **Depth of psoas invasion predicts nerve injury**: genitofemoral injury causes thigh paraesthesia, while femoral injury weakens hip flexion and knee extension.

KEY POINT

The operative problem is usually the neighbouring structure, not the tumour: RP tumours grow silently (resected sarcomas average **>15 cm**) and present with multi-organ involvement.

2 · Classification

The group decides first treatment.

Group	Entities	First treatment
Germ cell / gonadal	Seminoma, nonseminoma, sex cord stromal	Platinum chemotherapy (germ cell tumour, GCT); RP lymph node dissection, RPLND (sex cord stromal)
Mesodermal — malignant	Liposarcoma, leiomyosarcoma, undifferentiated pleomorphic sarcoma (UPS), synovial, Ewing, desmoplastic small round cell tumour (DSRCT), unclassified	Surgery; chemotherapy (Ewing, DSRCT)
Mesodermal — benign/intermediate	Lipoma, leiomyoma, solitary fibrous tumour (SFT), PEComa, desmoid, myelolipoma, GIST	Excision / observation
Neurogenic (~10%)	Schwannoma, neurofibroma, malignant peripheral nerve sheath tumour (MPNST), ganglioneuroma, neuroblastoma, paraganglioma	Surgery
Haematological (~1/3 of primary RP)	Lymphoma, plasmacytoma	Chemotherapy — biopsy only
Cystic	Lymphangioma, mucinous cystadenoma, cystic mesothelioma, cystic teratoma	Complete excision
Unknown primary	Poorly differentiated carcinoma encasing vessels	Histology-led

3 · Clinical evaluation

Tissue only if it changes treatment.

- Most are painless or incidental. The sequence is **imaging, then serology, then tissue**.

- **CT chest/abdomen/pelvis** comes first (NCCN). **Negative HU (fat)** suggests liposarcoma, angiomyolipoma or myelolipoma, but CT cannot tell origin or malignancy. **MRI** shows vascular involvement; PET-CT is selective.
- In a young man check **AFP and hCG**; in an older man, PSA. Add blood count and electrophoresis; † metanephrines if paraganglioma is possible.
- **Biopsy** only when it **changes first treatment**: lymphoma, GCT, Ewing, or high-grade sarcoma for neoadjuvant therapy. Use a **core (14–16 G coaxial) rather than FNA**; † take a retroperitoneal route, with the tract inside the future resection field.
- Avoid biopsy of purely cystic lesions, † an unblocked paraganglioma, or a young man with raised AFP/hCG.

4 · Mesenchymal tumours (sarcoma)

- Soft-tissue sarcoma is <1% of adult cancers; **15–20%** arise in the RP.

Entity	Key features	Treatment
Well-/de-differentiated (WD/DD) liposarcoma (commonest, **~60%**)	Fat density; **MDM2/CDK4 (12q) amplification**, ring chromosomes; S100+	Radical compartmental resection
Leiomyosarcoma (**~20%**)	From vessel media (IVC); no fat; SMA, desmin, h-caldesmon+	Negative margins + adjuvant
UPS (ex-MFH)	5th–6th decade; no fat, peripheral nodular enhancement	Wide resection
Unclassified (≤20%)	Exclusion; prior RT in a quarter	Screen MDM2
Synovial	Calcification; TLE1; **t(X;18)**	Chemosensitive
SFT	**NAB2-STAT6**; Doege-Potter hypoglycaemia (IGF)	En bloc; radiosensitive
Ewing	EWSR1-FLI1 t(11;22); infiltrative	Chemotherapy-based
DSRCT	Young men; peritoneal implants; WT1-EWSR1 t(11;22)	Multimodal; poor
PEComa	HMB-45 + muscle markers; TSC2	Excision; mTOR inhibitors if unresectable
Desmoid	Nuclear β-catenin; APC/CTNNB1; never metastasises; recurs in up to half	Surgery or observation

- **Liposarcoma**: **dense nodules in a fatty mass mean de-differentiation** and are the biopsy target. WD recurs locally in up to 40% but rarely metastasises; DD spreads distantly in up to 20%. Mean time to recurrence is ~**2 years**, so surveillance is lifelong.
- If unresectable, anthracycline or trabectedin offers modest benefit. **Pleomorphic liposarcoma** is MDM2-negative and metastasises early to lung.
- **Leiomyosarcoma** obstructs veins, urine and bowel; biopsy underestimates its aggressiveness.

KEY POINT

MDM2 amplified (WD/DD) means local disease, won or lost at the margin; MDM2 absent (pleomorphic) means systemic disease. Infiltrative types (Ewing, DSRCT) need chemotherapy because no margin is achievable.

- † **FNCLCC grade** adds differentiation (1–3), mitoses per 10 high-power fields (1: 0–9, 2: 10–19, 3: ≥20) and necrosis (0 none, 1 <50%, 2 ≥50%): **2–3 = G1, 4–5 = G2, 6–8 = G3**. Grade and histology, not size, predict distant spread.

5 · Neurogenic tumours

Tumour	Key features	Management
Schwannoma	Paravertebral; encapsulated; S100+; slow	Local resection curative (often enucleable)
Neurofibroma	NF-1 or sporadic; malignant change up to 20%	Surgical
MPNST	<10% of RP sarcoma; half NF-1; H3K27me3 loss; spreads along roots	Resect with **parent nerve**; 5-year ~70% R0 vs ~20% R1
Ganglioneuroma	3rd–4th decade; calcification up to 30%	Excision
Neuroblastoma	MYCN adverse; MIBG staging	Paediatric chemotherapy with or without surgery
Paraganglioma	Commonest at organ of Zuckerkandl (below IMA); T2-bright; secretory ~40%; metastases 20–50%; NF-1, MEN-2A, SDHB	Surgery after α-blockade† (see Ch 145)

6 · Germ cell, haematological, cystic, unknown primary

- **GCT**: an RP mass in a young man is GCT until proven otherwise; extragonadal disease is often a **burnt-out testicular primary**. Give platinum first, then resect all nonseminomatous residual masses, because teratoma is chemoresistant and marker-silent (see Ch 79–80).
- **Sex cord stromal** tumours resist chemo and radiotherapy, so do early RPLND in high-risk node-negative primaries.
- **Lymphoma** forms confluent nodal masses: biopsy, never resect.
- **Cystic** lesions (lymphangioma, mucinous cystadenoma, cystic mesothelioma) need **complete excision**, because aspiration or marsupialisation recur.
- **Unknown primary**: if midline or i(12p)-positive, give platinum; a single deposit is excised or irradiated.

KEY POINT

Histology decides first treatment: exclude chemotherapy-first entities (GCT, lymphoma, Ewing, midline unknown primary) before planning sarcoma resection.

7 · Surgery for RP sarcoma

The first operation decides cure.

- **Complete en bloc (R0) resection at the first operation** in an experienced centre is the best chance.

- **Compartmental resection** removes adherent uninvolved neighbours (kidney, colon, psoas fascia), which lowers local recurrence. Preserve duodenum, pancreatic head, major vessels and femoral nerve unless invaded. † **Never enucleate**, because the pseudocapsule contains tumour.
- Nephrectomy is needed in up to **55%** but rarely leads to dialysis; † get a split renogram first.
- After **bleomycin**, keep FiO_2 low and fluids restrictive.
- Through a midline laparotomy, **split and roll** over the great vessels and ligate lumbar vessels and retrocrural lymphatics to prevent chylous ascites.
- The **aorta is always reconstructed**. A chronically occluded **IVC** tolerates ligation; IVC grafts thrombose in 10–40%, so anticoagulate for ≥3 months.

8 · RPLND in germ cell tumours

*See the paper answer **Lymphatic drainage of testis and scrotum, RPLND templates and stage I seminoma** (Section 10, Ch 78).*

Also, from the condensed book (not in the paper answer):

- Primary RPLND gives survival ~100% with either template. In post-chemotherapy RPLND, a full bilateral infrahilar template is standard; unilateral only if the mass is **≤5 cm** in the landing zone, markers are normal, and IGCCCG risk is good or intermediate.
- Sparing the L1–L4 fibres and the superior hypogastric plexus gives antegrade ejaculation in **90–100%**.

9 · Complications

*See the paper answer **Lymphocele and postoperative lymphatic leak (chylous ascites)** (Section 15, Ch 91).*

Also, from the condensed book (not in the paper answer):

- Chylous ascites occurs in 2–7% after chemotherapy. Other risks are pancreatic leak (check drain amylase), VTE after caval reconstruction and femoral neuropraxia.

ALGORITHM

1. **Chylous ascites**:
2. Paracentesis and a medium-chain triglyceride (MCT) diet, with or without octreotide.
3. If it reaccumulates, place a drain.
4. Then nil by mouth and total parenteral nutrition.
5. If output stays >100 mL/day, then peritoneovenous (LeVeen) shunt or ligation.
6. Continue the MCT diet for 1–3 months after resolution.

10 · Adjuvant and neoadjuvant therapy

Neither is routine; surgery is the mainstay.

- **Radiotherapy**: pre-op is better tolerated than post-op. **STRASS** showed no gain, so it is **not standard**; † an exploratory benefit in low-grade liposarcoma keeps it an option.
- **Chemotherapy**: WD liposarcoma and leiomyosarcoma are largely resistant; myxoid liposarcoma, synovial sarcoma and Ewing respond. **Standard neoadjuvant chemotherapy is only for Ewing**.

11 · Recurrence and follow-up

- † Local recurrence kills in liposarcoma, whereas leiomyosarcoma fails distantly (lung, liver).
- † CT every 3–6 months for 2–3 years, 6-monthly to 5 years, then annually; lifelong for liposarcoma.
- Re-resect slow-growing resectable recurrence (WD liposarcoma).

12 · Landmark trials

Trial	Design	Result	Take-home
EORTC-62092 STRASS	Phase III, 266 patients: pre-op RT + surgery vs surgery	No abdominal recurrence-free survival difference (HR 1.01); more complications	Pre-op RT not standard
STRASS II (ongoing)	Neoadjuvant doxorubicin + ifosfamide/dacarbazine + surgery vs surgery	Awaited	Neoadjuvant chemotherapy unproven

★ Putting it together: adrenal and retroperitoneal decisions

A · Adrenal incidentaloma: two questions

ALGORITHM

1. **Malignant?** If unenhanced CT is ≤10 HU or MRI shows drop-out, then adenoma. If >10 HU, do washout CT (APW >60% / RPW >40% = adenoma). If >4 cm, suspicious or growing, consider resection.
2. **Functioning?** Test all with 1-mg DST and metanephrines, plus ARR if hypertensive or hypokalaemic.
3. If **silent, <4 cm and benign**, then no follow-up.

B · Functional tumours compared

	Aldosteronism	**Cushing**	**Phaeo**
Screen	ARR	1-mg DST, salivary or urinary cortisol	Metanephrines
Next	Confirm, then CT, then AVS	ACTH: low = adrenal; normal/high = pituitary/ectopic	CT/MRI, functional imaging; genetics
Pre-op	Correct K^+	Steroid cover	α then β; fluids, salt
Post-op	Stop K^+/spironolactone; hyperkalaemia	Steroids until axis recovers	Hypotension, hypoglycaemia
Medical	Mineralocorticoid antagonist (bilateral)	Steroidogenesis inhibitors	Long-term α-blockade

C · Adrenalectomy approach

KEY POINT

Use laparoscopy for benign tumours. Operate open for large suspected carcinoma or invasion, because rupture causes recurrence. On the right, control the short caval vein carefully. Partial adrenalectomy suits bilateral or hereditary disease.

D · Retroperitoneal mass

ALGORITHM

1. **Image**, then **exclude** paraganglioma (metanephrines), germ cell tumour (testis, markers) and lymphoma.
2. If it changes treatment, do a **core biopsy** by the retroperitoneal route.
3. **Sarcoma:** en bloc R0 resection at the first operation in a specialist centre; never enucleate. Follow liposarcoma with lifelong CT.

Rapid Revision Sheet · Section 14

Phaeochromocytoma: adrenal anatomy, hormones and perioperative management (LAQ)

- Right vein short into IVC; left vein into left renal vein.
- StAR is rate-limiting for steroids; tyrosine hydroxylase for catecholamines.
- Cortisol induces PNMT; COMT forms metanephrines inside the tumour.
- Metanephrines first; never biopsy.
- α-blockade 7–14 days for all; never β before α (α before β), plus volume expansion.
- Volume expansion prevents hypotension; watch glucose.
- Arterial line before induction; avoid histamine releasers and ephedrine.
- Surge: pause handling, nitroprusside or phentolamine; esmolol for tachycardia.
- After vein ligation: fluids, vasopressors, glucose.
- One-third hereditary; SDHB most often metastatic.

Adrenal mass evaluation, incidentaloma and adrenocortical carcinoma (LAQ)

- ≤10 HU = lipid-rich adenoma; APW >60%, RPW >40%.
- 1-mg DST: >1.8 µg/dL = MACS.
- Exclude phaeochromocytoma before biopsy or surgery.
- Surgery for incidentaloma: functioning, suspicious, >4 cm or growing.
- Feminisation in a man, >6 cm, >10 HU, poor washout: think ACC. Never biopsy before resection.
- Weiss ≥3/9; ENSAT stage IV = distant metastasis only.
- Open, R0, intact capsule, en bloc + lymphadenectomy.
- Mitotane level 14–20 mg/L; ADIUVO (low risk: no benefit); FIRM-ACT: EDP-M first line.

MIBG scan (SAQ)

- Noradrenaline analogue; block thyroid; stop labetalol.
- ^{123}I diagnostic, ^{131}I therapeutic.
- Best for extra-adrenal, metastatic and MEN-2 tumours; cold in SDHB and VHL.

Cushing syndrome and the dexamethasone suppression test (SAQ)

- Exclude exogenous steroids; 1-mg DST cortisol >1.8 µg/dL abnormal.
- 1 mg at 11 pm; 8 am cortisol ≤1.8 µg/dL excludes; >5.0 µg/dL confirms autonomy.
- Oral contraceptives and enzyme inducers cause false positives.
- ACTH low = adrenal; high = pituitary/ectopic (IPSS); ACTH separates adrenal from pituitary/ectopic.
- High-dose DST no longer routine.
- Steroid cover after adrenalectomy; Nelson syndrome after bilateral.

How will you evaluate and manage 38 yr male with uncontrolled hypertension (LAQ)

- Hypertension below 40 years: screen for secondary causes.
- ARR → saline suppression → CT → AVS (skip if <40 with adenoma >1 cm).
- Metanephrines for phaeochromocytoma; α before β.
- FMD: angioplasty; atherosclerotic RAS: medical (ASTRAL, CORAL).

Adrenal Sx. (SAQ)

- Function (any size) or malignancy risk (≥4 cm, >10 HU) indicates surgery.
- Control the vein early; right vein short into IVC.
- Open en bloc surgery for ACC. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

SECTION 15

Renal, Nephrology & Transplant

88 Surgical, Radiologic and Endoscopic Anatomy of the Kidney and Ureter

1 · Kidney: position and relations

The kidney's neighbours are what you injure at surgery.

- The kidney is ≈**11 cm** long, 125–170 g (male), at T12–L3. The **right lies 1–2 cm lower**; both descend ≈3 cm on inspiration or standing.

P1 *Embryology and anatomy of kidney; nephron-sparing surgery in small renal mass and renal mass — relations, coverings, Gerota open below, hilum order*

- Traction on the hepatorenal or splenorenal ligament tears the liver or splenic capsule.

LAQ Pelvicalyceal (renal collecting system) anatomy, its relevance to PCNL and endourology, and mini-PCNL

Paper answer · P1

Asked as:

- Describe anatomy of pelvicalyceal system in detail and its application in endourological surgeries — *Winter 2017 Paper I Q1 (LAQ)*
- Anatomy of renal collecting system and its relevance to PCNL — *Practice Set 1 Paper I Q3d (SAQ)* (also answered in 2022-Summer Paper 1, Q1 and 2017-Winter Paper 1, Q1)
- Describe anatomy of kidney in relation to PCNL and describe indication, contraindication, procedure and complications of mini-PCNL. — *Summer 2022 Paper I Q1 (LAQ)* (also answered in 2017-Winter Paper 1, Q1 (pelvicalyceal anatomy) and 2017-Winter Paper 4, Q3d (Miniperc))

Type: LAQ (25 marks); also asked as SAQ (10 marks)

Introduction

- Papilla → minor calyx → infundibulum → major calyx → renal pelvis → UPJ (ureteropelvic junction); lined by urothelium. Anatomy is very variable, so map it (CT (computed tomography) urogram or pyelogram) before every endourological procedure.
- Safe PCNL (percutaneous nephrolithotomy) access depends on the kidney's position, neighbours, vessels and calyces.

Surgical anatomy relevant to PCNL: position and orientation

- Lies T12–L3 (right 1–2 cm lower); descends about 3 cm on inspiration.
- It lies on the cone-shaped psoas, so the upper pole is medial and posterior and the lower pole lateral and anterior.
- Tract: skin, flank muscles, thoracolumbar fascia, paranephric fat, Gerota fascia, perirenal fat, cortex, then **pyramid and papilla** into the calyx.

Anatomy of the calyces (collecting system)

- **Minor calyx:** a cup around one papilla; its two side projections are the **fornices**; it drains by an infundibulum to a major calyx and pelvis. **Compound calyces** (2–3 papillae) are commonest at (cluster at) the poles. Average 7–9 calyces per kidney (range 4–19 or more).
- **Distribution:** upper pole usually 3 calyces, interpolar 3–4, lower pole 2–3.
- **Drainage pattern:** upper pole by a single midline infundibulum; lower pole by a single midline infundibulum or paired calyces; hilar region (hilar ones) by **anterior and posterior rows** of paired calyces.
- **Orientation:** the kidney lies on psoas, rotated 30–50° behind the coronal plane (posteriorly), so **posterior calyces point posterolaterally** (towards the prone surgeon). On a pyelogram anterior and posterior calyces (rows) overlap.
- **Brödel vs Hodson pattern (Kaye and Reinke):** in a Brödel-type kidney the posterior calyces lie lateral and the anterior calyces medial; in a Hodson-type kidney the reverse. Brödel type is commoner on the right, Hodson type on the left, so confirm calyx orientation with contrast/air or ultrasound before puncture.

Renal pelvis

- Pyramidal, apex to the ureter; capacity 3–10 mL. **Intrarenal** (in the sinus), **extrarenal** (larger, longer infundibula, mimics dilatation) or mixed.
- **Duplex (divided) pelvis:** upper and lower parts join at the hilum, with no direct connection between upper and lower calyces (does not link upper and lower calyces). Suspect it when one pole cannot be reached from a puncture.

Sampaio classification (Sampaio types; 140 resin endocasts)

Type	Frequency	Midzone (hilar) drainage
A-I	45%	Midzone calyces drain into the superior or inferior calyceal group (via a polar group)
A-II	17.2%	**Crossed calyces**: one to the superior and one to the inferior group (to both groups)
B-I	21.4%	Independent major calyceal group (major calyx) from the midzone
B-II	16.4%	1–4 minor calyces enter the pelvis directly (straight into the pelvis)

- **Perpendicular minor calyx (11.4%):** drains straight into the pelvis, hidden on a pyelogram (pyelography); narrow neck (≤4 mm) hinders clearance.
- **Crossed calyces:** the limb to the inferior group is **ventral in 87.5%**.

Vascular relations (the basis of safe access)

- Hilum, anterior to posterior: vein, artery, pelvis, **posterior segmental artery** (the only branch passing behind the pelvis).
- Five segmental **end arteries**; interlobar arteries run in the **columns of Bertin**, hugging the anterior and posterior faces of the infundibula (beside the infundibula).
- **Brödel line:** avascular plane between anterior and posterior arterial territories, just posterior to (behind) the lateral border, along the posterior calyces. Posterior calyces along it are the target.

Puncture route / site (Sampaio)	Vascular injury
Upper-pole infundibulum	67% (arterial 26%); posterior segmental artery
Mid-pole infundibulum	Arterial 23%

Puncture route / site (Sampaio)	Vascular injury
Lower-pole infundibulum	Infundibular artery (present) in 38%
Calyceal fornix	Venous <8%, **no arterial injury**

- So the needle passes **through the papilla along the calyx axis**, never into an infundibulum or the pelvis.

Relations (the hazards of access)

- **Relations:** pleura reaches the 12th rib posteriorly, lung the 10th intercostal space; liver and spleen descend in inspiration; **retrorenal colon** is commoner at the lower pole and when prone.

Structure	Relation	Practical point
Pleura, lung	Pleura reaches the 12th rib; lung the 10th intercostal space	Prefer subcostal or 11th-space access; risk rises above the 12th rib, highest above the 11th; puncture in expiration
Liver, spleen	Over the upper poles; descend in inspiration	Risk with supracostal access, inspiration, organomegaly
Colon	Retrorenal colon, mainly at the lower pole, commoner prone and on the left	Review CT; ultrasound shows bowel

Applications in endourology

1. PCNL (percutaneous nephrolithotomy) access

- Ideal target: a **posterior calyx with a short, wide infundibulum**, punctured (entered) **through the papilla and fornix** along the calyx axis, never into (through) the infundibulum or pelvis; identified by injected air, C-arm rotation or ultrasound. This passes through a pyramid and avoids interlobar arteries.
- **Upper-pole posterior access** gives the straightest route (best reach) to the pelvis, UPJ and most calyces (staghorn; staghorn branches). Chest risk (pneumothorax, hydrothorax) rises above the 12th rib, highest above the 11th. Puncture in expiration.
- **Lower-pole access** is chest-safe but risks (retrorenal) colon (left) and reaches the UPJ poorly (poor UPJ reach).
- Identifying (identify) posterior calyces: rainbowing the C-arm towards the surgeon shortens them (they shorten when the C-arm rotates towards the surgeon); injected air fills non-dependent (posterior, when prone) calyces (rises posteriorly when prone); ultrasound also shows bowel and pleura.
- **Also know:** traps: perpendicular or crossed calyces, duplex pelvis.

2. RIRS (retrograde intrarenal surgery)

- Entering the lower-pole infundibulum needs about **140° deflection** (104–175°); ≥270° scopes see it, but working there is harder.
- **Unfavourable lower-pole anatomy:** steep infundibulopelvic angle, long calyx (>10 mm), narrow infundibulum (<5 mm). Relocate stones to an upper calyx with a basket.
- Lower-pole stone >1 cm: PCNL preferred (**Lower Pole I, 2001:** stone-free 91% PCNL vs 21% SWL (shock wave lithotripsy)).

3. Endopyelotomy

- Incise the UPJ **laterally**, where crossing vessels are absent. A crossing vessel lowers success (42% vs 86%).
- Antegrade route: mid-posterior or superolateral calyx; not lower pole (acute angle to the UPJ).

4. **Infundibulotomy and calyceal diverticulum**

- Incise infundibular stenosis **superiorly or inferiorly**; never anteriorly.
- **Calyceal diverticulum:** cortical cavity joined to the fornix by a narrow neck; treat if symptomatic. Upper or interpolar: ureteroscopy, incise or dilate the neck. Lower pole: direct puncture onto the stone, mini-PCNL; fulguration optional.

5. **Open/anatrophic nephrolithotomy:** incision along the Brödel line.

- **Also know:** horseshoe kidney (medial access, upper pole usually subcostal); pelvic kidney (ultrasound or laparoscopic guidance); transplant kidneys; megacalycosis mimics obstruction; AV (arteriovenous) fistula or pseudoaneurysm → angioembolisation.

Mini-PCNL: definition

Term	Tract size
Standard	24–34 Fr
Mini	15–22 Fr
Ultra-mini	11–14 Fr
Micro-perc	<10 Fr

- First mini-perc: Jackman, 1998 (13 Fr).

Indications

- Renal stones 1–2 cm; lower-pole stones >1 cm; failed SWL (shock wave lithotripsy) or RIRS (retrograde intrarenal surgery).
- Calyceal diverticulum (lower pole), children, poor retrograde access (urinary diversion, ureteric stricture), selected stones >2 cm or extra tracts for staghorns.

Contraindications

- Untreated UTI (urinary tract infection), uncorrected coagulopathy, pregnancy (EAU, European Association of Urology).
- Tumour in the access tract or suspected malignant renal tumour; no safe window; unfit for anaesthesia.

Procedure

1. Urine culture, blood count, coagulation, creatinine; **non-contrast CT (computed tomography)**; antibiotic prophylaxis.
2. Lithotomy: ureteric catheter and retrograde pyelogram.
3. Prone, or supine (Galdakao-modified) for single-position retrograde access.
4. 18 G puncture of a posterior calyx under fluoroscopy (bull's-eye or triangulation) or ultrasound; urine return; guidewire and safety wire.
5. One-step, serial or balloon dilatation to 15–22 Fr; place the sheath.
6. Mini-nephroscope; holmium/thulium laser or pneumatic lithotripsy. Fragments wash out by the irrigation **"vacuum-cleaner" effect** or a **suction sheath**.
7. Flexible nephroscopy to clear calyces.

8. Exit: tubeless, small nephrostomy (≤10 Fr: less pain) or double-J stent.

Advantages and limitations

- Meta-analysis: **non-inferior stone-free rate**, smaller haemoglobin drop, **fewer transfusions**, shorter stay.
- Longer operation, poorer vision. A tight sheath raises **intrarenal pressure**, causing pyelovenous backflow and **fever or sepsis**. Advantages disappear against 24 Fr tracts.

Complications (grade by Clavien–Dindo)

Type	Points
Bleeding	Acute tract bleeding: tamponade sheath, balloon or Kaye catheter. Delayed: pseudoaneurysm or AV (arteriovenous) fistula, treated by angioembolisation
Infection	Fever, SIRS (systemic inflammatory response syndrome), urosepsis
Thoracic	Pneumothorax, hydrothorax: chest drain
Visceral	Colon (stent plus nephrostomy withdrawn into the colon), spleen, liver, duodenum
Collecting system	Perforation, extravasation, later infundibular stenosis; residual stones
Others	Irrigation absorption, venous gas embolism, DVT (deep vein thrombosis), neuropraxia, renal function loss, death

Diagram to draw: transverse section of kidney showing anterior and posterior calyces (posterior calyx), Brödel line and the needle through the posterior calyceal fornix.

> **Key points to remember:**
> - Sampaio: A-I 45%, A-II (crossed) 17.2%, B-I 21.4%, B-II 16.4%.
> - Fornix puncture: no arterial injury; upper-pole infundibulum 67% vascular injury.
> - Target a posterior calyx with a short, wide infundibulum; upper pole reaches most.
> - Lower pole: steep angle, long calyx >10 mm, infundibulum <5 mm unfavourable.
> - Endopyelotomy lateral; infundibulotomy superior or inferior, never anterior.
> - Pleura to 12th rib; retrorenal colon at the lower pole, commoner prone.
> - Mini 15–22 Fr (Jackman 1998): fewer transfusions, longer operation.
> - High intrarenal pressure in mini-PCNL causes fever and sepsis.

Also, from the condensed book (not in the paper answer):

- Upper-pole excision through the diaphragm risks pneumothorax.

2 · Coverings and compartments

- Each perirenal space is separate, so extravasation rarely crosses the midline. The **anterior pararenal space** (colon, duodenum, pancreas) does cross it; the posterior pararenal space holds fat only.
- Transperitoneally, open the **white line of Toldt**. Reach the right hilum by a **Kocher manoeuvre**; on the left, reflect the pancreatic tail, spleen and splenic vessels medially.

3 · Parenchyma and its imaging

- Cortex is ≈1 cm. **Columns of Bertin** carry interlobar arteries, so the safe track is **through pyramid and papilla**.
- Glomerular filtration rate (GFR) is ≈**125 mL/min**; ≈99% of filtrate is reabsorbed.

Modality	Normal kidney
Ultrasound	Cortex iso-/hypoechoic to liver (hyperechoic in newborn)
Computed tomography (CT)	Unenhanced 30–60 Hounsfield units (HU), cortex = medulla; enhanced 80–120 HU
Magnetic resonance (MRI)	T1 cortex brighter; T2 cortex slightly darker

CT phase	Timing	Use
Arterial	20–30 s	Vessels
Corticomedullary	30–70 s	Cortex enhances
Nephrographic	**80–120 s**	**Best for renal masses**
Excretory	>3 min	Collecting system, ureter

- On intravenous urography, no excretion at 24 h means a non-functioning kidney.
- **Corticomedullary differentiation** is lost in chronic kidney disease and acute kidney injury and returns with recovery.

Congenital anomalies

- **Malrotation** shows a central pelvis with calyces medial to it. In **ectopia**, poor drainage causes stasis, infection and stones.
- **Horseshoe**: lower poles fuse and ascent is arrested by the inferior mesenteric artery (**IMA**); it brings ureteropelvic junction obstruction (UPJO), stones, infection and multiple arteries.

4 · Renal vessels, lymphatics and nerves

End arteries make partial nephrectomy possible.

LAQ Renovascular hypertension: pathophysiology, evaluation and treatment

Paper answer · P1 · P2 · merged from 2

Asked as: P2 LAQ "Renovascular hypertension and ischaemic nephropathy: renal vascular anatomy, pathophysiology, evaluation and treatment" (asked as "Pathophysiology of renovascular hypertension (RVH) and iscemic nephropathy. Describe treatment of RVH in detail", Summer 2022 Paper II Q2; "Discuss evaluation of & management of reno-vascular hypertension", Practice Set 2 Paper II Q1; "Anat. of Renal Vasculature + Dx modalities & Rx optns for Renovasc. HTN.", Practice Set 4 Paper II Q2) · P1 LAQ "Pathophysiology of Renovascular hypertension" (Practice Set 8 Paper 1 Q2)

"Iscemic" is read as ischaemic nephropathy.

Anatomy of the renal vasculature

Renal arteries

- Arise from the aorta at the **L1–L2 disc**. The longer **right artery passes behind the IVC (inferior vena cava)**. Branches supply the adrenal, renal pelvis and proximal ureter.

- **Hilum, anterior to posterior:** renal vein, renal artery, pelvis; the posterior segmental artery lies behind the pelvis.
- **Anterior division** gives four segmental branches (apical, upper, middle, lower). **Posterior division** gives the posterior segmental artery: the first, most constant branch, passing behind the pelvis. If it crosses in front of the ureter, it can cause UPJO (ureteropelvic junction obstruction).
- **Five segmental arteries are end arteries** with no significant anastomosis. Occlusion causes segmental infarction; each segment can be resected separately.
- **Sequence of branching:** 1. Segmental 2. Lobar 3. Interlobar (columns of Bertin) 4. Arcuate (corticomedullary junction) 5. Interlobular 6. Afferent arteriole, glomerulus, efferent arteriole 7. Peritubular capillaries or vasa recta
- **Brödel line:** avascular plane between anterior and posterior territories, just behind the lateral border. Used for anatrophic nephrolithotomy and posterior calyceal access.
- **Variations (25–40%):** accessory arteries in 25–28% (T11–L4, rarely iliac or SMA [superior mesenteric artery]). They are end arteries, usually to the lower pole. Relevant to donor nephrectomy and endopyelotomy.

Renal veins

- Veins **anastomose freely** (venous collars, subcapsular plexus), so segmental vein ligation is safe. 3–5 segmental veins form the renal vein, anterior to the artery.

	Right renal vein	**Left renal vein**
Length	2–4 cm	6–10 cm
Tributaries	None	Left adrenal, left gonadal, lumbar
Course	Direct to IVC	Between SMA and aorta (nutcracker syndrome)
Anomalies	Accessory veins commoner	Circumaortic (2–16%); retroaortic

Relevance to RVH (renovascular hypertension): atheroma affects the **ostium and proximal third**; **FMD (fibromuscular dysplasia)** affects the **mid-distal artery and branches**; a stenosed accessory artery can cause segmental RVH.

Definitions

- **RVH:** hypertension caused by RAS (renal artery stenosis) that is relieved by revascularisation or nephrectomy. RAS is common in older (vascular) adults; RVH is proved only by cure.
- **Ischaemic nephropathy:** loss of GFR (glomerular filtration rate) from haemodynamically significant RAS affecting the **whole renal mass** (bilateral RAS or stenosis to a solitary kidney).

Causes

	Atherosclerotic RAS	**FMD**
Share	Most (about 90%)	Most of the rest
Patient	Older, smokers, diabetic, vascular disease	Young women
Site	**Ostium and proximal third** (aortic plaque)	**Mid and distal artery, branches**; "string of beads" (medial fibroplasia, commonest type)

	Atherosclerotic RAS	FMD
Course	Progresses to occlusion; renal function lost	Rarely occludes
Treatment	Medical first	**Balloon angioplasty without stent**; good BP (blood pressure) response

Also know: Takayasu arteritis, neurofibromatosis, midaortic syndrome (children), renal artery aneurysm, dissection, embolism, **Page kidney** (perinephric compression, e.g. subcapsular haematoma), transplant RAS.

The RAAS (renin–angiotensin–aldosterone system)

- **Renin** is released by juxtaglomerular cells of the afferent arteriole in response to: 1. low afferent arteriolar pressure (intrarenal baroreceptor); 2. low sodium chloride delivery to the **macula densa**; 3. sympathetic **β1** stimulation.
- Renin cleaves hepatic **angiotensinogen** to angiotensin I. **ACE (angiotensin-converting enzyme)**, mainly in lung endothelium, converts it to **angiotensin II**.
- **Angiotensin II (AT1 receptor) actions:** systemic arteriolar vasoconstriction; **aldosterone** release (sodium retention, potassium loss); proximal tubular sodium reabsorption; thirst and ADH (antidiuretic hormone); sympathetic facilitation; **efferent arteriolar constriction**, which maintains GFR; cardiac and vascular hypertrophy and fibrosis.
- Normally, rising BP and sodium suppress renin (negative feedback). In RVH the stenosed kidney "sees" low pressure, so this feedback fails.

Pathophysiology of RVH: Goldblatt models (basis of the tests)

- **Goldblatt (1934)** clipped the canine renal artery and produced sustained hypertension, reversed by early clip removal. Loop: low perfusion pressure releases juxtaglomerular renin, which produces angiotensin II (vasoconstriction) and aldosterone (sodium retention).

	Two-kidney, one-clip (2K1C)	One-kidney, one-clip (1K1C)
Human equivalent	Unilateral RAS, normal opposite kidney	Bilateral RAS or stenosed solitary kidney
Sodium	Normal contralateral kidney excretes the extra sodium (**pressure natriuresis**)	No normal kidney: sodium and water retained
Renin	**High** from stenotic side; contralateral suppressed	Rises, then returns to normal as volume expands
Type	**Renin (angiotensin)-dependent**	**Volume-dependent**
ACE inhibitor effect	BP falls; captopril test positive	Less BP effect; may cause **acute renal failure / acute kidney injury**

Phases of 2K1C hypertension

1. **Acute phase:** renin-dependent; ACE inhibitor or clip removal reverses it.
2. **Transition:** sodium retention and secondary hyperaldosteronism; renin falls towards normal.
3. **Chronic phase:** hypertension damages the contralateral kidney (nephrosclerosis) and resistance vessels. BP no longer depends on the stenosis, so **revascularisation cures less** (cure falls with duration). This explains poorer results in long-standing hypertension.

Other contributing mechanisms

- Renal afferent nerve signals raise **sympathetic activity**.
- Angiotensin II generates **oxidative stress** (reactive oxygen species), lowers **nitric oxide** and raises **endothelin**, worsening vasoconstriction.
- Inflammation and fibrosis; left ventricular hypertrophy and diastolic dysfunction.

Pathophysiology of ischaemic nephropathy

- The kidney receives about **25% of cardiac output** (P1: 20–25%), far more than its metabolic need. **Autoregulation** holds GFR until stenosis is severe (about **70% or more**, or a significant translesional pressure gradient).
- Beyond this, the stenotic kidney keeps its GFR by angiotensin II-mediated **efferent constriction**. Blocking it (ACE inhibitor or ARB [angiotensin receptor blocker]) drops GFR: hence the **creatinine rise >30%** after these drugs.
- Chronic hypoperfusion causes hypoxia, inflammation, microvascular rarefaction, **tubulointerstitial fibrosis** and atrophy. Atheroemboli and hypertensive injury add damage.
- Early disease is reversible ("hibernating" kidney); established fibrosis is not.
- **Flash pulmonary oedema:** bilateral RAS causes volume retention plus angiotensin II-driven diastolic dysfunction.

Evaluation: clinical clues (who to test)

- Onset <30 years (FMD) or severe hypertension after 55 years; **resistant** (≥3 drugs) or accelerated hypertension; sudden worsening of controlled BP.
- **Creatinine rise >30% after ACE inhibitor/ARB**; **flash pulmonary oedema**; kidneys differing in size by >1.5 cm; abdominal bruit; **hypokalaemia** (secondary hyperaldosteronism).

Evaluation: basic tests

- Creatinine and eGFR (estimated GFR), potassium, urinalysis and proteinuria, ECG; exclude primary aldosteronism and phaeochromocytoma.

Evaluation: imaging and functional tests

Test	Method and finding	Comment
Duplex Doppler US (ultrasound)	High PSV (peak systolic velocity), above about 180–200 cm/s; renal/aortic velocity ratio >3.5; **RI (resistive index) = (PSV − EDV)/PSV**	First line, no contrast; operator dependent. **RI ≥0.8 (>0.8) means parenchymal disease and predicts poor response**
CT angiography	Stenosis, ostial plaque, beads	Best non-invasive anatomy; contrast nephropathy
MR (magnetic resonance) angiography	Gadolinium-enhanced	Caution if eGFR <30 mL/min

Test	Method and finding	Comment
Captopril renography (DTPA, diethylenetriaminepentaacetic acid)	Day 1 after oral captopril: delayed uptake, low peak; repeat off ACE inhibitor and calcium channel blocker. Improvement of ≥10% = high probability RVH	Removing angiotensin II (efferent constriction) drops GFR in the stenosed kidney, proving renin-dependence; DTPA is cleared purely by filtration; use declining
Catheter angiography	Gold standard; **translesional (pressure) gradient** (systolic ≥20 or mean ≥10 mmHg = significant)	Done when intervention is planned
Renal vein renin	Ratio >1.5 lateralises	Before nephrectomy

Algorithm `Clinical clue -> Duplex US (or CTA/MRA if eGFR allows) -> no RAS: treat as essential hypertension -> RAS: FMD -> angioplasty Atherosclerotic -> optimal medical therapy -> high-risk features + viable kidney -> angiography + gradient -> stent`

Treatment 1: medical (first line for atherosclerotic RAS)

- **ACE inhibitor or ARB** controls renin-dependent BP; check creatinine and potassium at 1–2 weeks. Calcium channel blockers, diuretics.
- **Statin, antiplatelet, smoking cessation, diabetes control**: most deaths in these patients are cardiovascular.

Treatment 2: landmark trials (atherosclerotic RAS)

- **ASTRAL (2009):** stenting plus medical therapy versus medical therapy alone gave **no benefit** in renal function, BP, events or mortality; procedural complications.
- **CORAL (2014):** stenting versus optimal medical therapy gave **no reduction** in cardiovascular or renal events; only a small BP fall.
- Hence **no routine revascularisation** of atherosclerotic RAS (ESC [European Society of Cardiology] 2017).

Treatment 3: indications for revascularisation

- **FMD with hypertension** (first choice).
- Atherosclerotic RAS with **flash pulmonary oedema** or recurrent unexplained heart failure, **rapidly declining function** with bilateral or solitary-kidney RAS, or truly resistant hypertension.
- Kidney must be viable (not small and atrophic; RI <0.8).

Treatment 4: endovascular

- **PTRA (percutaneous transluminal renal angioplasty) alone for FMD**; stent only for dissection or failure.
- **Ostial atherosclerotic lesion: angioplasty plus stent**, because a balloon alone fails from elastic recoil of aortic plaque. Restenosis is the main late problem.
- Complications: contrast nephropathy, **atheroembolism**, dissection, rupture, thrombosis, access haematoma.

Treatment 5: surgical revascularisation

- Indications: failed endovascular therapy, **branch disease or aneurysm**, children, coexisting aortic surgery.

Technique	Use
Aortorenal bypass (saphenous vein; internal iliac artery in children; PTFE [polytetrafluoroethylene])	Standard
Transaortic endarterectomy	Bilateral ostial plaque
Splenorenal (left) / hepatorenal (right) bypass	Hostile (diseased) aorta
Ex vivo bench repair + autotransplantation	Branch FMD, aneurysms
Nephrectomy (partial or total)	Small non-functioning kidney with lateralising renin

Transplant RAS

- Raised velocity on Doppler; contrast imaging confirms; angioplasty or stent first, surgery if this fails.

Follow-up

- BP, creatinine, duplex at intervals for restenosis; cure is commoner in FMD than atherosclerosis.

Diagrams to draw

- 2K1C versus 1K1C with the renin–angiotensin loop (low pressure, renin, angiotensin II, aldosterone, hypertension).
- RAAS cascade from renin release to angiotensin II actions.
- Renal artery dividing into anterior (four segments) and posterior divisions, with the Brödel line.

> **Key points to remember:**
>
> - Five segmental end arteries; posterior segmental passes behind the pelvis; veins anastomose freely.
> - Left renal vein 6–10 cm, receives adrenal, gonadal and lumbar veins.
> - Goldblatt 1934: 2K1C = renin-dependent; 1K1C = volume-dependent.
> - Chronic phase: contralateral nephrosclerosis makes RVH stenosis-independent.
> - Angiotensin II keeps GFR by efferent constriction; ACE inhibitor causes acute kidney injury in bilateral RAS.
> - RVH is proved only by cure; duplex first, angiography with gradient before intervention.
> - ASTRAL 2009 and CORAL 2014: stenting adds nothing to medical therapy in atherosclerotic RAS; no routine stenting.
> - FMD gets balloon angioplasty without stent; ostial atheroma: stent.
> - Revascularise for flash pulmonary oedema, declining GFR in a viable kidney (RI <0.8), resistant hypertension.

Also, from the condensed book (not in the paper answer):

- **CT angiography** is the gold standard for vascular anatomy but misses vessels <2 mm.
- Right-sided lymph drains to interaortocaval/paracaval nodes but **may cross to left para-aortic** nodes.

5 · Percutaneous puncture and vessels

*See the paper answer **Pelvicalyceal (renal collecting system) anatomy, its relevance to PCNL and endourology, and mini-PCNL** (Section 15, Ch 88).*

Also, from the condensed book (not in the paper answer):

- A through-and-through puncture hits anterior segmental vessels and causes marked haemorrhage.

> **KEY POINT**
>
> Why the fornix? Arteries run on both faces of the infundibula, and a forniceal puncture through the papilla avoids them. If an infundibulum is unavoidable, choose the lower pole. Vascular injury causes haemorrhage, arteriovenous fistula, pseudoaneurysm and loss of function.

6 · Pelvicalyceal system

*See the paper answer **Pelvicalyceal (renal collecting system) anatomy, its relevance to PCNL and endourology, and mini-PCNL** (Section 15, Ch 88).*

Also, from the condensed book (not in the paper answer):

- Dilatation is mimicked by an extrarenal pelvis, **megacalycosis** and a calyceal diverticulum.

7 · Ureter: course and relations

Injuries occur at crossings; stones stick at narrowings.

- The adult ureter is 22–30 cm. Stones impact at three **narrowings**: ureteropelvic junction (UPJ), iliac crossing and ureterovesical junction (UVJ).
- It lies on psoas and crosses the **common iliac bifurcation** (landmark), with gonadal vessels in front. The **pelvic brim** is a commonest injury site.
- In females it passes under the **uterine artery 1.5–2 cm** from the cervix ("water under the bridge"). In males the vas loops over it.
- The **intramural** segment is 1.2–2.5 cm long. Bladder pressure closes this oblique tunnel, which prevents reflux.

8 · Radiologic anatomy of the ureter

- Because of **peristalsis**, the whole ureter is rarely opacified; a continuous column suggests obstruction or atony.
- A stricture needs **proximal dilatation**; crossing vessels and iliac indentation can mimic it.
- Incomplete duplication (one bud, one orifice) is commoner than complete duplication (two buds, two orifices).

KEY POINT

Weigert–Meyer rule: the upper-moiety orifice is **inferior and medial** and obstructs (ureterocele); the lower moiety refluxes (vesicoureteral reflux, VUR).

- Bilateral medial displacement suggests retroperitoneal fibrosis or pelvic lipomatosis.

9 · Ureteric blood supply, nerves, histology

Segment	Artery	Enters	Endoureterotomy
Proximal	Renal	Medially	**Lateral**
Middle	Common iliac	Posteriorly	**Anterior**
Distal	Superior vesical	Laterally	**Medial**

- The longitudinal anastomosis is fragile, so preserve adventitia and avoid lateral retraction to prevent ischaemic stricture.
- Peristalsis is **myogenic**, so denervation does not cause reflux. Pain (T11–L1) refers to flank, groin, scrotum or labia.
- In **UPJO**, the junction has excess longitudinal muscle and collagen.

10 · Endoscopic anatomy

- The trigone is the bladder's most vascular part. Orifices lie 1–2 cm from the midline and move laterally on filling.

Lyon axis	Grades
Position	A medial trigone (normal) · B · C lateral
Configuration	0 volcano (normal) · 1 stadium · 2 horseshoe · 3 golf-hole

- A higher grade means more reflux. The **intramural ureter is narrowest** (3–4 mm) and may need dilatation.
- The lower-pole infundibulum needs ≈**140°** deflection.

ALGORITHM

1. Infundibulotomy: incise superior, then inferior, then posterior quadrants.
2. **Never anterior**.

90 Fundamentals of Nephrology

1 · Why a urologist needs nephrology

- Each kidney has ≈**1 million nephrons**, fixed at birth. When nephron mass falls (nephrectomy, agenesis), the rest **hyperfilter**.
- Interstitial fibroblasts make **erythropoietin**; the proximal tubule makes **calcitriol** by 1α-hydroxylation.

> **KEY POINT**
>
> The glomerulus is a pressure filter set by arteriolar tone; the tubule is energy-hungry and near-hypoxic. Most disorders fail one or the other.

2 · Measuring kidney function

Serum creatinine (SCr) often misleads in surgical patients.

SAQ GFR estimation in prospective renal donor

Paper answer · P1 · P2 · merged from 2

Asked as: P1 SAQ "GFR estimation in a prospective renal donor and creatinine clearance" (asked as "Describe methods to estimate GFR in a prospective renal donor", Summer 2019 Paper I Q3a; "Creatinine clearance", Practice Set 4 Paper I Q3d) · P2 SAQ "GFR estimation: clinical assessment and in the prospective transplant donor" (asked as "Methods for GFR estimation in prospective tx donor", Practice Set 9 Paper II Q3a; "GFR clinical assessment", Winter 2023 Paper II Q3a)

Assumed meaning: tx = transplant (prospective living kidney donor).

Definition

- GFR (glomerular filtration rate) is the plasma volume filtered by both kidneys per minute; normal about 120 mL/min/1.73 m². It is the standard index of kidney function.
- The ideal marker is freely filtered and neither secreted nor reabsorbed.

Why it matters in the donor

- The donor loses about half the nephron mass; function settles at 60–80% of the pre-donation level.
- **KDIGO (Kidney Disease: Improving Global Outcomes) 2017** sets GFR thresholds and asks for **measurement, not just estimation**, at a decision point: GFR is assessed by at least two methods, **including a measured one**.

Method 1: serum creatinine and eGFR (estimated GFR)

- Creatinine comes from muscle and is freely filtered (not reabsorbed), but generation depends on **muscle mass**, diet (cooked meat, creatine supplements) and age. It is also secreted, so it **overestimates GFR**, increasingly as function falls. Trimethoprim blocks secretion.
- **CKD-EPI (Chronic Kidney Disease Epidemiology Collaboration) 2021 creatinine equation:** age and sex, **no race term**; currently recommended; used for staging. MDRD (Modification of Diet in Renal Disease) and Cockcroft–Gault are older.
- Less accurate above 60 mL/min, where donors lie: a screening test only.

Method 2: cystatin C

- 13-kDa protein made by all nucleated cells; freely filtered, then fully reabsorbed and broken down. Independent of muscle mass; affected by thyroid state, steroids, inflammation.
- The **combined creatinine–cystatin C equation** is more accurate than either alone; it is the most accurate estimate.

Method 3: measured 24-hour creatinine clearance

- **Definition:** creatinine clearance (CrCl) is the volume of plasma cleared of creatinine per minute. It estimates GFR because creatinine is freely filtered and not reabsorbed.
- Normal about 120 mL/min/1.73 m^2 in young men, slightly lower in women; falls with age.
- **CrCl = (U × V) / P**: U = urine creatinine, V = urine flow (mL/min), P = plasma (serum) creatinine.
- **Method:** discard the first morning void, collect all urine for 24 hours including the next morning's first void; take serum creatinine during the collection.
- **Completeness check:** creatinine excretion should be **20–25 mg/kg/day in men, 15–20 mg/kg/day in women**.

Limitations of creatinine clearance

- **Tubular secretion** of creatinine is small in health but rises as function falls, so CrCl **overestimates GFR** in CKD (chronic kidney disease).
- Collection errors are common. Creatinine generation depends on muscle mass, diet and age.
- Drugs blocking secretion (**trimethoprim, cimetidine**) raise serum creatinine without real injury.
- Invalid in AKI (acute kidney injury).
- **Also know:** cimetidine pretreatment blocks secretion to improve accuracy; averaging urea and creatinine clearance helps in advanced CKD.

Estimated clearance: Cockcroft–Gault

- **CrCl = (140 − age) × weight (kg) / (72 × serum creatinine mg/dL)**, × 0.85 in women.
- Used for **drug dosing**; not normalised to body surface area; inaccurate in obesity.

Method 4: mGFR (measured GFR), the reference standard

Marker	Method / comment
Inulin	Urinary clearance; gold standard, research only
Iohexol	Plasma clearance; safe and accurate (another source: plasma or urinary clearance)
Iothalamate	Urinary or plasma clearance
^{51}Cr-EDTA (ethylenediaminetetraacetic acid)	Plasma clearance
^{99m}Tc-DTPA (diethylenetriaminepentaacetic acid)	Plasma sampling, or gamma camera (Gates method); purely filtered; also gives **split function**

Method 5: split function

- DTPA renogram gives each kidney's share (differential function); MAG3 or DMSA (dimercaptosuccinic acid) are alternatives.
- A difference >10% (beyond 45:55) matters: **leave the better kidney with the donor**.

KDIGO 2017 donor decision

- Initial **eGFRcr**; confirm with **mGFR, measured creatinine clearance or eGFRcr-cys** (creatinine–cystatin C eGFR).

GFR (mL/min/1.73 m²)	Decision
≥90	Accept
60–89	Individualise (age, lifetime risk)
<60	Decline

- Indian practice: **DTPA GFR** with split function plus CT (computed tomography) angiography is standard.

Clinical uses of GFR and creatinine clearance

- **CKD staging (KDIGO):** G1 ≥90, G2 60–89, G3a 45–59, G3b 30–44, G4 15–29, G5 <15 mL/min/1.73 m², always with albuminuria (ACR, albumin-to-creatinine ratio).
- **Living donor evaluation:** KDIGO 2017 table above; confirm with measured GFR.
- **Urology:** drug and contrast dosing; **cisplatin eligibility** (CrCl <60 mL/min usually unfit); continent diversion (>35 mL/min); partial versus radical nephrectomy.
- Extremes of body size, amputees and pregnancy, where eGFR is unreliable: measure creatinine clearance.
- **Acute change:** eGFR is invalid when function is unstable; AKI uses creatinine change and urine output. NGAL (neutrophil gelatinase-associated lipocalin) is emerging.

Diagrams to draw

- Plasma tracer clearance curve after a single injection.
- Nephron showing filtration plus a small secreted fraction of creatinine.

Key points to remember:

- Creatinine overestimates GFR (secretion); CKD-EPI 2021 is race-free.
- Cystatin C adds accuracy; creatinine–cystatin C combined equation is most accurate.
- Inulin is the gold standard; iohexol or DTPA in practice; DTPA gives GFR and split function.
- KDIGO 2017: ≥90 accept, 60–89 individualise, <60 decline.
- Better kidney stays with the donor.
- CrCl = UV/P; check completeness by creatinine excreted per kg.
- Secretion makes CrCl overestimate GFR as function falls.
- Cockcroft–Gault: (140 − age) × weight / (72 × SCr), × 0.85 women.

Also, from the condensed book (not in the paper answer):

- SCr is falsely low with small muscle mass, and falsely raised by **trimethoprim and cimetidine**, which block secretion. Cystatin C is raised by hyperthyroidism, steroids and inflammation.

KEY POINT

Trap: after cystectomy or bladder repair, a rising SCr with ascites may be an **intraperitoneal urine leak** reabsorbed across the peritoneum, not kidney injury.

*See the paper answer **Proteinuria** (Section 1, Ch 3).*

Also, from the condensed book (not in the paper answer):

- **eGFR is invalid when function is changing**, so acute kidney injury (AKI) is defined by SCr change and urine output.

3 · Acute kidney injury

Glomerular haemodynamics

- Net filtration pressure ≈**10 mm Hg** = glomerular hydrostatic 60 − Bowman's 18 − oncotic 32.

Arteriolar change	Renal blood flow	GFR
Afferent constriction	Falls	Falls
Afferent dilatation	Rises	Rises
Efferent constriction	Falls	Rises
Efferent dilatation	Rises	Falls

Angiotensin II blockade therefore drops GFR when perfusion is low.

Definition and staging (KDIGO)

Stage	Serum creatinine	Urine output
1	Rise ≥**0.3 mg/dL** (within 48 h†) or ×1.5 baseline (within 7 days†)	<0.5 mL/kg/h for 6 h
2	×2 baseline	<0.5 mL/kg/h for 12 h
3	×3, or SCr ≥4 mg/dL with acute rise ≥0.5 (†or dialysis started)	<0.3 mL/kg/h for 24 h, or anuria 12 h

LAQ Acute renal failure (AKI) and acute tubular necrosis

Paper answer · P2

Asked as:

- Etiology, pathogenesis and management of acute renal failure. — *Winter 2024 Paper II Q2 (LAQ)* (acute tubular necrosis also answered in 2022-Summer Paper 2, Q3c)
- Acute tubular necrosis. — *Summer 2022 Paper II Q3c (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

Acute renal failure is now called AKI (acute kidney injury).

Definition and staging: KDIGO (Kidney Disease: Improving Global Outcomes)

- AKI (per KDIGO) = rise in serum creatinine **or** fall in urine output; either alone is enough.

Stage	Serum creatinine	Urine output
1	Rise ≥0.3 mg/dL in 48 h, or ×1.5 baseline in 7 days	<0.5 mL/kg/h for 6 h
2	×2 baseline	<0.5 mL/kg/h for 12 h
3	×3, or ≥4 mg/dL, or RRT (renal replacement therapy)	<0.3 mL/kg/h for 24 h, or anuria 12 h

- One-fifth to one-third of inpatients; mortality about 25%, about 50% (about half) if RRT is needed.

- **ATN definition:** ATN is the commonest **intrinsic** cause of AKI (acute kidney injury): tubular injury from **ischaemia or toxins**. A misnomer: cell death is mainly **apoptosis**.

Etiology

Type	Causes
Prerenal	Hypovolaemia (bleeding, vomiting, burns), heart failure, sepsis, hepatorenal syndrome; worsened by NSAIDs (non-steroidal anti-inflammatory drugs) and ACE (angiotensin-converting enzyme) inhibitors
Intrinsic	**ATN (acute tubular necrosis):** ischaemia, contrast, aminoglycosides, cisplatin, myoglobin. **AIN (acute interstitial nephritis):** β-lactams, NSAIDs, PPIs (proton pump inhibitors), checkpoint inhibitors. Glomerulonephritis; vascular (renal artery or vein thrombosis, atheroemboli)
Postrenal	BOO (bladder outlet obstruction), bilateral ureteric obstruction or obstructed single kidney: stones, tumours, retroperitoneal fibrosis, ureteric injury at surgery

- Prerenal AKI and ATN cause about three-quarters of hospital AKI.

Causes of ATN

Ischaemic	Toxic
Prolonged prerenal state: shock, sepsis, haemorrhage, burns, cardiac surgery, intraoperative hypotension, renal artery clamping, **transplant (delayed graft function)**	**Exogenous:** aminoglycosides, amphotericin, cisplatin, iodinated contrast. **Endogenous:** myoglobin (rhabdomyolysis, e.g. prolonged flank position), haemoglobin, light chains

Pathogenesis (pathophysiology)

- **Autoregulation** holds GFR (glomerular filtration rate) by afferent dilatation (prostaglandins, myogenic reflex) and efferent constriction (angiotensin II, tubuloglomerular feedback).
- **Prerenal:** low perfusion lowers glomerular pressure. Tubules are intact, so they **conserve sodium and water**. NSAIDs block afferent dilatation; ACE inhibitors block efferent constriction. Prolonged prerenal state becomes ATN.
- **Ischaemic ATN:**
- **Why tubules:** the medulla works near hypoxia (medullary oxygen tension is only 10–20 mmHg), while the **S3 proximal tubule (segment)** and **medullary thick ascending limb** consume much oxygen and are most vulnerable.
- **Loss of polarity:** Na^+/K^+-ATPase loses its basolateral position (moves off the basolateral membrane) → sodium reabsorption fails → high sodium at macula densa → **tubuloglomerular feedback** constricts afferent arteriole (afferent constriction) → GFR falls (protective).
- Cells die (mostly apoptosis) and slough (dead cells slough) → **muddy-brown casts** block (obstruct) tubules; filtrate leaks back; endothelial injury and inflammation worsen hypoxia; autoregulation is lost.
- Phases (clinical course): initiation → **maintenance** (oliguria, rising creatinine) → **recovery** (polyuria: tubules regenerate before concentrating ability returns).
- **Toxic ATN:** vasoconstriction plus direct tubular injury. **Contrast:** risk with age, diabetes, CKD (chronic kidney disease). **Rhabdomyolysis** (long lateral positioning): myoglobin causes toxicity, cast obstruction and hypoperfusion.

- **Postrenal:** raised tubular pressure, then afferent vasoconstriction; recovery depends on duration (dog studies of complete obstruction: full at 1 week, 70% at 2 weeks, none at 6 weeks).

Evaluation (diagnosis: ATN versus prerenal)

- History (fluid loss, hypotension, drugs, contrast), volume status, bladder scan.
- **Ultrasound early** in unexplained AKI to exclude obstruction.

Index	Prerenal	ATN
Urine sodium	<20 mmol/L	>20 mmol/L
FeNa (fractional excretion of sodium)	<1%	**>1%**
FeUrea (on diuretics)	<35%	>35%
Specific gravity	>1.015	<1.010
Sediment	Bland	**Muddy-brown granular casts**, tubular cells
Fluid challenge (response to fluids)	Quick (rapid) recovery	Lag of days to weeks

- Creatine kinase; white cells (AIN); red cell casts (glomerulonephritis); biopsy only if recovery lags (fails).

Management

General

- Treat (remove) the cause; **avoid hypotension** (MAP, mean arterial pressure, >65 mmHg or near baseline) and nephrotoxins; stop NSAIDs and ACE inhibitors or ARBs (angiotensin receptor blockers) (RAAS (renin–angiotensin–aldosterone system) blockers); adjust drug doses.
- Isotonic crystalloid (fluids) for hypovolaemia; aggressive fluids in rhabdomyolysis; intravenous fluid before contrast (pre-contrast hydration).
- **Postrenal:** urethral or suprapubic catheter; **double-J stent or PCN (percutaneous nephrostomy)**; infected obstructed kidney is an emergency.
- AIN: stop drug ± steroids; glomerulonephritis: nephrology and biopsy.

Complications

- **Hyperkalaemia:** calcium gluconate 10 mL of 10% (protects heart) → insulin 10 U with 50 mL 50% dextrose, salbutamol nebuliser (shift) → binders, diuretics or dialysis (remove).
- **Volume overload:** loop diuretics in escalating doses. **Diuretics (only for volume overload) do not treat ATN.**
- Acidosis: bicarbonate. Uraemic bleeding: desmopressin 0.3 µg/kg.

Dialysis indications (AEIOU), refractory to medical therapy

	Indication
A	Acidosis
E	Electrolytes, mainly hyperkalaemia
I	Intoxications / ingestions (lithium, salicylate, alcohols)
O	Overload with respiratory distress

	Indication
U	Uraemia: encephalopathy, pericarditis, bleeding

- Intermittent haemodialysis via 10–12 Fr jugular or femoral catheter; **CRRT (continuous RRT)** if haemodynamically unstable. **STARRT-AKI (2020):** early start gave no survival benefit.

Post-obstructive diuresis

- Causes: retained urea (osmotic diuresis), volume expansion with ANP (atrial natriuretic peptide), and down-regulated aquaporin-2 and sodium transport.
- Up to 300 mL/h for about 2 days. Free oral fluids; if unable to drink, replace about 75% of output with 0.45% saline; electrolytes every 4–6 hours. Gradual bladder decompression has no role.

Outcome

- Full or partial recovery, or dialysis dependence; persistence ≥90 days = CKD. ATN recovery takes days to weeks.
- **Also know:** (early) biomarkers NGAL (neutrophil gelatinase-associated lipocalin), KIM-1 (kidney injury molecule-1); AKD (acute kidney disease, under 3 months); gadolinium (groups I and III) avoided in AKI.

Diagram to draw: tubular cell losing polarity → sloughing → casts and back-leak.

> **Key points to remember:**
> - KDIGO: creatinine ≥0.3 mg/dL or ×1.5; urine <0.5 mL/kg/h for 6 h.
> - FeNa <1% prerenal, >1% ATN; specific gravity <1.010, muddy-brown casts in ATN; ultrasound to exclude obstruction.
> - S3 segment and thick ascending limb are most vulnerable.
> - Diuretics only for overload (do not treat ATN); dialysis for refractory AEIOU.
> - Post-obstructive diuresis: oral fluids; replace 75% with 0.45% saline if needed.

Also, from the condensed book (not in the paper answer):

- **3 months** separates acute kidney disease (AKD) from CKD.

Prerenal AKI

- Urine is bland, fractional excretion of sodium (FeNa) is low, and function **recovers immediately** on correction.
- **Autoregulation**: the **myogenic reflex** dilates the afferent arteriole via prostaglandins, so it is **disabled by NSAIDs** (non-steroidal anti-inflammatory drugs).
- In **tubuloglomerular feedback (TGF)**, angiotensin II constricts the efferent arteriole, so it is **disabled by renin–angiotensin–aldosterone system (RAAS) blockade**.
- **Modifiable** factors are hypotension, anaemia, sepsis, nephrotoxins, rhabdomyolysis and hypercalcaemia.

*See the paper answer **Acute renal failure (AKI) and acute tubular necrosis** (Section 15, Ch 90).*

Also, from the condensed book (not in the paper answer):

- High FeNa after diuretics does not prove ATN; use fractional excretion of urea (FeUrea). **Response to correction is the single most important test.**
- Keep mean arterial pressure (MAP) >**65 mm Hg**. **Fluids** (mOsm/L): plasma and 0.9% saline 308; Plasmalyte 294; Ringer's lactate 273; 0.45% saline 154; 5% dextrose (D5) 0, so D5 is free water.

Intrinsic AKI: the cast localises the compartment

	Acute tubular necrosis (ATN)	Acute glomerulonephritis (GN)	Acute interstitial nephritis (AIN)
Cause	Ischaemia, toxins	Immune complex, anti-glomerular basement membrane (anti-GBM), cell-mediated	Drug, infection, systemic
Urine	Granular casts	Dysmorphic red cells, **red cell casts**, proteinuria	White cells, **white cell casts**, with or without eosinophils

Acute tubular necrosis

- **Rhabdomyolysis** (obese patient, long flexed lateral robotic case) needs **early aggressive isotonic fluid**; alkalinisation is unproven. †Prevent it with padding and limited flexion time.
- **Give diuretics only for symptomatic overload; they do not treat ATN.**

Postcontrast AKI

- This is an SCr rise within **48 h** of intravenous (IV) iodinated contrast. Risks include CKD, diabetes and the arterial route.
- **IV fluid before contrast is the only effective prevention** (†for eGFR <30 or AKI); use the lowest dose.
- **Gadolinium** groups I and III cause nephrogenic systemic fibrosis (**NSF**), so avoid them in AKI, eGFR <**30** and dialysis. Group II (gadobenate, gadoteridol, gadobutrol, gadoterate) is allowed. **Group III (gadoxetate) is treated as group I.**

Acute interstitial nephritis

- **Drugs cause most**, notably β-lactams, NSAIDs, proton pump inhibitors and **checkpoint inhibitors**.
- Suspect it in unexplained hospital AKI with **sterile pyuria**. Urine eosinophils are insensitive and non-specific MCQ.
- **Stop the drug**; early steroids may help, and about half recover fully.

Glomerulonephritis

- In microhaematuria, a rising SCr, proteinuria, red cell casts, hypertension or purpura call for **urgent nephrology and biopsy**, not cystoscopy first.

Complications of AKI

- **Overload** is urgent only with respiratory distress. Give a high-dose IV loop diuretic and add metolazone or acetazolamide (**sequential blockade**).
- For uraemic bleeding, give desmopressin (**DDAVP**) **0.3 µg/kg** IV/IM 30–60 min before the procedure.

Renal replacement therapy

- AEIOU (acidosis, electrolytes, ingestions, overload, uraemia) means failed medical management, not a biochemical threshold. **Early dialysis gives no benefit.** Continuous therapy is not superior.
- Non-recovery for ≥**90 days** means CKD or end-stage kidney disease (ESKD).

4 · Drug dosing in reduced kidney function

- In AKI, eGFR overestimates function, so reassess doses daily.
- †**Cockcroft–Gault** = (140 − age) × weight (kg) / (72 × SCr mg/dL), × 0.85 if female.
- Avoid NSAIDs; †withhold RAAS blockers in dehydration or perioperative hypotension ("sick-day" rule).
- †Avoid nitrofurantoin if eGFR <30 (UK guidance allows short courses at 30–44). †Stop metformin at eGFR <30, and withhold it before iodinated contrast in AKI or eGFR <30.

5 · Chronic kidney disease

- **KDIGO** (Kidney Disease: Improving Global Outcomes) defines CKD as abnormal structure or function for ≥**3 months**. Classify by **cause, GFR and albuminuria**.

GFR (mL/min/1.73 m²)	A1 albumin-to-creatinine ratio (ACR) <30 mg/g (<3 mg/mmol)	A2 30–300	A3 >300 (>30 mg/mmol)
G1 ≥90	Low*	Moderate	High
G2 60–89	Low*	Moderate	High
G3a 45–59	Moderate	High	Very high
G3b 30–44	High	Very high	Very high
G4 15–29	Very high	Very high	Very high
G5 <15 (kidney failure)	Very high	Very high	Very high

*G1–G2 count as CKD only with albuminuria or structural damage. Albuminuria can shift risk two bands: eGFR 43 with ACR 1100 mg/g is **G3bA3, very high**.

- **Each perioperative AKI is a modifiable progression factor**.

SAQ Metabolic abnormalities in CKD

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 3 Paper 2, Q3e

Definition

- **CKD (chronic kidney disease), KDIGO (Kidney Disease: Improving Global Outcomes):** abnormal kidney structure or function for **≥3 months**. Staged by GFR (glomerular filtration rate) G1–G5 (G3a 45–59, G3b 30–44, G4 15–29, G5 <15 mL/min/1.73 m²) and albuminuria A1–A3.
- Each abnormality is a lost kidney function: excretion, acid–base balance, hormone production.

Abnormalities

Abnormality	Mechanism	Management
Sodium and water	Fixed urine concentration; sodium retention → overload, hypertension	Salt restriction; diuretics (chlorthalidone works even in advanced CKD)
Hyperkalaemia	Less distal sodium delivery; RAAS (renin–angiotensin–aldosterone system) blockers lower aldosterone	Potassium binders and diuretics, so RAAS blockers can continue
Metabolic acidosis	Less ammonium excretion; retained sulphate and phosphate (anion gap rises late); bicarbonate loss	Sodium bicarbonate or citrate to keep bicarbonate **≥22 mEq/L**; slows progression, protects bone and muscle
CKD-MBD (mineral and bone disorder)	Phosphate retention → FGF-23 (fibroblast growth factor 23) rises → less **calcitriol** → hypocalcaemia → **secondary hyperparathyroidism**	Phosphate restriction, binders with meals, active vitamin D; calcimimetic; parathyroidectomy if tertiary
Anaemia	Erythropoietin falls, mainly below GFR 30	Iron first, then erythropoietin; Hb (haemoglobin) target 10–11.5 g/dL (harm above 13)
Hyperuricaemia	Reduced excretion	Treat gout; asymptomatic lowering unproven (CKD-FIX)
Lipids	High triglycerides, low HDL (high-density lipoprotein)	Statin ± ezetimibe if ≥50 years, not on dialysis (SHARP)
Glucose	Insulin resistance, but slower insulin clearance → hypoglycaemia	Lower insulin doses

Bone disease (renal osteodystrophy)

- **High turnover:** osteitis fibrosa (PTH [parathyroid hormone] excess).
- **Low turnover:** adynamic bone (oversuppressed PTH), osteomalacia.
- **Tertiary hyperparathyroidism:** autonomous gland, hypercalcaemia; persists after transplant.
- **Vascular calcification** adds cardiovascular death.

Other endocrine and uraemic effects

- **Also know:** hypogonadism, hyperprolactinaemia, erectile dysfunction (better after transplant); growth failure in children; protein-energy wasting; dialysis amyloid; uraemic platelet dysfunction (desmopressin).
- **Urological relevance:** dose drugs by GFR; avoid gadolinium if eGFR (estimated GFR) <30; each AKI (acute kidney injury) speeds progression.

Diagram to draw: CKD-MBD cascade: phosphate ↑ → FGF-23 ↑ → calcitriol ↓ → calcium ↓ → PTH ↑ → bone disease.

> **Key points to remember:**
> - FGF-23 rises first; secondary then tertiary hyperparathyroidism.
> - Bicarbonate ≥22 mEq/L; Hb 10–11.5 g/dL.
> - Hyperkalaemia: binders rather than stopping RAAS blockade.

Also, from the condensed book (not in the paper answer):

- Tertiary hyperparathyroidism (HPT) is treated with a calcimimetic, then subtotal parathyroidectomy.

Slowing progression

- Target glycated haemoglobin <7.5% and blood pressure <**130/80**; the agent matters only in proteinuric disease (RAAS blockade).
- **Sodium–glucose cotransporter-2 (SGLT2) inhibitors** protect kidney and heart and cut mortality, with or without diabetes. Add **finerenone** in CKD with type 2 diabetes.

6 · Haematuria and proteinuria: when to involve a nephrologist

- †Microhaematuria is ≥**3** red blood cells per high-power field (RBC/HPF) on microscopy; a positive dipstick needs confirmation.
- **Glomerular clues** are dysmorphic red cells, red cell casts, raised ACR or protein-to-creatinine ratio (PCR), rising SCr, hypertension and systemic disease.

ALGORITHM

1. Haematuria with or without proteinuria: send urine microscopy, ACR, SCr/eGFR.
2. If there are casts, dysmorphic cells, AKI or systemic features, refer for **urgent nephrology**.
3. If isolated non-glomerular haematuria, do a urological evaluation.
4. If isolated albuminuria, stage CKD and refer per criteria.

- †KDIGO referral criteria: AKI or an abrupt GFR fall; GFR <30; ACR ≥300 mg/g (≥30 mg/mmol); a fall >5 mL/min/1.73 m²/year; red cell casts or haematuria >20 RBC/HPF; BP refractory to ≥4 drugs; persistent potassium abnormality; recurrent stones; hereditary disease.

7 · Electrolyte and acid–base disorders

Water and sodium

- **Sodium disorders are water disorders**. **Volume depletion overrides hypo-osmolality** in driving antidiuretic hormone (ADH).
- Of 1 L, D5 leaves **83 mL** in the circulation, 0.9% saline 250 mL and 5% albumin 1000 mL.

Hyponatraemia

- †Transurethral resection (TUR) syndrome is absorption of hypotonic glycine or sorbitol during monopolar resection.

Volume	Urine	Causes
Hypovolaemic	Na >20	Diuretics, mineralocorticoid deficiency, cerebral salt wasting
Hypovolaemic	Na <20	Vomiting, diarrhoea
Euvolaemic	Osm > serum	Syndrome of inappropriate ADH (SIADH), cortisol deficiency, hypothyroidism, pain, drugs
Euvolaemic	Osm low	Polydipsia, low solute ("tea and toast", beer potomania)
Hypervolaemic	Na >20	Kidney failure

Volume	Urine	Causes
Hypervolaemic	Na <20	Heart failure, cirrhosis, nephrotic

- Treat SIADH with water restriction, then vaptans, salt with furosemide, or urea.
- **Rate**: in chronic hyponatraemia, raise sodium ≤**8–12 mEq/L per 24 h**, or **osmotic demyelination** follows. Treat overcorrection with DDAVP and free water.
- With severe symptoms, give **3% saline to raise sodium 4–6 mEq/L**, then slow down.

Hypernatraemia

- Nephrogenic diabetes insipidus (DI) follows lithium, hypercalcaemia, hypokalaemia and **relief of obstruction**.
- Give free water. In chronic cases, lower sodium ≤**0.5 mEq/L/h**, or cerebral oedema follows.

Potassium

SAQ Management of acute hyperkalaemia

Paper answer · P1

Asked as:

- Management of Acute Hyperkalemia — *Winter 2025 Paper I Q3e (SAQ)*
- Mgt. of acute hyperkalemia. — *Practice Set 6 Paper I Q3e (SAQ)*

Type: SAQ (10 marks)

(also answered in 2025-Winter Paper 1, Q3e)

Definition and physiology

- Serum potassium **>5.5 mmol/L**; mild 5.5–5.9, moderate 6.0–6.4, **severe ≥6.5** (UK Kidney Association). Above 6.0 is generally urgent.
- About 98% of potassium is intracellular; the ratio sets the resting membrane potential, so a high level (high levels) causes **arrhythmia and cardiac arrest**.
- Defence: rapid shift into cells (insulin, β_2-agonists via Na^+/K^+-ATPase), then renal excretion (needs distal sodium delivery, urine flow, aldosterone).

Causes

Mechanism	Examples
Pseudohyperkalaemia	Haemolysis, tourniquet, fist clenching, thrombocytosis
Reduced excretion	AKI (acute kidney injury), CKD (chronic kidney disease), **obstructive uropathy** (type 4 RTA [renal tubular acidosis]), adrenal insufficiency; drugs: ACE (angiotensin-converting enzyme) inhibitors, ARBs (angiotensin receptor blockers), spironolactone, NSAIDs (non-steroidal anti-inflammatory drugs), tacrolimus
Shift out of cells	Acidosis, insulin deficiency, β-blockers, suxamethonium
Cell release	**Rhabdomyolysis** (prolonged lithotomy or flank position), tumour lysis, crush
Urological	**Jejunal conduit** (reabsorbs urinary potassium)

Assessment

- Repeat the sample to **exclude pseudohyperkalaemia** if the ECG (electrocardiogram) is normal.

- **ECG changes:** peaked T (earliest) → prolonged PR, wide QRS → loss of P → sine wave → ventricular fibrillation (VF). Ascending weakness.
- Creatinine, glucose, blood gas, CK (creatine kinase); stop potassium sources.

Emergency treatment (in order)

Step	Drug and dose	Onset and effect
1. Protect the heart	**Calcium gluconate 10%, 10 mL IV (intravenous)** over 5–10 min; repeat if ECG unchanged	1–3 min; lasts 30–60 min; does not lower potassium
2. Shift into cells	**Insulin 10 units regular IV with 50 mL of 50% dextrose**	15–30 min; lowers about 0.6–1 mmol/L; check (monitor) glucose (hypoglycaemia)
	Salbutamol 20 mg nebulised	Additive to insulin
	Sodium bicarbonate	Only with metabolic acidosis
3. Remove	Loop diuretic if passing urine; **binders**: sodium zirconium cyclosilicate (about 1 h), patiromer (about 7 h), sodium polystyrene sulfonate (slow; colonic necrosis)	Hours
	Haemodialysis	Refractory or oliguric, ESKD (end-stage kidney disease)

- **Relieve obstruction** (catheter, nephrostomy); watch for post-obstructive diuresis.
- Monitor ECG, potassium and glucose; treat the cause.

Diagram to draw: flowchart: ECG change or K^+ ≥6.5 → calcium → shift → remove.

> **Key points to remember:**
> - Calcium stabilises the membrane but does not lower potassium.
> - Insulin 10 units + 50 mL 50% dextrose; salbutamol 20 mg nebulised.
> - Binders, diuretics and dialysis remove potassium; relieve obstruction.

ALGORITHM

1. **Stabilise** if there is any ECG change: calcium gluconate 10%, 10 mL IV.
2. **Shift**: insulin 10 U with 50 mL of 50% dextrose; nebulised salbutamol 20 mg.
3. **Remove**: haemodialysis if oliguric or ESKD; loop diuretic if non-oliguric; bicarbonate if acidotic; a binder.
4. Stop culprit drugs.

- **Sodium polystyrene sulfonate (SPS)** is slow and causes **colonic necrosis**, so it is not first-line.
- **Hypokalaemia** from **vomiting** is caused by kidney loss, not gastric loss. Replace potassium chloride, **oral preferred**, 20–40 mEq repeated.

Calcium

- Saline and loop diuretics raise urine calcium; **thiazides lower it**.
- **Hypercalcaemia in a stone former**: non-suppressed PTH means primary HPT; if suppressed, check PTH-related peptide (renal, bladder, squamous cancer) and 1,25-vitamin D (sarcoid, lymphoma). Low urine calcium suggests familial hypocalciuric hypercalcaemia.

- Treat if symptomatic or ≥**14 mg/dL**: **isotonic saline** 200 mL/h to urine 100–150 mL/h; loop diuretic only if overloaded; calcitonin; IV bisphosphonate; denosumab.

Acid–base

- Anion gap (AG) = Na − (Cl + HCO_3); it is raised if >12. If the AG rises less than the bicarbonate falls, a non-AG acidosis coexists.
- **Urine AG** = (Na + K) − Cl. Negative means ammonium excretion, so gut loss; positive means a renal cause (renal tubular acidosis, RTA).

*See the paper answer **Renal tubular acidosis (distal and type 2)** (Section 11, Ch 95).*

Also, from the condensed book (not in the paper answer):

- **Topiramate and zonisamide** cause calcium phosphate stones. MCQ Laxative abuse leads to **ammonium acid urate stones**.
- †Bowel conduits and neobladders absorb ammonium chloride, causing hyperchloraemic non-AG acidosis.

8 · Urinary tract obstruction and post-obstructive diuresis

Relief of obstruction can cause dangerous fluid loss.

Term	Meaning
Hydronephrosis	Pelvicalyceal dilatation, with or without obstruction
Obstructive uropathy (OU)	Obstruction at any level
Obstructive nephropathy	Irreversible parenchymal loss

- OU is **curable**, so exclude it (†by ultrasound) in any unexplained AKI or CKD.

*See the paper answer **Post-obstructive diuresis** (Section 4, Ch 51).*

ALGORITHM

1. Relieve the obstruction, then allow free oral fluids; do not give routine IV fluid.
2. If output stays high, or there is hypovolaemia, or the patient is obtunded or nil by mouth, then check a panel every **4–6 h** (Na, K, phosphate, Ca, Mg).
3. Replace ≈**75%** of the output with **0.45% saline** plus electrolytes.

KEY POINT

Why not replace all output? The diuresis is mostly appropriate excretion of retained salt, water and urea; †full replacement sustains it.

Recovery after obstruction

Canine study	Model	Recovery
Vaughan and Gillenwater	Complete unilateral ureteral obstruction	100% at 7 days; 70% at 14 days; 30% at 4 weeks; **none at 6 weeks**

- Duration, not dilatation, decides the injury. A small stone with low-grade obstruction can be observed safely for **2–4 weeks**; beyond 4 weeks the outcome is unpredictable.
- Keep a **low threshold for a stent or nephrostomy**. †Infection with obstruction needs urgent drainage.

9 · Landmark trials

Trial	Design	Result	Take-home
†PRESERVE	Bicarbonate vs saline; N-acetylcysteine (NAC) vs placebo	No difference	Saline; no NAC
†STARRT-AKI	Accelerated vs standard renal replacement therapy	No survival benefit; more dialysis dependence	Dialyse for AEIOU, not early
†DAPA-CKD / EMPA-KIDNEY	SGLT2 inhibitor, CKD with or without diabetes	Less progression	Standard in CKD
†FIDELIO-DKD	Finerenone, CKD with type 2 diabetes	Fewer kidney and cardiovascular events	Add to RAAS blockade
†CHOIR / TREAT	High vs lower haemoglobin target	Harm with high target	Avoid Hb >13 g/dL

91 Renal Transplantation

1 · End-stage disease and why transplant

Transplant beats dialysis, so plan it early.

- KDIGO stage **G5 (eGFR <15 mL/min/1.73 m²)** is end-stage renal disease (ESRD).
- **Transplant is optimal therapy**, and **pre-emptive** transplant (before dialysis) is best. It may triple life expectancy, but death risk is **higher for the first 3–6 months**, then much lower (Wolfe).

2 · Recipient evaluation

> **KEY POINT**
>
> Two questions: will the patient survive the operation, and will immunosuppression unmask something dormant (infection, cancer)?

- Deceased-donor listing is allowed at eGFR ≤**20**.

Recurrence risk	Diseases	Implication
Very low	Autosomal dominant polycystic kidney disease (ADPKD), dysplasia, Alport, **genetic** focal segmental glomerulosclerosis (FSGS)	None
Higher	**Primary FSGS**, membranoproliferative GN, membranous	Counsel; monitor
Histological often, clinical rarely	IgA nephropathy	—
Metabolic	Primary hyperoxaluria	Combined liver–kidney

Disease coded in native-kidney genes cannot recur in the graft; disease from circulating or immune factors can.

- **Cardiac**: coronary artery disease is the main cause of death; screen high-risk patients.
- **Vascular**: heavy iliac calcification may prevent anastomosis (non-contrast CT). ADPKD with a family history of intracranial bleed needs an aneurysm screen.
- **Obesity** is a relative contraindication that increases wound complications, delayed graft function (DGF) and rejection.
- **Infection**: active tuberculosis contraindicates until treated. Screen for hepatitis B, C and HIV, and **vaccinate before** transplant, because live vaccines are barred afterwards.

Malignancy	Transplant
Prostate Gleason ≤6; superficial non-melanoma skin cancer; incidental renal tumour <1 cm	Without prior treatment
Basal/squamous skin, melanoma in situ, renal cell carcinoma <3 cm, small low-grade thyroid, superficial bladder cancer	No wait after cure
All other	After treatment, individualised wait

SAQ Pre transplant nephrectomy

Paper answer · P2

Type: SAQ (10 marks)
Source: 2022-Summer Paper 2, Q3e

Assumed meaning: removal of the native kidney(s) before or at kidney transplantation.

Principle

- **Default: leave native kidneys.** They give residual urine (less fluid restriction), erythropoietin and vitamin D activation; nephrectomy adds morbidity.
- Remove them only when they will threaten the patient or graft under **immunosuppression**.

Indications

Group	Indication	Reason
Infection	Chronic pyelonephritis, infected stones, recurrent upper UTI (urinary tract infection) with **high-grade VUR (vesicoureteral reflux)** or obstruction, XGP (xanthogranulomatous pyelonephritis)	Sepsis under immunosuppression
ADPKD (autosomal dominant polycystic kidney disease)	No room for graft (kidney below iliac crest), recurrent cyst infection or bleeding, pain, early satiety	Space and symptoms
Malignancy	Suspicious mass or complex cyst (acquired cystic disease), high tumour risk (e.g. Denys–Drash)	Most RCC (renal cell carcinoma) in recipients arises in native kidneys
Hypertension	Uncontrolled despite multiple drugs	Renin-driven
Heavy proteinuria	Nephrotic syndrome with malnutrition or thrombosis (e.g. congenital nephrotic syndrome)	Hides graft proteinuria, hypercoagulability
Others	Massive polyuria, large stone burden	Graft fluid balance, infection

Timing

- **Before transplantation:** infection or tumour; reconstruction 6–12 weeks earlier.
- **Simultaneous:** ADPKD needing space, ipsilateral through the transplant incision; avoided in small children.
- **After transplantation:** symptomatic ADPKD once the graft works.

Technique

- **Laparoscopic** (transperitoneal or retroperitoneal), open, or **embolisation** in high-risk patients.
- **PD (peritoneal dialysis) patients:** retroperitoneal route keeps PD going; transperitoneal surgery needs temporary haemodialysis.
- Bilateral: prone approach for small kidneys; morcellation or open for large ADPKD kidneys.
- **Ureter:** transect high and leave a normal ureter (for later ureteroureterostomy); **remove a dilated refluxing ureter completely**; keep it as a possible continent stoma if catheterisation may be needed.

Complications

- Bleeding, transfusion (sensitisation), infection, bowel injury, hypotension after bilateral removal, loss of residual function.

Also know: pre-transplant urodynamics; bladder augmentation if capacity <75% expected or pressure >40 cmH_2O.

Diagram to draw: enlarged ADPKD kidney reaching the iliac fossa beside the planned graft site.

> **Key points to remember:**
> - Default: leave native kidneys.
> - Indications: infection/VUR, ADPKD space, tumour, hypertension, proteinuria.
> - Retroperitoneal route for PD; leave a normal ureter.

3 · Immunological compatibility

LAQ Discuss the role of tissue typing and cross match in live and cadaveric

Paper answer · P3

Type: LAQ (25 marks)
Source: 2018-Summer Paper 3, Q2

The question is incomplete; it is taken to mean "... live and cadaveric (deceased-donor) renal transplantation".

Introduction

- Two immune barriers decide compatibility: **preformed antibodies** (anti-A/B isoagglutinins and anti-HLA (human leukocyte antigen) antibodies) and **T-cell responses** to mismatched HLA.
- **Tissue typing** measures how well donor and recipient HLA match. **Antibody screening and crossmatch** detect preformed antibodies before the kidney meets them. Together they prevent **hyperacute rejection**, now practically eliminated.

The HLA system

- HLA is the human MHC (major histocompatibility complex), a gene cluster on **chromosome 6 (short arm)**. Genes are inherited as two **haplotypes** (one from each parent) and expressed codominantly.

	Class I	**Class II**
Loci	HLA-A, -B, -C	HLA-DR, -DQ, -DP
Expression	All nucleated cells	Antigen-presenting cells, endothelium
Presents to	CD8 (cytotoxic) T cells	CD4 (helper) T cells

- **Sibling** chance: 25% two-haplotype identical, 50% one-haplotype, 25% none. A parent or child always shares one haplotype.

Tissue typing methods

Method	Principle	Note
Serological (microlymphocytotoxicity)	Lymphocytes + typed antisera + complement; dead cells take up dye	Low resolution; largely replaced
Molecular (DNA-based)	PCR (polymerase chain reaction) with SSP (sequence-specific primers), PCR-SSO (sequence-specific oligonucleotide probes), sequence-based typing, **NGS** (next-generation sequencing)	High (allele-level) resolution; now standard

- **Matching:** usually reported as mismatches at **A, B and DR** (0–6). **DR matters most**, then B, then A; DQ increasingly typed. Fewer mismatches mean better graft survival and less sensitisation for a future transplant (Collaborative Transplant Study, Opelz). The effect is larger in deceased-donor grafts.

ABO compatibility

- Same rules as transfusion: **O universal donor, AB universal recipient**.
- **A2 (non-A1) donors** can go to O or B recipients with low anti-A titre.
- **ABO-incompatible** living transplant is possible: lower titres (plasmapheresis or immunoadsorption, rituximab, IVIG (intravenous immunoglobulin)) to a centre target before surgery.

Sensitisation and antibody screening

- Anti-HLA antibodies come from **pregnancy, blood transfusion, previous transplant** (and infection).
- **PRA (panel reactive antibody):** recipient serum tested against a panel of lymphocytes; result as % of panel killed.
- **Solid-phase (Luminex) single-antigen beads:** identify each antibody and its strength (MFI, mean fluorescence intensity). Antibodies against a donor are **DSA (donor-specific antibodies)**.
- **cPRA (calculated PRA):** % of donors the patient would be incompatible with, calculated from listed "unacceptable antigens". It replaced cell-panel PRA for allocation; very high cPRA gets allocation priority.

Crossmatch

Type	Method	Meaning if positive
CDC (complement-dependent cytotoxicity)	Donor T and B lymphocytes + recipient serum + complement; cell death scored. **DTT** (dithiothreitol) removes IgM autoantibody; AHG (antihuman globulin) increases sensitivity	**Positive T-cell CDC with IgG DSA = absolute contraindication** (hyperacute rejection; Patel and Terasaki, 1969)
Flow cytometry crossmatch	Antibody binding on donor cells detected by fluorescent anti-IgG	More sensitive; CDC-negative/flow-positive = higher risk of AMR (antibody-mediated rejection); desensitise or exchange
Virtual crossmatch	Compare recipient's bead-defined antibodies with donor HLA type	Speeds deceased-donor allocation; confirm by cell-based crossmatch

- **B-cell positive** alone: class II antibody or non-specific binding; interpret with bead results.

- Crossmatch uses a **recent serum** plus historical peak sera.

Living versus deceased donor

	Living donor	Deceased (cadaveric) donor
Typing	ABO and HLA early in donor work-up; choose best of several donors	ABO and HLA at donor referral
Crossmatch	At work-up and a **final crossmatch shortly before surgery**	At the time of offer, often virtual first; done against the list rapidly (cold ischaemia)
Role of matching	Less critical: even unrelated living donors do better than well-matched deceased donors (Terasaki, 1995)	Allocation uses HLA match, sensitisation (cPRA), waiting time and donor quality (KDPI, Kidney Donor Profile Index)
If incompatible	**Paired (swap) exchange**, desensitisation, ABO-incompatible protocol	Patient waits; acceptable-mismatch programmes

- **India:** swap (paired) donation is legal under the Transplantation of Human Organs (THOA) amendment, 2011.

Managing the incompatible or sensitised recipient

- **Kidney paired donation**, including chains with altruistic donors.
- **Desensitisation:** plasmapheresis + low-dose IVIG, or high-dose IVIG, ± rituximab; **imlifidase** (cleaves IgG) for highly sensitised deceased-donor recipients (conditional EU approval). Desensitised HLA-incompatible living transplant gave better survival than waiting on dialysis (Montgomery, 2011).

After transplant

- Monitor for **de novo DSA** (especially class II) with dd-cfDNA (donor-derived cell-free DNA) and biopsy; de novo DSA predicts graft loss.

Recent advances

- **Eplet (epitope) matching**, high-resolution NGS typing, virtual crossmatch, imlifidase.

Diagram to draw: CDC crossmatch: donor lymphocytes + recipient serum + complement, dead cells take up dye (positive).

> **Key points to remember:**
> - HLA on chromosome 6; match A, B, DR; DR matters most.
> - Positive T-cell CDC crossmatch (IgG DSA) = no transplant: hyperacute rejection.
> - Flow and bead assays are more sensitive; cPRA guides allocation.
> - Incompatible pair: paired exchange (legal in India since 2011) or desensitisation.

4 · Deceased donation

LAQ Deceased donor kidney: selection, maintenance and retrieval techniques

Paper answer · P1 · P2 · merged from 2

Asked as: P2 LAQ "Deceased (cadaver) kidney donor: selection, identification, maintenance, distribution and retrieval" (asked as "Describe selection criteria for cadaver renal donor. Describe in details kidney retrieval surgery", Summer 2021 Paper II Q2; "Describe procedure of

identification, maintenance, distribution and retrieval of kidney in a 25 yr old brain dead pt", Practice Set 3 Paper II Q1) · P1 SAQ "Kidney retrieval techniques for organ donation" (2023-Winter Paper 1 Q3c)

Assumed meaning: the Indian deceased-donor pathway: identification and brain-stem death certification, donor maintenance, allocation and retrieval. Marginal donor: see 2016-Summer Paper 3, Q3f; brain death certification: 2018-Summer Paper 4, Q3c; also answered in 2022-Summer Paper 4, Q3b.

Types of donor

- **Deceased (cadaver): DBD** (donation after brain death: circulation maintained until cold perfusion) or **DCD** (donation after circulatory death: warm ischaemia).
- **Living:** related or unrelated (THOA, Transplantation of Human Organs Act, in India).
- A 25-year-old brain-dead patient is an ideal **standard-criteria DBD** donor: circulation continues until cold perfusion, so warm ischaemia is minimal.
- India: **THOA 1994**, amended 2011, **Rules 2014**, makes brain-stem death legal death; **NOTTO** (National Organ and Tissue Transplant Organisation) allocates organs.

Identification and selection criteria

Potential donor: ventilated patient with catastrophic brain injury (head injury, intracranial bleed, hypoxia). ICU (intensive care unit) informs the **transplant coordinator**; **mandatory request** to the family.

Donor evaluation

- Cause of death, medical and social (high-risk behaviour) history; kidney: terminal creatinine, urine output, hypertension, diabetes, ultrasound.
- Laboratory: ABO and HLA (human leucocyte antigen) typing; serology and NAT (nucleic acid testing) for HIV (human immunodeficiency virus), hepatitis B and C, CMV (cytomegalovirus), EBV (Epstein–Barr virus), syphilis, toxoplasma.

Absolute contraindications

- Active or metastatic cancer, haematological malignancy, past melanoma, untreatable viral infection.
- Treated bacterial infection (even bacterial meningitis) is acceptable; viral encephalitis needs care.

Standard versus marginal donor

Category	Definition
ECD (expanded criteria donor), UNOS (United Network for Organ Sharing)	Age ≥60, or 50–59 with 2 of: hypertension, creatinine >1.5 mg/dL, death from CVA (cerebrovascular accident)
KDPI (Kidney Donor Profile Index)	Replaced ECD in the US (2014); >85% = marginal; longevity matching with EPTS (Estimated Post-Transplant Survival)
Paediatric <15 kg	En bloc transplant

- **Remuzzi biopsy score** for marginal kidneys: 0–3 single, 4–6 dual, ≥7 discard.
- **Also know:** HCV (hepatitis C virus)-viraemic donors with antivirals.

Legal death and consent: certification of brain-stem death (THOA)

- **DBD preconditions:** apnoeic coma with a known irreversible cause; exclude drugs (sedatives, relaxants), hypothermia, metabolic causes and shock.

- **Board of four (4-member Board):** doctor in charge of the hospital; independent specialist from an approved panel; neurologist or neurosurgeon (else a nominated surgeon, physician, anaesthetist or intensivist); treating doctor. **None from the transplant team.**

Brainstem reflex (all absent)	Method	Cranial nerves
Pupillary	Bright light: pupils fixed, no reaction	II, III
Corneal	Touch cornea: no blink	V, VII
Oculocephalic (doll's eye)	Turn head: eyes move with the head	VIII, III, VI
Oculovestibular (caloric)	Ice-cold water into each ear (drum seen): no eye movement	VIII, III, VI
Cranial motor response	Supraorbital pressure: no grimace	V, VII
Gag and cough	Touch pharynx; suction catheter in trachea: no response	IX, X

- **Apnoea test (last):** pre-oxygenate with 100% oxygen; disconnect the ventilator with oxygen insufflation; **no breathing effort at $PaCO_2$ (arterial carbon dioxide tension) ≥60 mmHg**. Abort for hypotension, desaturation or arrhythmia.
- **Two sets of tests at least 6 hours apart**; **Form 10**. Ancillary tests (angiography, perfusion scan) only if examination is impossible.
- **DCD (Maastricht):** I dead on arrival; II failed resuscitation (both uncontrolled); **III arrest after withdrawal of support (controlled, commonest)**; IV arrest in a brain-dead donor; V euthanasia.
- **Consent:** next-of-kin (near relative) consent after a **mandatory request** by ICU staff; in medico-legal cases inform police.

Donor maintenance

- **Why:** herniation causes a **catecholamine storm**, then loss of sympathetic tone (hypotension), **diabetes insipidus (DI)**, hypothermia, DIC (disseminated intravascular coagulation), low thyroid hormone and cortisol, and pulmonary oedema.
- **"Rule of 100s":** systolic pressure >100 mmHg, urine >100 mL/h, PaO_2 (arterial oxygen tension) >100 mmHg, haemoglobin >100 g/L.
- **Measures:** fluids guided by CVP (central venous pressure), no starch colloids (nephrotoxic); **vasopressin** first (pressor and treats DI), then noradrenaline; methylprednisolone 15 mg/kg, thyroid hormone, insulin; lung-protective ventilation; warming; correct sodium, potassium and clotting.

Distribution (allocation)

- **NOTTO**, New Delhi, with **ROTTOs** (regional) and **SOTTOs** (state organisations), keeps registries and waiting lists.
- **Sequence:** hospital and state list, then region, then national.
- **Kidney criteria:** ABO compatibility, waiting time on dialysis, HLA match, priority for children and previous living donors, **negative crossmatch** (donor spleen or nodes). Many states let the retrieving hospital keep one kidney.
- 2023 policy removed the domicile requirement and upper age limit for registration.

Deceased multi-organ kidney retrieval (DBD)

1. **Checks:** death certificate, consent, blood group; coordinate thoracic, liver, pancreas and kidney teams.
2. **Incision:** midline xiphoid-to-pubis laparotomy with median sternotomy.
3. **Exposure:** mobilise right colon and small-bowel mesentery off the retroperitoneum (Cattell–Braasch manoeuvre); expose aorta and IVC (inferior vena cava).
4. **Prepare cannulation:** encircle the distal aorta above the bifurcation; ligate the inferior mesenteric artery; prepare the supracoeliac aorta for clamping. **Look for and preserve aberrant (accessory) renal arteries**; keep accessory arteries from the iliacs.
5. **Cold perfusion:** systemic heparin; cannulate the distal aorta; clamp the supracoeliac aorta; **vent** the IVC (or right atrium); flush cold UW (University of Wisconsin) or HTK (histidine–tryptophan–ketoglutarate) solution; crushed (surface) ice; open Gerota's fascia laterally to cool the kidneys.
6. **Remove heart, lungs, liver and pancreas first.**
7. **Kidneys en bloc (last):** lift both with aorta, IVC, perinephric fat and ureters off the posterior wall; keep periureteric tissue; divide ureters low. Divide aorta and IVC (upper limit set by the liver team, lower at the bifurcation); lift everything anteriorly and divide prevertebral tissue.
8. **Extras:** iliac vessels for reconstruction; spleen and lymph nodes for crossmatch.

DCD: after death and a no-touch period, **super-rapid** laparotomy, direct aortic cannulation and IVC venting limit warm ischaemia; kidneys usually go on machine perfusion.

Back-bench (back-table) preparation

- Separate the kidneys: open the aorta posteriorly to see the renal ostia and divide it between them, keeping an aortic **Carrel patch** for each.
- Keep the IVC with the right kidney to lengthen the short right renal vein.
- Ligate tributaries and lymphatics (prevents lymphocele); trim fat without decapsulation; spare the lower pole–ureter triangle.
- Document anatomy and injuries; biopsy marginal kidneys; triple-bag in cold solution on ice at 4 °C; label.

Living donor nephrectomy

- **Principles:** donor safety first; maximal vessel length; keep periureteric tissue and the lower pole–ureter triangle; minimal handling; short warm ischaemia.
- **Left kidney preferred** (longer vein).

Technique	Features
Open (flank or mini-incision)	Complex anatomy, more often right side
Laparoscopic (Ratner, 1995)	Preferred; less pain, faster recovery
Hand-assisted	About two-thirds of US cases; tactile control
Retroperitoneoscopic, LESS (laparoendoscopic single-site), robotic	Selected centres

- **Sequence:** 1. Hilar dissection. 2. Ureter divided. 3. Artery then vein stapled or clipped. 4. Rapid extraction. 5. Cold flush.

- **Risks:** complications about 16%; mortality 3 per 10,000; conversion 0.26–3.34%.

Preservation

- Cooling to 4 °C slows metabolism. **Hypothermic machine perfusion** reduces DGF (delayed graft function) versus static cold storage (Moers, 2009), but not long-term graft loss.
- Keep cold ischaemia short: cold ischaemia >24 hours raises thrombosis risk.
- Normothermic perfusion is experimental.

Pitfalls

- Polar artery or intimal injury, short right vein, ureteric devascularisation, capsular tears.

Diagram to draw

- Aorta and IVC with distal aortic cannula, supracoeliac clamp, IVC vent and en bloc kidneys.

Key points to remember:

- THOA: 4-member Board (Board of four), none from the transplant team; two tests ≥6 h apart; Form 10.
- Apnoea test: no breathing at $PaCO_2$ ≥60 mmHg.
- ECD: ≥60, or 50–59 with 2 of hypertension, creatinine >1.5, CVA death; KDPI >85%.
- Rule of 100s; vasopressin for hypotension and DI.
- Cannulate, clamp, vent, cool: distal aortic cannula, supracoeliac clamp, IVC vent, cold UW/HTK and ice.
- Kidneys removed last, en bloc; Carrel patch, IVC for right vein.
- Left kidney preferred for living donation.
- Machine perfusion reduces DGF.

Also, from the condensed book (not in the paper answer):

- En bloc kidneys go to an adult with body mass index <30 and low immunological risk; thrombosis is 10–15%.

	Donation after brain death (DBD)	**Donation after circulatory death (DCD)**
Warm ischaemia	Minimal	Present, so rapid aortic cannulation and cooling are needed
Preservation	Static cold storage	Usually machine perfusion (reduces DGF)

ALGORITHM

1. Midline incision and sternotomy; mobilise bowel off the retroperitoneum.
2. Control the aorta and inferior vena cava (IVC).
3. Find aberrant or accessory arteries **before cannulation**.
4. Perfuse; ice via opened Gerota fascia.
5. Remove kidneys en bloc with **periureteric tissue preserved**.
6. Separate on the back table; store cold.

5 · Living donation

SAQ Evaluation and selection criteria of the living renal donor

Paper answer · P1 · P2 · P3 · merged from 3

Asked as: P2 SAQ "Pre-operative evaluation of the living (related and unrelated) renal donor" (asked as "Pre operative investigations for renal donor", Summer 2017 Paper II Q3b; "Pre-op assessment of unrelated donor", Practice Set 8 Paper II Q3a) · P3 SAQ "Assessment criteria for kidney donor." (2022-Summer Paper 3 Q3a) · P1 SAQ "Extended Criteria for Selection of Live Renal Donor" (2025-Winter Paper 1 Q3a)

Assumed meaning: evaluation and acceptance criteria for a living kidney donor; for the unrelated question, a living unrelated kidney donor (Indian context); "extended criteria" read as the expanded (marginal, "medically complex") living donor. Related: 2023-Winter Paper 4, Q3e; marginal donor in 2024-Winter Paper 4, Q3e.

Principles

- **Donor safety first. KDIGO (Kidney Disease: Improving Global Outcomes) 2017** replaced fixed cut-offs with a **risk-based approach**: the donor's estimated lifetime ESKD (end-stage kidney disease) risk is compared with the centre's acceptable threshold.
- Multidisciplinary team with an **independent donor advocate**.
- For an unrelated donor the medical work-up equals that of a related donor; the extra step is **legal and ethical**: proof of genuine, unpaid motive.
- India: **THOA (Transplantation of Human Organs Act) 1994, amended 2011** (Rules 2014): near relatives may donate; others need authorisation committee approval; swap donation allowed.

Legal assessment of the unrelated donor (THOA)

- A donor "other than near relative" donates for **affection or attachment**: **Authorisation Committee approval is mandatory**.
- Committee: hospital-based if >25 transplants a year, otherwise district or state; members are not from the transplant team.
- Process: joint application; **video-recorded interview** of donor and recipient; verify identity, documents, financial status and long association; exclude middlemen and payment; confirm free, informed consent.
- Foreign recipient with unrelated Indian donor not approved; appeal within 30 days.

Medical assessment sequence

1. Compatibility (immunological)

- ABO group, HLA (human leukocyte antigen) typing, **crossmatch** (CDC [complement-dependent cytotoxicity] and flow), DSA (donor-specific antibody). Incompatible pairs: paired exchange.
- Unrelated donor: more HLA mismatch is expected, yet living unrelated grafts outperform deceased-donor grafts.

2. Kidney function and urine

- GFR (glomerular filtration rate) by two methods, one measured (e.g. DTPA [diethylenetriamine pentaacetic acid] GFR or creatinine clearance).

- Urinalysis, culture; persistent microscopic haematuria: cystoscopy, imaging, ± kidney biopsy.

3. Medical and metabolic

- Blood pressure (ambulatory if borderline), fasting glucose, **HbA1c (glycated haemoglobin)**, OGTT (oral glucose tolerance test) if at risk; lipids, liver tests, coagulation, BMI (body mass index).
- Active cancer, chronic infection and cardiovascular disease exclude.

4. Infection screen

- HIV (human immunodeficiency virus), HBsAg (hepatitis B surface antigen), anti-HBc, HCV (hepatitis C virus), CMV (cytomegalovirus), syphilis; **tuberculosis** screen.

5. Cardiopulmonary: ECG (electrocardiogram), chest X-ray, echo; stress test if indicated.

6. Age-appropriate cancer screening: Pap smear, mammogram, PSA (prostate-specific antigen), stool test or colonoscopy.

7. Urological: stones (metabolic work-up), haematuria, malignancy, recurrent UTI (urinary tract infection).

8. Anatomy

- **CT (computed tomography) angiography:** number of vessels, early branching, stones, masses.
- **Split function** (DTPA or DMSA [dimercaptosuccinic acid]) if asymmetric: the **better kidney stays with the donor**; left preferred (longer vein).

9. Psychosocial: voluntary, no coercion or payment; independent donor advocate.

Acceptance criteria: standard, extended and exclusion

Factor	Standard (accept)	Extended (individualise)	Exclude (decline)
Age	Adult	Older donors (often >60) acceptable; no upper limit	Under legal age
GFR, mL/min/1.73 m²	≥90	**60–89**	<60
Albuminuria	<30 mg/g ACR (P1: <30 mg/day)	**30–100** mg/g ACR (P1: mg/day)	>100 mg/g ACR (P1: mg/day)
Hypertension	None	Controlled on 1–2 drugs, no target-organ damage	Uncontrolled, organ damage
Obesity	BMI <30	**30–35**; individualised risk	Many centres: >35
Glucose	Normal	Prediabetes; selected type 2 diabetes	Type 1 diabetes
Stones	None	Single past stone, low recurrence risk; small stone removed ex vivo	Bilateral or recurrent; cystinuria, primary hyperoxaluria
Anatomy	Single vessels	Multiple arteries, simple cyst, mild FMD (fibromuscular dysplasia)	Major anomaly

- ACR = albumin–creatinine ratio.

- P2 (unrelated-donor answer) states that diabetes, significant obesity and uncontrolled hypertension exclude.

Extended criteria (marginal, medically complex living donor)

- A living donor with a risk factor that would once have excluded donation; used because of organ shortage.
- **Why older donors are acceptable:** their remaining lifespan is short, so lifetime ESKD risk is low. Young donors carry the highest lifetime risk. Healthy older donors are acceptable.
- **Also know:** isolated microscopic haematuria after negative work-up; treated low-risk cancer; APOL1 (apolipoprotein L1) high-risk genotype (test in African ancestry); future pregnancy (more pre-eclampsia in later pregnancy).
- **Safeguards:** measured GFR and split function (better kidney stays with the donor); CT angiography; laparoscopic retrieval; THOA authorisation; independent donor advocate.
- **Outcome:** grafts from older living donors do about as well as standard deceased-donor kidneys.

Counselling figures (Muzaale 2014)

- Function settles at 60–80% of baseline.
- Lifetime ESKD (ESRD, end-stage renal disease) 90 per 10,000 donors vs 14 in matched healthy non-donors; mortality 3 per 10,000.

Diagrams to draw

- Donor work-up flowchart: 1. Legal approval. 2. Compatibility. 3. Medical. 4. Anatomy. 5. Surgery.
- Risk-threshold scale (accept, individualise, decline).

> **Key points to remember:**
>
> - KDIGO 2017: risk-based lifetime ESKD threshold; GFR ≥90 accept, <60 decline; ACR >100 decline.
> - GFR 60–89 and albuminuria 30–100 are individualised.
> - Controlled hypertension on 1–2 drugs may donate; type 1 diabetes and bilateral stones exclude.
> - ABO and crossmatch first; CT angiography for anatomy.
> - Better kidney stays with the donor.
> - THOA: near relatives, else authorisation committee; unrelated donor needs Authorisation Committee approval.
> - Interview video-recorded; no commerce.

Also, from the condensed book (not in the paper answer):

- Donor mortality is **3/10,000**. An independent donor advocate is mandatory.
- **Laparoscopic donor nephrectomy** is standard.

6 · The operation

- On the back table, ligate hilar lymphatics (lymphocele). **Never dissect the triangle between lower pole and ureter.** Extend a short right vein with donor IVC.

- **Multiple arteries** (18–43%) mean more DGF but similar survival; use a Carrel patch, conjoining or separate anastomoses.
- **Always keep the lower pole artery**, because it supplies the ureter; upper pole branches ≤2 mm may be ligated.
- The graft goes in the retroperitoneal iliac fossa. **Contralateral placement** brings the pelvis and ureter anterior.

	Gibson	Hockey stick
Course	Oblique, 2 cm above pubis to near the anterior superior iliac spine (ASIS)	Midline suprapubic, up the rectus edge
Access	Iliac vessels	Better to common iliac, IVC
Hernia	Lower	Higher; rectus denervation

ALGORITHM

1. **Vein first**: end-to-side to the external iliac vein, 5-0/6-0 polypropylene.
2. Artery end-to-side to the external iliac artery (punch arteriotomy; avoid calcified clamp sites).
3. Reperfuse, then position the kidney without kink or stretch.

- A flaccid graft suggests poor inflow and a tense, engorged graft poor outflow; check with intraoperative ultrasound.

ALGORITHM

1. **Lich–Gregoir** extravesical: fill the bladder; incise the detrusor down to mucosa.
2. Open the mucosa distally and spatulate the ureter.
3. Suture mucosa to mucosa with 5-0/6-0 absorbable, with or without a stent.
4. Close the detrusor loosely as a short tunnel.

- Document stent removal, because forgotten stents encrust.

7 · Surgical complications

Early vascular problems lose grafts within hours.

SAQ Vascular complications after renal transplantation, including transplant renal artery stenosis

Paper answer · P1

Asked as:

- Vascular Complications after Renal Transplantation in Recipient — *Winter 2025 Paper I Q3d (SAQ)*
- Transplant renal artery stenosis — *Practice Set 20 SAQ Q3d (SAQ)*

Type: SAQ (10 marks)

(vascular complications of transplantation also answered in 2025-Winter Paper 1, Q3d)

Introduction

- Uncommon but a major cause of **early graft loss**. Doppler ultrasound is the first test.

- **Concept:** no inflow gives anuria; blocked outflow gives a swollen, tender graft with haematuria; reduced inflow gives hypertension and fluid retention.

Main complications

	Arterial thrombosis	Venous thrombosis	TRAS (transplant renal artery stenosis)
Incidence	0.2–5%	0.1–8.2%	1–23% (commonest)
Timing	First days	First week	Months (usually 3 months–2 years)
Presentation	Sudden anuria	Painful swollen graft, haematuria, oliguria	Refractory hypertension, rising creatinine, flash pulmonary oedema
Doppler	No arterial flow	**Reversed diastolic flow**, clot in vein	High velocity at stenosis; **parvus-tardus** waveform distally
Treatment	Immediate exploration; thrombectomy (catheter thrombolysis described); nephrectomy if non-viable	Immediate exploration, thrombectomy; nephrectomy if necrotic	**Angioplasty or stenting** (first choice); surgical revision if that fails

- **Thrombosis risks:** extreme donor or recipient age, **cold ischaemia >24 hours**, multiple arteries, technical error (kink, intimal flap), hypovolaemia, rejection, thrombophilia (factor V Leiden, antiphospholipid antibodies).
- **TRAS causes:** anastomotic technique, clamp injury, recipient atherosclerosis (iliac inflow stenosis mimics it). Contrast angiography confirms.

Transplant renal artery stenosis (TRAS) in detail

Definition and incidence

- TRAS (transplant renal artery stenosis): narrowing of the graft artery or its inflow; **commonest vascular complication (1–23%)**, usually 3 months to 2 years after transplant.

Types and causes

- **Anastomotic:** suture technique, end-to-end anastomosis to the internal iliac artery, no aortic (Carrel) patch.
- **Post-anastomotic (distal):** clamp or perfusion-cannula intimal injury, kinking, rejection.
- **Pre-anastomotic ("pseudo-TRAS"):** recipient iliac atherosclerosis.
- **Also know:** CMV (cytomegalovirus) infection, delayed graft function, deceased donor.

Pathophysiology and clinical features

- Reduced graft perfusion activates the **renin–angiotensin–aldosterone system**.
- **Refractory hypertension**, rising creatinine, fluid retention, **flash pulmonary oedema**; graft bruit.
- Clue: creatinine rises sharply after an ACE (angiotensin-converting enzyme) inhibitor.

Evaluation

- **Doppler ultrasound first:** high peak velocity at stenosis (about >200–250 cm/s), and distal **parvus-tardus** waveform with low resistive index.

- MRA (magnetic resonance angiography) or CO_2 angiography to spare contrast.
- **Contrast angiography is the gold standard** and allows treatment at once. Iodinated contrast risks nephrotoxicity; gadolinium is avoided at low GFR (glomerular filtration rate).
- **Differential of graft dysfunction with hypertension:** rejection, calcineurin-inhibitor toxicity, ureteric obstruction.

Management

- **Medical:** for mild stenosis with stable function: blood pressure control (calcium-channel blockers), statin, antiplatelet; avoid or monitor ACE inhibitors.
- **Percutaneous angioplasty with or without stent: treatment of choice** for significant stenosis (usually >50–60%). High technical success; reverses hypertension and graft dysfunction. Stents reduce restenosis.
- **Surgical revision** (resection and re-anastomosis, interposition graft, bypass) if angioplasty fails or anatomy unsuitable; risks graft loss.
- Follow-up Doppler after angioplasty detects restenosis; untreated TRAS ends in graft loss.

Other vascular complications

- **Haemorrhage:** hilar vessels or anastomosis; re-explore if unstable.
- **Pseudoaneurysm (<1%):** technical or **mycotic**; rupture risk; endovascular or open repair, often graft nephrectomy, prolonged antibiotics.
- **Post-biopsy AV (arteriovenous) fistula:** persistent haematuria; usually closes; large or with steal: **coil embolisation**. Subcapsular haematoma can cause a **Page kidney**.
- **Graft torsion:** intraperitoneal grafts; prevent with nephropexy.
- **DVT (deep vein thrombosis)** of the iliac vein; **lymphocele** compressing the iliac or renal vein.
- **Limb ischaemia** from iliac dissection; internal iliac use: gluteal claudication, erectile dysfunction.

Prevention

- Anastomose away from atheroma; check the lie for kinking; avoid clamp injury and kinking; use a Carrel patch where possible.
- Extend a short right renal vein with donor IVC (inferior vena cava).
- Short cold ischaemia; anticoagulate thrombophilic patients; avoid hypovolaemia.
- At reperfusion, a soft kidney suggests poor inflow; a tense, engorged kidney suggests outflow block.

Diagram to draw: Doppler waveforms: normal, parvus-tardus (TRAS; distal to stenosis) and reversed diastolic flow (venous thrombosis).

> **Key points to remember:**
>
> - TRAS is the commonest vascular complication; presents months later; angioplasty or stent first (first-line); angiography is gold standard.
> - Refractory hypertension plus rising creatinine: Doppler first.
> - Venous thrombosis: reversed diastolic flow, swollen tender graft.
> - Arterial thrombosis: sudden anuria; explore immediately.

Also, from the condensed book (not in the paper answer):

- Bleeding (brisk drain, falling haematocrit) needs transfusion and re-exploration.

SAQ Lymphocele and postoperative lymphatic leak (chylous ascites)

Paper answer · P1

Asked as:

- Lymphocele — *Practice Set 8 Paper I Q3a (SAQ)*
- Postoperative lymphatic leak — *Practice Set 14 Paper I Q3b (SAQ)*

Type: SAQ (10 marks)

Postoperative lymphatic leak taken as lymphocele and chylous ascites after RPLND (retroperitoneal lymph node dissection), nephrectomy, pelvic node dissection and transplantation.

Definition and types

- **Lymphocele:** a collection of lymph **without an epithelial lining** (a pseudocyst with a fibrous wall), caused by divided lymphatics that keep leaking. It is the **commonest complication of PLND (pelvic lymph node dissection)**.
- **Chylous ascites:** lipid-rich lymph in the peritoneum from the cisterna chyli or major trunks.

Settings and incidence

- **PLND** (radical prostatectomy, cystectomy): radiological in up to 60%, clinically significant 2–5%. An extended template roughly doubles the rate.
- **Renal transplantation:** from lymphatics around the recipient iliac vessels or donor hilum. Radiological 0.6–33.9% (symptomatic 0.03–26%); most appear within 6 months, peaking at 6 weeks.
- Also after RPLND (retroperitoneal lymph node dissection; lymphocele ≤1.7%), inguinal dissection, nephrectomy.
- **Chylous ascites:** RPLND: 0.2–2.1% primary, **2–12% post-chemotherapy**; left nephrectomy with hilar dissection.

Type	Definition	Setting and frequency
Lymphocele	Lymph collection **without epithelial lining**	**PLND (pelvic lymph node dissection):** radiological up to 60%, significant 2–5%. **Transplant:** symptomatic 0.03–26%, peak 6 weeks. **RPLND:** ≤1.7%
Chylous ascites	Lipid-rich lymph in the peritoneum from the cisterna chyli or major trunks	RPLND: 0.2–2.1% primary, **2–12% post-chemotherapy**; left nephrectomy with hilar dissection

- **Risk factors:** extensive dissection, extended templates, suprahilar dissection near the cisterna chyli (L1–L2), post-chemotherapy surgery, obesity, heparin prophylaxis, mTOR (mammalian target of rapamycin) inhibitors, acute rejection, retransplantation (re-transplant).

Clinical features

- **Lymphocele:** most are silent (usually silent). Large ones compress:
- iliac vein: leg oedema, **DVT (deep vein thrombosis)**;
- bladder: frequency, urgency, pelvic pressure;

- ureter or graft: hydronephrosis, rising creatinine (graft dysfunction).
- Infection (may become infected): fever, rigors, sepsis.
- **Chylous ascites:** rising girth, early satiety, dyspnoea; milky drain fluid.

Diagnosis

- **Ultrasound:** anechoic, often septated, thin-walled collection. **CT (computed tomography)** maps it before intervention. Air or rim enhancement means infection.
- **Aspirate:** lymph has creatinine equal to (equals) serum (urinoma: much higher), lymphocytes; culture excludes abscess. Chyle: **triglyceride 2–8 × serum**, protein >3 g/dL.

Management

Lymphocele (treat by symptoms, not size alone)

Situation	Treatment
Asymptomatic, <5 cm	Observe; most resolve
Symptomatic	**Percutaneous drain** ± sclerotherapy (povidone-iodine, doxycycline)
Recurrent or thick-walled	**Laparoscopic fenestration (marsupialisation)** into a peritoneal window (best)
Infected	Antibiotics + percutaneous drainage; exclude urine leak
Laparoscopy unsafe	Open drainage

`Asymptomatic, <5 cm → observe Symptomatic → percutaneous drain (± sclerotherapy) Recurrent / thick-walled → laparoscopic peritoneal fenestration (best) Infected → drainage + antibiotics; exclude urine leak`

- **Success:** laparoscopic fenestration best, open next, drain or sclerotherapy intermediate, **aspiration alone worst** (reaccumulates, recurs, risks infection).
- Sclerosants (povidone-iodine, doxycycline) may hinder later fenestration.
- Fenestration lets the peritoneum absorb lymph (internal drainage): the definitive cure.
- **Lymphangiography with embolisation:** high success; long-term data awaited.

Chylous ascites (stepwise)

1. Paracentesis; drain if it reaccumulates.
2. **Very low-fat, high-protein diet with MCT (medium-chain triglycerides)**, which are absorbed directly into portal blood, bypassing lacteals; plus **octreotide**.
3. Nil by mouth with **TPN (total parenteral nutrition)**.
4. Refractory: lymphangiography with embolisation, peritoneovenous shunt or surgical ligation. - Continue a low-fat diet for **≥4 weeks** after resolution.

Prevention

- **Clip or ligate visible lymphatics** (clips often preferred; no robust data) (cephalad at the renal hilum in RPLND; perivascular at transplant). Spare lymphatics over the external iliac artery (leg lymph). Avoid unneeded suprahilar dissection.
- **Peritoneal interposition flap** fixed to the bladder (peritoneal flap after robotic PLND) reduces clinically significant lymphocele. Transperitoneal access is **not** protective (fluid loculates within 48–72 h).
- Drain after transplantation and major nodal dissections.

Diagram to draw: pelvic lymphocele compressing the external iliac vein and bladder, with a laparoscopic peritoneal window.

> **Key points to remember:**
> - Lymphocele: commonest PLND complication; radiological up to 60%, significant 2–5%.
> - Lymph creatinine = serum; urinoma creatinine higher.
> - Laparoscopic fenestration best (is best); aspiration alone worst.
> - Chylous ascites: 2–12% after post-chemotherapy RPLND; triglyceride 2–8 × serum.
> - Ladder: diet with MCT and octreotide → TPN → embolisation or shunt.

Transplant nephrectomy

- Indications are **graft intolerance syndrome** (pain, swelling, haematuria, fever; steroids first), infection and tumour, not an asymptomatic failed graft.
- **Early (<6–8 weeks), go extracapsular**; later, go **subcapsular**, except for tumour.

8 · Rejection

Chronic antibody injury now limits graft survival.

- The usual regimen is induction, then tacrolimus and mycophenolate, with or without steroids.

LAQ Allograft rejection and immunobiology of renal transplantation

Paper answer · P1 · P4 · merged from 2

Asked as: P4 LAQ "Allograft rejection, immunobiology of renal transplant, marginal kidney donor and newer immunosuppressant drugs" (asked as "Discuss pathophysiology of allograft rejection. Discuss newer immunosuppressive drugs in renal transplant", Winter 2017 Paper IV Q2; "Describe immunobiology of renal transplant and newer immunosuppressant drugs", Summer 2020 Paper IV Q1; "Immunobiology of renal transplant, describe. Also, describe newer immunosuppressant drugs", Practice Set 3 Paper IV Q1; "Discuss marginal kidney donor and describe newer immunosuppressant drugs in renal transplantation", Summer 2015 Paper IV Q2) · P1 SAQ "Renal allograft rejection." (2020-Summer Paper 1 Q3d)

Definition and introduction

- Rejection is the recipient's immune attack on donor (allo) antigens, which the recipient's immune system recognises as foreign: **HLA (human leucocyte antigen)** class I (A, B, C) and II (DR, DQ, DP), ABO and non-HLA antigens.
- Immunosuppression has cut acute rejection from over 60% (pre-ciclosporin era) to **below 10%**. Slow, subclinical **chronic antibody-mediated injury** is now the main problem and the main cause of late graft loss.
- Organ shortage drives use of **marginal donors**. Their kidneys carry more DGF (delayed graft function) and shorter graft survival, but transplantation still beats dialysis.

Transplant (target) antigens

Antigen	Features
HLA = MHC (major histocompatibility complex)	Class I (A, B, C) on all nucleated cells; class II (DR, DQ, DP) on antigen-presenting and endothelial cells. Class II mismatch is most immunogenic

Antigen	Features
ABO blood group	On endothelium; incompatibility causes hyperacute rejection unless desensitised
Minor and non-HLA antigens	Rejection despite HLA identity

- **Sensitisation:** transfusion, pregnancy or previous transplant creates preformed anti-HLA antibodies.
- **Testing:** CDC (complement-dependent cytotoxicity) and flow crossmatch (donor lymphocytes with recipient serum); **Luminex single-antigen beads** define each recipient's antibodies and allow a **virtual crossmatch**; cPRA (calculated panel-reactive antibody) expresses sensitisation.

Innate immunity: the first trigger

- **Ischaemia–reperfusion injury** releases danger signals (DAMPs, damage-associated molecular patterns). These activate **Toll-like receptors**, complement and macrophages, and up-regulate MHC on the graft. Hence DGF raises rejection risk.
- **Dendritic cells** bridge innate and adaptive immunity: activated, they up-regulate MHC and costimulatory B7 and migrate to lymph nodes.

Allorecognition (how T cells see the graft)

Pathway	Mechanism	Main role
Direct	Recipient T cells recognise **intact donor MHC** on donor "passenger" dendritic cells (donor antigen-presenting cells)	Very strong (high T-cell frequency); early acute TCMR (T-cell-mediated rejection)
Indirect	Recipient APCs (antigen-presenting cells) present **processed donor peptides on self-MHC**	Lifelong; helps B cells make DSA (donor-specific antibody), so alloantibody and chronic rejection
Semi-direct	Recipient APCs acquire **intact donor MHC** (membrane transfer) and present it	Links both pathways

Three-signal model of T-cell activation

1. **Signal 1 (antigen):** TCR (T-cell receptor) binds MHC–peptide; **calcineurin** dephosphorylates NFAT (nuclear factor of activated T cells), which drives IL-2 (interleukin-2) transcription.
2. **Signal 2 (costimulation): CD28 binds CD80/86 (B7)**. Signal 1 without signal 2 causes **anergy**. CTLA-4 (cytotoxic T-lymphocyte antigen-4) is the natural brake.
3. **Signal 3:** IL-2 binds **CD25** (IL-2 receptor) and signals through **mTOR (mammalian target of rapamycin)**, moving the cell cycle from G1 to S; clonal proliferation follows, which needs de novo purine synthesis.

Effector mechanisms

- **Cellular:** CD4 Th1 cells (IL-2, IFN-γ [interferon-γ]) recruit macrophages; CD8 cytotoxic T cells kill by perforin and granzyme, causing **tubulitis and endarteritis (arteritis)**.
- **Humoral:** B cells with T-cell help become plasma cells that make **DSA**. DSA binds endothelium and fixes complement, leaving **C4d** on peritubular capillaries, with **microvascular inflammation** (glomerulitis, capillaritis). NK (natural killer) cell injury is complement-independent, so AMR (antibody-mediated rejection) may be C4d-negative.

- **Regulation: Tregs** (regulatory T cells, FoxP3+) suppress responses; the basis of tolerance research.

Types of rejection

Type	Timing	Mechanism and features
Hyperacute	Minutes–hours	Preformed ABO/HLA antibody; complement, thrombosis; graft turns blue; **prevented by crossmatch**
Accelerated acute	2–5 days	Memory (anamnestic) response in sensitised recipients
Acute TCMR	Days–months (mostly first 6 months)	Interstitial infiltrate, tubulitis, arteritis
Acute AMR	Any time	DSA, C4d, microvascular inflammation (capillaritis)
Chronic active AMR	Months–years	**Transplant glomerulopathy** (GBM [glomerular basement membrane] double contours), arterial intimal fibrosis; commonest (main) cause of late graft loss
Subclinical	—	Histological rejection, stable function

- **Risk factors:** HLA mismatch, DSA, ABO incompatibility, long ischaemia and DGF, **non-adherence**, retransplant, previous rejection.

Diagnosis and monitoring

- **Clinical:** rising creatinine, oliguria; fever and graft tenderness are now rare.
- **Exclude:** obstruction, urine leak, lymphocele, renal artery stenosis or thrombosis (ultrasound with Doppler), CNI (calcineurin inhibitor) toxicity, BK virus nephropathy, recurrent disease.
- **Monitoring: de novo DSA** predicts graft loss; **dd-cfDNA (donor-derived cell-free DNA)** has negative predictive value >90% and avoids unnecessary biopsy; gene-expression panels.
- **Biopsy is the gold standard;** Banff grades the findings and classifies TCMR and AMR.

Banff classification

Category	Key criteria
Borderline	Foci of tubulitis with mild interstitial inflammation
TCMR IA / IB	Interstitial inflammation >25% with moderate (IA) or severe (IB) tubulitis
TCMR IIA / IIB	Mild-moderate (IIA) or severe (IIB) intimal arteritis
TCMR III	Transmural arteritis or fibrinoid necrosis
Active AMR	All three: (1) tissue injury (microvascular inflammation, arteritis, TMA [thrombotic microangiopathy]); (2) antibody–endothelium interaction (C4d, microvascular inflammation or gene transcripts); (3) **circulating DSA**

Treatment of rejection

- **TCMR:** IV (intravenous) pulse methylprednisolone; **rATG (rabbit antithymocyte globulin) / ATG** for Banff II–III or steroid-resistant rejection.

- **AMR** (not standardised): steroids, **plasmapheresis, IVIG (intravenous immunoglobulin)**, rituximab; eculizumab in selected patients; anti-IL-6 (interleukin-6) for chronic active AMR.
- Optimise maintenance (tacrolimus, mycophenolate, steroids); address adherence.

Marginal kidney donor: categories

1. ECD (expanded criteria donor), UNOS (United Network for Organ Sharing) 2002: age **≥60**, or **50–59** with **≥2** of **hypertension**, **creatinine >1.5 mg/dL**, **death from CVA (cerebrovascular accident)**. About 70% higher risk of graft loss.

2. KDPI (Kidney Donor Profile Index): percentile of a 10-factor risk index. **KDPI >85%** replaced ECD in the US (December 2014). Low-KDPI kidneys go to recipients with the longest EPTS (Estimated Post-Transplant Survival): **longevity matching**.

3. DCD (donation after circulatory death): Maastricht

Category	Description	Type
I / II	Dead on arrival / failed resuscitation	Uncontrolled
III	Arrest after withdrawal of therapy	**Controlled (commonest)**
IV	Arrest in a brain-dead donor	Either
V	Euthanasia	Controlled

- Warm ischaemia needs rapid cannulation and cooling, and DCD kidneys usually go on machine perfusion. DGF is **higher (roughly 40–50%)**, but long-term survival is similar to DBD (donation after brain death).

4. Others: AKI (acute kidney injury) donors; **HCV (hepatitis C virus)-viraemic** donors into negative recipients with DAAs (direct-acting antivirals); **paediatric donors <15 kg** (en bloc); older or hypertensive living donors (KDIGO [Kidney Disease: Improving Global Outcomes] 2017 risk-based approach).

- **Absolute exclusions:** metastatic or haematological cancer, melanoma, untreatable viral infection. Treated bacterial infection is acceptable; screen by serology and nucleic acid testing.

Marginal kidney donor: assessment and optimisation

- **Remuzzi biopsy score** (glomeruli, tubules, interstitium, arteries, each 0–3): **0–3 single**, **4–6 dual kidney transplant**, **≥7 discard**.
- **Cold storage** at 4 °C in UW (University of Wisconsin) or HTK (histidine–tryptophan–ketoglutarate) solution slows metabolism.
- **HMP (hypothermic machine perfusion)** reduces DGF (**Moers, NEJM 2009**: about 21% vs 27%; greatest benefit in ECD kidneys); Campbell notes no long-term survival difference. Normothermic perfusion allows viability testing (limited experience).
- **ESP (Eurotransplant Senior Program):** donors **≥65** to recipients **≥65**, prioritising short cold ischaemia over HLA matching. Informed consent is essential.
- **Ojo (JASN 2001):** ECD transplantation improves survival over dialysis, although graft survival is lower than with standard kidneys.
- With DGF, give antibody induction and delay nephrotoxic CNI therapy.

Standard immunosuppression (current practice)

- **Induction** (over 90% receive it) plus **tacrolimus + mycophenolate** ± steroids (mycophenolate: GI [gastrointestinal] upset, teratogenic).
- Acute rejection is below 10%; chronic AMR is now the main problem. Newer drugs aim to avoid CNI nephrotoxicity, improve adherence and treat AMR.

Newer immunosuppressive drugs

Drug	Mechanism (site)	Key points
Belatacept	**CTLA4-Ig**: binds CD80/86, blocks **CD28 costimulation (signal 2)**	FDA (Food and Drug Administration) 2011; IV 10 mg/kg induction, then **5 mg/kg 4-weekly** (monthly); CNI-free. **BENEFIT**: better GFR (glomerular filtration rate), more early acute rejection; 7-year better patient and graft survival. **Contraindicated if EBV (Epstein–Barr virus)-seronegative** (PTLD [post-transplant lymphoproliferative disorder])
Extended-release tacrolimus	CNI: binds FKBP12 and inhibits calcineurin (**signal 1**), so less IL-2 transcription	Once daily, better adherence, steadier levels; nephrotoxicity, NODAT (new-onset diabetes after transplant), tremor
Mycophenolate	Inhibits IMPDH (inosine monophosphate dehydrogenase), blocking lymphocyte purine synthesis	Standard partner of tacrolimus
mTOR inhibitors (sirolimus, everolimus)	Bind FKBP12 and block mTOR (**signal 3**), arresting G1 to S	Not nephrotoxic; fewer CMV (cytomegalovirus)/BK infections; antitumour effect. **TRANSFORM (2018)**: everolimus + reduced (low-dose) CNI non-inferior. Poor wound healing, **lymphocele**, proteinuria, dyslipidaemia
Basiliximab	Anti-CD25 (IL-2 receptor), non-depleting (**signal 3**)	Low-risk induction, **20 mg IV days 0 and 4**
rATG; **alemtuzumab**	T-cell depletion; anti-CD52	High-risk induction, DGF; infection (CMV), PTLD
Rituximab; bortezomib	Anti-CD20 (B cells); proteasome inhibitor (plasma cells)	AMR, desensitisation
Eculizumab	Anti-C5 (complement)	Selected AMR; aHUS (atypical haemolytic uraemic syndrome) recurrence
Imlifidase	Enzyme that **cleaves IgG**	**EMA (European Medicines Agency) 2020**: crossmatch-positive, highly sensitised patients
Anti-IL-6 therapy (tocilizumab, clazakizumab)	Blocks IL-6 pathway	Chronic active AMR

- **Also know:** anti-CD40 costimulation blockade is under trial. **Voclosporin** is approved for lupus nephritis, **not transplantation**.

Recent advances

- **Tolerance induction** by mixed chimerism (combined kidney and donor stem-cell transplant) and Treg therapy; normothermic repair; **pig-kidney xenotransplant** into a living human (March 2024).

Diagrams to draw

- Three-signal T-cell activation (APC–T-cell synapse showing signals 1–3) with drug sites: CNI at signal 1, belatacept at 2, basiliximab and mTOR inhibitors at 3, mycophenolate at DNA synthesis.
- Banff biopsy features: tubulitis, arteritis, peritubular capillary C4d.

Key points to remember:

- Direct pathway drives early TCMR; indirect pathway drives antibody (DSA) and chronic rejection.
- Signal 1 calcineurin, signal 2 CD28–B7, signal 3 IL-2R–mTOR.
- Hyperacute rejection prevented by crossmatch.
- Banff AMR = tissue injury + antibody–endothelium interaction (C4d) + DSA.
- TCMR: pulse steroids, ATG; AMR: plasmapheresis, IVIG.
- Chronic active AMR (transplant glomerulopathy) is the main cause of late graft loss.
- Belatacept blocks signal 2; better GFR, contraindicated if EBV-negative; imlifidase (IgG cleavage) for highly sensitised patients; dd-cfDNA NPV >90%.
- ECD: ≥60, or 50–59 with ≥2 of hypertension, creatinine >1.5, CVA death; KDPI >85% since 2014.
- Remuzzi 0–3 single, 4–6 dual, ≥7 discard; HMP reduces DGF; ESP ≥65 to ≥65.

Also, from the condensed book (not in the paper answer):

- De novo donor-specific antibody (DSA) is linked to class II mismatch and under-immunosuppression.
- Hyperacute rejection is †irreversible, so the graft is removed.

9 · Landmark trials

Trial	Design	Result	Take-home
Wolfe 1999	Registry: transplant vs wait-listed dialysis	Risk 2.84 early; equal at 106 days; long-term 0.32	Early peril, durable survival gain

★ Putting it together: kidney function and transplant decisions

A · Rising creatinine after urological surgery

ALGORITHM

1. **Postrenal first:** check the catheter and scan. After cystectomy, a rise with ascites may be a urine leak (drain creatinine ≫ serum).
2. **Prerenal vs intrinsic:** FeNa <1% with bland urine means volume depletion; >1% with granular casts means acute tubular necrosis. Stop nephrotoxins, RAAS blockers and metformin.
3. **Dialysis** only for refractory "AEIOU": acidosis, electrolytes (K), intoxication, overload, uraemia.

B · Post-obstructive diuresis

KEY POINT

After bilateral or solitary-kidney relief, monitor output and electrolytes. Diuresis is mostly physiological; †it is pathological if it persists after euvolaemia. †Do not replace every millilitre; if the patient cannot drink, replace ≈75% with 0.45% saline.

C · Transplant ureteric complications

Problem	Clue	Management
Urine leak (early)	High drain output; fluid creatinine ≫ serum	Foley and nephrostomy ± stent; explore if large or persistent
Ureteric stricture	Falling function ± mild hydronephrosis	Nephrostomy, then dilation or endoureterotomy; open repair is most durable
Lymphocele	Fluid creatinine ≈ serum; may obstruct ureter or vessels	Treat only if symptomatic (<15%); laparoscopic window is best
Stent infection	Colonised stent	Remove at 14–21 days (earlier risks leak; later risks infection)

Prevent them with a stent and by sparing the lower pole–ureter triangle.

D · Rejection at a glance

Type	Cause	Point
Hyperacute	Preformed donor-specific antibody (DSA)	Crossmatch prevents it; †irreversible, so remove graft
Acute	T-cell-mediated (TCMR) or antibody-mediated (AMR), Banff	Biopsy. TCMR: IV steroids, then antithymocyte globulin (ATG). AMR: steroids, IV immunoglobulin, plasmapheresis, rituximab
Chronic	Mainly antibody-mediated	Main late loss; subclinical

E · Child going for transplant

Fix the bladder before transplant.

ALGORITHM

1. Most have congenital anomalies (CAKUT), chiefly valves.
2. Make the bladder safe: age-appropriate capacity at detrusor pressure <40 cm H_2O (antimuscarinics, catheterisation, augmentation).
3. Remove refluxing ureters or kidneys causing infection, polyuria or hypertension.
4. A poor bladder destroys the graft.

Other paper questions in this section

Asked in the papers but not tied to a specific place in the book above.

LAQ Vascular anastomosis and vascular access for dialysis

Paper answer · P2 · P4 · merged from 2

Asked as: P2 LAQ "Principles of vascular anastomosis and vascular access surgery for HD" (asked as "Describe principles of vascular anastomosis. Enumerate and describe options for vascular access surgery for HD", Practice Set 5 Paper II Q2 LAQ and Practice Set 12 Paper II Q3a SAQ) · P4 SAQ "Vascular access for dialysis." (2020-Summer Paper 4 Q3b)

Assumed meaning: HD = haemodialysis. Vascular access also answered in 2025-Winter Paper 2, Q3c.

Introduction

- A vascular anastomosis joins two vessels so that flow is smooth, leak-free and lasting (transplantation, autotransplantation, caval repair, dialysis access).
- **Types:** end-to-end, **end-to-side** (commonest: transplant, fistula), side-to-side, and patch (Carrel aortic patch).
- Haemodialysis needs a blood flow of about **250–350 mL/min**; access failure is a major cause of morbidity in ESKD (end-stage kidney disease).

Principles of vascular anastomosis

Principle	How and why
Exposure and control	Proximal and distal control (vessel loops, Rummel tourniquets, bulldog or Satinsky clamps)
Anticoagulation	Systemic heparin before clamping; heparinised saline flush; prevents thrombosis in the static column
Atraumatic handling	Hold the adventitia only, never the intima; clamp clear of atheroma, because clamp injury causes intimal dissection and stenosis
Healthy ends	Trim diseased ends; resect venous valve cusps at the suture line; prevents stenosis and leak
No tension, twist or kink	Adequate length; trial the final position before suturing
Intima-to-intima eversion	Smooth endothelium faces the lumen; adventitia and rough surfaces stay outside, as they are thrombogenic
Wide opening	Spatulate or bevel small vessels, or use end-to-side with an elliptical arteriotomy (aortic punch), to avoid stenosis (stricture)
Suture	**Non-absorbable monofilament polypropylene**, double-armed: smooth, low drag, low thrombogenicity. IVC (inferior vena cava) or aorta 3-0/4-0; transplant vessels 5-0/6-0; wrist fistula about 6-0/7-0
Needle direction	Equal, small, close bites; pass **inside-out on the artery** so plaque and intima are tacked down, not lifted into a flap
Continuous or interrupted	Continuous is quick and haemostatic; **interrupted** for small vessels and children, allowing growth
Technique	**Carrel triangulation** (three stay sutures keep the back wall clear); **parachute** (sew loosely, then draw up; one knot) or tack both apices and run to the midpoint

Principle	How and why
Flush and release	Before the last knot, briefly release clamps to flush air, clot and debris; leave slack ("growth factor") to avoid purse-stringing; release venous/distal clamp first, then arterial
Confirm	Haemostasis, pulse, thrill, Doppler

- **Failure:** early from technical error (thrombosis, leak, stenosis); late from **intimal hyperplasia**.
- **Also know:** vascular staplers; grafts (autologous vein for small or contaminated fields; PTFE [polytetrafluoroethylene] for large vessels).

Options for vascular access surgery for HD

Option	Construction / description	Ready	Comment
1. Non-tunnelled CVC (central venous catheter)	Dual-lumen 10–12 Fr in internal jugular or femoral vein	At once	AKI (acute kidney injury) and emergencies; acute dialysis, days only
2. Tunnelled cuffed CVC	Right internal jugular; cuff blocks bacterial migration	At once	Bridge or last resort; infection, central stenosis
3. Native AVF (arteriovenous fistula)	Artery joined to a superficial vein, usually vein end to artery side	6–8 weeks (matures)	**First choice:** best patency, fewest infections
a. Snuffbox or **radiocephalic (Brescia–Cimino, 1966)**	Radial artery to cephalic vein at wrist	6–8 weeks	Most distal; preserves proximal sites; higher failure to mature (most fail to mature)
b. Brachiocephalic	Brachial artery to cephalic vein at elbow	6–8 weeks	Higher flow; more steal
c. Brachiobasilic transposition	Deep basilic vein mobilised and moved superficially	Longer, often two stages	When the cephalic vein is unusable
4. AVG (arteriovenous graft)	PTFE loop or straight: forearm loop, upper-arm or thigh	2–3 weeks (usable)	Poor veins; more stenosis (venous anastomosis), thrombosis, infection
5. HeRO (Haemodialysis Reliable Outflow) graft	Graft outflow bypasses occluded central veins into the right atrium		Central vein occlusion
6. Endovascular AVF (WavelinQ, Ellipsys)	Percutaneous radiofrequency or thermal anastomosis between adjacent forearm vessels		No incision; selected anatomy

- **Also know:** lower-limb femoral vein transposition when arms are exhausted; peritoneal dialysis needs no vascular access.

Planning

- **Hierarchy:** AVF, then AVG, then CVC; **distal before proximal; non-dominant arm first**; individualised (KDOQI [Kidney Disease Outcomes Quality Initiative] 2019 "ESKD Life-Plan").
- **Timing:** refer at eGFR (estimated glomerular filtration rate) <30; create the AVF about **6 months** before dialysis.

- **Vein preservation:** preserve arm veins; no venepuncture, cannula or PICC (peripherally inserted central catheter) in that arm; avoid subclavian lines.
- **Assessment:** pulses, **Allen test** (ulnar collateral flow), **Doppler vein mapping** (vessels about ≥2 mm).
- **Maturity "rule of 6s":** flow >600 mL/min, diameter >6 mm, depth <6 mm, at 6 weeks.

Complications of access

Complication	Treatment
Failure to mature (juxta-anastomotic stenosis)	Angioplasty (PTA), ligate side branches
Stenosis (neointimal hyperplasia)	PTA (percutaneous transluminal angioplasty), stent graft
Thrombosis (usually on a stenosis)	Thrombectomy, then treat the stenosis
Steal syndrome (hand ischaemia)	Banding, DRIL (distal revascularisation–interval ligation)
Aneurysm; infection (CVC > AVG > AVF)	Repair; antibiotics, excise graft, remove catheter

- **Also know:** venous hypertension; high-output cardiac failure.

Monitoring

- Thrill and bruit, arm-elevation test, prolonged bleeding after needling, rising venous pressure; duplex when abnormal.

Recent advances

- **Endovascular AVF** (WavelinQ, Ellipsys); drug-coated balloons; bioengineered vessels.

Diagram to draw

- Radiocephalic end-to-side fistula at the wrist with Carrel triangulation stay sutures.

> **Key points to remember:**
> - Intima-to-intima eversion, monofilament polypropylene, inside-out on the artery, flush before tying.
> - AVF first, distal, non-dominant arm, about 6 months ahead.
> - Rule of 6s: flow >600 mL/min, >6 mm wide, <6 mm deep, 6 weeks.
> - Thrombosis sits on a stenosis; steal: banding or DRIL.

LAQ Describe in details basic patho-physiology of urine production and transport

Paper answer · P1

Type: LAQ (25 marks)
Source: 2015-Summer Paper 1, Q1

Introduction

- Urine is formed by filtration, reabsorption, secretion and concentration, then propelled to a low-pressure bladder.

A. Functional anatomy

- Glomerular barrier (fenestrated endothelium, basement membrane, podocytes) restricts by size and negative charge, so albumin is excluded.
- **Juxtamedullary nephrons** (long loops) build the medullary gradient.

B. Renal blood flow (RBF) and autoregulation

- RBF ~25% of cardiac output. The medulla is hypoxic (10–20 mmHg), endangering the thick ascending limb and S3 segment.
- **Autoregulation** keeps RBF and GFR (glomerular filtration rate) constant at mean arterial pressure 80–180 mmHg:
- **Myogenic:** the afferent arteriole constricts when stretched and dilates (prostaglandins) when pressure falls; NSAIDs (non-steroidal anti-inflammatory drugs) abolish this.
- **Tubuloglomerular feedback:** low NaCl at the macula densa releases renin; angiotensin II constricts the efferent arteriole. Angiotensin blockade abolishes this.

C. Glomerular filtration

- GFR ~125 mL/min (180 L/day); filtration fraction ~20%.
- **Net filtration pressure** ≈ glomerular hydrostatic 60 – Bowman's space 18 – oncotic 32 = **~10 mmHg**.
- Efferent constriction lowers RBF but **raises** GFR; afferent constriction lowers both. Obstruction raises Bowman's space pressure, so GFR falls.
- **Also know:** GFR estimated from creatinine or cystatin C (race-free 2021 equations).

D. Tubular handling

Segment	Key transporter	Handling	Drug/disease link
Proximal tubule	Na^+/H^+ exchanger, SGLT2 (sodium–glucose cotransporter 2)	~**65%** Na^+ and water (iso-osmotic); all glucose; most bicarbonate and phosphate; secretes drugs	SGLT2 inhibitors
Thin descending limb	Aquaporin-1	Water out, solute stays: fluid concentrates	–
Thick ascending limb	NKCC2 (Na^+-K^+-$2Cl^-$)	~**25%** NaCl; water-impermeable "diluting segment"	Loop diuretics
Distal convoluted tubule	NCC (Na^+-Cl^- cotransporter)	5–7% Na^+; Ca^{2+} (parathyroid hormone)	Thiazides lower urinary Ca^{2+}
Collecting duct	ENaC (epithelial Na^+ channel), AQP2	Principal cells: Na^+/K^+ (aldosterone), water (ADH); intercalated cells: acid	Amiloride, spironolactone; distal RTA (renal tubular acidosis)

E. Concentration

- **Countercurrent multiplication:** ascending limbs absorb NaCl without water, raising medullary osmolality from 300 to ~**1200 mOsm/kg**; urea recycling adds about half. Vasa recta exchange without washing out the gradient.
- **Antidiuretic hormone (ADH)** is released by high osmolality or low volume. Signal chain: V2 receptor → cAMP (cyclic adenosine monophosphate) → protein kinase A → aquaporin-2 (AQP2) inserted into the apical membrane → water reabsorbed. Concentration needs both a hypertonic medulla and AQP2; urine osmolality ranges ~50–1200 mOsm/kg.

Hormone	Site	Effect
Angiotensin II, aldosterone	Efferent arteriole, proximal tubule; collecting duct	Na^+ retention, K^+/H^+ loss, GFR maintained
Atrial natriuretic peptide	Afferent arteriole, collecting duct	Natriuresis, lowers renin
Parathyroid hormone	Proximal and distal tubule	Phosphaturia, Ca^{2+} retention

F. Ureteric transport

- **Pacemaker:** atypical smooth muscle cells at the minor calyces/pelvicalyceal border.
- Peristalsis is **myogenic**: the excised or denervated ureter keeps contracting. Nerves (T11–L1, S2–4) carry only pain.
- Calcium entry triggers action potentials spread by gap junctions; calcium–calmodulin then activates myosin light-chain kinase.
- Walls coapt behind each bolus: 2–6 waves/min at 20–80 cmH_2O. Very high flow makes boluses coalesce, dilating the ureter **without obstruction**.
- Ureterovesical tunnel length:diameter 5:1 prevents reflux; bladder pressure >40 cmH_2O stops efflux.

Factor	Effect on transport
Obstruction	Acute: stronger contractions; chronic: dilatation, lost coaptation
Infection	Gram-negative endotoxin inhibits peristalsis (dilatation without obstruction)
Pregnancy	Right > left dilatation: compression at pelvic brim, progesterone
Drugs	α1-blockers and calcium-channel blockers relax (medical expulsive therapy); NSAIDs lower pelvic pressure in colic

G. Bladder storage and voiding

- **Storage** below ~10 cmH_2O: compliance, sympathetic β3 detrusor relaxation, α1 bladder-neck and pudendal sphincter tone.
- **Voiding:** pelvic parasympathetic (S2–4) acetylcholine on M3 receptors contracts the detrusor; nitric oxide relaxes the urethra.

H. Pathophysiology of obstruction (unilateral: triphasic)

`Phase 1 (0–1.5 h): RBF up, pressure up (afferent dilatation: prostaglandins, nitric oxide) Phase 2 (1.5–5 h): RBF down, pressure up (efferent constriction) Phase 3 (>5 h): RBF down, pressure down (afferent constriction: angiotensin II, thromboxane A2, endothelin)`

- Ischaemic injury starts **within 2 hours** of complete block.
- **Tubules:** down-regulated aquaporins and Na^+ transporters cause concentrating defect, salt wasting and distal RTA.
- **Fibrosis** (final path to failure): stretch, inflammation, TGF-β (transforming growth factor-β), angiotensin II, microRNAs.
- **Bilateral obstruction or solitary kidney:** pressure stays high longer and natriuretic peptides and urea accumulate, so **post-obstructive diuresis** follows relief in ~50% (vs ~2% unilateral).
- **Clinical:** hypertension (~77% bilateral, ~20% unilateral); contralateral hypertrophy; pyelorenal backflow acts as a pop-off; forniceal rupture causes urinoma.

- **Also know:** immature kidneys fail to develop; outlet obstruction causes detrusor hypertrophy then decompensation; diuretic MAG3 (mercaptoacetyltriglycine) renogram (half-time >40 min = obstructed); Whitaker test (>15–20 cmH_2O = obstructed); tolvaptan in polycystic disease.

Diagram to draw: Nephron with transporters; triphasic graph.

> **Key points to remember:**
> - RBF 25% of cardiac output; GFR 125 mL/min; net filtration pressure ~10 mmHg.
> - Proximal 65%, loop 25%, distal 5–7% of Na^+; ADH → V2 → AQP2.
> - Myogenic peristalsis from calyceal pacemaker; post-obstructive diuresis 2% vs 50%.

LAQ Lymphatic drainage of the genitourinary system and chyluria (etiopathogenesis and management)

Paper answer · P1

Asked as:

- Describe anatomy of lymphatic drainage of genitourinary system, etiopathogenesis and management of Chyluria — *Summer 2018 Paper I Q1 (LAQ)*
- Describe lymphatic system of urinary tract with reference to etiopathogenesis of chyluria. — *Practice Set 8 Paper I Q1 (LAQ)*
- Management of chyluria — *Practice Set 10 Paper I Q1 (LAQ)*

Type: LAQ (25 marks)
(also answered in 2018-Summer Paper 1, Q1 and 2024-Winter Paper 3, Q2)

General plan of lymphatic drainage

- Lymph flows **cephalad and right → left** and returns to the left innominate (brachiocephalic) vein.
- Pelvic nodes (external iliac, internal iliac, obturator, presacral) → common iliac → lumbar (para-aortic, interaortocaval, paracaval) nodes → **lumbar trunks**. These join the intestinal trunk at the **cisterna chyli** (in front of L1–L2, behind the right side of the aorta). The **thoracic duct** rises behind the aorta to the left innominate vein.
- Pelvic lymph crosses freely between right and left (the two sides). Organs supplied by lateral aortic branches (kidney, adrenal, ureter, gonad) drain to lumbar nodes.
- **Lumbar groups:** left lumbar (preaortic, left para-aortic, retroaortic), interaortocaval, and right lumbar (precaval, paracaval, retrocaval). They drain organs supplied by lateral aortic branches: kidneys, adrenals, ureters and gonads.
- Preaortic nodes (coeliac, superior and inferior mesenteric) form the **intestinal trunk**, which carries **chyle** from the intestinal lacteals.
- Lumbar and intestinal trunks join at the **cisterna chyli** (in front of L1–L2, behind the right side of the aorta). It continues as the **thoracic duct**, which rises behind the aorta to the left innominate vein.
- **Lymphatic anatomy (summary):** intestinal lacteals → intestinal trunk → **cisterna chyli** (L1–L2) → **thoracic duct** → left subclavian–jugular junction. Renal lumbar lymphatics join the cisterna; valves normally prevent backflow.

Drainage by organ

Organ	Primary nodes	Notes
Left kidney	Left lateral para-aortic	Capsular and deep hilar networks drain to renal vein nodes; may go retrocrural
Right kidney	Right paracaval, interaortocaval	May cross to left para-aortic
Ureter	Abdominal: para-aortic (left), paracaval/interaortocaval (right); middle: common iliac; pelvic: common, external, internal iliac	Plexus in muscle and adventitia
Bladder	Mainly **external iliac**; also internal iliac, obturator	Begins in lamina propria; bilateral cross-drainage
Prostate	**Obturator and internal iliac**; presacral	Three routes: superolateral (side wall), posterior (presacral), apical (internal pudendal)
Testis	Left: left para-aortic; right: interaortocaval, then right paracaval	Follows gonadal vessels; crossover only right → left. Epididymal cauda and vas → external iliac
Penis	Skin → superficial inguinal (both groins); glans and corpora → superficial and deep inguinal, then pelvic	Shaft lymphatics converge dorsally
Scrotum	Ipsilateral superficial inguinal	Does not cross the raphe
Urethra (male)	Penile → inguinal; bulbomembranous and prostatic → internal/external iliac, obturator	Female: proximal → pelvic; distal → inguinal

Lymphatic drainage of the urinary tract (intrinsic lymphatics and regional nodes)

Organ	Intrinsic lymphatics	Regional nodes
Kidney	Superficial capsular and deep hilar networks along the interlobular, arcuate and interlobar arteries, to nodes at the renal vein	**Left:** left lateral para-aortic (sometimes retrocrural, or thoracic duct above the diaphragm). **Right:** interaortocaval and right paracaval (may cross to left para-aortic)
Ureter	Plexuses in the muscle and adventitia	Abdominal: left para-aortic (left); paracaval and interaortocaval (right). Middle: common iliac. Pelvic: common, external and internal iliac
Bladder	Begins in the lamina propria; a few perivesical nodes	Mainly **external iliac**; also internal iliac and obturator; cross-drainage to both sides
Prostate	Periprostatic plexus	Obturator and internal iliac; presacral; external iliac
Urethra	Submucosal plexus	Male prostatic and membranous: internal iliac, obturator; penile: inguinal, then external iliac. Female: proximal to pelvic, distal to inguinal nodes

- **Applied:** upper tract cancers spread to lumbar nodes; bladder and prostate cancers to pelvic nodes. Renal drainage is asymmetric: right renal lymph can cross left; left stays left.
- Valves in the renal hilar lymphatics normally stop backflow from the cisterna chyli. **Failure of these valves is the basis of chyluria.**

Chyluria: definition

- Passage of **chyle** (intestinal lymph rich in chylomicrons, triglyceride and protein) in urine, giving milky urine (making it milky; urine looks milky). Caused by an abnormal **lymphourinary (chylourinary) fistula**, usually at the renal fornices / a renal calyceal fornix (**pyelolymphatic fistula**).

Etiology

Parasitic (commonest; India, China)	Non-parasitic
Wuchereria bancrofti (lymphatic filariasis; *Culex* mosquito)	Congenital lymphangiectasia, renal lymphangioma
Brugia (lymphatic filariasis that spares the genitals; rarely urological)	Trauma, surgery (e.g. partial nephrectomy, RPLND (retroperitoneal lymph node dissection))
	Retroperitoneal tumour or fibrosis, tuberculosis, pregnancy

Pathogenesis (obstructive theory)

- **Life cycle:** mosquito-borne larvae mature into adult worms that live for years in retroperitoneal and genital lymphatics. Females release microfilariae with **nocturnal periodicity**.
- Adult worms live in retroperitoneal lymphatics and cause lymphangitis and fibrosis. This **obstructs** flow between the intestinal lymph trunks and the thoracic duct.
- Proximal lymphatics dilate and their valves fail, so chyle **refluxes** into renal hilar lymphatics.
- Raised pressure ruptures them into a calyceal fornix → **pyelolymphatic fistula**. Rarely the fistula is in the ureter or bladder.
- **Obstructive (reflux) theory:** `Adult worms + host immune reaction -> lymphangitis and fibrosis -> obstruction between intestinal trunks/cisterna chyli and thoracic duct -> dilatation of retroperitoneal lymphatics, valve incompetence -> reflux of chyle into renal hilar and peripelvic lymphatics -> raised pressure ruptures thin-walled lymphatics into a calyceal fornix -> pyelolymphatic fistula -> chyluria`
- **Pathogenesis (filarial):** adult *Wuchereria bancrofti* worms and the host reaction cause **lymphangitis and fibrosis** between the intestinal trunks, cisterna chyli and thoracic duct. Lymphatics below the block dilate and their valves fail. Chyle refluxes into renal hilar lymphatics, which rupture into a calyx.
- The fistula is usually renal (hilar lymphatics lie against thin fornices); rarely ureteric or vesical.
- Chyluria appears **earlier than genital elephantiasis**, is usually **intermittent** and may remit spontaneously. It worsens after fatty meals, when lacteal flow is highest. So every treatment is judged against natural remission.
- **Non-parasitic causes** work the same way: lymphatic obstruction (tumour, fibrosis), direct disruption (surgery, trauma) or congenital lymphangiectasia.
- **Effects:** loss of fat, protein and lymphocytes causes weight loss, hypoproteinaemia and lowered immunity. Fibrin in chyle forms clots.

Clinical features

- Milky urine, worse after fatty meals; **chylous clots** (retention, colic); **haematochyluria**.

- Loss of fat and protein → weight loss, **hypoproteinaemia**, oedema, anaemia, fatigue, lowered immunity.
- **Severity:** mild (intermittent, no systemic effect); moderate (persistent, clots); severe (haematochyluria, clot retention, weight loss, hypoalbuminaemia).

Grade	Features
Mild	Intermittent; no systemic effect
Moderate	Persistent; clots
Severe	Haematochyluria, clot retention, weight loss, hypoalbuminaemia

Evaluation (before treatment)

1. Confirm chyle

- **Ether (chloroform) test**: milky urine shaken with ether clears because fat dissolves. **Urine triglycerides** (and protein) positive/raised; Sudan III stains fat globules.
- Distinguish from / exclude **phosphaturia** (clears with acetic acid) and **pyuria** (leucocytes, no clearing).

2. Filarial status (filarial tests)

- Night blood smear (often negative at this stage / late), **circulating filarial antigen** (card) test, eosinophilia; PCR (polymerase chain reaction).

3. Lateralise the side and localise the fistula

- **Cystoscopy after a fatty meal** shows milky efflux from the ureteric orifice (one, sometimes both).
- **Retrograde pyelography** shows **pyelolymphatic backflow**.
- **Define fistula: lymphoscintigraphy** (non-invasive, shows tracer in kidney and bladder), pedal lymphangiography (historical gold standard), MR (magnetic resonance) lymphangiography or CT.

4. Baseline: serum albumin, haemoglobin, renal function; urine culture before instillation.

Management (outline / algorithm)

`Mild → diet + DEC Persistent / moderate → add sclerotherapy (can repeat) Severe or failed sclerotherapy → renal hilar lymphatic disconnection Mild -> low-fat, high-protein diet + MCT oil + DEC Moderate/persistent -> endoscopic sclerotherapy (silver nitrate 0.5-1%, povidone-iodine) Severe or failed -> renal hilar lymphatic disconnection (open, laparoscopic, robotic) Confirm chyle -> grade severity Mild -> diet + antifilarial drugs; observe Moderate/persist -> add endoscopic sclerotherapy (repeatable) Severe or failed -> renal hilar lymphatic disconnection`

1. Conservative (all patients)

- **Low-fat, high-protein diet with MCT (medium-chain triglyceride) oil**, plenty of fluids.
- **Why MCT:** MCTs (medium-chain fats) are absorbed straight/directly into the portal blood, not into lacteals, so chyle formation falls (less chyle is formed).
- **DEC (diethylcarbamazine)** 6 mg/kg/day for 12 days kills microfilariae (another source: 2 mg/kg orally three times daily, traditionally 12 days; a 1-day course is as effective). It has only modest effect on adults.

- **Doxycycline** 200 mg daily for 4–8 weeks kills *Wolbachia* (endosymbiont; anti-*Wolbachia*) and so sterilises or shortens the life of adult worms. Albendazole and ivermectin are alternatives.
- Correct anaemia and protein deficit. Most mild cases settle on this, and many remit spontaneously.

2. Endoscopic sclerotherapy (renal pelvic instillation sclerotherapy)

- Ureteric catheter on the affected (leaking) side, head-down position. Instil **silver nitrate** (0.5–1%) or **povidone-iodine**, repeated over several sessions.
- **Mechanism:** chemical lymphangitis and fibrosis seal the fistula.
- **Contraindications:** active UTI (urinary tract infection), bilateral simultaneous instillation (risk of anuria).
- Good initial success/response; recurrence is common, but it is repeatable and avoids surgery.
- **Complications:** flank pain, haematuria, papillary necrosis, ureteric stricture, anuria; rarely renal failure or death (silver nitrate).

3. Surgery: renal hilar lymphatic disconnection (chylolymphatic disconnection)

- **Indications:** severe disease, failed sclerotherapy, haematochyluria with clot retention, marked weight loss.
- **Steps:** 1. Full renal mobilisation (nephrolysis). 2. **Skeletonise the renal artery and vein**, ligating, clipping or sealing every/all hilar lymphatics. 3. Strip the upper ureter; remove perinephric fat. 4. Nephropexy ± optional **omental wrap** around the hilum as a barrier. 5. **On-table cystoscopy** confirms clear urine/efflux.
- Reflux occurs only through **hilar lymphatics**, so upper-pole dissection can be limited.
- **Approach:** open, laparoscopic (transperitoneal or retroperitoneal) or robotic. **Retroperitoneoscopic** ligation beats open in meta-analysis: shorter operation and stay, less blood loss, faster bowel recovery, fewer complications.
- **Complications:** lymphocele, renal vein injury, recurrence.
- **Also know:** lymphovenous anastomosis; autotransplantation (rare); nephrectomy (now) obsolete; prevent by clipping lymphatics during renal and nodal surgery.

4. Follow-up and prevention

- Urine triglycerides and weight; recurrence needs repeat sclerotherapy or surgery.
- Mass drug administration (albendazole with DEC or ivermectin), bednets; clip lymphatics during renal and nodal surgery.

Diagram to draw: Retroperitoneal lymphatics with cisterna chyli, obstruction (a block above it / obstructed thoracic duct), dilated hilar lymphatics, and reflux to a pyelolymphatic fistula at a calyceal fornix.

Key points to remember:

- Lymph flows right → left; cisterna chyli at L1–L2, behind the right side of the aorta.
- Bladder drains mainly to external iliac nodes; left kidney to left para-aortic nodes.
- Chyluria = pyelolymphatic fistula from lymphatic obstruction and valve failure, usually filarial (*W. bancrofti*); often intermittent.
- Ether test and (urine) triglycerides confirm (ether test clears milky urine); cystoscopy after (a) fatty meal lateralises.
- Diet (MCT) + DEC (diet with MCT plus DEC) → (silver nitrate) sclerotherapy → laparoscopic hilar lymphatic disconnection; disconnection surgery for severe or refractory cases.
- Retroperitoneoscopic disconnection beats open.

LAQ Haemodialysis principles, dialysis techniques and dialysis access (vascular and PD)

Paper answer · P2

Asked as:

- Discuss principles of hemodialysis and vasular access in details. — *Winter 2017 Paper II Q1 (LAQ)*
- Vascular access for hemodialysis — *Winter 2025 Paper II Q3c (SAQ)*
- Hemodialysis access. — *Practice Set 10 Paper II Q3c (SAQ)*
- Access techniques in dialysis — *Practice Set 12 Paper II Q3f (SAQ)*
- Discuss techniques of dialysis — *Practice Set 5 Paper II Q3e (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

(Winter 2025 Paper II Q3c: also answered in 2020-Summer Paper 4, Q3b) (Practice Set 10 Paper II Q3c: also answered in 2020-Summer Paper 4, Q3b and 2025-Winter Paper 2, Q3c) (Practice Set 12 Paper II Q3f: CAPD also answered in Practice Set 1 Paper 2, Q3a) (Practice Set 5 Paper II Q3e: HD principles also answered in 2017-Winter Paper 2, Q1; CAPD in 2018-Summer Paper 2, Q3e)

Assumed meaning (Winter 2017): "vasular" is a typo for "vascular" access for haemodialysis. Assumed meaning (Practice Set 12): access for PD (peritoneal dialysis) and HD (haemodialysis); emphasis on PD catheter insertion, since HD vascular access is answered in full in Practice Set 12 Paper 2, Q3a. Assumed meaning (Practice Set 5): principles and techniques of HD (haemodialysis) and PD (peritoneal dialysis), with continuous therapies.

Introduction

- Haemodialysis (HD) removes solutes and water from blood across a semipermeable membrane in an extracorporeal circuit. It is the commonest KRT (kidney replacement therapy) for ESKD (end-stage kidney disease) and severe AKI (acute kidney injury). It needs an access giving about **250–350 mL/min** of blood flow.
- Access failure is a major cause of morbidity in ESKD (end-stage kidney disease).

Principles of haemodialysis (physical principles)

Process	Driving force	What it removes
Diffusion	Concentration gradient across the membrane	Small solutes (urea, creatinine, potassium); bicarbonate enters blood
Convection	Transmembrane pressure; solute is dragged with water	Middle molecules (e.g. β2-microglobulin)
Ultrafiltration	Transmembrane (hydrostatic) pressure (HD) or osmotic (PD) gradient	Excess water, down to "dry weight"

- **Countercurrent flow:** dialysate runs opposite to blood (**countercurrent**), keeping (to keep) a gradient along the whole dialyser. Clearance rises with flows, membrane area and time.
- **Dialyser:** blood (250–350 mL/min) passes through hollow fibres (hollow-fibre) of **synthetic membrane**; high-flux membranes clear more middle molecules.
- **Dialysate:** ultrapure (reverse-osmosis) water with sodium about 140 mmol/L, potassium about 2 mmol/L, bicarbonate and calcium, adjusted to the patient.
- **Anticoagulation:** heparin; heparin-free or regional citrate if bleeding risk is high (e.g. after surgery).
- **Schedule:** about 4 h (hours) three times (3 times) a week. **Also know:** haemofiltration, haemodiafiltration (HDF), CRRT (continuous renal replacement therapy).
- **Access:** AVF (arteriovenous fistula), AVG (arteriovenous graft) or CVC (central venous catheter).

Adequacy

- **Kt/V** (dialyser urea clearance × time ÷ urea distribution volume): target 1.4, minimum 1.2 (Kt/V ≥1.2) per thrice-weekly session (KDOQI [Kidney Disease Outcomes Quality Initiative] 2015). **URR (urea reduction ratio) ≥65%.**

Indications

- **Acute ("AEIOU"), refractory to medical treatment:** Acidosis; Electrolytes (hyperkalaemia); Ingestions (lithium, salicylates, alcohols); Overload with respiratory distress; Uraemia (encephalopathy, pericarditis, bleeding).
- **Chronic:** symptomatic ESKD, usually at eGFR (estimated glomerular filtration rate) 5–10; IDEAL (2010): early start gave no benefit.

Complications of haemodialysis

- **During dialysis:** hypotension (commonest), cramps, arrhythmia, air embolism, bleeding. **Disequilibrium syndrome:** rapid urea removal in early sessions shifts water into the brain (cerebral oedema); prevent with short, gentle first sessions.
- **Long term:** cardiovascular disease (main cause of death), mineral–bone disorder, calciphylaxis, β2-microglobulin amyloidosis (carpal tunnel), infection, hepatitis B and C; acquired cystic kidney disease (about 80% in ESKD; RCC [renal cell carcinoma] in 1–2% of these).
- **Access-related:** access infection, thrombosis, steal.

Peritoneal dialysis

- The **peritoneum** is the membrane: diffusion for solutes; **glucose or icodextrin** draws water osmotically.
- **Tenckhoff catheter** (two Dacron cuffs), placed open, laparoscopically (omentopexy) or percutaneously (see PD access below).
- **CAPD (continuous ambulatory PD):** about 4 exchanges of 2 L daily. **APD (automated PD):** night cycler.
- **Adequacy:** weekly Kt/V ≥1.7; PET (peritoneal equilibration test) classifies transport.
- **Complications:** peritonitis, exit-site infection, catheter malfunction, hernia, hyperglycaemia, peritoneal fibrosis.

Continuous and hybrid therapies

- **CRRT (continuous renal replacement therapy):** CVVH (continuous veno-venous haemofiltration; convection), CVVHD (haemodialysis; diffusion), CVVHDF (both). For haemodynamically unstable AKI (acute kidney injury).
- **Also know:** SLED (sustained low-efficiency dialysis); HDF (haemodiafiltration: CONVINCE 2023 lowered mortality versus high-flux HD); home and nocturnal HD.

HD versus PD

	HD	PD
Setting	Centre or home	Home, continuous
Haemodynamics	Rapid shifts	Gentle; preserves residual function
Favoured in	Abdominal surgery, failed PD	Children, cardiac instability, no vascular access

Types of vascular access

Access	Construction	Ready	Comment
Radiocephalic AVF (arteriovenous fistula; Brescia–Cimino, 1966)	Radial artery to cephalic vein at the wrist, end-to-side (now usually vein end to artery side)	≥6–8 weeks	**First choice**: best patency, fewest infections; highest failure to mature
Brachiocephalic AVF	Brachial artery to cephalic vein at the elbow	≥6–8 weeks	Higher flow; more steal
Brachiobasilic (transposition) AVF	Deep basilic vein transposed superficially	Longer	When the cephalic vein is unusable
AVG (arteriovenous graft)	PTFE (polytetrafluoroethylene), forearm loop or upper-arm straight	2–3 weeks	More stenosis, thrombosis, infection
Tunnelled cuffed CVC (central venous catheter)	Right internal jugular; cuff blocks bacterial migration	At once	Bridge or last resort; highest infection and central stenosis

- Non-tunnelled 10–12 Fr jugular or femoral catheters are for acute dialysis only.

Planning the access

- **Hierarchy:** AVF, then AVG, then CVC; **distal before proximal; non-dominant arm first** (US **Fistula First**, 2003). **KDOQI 2019** asks for (favours) an individual "ESKD Life-Plan": the right access for the right patient at the right time.

- **Timing:** refer at eGFR <30; create an AVF at least (about) **6 months** before dialysis is expected.
- **Vein preservation:** no venepuncture, cannula or PICC (peripherally inserted central catheter) in the non-dominant arm (that arm); avoid subclavian catheters (lines).
- **Assessment:** pulses, **Allen test** (ulnar collateral flow), **Doppler vein mapping** (vessels about ≥2 mm), venography if central stenosis is suspected.
- **Maturity, "rule of 6s":** flow >600 mL/min, diameter >6 mm, depth <6 mm, by (at) 6 weeks.

Access complications

Complication	Key point	Treatment
Failure to mature	Juxta-anastomotic stenosis; side veins	Angioplasty; ligate (side) branches
Stenosis	Neointimal hyperplasia; AVG at venous anastomosis	PTA (percutaneous transluminal angioplasty), stent graft, revision
Thrombosis	Usually on an underlying stenosis	Thrombectomy, then treat (fix) the stenosis
Steal syndrome	Hand ischaemia; diabetes, brachial-based access	Banding, DRIL (distal revascularisation–interval ligation), ligation
Aneurysm	Repeated needling at one site	Repair if skin is thin
Infection	CVC > AVG > AVF	Antibiotics; excise graft; remove catheter

- **Also know:** brachiobasilic transposition AVF; venous hypertension (central stenosis); high-output cardiac failure (banding); ischaemic monomelic neuropathy; graft seroma.

Monitoring

- Each session: thrill and bruit, arm-elevation test (a healthy fistula collapses), prolonged bleeding after needling, rising venous pressure, falling Kt/V. Duplex scan when abnormal; treat a stenosis causing clinical change. Rotate needle sites ("rope-ladder").

Peritoneal dialysis access

Catheter: Tenckhoff silicone catheter with **two Dacron cuffs**. The deep cuff sits in the rectus muscle; the superficial cuff about 2–4 cm from the exit. The tip lies in the **pelvis (rectovesical or rectouterine pouch)**; the exit site points **downwards** to reduce infection.

Insertion techniques

Technique	Steps	Comment
Open (mini-laparotomy)	Paramedian incision → split rectus → open posterior sheath and peritoneum → catheter to pelvis on a stylet → purse-string round deep cuff → tunnel → exit site	Simple; general or local anaesthesia
Laparoscopic (advanced)	Rectus sheath tunnel; **omentopexy**, adhesiolysis, hernia repair as needed	Best function; fixes causes of outflow failure
Percutaneous (Seldinger)	Needle, guidewire, dilator and peel-away sheath under local anaesthesia, ± ultrasound or fluoroscopy	Bedside; avoid after major abdominal surgery

Technique	Steps	Comment
Peritoneoscopic	Small-scope guided placement	Variant of percutaneous
- **Break-in:** delay full-volume exchanges about 2 weeks to prevent leak; urgent start uses small supine volumes.		
- **Complications:** outflow failure (constipation, omental wrap, migration), leak, hernia, exit-site or tunnel infection, peritonitis, bowel perforation.		
- **Contraindications:** peritoneal scarring, uncorrectable hernia, inability to perform exchanges.		

Haemodialysis catheter access (brief, insertion)

- **Non-tunnelled CVC (central venous catheter):** dual-lumen 10–12 Fr, internal jugular or femoral vein by **ultrasound-guided Seldinger**; for AKI (acute kidney injury); avoid subclavian (central stenosis).
- **Tunnelled cuffed CVC:** right internal jugular, tip at the cavoatrial junction under fluoroscopy; bridge to AVF.
- **AVF (arteriovenous fistula)** first choice, then **AVG (arteriovenous graft)**; HeRO and endovascular AVF in selected patients (see Practice Set 12 Paper 2, Q3a).

Recent advances

- **Endovascular AVF** (percutaneous creation: WavelinQ, Ellipsys); drug-coated balloons; bioengineered vessels; home and nocturnal HD.
- **CONVINCE (2023):** high-dose HDF lowered all-cause mortality compared with high-flux HD.

Diagram to draw: dialyser with countercurrent blood and dialysate (beside a PD catheter in the pelvis), and a radiocephalic end-to-side fistula at the wrist; Tenckhoff catheter with two cuffs, tunnel, downward exit and pelvic tip.

> **Key points to remember:**
> - Diffusion clears small solutes, convection middle molecules, ultrafiltration water.
> - HD Kt/V minimum 1.2 (target 1.4); URR ≥65%; PD weekly Kt/V ≥1.7.
> - AVF first, distal, non-dominant arm, ≥6 months ahead; rule of 6s: >600 mL/min, >6 mm, <6 mm deep, 6 weeks.
> - Thrombosis usually sits on a stenosis; steal is treated by (→) banding or DRIL.
> - Tenckhoff: two cuffs, pelvic tip, downward exit.
> - Laparoscopic placement with omentopexy gives best function.
> - Avoid subclavian lines; AVF first for long-term HD.
> - CRRT for unstable AKI; PD preferred in children.

SAQ Brain-stem death: pathophysiology, guidelines and declaration

Paper answer · P1 · P2 · P4 · merged from 3

Asked as: P2 SAQ "Brainstem death: biology and clinical implications" (asked as "Brain stem Death: Biology and clinical implications", Winter 2024 Paper II Q3d; "Brainstem death", Practice

Set 13 Paper II Q3c) · P1 SAQ "Patho-physiology of Brain stem death" (2015-Summer Paper 1 Q3d) · P4 SAQ "Brain-stem death: guidelines for declaration and the Board of Medical Experts (brain dead committee)" (asked as "Brain dead committee", Summer 2018 Paper IV Q3c; "Guidelines for declaration of brain stem death", Summer 2022 Paper IV Q3b)

Assumed meaning (P4): the Board of Medical Experts that certifies brain-stem death under Indian law.

Definition and legal basis

- **Brain-stem death:** irreversible loss of the capacity for consciousness and of the capacity to breathe, from permanent brainstem failure; equivalent to death of the person (legally death).
- **India: THOA (Transplantation of Human Organs Act) 1994** legalised brain-stem death; **amended 2011** (Transplantation of Human Organs and Tissues Act); **Rules 2014**.
- Internationally: Harvard criteria (1968), AAN (American Academy of Neurology) criteria (updated 2023).
- Donation after brain death (DBD) keeps the circulation until procurement, so warm ischaemia is minimal.

Biology (pathophysiology)

Sequence

1. Brain injury causes oedema and raised ICP (intracranial pressure).
2. Cerebral perfusion falls, leading to herniation ("coning").
3. Pons: Cushing reflex (hypertension, bradycardia).
4. Medulla: catecholamine storm (hypertension, tachyarrhythmia).
5. Loss of sympathetic tone: vasodilatation, hypotension.

System	Consequence
Heart and vessels	Storm causes myocardial injury and arrhythmia, then **hypotension**
Lungs	Neurogenic pulmonary oedema
Pituitary	**Diabetes insipidus (DI)** (no ADH, antidiuretic hormone): polyuria, hypernatraemia; low thyroid hormone and cortisol
Hypothalamus	**Hypothermia** (poikilothermia)
Blood	DIC (disseminated intravascular coagulation) from brain thromboplastin
Whole body	Cytokine injury to organs; hyperglycaemia

Composition of the Board of Medical Experts (India; Rules 2014)

Member	Role
1. Registered medical practitioner **in charge of the hospital**	Hospital authority
2. **Independent specialist**, nominated by the in-charge from a panel approved by the Appropriate Authority	Independent check
3. **Neurologist or neurosurgeon** from the approved panel (if unavailable: surgeon, physician, anaesthetist or intensivist)	Expert
4. Medical practitioner **treating the patient**	Clinical history

- **No member may belong to the transplant team.**

Certification: preconditions

- Apnoeic coma on a ventilator with a **known irreversible cause**.
- Exclude drugs (**sedatives, relaxants**), hypothermia (35–36 °C by protocol), metabolic or endocrine causes, shock.

Certification: brainstem reflexes (all must be absent)

Brainstem reflex	Method	Cranial nerves
Pupillary light reflex	Bright light: pupils fixed, no reaction	II, III
Corneal	Touch cornea: no blink	V, VII
Oculocephalic (doll's eye)	Turn head: eyes move with the head	VIII, III, VI
Oculovestibular (caloric)	~50 mL ice-cold water into each ear (drum visible): no eye movement	VIII, III, VI
Motor response in cranial nerve area	Central painful stimulus (supraorbital pressure): no facial grimace	V, VII
Gag and cough	Touch pharynx; pass suction catheter down trachea: no response	IX, X

Certification: apnoea test and timing

- **Apnoea test (last):** pre-oxygenate with 100% oxygen; disconnect the ventilator with oxygen insufflation; watch for breathing effort; **no breathing effort at $PaCO_2$ (arterial carbon dioxide tension) ≥60 mmHg** (or ≥20 mmHg above baseline) is a positive test. Abort for hypotension, desaturation or arrhythmia.
- **Two sets of tests at least 6 hours apart**, recorded on **Form 10**.
- **Ancillary tests** are not mandatory; used when clinical examination cannot be completed (e.g. facial trauma): EEG (electroencephalogram), cerebral angiography, perfusion scan, transcranial Doppler.

Clinical implications: linked duties after declaration

- **Mandatory request:** ICU (intensive care unit) staff must inform relatives of the option to donate; a **transplant coordinator** counsels; next of kin give consent.
- **NOTTO** (National Organ and Tissue Transplant Organisation), with regional and state bodies (ROTTO, SOTTO: regional and state organ and tissue transplant organisations), allocates organs and keeps the registry.

Clinical implications: donor management

- **Rule of 100s:** systolic pressure >100 mmHg, urine >100 mL/h, PaO_2 (arterial oxygen tension) >100 mmHg, haemoglobin >100 g/L.
- **Vasopressin** preferred (pressor, also treats DI); thyroid hormone, methylprednisolone, insulin; warming; protective ventilation.
- **Also know:** cold storage at 4 °C (University of Wisconsin solution); donor exclusions (metastatic cancer, melanoma, untreatable viral infection).

Recent advances

- Hypothermic machine perfusion reduces delayed graft function.

Diagrams to draw

- Herniation sequence leading to storm, DI, hypothermia, DIC.
- Certification flowchart: 1. Coma. 2. Preconditions. 3. Test 1. 4. 6 hours. 5. Test 2. 6. Form 10. 7. Consent. 8. NOTTO.

> **Key points to remember:**
>
> - Raised ICP, Cushing reflex, catecholamine storm, then hypotension, DI, hypothermia, DIC.
> - Six brainstem reflexes absent; apnoea test: no breathing at $PaCO_2$ ≥60 mmHg.
> - THOA: four-member Board; no transplant-team member.
> - Two tests ≥6 h apart; Form 10; Rule of 100s.

SAQ Immunosuppression in renal transplantation: protocols, induction and steroid-free regimens

Paper answer · P1 · P2 · P4 · merged from 3

Asked as: P1 SAQ "Immunosuppressants in renal transplantation (classification, protocols, ciclosporin)" (asked as "Immunosuppressants", Summer 2021 Paper I Q3f; "Classify immunosuppresants. Describe few common protocols", Practice Set 3 Paper I Q3b; "Cyclosporins", Practice Set 4 Paper I Q3f) · P2 SAQ "Pre transplant Immunosupression treatment options" (2023-Winter Paper 2 Q3e) · P4 SAQ "Steroid-free immunosuppression in renal transplantation" (Practice Set 3 Paper 4 Q3f)

Taken as immunosuppressant drugs in renal transplantation; "Cyclosporins" taken as ciclosporin (cyclosporin A); "pre-transplant" taken as induction immunosuppression and desensitisation given before or at transplantation. Also answered in 2020-Summer Paper 4, Q1; tacrolimus in 2017-Winter Paper 3, Q3c.

Principle

- T-cell activation needs three signals: **signal 1** (TCR [T-cell receptor] binds donor antigen; calcineurin then drives IL-2 [interleukin-2] transcription), **signal 2** (costimulation, CD28–CD80/86) and **signal 3** (IL-2 receptor signalling through mTOR [mammalian target of rapamycin], driving proliferation). Each drug blocks a step.
- Rejection risk is highest in the first weeks.

Phases

- **Induction** (peri-transplant intense immunosuppression), **maintenance** (lifelong), **rescue** (treatment of rejection).
- **Desensitisation** removes preformed antibody in HLA (human leucocyte antigen)- or ABO-incompatible recipients.

Classification

Class	Drug(s)	Mechanism / action	Main toxicity
CNI (calcineurin inhibitors)	**Tacrolimus**, ciclosporin	Block calcineurin (signal 1)	**Nephrotoxicity**, diabetes, tremor, hypertension; ciclosporin: gum hypertrophy, hirsutism

Class	Drug(s)	Mechanism / action	Main toxicity
Antimetabolites (antiproliferatives)	**Mycophenolate**, azathioprine	Block purine synthesis (IMPDH [inosine monophosphate dehydrogenase])	Diarrhoea, leucopenia
mTOR inhibitors	Sirolimus, everolimus	Block signal 3	**Lymphocele, poor wound healing**, proteinuria, dyslipidaemia; not nephrotoxic, antitumour
Corticosteroids	Prednisolone, methylprednisolone	Block cytokine transcription (IL-1, IL-2) and redistribute lymphocytes	Diabetes, bone loss, cataract
Costimulation blocker	**Belatacept**	CTLA4-Ig binds CD80/86 (signal 2)	**PTLD** (post-transplant lymphoproliferative disorder); avoid if EBV (Epstein–Barr virus)-negative
Depleting antibodies	rATG (rabbit antithymocyte globulin), alemtuzumab	T-cell depletion	Infection, PTLD
Non-depleting antibody	**Basiliximab**	Anti-CD25 (IL-2 receptor)	Few
Antibody / B-cell agents	Rituximab, IVIG (intravenous immunoglobulin), plasmapheresis, bortezomib, eculizumab, imlifidase	Antibody removal or blockade; antibody-mediated rejection, desensitisation	—

Pre-transplant assessment of immunological risk

- cPRA (calculated panel-reactive antibody), DSA (donor-specific antibody) on Luminex beads, crossmatch (CDC [complement-dependent cytotoxicity] and flow).

Induction agents

Agent	Mechanism	Use and drawbacks
Basiliximab	Anti-CD25 (IL-2 receptor); non-depleting	Low-risk recipients; 20 mg IV (intravenous) on days 0 and 4; few side effects
rATG	Polyclonal T-cell depletion	High-risk (sensitised, retransplant, delayed graft function); less rejection than basiliximab; cytokine release, leucopenia, infection, PTLD
Alemtuzumab	Anti-CD52; profound depletion	Allows steroid avoidance; off-label
IV methylprednisolone	Cytokine suppression	Given intraoperatively with all regimens

- **Over 90%** receive induction, then **tacrolimus + mycophenolate** ± steroids; steroids in more than two-thirds. Acute rejection has fallen from ≥60% to <10%.

- In living-donor transplants, tacrolimus and mycophenolate often start days before surgery.

Common protocols

1. **Standard-risk (most recipients):** basiliximab 20 mg IV on days 0 and 4 + intraoperative methylprednisolone; then **tacrolimus + mycophenolate + tapering prednisolone**.
2. **High-immunological-risk** (sensitised, retransplant, DGF [delayed graft function] expected): **rATG** induction, then triple therapy.
3. **Steroid-avoidance or early withdrawal:** low-risk recipients, usually after depleting induction (see below).
4. **CNI-sparing:** belatacept- or mTOR-based regimens to spare nephrotoxicity; TRANSFORM: everolimus with reduced CNI non-inferior.
5. **ABO-incompatible or HLA-sensitised:** desensitisation with rituximab, plasmapheresis and IVIG; imlifidase for crossmatch-positive (see below).
6. **Rejection:** T-cell-mediated: pulse IV methylprednisolone, then rATG (for Banff grade II–III); antibody-mediated: steroids, plasmapheresis, IVIG, rituximab.

- **Monitoring: tacrolimus trough levels** (higher early, lower later); CYP3A4 (cytochrome P450 3A4) interactions (azoles, macrolides raise levels; rifampicin lowers).

Desensitisation: HLA-incompatible

- **Plasmapheresis** (removes antibody) plus **low-dose IVIG** after each session; or **high-dose IVIG** (immunomodulation).
- **Rituximab** (anti-CD20) reduces B cells that renew antibody. Bortezomib (plasma cells) and eculizumab (anti-C5) in selected cases.
- **Imlifidase:** bacterial enzyme that **cleaves all IgG** within hours, turning a positive crossmatch negative; EMA (European Medicines Agency) conditional approval 2020 for highly sensitised patients.
- Desensitised live-donor transplant gives better survival than staying on dialysis (Montgomery, 2011).

Desensitisation: ABO-incompatible

- **Rituximab** 2–4 weeks before; **plasmapheresis or immunoadsorption** to lower anti-A/B titres to a centre target (commonly ≤1:8); IVIG; tacrolimus–mycophenolate started before surgery. Splenectomy is now obsolete.
- Post-transplant titre monitoring; A2 donors carry lower risk.
- **Alternatives: kidney paired exchange** avoids desensitisation; acceptable-mismatch programmes.

Steroid-free immunosuppression

Why avoid steroids

- **Metabolic:** PTDM (post-transplant diabetes mellitus), hypertension, dyslipidaemia, weight gain, hence **cardiovascular death** (leading cause of death with a functioning graft).
- **Bone:** osteoporosis, avascular necrosis; **growth failure** in children.
- Cataract, infection, poor healing.

Strategies

Strategy	Timing
Steroid avoidance	No steroids, or perioperative doses only

Strategy	Timing
Early (rapid) withdrawal	Stopped within the first week; most studied
Late withdrawal	Months later; more rejection, less favoured

Regimen

- **Induction is essential:** depleting rATG or alemtuzumab; basiliximab in low risk.
- **Maintenance:** tacrolimus at target levels plus MMF (mycophenolate mofetil).

Selection

- **Suitable:** low immunological risk: first transplant, non-sensitised, no DSA, prompt graft function, good adherence.
- **Unsuitable:** highly sensitised, DSA, retransplant after immunological loss, DGF, early rejection, **recurring steroid-responsive disease** (IgA nephropathy, lupus), steroids needed for other illness.

Evidence

- **Woodle et al. (2008):** withdrawal after day 7 versus low-dose maintenance: similar 5-year outcomes; **more mild, steroid-sensitive rejection**.
- **HARMONY (Lancet 2016):** steroids stopped on day 8: similar rejection and graft survival, **less PTDM**.
- **Cochrane review (2016):** more acute rejection, but no difference in patient or graft survival up to 5 years.
- **KDIGO (Kidney Disease: Improving Global Outcomes) 2009:** low-risk recipients with induction may stop steroids within the first week; if continued beyond it, continue rather than withdraw.
- **Practice:** over two-thirds still take maintenance steroids (Chapter 91); steroid-free maintenance is increasing in low-risk children.

Risks and safeguards

- Early rejection; possibly more chronic injury and recurrence. Safeguards: tacrolimus levels, adherence, DSA monitoring, biopsy for rising creatinine.

Ciclosporin (cyclosporin A)

Introduction

- Cyclic peptide of 11 amino acids from the fungus *Tolypocladium inflatum*; the first **CNI**.
- Its introduction in the early 1980s cut acute rejection from at least 60% (pre-ciclosporin era) towards today's <10%. Tacrolimus has now largely replaced it.

Mechanism

1. Ciclosporin binds cyclophilin in the cytoplasm.
2. The complex inhibits calcineurin.
3. NFAT (nuclear factor of activated T cells) stays phosphorylated.
4. No IL-2 transcription, so T-cell activation is blocked (signal 1).

Pharmacokinetics and monitoring

- Variable oral absorption; the **microemulsion form** absorbs more predictably.
- Metabolised by **CYP3A4** and P-glycoprotein; narrow therapeutic index, so whole-blood **trough (C0) or 2-hour (C2) levels** guide dosing.

Uses

- **Transplantation:** maintenance with an antimetabolite and steroids, when tacrolimus is not tolerated.
- **IC/BPS (interstitial cystitis/bladder pain syndrome):** refractory **Hunner lesions**, 1–3 mg/kg/day; better than pentosan polysulfate.
- **Others:** retroperitoneal fibrosis after steroid failure; frequently relapsing nephrotic syndrome; severe psoriasis and other dermatoses.

Toxicity

System	Effect
Kidney	**Nephrotoxicity**: acute afferent arteriolar vasoconstriction (reversible); chronic striped fibrosis, arteriolar hyalinosis; TMA (thrombotic microangiopathy)
Metabolic	Hypertension, hyperkalaemia, hypomagnesaemia, **hyperuricaemia** (gout), dyslipidaemia
Cosmetic	**Gingival hyperplasia, hirsutism**
Neurological	Tremor, PRES (posterior reversible encephalopathy syndrome)
Immune	Infection, skin cancer, PTLD

Interactions

- **Levels rise:** azoles, erythromycin and clarithromycin, diltiazem, verapamil, grapefruit.
- **Levels fall:** rifampicin, phenytoin, carbamazepine, St John's wort.
- More nephrotoxicity with NSAIDs (non-steroidal anti-inflammatory drugs) and aminoglycosides.

Versus tacrolimus

- Ciclosporin: more hirsutism, gum hypertrophy, hypertension and hyperlipidaemia; less diabetes and tremor.
- **ELITE-Symphony (2007):** low-dose tacrolimus gave less rejection and better GFR (glomerular filtration rate) than ciclosporin regimens.

Urological relevance

- **Infection:** UTI (urinary tract infection), **BK virus** (ureteric stricture, nephropathy), CMV (cytomegalovirus).
- **Malignancy:** skin cancer, PTLD, RCC (renal cell carcinoma, mostly native kidneys), **bladder cancer about 3-fold** and more aggressive.
- **Surgery:** lymphocele and wound problems with mTOR inhibitors; stones (CNI, steroids): CNIs promote transplant stones (hyperuricaemia, hyperoxaluria, hypocitraturia).
- **Also know:** belatacept (BENEFIT trial: better GFR), extended-release tacrolimus, TRANSFORM (everolimus with reduced CNI non-inferior).

Diagrams to draw

- APC (antigen-presenting cell)–T cell synapse with signals 1–3 and the drug acting at each.
- T cell with the cyclophilin–calcineurin–NFAT–IL-2 pathway.
- Antibody titre falling with plasmapheresis sessions to target before transplant.
- Timeline: induction, steroids stopped by day 7, tacrolimus–MMF maintenance.

Key points to remember:

- Tacrolimus + mycophenolate ± steroids (maintenance) after induction (basiliximab or rATG).
- Basiliximab for low risk; rATG for high immunological risk.
- CNI nephrotoxic; mTOR inhibitors cause lymphocele and poor healing.
- Belatacept blocks signal 2; avoid if EBV-negative.
- Rejection: steroids for T-cell-mediated; plasmapheresis, IVIG, rituximab for antibody-mediated.
- HLA-incompatible: plasmapheresis + IVIG ± rituximab; imlifidase cleaves IgG.
- ABO-incompatible: rituximab + plasmapheresis to target titre.
- Steroid-free: avoidance, early (first week) or late withdrawal; needs induction plus tacrolimus–MMF; low-risk recipients only.
- More mild rejection, same survival; less PTDM (HARMONY). KDIGO 2009: stop within week 1, or continue.
- Ciclosporin: binds cyclophilin, inhibits calcineurin, NFAT stays phosphorylated, IL-2 blocked.
- Ciclosporin: nephrotoxic; gum hyperplasia and hirsutism; CYP3A4 interactions.
- Ciclosporin urology uses: transplant, refractory Hunner lesions, retroperitoneal fibrosis.

SAQ CAPD and CAPD peritonitis

Paper answer · P1 · P2 · merged from 2

Asked as: P2 SAQ "CAPD (Continuous Ambulatory Peritoneal Dialysis)" (Summer 2018 Paper II Q3e; Practice Set 1 Paper II Q3a) · P1 SAQ "Management of CAPD peritonitis" (Practice Set 21 SAQ Sheet Q3d)

Definition and principle

- CAPD (continuous ambulatory peritoneal dialysis): home dialysis using the **peritoneum** (about 1–2 m^2) as the membrane; dialysate dwells in the abdomen.
- **Diffusion:** urea, creatinine and potassium move into dialysate down a concentration gradient.
- **Ultrafiltration:** an osmotic agent draws water by osmosis: **glucose** (1.5%, 2.5%, 4.25%) or **icodextrin** (for the long dwell).
- **CAPD regimen:** about **4 exchanges of 2 L daily**, 4–6-hour dwells, overnight long dwell. **APD** (automated peritoneal dialysis) uses a night cycler.

Catheter

- **Tenckhoff catheter:** silicone with **two Dacron cuffs** (deep cuff in rectus muscle; superficial cuff about 2 cm from the exit), exit site pointing downward, tip in the pelvis.
- **Insertion:** open; **laparoscopic** (rectus sheath tunnelling, omentopexy, adhesiolysis); percutaneous Seldinger. Delay full volumes about 2 weeks to prevent leaks.

Adequacy

- Weekly **Kt/V urea ≥1.7** (peritoneal plus residual renal urea clearance × time ÷ urea distribution volume), with clinical well-being.

- **PET** (peritoneal equilibration test) identifies high transporters, who do better on short dwells (APD).

Advantages over haemodialysis

- Home-based, no vascular access, **haemodynamically gentle**, **preserves residual renal function**. Favoured in children, poor vascular access and unstable heart disease.

Contraindications

- **Absolute:** peritoneal adhesions or fibrosis (scarring), loss of membrane function (membrane failure), uncorrectable hernia, nobody able to do exchanges.
- **Relative:** recent abdominal surgery, stoma, abdominal wall infection, massive polycystic kidneys, morbid obesity, severe malnutrition.

Complications (overview)

- **Peritonitis** (see below).
- **Exit-site and tunnel infection:** prevent with topical mupirocin or gentamicin.
- **Mechanical:** outflow failure (constipation, omental wrap, migration), pericatheter **leak**, **hernias** (raised abdominal pressure), hydrothorax, scrotal oedema (patent processus vaginalis).
- **Metabolic:** hyperglycaemia, weight gain, hypokalaemia, protein loss.
- **Long term:** ultrafiltration failure; **EPS (encapsulating peritoneal sclerosis)**: rare, presents as bowel obstruction.
- **Also know:** the PD (peritoneal dialysis) catheter is removed after a successful transplant; retroperitoneal laparoscopy protects the peritoneum for PD patients.

CAPD peritonitis: introduction

- The leading cause of technique failure and transfer to haemodialysis. Commonest organism: **coagulase-negative *Staphylococcus*** (touch contamination).
- ISPD (International Society for Peritoneal Dialysis) 2022 target: centre rate **≤0.40 episodes per patient-year**.

CAPD peritonitis: diagnosis (any 2 of 3)

1. Abdominal pain and/or **cloudy effluent**.
2. Effluent **WCC (white cell count) >100/µL with >50% neutrophils** (dwell ≥2 hours).
3. Positive effluent culture.

CAPD peritonitis: initial steps

- Effluent cell count, Gram stain, **culture in blood-culture bottles** (culture-negative rate should be <15%).
- Examine exit site and tunnel; analgesia; heparin in bags if fibrin.

CAPD peritonitis: empirical antibiotics

- **Intraperitoneal route preferred** (intravenous if septic); start immediately after samples.
- Cover **Gram-positives** (vancomycin or first-generation cephalosporin, e.g. cefazolin) **and Gram-negatives** (third-generation cephalosporin, e.g. ceftazidime, or aminoglycoside). Choice follows centre sensitivities.
- Intermittent (once daily in the long dwell) or continuous dosing; avoid prolonged aminoglycoside (residual function, ototoxicity).

- **Antifungal prophylaxis** (nystatin or fluconazole) during antibiotics.

CAPD peritonitis: organism-specific duration

Organism	Duration
Coagulase-negative *Staphylococcus*, streptococci, culture-negative	2 weeks
S. aureus, enterococcus, Gram-negatives	3 weeks
Pseudomonas	3 weeks, two antipseudomonal drugs
Polymicrobial enteric	Add metronidazole; surgical review (bowel perforation)
Fungal	**Immediate catheter removal** + antifungal

CAPD peritonitis: catheter removal

- **Refractory** (effluent not clear after 5 days of the right antibiotics), relapsing, fungal, refractory exit-site or tunnel infection.
- Haemodialysis meanwhile; new catheter after at least 2 weeks.

CAPD peritonitis: definitions

- **Relapse:** same organism (or sterile) within 4 weeks of finishing therapy.
- **Recurrent:** different organism within 4 weeks.
- **Repeat:** same organism after 4 weeks.

CAPD peritonitis: prevention

- Training, exit-site mupirocin or gentamicin, prophylaxis before colonoscopy or gynaecological procedures.

Diagrams to draw

- Tenckhoff catheter: two cuffs (deep and superficial), tunnel, downward exit site, pelvic tip.
- Peritonitis management flowchart from cloudy bag to catheter removal.

> **Key points to remember:**
> - Peritoneum is the membrane; glucose or icodextrin drives ultrafiltration.
> - Tenckhoff catheter with two cuffs; Kt/V ≥1.7 per week.
> - Peritonitis: two of cloudy fluid or pain, WCC >100/µL with >50% neutrophils, positive culture.
> - Intraperitoneal Gram-positive plus Gram-negative cover.
> - Remove catheter for fungal or refractory (5 days) peritonitis.

SAQ THOA/THOTA legislation and hospital authorization committee

Paper answer · P2 · P4 · merged from 2

Asked as: P2 SAQ "THOA 1994 and its 2011 amendment (THOTA) and 2014 Rules" (Amendments to THOA: 2011; HOTA and its modification) · P4 SAQ "Hospital authorization committee for live related kidney donation."

Indian law; not covered in the chapter notes. Assumed meaning (Practice Set 11): India's Transplantation of Human Organs Act (THOA) 1994, its 2011 amendment and 2014 Rules.

Legal framework and background: THOA 1994 aims

- **THOA (Transplantation of Human Organs Act) 1994**; amended 2011 as **THOTA (Transplantation of Human Organs and Tissues Act)**; **Rules 2014**.
- THOA 1994 regulates removal, storage and transplantation of organs, **prevents commercial dealing** (banned commercial dealing) and **recognised brain-stem death as legal death**, allowing deceased donation.
- Purpose: **prevent organ trade** while allowing genuine donation.
- Organ trade continued and deceased donation stayed low. The **Transplantation of Human Organs (Amendment) Act 2011** responded; **Rules followed in 2014**.

Main provisions of THOA 1994

- **Living donors: near relatives** (spouse, parents, children, siblings) after verification. Others, donating "by reason of affection or attachment", need **Authorisation Committee** approval.
- **Appropriate Authority** registers and inspects hospitals; only registered centres may transplant.
- **Brain-stem death certification by a Board of four:** doctor in charge of the hospital; independent specialist from an approved panel; neurologist or neurosurgeon; treating doctor. **None from the transplant team.** Two sets of tests ≥6 hours apart.
- **Organ sale is a punishable offence.**

2011 amendment (THOTA)

Area	Change	Purpose
Scope	**Tissues** added; renamed the **Transplantation of Human Organs and Tissues Act (THOTA)**; **tissue banks** registered	Regulates tissue banks
Near relatives	**Grandparents and grandchildren** added to spouse, children, parents and siblings	Wider related donor pool
Swap donation	**Paired exchange** (swap donation) between near-relative pairs allowed, with Authorisation Committee approval	Overcomes ABO or crossmatch incompatibility
Retrieval centres	Hospitals with ICU (intensive care unit) facilities can register as **non-transplant organ retrieval centres**	
Transplant coordinator	Mandatory in every registered transplant hospital	Organises donation
Mandatory ("required") request	ICU doctor or staff must ask families of a potential brain-dead donor about donation and inform the retrieval centre	More deceased donors
Brain-death board	If a neurologist or neurosurgeon is unavailable, a surgeon, physician, anaesthetist or intensivist nominated by the hospital can serve	Smaller hospitals can certify
Eye donation	Trained technicians may remove corneas	
Registry and network	National human organ and tissue removal and storage network and **national registry** of donors and recipients (now run by NOTTO, the National Organ and Tissue Transplant Organisation)	Fair allocation

Area	Change	Purpose
Penalties	**Higher penalties** for organ trade (up to 10 years' imprisonment; fine up to ₹1 crore) (see below)	Deterrence

Penalties under the amendment

- **Removal of organs without authority:** imprisonment up to **10 years** and fine up to ₹20 lakh.
- **Commercial dealing in organs:** imprisonment **5–10 years** and fine **₹20 lakh to ₹1 crore** (P4: imprisonment up to 10 years and fine up to ₹1 crore).
- Registered medical practitioner: name removed from the register (doctors' registration removed) for **3 years** for a first offence and **permanently** for a later offence.

Rules 2014 (THOT Rules)

- Standard forms for consent and certification (**Form 10** for brain-stem death); brainstem reflexes and apnoea test set out.
- Authorisation Committee interviews donor and recipient to exclude coercion or payment.
- **Also know:** regional and state networks (ROTTO, SOTTO); 2023 guidelines removed the domicile rule and age limit for deceased-donor listing.

Living donor categories and approval

Donor	Approval
Near relative: spouse, son, daughter, father, mother, brother, sister, grandparents, grandchildren	Verified by the hospital's competent authority
Other than near relative (affection or attachment)	**Authorisation Committee approval mandatory**
Swap (paired) donation between near-relative pairs (2011)	Authorisation Committee
Foreign donor or recipient	Authorisation Committee; embassy certifies relationship

Hospital authorisation committee: constitution

- **Hospital-based** if a centre does more than **25 transplants a year**; otherwise **district or state committee**.
- **Members:** medical director or superintendent (chair); two senior doctors of the hospital **not in the transplant team**; two persons of high integrity and social standing (preferably one woman), nominated by the State; state committees add health-department nominees.

Hospital authorisation committee: functions and procedure

1. Receive the **joint donor–recipient application** with consent forms, fitness certificates and identity and relationship documents.
2. **Personal interview** of donor, recipient and families; proceedings **video-recorded**.
3. Verify **relationship** (documents; **HLA [human leukocyte antigen] or DNA testing** if doubtful).
4. Exclude **commercial dealing**: financial status, reasons, long association, middlemen.
5. Confirm donor understands risks and consents freely, without coercion.
6. Check **domicile** (other-state donors).

7. Decide promptly; decision recorded and displayed; reasons given if refused. **Appeal** to the State Government within **30 days**.

Safeguards and restrictions

- Unrelated donors and foreigners need **Authorisation Committee** approval.
- **Minors and mentally ill persons** cannot be living donors except in exceptional, approved cases.
- **Foreign recipient with unrelated Indian donor** not approved.

Diagrams to draw

- Deceased donation flowchart: potential donor (brain-dead patient) in ICU → required request → transplant coordinator → brain-stem death board (Board of four) → family consent → retrieval → national registry and allocation (NOTTO allocation).
- Living donor approval flowchart: application → near relative (competent authority) or other (committee) → interview, verify → decision → appeal.

> **Key points to remember:**
>
> - THOA 1994: brain-stem death legal; anti-commercial; Authorisation Committee for unrelated donors; amended 2011; Rules 2014.
> - 2011: tissues added; grandparents and grandchildren became near relatives.
> - Swap donation, retrieval centres, transplant coordinators, required request, higher penalties.
> - Commercial dealing: 5–10 years and ₹20 lakh to ₹1 crore.
> - Rules 2014: Form 10; Board of four; tests ≥6 hours apart.
> - Hospital-based committee if >25 transplants a year; no transplant team member.
> - Non-relatives, swap and foreigners need committee approval.
> - Interview video-recorded; appeal within 30 days.

SAQ Marginal kidney donor and Remuzzi score

Paper answer · P2 · P3 · P4 · merged from 3

Asked as: P4 SAQ "Marginal donor and Remuzzi score for pre-implantation biopsy" (Define Marginal donor. Mention components of Remuzzi scoring system for preemptive renal biopsy of marginal donor kidneys.) · P2 SAQ "Marginal kidney donor" · P3 SAQ "Marginal Donor"

"Preemptive" is read as pre-implantation (procurement) biopsy.

Definition

- A **marginal (extended) donor** is one whose kidney carries a **higher risk of DGF (delayed graft function) and shorter graft survival** than a standard donor. Such kidneys are used because of organ shortage.
- **Ojo (2001):** ECD transplantation still gives better survival than staying on dialysis.

Deceased marginal donors

1. ECD (expanded criteria donor), UNOS (United Network for Organ Sharing) 2002

- Age **60 or over**, or
- Age **50–59 with 2 or more of:** hypertension, **creatinine >1.5 mg/dL**, death from **CVA (cerebrovascular accident)**.

- Graft-loss risk about 1.7 times that of a standard donor (about 70% higher).

2. KDPI (Kidney Donor Profile Index) >85%

- Replaced ECD in the US (2014); 10 donor factors (age, height, weight, ethnicity, hypertension, diabetes, cause of death, creatinine, HCV [hepatitis C virus], DCD).
- **Longevity matching:** lowest-KDPI kidneys go to recipients with the best EPTS (Estimated Post-Transplant Survival).

3. DCD (donation after circulatory death), Maastricht

Category	Description	Type
I / II	Dead on arrival / failed resuscitation	Uncontrolled
III	Arrest after withdrawal of support	**Controlled (commonest)**
IV	Arrest in a brain-dead donor	Either
V	Euthanasia	Controlled
- Warm ischaemia raises DGF (roughly 40–50%); long-term survival is similar to DBD (donation after brain death).		

4. Others

- Donor AKI (acute kidney injury).
- **HCV-positive (HCV-viraemic)** donors with antiviral treatment.
- **Paediatric donors under 15 kg (en bloc).**
- **Excluded:** metastatic or haematological cancer, melanoma, untreatable viral infection.

Living marginal donors

- Older donors, controlled hypertension (older or hypertensive living donors), obesity, small stones, multiple renal arteries. Acceptable after careful evaluation and consent.

Remuzzi score (Karpinski/Remuzzi)

- Pre-implantation biopsy (ideally at least 25 glomeruli), read on rapid sections.
- **Four components (glomeruli, tubules, interstitium, vessels), each 0–3: total 0–12.**

Component	0	1	2	3
Glomerular global sclerosis	None	<20%	20–50%	>50%
Tubular atrophy	None	<20%	20–50%	>50%
Interstitial fibrosis	None	<20%	20–50%	>50%
Arterial and arteriolar narrowing	None	Wall thickness < lumen diameter	Wall = or slightly > lumen	Wall far > lumen, severe narrowing or occlusion

Allocation by score

Total	Grade	Use
0–3	Mild	**Single** kidney transplant
4–6	Moderate	**Dual** kidney transplant

Total	Grade	Use
7–12	Severe	**Discard**

- Remuzzi (NEJM 2006): biopsy-guided older-donor kidneys, single or dual, had graft survival similar to young-donor kidneys.
- **Limitations:** sampling error; wedge biopsy overestimates sclerosis; interobserver variation.

Optimising use of marginal kidneys

- Short cold ischaemia.
- **HMP (hypothermic machine perfusion)** reduces DGF versus cold storage (Moers, 2009), without changing long-term survival; normothermic perfusion tests viability.
- **Old-for-old** matching: **ESP (Eurotransplant Senior Program)**, donors 65 or over to recipients 65 or over, with short cold ischaemia.
- Recipient: informed consent; induction; delayed CNI (calcineurin inhibitor) if DGF.

Urological relevance

- Older and ECD kidneys have more **urine leak and ureteric stricture** (ischaemic distal ureter), as do multiple arteries and DGF.

Diagrams to draw

- Remuzzi grid with four compartments scored 0–3.
- Maastricht categories on a timeline of circulatory arrest.

> **Key points to remember:**
> - ECD: ≥60, or 50–59 with 2 of hypertension, creatinine >1.5, CVA death.
> - KDPI >85% replaced ECD (2014).
> - Remuzzi: glomeruli, tubules, interstitium, vessels; 0–3 each.
> - 0–3 single, 4–6 dual, ≥7 discard.
> - HMP reduces DGF; Maastricht III is the commonest DCD.

SAQ Renin-angiotensin system

Paper answer · P1

Type: SAQ (10 marks)
Source: 2015-Summer Paper 1, Q3c

Definition

- Hormonal cascade controlling **blood pressure, extracellular volume, Na^+/K^+ balance and GFR (glomerular filtration rate)**.

Cascade

```
Angiotensinogen (liver) ↓ renin (juxtaglomerular cells) — rate-limiting
Angiotensin I ↓ ACE (angiotensin-converting enzyme), mainly lung Angiotensin II
→ AT1 / AT2 receptors ↓ ACE2 Angiotensin (1-7) → Mas receptor (protective)
```

Juxtaglomerular apparatus and renin release

- Granular cells in the afferent arteriole secrete renin and act as baroreceptors; the **macula densa** senses tubular NaCl.

Renin stimulated by	Renin inhibited by
Low renal perfusion pressure (baroreceptor)	High perfusion pressure
Low NaCl at macula densa	High NaCl at macula densa
Sympathetic β1 activity	Angiotensin II (negative feedback)
Prostaglandins, hypovolaemia	Atrial natriuretic peptide, β-blockers, NSAIDs (non-steroidal anti-inflammatory drugs)

Actions of angiotensin II

- **AT1 receptor:** vasoconstriction; **aldosterone** (distal Na^+ retention, K^+ loss); direct proximal Na^+ reabsorption; thirst and ADH (antidiuretic hormone); **fibrosis**.
- **AT2 receptor:** vasodilatation; fetal urinary tract development.
- **Intrarenal:** efferent constriction maintains GFR when perfusion falls, so ACE inhibitors and ARBs (angiotensin receptor blockers) drop GFR in bilateral or solitary-kidney renal artery stenosis.

Urological relevance

- **Renovascular hypertension (Goldblatt kidney):** atherosclerosis (older men, ostial) or fibromuscular dysplasia (young women). Diagnose by duplex ultrasound, computed tomography or magnetic resonance angiography, captopril renography. **Renal vein renin ratio >1.5** with contralateral suppression predicts cure by revascularisation or nephrectomy.
- **Page kidney:** subcapsular haematoma compresses parenchyma; ACE inhibitor/ARB, then drainage.
- **Reninoma (juxtaglomerular cell tumour):** young women, ≤3 cm mass, refractory hypertension, **hypokalaemia**; excision cures most.
- **Obstruction:** hypertension (~77% bilateral, ~20% unilateral); angiotensin II drives fibrosis.
- **Also know:** post-traumatic hypertension (arteriovenous fistula, scarring); reflux nephropathy (renin-guided segmental resection); hyperkalaemia with angiotensin blockade in CKD (chronic kidney disease).
- **Drugs:** ACE inhibitors (cough, angio-oedema from bradykinin), ARBs, aliskiren (direct renin inhibitor), aldosterone antagonists; all contraindicated in pregnancy.

Diagram to draw: Juxtaglomerular apparatus and cascade.

Recent advances

- Renal denervation (US approval 2023) for resistant hypertension.

> **Key points to remember:**
> - Renin is rate-limiting; released by low pressure, low NaCl, β1.
> - Efferent constriction maintains GFR, so ACE inhibitors risk renal failure in bilateral stenosis.
> - Renin ratio >1.5 lateralises; reninoma excision cures most.

SAQ Drugs associated with Acute Renal Failure.

Paper answer · P1

Type: SAQ (10 marks)
Source: 2016-Summer Paper 1, Q3c

Introduction

- AKI (acute kidney injury), KDIGO (Kidney Disease: Improving Global Outcomes): creatinine rise ≥0.3 mg/dL in 48 hours or ≥1.5 × baseline in 7 days, or urine <0.5 mL/kg/h for 6 hours.

Classification by mechanism

Mechanism	Drugs	How
Haemodynamic (prerenal)	**NSAIDs** (non-steroidal anti-inflammatory drugs)	Block prostaglandin afferent dilatation
	ACE (angiotensin-converting enzyme) inhibitors, ARBs (angiotensin receptor blockers)	Remove efferent constriction that holds GFR (glomerular filtration rate)
	Ciclosporin, tacrolimus, contrast; diuretics	Afferent vasoconstriction; volume loss
ATN (acute tubular necrosis)	**Aminoglycosides**, amphotericin B, vancomycin, cisplatin, ifosfamide, contrast	Direct tubular toxicity
AIN (acute interstitial nephritis)	**β-lactams**, sulfonamides, quinolones, rifampicin, NSAIDs, **PPIs (proton pump inhibitors)**, allopurinol, **checkpoint inhibitors**	Cell-mediated hypersensitivity
Crystals	Aciclovir, methotrexate, indinavir, sulfadiazine; tumour lysis	Intratubular obstruction
Microangiopathy	Gemcitabine, VEGF (vascular endothelial growth factor) inhibitors	Endothelial injury
Rhabdomyolysis	Statins (with fibrates)	Myoglobin toxicity, precipitation

- **"Triple whammy":** NSAID + ACE inhibitor/ARB + diuretic in a dehydrated patient.
- **False creatinine rise:** trimethoprim and cimetidine block creatinine secretion.

Recognition

Type	Urine
Prerenal	FeNa (fractional excretion of sodium) <1%, bland
ATN	FeNa >1%, muddy-brown casts
AIN	Sterile pyuria, white cell casts, ± rash, fever, eosinophilia
- AIN appears days to weeks after the drug starts; it is the commonest biopsy lesion with checkpoint inhibitors. Urine eosinophils are unreliable.	

Prevention and management

- Identify risk (CKD (chronic kidney disease), age, diabetes). Dose by eGFR (estimated GFR); monitor drug levels; once-daily aminoglycoside.
- Hydrate before contrast or cisplatin; alkalinise for methotrexate.

- **Stop the culprit**, withhold NSAIDs and ACE inhibitors, restore volume. **Early steroids for AIN** (only about half of antibiotic AIN recovers fully).
- Dialysis for refractory AEIOU: acidosis, electrolytes, intoxication, overload, uraemia.

Diagram to draw: glomerulus with afferent (prostaglandin) and efferent (angiotensin II) control.

> **Key points to remember:**
> - NSAIDs block afferent dilatation; ACE inhibitors/ARBs block efferent constriction.
> - ATN: aminoglycosides, amphotericin, cisplatin, contrast.
> - AIN: β-lactams, PPIs, NSAIDs, checkpoint inhibitors; stop and give steroids.
> - Crystals: aciclovir, methotrexate, indinavir.

SAQ Hypokalemia

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 5 Paper 1, Q3f

Definition and physiology

- **Hypokalaemia:** serum potassium **<3.5 mEq/L**; severe <2.5 mEq/L.
- About **98% of body potassium is intracellular**; the ratio sets the resting membrane potential.
- Two defences: **shift into cells** (insulin, β2-adrenergic stimulation, alkalosis) and **renal excretion** (distal nephron). Distal secretion rises with high urine flow, distal sodium delivery, **aldosterone** and alkalosis.

Causes

Mechanism	Causes
Low intake	Malnutrition
Shift into cells	Insulin, catecholamines (stress, β-agonists), alkalosis, refeeding, periodic paralysis
GI (gastrointestinal) loss	Diarrhoea, villous adenoma, **ureterosigmoidostomy and colonic diversions** (colon secretes potassium, ileum absorbs it)
Renal loss, high BP (blood pressure)	**Primary hyperaldosteronism (Conn)**, renin-secreting juxtaglomerular tumour, Cushing, Liddle, **abiraterone** (mineralocorticoid excess)
Renal loss, normal BP	Loop and thiazide diuretics, **postobstructive diuresis**, osmotic diuresis, **RTA (renal tubular acidosis) types 1 and 2**, Bartter, Gitelman, hypomagnesaemia, amphotericin, cisplatin

- **Vomiting:** potassium is lost mainly in urine (alkalosis → kaliuresis), not in gastric juice.

Clinical features

- Muscle weakness, cramps, ileus, **flaccid paralysis**, rhabdomyolysis.
- Arrhythmias; potentiates digoxin toxicity.
- Polyuria (nephrogenic diabetes insipidus); **hypocitraturia** → calcium stones.
- **ECG (electrocardiogram):** flat T wave, **prominent U wave**, ST depression, prolonged QU; ventricular arrhythmias.

Evaluation

- Repeat potassium; magnesium, bicarbonate, glucose.
- **24-h urine potassium:** <20 mmol/day suggests extrarenal loss; higher suggests renal loss.
- Renal loss: check blood pressure and acid–base; **aldosterone–renin ratio** if hypertensive (only a minority of Conn patients are hypokalaemic).

Management

- Treat the cause; stop offending drugs.
- **Potassium chloride** preferred (potassium citrate if acidotic, phosphate if hypophosphataemic). **Oral** 20–40 mEq repeated when possible.
- **Intravenous** for severe or symptomatic cases: usually ≤10 mEq/h peripherally; faster only centrally with ECG monitoring. Never as a bolus.
- **Correct magnesium**, or potassium will not correct.
- **Urinary diversion:** replace potassium **before or with bicarbonate**; correcting acidosis alone drives potassium into cells → paralysis.
- Correct before abiraterone and adrenalectomy for Conn syndrome (spironolactone preoperatively).

Diagram to draw: ECG strip showing flat T and U wave.

> **Key points to remember:**
> - Shift or loss; urine potassium separates renal from extrarenal.
> - ECG: flat T, U wave.
> - Diversion acidosis: give potassium with or before bicarbonate.
> - Correct magnesium first.

SAQ Hyperfiltration injury in AKI and CKD

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 14 Paper 1, Q3c

Definition

- **Hyperfiltration:** a rise in **SNGFR (single-nephron glomerular filtration rate)** in surviving nephrons when nephron mass falls; whole-kidney GFR (glomerular filtration rate) may look normal.
- **Brenner hypothesis (1982):** the adaptation itself damages glomeruli.

Physiology

- Net filtration pressure ≈ 10 mmHg (hydrostatic 60 − Bowman's 18 − oncotic 32).
- Afferent dilatation raises flow and filtration; **efferent constriction (angiotensin II) raises filtration pressure**.

Mechanism

Nephron loss → afferent dilatation + efferent constriction → raised glomerular capillary pressure and flow (hyperfiltration) → glomerular hypertrophy; podocytes cannot divide → stretch, detachment → proteinuria → FSGS (focal

segmental glomerulosclerosis) → tubulointerstitial inflammation, fibrosis → more nephron loss (vicious cycle)

In AKI (acute kidney injury)

- Incomplete repair leaves fewer nephrons, which hyperfilter; creatinine may normalise while **renal reserve is lost**.
- This drives the **AKI-to-CKD (chronic kidney disease) transition**; AKI persisting ≥90 days becomes CKD.
- Each perioperative AKI episode is a **modifiable risk factor** for CKD progression.

In CKD

- Final common pathway of progression; marked by **albuminuria (KDIGO (Kidney Disease: Improving Global Outcomes) A stages)**, hypertension, falling eGFR (estimated GFR).

Urological settings

Setting	Point
Nephrectomy or donation	Compensatory hypertrophy 20–30%; donor function 60–80% of pre-donation
Partial nephrectomy, solitary kidney	>50% parenchymal loss → proteinuria, FSGS; marked when total nephron mass falls >80%; ESRD (end-stage renal disease) if <20–30% of one kidney remains
Congenital solitary kidney, MCDK (multicystic dysplastic kidney)	Monitor BP (blood pressure), proteinuria, yearly ultrasound

- **Also know:** reflux nephropathy, infundibulopelvic stenosis, diabetes, obesity, pregnancy, low birth weight (low nephron endowment).

Prevention and treatment

- Nephron sparing; avoid NSAIDs (non-steroidal anti-inflammatory drugs) and nephrotoxins.
- **BP <130/80 mmHg**; **ACE (angiotensin-converting enzyme) inhibitor or ARB (angiotensin receptor blocker)** in proteinuric disease: efferent dilatation lowers glomerular pressure.
- **SGLT2 (sodium–glucose cotransporter-2) inhibitors:** restore tubuloglomerular feedback, causing afferent constriction. **DAPA-CKD** and **EMPA-KIDNEY** slowed progression with or without diabetes.
- Finerenone (type 2 diabetic CKD); dietary protein restriction.

Diagram to draw: glomerulus with dilated afferent and constricted efferent arterioles.

> **Key points to remember:**
> - Hyperfiltration = raised SNGFR; Brenner 1982.
> - Leads to FSGS, proteinuria and the AKI-to-CKD transition.
> - ACE inhibitors and SGLT2 inhibitors lower glomerular pressure.

SAQ UW solution and organ preservation solutions

Paper answer · P2

Asked as:

- UW solution. — *Summer 2017 Paper II Q3c (SAQ)*

- Organ preservation solution — *Practice Set 7 Paper II Q3b (SAQ)*

Type: SAQ (10 marks)

Assumed context: preservation of the kidney for transplantation.

Introduction

- **University of Wisconsin solution** (Belzer, late 1980s): the **gold standard** cold-storage solution for kidney, liver and pancreas.

Principles of cold preservation

- Cooling / **Hypothermia** to **4 °C** cuts (slows) metabolism to roughly a tenth, but does not stop it, so injury accumulates with time.
- Ischaemia causes ATP (adenosine triphosphate) loss (depletion) → **Na^+/K^+-ATPase pump failure** → **cell swelling**; anaerobic glycolysis → **acidosis**; calcium influx. On reperfusion, hypoxanthine is converted by xanthine oxidase to **oxygen free radicals** → ischaemia–reperfusion injury, seen clinically as (clinically this appears as) **DGF (delayed graft function)**.
- A solution must therefore prevent swelling and oedema, buffer acid, scavenge radicals and supply ATP precursors. It therefore needs:
- **Impermeants** (lactobionate, raffinose, mannitol, glucose) to prevent cell swelling.
- **Buffers** (phosphate, histidine) against acidosis.
- **Colloid** (hydroxyethyl starch, polyethylene glycol) against interstitial oedema.
- **Antioxidants and energy precursors** (glutathione, allopurinol, adenosine).
- Electrolytes: "intracellular" (high K^+, low Na^+) or "extracellular" (low K^+, high Na^+) type.

UW composition (per litre)

Component	Amount	Purpose
Lactobionate (K^+ salt)	100 mmol	Impermeant anion; prevents cell swelling; chelates calcium
Raffinose	30 mmol	Impermeant trisaccharide
Hydroxyethyl starch	50 g	Colloid; prevents interstitial oedema
KH_2PO_4	25 mmol	Buffer
$MgSO_4$	5 mmol	Membrane stability
Adenosine	5 mmol	ATP precursor
Glutathione	3 mmol	Antioxidant
Allopurinol	1 mmol	Xanthine oxidase inhibitor; fewer free radicals
Insulin, dexamethasone, antibiotic	Small	Additives
- **Intracellular-type**: K^+ about 125 mmol/L, Na^+ about 30 mmol/L; osmolality ~320 mOsm/kg; pH 7.4.		

UW drawbacks

- **Viscous** (starch): slow flush. **High K^+**: flush before reperfusion (hyperkalaemia, vasospasm). May crystallise (filter).

Comparison of solutions

Feature	UW	HTK (Custodiol)	Euro-Collins
Type	Intracellular (high K^+)	Low K^+, low Na^+	Intracellular
Impermeant	Lactobionate, raffinose	Mannitol	Glucose (enters cells; weaker)
Buffer	Phosphate	**Histidine** (strong)	Phosphate
Colloid	HES (hydroxyethyl starch)	None (low viscosity)	None
Others	Antioxidants, adenosine	Tryptophan, ketoglutarate	—

Solution	Type and key components	Points
UW (University of Wisconsin, Belzer)	Intracellular; lactobionate, raffinose, hydroxyethyl starch, glutathione, adenosine, allopurinol	Standard for static storage; viscous; flush out high K^+ before reperfusion
HTK (histidine–tryptophan–ketoglutarate, Custodiol)	Low K^+; histidine buffer, tryptophan (membrane stabiliser), ketoglutarate (substrate), mannitol	Low viscosity, fast cooling; large volume; outcomes similar to UW
Euro-Collins	Intracellular; glucose impermeant, phosphate	First generation; largely abandoned
Celsior	Extracellular; lactobionate, mannitol, glutathione, histidine	Designed for heart; also abdominal organs
IGL-1 (Institut Georges Lopez)	UW-like, but polyethylene glycol and reversed Na^+/K^+	Extracellular; less viscous
- Kidney outcomes with UW and HTK (histidine–tryptophan–ketoglutarate) are similar; Euro-Collins largely abandoned. **Also know:** Celsior, IGL-1, Marshall's (hypertonic) citrate.		

Practical use and methods

- Living donor, very short cold ischaemia: cold lactated Ringer's suffices; deceased donor or exchange: UW or HTK.
- **Static cold storage:** in situ aortic flush, IVC (inferior vena cava) venting, ice slush; storage at 4 °C. Keep (aim) cold ischaemia short (ideally <24 h / hours): DGF and thrombosis rise with time (beyond).
- **Hypothermic machine perfusion** (Belzer MPS (machine perfusion solution), gluconate-based): pulsatile cold perfusion; flow and resistance help assess viability. Reduces DGF vs static storage; **Moers (2009)**: DGF 20.8% vs 26.5%, 1-year graft survival 94% vs 90%, without long-term survival gain (the chapter notes no difference in long-term survival). Used especially for DCD (donation after circulatory death) kidneys.
- **Normothermic perfusion:** experimental; allows assessment and repair; limited experience.

Diagram to draw: ischaemia(–reperfusion) cascade with the UW / solution component acting at (that blocks) each step.

Key points to remember:

- Hypothermia slows, not stops, metabolism.
- Lactobionate and raffinose (impermeants), HES / starch (colloid).
- Allopurinol, glutathione (antioxidants); adenosine (ATP).
- High K^+: flush before reperfusion.
- HTK: histidine buffer, tryptophan, ketoglutarate, low viscosity, low K^+.
- Machine perfusion reduces DGF.

SAQ Post transplant malignancy

Paper answer · P2

Type: SAQ (10 marks)
Source: 2023-Winter Paper 2, Q3c

Introduction

- Cancer risk after kidney transplantation is about 2–3 times the general population; a leading cause of late death with a functioning graft.

Causes

- **Lost immune surveillance** from immunosuppression (dose and duration matter; depleting antibodies add risk).
- **Oncogenic viruses:** EBV (Epstein–Barr virus) → PTLD (post-transplant lymphoproliferative disorder); HHV-8 (human herpesvirus-8) → Kaposi sarcoma; HPV (human papillomavirus) → skin and anogenital cancer; hepatitis B/C → liver cancer.
- **Direct drug effects:** azathioprine sensitises skin to UV (ultraviolet) light; calcineurin inhibitors promote tumour growth. mTOR (mammalian target of rapamycin) inhibitors are **antitumour**.
- **Pre-existing risk:** ESRD (end-stage renal disease), acquired cystic disease, analgesic nephropathy; rarely donor-transmitted.

Common cancers

Cancer	Key facts
Non-melanoma skin	Commonest; squamous exceeds basal cell (reverse of general population); multiple, aggressive
PTLD	Mostly B-cell, EBV-driven; highest in first year, children and **EBV-negative recipient of EBV-positive donor**; may present as graft mass
Kaposi sarcoma	HHV-8; regresses with less immunosuppression and switch to sirolimus
RCC (renal cell carcinoma)	4.6% of post-transplant cancers; **90% in native kidneys**; graft tumours mostly papillary, low grade
Bladder cancer	About **3-fold** risk; more aggressive
Prostate	Same incidence as general population

Management

- **Reduce immunosuppression** and switch to an mTOR inhibitor; sirolimus conversion reduced new skin squamous cancers (Euvrard, 2012).
- **PTLD:** reduce immunosuppression, rituximab, then chemotherapy (R-CHOP).

- RCC, prostate and bladder: treat as in the general population; protect graft vessels and ureter at surgery. BCG (bacillus Calmette–Guérin) is usable in selected bladder cancer.

Prevention and screening

- Pre-transplant cancer screening and waiting periods (none after curative treatment of small RCC <3 cm, Gleason 6 prostate cancer, superficial bladder cancer).
- Sun protection, yearly skin examination, age-appropriate screening.
- KDIGO (Kidney Disease: Improving Global Outcomes) does **not** endorse routine native-kidney screening; screen only high-risk recipients.

Diagram to draw: pie chart of post-transplant cancers (skin largest).

> **Key points to remember:**
> - Skin cancer commonest; squamous more than basal cell.
> - PTLD: EBV-negative recipient of EBV-positive donor, first year.
> - RCC mostly in native kidneys; bladder cancer 3-fold and aggressive.
> - Reduce immunosuppression and switch to mTOR inhibitor.

SAQ Mycophenolate Mofetil.

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 2 Paper 2, Q3b (immunosuppressants also answered in 2021-Summer Paper 1, Q3f and 2020-Summer Paper 4, Q1)

Introduction

- **MMF (mycophenolate mofetil):** an antimetabolite immunosuppressant; morpholinoethyl ester **prodrug** of MPA (mycophenolic acid). With tacrolimus it forms maintenance therapy in **over 90%** of kidney recipients.

Mechanism

- MPA is a reversible, non-competitive inhibitor of **IMPDH (inosine monophosphate dehydrogenase)**, the key enzyme of **de novo guanosine nucleotide synthesis**.
- **Why lymphocyte-selective:** T and B lymphocytes depend on the de novo pathway; other cells use the salvage pathway. MPA also inhibits the type II isoform found in activated lymphocytes.
- Result: arrest of T- and B-cell proliferation and reduced antibody formation.

Pharmacology

Feature	Detail
Dose (kidney)	**1 g twice daily** oral or IV; EC-MPS (enteric-coated mycophenolate sodium) 720 mg twice daily is equivalent
Metabolism	Hydrolysed to MPA; liver glucuronidation to inactive MPAG (MPA glucuronide); **enterohepatic recirculation**
Excretion	MPAG in urine; accumulates in renal failure
Interactions	**Ciclosporin lowers MPA** (blocks enterohepatic recirculation), tacrolimus does not; antacids, cholestyramine reduce absorption; never with azathioprine

Uses

- Kidney transplant maintenance (tacrolimus + MMF ± steroids); other solid organs.
- Lupus nephritis, vasculitis; steroid-refractory immune-related adverse events of checkpoint inhibitors.

Adverse effects

- **GI (gastrointestinal):** diarrhoea, nausea, colitis (commonest; dose split or EC-MPS).
- **Marrow:** leucopenia, anaemia, thrombocytopenia: blood count weekly in month 1.
- **Infection:** CMV, BK virus (ureteric stricture), rarely PML (progressive multifocal leucoencephalopathy).
- **Malignancy:** skin cancer, PTLD (post-transplant lymphoproliferative disorder).
- **Teratogenic:** miscarriage, ear and facial malformations; contraception mandatory; switch to azathioprine before conception.
- Advantage: **not nephrotoxic**; no hypertension, diabetes or dyslipidaemia.

Evidence

- Three mid-1990s randomised trials (US, European, Tricontinental) showed fewer early acute rejections than azathioprine or placebo, so MMF replaced azathioprine.
- **ELITE-Symphony (2007):** low-dose tacrolimus + MMF + daclizumab + steroids gave the lowest rejection and best GFR.

Diagram to draw: purine synthesis pathway showing IMPDH block (de novo) and the salvage pathway spared.

> **Key points to remember:**
> - Prodrug of MPA; IMPDH inhibitor; lymphocytes need de novo purine synthesis.
> - 1 g twice daily; ciclosporin lowers MPA levels.
> - Toxicity: diarrhoea, leucopenia, CMV/BK, teratogenic; not nephrotoxic.

SAQ Renal transplant in neurogenic bladder

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 4 Paper 2, Q3a

Why the bladder matters

- A neurogenic bladder with **poor compliance, low capacity or detrusor overactivity** transmits high pressure to the graft even without reflux. Poor emptying adds infection. Recurrent pyelonephritis, not augmentation, causes graft loss.
- **Five goals:** unimpeded drainage, low-pressure capacious storage, effective emptying, no infection, fewest procedures.

Pretransplant evaluation

- Renal ultrasound, VCUG (voiding cystourethrogram) and **urodynamics** (capacity, compliance, overactivity, emptying, sphincter). Repeat them; the bladder changes over time.
- **Target:** storage to expected capacity at detrusor pressure **<40 cm H_2O**.

- **Defunctionalised bladder** (anuria or diversion): **bladder cycling** (saline fills of increasing volume via catheter), then urodynamics.

Pretransplant optimisation (stepwise)

`Antimuscarinic (oxybutynin), mirabegron, botulinum toxin → CIC (clean intermittent catheterisation) ± overnight drainage → bladder cycling if defunctionalised → still >40 cm H₂O, <75% expected capacity or CIC ≤3-hourly → augmentation`

- **Augmentation:** ileum or sigmoid most used; dilated ureter (ureterocystoplasty) preferred when available (no mucus or metabolic effects). **Stomach avoided** (haematuria–dysuria syndrome in anuric patients). Augmentation does not clearly raise transplant complications.
- **Mitrofanoff** channel if urethral CIC is difficult; site it **medial** to the planned transplant incision.
- Do reconstruction **before** transplant; allow **6–12 weeks** to heal. Neuropathic bladders may deteriorate under high graft output, so earlier reconstruction is justified.
- **Native nephrectomy** only for named reasons: recurrent upper-tract infection, massive reflux, stones, uncontrolled hypertension.

At transplant

- Extravesical non-refluxing ureteroneocystostomy into bladder or augment (know the augment pedicle). Ureteroureterostomy to a native ureter is an option.
- Ileal conduit drainage if the bladder is unusable.

Post-transplant care

- Continue CIC; colonisation is expected, so treat only symptomatic UTI (urinary tract infection).
- **Creatinine rise with hydronephrosis:** catheter drainage for 1–2 weeks first. If it improves, the bladder is the cause; otherwise stent or nephrostomy and consider rejection.
- Febrile UTI with graft reflux: reimplant after optimising the bladder.
- Lifelong surveillance of augments (malignancy, stones).

Diagram to draw: stepwise bladder optimisation ladder.

> **Key points to remember:**
> - Target <40 cm H_2O storage before transplant.
> - Drugs → CIC → cycling → augmentation; reconstruct 6–12 weeks before.
> - Pyelonephritis, not augmentation, loses grafts.

SAQ Pre-emptive renal transplant

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 6 Paper 2, Q3a

Definition

- Kidney transplantation **before starting maintenance dialysis** in a patient with advanced CKD (chronic kidney disease).

Rationale

- Every year on dialysis adds cardiovascular disease, vascular calcification and bone–mineral damage that the graft later inherits. Pre-emptive transplantation **avoids this exposure**.
- Transplantation beats dialysis: Wolfe (1999) showed early risk is higher, equal at 106 days, then 68% lower long term.

Advantages

- Better patient and graft survival than transplantation after dialysis; longer dialysis time worsens outcome (registry data).
- No dialysis access surgery, access infection or catheter complications.
- Fewer transfusions, so less HLA (human leucocyte antigen) sensitisation.
- Better quality of life, employment; lower cost.
- Children: better growth and development.

Frequency

- 28.7% of US transplants in 2021 (living donor 51%, deceased donor 21%, children 56%). Commonest with a **living donor**.

Timing and evaluation

- Evaluate early, ideally before dialysis. KDIGO (Kidney Disease: Improving Global Outcomes) 2020: refer when eGFR (estimated glomerular filtration rate) is <30 mL/min/1.73 m², 6–12 months before expected dialysis. Living-donor transplant is usually done at eGFR <20 with progressive decline.
- Deceased-donor listing (US) at eGFR ≤20.
- Full recipient work-up: cardiac, vascular (iliac calcification), infection (TB [tuberculosis], hepatitis B and C, HIV [human immunodeficiency virus]), malignancy, psychosocial; recurrence risk (primary FSGS [focal segmental glomerulosclerosis]).
- **Urological:** bladder function and lower tract (valves, neurogenic bladder); native nephrectomy only for specific indications.
- India: mainly living related donors under THOA (Transplantation of Human Organs Act).

Limitations

- Needs early referral and a ready donor.
- Transplant may come earlier than needed, using up dialysis-free native function; adherence untested.
- In advanced CKD, bowel augmentation may speed decline.

Also know: ABO-incompatible or paired exchange programmes widen pre-emptive options.

Diagram to draw: Wolfe curve: relative risk of death after transplant vs dialysis over time.

> **Key points to remember:**
> - Transplant before dialysis = best outcome.
> - Refer at eGFR <30; transplant usually at eGFR <20.
> - Commonest with living donors and children.

SAQ Tacrolimus.

Paper answer · P3

Type: SAQ (10 marks)
Source: 2017-Winter Paper 3, Q3c

Introduction

- **Tacrolimus (FK506)**: macrolide CNI (calcineurin inhibitor) from *Streptomyces tsukubaensis*; about 10–100 times more potent than ciclosporin. With mycophenolate it is maintenance in **over 90%** of kidney recipients.

Mechanism

`Tacrolimus + FKBP12 (FK506-binding protein 12) → complex inhibits calcineurin (Ca²⁺/calmodulin phosphatase) → NFAT (nuclear factor of activated T cells) stays phosphorylated, cannot enter nucleus → no IL-2 (interleukin-2) transcription → T-cell activation blocked (signal 1)`

- Ciclosporin binds **cyclophilin** instead; same end result.

Pharmacokinetics

- Poor, variable oral bioavailability; reduced by food. Binds red cells, so **whole-blood trough (C0)** levels are monitored (narrow therapeutic index).
- Metabolised by **CYP3A4/3A5**; effluxed by **P-gp (P-glycoprotein)**. CYP3A5 expressers need higher doses.

Formulations

- Immediate-release (twice daily); prolonged-release (Advagraf, once daily); LCP-tacrolimus (Envarsus, once daily, higher bioavailability); intravenous; topical. Once-daily forms aid adherence.

Dosing and monitoring

- Started about 0.1–0.2 mg/kg/day in two doses, then adjusted to trough. Targets vary by centre and risk; **ELITE-Symphony low-dose arm: 3–7 ng/mL**.

Toxicity

System	Effect
Kidney	**Nephrotoxicity**: acute afferent vasoconstriction (reversible); chronic arteriolar hyalinosis, striped fibrosis; TMA (thrombotic microangiopathy)
Metabolic	**NODAT/PTDM (new-onset/post-transplant diabetes)**, more than ciclosporin; **hyperkalaemia**, **hypomagnesaemia**, hyperuricaemia
Neurological	**Tremor**, headache, paraesthesia, seizures, PRES (posterior reversible encephalopathy syndrome)
Others	Hypertension, GI (gastrointestinal) upset, alopecia, infection (CMV (cytomegalovirus), BK polyomavirus), malignancy (skin, PTLD (post-transplant lymphoproliferative disorder))

Drug interactions

- **Raise levels (CYP3A4 inhibitors):** azoles, clarithromycin/erythromycin, diltiazem, verapamil, protease inhibitors, grapefruit; diarrhoea.
- **Lower levels (inducers):** rifampicin, phenytoin, carbamazepine, St John's wort.
- Additive nephrotoxicity: NSAIDs (non-steroidal anti-inflammatory drugs), aminoglycosides; ACE (angiotensin-converting enzyme) inhibitors add to hyperkalaemia.

Versus ciclosporin

	Tacrolimus	Ciclosporin
Acute rejection	Lower	Higher
Diabetes, tremor, alopecia	More	Less
Hirsutism, gum hypertrophy, hyperlipidaemia, hypertension	Less	More
- **ELITE-Symphony (2007):** low-dose tacrolimus + mycophenolate + daclizumab + steroids gave the **lowest rejection, best GFR (glomerular filtration rate) and graft survival** at 1 year versus ciclosporin or sirolimus regimens.		
- **Also know:** CNI minimisation with belatacept or everolimus; topical tacrolimus in genital dermatoses (lichen planus, Zoon balanitis).		

Diagram to draw: T-cell showing FKBP12–calcineurin–NFAT–IL-2 pathway.

> **Key points to remember:**
> - FKBP12 → calcineurin → NFAT → IL-2 blocked.
> - CYP3A4/P-gp; trough-guided; many interactions.
> - Nephrotoxicity, NODAT, tremor, hyperkalaemia, hypomagnesaemia.
> - ELITE-Symphony: low-dose tacrolimus best.

SAQ Urological Complications post Kidney Transplantation.

Paper answer · P3

Type: SAQ (10 marks)
Source: 2023-Winter Paper 3, Q3d

Introduction

- Urological complications are the **commonest surgical complications** (8–9.5%). Most arise from the **ureter and ureterovesical anastomosis**.
- The distal donor ureter depends on a single renal-artery blood supply, so ischaemia is the root cause. **Denervation** masks pain.
- A **routine stent** reduces leak and stricture; remove at **14–21 days** (earlier increases leak; later increases UTI [urinary tract infection]).

Complications

Complication	Key facts	Management
Urine leak (1.2–8.9%; commonest early)	Distal ureteric necrosis; lost lower-pole artery; technical	Fluid creatinine far above serum. Foley + PCN (percutaneous nephrostomy) + antegrade stent; early reimplant or native ureteroureterostomy if large
Ureteric stricture (0.5–9%; commonest beyond 3 months)	Ischaemia; long cold ischaemia, rejection, BK (BK polyomavirus) and CMV (cytomegalovirus)	Nephrostomy first; balloon or laser endoureterotomy; **open reimplant, native ureteroureterostomy or Boari flap best long-term**
Lymphocele (symptomatic 0.03–26%; peak 6 weeks)	Divided lymphatics; mTOR (mammalian target of rapamycin) inhibitors	Treat only if symptomatic; **laparoscopic peritoneal fenestration** best

Complication	Key facts	Management
VUR (vesicoureteric reflux)	50–86% radiological; <1% significant	Treat cause; endoscopic injection low grade; reimplant
Haematuria	Anastomotic; persistent after biopsy = AV fistula	Self-limiting; selective embolisation
Stones (about 1%)	Hyperparathyroidism; calcineurin inhibitors	As native kidney; watch silent obstruction
UTI (commonest infection)	Stent, reflux, steroids	Treat 14–21 days if pyelonephritis

- **Obstruction:** early is technical, oedema or compression (haematoma, lymphocele); late is stricture, stone or lymphocele.
- **Encrusted stent:** CT, nephrostomy, then PCNL (percutaneous nephrolithotomy), RIRS (retrograde intrarenal surgery) or ESWL (shock-wave lithotripsy).
- **Lower tract:** bladder capacity recovers; BPH (benign prostatic hyperplasia) treated by TURP (transurethral resection of prostate) after 3 weeks, never with a stent in place.
- **Also know:** ED (erectile dysfunction) persists in 20–50%; bladder cancer risk about threefold; RCC (renal cell carcinoma) mostly in native kidneys; renal artery stenosis and thrombosis.

Diagram to draw: transplant kidney in iliac fossa showing leak, stricture and lymphocele sites.

> **Key points to remember:**
> - Ureter is the weak point; ischaemia is the cause.
> - Leak early, stricture late; open repair best.
> - Stent out at 2–3 weeks; laparoscopic fenestration for lymphocele.

SAQ Female kidney donors: special precautions

Paper answer · P4

Asked as:

- Female kidney donors: enumerate special precautions — *Winter 2023 Paper IV Q3d (SAQ)*
- Female kidney donors: enumerate special precautions — *Practice Set 4 Paper IV Q3a (earlier version) (SAQ)*

Type: SAQ (10 marks)
(also answered in 2023-Winter Paper 4, Q3d)

Introduction

- Most living donors, especially in India, are **women** (wives, mothers). Guidance: **KDIGO (Kidney Disease: Improving Global Outcomes) 2017** living donor guideline.

Before donation

1. **Free, uncoerced consent:** interview the donor alone; independent donor advocate; family pressure and gender inequity screened. In India the THOA (Transplantation of Human Organs Act) authorisation committee verifies this.
2. **Pregnancy:** pregnancy test before contrast CT (computed tomography) and surgery; **reliable contraception** during work-up. Pregnant women do not donate.

3. **Obstetric history:** previous **pre-eclampsia, gestational hypertension or gestational diabetes** signals higher future hypertension, diabetes and kidney risk; evaluate and counsel.
4. **Future pregnancy counselling (KDIGO):** donation **increases gestational hypertension and pre-eclampsia** (Garg et al., NEJM 2015: about 11% vs 5% in non-donors). Maternal and fetal outcomes are otherwise similar (Chapter 91).
5. **Oestrogen therapy:** combined OCPs (oral contraceptive pills) and HRT (hormone replacement therapy) raise **VTE (venous thromboembolism)** risk; stop them some weeks before surgery and give barrier or progestogen-only contraception; VTE prophylaxis.
6. **Gynaecological and cancer screening:** cervical cytology, breast examination or mammography (age-appropriate); pelvic ultrasound if indicated.
7. **Urine tests:** repeat urinalysis **outside menstruation** to avoid false haematuria; exclude recurrent UTI (urinary tract infection).
8. **GFR (glomerular filtration rate):** low muscle mass makes creatinine-based estimates unreliable; **measured GFR** preferred. Note nephron mismatch with a large recipient.
9. Correct **iron-deficiency anaemia**.

Operative

- Laparoscopic donor nephrectomy; **Pfannenstiel** extraction for cosmesis; transvaginal extraction possible.

After donation

- **Defer pregnancy** until recovery and stable function; many programmes advise at least 6–12 months (another source: often 6–12 months).
- In pregnancy: inform obstetrician; **high-risk antenatal care**; monitor BP (blood pressure), proteinuria and creatinine; low-dose aspirin may be considered.
- **Lifelong annual BP, creatinine and proteinuria**; avoid NSAIDs (non-steroidal anti-inflammatory drugs).

Diagram to draw: flowchart: consent → pregnancy test → hormone review → GFR → surgery → pregnancy follow-up.

> **Key points to remember:**
> - Ensure free consent (Indian gender gap).
> - Pregnancy test; contraception; stop oestrogens.
> - Donors: gestational hypertension or pre-eclampsia about 11% vs 5%.
> - Defer pregnancy; high-risk antenatal follow-up.

SAQ Laparoscopic donor nephrectomy (including pros and cons)

Paper answer · P4

Asked as:

- Laparoscopic donor nephrectomy — *Practice Set 2 Paper IV Q3a (SAQ)*
- Pros and cons of lap donor nephrectomy — *Practice Set 2 Paper IV Q3d (SAQ)*

Type: SAQ (10 marks)
(living donation also in 2023-Winter Paper 1, Q3c; the pros and cons are asked again in Practice Set 2 Paper 4, Q3d) **See Practice Set 2 Paper 4, Q3a (full answer).**

Introduction

- **Ratner (1995)**; now the preferred approach (open under 2% in the United States). **Donor safety comes first.**
- Work-up includes measured GFR (glomerular filtration rate) and CT (computed tomography) angiography.

Side selection

- **Left preferred:** longer renal vein, simpler dissection.
- **Right** if the left has multiple arteries or complex veins, or the right is smaller or abnormal: the donor keeps the better kidney. The right vein is short.

Techniques

Technique	Features
Pure transperitoneal laparoscopic	3–4 ports; colon reflected; Pfannenstiel extraction
Hand-assisted (about two-thirds of US cases)	Hand port is also the extraction site; tactile feedback, quick haemostasis
Retroperitoneoscopic	Balloon-made space; no bowel mobilisation; direct hilum; less room
Robotic; LESS (laparoendoscopic single-site surgery)	Wristed dissection; single umbilical incision; cost

Key steps

1. Reflect colon; divide gonadal, adrenal and lumbar veins (left).
2. Keep periureteric tissue and the gonadal–ureter triangle (ureteric blood supply).
3. Hydration and diuretic before vessel division.
4. Divide ureter, artery, then vein (stapler; see Q3f on Hem-o-lok).
5. Extract quickly; cold flush.

Warm ischaemia

- Arterial occlusion to cold flush: a few minutes, slightly longer than open, without effect on long-term graft function.

Pros and cons (versus open)

Pros	Cons
Less pain, shorter stay, faster (earlier) return to work	Longer operation; learning curve
Better cosmesis; **more people volunteer to donate** (more willing donors)	Slightly longer warm ischaemia
Magnified view of hilum	Shorter vessels, especially right vein (on the right)
Equal graft function and survival (graft function equals open)	Pneumoperitoneum lowers renal blood flow; cost

- **Risks (safety):** complications about 16% (mostly gastrointestinal); mortality **3 per 10,000**; conversion 0.26–3.34%. Lifetime ESRD (end-stage renal disease) 90 per 10,000 versus 14 in matched non-donors.

Diagram to draw: left hilum with gonadal, adrenal and lumbar veins and the stapled artery and vein.

> **Key points to remember (including pros and cons angle):**
> - Ratner 1995; left kidney preferred (longer vein).
> - Hand-assisted in about two-thirds of US cases.
> - Pros: less pain, shorter stay, earlier return to work, better cosmesis, and more willing donors; graft function equals open.
> - Cons: longer operation, learning curve, slightly longer warm ischaemia, shorter vessels (especially on the right), cost.
> - Safety: mortality 3 per 10,000; conversion 0.26–3.34%.
> - Never rely on Hem-o-lok alone on the donor renal artery (do not use Hem-o-lok clips alone).

SAQ Artificial kidney

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 6 Paper 4, Q3e

Read as devices that replace kidney function: the dialyser (classic "artificial kidney") and newer wearable, implantable and bioartificial kidneys.

Definition

- An **artificial kidney** removes solutes and water from blood across a semipermeable membrane. The haemodialysis dialyser is the standard example.
- Pioneered by **Kolff (1943)**; long-term use followed the Scribner shunt (1960) and Brescia–Cimino AV (arteriovenous) fistula (1966).

Principles

Process	Mechanism	Removes
Diffusion	Concentration gradient across membrane	Small solutes (urea, potassium)
Convection (haemofiltration)	Solute dragged with water	Middle molecules
Ultrafiltration	Hydrostatic pressure gradient	Water
Adsorption	Binding to membrane or sorbent	Some proteins, toxins

- **Hollow-fibre dialyser**; blood and dialysate flow **countercurrent** to keep the gradient.
- **Schedule:** blood flow 250–350 mL/min, 3–5 hours, 3–5 times a week; adequacy by Kt/V (urea clearance × time ÷ volume), at least 1.2 per session.
- **Peritoneal dialysis** uses the peritoneum as membrane (glucose or icodextrin draws water).

Limitations of current dialysis

- Intermittent: fluid and solute swings cause cramps.
- No endocrine (erythropoietin, vitamin D activation) or tubular metabolic function.
- Access thrombosis, infection, steal; cardiovascular disease, vascular calcification.

- **Transplantation remains the best renal replacement therapy.**

Newer artificial kidneys

Device	Principle	Status
WAK (wearable artificial kidney)	Belt-worn; sorbent regenerates a small volume of dialysate	Early human trials
Wearable or portable peritoneal dialysis	Continuous sorbent-regenerated peritoneal fluid	Development
Implantable bioartificial kidney	Silicon nanopore haemofilter plus bioreactor of renal tubule cells; driven by blood pressure, no pump	Preclinical
RAD (renal assist device)	Human proximal tubule cells in a hollow-fibre cartridge with continuous dialysis	Phase II in acute kidney injury; not adopted
Decellularised or organoid kidneys; pig kidney xenotransplant	Regenerative and xenograft routes	Experimental (see Practice Set 6 Paper 4, Q2)

Urologist's role

- AV fistula and graft work-up, laparoscopic PD (peritoneal dialysis) catheter placement with omental fixation, transplant.

Diagram to draw: hollow-fibre dialyser with countercurrent blood and dialysate flow.

> **Key points to remember:**
> - Diffusion, convection, ultrafiltration across a semipermeable membrane.
> - Dialysis lacks endocrine and tubular function; transplant is best.
> - Wearable, implantable bioartificial and RAD devices are investigational.

SAQ Robotic kidney transplantation

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 6 Paper 4, Q3f

Introduction

- **RAKT (robot-assisted kidney transplantation)**: the graft is implanted intracorporeally on the iliac vessels using the robot.
- Milestones: Hoznek (2002, robot-assisted anastomosis); Giulianotti (2010, obese recipient); **Menon and Ahlawat (Vattikuti–Medanta, 2014)** standardised the transperitoneal technique with **regional hypothermia**, assessed by the IDEAL (idea, development, exploration, assessment, long-term study) framework.

Indications and selection

- Living-donor grafts mainly; increasingly deceased donors.
- **Obese recipients** gain most, as open transplant carries more wound complications.
- **Contraindications (relative):** heavy iliac calcification on non-contrast CT (computed tomography), multiple graft vessels early in experience, previous transplant on that side, dense adhesions.

Technique (Vattikuti–Medanta)

1. Lithotomy, steep Trendelenburg; transperitoneal ports.
2. Periumbilical **GelPOINT** incision for graft and ice insertion.
3. Peritoneal flap; expose the **external iliac vessels**.
4. Graft wrapped in gauze with ice slush; **ice slush in the iliac fossa** for regional hypothermia.
5. **Vein first:** renal vein end-to-side to external iliac vein, to shorten arterial ischaemia.
6. Renal artery end-to-side to external iliac artery; unclamp and reperfuse.
7. Extravesical **Lich–Gregoir** ureteroneocystostomy over a double-J stent.
8. Graft retroperitonealised with the peritoneal flap to prevent torsion.

Advantages

- Small incision: **fewer wound infections, less lymphocele**, less pain, faster recovery, better cosmesis.
- Magnified 3D (three-dimensional) view and precise suturing in deep pelvis or obese patients.

Disadvantages

- **Longer rewarming (anastomosis) time**; cost; learning curve; pneumoperitoneum and Trendelenburg effects.
- No tactile feedback for calcified vessels.

Outcomes

- Multicentre and ERUS (European Association of Urology Robotic Urology Section) series: graft function and survival comparable with open transplant, with lower surgical-site complications.

Complications

- Graft thrombosis, haemorrhage, urine leak, ureteric stricture, lymphocele (less), port-site hernia; DGF (delayed graft function).

Diagram to draw: pelvis with graft on external iliac vessels and ureter to bladder dome; ice slush via GelPOINT.

> **Key points to remember:**
>
> - Vattikuti–Medanta technique with regional hypothermia (2014).
> - Vein first, then artery, then Lich–Gregoir.
> - Best for obese recipients; graft function equals open.

Rapid Revision Sheet · Section 15

Pelvicalyceal (renal collecting system) anatomy, its relevance to PCNL and endourology, and mini-PCNL (LAQ)

- Sampaio: A-I 45%, A-II (crossed) 17.2%, B-I 21.4%, B-II 16.4%.
- Fornix puncture: no arterial injury; upper-pole infundibulum 67% vascular injury.
- Target a posterior calyx with a short, wide infundibulum; upper pole reaches most.
- Lower pole: steep angle, long calyx >10 mm, infundibulum <5 mm unfavourable.
- Endopyelotomy lateral; infundibulotomy superior or inferior, never anterior.
- Pleura to 12th rib; retrorenal colon at the lower pole, commoner prone.
- Mini 15–22 Fr (Jackman 1998): fewer transfusions, longer operation.
- High intrarenal pressure in mini-PCNL causes fever and sepsis.

Renovascular hypertension: pathophysiology, evaluation and treatment (LAQ)

- Five segmental end arteries; posterior segmental passes behind the pelvis; veins anastomose freely.
- Left renal vein 6–10 cm, receives adrenal, gonadal and lumbar veins.
- Goldblatt 1934: 2K1C = renin-dependent; 1K1C = volume-dependent.
- Chronic phase: contralateral nephrosclerosis makes RVH stenosis-independent.
- Angiotensin II keeps GFR by efferent constriction; ACE inhibitor causes acute kidney injury in bilateral RAS.
- RVH is proved only by cure; duplex first, angiography with gradient before intervention.
- ASTRAL 2009 and CORAL 2014: stenting adds nothing to medical therapy in atherosclerotic RAS; no routine stenting.
- FMD gets balloon angioplasty without stent; ostial atheroma: stent.
- Revascularise for flash pulmonary oedema, declining GFR in a viable kidney (RI <0.8), resistant hypertension.

GFR estimation in prospective renal donor (SAQ)

- Creatinine overestimates GFR (secretion); CKD-EPI 2021 is race-free.
- Cystatin C adds accuracy; creatinine–cystatin C combined equation is most accurate.
- Inulin is the gold standard; iohexol or DTPA in practice; DTPA gives GFR and split function.
- KDIGO 2017: ≥90 accept, 60–89 individualise, <60 decline.
- Better kidney stays with the donor.
- CrCl = UV/P; check completeness by creatinine excreted per kg.
- Secretion makes CrCl overestimate GFR as function falls.
- Cockcroft–Gault: (140 − age) × weight / (72 × SCr), × 0.85 women.

Acute renal failure (AKI) and acute tubular necrosis (LAQ)

- KDIGO: creatinine ≥0.3 mg/dL or ×1.5; urine <0.5 mL/kg/h for 6 h.
- FeNa <1% prerenal, >1% ATN; specific gravity <1.010, muddy-brown casts in ATN; ultrasound to exclude obstruction.
- S3 segment and thick ascending limb are most vulnerable.
- Diuretics only for overload (do not treat ATN); dialysis for refractory AEIOU.
- Post-obstructive diuresis: oral fluids; replace 75% with 0.45% saline if needed.

Metabolic abnormalities in CKD (SAQ)

- FGF-23 rises first; secondary then tertiary hyperparathyroidism.
- Bicarbonate ≥22 mEq/L; Hb 10–11.5 g/dL.
- Hyperkalaemia: binders rather than stopping RAAS blockade.

Management of acute hyperkalaemia (SAQ)

- Calcium stabilises the membrane but does not lower potassium.
- Insulin 10 units + 50 mL 50% dextrose; salbutamol 20 mg nebulised.
- Binders, diuretics and dialysis remove potassium; relieve obstruction.

Pre transplant nephrectomy (SAQ)

- Default: leave native kidneys.
- Indications: infection/VUR, ADPKD space, tumour, hypertension, proteinuria.
- Retroperitoneal route for PD; leave a normal ureter.

Discuss the role of tissue typing and cross match in live and cadaveric (LAQ)

- HLA on chromosome 6; match A, B, DR; DR matters most.
- Positive T-cell CDC crossmatch (IgG DSA) = no transplant: hyperacute rejection.
- Flow and bead assays are more sensitive; cPRA guides allocation.
- Incompatible pair: paired exchange (legal in India since 2011) or desensitisation.

Deceased donor kidney: selection, maintenance and retrieval techniques (LAQ)

- THOA: 4-member Board (Board of four), none from the transplant team; two tests ≥6 h apart; Form 10.
- Apnoea test: no breathing at $PaCO_2$ ≥60 mmHg.
- ECD: ≥60, or 50–59 with 2 of hypertension, creatinine >1.5, CVA death; KDPI >85%.
- Rule of 100s; vasopressin for hypotension and DI.
- Cannulate, clamp, vent, cool: distal aortic cannula, supracoeliac clamp, IVC vent, cold UW/HTK and ice.
- Kidneys removed last, en bloc; Carrel patch, IVC for right vein.
- Left kidney preferred for living donation.
- Machine perfusion reduces DGF.

Evaluation and selection criteria of the living renal donor (SAQ)

- KDIGO 2017: risk-based lifetime ESKD threshold; GFR ≥90 accept, <60 decline; ACR >100 decline.
- GFR 60–89 and albuminuria 30–100 are individualised.
- Controlled hypertension on 1–2 drugs may donate; type 1 diabetes and bilateral stones exclude.
- ABO and crossmatch first; CT angiography for anatomy.
- Better kidney stays with the donor.
- THOA: near relatives, else authorisation committee; unrelated donor needs Authorisation Committee approval.
- Interview video-recorded; no commerce.

Vascular complications after renal transplantation, including transplant renal artery stenosis (SAQ)

- TRAS is the commonest vascular complication; presents months later; angioplasty or stent first (first-line); angiography is gold standard.
- Refractory hypertension plus rising creatinine: Doppler first.
- Venous thrombosis: reversed diastolic flow, swollen tender graft.
- Arterial thrombosis: sudden anuria; explore immediately.

Lymphocele and postoperative lymphatic leak (chylous ascites) (SAQ)

- Lymphocele: commonest PLND complication; radiological up to 60%, significant 2–5%.
- Lymph creatinine = serum; urinoma creatinine higher.
- Laparoscopic fenestration best (is best); aspiration alone worst.
- Chylous ascites: 2–12% after post-chemotherapy RPLND; triglyceride 2–8 × serum.
- Ladder: diet with MCT and octreotide → TPN → embolisation or shunt.

Allograft rejection and immunobiology of renal transplantation (LAQ)

- Direct pathway drives early TCMR; indirect pathway drives antibody (DSA) and chronic rejection.
- Signal 1 calcineurin, signal 2 CD28–B7, signal 3 IL-2R–mTOR.
- Hyperacute rejection prevented by crossmatch.
- Banff AMR = tissue injury + antibody–endothelium interaction (C4d) + DSA.
- TCMR: pulse steroids, ATG; AMR: plasmapheresis, IVIG.
- Chronic active AMR (transplant glomerulopathy) is the main cause of late graft loss.
- Belatacept blocks signal 2; better GFR, contraindicated if EBV-negative; imlifidase (IgG cleavage) for highly sensitised patients; dd-cfDNA NPV >90%.
- ECD: ≥60, or 50–59 with ≥2 of hypertension, creatinine >1.5, CVA death; KDPI >85% since 2014.
- Remuzzi 0–3 single, 4–6 dual, ≥7 discard; HMP reduces DGF; ESP ≥65 to ≥65.

Vascular anastomosis and vascular access for dialysis (LAQ)

- Intima-to-intima eversion, monofilament polypropylene, inside-out on the artery, flush before tying.
- AVF first, distal, non-dominant arm, about 6 months ahead.
- Rule of 6s: flow >600 mL/min, >6 mm wide, <6 mm deep, 6 weeks.
- Thrombosis sits on a stenosis; steal: banding or DRIL.

Describe in details basic patho-physiology of urine production and transport (LAQ)

- RBF 25% of cardiac output; GFR 125 mL/min; net filtration pressure ~10 mmHg.
- Proximal 65%, loop 25%, distal 5–7% of Na^+; ADH → V2 → AQP2.
- Myogenic peristalsis from calyceal pacemaker; post-obstructive diuresis 2% vs 50%.

Lymphatic drainage of the genitourinary system and chyluria (etiopathogenesis and management) (LAQ)

- Lymph flows right → left; cisterna chyli at L1–L2, behind the right side of the aorta.
- Bladder drains mainly to external iliac nodes; left kidney to left para-aortic nodes.
- Chyluria = pyelolymphatic fistula from lymphatic obstruction and valve failure, usually filarial (*W. bancrofti*); often intermittent.

- Ether test and (urine) triglycerides confirm (ether test clears milky urine); cystoscopy after (a) fatty meal lateralises.
- Diet (MCT) + DEC (diet with MCT plus DEC) → (silver nitrate) sclerotherapy → laparoscopic hilar lymphatic disconnection; disconnection surgery for severe or refractory cases.
- Retroperitoneoscopic disconnection beats open.

Haemodialysis principles, dialysis techniques and dialysis access (vascular and PD) (LAQ)

- Diffusion clears small solutes, convection middle molecules, ultrafiltration water.
- HD Kt/V minimum 1.2 (target 1.4); URR ≥65%; PD weekly Kt/V ≥1.7.
- AVF first, distal, non-dominant arm, ≥6 months ahead; rule of 6s: >600 mL/min, >6 mm, <6 mm deep, 6 weeks.
- Thrombosis usually sits on a stenosis; steal is treated by (→) banding or DRIL.
- Tenckhoff: two cuffs, pelvic tip, downward exit.
- Laparoscopic placement with omentopexy gives best function.
- Avoid subclavian lines; AVF first for long-term HD.
- CRRT for unstable AKI; PD preferred in children.

Brain-stem death: pathophysiology, guidelines and declaration (SAQ)

- Raised ICP, Cushing reflex, catecholamine storm, then hypotension, DI, hypothermia, DIC.
- Six brainstem reflexes absent; apnoea test: no breathing at $PaCO_2$ ≥60 mmHg.
- THOA: four-member Board; no transplant-team member.
- Two tests ≥6 h apart; Form 10; Rule of 100s.

Immunosuppression in renal transplantation: protocols, induction and steroid-free regimens (SAQ)

- Tacrolimus + mycophenolate ± steroids (maintenance) after induction (basiliximab or rATG).
- Basiliximab for low risk; rATG for high immunological risk.
- CNI nephrotoxic; mTOR inhibitors cause lymphocele and poor healing.
- Belatacept blocks signal 2; avoid if EBV-negative.
- Rejection: steroids for T-cell-mediated; plasmapheresis, IVIG, rituximab for antibody-mediated.
- HLA-incompatible: plasmapheresis + IVIG ± rituximab; imlifidase cleaves IgG.
- ABO-incompatible: rituximab + plasmapheresis to target titre.
- Steroid-free: avoidance, early (first week) or late withdrawal; needs induction plus tacrolimus–MMF; low-risk recipients only.
- More mild rejection, same survival; less PTDM (HARMONY). KDIGO 2009: stop within week 1, or continue.
- Ciclosporin: binds cyclophilin, inhibits calcineurin, NFAT stays phosphorylated, IL-2 blocked.
- Ciclosporin: nephrotoxic; gum hyperplasia and hirsutism; CYP3A4 interactions.
- Ciclosporin urology uses: transplant, refractory Hunner lesions, retroperitoneal fibrosis.

CAPD and CAPD peritonitis (SAQ)

- Peritoneum is the membrane; glucose or icodextrin drives ultrafiltration.
- Tenckhoff catheter with two cuffs; Kt/V ≥1.7 per week.

- Peritonitis: two of cloudy fluid or pain, WCC >100/µL with >50% neutrophils, positive culture.
- Intraperitoneal Gram-positive plus Gram-negative cover.
- Remove catheter for fungal or refractory (5 days) peritonitis.

THOA/THOTA legislation and hospital authorization committee (SAQ)

- THOA 1994: brain-stem death legal; anti-commercial; Authorisation Committee for unrelated donors; amended 2011; Rules 2014.
- 2011: tissues added; grandparents and grandchildren became near relatives.
- Swap donation, retrieval centres, transplant coordinators, required request, higher penalties.
- Commercial dealing: 5–10 years and ₹20 lakh to ₹1 crore.
- Rules 2014: Form 10; Board of four; tests ≥6 hours apart.
- Hospital-based committee if >25 transplants a year; no transplant team member.
- Non-relatives, swap and foreigners need committee approval.
- Interview video-recorded; appeal within 30 days.

Marginal kidney donor and Remuzzi score (SAQ)

- ECD: ≥60, or 50–59 with 2 of hypertension, creatinine >1.5, CVA death.
- KDPI >85% replaced ECD (2014).
- Remuzzi: glomeruli, tubules, interstitium, vessels; 0–3 each.
- Remuzzi 0–3 single, 4–6 dual, ≥7 discard.
- HMP reduces DGF; Maastricht III is the commonest DCD.

Renin-angiotensin system (SAQ)

- Renin is rate-limiting; released by low pressure, low NaCl, β1.
- Efferent constriction maintains GFR, so ACE inhibitors risk renal failure in bilateral stenosis.
- Renin ratio >1.5 lateralises; reninoma excision cures most.

Drugs associated with Acute Renal Failure. (SAQ)

- NSAIDs block afferent dilatation; ACE inhibitors/ARBs block efferent constriction.
- ATN: aminoglycosides, amphotericin, cisplatin, contrast.
- AIN: β-lactams, PPIs, NSAIDs, checkpoint inhibitors; stop and give steroids.
- Crystals: aciclovir, methotrexate, indinavir.

Hypokalemia (SAQ)

- Shift or loss; urine potassium separates renal from extrarenal.
- ECG: flat T, U wave.
- Diversion acidosis: give potassium with or before bicarbonate.
- Correct magnesium first.

Hyperfiltration injury in AKI and CKD (SAQ)

- Hyperfiltration = raised SNGFR; Brenner 1982.
- Leads to FSGS, proteinuria and the AKI-to-CKD transition.
- ACE inhibitors and SGLT2 inhibitors lower glomerular pressure.

UW solution and organ preservation solutions (SAQ)

- Hypothermia slows, not stops, metabolism.
- Lactobionate and raffinose (impermeants), HES / starch (colloid).

- Allopurinol, glutathione (antioxidants); adenosine (ATP).
- High K^+: flush before reperfusion.
- HTK: histidine buffer, tryptophan, ketoglutarate, low viscosity, low K^+.
- Machine perfusion reduces DGF.

Post transplant malignancy (SAQ)

- Skin cancer commonest; squamous more than basal cell.
- PTLD: EBV-negative recipient of EBV-positive donor, first year.
- RCC mostly in native kidneys; bladder cancer 3-fold and aggressive.
- Reduce immunosuppression and switch to mTOR inhibitor.

Mycophenolate Mofetil. (SAQ)

- Prodrug of MPA; IMPDH inhibitor; lymphocytes need de novo purine synthesis.
- 1 g twice daily; ciclosporin lowers MPA levels.
- Toxicity: diarrhoea, leucopenia, CMV/BK, teratogenic; not nephrotoxic.

Renal transplant in neurogenic bladder (SAQ)

- Target <40 cm H_2O storage before transplant.
- Drugs → CIC → cycling → augmentation; reconstruct 6–12 weeks before.
- Pyelonephritis, not augmentation, loses grafts.

Pre-emptive renal transplant (SAQ)

- Transplant before dialysis = best outcome.
- Refer at eGFR <30; transplant usually at eGFR <20.
- Commonest with living donors and children.

Tacrolimus. (SAQ)

- FKBP12 → calcineurin → NFAT → IL-2 blocked.
- CYP3A4/P-gp; trough-guided; many interactions.
- Nephrotoxicity, NODAT, tremor, hyperkalaemia, hypomagnesaemia.
- ELITE-Symphony: low-dose tacrolimus best.

Urological Complications post Kidney Transplantation. (SAQ)

- Ureter is the weak point; ischaemia is the cause.
- Leak early, stricture late; open repair best.
- Stent out at 2–3 weeks; laparoscopic fenestration for lymphocele. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Female kidney donors: special precautions (SAQ)

- Ensure free consent (Indian gender gap).
- Pregnancy test; contraception; stop oestrogens.
- Donors: gestational hypertension or pre-eclampsia about 11% vs 5%.
- Defer pregnancy; high-risk antenatal follow-up.

Laparoscopic donor nephrectomy (including pros and cons) (SAQ)

- Ratner 1995; left kidney preferred (longer vein).
- Hand-assisted in about two-thirds of US cases.
- Pros: less pain, shorter stay, earlier return to work, better cosmesis, and more willing donors; graft function equals open.

- Cons: longer operation, learning curve, slightly longer warm ischaemia, shorter vessels (especially on the right), cost.
- Safety: mortality 3 per 10,000; conversion 0.26–3.34%.
- Never rely on Hem-o-lok alone on the donor renal artery (do not use Hem-o-lok clips alone).

Artificial kidney (SAQ)

- Diffusion, convection, ultrafiltration across a semipermeable membrane.
- Dialysis lacks endocrine and tubular function; transplant is best.
- Wearable, implantable bioartificial and RAD devices are investigational.

Robotic kidney transplantation (SAQ)

- Vattikuti–Medanta technique with regional hypothermia (2014).
- Vein first, then artery, then Lich–Gregoir.
- Best for obese recipients; graft function equals open. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

SECTION 16

BPH

148 Anatomy and Development of the Prostate

1 · Development (embryology)

The androgen pathways that build the prostate also drive BPH.

- **Origin**: the urogenital sinus (UGS), endoderm in mesoderm. Incomplete cloacal division (imperforate anus, urorectal fistula) impairs development.
- **Dihydrotestosterone (DHT)**: †5α-reductase type 2 in UGS mesenchyme converts testosterone to DHT, the more potent androgen receptor (AR) ligand. **SRD5A2** loss leaves the prostate incomplete; castration or inactive AR leaves it absent. †The same enzyme is the target of 5α-reductase inhibitors (5-ARIs) in BPH.
- **Androgen is not rate-limiting**: testosterone is present at week 9 but budding starts at week 10, gated by **AR demethylation** in the UGS.

Week	Event
6–7	Cloaca divides into the ventral UGS and the dorsal anorectum
≈9	Testosterone activates **mesenchymal** AR, whose andromedins instruct the epithelium
10	Epithelial buds invade the mesenchyme; this fixes the adult ductal pattern

Mesenchyme instructs, epithelium responds

- Budding and branching need **mesenchymal (stromal)** AR, not epithelial AR; UGS mesenchyme can even turn bladder epithelium into prostate (Cunha).
- **Andromedins** are androgen-induced stromal signals, chiefly **fibroblast growth factor 10 (FGF10)**; Nkx3-1 specifies the epithelium.

> **KEY POINT**
>
> McNeal: **re-awakening of embryonic induction** in ageing transition-zone stroma may initiate new glands, one theory of BPH.

- Gestational diethylstilboestrol, phthalates and dioxin disrupt development; dioxin-induced hyperinnervation may explain variable α-blocker response.

2 · Zonal anatomy (McNeal)

LAQ Prostate anatomy, neurovascular bundles and surgical concepts in localised CA prostate

Paper answer · P1 · P3 · P4 · merged from 3

Asked as: P1 LAQ "Anatomy of prostate and neurovascular bundles: clinical and surgical concepts in localised prostate cancer" (Anatomy of prostate and its neurovascular bundles with its clinical details; Describe in detail, anatomy of prostate with special reference to concepts in surgical management if localised CA prostate) · P4 LAQ "Prostate anatomy, newer surgical concepts in CA prostate and neurovascular bundle preservation in radical prostatectomy"

(Discuss in detail anatomy of prostate with special reference to newer concepts in surgical management of CA prostate; also SAQ: Neurovascular bundle preservation in Radical prostatectomy: recent techniques) · P3 SAQ "Describe in details anatomy of prostate with special reference to newer concept in surgical management of localized CA prostate"

"if localised" is read as "of localised" prostate cancer.

Introduction

- A fibromuscular, glandular organ of about 20 g around the proximal urethra, from bladder neck (**base**) to urogenital diaphragm (**apex**), in front of the rectum.
- Buds from the urogenital sinus at week 10, driven by androgen acting on the **mesenchymal** AR (androgen receptor).
- RP (radical prostatectomy) aims at the "trifecta" of cancer control, continence and potency. Modern RP is anatomical: each anatomical step or new technique either secures a clear margin or saves a named structure (nerves, sphincter, anterior support).

Relations

- **Anterior:** retropubic space (of Retzius), puboprostatic ligaments, DVC (dorsal venous complex).
- **Posterior:** Denonvilliers fascia, then rectal ampulla, hence DRE (digital rectal examination) and TRUS (transrectal ultrasound).
- **Lateral:** levator ani with its fascia; NVB (neurovascular bundle) posterolaterally.
- **Base:** bladder neck; seminal vesicles behind. **Inside:** the urethra bends about 35° at the **verumontanum**, where the ejaculatory ducts open beside the utricle.

Zonal anatomy (McNeal 1981)

Zone	Share of glands	Position	Clinical / surgical relevance
Peripheral (PZ)	About 70%	Posterior and lateral; longest ducts	**70% of cancers**; felt on DRE; biopsy target; posterolateral, so ECE (extracapsular extension) runs along the nerves
Central (CZ)	About 25%	Cone around the ejaculatory ducts at the base	Few cancers; mimics HGPIN (high-grade prostatic intraepithelial neoplasia)
Transition (TZ)	About 5%	Two lobes beside the preprostatic sphincter; shortest ducts	**BPH (benign prostatic hyperplasia)**; anterior cancers (about 15%) missed by DRE
Periurethral glands	Small	Within the preprostatic sphincter	Middle-lobe BPH
Anterior fibromuscular stroma (AFMS)	No glands	Anterior	Anterior tumours reach it; anterior margin risk in posterior approaches

- **Capsule:** there is **no true capsule**, so cancer breaching the gland shows "extraprostatic extension". In BPH the compressed PZ forms the **surgical capsule**, the plane of enucleation. The capsule transmits growth pressure to the urethra, so TUIP (transurethral incision) relieves obstruction.

Fascial investments and dissection planes

Layer	Position	Surgical rule
Prostatic fascia	On the gland; carries DVC tributaries anteriorly	Stays on the prostate in nerve sparing
Levator fascia	Covers levator; fuses with prostatic fascia as the lateral pelvic fascia	Incised to reach the NVB
Denonvilliers fascia	Between prostate and rectum; thickest at base and vesicles	Excised completely for the posterior margin

- **Denonvilliers fascia** stays on the rectum when ECE risk is low; it is excised when posterolateral (posterior) ECE is suspected.

Nerve-sparing plane	Lies	Trade-off
Intrafascial	Under the prostatic fascia (between capsule and prostatic fascia)	Most nerves kept; more positive margins
Interfascial	Between prostatic and levator fascia	**Standard nerve-sparing plane** for organ-confined disease
Extrafascial	Outside levator and Denonvilliers fascia	Widest margin; ED (erectile dysfunction) likely

- Chapter 160 (Walsh) calls intrafascial "never used"; Chapter 161 accepts it in selected men.

Blood supply and lymphatics

- **Inferior vesical artery** ends as: **urethral group** (enters at the vesicoprostatic junction, 5 and 7 o'clock; supplies the adenoma) and **capsular group** (posterolateral, in the lateral pelvic fascia, marking the cavernous nerves).
- **Accessory pudendal arteries** run over the anterolateral prostate; dividing them reduces penile inflow. Preserve them; they may be the main penile supply.
- **Veins:** the deep dorsal vein passes under the pubis into a superficial branch and lateral (**Santorini**) plexuses, draining to the internal iliac vein. They anastomose freely, so tears bleed heavily.
- DVC: CO_2 pneumoperitoneum tamponades it, so robotic blood loss is low.
- **Lymph:** obturator, internal and external iliac, common iliac, presacral nodes.

Neurovascular bundle

- **Pelvic plexus:** parasympathetic S2–S4 plus sympathetic (T10–L2) via the hypogastric nerves; beside the rectum, centred on the seminal vesicle tip. It gives the **cavernous nerves** (parasympathetic).
- **NVB of Walsh (Walsh and Donker, 1982):** capsular vessels plus **cavernous nerves**. The nerves are invisible; the capsular vessels mark them.
- It runs posterolaterally in the lateral pelvic fascia **between prostatic and levator fascia**, **3–5 mm from the capsule**, outside Denonvilliers fascia.
- At the apex they lie at **5 and 7 o'clock**, posterolateral to the sphincter. The nerves then swing to about **3 and 9 o'clock** at the mid-urethra and enter the corpora behind the dorsal penile artery. The apex is where most nerve injuries occur.

- Distribution varies (is diffuse): main posterolateral bundle, posterior "hammock", anterolateral fibres (the basis of high anterior release).
- **Injury mechanisms:** incision, entrapment (clip or suture), **heat** (thermal), **traction**; so dissect athermally without traction.
- **Striated sphincter:** a vertical tube over the apex (Oelrich), pudendal nerve; passive continence.

Continence anatomy

Passive	Active
Smooth muscle of bladder neck and urethra; anterior support / suspensory system (endopelvic fascia, detrusor apron, puboprostatic ligaments, DVC)	Horseshoe striated sphincter at the apex, supplied by pudendal nerve branches that enter it at the apex

Clinical details

- **BPH surgery:** TURP (transurethral resection of the prostate) stays proximal to the verumontanum (sphincter lies just distal); urethral-group vessels at 5 and 7 o'clock bleed.
- **Cancer:** PZ cancer exits posterolaterally along nerves; anterior TZ tumours are missed by DRE and biopsy, so MRI (magnetic resonance imaging) helps.
- **Radical prostatectomy (RP) steps:** 1. Incise endopelvic fascia; spare the pubourethral ligament. 2. Ligate the DVC over the urethra before division. 3. Preserve accessory pudendal arteries. 4. Excise Denonvilliers fascia; the apex is the commonest positive-margin site.
- Excise an NVB for palpable extracapsular disease; margins are equal with or without nerve sparing.
- **Also know:** rectal injury at the ampulla; PDE5 (phosphodiesterase type 5) inhibitor rehabilitation.

Concepts and newer concepts in surgical management of localised cancer

1. **Selection:** RP suits localised disease (mainly intermediate and selected high risk) with a life expectancy of more than 10 years; low risk is usually offered active surveillance.
2. **RARP (robot-assisted RP):** 3D, 10× magnified view and wristed instruments removed the suturing barrier. **Yaxley 2016** RCT (randomised controlled trial): equal early function and early oncology versus open; less blood loss (300 versus 1400 mL) and shorter stay. Surgeon matters more than platform.
3. **Risk-graded (risk-stratified) nerve sparing:** each side is graded from full (intrafascial) through partial to none (for example the Tewari four-grade scale), using grade, positive cores, mpMRI (multiparametric magnetic resonance imaging) capsular contact and nomograms, side by side.
4. **NeuroSAFE:** intraoperative frozen section of each (the whole) posterolateral surface; if positive, that NVB is resected. It aims to allow more nerve sparing without more positive margins; the UK NeuroSAFE PROOF RCT tested its effect on erectile function.
5. **Anterior-support preservation:** - **Retzius-sparing** (Bocciardi, 2010): fully posterior through the pouch of Douglas. Earlier continence at 3 months, equal at 12; more positive margins (anteriorly), so avoid for anterior tumours. - **Hood technique** (Tewari, 2021): anterior approach keeping detrusor apron, puboprostatic ligaments and DVC; early data. - Retzius-sparing and Hood are aimed at continence; potency equal at 12 months.

6. **Apical dissection:** the apex is the commonest positive-margin site. Keep maximal urethral length, watch for an apical lip, divide sharply without cautery (cold division); control the DVC distally or late. Posterior (Rocco) reconstruction may aid early continence.
7. **PLND (pelvic lymph node dissection) templates:**

Template	Nodes	Drainage captured
Limited	Obturator	Least
Standard	+ External iliac	About 38%
Extended	+ Internal iliac	About 63%
+ Common iliac	+ Common iliac	About 75%

- Extended PLND when nodal risk exceeds 5% (EAU [European Association of Urology]). It stages but does not improve survival (staging, not therapeutic); lymphocele is the commonest complication. 8. **Single-port robot (da Vinci SP, 2018):** one 2.5-cm cannula; extraperitoneal or transvesical; same-day discharge. 9. **Focal therapy anatomy:** ablate the **index lesion**. Transrectal HIFU (high-intensity focused ultrasound) suits posterior PZ lesions; anterior lesions need transperineal cryotherapy or TULSA (transurethral ultrasound ablation). Spare sphincter, NVB and rectum.
- **Also know:** PSMA PET (prostate-specific membrane antigen positron emission tomography) staging; salvage robotic RP; new platforms; ERAS (enhanced recovery after surgery).

Nerve-sparing techniques (NVB preservation): recent techniques

- **Walsh anatomic (open):** release apex to base from the rectal surface; small clips; **no thermal energy**.
- **High anterior release** (open) and **Veil of Aphrodite** (robotic): lateral prostatic fascia lifted high anteriorly as a "veil". Select by **Tsuzuki criteria**: ≥2 of PSA (prostate-specific antigen) >10 ng/mL, Gleason >6, abnormal DRE, perineural invasion, high core involvement → >10% risk, so avoid.
- **Antegrade release (robotic):** incise levator fascia, sweep NVB posterolaterally, clip pedicle at the "groove".
- **Athermal, traction-free:** clips or bulldogs, cold scissors.
- Graded nerve sparing, NeuroSAFE, Retzius-sparing and Hood: see above.

Adjuncts

- **Penile rehabilitation:** PDE5i (phosphodiesterase type 5 inhibitor), then injections.
- **Also know:** sural nerve grafts and nerve stimulation tests (abandoned); amniotic membrane wraps, fluorescence nerve imaging (experimental).

Diagram to draw

- Axial prostate showing the three fasciae, the NVBs at 5 and 7 o'clock and the intra-, inter- and extrafascial planes.

Key points to remember:

- PZ gives 70% of cancers; TZ gives BPH; CZ around ejaculatory ducts; no true capsule.
- NVB between prostatic and levator fascia, 5 and 7 o'clock; apex is the commonest margin and nerve-injury site.
- Interfascial plane is the standard nerve-sparing plane; avoid heat and traction; clips, cold scissors.
- Ligate DVC; excise Denonvilliers fascia; spare accessory pudendals.
- Veil and high release for low-risk sides (Tsuzuki).
- NeuroSAFE frozen section: more sparing, safe margins.
- Retzius-sparing: earlier continence, more positive margins.
- Extended PLND captures about 63% of drainage.

Also, from the condensed book (not in the paper answer):

- The urethra bends **≈35°** at the verumontanum, where 13–15 ducts open. This is the distal limit of transurethral resection (TURP) and enucleation.
- The transition zone (TZ) has the shortest ducts and gives †≈20% of cancers; the peripheral zone (PZ) has the longest.

KEY POINT

Trap: the central zone (CZ), around the ejaculatory ducts, is not the TZ. Short proximal ducts **enlarge** (BPH); long distal ducts **transform** (cancer). The enlarging TZ compresses PZ into the **surgical (pseudo)capsule**, the enucleation plane (see Ch 152).

3 · Fascia, vessels †

- †The capsule is incomplete anteriorly and apically. The neurovascular bundles lie outside it, so enucleation spares them.
- †Prostatic veins join the vertebral (**Batson**) plexus, a route to spinal metastases.

4 · Histology and cell types

Feature	Luminal (secretory)	Basal
Keratins	8, 18	5, 14
Marker	Prostate-specific antigen (PSA), NKX3-1	KRT5/14
AR	High	Low
Secretes	PSA, prostatic acid phosphatase (PAP)	Almost none

- The urethra and proximal ducts carry club, hillock and neuroendocrine cells; BPH stroma contains prominent lymphocytes.

5 · Secretions and clinical markers

- Prostatic fluid is **37–44%** of semen and acidic (pH ≈6.3); seminal vesicle fluid is the alkaline majority.
- Kallikreins liquefy semen within **20–30 min**; incomplete liquefaction at 60 min is nonliquefaction.

- **Ion trapping**: macrolides concentrate in acidic fluid; inflammation alkalinises it.
- PSA, PSMA and PAP all occur outside the prostate, so "prostate-specific" is a misnomer.

Product	Key facts
PSA (KLK3)	Serine protease (19q13.4); liquefies semen
PSMA	Folate hydrolase; rarely deleted in cancer, hence PSMA-PET and ^{177}Lu-PSMA
PAP	First tumour marker (Huggins and Hodges, 1941) MCQ
Zinc	Highest of any tissue; inhibits mitochondrial aconitase, so citrate is secreted

KEY POINT

Benign cells: high zinc blocks aconitase, so citrate is exported. Cancer: **zinc is lost** and citrate is oxidised, so citrate is low (†the basis of MR spectroscopy).

149 Benign Prostatic Hyperplasia: Aetiology, Pathophysiology, Epidemiology and Natural History

1 · Definitions and the four overlapping entities

Histology, size, obstruction and symptoms differ; treatment depends on which.

*See the paper answer **Pathophysiology, classification and management of LUTS.** (Section 7, Ch 104).*

Also, from the condensed book (not in the paper answer):

- †BOOI = PdetQmax − 2Qmax: **>40 obstructed, 20–40 equivocal, <20 unobstructed**.
- "Hypertrophy" is wrong because BPH raises cell **number**, not size; "prostatism" is obsolete (Abrams coined "LUTS") MCQ.

Term	How measured
Benign prostatic hyperplasia (BPH): a histological increase in epithelial and stromal cell number in the periurethral/transition zone (TZ)	Histology
Benign prostatic enlargement (BPE): enlargement caused by BPH	Digital rectal examination (DRE), transrectal ultrasound (TRUS), MRI
Bladder outlet obstruction (BOO): obstruction from any cause (BPH, stricture, cancer, bladder neck sclerosis)	Pressure–flow (best); flow rate (indirect)
Benign prostatic obstruction (BPO): BOO caused by BPE	Pressure–flow
Lower urinary tract symptoms (LUTS): storage, voiding and post-micturition; not prostate-specific	International Prostate Symptom Score (IPSS)

- The four entities overlap only **partially**; BOO may exist without BPE (for example, a stricture). Detrusor ageing, polyuria, sleep disorders and impaired contractility can all mimic BPH.

> **KEY POINT**
>
> Viva: "Is every man with LUTS and a big prostate obstructed?" No. Size, symptoms, flow and obstruction are separate; obstruction is diagnosed only by pressure–flow.

2 · Epidemiology

- At autopsy there is **no BPH under 30**; prevalence is ≈ **50% in the 50s**† and **88% in the 80s**. It is **similar across all ethnic and geographic groups**.
- No clinical definition is globally accepted. BPH is described by **symptoms** (IPSS), **enlargement** and **obstruction**, plus bother. The proposed volume cut-off is **>30 mL**; a community composite is IPSS >7, maximum flow rate (Qmax) <15 mL/s and volume >20 mL.

SAQ IPSS score

Paper answer · P3

Type: SAQ (10 marks)
Source: 2022-Summer Paper 3, Q3e

Definition

- **IPSS (International Prostate Symptom Score):** validated, self-administered questionnaire grading male LUTS (lower urinary tract symptoms) over the past month.
- Based on the **AUA-SI (American Urological Association Symptom Index, 1992)**; adopted by WHO (World Health Organization) in 1993 by adding one quality-of-life question.

Items

No.	Symptom	Phase
1	Incomplete emptying	Voiding
2	Frequency (again within 2 hours)	Storage
3	Intermittency	Voiding
4	Urgency	Storage
5	Weak stream	Voiding
6	Straining	Voiding
7	Nocturia	Storage

- **Scoring:** items 0–5. 0 not at all; 1 less than 1 time in 5; 2 less than half the time; 3 about half; 4 more than half; 5 almost always. **Nocturia scored by number of times** (0 to ≥5).
- **Total 0–35:** voiding subscore 4 items, storage 3 (not validated).

Severity	Score
Mild	0–7
Moderate	8–19
Severe	20–35

- **QoL (quality of life)/bother question:** "If you were to spend the rest of your life with your urinary condition as it is now, how would you feel?" Scored 0 (delighted) to 6 (terrible); reported separately, outside the 35.

Uses

- First step in LUTS work-up (AUA and EAU [European Association of Urology]); grades severity and bother.
- Guides treatment: mild, non-bothersome → watchful waiting; bothersome → medical therapy or surgery.
- Monitors response: **a 3-point fall is the minimum perceptible change**.
- Standard end point in trials.

Limitations

- **Not disease-specific:** women score similarly; does not diagnose BOO (bladder outlet obstruction) or BPH (benign prostatic hyperplasia); correlates poorly with urodynamics, prostate size and flow.
- No incontinence, pain or post-micturition items.
- Needs literacy.

Alternatives

- **VPSS (Visual Prostate Symptom Score):** pictures, for low literacy.
- BPH Impact Index; ICIQ-MLUTS (International Consultation on Incontinence Questionnaire–Male LUTS); DAN-PSS (Danish Prostate Symptom Score); OAB-q (overactive bladder questionnaire); LURN (Lower Urinary Tract Research Network) SI-29 adds incontinence and pain.

Diagram to draw: IPSS chart with scoring columns.

> **Key points to remember:**
> - 7 items × 0–5 = 0–35; bands 0–7, 8–19, 20–35.
> - QoL 0–6 is separate; IPSS = AUA-SI + QoL.
> - 3-point change is noticeable; IPSS does not diagnose BOO.

SAQ Methods to estimate prostate size

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 14 Paper 1, Q3f (TRUS also answered in 2017-Winter Paper 1, Q3e)

Why measure

- Chooses treatment: 5ARI (5α-reductase inhibitor) if >30 mL (AUA (American Urological Association)) or >40 mL (EAU (European Association of Urology)); type of BPH (benign prostatic hyperplasia) surgery.
- **PSAD (PSA density)** = PSA (prostate-specific antigen) ÷ volume; >0.15 suggests cancer. Also AUR (acute urinary retention) risk, brachytherapy planning.
- Normal adult gland 20–25 g; about 48 g at 60–70 years. **1 cm³ ≈ 1 g**.

Methods

Method	Technique	Accuracy and use
DRE (digital rectal examination)	Estimate size, consistency	**Underestimates** above 30 mL (lateral lobes impalpable); flat gland predicts <30 mL
PSA as surrogate	Age-specific cut-offs for **>40 mL:** >1.6 ng/mL (50s), >2.0 (60s), >2.3 (70s)	AUC (area under curve) 0.76–0.78; within ±20% of TRUS in most
TAUS (transabdominal ultrasound)	Full bladder; transverse width and height, sagittal length	Less precise than TRUS; also PVR (post-void residual) and **IPP (intravesical prostatic protrusion)**
TRUS (transrectal ultrasound)	Ellipsoid formula; planimetry	**Method of choice**
MRI (magnetic resonance imaging)	T2-weighted ellipsoid or segmentation (manual or automated)	Most accurate and reproducible
CT (computed tomography)	Axial images	Overestimates; poor capsule definition
Cystoscopy	Urethral length, lobes, median lobe	Rough; guides procedure choice

TRUS formulas

Shape assumed	Formula
Ellipsoid (standard)	**π/6 (0.52) × transverse × AP (anteroposterior) × length**
Sphere	π/6 × transverse³
Prolate spheroid	π/6 × transverse² × AP
Planimetry	Sum of serial transverse areas

- Transverse and AP at the widest axial section; length sagittally. All formulas correlate >0.90 with specimen weight.
- **Planimetry:** probe on a stepper, images every 3–5 mm, areas summed; most precise; used for brachytherapy.

IPP grading (suprapubic, sagittal, bladder 100–200 mL)

- Grade I <5 mm, II 5–10 mm, III >10 mm; ≥10 mm predicts BOO (bladder outlet obstruction) and failed trial without catheter.

Diagram to draw: TRUS transverse and sagittal views with three measured diameters.

> **Key points to remember:**
> - Ellipsoid volume = 0.52 × T × AP × L; 1 mL ≈ 1 g.
> - DRE underestimates above 30 mL; PSA >1.6/2.0/2.3 by decade predicts >40 mL.
> - Planimetry most precise; MRI most reproducible.

Also, from the condensed book (not in the paper answer):

- Probability of obstruction by Qmax: <10 mL/s high, 10–15 intermediate, >15 low. Qmax falls almost linearly with age (Olmsted median 20.3 mL/s at 40–44 versus 11.3 at 75–79).
- DRE underestimates size by 25% to ≥50%.

ALGORITHM

1. Total prostate volume (TPV) ≈ **25 mL in the 30s → 35–45 mL in the 70s**.
2. Transition zone volume (TZV) ≈ 15 → 25 mL.

3 · Aetiology

> **KEY POINT**
>
> BPH is an **accumulation** disease: volume = proliferation minus death. Established BPH shows equal or reduced replication and raised bcl-2. Androgens act chiefly by **suppressing cell death**, so withdrawal shrinks the gland.

- New glands otherwise form only in the fetus, hence the theory of **embryonic reawakening** of stromal induction.

3.1 Androgens: permissive, not causal

- Prepubertal castrates and men with androgen defects **never develop BPH**; withdrawal gives only **partial** involution. The two established determinants are **age** and **functioning testes**.

- Serum testosterone falls with age, but **intraprostatic dihydrotestosterone (DHT) and androgen receptor (AR) stay high**. DHT is **maintained, not elevated**; old "raised DHT" reports used cadaveric controls MCQ.
- Serum testosterone does not predict volume or PSA. Androgens act via growth factors rather than as direct mitogens, and they are anti-inflammatory.

ALGORITHM

1. Testosterone enters the stromal cell.
2. **Type 2 5α-reductase** (nuclear membrane, interstitial fibroblast) converts it to DHT.
3. DHT acts on the stroma (autocrine) or diffuses to the epithelium (paracrine).
4. The DHT–AR complex binds DNA and switches on growth factors.
5. Growth factors drive epithelial survival and growth.

- ≈ **90%** of prostatic androgen is DHT of testicular origin; ≈10% is adrenal. DHT has higher AR affinity and forms a more stable complex.

Feature	Type 1	Type 2
Site	Skin, liver	Prostate **stromal fibroblasts**, some basal cells
5α-reductase deficiency	Normal	**Mutated**
Finasteride / dutasteride	Not inhibited / inhibited	Inhibited / inhibited
Role	Possibly cancer	Development and BPH growth

- Finasteride shrinks the gland as much as dual inhibition, so type 1 is unimportant in BPH.
- SRD5A2 lies on **chromosome 2p23** MCQ; its polymorphisms link inconsistently to BPH.

3.2 5α-reductase inhibitor (5-ARI) resistance

- 5-ARIs are the **only medical therapy that reduces volume**.
- Resistance is **not** failure to lower DHT but focal lack of luminal atrophy in nodules, or stromal hyperplasia, often linked to inflammation (NF-κB-driven AR variants, SRD5A2 silencing). The proposed definition is unchanged volume and no atrophy despite lower DHT.

3.3 Oestrogens

- In dogs, androgen plus oestradiol produces BPH; oestrogen induces AR and increases stroma. Oestrogen rises with age relative to testosterone, and adipose tissue aromatises.
- **Oestrogen receptor α (ERα) is stromal** and proliferative; **ERβ is epithelial** and antiproliferative. An ERβ agonist failed in a trial for LUTS.

3.4 Cell death; stromal–epithelial interaction

- Apoptosis after castration is active and probably mediated by transforming growth factor-β (TGF-β).
- Hypothesis: a **stromal defect removes a brake** on epithelial proliferation. Periurethral interstitial fibroblasts (carrying type 2 5α-reductase) increase in BPH and expand the TZ; **no drug targets them**.
- Chemokines such as interleukin-8 (IL-8) turn fibroblasts into collagen-producing myofibroblasts, and the resulting fibrosis contributes to LUTS.

3.5 Growth factors

Androgen permits; growth factors signal and are the likelier primary defect. The first prostatic growth factor found was basic fibroblast growth factor (FGF-2) MCQ.

Factor	Source and target	Role in BPH
FGF-2	Stroma (autocrine)	Raised 2–3-fold; induced by TGF-β and hypoxia
FGF-7 (keratinocyte growth factor, KGF)	Made by stroma; receptor **only on epithelium**	Leading stromal mediator of hormonal control
FGF-10	Smooth muscle to epithelium	Raised by androgens
TGF-β1	Stroma	**Inhibits epithelium** but is **mitogenic for fibroblasts**; BPH stroma shows a blunted response
Insulin-like growth factors (IGF-1, -2)	—	Transgenic IGF-1 causes enlargement; serum IGF-1 is unrelated to volume
Others	—	Epidermal growth factor (EGF), vascular endothelial growth factor (VEGF), FGF-17

- **Hypoxia hypothesis**: hypoxia induces HIF-1, which raises FGF-2 and FGF-7, so growing nodules become more hypoxic; vascular disease such as diabetes may add risk.

3.6 Inflammation

- **Activated T cells** secrete growth factors, and cytokines drive the stroma. Autoimmune disease increases BPH, whereas TNF antagonists are associated with less.
- **Inflammation correlates with enlargement and 5-ARI resistance**, but causality is unproven.

3.7 Metabolic syndrome, lifestyle

- Metabolic syndrome causes prostatic inflammation and LUTS via aromatisation, **hyperinsulinaemia driving sympathetic overactivity**, and low HDL.
- **Waist circumference** predicts prostatectomy and LUTS independently of BMI. BMI >35 versus <25 gives an odds ratio (OR) of **3.5** for a prostate ≥40 mL.

Factor	Direction
Diabetes, fasting glucose >110 mg/dL, obesity, central adiposity, hypertension (LUTS)	Increases risk
Physical activity; moderate alcohol	Lowers risk (alcohol lowers BPH, **not LUTS**)
Smoking, diet, vasectomy, sexual activity	No consistent effect (**vasectomy is not a risk factor**)
Jewish religion	More prostatectomy, not more BPH (access to care) MCQ

- **Depression** is strongly linked to LUTS, and LUTS severity correlates with sexual dysfunction.
- **Drugs that worsen LUTS**: α-sympathomimetic cold remedies, antidepressants, antihistamines and bronchodilators; anticholinergic and adrenergic drugs can precipitate retention.

3.8 Genetics and others

- In young men with large glands (>37 g), first-degree relatives have ≈ **4-fold** risk of surgery. Inheritance is **autosomal dominant**; ≈ **50%** of men operated on before 60 have heritable disease, versus ≈9% after 60.
- **Familial BPH** shows early onset, a **large gland**, normal androgens and a normal 5-ARI response. Only **dogs and humans** get BPH.

4 · Pathology

- **Every nodule arises in the TZ or the periurethral region**. The TZ lies outside the preprostatic sphincter; the periurethral zone lies inside it, with glands parallel to the urethra.
- The **middle lobe** is **periurethral**, because there is no TZ at the bladder neck MCQ. Patterns are middle, lateral, trilobar and posterior commissural (median bar).

> **KEY POINT**
>
> Only humans have a **capsule**, transmitting expansion onto the urethra; dogs get BPH but rarely obstruct. Transurethral incision of the prostate (TUIP) relieves obstruction without changing volume. Hence **size does not correlate with obstruction**.

Feature	Periurethral nodules	TZ nodules
Composition	Purely stromal (early), like **embryonic mesenchyme**	Glandular, tall epithelium
Origin	Interstitial fibroblasts	New duct branches forming a new ductal system

- **Two phases**: for ≈20 years nodule number rises; then glandular nodules grow.
- Small glands are mainly stroma and large glands mainly epithelial nodules. This variable stromal:epithelial ratio explains variable drug response, since α-blockers target smooth muscle and 5-ARIs target epithelium.

5 · Pathophysiology of obstruction

Many symptoms come from the bladder, so some persist after surgery.

ALGORITHM

1. Hyperplasia raises urethral resistance (static and dynamic components).
2. The detrusor compensates, at a cost to storage function.
3. Storage LUTS appear, with or without impaired emptying.
4. In a minority, the detrusor decompensates.

- The static component (bulk under the capsule, elastic ECM) is **not** reduced by α-blockade. The dynamic component is smooth muscle tone via **α1a** receptors (the most abundant subtype).
- Other regulators include endothelin and **phosphodiesterases 4 and 5 (PDE4/5)**, which is why PDE5 inhibitors help LUTS.
- Many LUTS arise from **obstruction-induced bladder change**. ≈ **One-third** of men keep voiding dysfunction (mostly storage) after relief.

- Storage changes are detrusor overactivity (DO) and lower compliance; emptying changes are reduced contractility and, in a minority, detrusor failure.

ALGORITHM

1. **Three-stage remodelling**:
2. Smooth muscle hypertrophy.
3. Compensation.
4. Decompensation (underactive bladder).

- Stretch, pressure and **hypoxia/ischaemia** switch the detrusor to a **secretory phenotype** that lays down ECM.
- **Trabeculation** reflects increased detrusor **collagen** and progresses to cellules, diverticula and failure; there is no accepted grading.

KEY POINT

Storage LUTS persist after transurethral resection of the prostate (TURP) because the collagen-rich, overactive detrusor does not reverse, so counsel men with DO before surgery. Acute urinary retention (AUR) is **not** the inevitable end: many men in AUR have an adequate detrusor and a precipitant.

6 · Natural history

Who progresses decides watchful waiting versus treatment.

- **Regression to the mean**: entry thresholds select men at a temporary extreme, so they seem to improve. It is part of every placebo effect.
- Within trials, IPSS, Qmax and volume barely correlate (restriction of range); only PSA and volume still correlate.
- ≈ **40%** improve without treatment (watchful waiting behaves like placebo); active drug gives about twice the placebo improvement.
- **Olmsted County**: IPSS rises ≈ **0.2–0.34 points/year**; Qmax falls ≈ **2.1%/year** (faster if age ≥70 or Qmax <10); volume grows ≈ **0.6 mL/year** (faster with higher PSA or volume). Volume >50 mL gives ≈3.5-fold odds of moderate–severe LUTS.
- A perceptible improvement is ≈ −2 IPSS points at low baseline versus −6 at high baseline.

Predictor of progression	Predicts
Age	Everything; AUR rises linearly from age 40 to 80+
PSA ≥1.4–1.5 ng/mL	Growth, symptoms and flow, AUR, surgery
Volume >30–40 mL	AUR (≈3-fold), surgery, new LUTS
IPSS ≥8 or high bother	AUR, surgery
Qmax <10–12 mL/s	AUR, flow decline, treatment
Raised post-void residual (PVR)	Treatment failure, AUR

- **MTOPS** (Medical Therapy of Prostatic Symptoms) placebo arm: progression **4.5 per 100 patient-years** (17% at 4 yr), mostly an **IPSS rise ≥4** (14%); AUR 2%; invasive therapy 5%; **no renal insufficiency**.
- **PLESS** (Proscar Long-term Efficacy and Safety Study) placebo arm: symptoms drifted back to baseline. Surgery was 10% at 4 yr, rising from ≈6% to ≈15% across PSA and volume

tertiles.

- **REDUCE** placebo arm: volume 40–80 mL and Qmax <10 each gave a hazard ratio of 1.67 for new LUTS.

> **KEY POINT**
>
> PSA is a cheap surrogate for volume, the main driver of AUR and surgery; models with >100 predictors barely beat PSA alone.

7 · Complications

These are the absolute indications for surgery.

> **KEY POINT**
>
> Death, renal failure, stones and infection are now rare. The common events are **AUR and surgery**, predicted by age, symptoms, low flow and a large gland (or high PSA).

7.1 Acute urinary retention

- Precipitants include infection, overdistension, fluid load, alcohol and bed rest; prostatic infarction is **not** proven.

	Spontaneous AUR	**Precipitated AUR**
Trigger	None	Surgery, anaesthesia, catheterisation, sympathomimetics, anticholinergics, antihistamines
Later spontaneous AUR	15%	9%
Later surgery	**75%**	**26%**

- AUR incidence is ≈ **4.5–6.8/1000 person-years** in the community, 18 in PLESS placebo and 6 in MTOPS. The hazard is constant over time.
- **Age and symptoms multiply risk**: in Olmsted, men in their 70s with IPSS >7 had 34.7/1000 person-years. Relative risks: IPSS ≥8 3.2; Qmax <12 3.9; volume >30 mL 3.0; PSA >1.4 several-fold.

7.2 Other complications

- **Renal**: "**silent prostatism**" is asymptomatic renal failure from BOO; MTOPS had none in >3000 men. †Catheterise first, watch for post-obstructive diuresis, let creatinine stabilise, then operate.
- **Bladder stones** are ≈ **8×** commoner because of stasis; renal stones are not.
- **Haematuria** reflects increased microvessel density; **finasteride** is first-line because it lowers VEGF. †Exclude cancer and stones first.
- **UTI** is rare (MTOPS 0.1/100 patient-years). **Incontinence** is overflow or urge (DO in up to half).
- **Surgery** is a soft end point and AUR a hard one. In Olmsted, moderate–severe symptoms, Qmax <12, volume >30 and PSA ≥1.4 each gave ≈4× treatment.

8 · Landmark trials

Trial	Design	Result	Take-home
VA Cooperative Study	TURP vs watchful waiting, moderate symptoms	Failure RR 0.48; 5-yr 10% vs 21%; ≈36% crossed over; delayed TURP smaller gains	Watchful waiting safe; delay may reduce benefit
PLESS	Finasteride vs placebo, 4 yr	Placebo returns to baseline; AUR/surgery rise with PSA/volume	PSA/volume predict progression
MTOPS	Placebo vs doxazosin vs finasteride vs combination	Placebo progression 4.5/100 patient-years, mostly IPSS ≥4; no renal failure	Defined "clinical progression"
Olmsted County Study	Population cohort, 40–79	IPSS +0.2–0.34/yr, Qmax −2.1%/yr, volume +0.6 mL/yr; AUR 6.8/1000 person-years	Best natural history; age, IPSS, Qmax, volume predict AUR

150 Evaluation and Nonsurgical Management of Benign Prostatic Hyperplasia

1 · Definitions and the diagnostic problem

Not every man with urinary symptoms is obstructed; find the cause first.

- Benign prostatic hyperplasia (BPH) is **histological** proliferation in the **transition zone (TZ)**. It can lead to benign prostatic enlargement (BPE), benign prostatic obstruction (BPO, a form of bladder outlet obstruction, BOO) and lower urinary tract symptoms (LUTS).
- LUTS are multifactorial, including detrusor overactivity (DO) or underactivity (DUA), stricture and infection (UTI).

> **KEY POINT**
>
> Bothersome LUTS are not BPH-related BOO until assessed. Aims: **grade severity and bother** and **identify the cause**. Symptoms, flow, post-void residual (PVR), volume and cystoscopy cannot separate obstruction from bladder dysfunction; only the **pressure–flow study** can (gold standard, reserved for when it changes surgery).

Guidelines: American Urological Association (AUA) BPH (2021); European Association of Urology (EAU) non-neurogenic male LUTS (2022).

2 · History and symptom assessment

- **Comorbidities**: diabetes mellitus or insipidus, renal disease, heart failure, sleep apnoea, neurological disease, haematuria and prior lower-tract surgery.
- **Drugs**: loop diuretics, β-blockers, excess fluid and caffeine worsen storage; anticholinergics impair contractility.
- Assess **sexual function**. LUTS do **not** raise prostate cancer (PCa) risk.
- Storage symptoms (frequency, urgency, urge incontinence, nocturia) form **overactive bladder (OAB)**.

*See the paper answer **IPSS score** (Section 16, Ch 149).*

Also, from the condensed book (not in the paper answer):

- The AUA Symptom Index (AUASI) and International Prostate Symptom Score (IPSS) total **0–35**: mild **0–7**, moderate **8–19**, severe **20–35**.
- The minimum perceptible change is **3 points**; with a severe baseline, **percentage** reduction is more reliable.

3 · Examination and DRE

- Look for a palpable bladder, meatal stenosis and phimosis, and check **anal tone** and lower-limb neurology.
- **Digital rectal examination (DRE)** estimates volume and screens for PCa. It **underestimates prostate volume (PV) above 30 mL**; a flat gland suggests <30 mL.
- PV does not correlate with symptoms or obstruction, but the **30 mL** and **50 mL** cut-offs predict treatment response and progression.

4 · Frequency–volume charts and nocturia

- A frequency–volume chart (FVC) records times and volumes; a bladder diary adds intake and urgency. **Indication**: predominant **storage LUTS or nocturia**. **Duration**: **3 days**, since adherence falls by 7 days.
- **Nocturia** is waking ≥1 time to void. **Nocturnal polyuria** is night output **>20%** of 24-h output (young) or **>33%** (over 65).

ALGORITHM

1. **Nocturia**
2. Complete an FVC.
3. If †global polyuria (>40 mL/kg/24 h), look for diabetes or excess intake.
4. If nocturnal polyuria, look for heart failure, sleep apnoea, evening fluids or diuretic timing.
5. If reduced capacity, think of OAB or BOO.
6. Treat the cause. If nocturia persists, consider **sleep apnoea** (AUA).

Nocturnal polyuria: no fluid 2 h before bed, no evening alcohol or caffeine, retimed diuretics, with or without **desmopressin**. †It risks **hyponatraemia**, so check sodium at baseline, 1 week and 1 month.

5 · Laboratory tests

- **Urinalysis is mandatory** for UTI, microhaematuria, glycosuria and proteinuria. Gross haematuria needs exclusion of bladder cancer; **cytology** is for severe storage symptoms with dysuria, or smokers.
- **Prostate-specific antigen (PSA)** gives PCa risk, a volume surrogate and progression risk. Test only if life expectancy is **>10 years** and it changes management.
- PSA predicting **PV >40 mL**: **>1.6** (50s), **>2.0** (60s), **>2.3 ng/mL** (70s). Baseline PSA also predicts acute urinary retention (AUR), surgery, and change in symptoms and maximum flow rate (Qmax).
- **5α-reductase inhibitors (5-ARIs) lower PSA by 40–50% by 12 months**, so take a baseline first. †Double the PSA after ≥6 months; any rise from the nadir needs evaluation.
- **Creatinine and eGFR** are checked (EAU) if renal impairment or retention is suspected.

6 · Instrumental investigations

PVR

- A PVR **<30 mL** is insignificant; persistently **>50 mL** is important; **200–300 mL** may mean DUA and a poorer response; **>300 mL** is significant chronic retention.
- High PVR means BOO **or DUA**; it predicts symptom deterioration (MTOPS >39 mL; ALTESS >93 mL), **not surgery**.

Uroflowmetry

- Record **Qmax**, voided volume and pattern. Repeat if voided volume is **<150 mL**.
- It **cannot distinguish BOO from DUA** (DUA in 9–48%). Qmax **10 mL/s** is more specific; **15 mL/s** is more sensitive.
- Optional at first; **recommended before active treatment** and for monitoring.

SAQ Role of UDS before prostate surgery

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 14 Paper 2, Q3a

Why consider UDS (urodynamic study)

- Symptoms, flow rate, PVR (post-void residual), prostate volume and cystoscopy **cannot separate BOO (bladder outlet obstruction) from DUA (detrusor underactivity)**: low Qmax (maximum flow rate) occurs in both. DUA is found in 9–48% of men with LUTS (lower urinary tract symptoms) studied.
- Surgery helps obstruction, not a weak bladder. **PFS (pressure–flow study) is the gold standard** for BOO.

Components

- **Filling cystometry:** sensation, compliance, **DO (detrusor overactivity)**.
- **PFS:** detrusor pressure at Qmax plotted against flow (ICS (International Continence Society) nomogram).

Index	Formula	Interpretation
BOOI (bladder outlet obstruction index)	PdetQmax − 2Qmax	>40 obstructed; 20–40 equivocal; <20 unobstructed
BCI (bladder contractility index)	PdetQmax + 5Qmax	>150 strong; 100–150 normal; <100 weak

- High pressure, low flow = BOO; low pressure, low flow = DUA.
- **Videourodynamics** adds anatomy (diverticula, reflux).

Indications before BPH (benign prostatic hyperplasia) surgery (EAU (European Association of Urology))

- Previous failed invasive treatment.
- Cannot void >150 mL (unreliable flow).
- PVR >300 mL.
- Age >80 years with mainly voiding LUTS.
- Age <50 years with mainly voiding LUTS.
- **Also:** suspected neurogenic bladder, prior pelvic surgery or radiation, Qmax >10 mL/s (AUA (American Urological Association): diagnostic uncertainty).

How results change management

- **Unobstructed or DUA:** poorer surgical results; counsel, consider CIC (clean intermittent catheterisation).
- **DO with BOO:** storage symptoms persist in about one-third after TURP (transurethral resection of the prostate); warn the patient.
- **Low compliance:** the only "absolute" urodynamic indication to relieve BOO (protects upper tracts).

Evidence and limitations

- Not routine. **UPSTREAM RCT (randomised controlled trial):** routine UDS did not change IPSS (International Prostate Symptom Score) or surgery rates.
- Invasive: UTI (urinary tract infection), discomfort, cost. Before **radical prostatectomy**, not routine.

Diagram to draw: ICS nomogram with obstructed, equivocal and unobstructed zones.

> **Key points to remember:**
> - PFS is the only test that separates BOO from DUA.
> - BOOI >40 obstructed; BCI <100 weak.
> - EAU indications: failed surgery, <150 mL voided, PVR >300 mL, age <50 or >80.
> - UPSTREAM: routine UDS did not change outcomes.

Also, from the condensed book (not in the paper answer):

- Pressure–flow study (PFS) indications (EAU): failed invasive treatment; cannot void **>150 mL**; PVR **>300 mL**; age **>80** or **<50 years** with voiding LUTS; neurological disease or pelvic surgery. AUA: before surgery if **Qmax >10 mL/s**.
- DUA predicts long-term surgical failure.

Imaging and endoscopy

*See the paper answer **Methods to estimate prostate size** (Section 16, Ch 149).*

Also, from the condensed book (not in the paper answer):

- **Intravesical prostatic protrusion (IPP) ≥10 mm** is high grade; it predicts need for therapy and a **failed trial without catheter (TWOC)**. Bladder wall thickness and other non-invasive tests are accurate, but **none replaces PFS**.
- **Transrectal ultrasound (TRUS)** is the method of choice for PV, needed before a 5-ARI and **any BPH surgery**; a median lobe guides the choice of minimally invasive therapy.
- **Cystoscopy** neither diagnoses BOO nor decides treatment. Indications include haematuria, recurrent UTI, suspected stricture, and **excluding a median lobe before prostatic urethral lift (PUL)**.

7 · Evaluation algorithms

ALGORITHM

1. **EAU (men ≥40 yr)**
2. History (including sexual), symptom score, examination with DRE, urinalysis, PVR, and PSA if life expectancy >10 yr.
3. If DRE, PSA or urinalysis is abnormal, or neurological disease, follow the relevant guideline.
4. If not bothersome, no further work-up.
5. If PVR is significant, renal ultrasound with or without renal function.
6. If bothersome, FVC (storage or nocturia), then prostate ultrasound and uroflow if treating, then the medical algorithm.
7. Before surgery, endoscopy if it alters the choice; PFS for specific indications.

ALGORITHM

1. **AUA**
2. History, examination, IPSS, urinalysis, with or without PVR and flow.
3. If **PVR >300 mL**, manage as non-neurogenic chronic retention.
4. Counsel: behavioural, medical, procedural.
5. Reassess at **4–12 weeks**. If improved, follow up. If nocturia predominates, FVC and a drug. If incomplete, FVC, PVR and flow, urodynamics and cystoscopy, then the OAB pathway, surgery, or both.

8 · Watchful waiting (WW)

- **Indication**: mild–moderate uncomplicated LUTS, minimal bother. **Excluded**: gross haematuria, UTI, raised PSA or PVR, and neurological disease.
- WW is reassurance, education, lifestyle and **yearly** review; in the VA trial ≈a quarter crossed over because of **bother**.
- Lifestyle: fluids **1.5–2 L/day**, none 2 h before bed; less caffeine and evening alcohol; retime diuretics; **double voiding** and **urethral milking**.

> **KEY POINT**
>
> Treatment is driven by **bother and complicating features**, not by score or size alone.

9 · Medical therapy: choice

- Drugs are first line without an **absolute surgical indication** (refractory or recurrent AUR, recurrent UTI, renal insufficiency, stones, recurrent haematuria; see Ch 151).

LAQ BPH: medical management and surgical modalities

Paper answer · P2

Asked as:

- Describe medical management of BPH and role of various surgical modalities in its management. — *Summer 2015 Paper II Q1 (LAQ)*
- Medical treatment of BPH — *Practice Set 6 Paper II Q3f (SAQ)* (also answered in 2015-Summer Paper 2, Q1)

Type: LAQ (25 marks); also asked as SAQ (10 marks)

Introduction, pathophysiology and rationale

- **BPH (benign prostatic hyperplasia)** is stromal and epithelial hyperplasia of the transition zone; it may cause BPO (benign prostatic obstruction) and **LUTS (lower urinary tract symptoms)**.
- Obstruction has a **static** component (gland bulk, DHT (dihydrotestosterone)-dependent / DHT-driven gland bulk) and a **dynamic** component (dynamic part: α1A-mediated smooth-muscle tone; static part: DHT-driven gland bulk). Drugs target each. The bladder responds with detrusor overactivity or later underactivity.

Evaluation (before treatment)

- **IPSS (International Prostate Symptom Score)** 0–35: mild 0–7, moderate 8–19, severe 20–35.

- DRE (digital rectal examination), urinalysis, **PSA (prostate-specific antigen)**, frequency-volume chart, Qmax (maximum flow), PVR (post-void residual); PVR >300 mL is managed as chronic retention (PVR >300 mL = chronic retention).
- Measure prostate volume before surgery (AUA (American Urological Association)); pressure-flow study if unclear.

Medical management

Option / class	Doses	Key facts
Watchful waiting	—	Mild or non-bothersome LUTS; lifestyle advice
α1-blockers (first line)	Tamsulosin 0.4 mg; alfuzosin 10 mg; silodosin 8 mg; doxazosin 4–8 mg (titrate)	First line; act within days; no effect on gland size, AUR (acute urinary retention) or surgery (do not prevent AUR or surgery). Hypotension, ejaculatory dysfunction; silodosin causes most anejaculation. **IFIS (intraoperative floppy iris syndrome)**
5-ARIs (5α-reductase inhibitors)	Finasteride 5 mg; dutasteride 0.5 mg	For glands (prostate) >30–40 mL or PSA >1.5 ng/mL. Shrink the gland about 18–25% (another source: ~20%) (maximal at 6 months; act after 6 months), halve PSA. PLESS (Proscar Long-Term Efficacy and Safety Study): AUR or surgery down (↓) 51%. Low libido, ED (erectile dysfunction). Also BPH haematuria
Combination	—	For high-risk men. The α-blocker gives early relief; the 5-ARI prevents hard events
Antimuscarinic or β3-agonist (mirabegron, vibegron)	Add-on	Add for persisting storage symptoms; trials included only men with low PVR (PVR <150 mL)
Tadalafil 5 mg daily	PDE5 (phosphodiesterase-5) inhibitor	LUTS with or without (±) ED; not with nitrates
Desmopressin	—	Nocturnal polyuria (night urine >33% of 24-h output); check (monitor) sodium, especially ≥65 years

Combination therapy

- **MTOPS (Medical Therapy of Prostatic Symptoms)**: progression fell (↓) 39% (doxazosin), 34% (finasteride) and **66% (combination)**; only finasteride or combination (finasteride-containing arms) reduced AUR and surgery.
- **CombAT (Combination of Avodart and Tamsulosin)**: combination cut AUR or surgery by about 66% versus tamsulosin.
- Combine when **PV (prostate volume) >30–40 mL and PSA >1.5 ng/mL**; α-blocker may stop (withdrawable) after 6 months of combination (SMART (Symptoms Management After Reducing Therapy) trial).
- Tadalafil + finasteride fixed dose approved (FDA [Food and Drug Administration] 2021).

Side effects and important cautions

- **Side effects**: α-blockers cause dizziness, postural hypotension, ejaculatory dysfunction and **IFIS (intraoperative floppy iris syndrome)**, which persists after stopping, so avoid starting them before planned cataract surgery. 5-ARIs: low libido, ED, gynaecomastia.

- **IFIS:** billowing iris, prolapse, miosis; stopping the drug does not prevent it.
- 5-ARIs: double the PSA; not for cancer prevention (more high-grade tumours in PCPT [Prostate Cancer Prevention Trial] and REDUCE).
- **AUR**: catheterise (catheter), then TWOC (trial without catheter) on an α-blocker (ALFAUR (Alfuzosin in Acute Urinary Retention): success 62% vs 48%).
- **Also know:** phytotherapy (saw palmetto) not recommended by guidelines.

Surgical management

Indications: refractory retention, recurrent UTI (urinary tract infection) or haematuria, bladder stones, renal insufficiency from BPO, failed medical therapy (surgery if refractory).

Modality	Size (AUA 2023)	Key feature
TURP (transurethral resection), monopolar or bipolar	30–80 mL	Reference standard; retrograde ejaculation ~two-thirds
TUIP (transurethral incision)	≤30 mL	Less retrograde ejaculation; more reoperation
PVP (photoselective vaporization)	≤80 mL	Safe when anticoagulation cannot stop; no histology
HoLEP/ThuLEP (laser enucleation)	Any size	Lowest retreatment, least bleeding
Simple prostatectomy (open or robotic)	>80 mL	Complete removal; diverticula or stones
Aquablation	30–80 mL (consider 80–150)	Heat-free; preserves ejaculation
PUL (prostatic urethral lift), Rezūm (water vapour)	30–80 mL (PUL: no median lobe)	Preserve sexual function; higher retreatment
iTind (temporary implant)	25–75 mL, no median lobe	Removed after a few days; modest efficacy
PAE (prostate artery embolisation)	Larger glands (AUA 2023: may be offered)	Local anaesthesia; less effective than TURP; radiation, non-target embolisation

- **TUR syndrome** (monopolar TURP): hypo-osmolar glycine or water is absorbed through open sinuses, causing dilutional hyponatraemia, fluid overload and confusion. Risk rises with gland >45 mL and resection >90 minutes. Treat with diuretics and, if severe, 3% saline, correcting slowly. **Bipolar TURP** uses saline and eliminates it.
- **Complications**: bleeding, stricture, bladder neck contracture, retreatment.
- **GOLIATH**: 180 W PVP non-inferior to TURP. **WATER (Waterjet Ablation Therapy for Endoscopic Resection)**: Aquablation non-inferior, with less retrograde ejaculation (10% vs 36%).
- **Also know:** TUVP (transurethral vaporization); TUMT (transurethral microwave therapy) dropped as legacy; Optilume paclitaxel-coated balloon (2023).

Follow-up

- Reassess at 4–12 weeks (α-blocker), 3–6 months (5-ARI) (another source: 6 months for 5-ARIs), then yearly; surgery if refractory.

Diagram to draw: Size-based surgical algorithm (≤30, 30–80, >80 mL); algorithm: LUTS → size, PSA, storage symptoms → drug choice.

Key points to remember:

- α-blockers: rapid relief, no prevention.
- 5-ARIs: prevent AUR and surgery in glands >30–40 mL.
- MTOPS: combination cut progression 66% (↓66%); only 5-ARI-containing arms cut AUR and surgery.
- AUA sizes: TUIP ≤30; TURP/PVP/MISTs (minimally invasive surgical therapies) 30–80; HoLEP any size; simple prostatectomy >80 mL.
- TUR syndrome: gland >45 mL, resection >90 min; bipolar TURP avoids it.

Also, from the condensed book (not in the paper answer):

- Antimuscarinics need PVR <200 mL; avoid oxybutynin in the elderly; hexane-extract Serenoa is an EAU option.

ALGORITHM

1. **EAU**
2. Not bothersome: WW.
3. Nocturnal polyuria: lifestyle with or without desmopressin.
4. Storage predominant: antimuscarinic or β3-agonist.
5. If **PV >40 mL and long-term treatment** intended: 5-ARI with or without an α1-blocker or phosphodiesterase-5 inhibitor (PDE5I).
6. Otherwise α1-blocker or PDE5I; if storage symptoms remain, add an antimuscarinic or β3-agonist. **AUA**: α-blocker (PDE5I if erectile dysfunction, ED) → **prostate >30 mL** → add 5-ARI → PDE5I trial → surgery.

KEY POINT

Reading trials: placebo improves scores and flow, so judge by a 3-point or percentage threshold; the goal includes **preventing progression**.

10 · α1-Blockers

SAQ Tamsulosin

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 12 Paper 2, Q3d

Introduction

- A **uroselective α1-adrenergic antagonist**, first-line drug for bothersome male LUTS (lower urinary tract symptoms) due to BPH (benign prostatic hyperplasia) (AUA [American Urological Association], EAU [European Association of Urology]).

Receptor basis

Subtype	Site	Effect of blockade
α1a (about 70% in prostate)	Prostate, bladder neck, prostatic urethra, iris dilator	Relieves dynamic obstruction; ejaculatory dysfunction; IFIS (intraoperative floppy iris syndrome)

Subtype	Site	Effect of blockade
α1b	Blood vessels	Hypotension (tamsulosin has 10-fold less affinity)
α1d	Detrusor, spinal afferents	Storage symptom relief

Mechanism

- Blocks noradrenaline at α1a receptors → relaxes prostatic and bladder-neck smooth muscle → lowers the **dynamic component** of BOO (bladder outlet obstruction). It does not shrink the prostate.

Pharmacology and dose

- **0.4 mg once daily** after the same meal (up to 0.8 mg); no titration.
- Modified release; hepatic metabolism (CYP3A4, CYP2D6); avoid strong CYP3A4 inhibitors (ketoconazole).
- Flow rises within 4–8 hours, symptoms by 1 week; 0.2 mg is not better than placebo.

Indications

1. **BPH/LUTS:** IPSS (International Prostate Symptom Score) falls about 30–40%; Qmax rises; storage and nocturia improve.
2. **Combination with 5ARI** (5α-reductase inhibitor) for prostate >30–40 mL: **CombAT** (dutasteride + tamsulosin) and MTOPS (combination cut progression by 66%).
3. **MET (medical expulsive therapy):** distal ureteric stones 5–10 mm; SUSPEND (2015) negative overall but included all sizes.
4. **AUR (acute urinary retention):** before trial without catheter.
5. **Also know:** ureteric stent symptoms; female functional outlet obstruction.

Adverse effects

- **Ejaculatory dysfunction** (dose-dependent; less than silodosin): retrograde ejaculation from bladder-neck relaxation, or anejaculation (emission failure).
- **IFIS:** floppy, billowing iris, iris prolapse, progressive miosis during cataract surgery; **highest risk with tamsulosin**. Stopping the drug does not remove the risk, so warn the ophthalmologist.
- Dizziness, rhinitis, headache; less orthostatic hypotension than non-selective agents. Caution: sulfonamide allergy.

Comparison

	Tamsulosin	Silodosin	Alfuzosin
Selectivity	α1a/α1d	Most α1a	Clinical only
Ejaculatory effect	Moderate	Highest	Lowest

Diagram to draw: prostate cross-section showing α1a receptors in stroma and bladder neck.

> **Key points to remember:**
>
> - 0.4 mg once daily; α1a/α1d selective; no titration.
> - IFIS highest with tamsulosin; persists after stopping.
> - MET benefit in distal stones 5–10 mm.

SAQ Silodosin

Paper answer · P2 · P4 · merged from 2

Asked as: P4 SAQ "Silodosin" (Silodosin.) · P2 SAQ "Silodosin."

Definition and pharmacology

- **Uroselective α1-blocker**, the **most α1A-selective** agent (well over 100-fold vs α1B, or by another source several hundredfold over α1B; about 50-fold vs α1D), for LUTS (lower urinary tract symptoms) due to BPH (benign prostatic hyperplasia).
- Sparing vascular α1B gives **minimal hypotension** (few blood-pressure effects). Blocking α1A in the vas and seminal vesicles causes **failure of emission**.
- Onset: flow improves within 2–6 hours, symptoms within 3–4 days. Metabolised by CYP3A4.

Mechanism of action

- **α1A receptors** (about 70% of prostatic α1 receptors) dominate in the prostate, bladder neck and prostatic urethra. Noradrenaline on them causes the **dynamic component** of BOO (bladder outlet obstruction).
- Blockade relaxes this smooth muscle.

Dose and contraindications

Item	Detail
Dose	**8 mg once daily with a meal**; no titration
Renal: CrCl (creatinine clearance) 30–50 mL/min	**4 mg daily**
Contraindicated	CrCl <30; severe liver (hepatic) impairment; strong CYP3A4 inhibitors (e.g. ketoconazole)
Caution	Other α-blockers (avoid), PDE5 (phosphodiesterase 5) inhibitors, planned cataract surgery

Indications and efficacy

Indication	Evidence
LUTS due to BPE (benign prostatic enlargement)	Japanese, American and European 12-week RCTs (randomised controlled trials): IPSS (International Prostate Symptom Score) and flow better than placebo; **non-inferior to tamsulosin 0.4 mg** (**Chapple 2011**, European trial), not better; effective in severe LUTS
MET (medical expulsive therapy)	Distal ureteric stones 5–10 mm; α-blockers reduce ureteric spasm; meta-analyses show higher, faster expulsion than tamsulosin; **SUSPEND (2015)** found no overall tamsulosin benefit
CP/CPPS (chronic prostatitis/chronic pelvic pain syndrome)	α-blocker-naive men with voiding symptoms; of recent onset

- **Greatest fall in BOOI (bladder outlet obstruction index)** of all α1-blockers (subgroup data).
- Does not shrink the prostate; add a 5-ARI (5α-reductase inhibitor) for large glands.
- **Also know:** before ureteroscopy (easier access); with propiverine for storage symptoms; nocturia (modest).

Adverse effects

- **Ejaculatory dysfunction / anejaculation (signature effect):** over a quarter of men in registration trials (about 14% vs 2% with tamsulosin head-to-head), yet few stop the drug; men with anejaculation had better responses. Mechanism: **emission failure** (vas and seminal vesicle α1A blockade) and bladder-neck relaxation. Reversible. Counsel young men.
- **Cardiovascular:** dizziness; orthostatic hypotension rare; no QT change; no excess hypotension in the elderly or with antihypertensives.
- Falls in older men: falls and fractures rise in men ≥66 on any α-blocker.
- **IFIS (intraoperative floppy iris syndrome):** not specifically studied but reported; class effect; warn the ophthalmologist before cataract surgery (stopping the drug does not remove risk).

Comparison with other α-blockers

Drug	Selectivity	Dose	Hypotension	Ejaculatory dysfunction	Efficacy in LUTS
Silodosin	α1A (highest); α1A ≫ α1D, α1B	8 mg OD	Lowest	**Highest**	Equal
Tamsulosin	α1A, α1D > α1B	0.4 mg OD	Low	Moderate	Equal
Alfuzosin	Clinically uroselective	10 mg OD	Low	Lowest	
Naftopidil	α1D	25–75 mg	Some	Less	
Doxazosin, terazosin	Non-selective; titrate	1–8 mg, 1–10 mg	Highest	Low	

Diagram to draw

- α1A in prostate and vas, α1B in vessels, α1D in bladder; prostate and bladder neck showing α1A receptor sites blocked by silodosin.

> **Key points to remember:**
> - Most α1A-selective α-blocker; 8 mg with a meal; no titration.
> - 4 mg if CrCl 30–50; avoid if <30.
> - Equal efficacy to tamsulosin 0.4 mg; greatest BOOI fall; better for MET.
> - Anejaculation is the signature effect; least hypotension.

Also, from the condensed book (not in the paper answer):

- **No α1-blocker is clearly more effective**; they differ in adverse effects (AEs). Doxazosin cut progression 39% (MTOPS), **not AUR or surgery**.
- **ALLHAT**: the doxazosin arm stopped for heart failure, so never use it as sole antihypertensive; **never use two α-blockers**.

Class	Drug	Dose	Notes
Non-selective	Phenoxybenzamine	10 mg bd	α1+α2; abandoned
Short-acting	Prazosin, indoramin	bd	Not used for LUTS
Long-acting	Terazosin	5–10 mg od	Start **1 mg at bedtime**, titrate
	Doxazosin immediate release (IR)	2–8 mg od	Start 2 mg at bedtime, titrate

Class	Drug	Dose	Notes
	Doxazosin GITS (sustained release)	4–8 mg od	Start **4 mg**, **no titration**
Uroselective	Alfuzosin extended release (ER)	10 mg od	Clinically uroselective
	Tamsulosin	0.4 mg od (to 0.8)	α1a/α1d
	Silodosin	8 mg od	Most α1a-selective
	Naftopidil	25–75 mg/day	Only α1d-selective; Asia

KEY POINT

Titration: α1b blockade causes **first-dose hypotension** and syncope; a bedtime start and titration avoid it. Uroselective drugs are fixed-dose; doxazosin GITS starts at 4 mg without titration.

- They double the placebo symptom gain with **durable** benefit but do **not** shrink the gland; tamsulosin plateaus at **0.4 mg**.
- **AEs**: dizziness, asthenia, orthostatic hypotension and anejaculation. In the **elderly**, dizziness causes falls; severe hypotension is mostly in the **first 8 weeks**.
- **Sexual**: ED equals placebo. Ejaculatory dysfunction is mostly **anejaculation** (emission failure): silodosin (odds ratio ≈32) > tamsulosin (≈9) > alfuzosin.

KEY POINT

Intraoperative floppy iris syndrome (IFIS): **billowing iris, iris prolapse, progressive miosis**. α1a blockade causes dilator atrophy, so IFIS occurs **years after stopping**; stopping pre-op does **not** help. Highest with **tamsulosin**, then alfuzosin, doxazosin and naftopidil. †If cataract surgery is awaited, defer the α1-blocker.

Drug	Vascular	Ejaculatory	IFIS
Terazosin	High	Low	Reported
Doxazosin	High (GITS lower)	Low	Lower
Alfuzosin	Intermediate	Lowest uroselective	Intermediate
Tamsulosin	Low	Dose-dependent	**Highest**
Silodosin	Lowest	**Highest**	Not well studied

KEY POINT

One efficacy, three toxicities: **α1b blockade gives vascular effects** (terazosin, doxazosin); **α1a blockade affects ejaculation and causes IFIS** (silodosin, tamsulosin); **α1d blockade gives storage benefit** (naftopidil).

11 · 5α-Reductase inhibitors

The only drugs that prevent retention and surgery.

SAQ Dutasteride.

Paper answer · P2

Type: SAQ (10 marks)

Source: 2018-Summer Paper 2, Q3c

Pharmacology

- A **dual 5ARI (5α-reductase inhibitor)**: it blocks **type 1** (liver, skin) and **type 2** (prostate stroma, genital tissue) isoenzymes. Finasteride blocks type 2 only.
- **Mechanism:** blocks conversion of testosterone to DHT (dihydrotestosterone), the main prostatic androgen; serum DHT falls about **90%**. Epithelial atrophy shrinks the gland and relieves the **static** component of obstruction.
- **Dose: 0.5 mg once daily**; hepatic metabolism; **half-life about 5 weeks**.

Effects

- **Prostate volume falls about 25%** (maximum by 6 months).
- Symptoms and Qmax (maximum flow rate) improve modestly and slowly (**at least 6 months**).
- **Changes natural history:** pivotal trials: AUR (acute urinary retention) down 57%, BPH (benign prostatic hyperplasia) surgery down 48%.
- **PSA (prostate-specific antigen) halves** by 6–12 months: **double the PSA** to interpret it. A rise from the nadir on treatment needs evaluation.

Indications

- Moderate-to-severe LUTS (lower urinary tract symptoms) with an enlarged prostate: **>40 mL (EAU [European Association of Urology])** or PSA about >1.5 ng/mL; AUA (American Urological Association): >30 mL.
- **Combination with tamsulosin** for men at risk of progression (fixed-dose capsule available).
- Off-label: BPH haematuria (lowers prostatic microvessel density); before TURP (transurethral resection of the prostate) to reduce bleeding (inconsistent evidence).

Landmark trials

Trial	Result
CombAT (Combination of Avodart and Tamsulosin, 4 years; PV [prostate volume] ≥30 mL, PSA ≥1.5)	Combination best for symptoms and progression; dutasteride (and combination) beat tamsulosin for AUR and surgery
REDUCE (Reduction by Dutasteride of Prostate Cancer Events, 4 years)	Prostate cancer down about 23%, but slightly more Gleason 8–10; **not licensed for prevention**

Side effects

- **Sexual:** ED (erectile dysfunction), low libido, ejaculatory disorders; lessen with time.
- Gynaecomastia and breast tenderness; depression (FDA [Food and Drug Administration] warning).
- **No IFIS** (intraoperative floppy iris syndrome) and no hypotension, unlike α-blockers.

Precautions

- **Teratogenic:** pregnant women must not handle leaking capsules. **No blood donation for 6 months** after the last dose.

Diagram to draw: testosterone to DHT pathway with 5α-reductase types 1 and 2 blocked by dutasteride.

Key points to remember:

- Dual type 1 + 2 inhibitor; 0.5 mg; DHT down 90%; half-life about 5 weeks.
- Volume down 25%; AUR and surgery down about half; needs 6 months.
- Double the PSA; CombAT and REDUCE.

Also, from the condensed book (not in the paper answer):

- **Finasteride 5 mg** (type 2) lowers dihydrotestosterone (DHT) by †~70% and PV by 18–20%; in PLESS, AUR fell 57% and surgery 55%. It performs the same as dutasteride head to head.
- Indication: moderate–severe LUTS with PV >40 mL (EAU) or >30 mL (AUA).
- They relieve the **static component**. Benefit is **slow** (maximum shrinkage at **6 months**) and **volume dependent**, best at **>40 mL**.
- They are the **only class altering natural history**.
- PCPT and REDUCE showed fewer cancers but slightly more Gleason ≥8, so they are **not licensed for prevention**.
- **AEs**: ED, lower libido and lower ejaculate volume (≈2× placebo, waning after 1–2 yr), **depression** (FDA 2011) and rarely gynaecomastia. **Post-finasteride syndrome** is persistence of symptoms after stopping.

KEY POINT

5-ARIs treat the gland: slow, volume-dependent and modest for symptoms, but they prevent AUR and surgery, and they halve PSA (both a trap and a tool).

12 · Antimuscarinics

- They block detrusor **M2/M3** receptors, mainly on **afferent** pathways.
- **Indication**: predominant storage LUTS, **PVR <200 mL** and no prior AUR.
- Agents: darifenacin 7.5/15 mg, fesoterodine 4/8 mg, oxybutynin (IR 5 mg 3–4×/day; ER), propiverine, solifenacin 5/10 mg, tolterodine ER 4 mg and trospium.
- In BOO, PVR rises slightly and AUR equals placebo (trials excluded PVR ≥200).
- AEs: **dry mouth**, constipation and cognitive impairment. **Oxybutynin** has most brain penetration, so avoid it in the elderly; trospium least†.

KEY POINT

Antimuscarinics treat the bladder, so every recommendation is anchored to a PVR ceiling (200 mL).

13 · β3-Agonists

- β3 stimulation gives **storage relaxation without weakening voiding** (Qmax unchanged), with less dry mouth than antimuscarinics.
- **Mirabegron** starts at **25 mg** and rises to **50 mg**; in BOO, PVR rose only at 100 mg.
- AEs: nasopharyngitis, headache, blood pressure rise ≤1 mm Hg. **Contraindicated** in uncontrolled hypertension **≥180/110 mm Hg**; it is a moderate CYP2D6 inhibitor†.
- **Vibegron** 75 mg has no CYP2D6 inhibition; data are mainly in women.

14 · PDE5 inhibitors

- **Tadalafil 5 mg daily** is the only licensed agent (2.5 mg is ineffective). It lowers IPSS by ≈6 (placebo 3.6), **independent of ED**, and reduces nocturia. **Qmax is unchanged**.
- First choice for LUTS with ED (AUA). AEs: headache, back pain, dyspepsia, flushing. †Contraindicated with nitrates and riociguat.

15 · Combination therapy

- **MTOPS** (Medical Therapy of Prostatic Symptoms): progression = AUASI rise **≥4**, AUR, UTI or urosepsis, incontinence, or creatinine rise ≥1.5 mg/dL. Placebo 17%. Number needed to treat (NNT) for combination 8.4 (≈5 if PSA >4 and PV >40).
- **CombAT** (PV ≥30, PSA ≥1.5): AUR or surgery fell two-thirds versus tamsulosin but not significantly (NS) versus dutasteride.
- **Indication**: high risk, PV **>30–40 mL** and PSA **>1.5 ng/mL**. **SMART**: stop the α-blocker after 6 months.

MTOPS reduction vs placebo	Doxazosin	Finasteride	Combination
Overall progression	39%	34%	66%
≥4-point symptom rise	45%	30%	64%
AUR	NS	68%	81%
Invasive therapy	NS	64%	67%

KEY POINT

α-blockers prevent symptomatic progression only; 5-ARIs prevent AUR and surgery. The α-blocker gives fast dynamic relief; the 5-ARI gives slow static shrinkage.

Combination	Indication	Caution
α1-blocker + antimuscarinic	Persistent storage LUTS (tolterodine 4-arm; NEPTUNE)	PVR <150–200 mL only; AUR <1–3%
α1-blocker + β3-agonist	Persistent storage (MATCH)	PVR rise; limited data
α1-blocker + PDE5I	LUTS with or without ED; small gain	Hypotension; not recommended; start lowest dose
PDE5I + 5-ARI	Large prostate with or without ED	Single trial; FDA capsule 2021

16 · Phytotherapy

- Evidence is product-specific. Most **Serenoa repens** extracts show no benefit, but the **hexane extract (Permixon) 320 mg daily with food** reduced nocturia and raised Qmax and is an EAU option. AUA/EAU make **no general recommendation** for phytotherapy.

17 · Follow-up

- Worsening symptoms and rising PVR predict AUR.

ALGORITHM

1. Review at **4–12 weeks** (α-blocker response, AEs).
2. Review at **6 months** (5-ARI).
3. Then **annually**: score, flow, PVR; PSA on a 5-ARI if life expectancy >10 yr. WW: yearly.

18 · Acute urinary retention

SAQ Catheter free trial

Paper answer · P2

Type: SAQ (10 marks)

Source: 2016-Summer Paper 2, Q3e

Assumed meaning: trial without catheter (TWOC) after acute urinary retention.

Definition

- **TWOC (trial without catheter)**: catheter removal after decompression to test whether the patient voids adequately.
- **AUR (acute urinary retention)**: sudden, painful inability to void; commonest in older men with LUTS (lower urinary tract symptoms).

Types of AUR

Spontaneous	Precipitated
BPH (benign prostatic hyperplasia) progression	Anaesthesia, surgery, UTI (urinary tract infection), overdistension, anticholinergic or sympathomimetic drugs
TWOC fails more often; about 75% later need surgery	Better TWOC outcome; about a quarter need surgery

Initial management

- Catheterise; record drained volume; check creatinine; treat the trigger.
- Start an **α1-blocker** (alfuzosin or tamsulosin).

Rationale for α-blockers

- Sympathetic bladder-neck tone (dynamic component) adds to retention; α1-blockers relax it.
- **ALFAUR (Alfuzosin in Acute Urinary Retention)**: alfuzosin 10 mg for 3 days → TWOC success **62% vs 48%**.
- Meta-analysis: success relative risk about 1.5; fewer recurrences.

Procedure

- Remove the catheter after **2–3 days of α-blocker**, usually in the morning (no evidence favours short or long catheterisation).
- Encourage fluids; record voided volumes and **PVR (post-void residual)** by bladder scan.
- **Success**: comfortable voiding, low residual (unit threshold). **Failure**: recatheterise or start CISC (clean intermittent self-catheterisation).

Predictors of failure

- Advanced age; **drained volume >1 L**; spontaneous AUR; severe LUTS.
- Ultrasound: **IPP (intravesical prostatic protrusion) ≥10 mm**; thick bladder wall.

Results

- Success about **60%**; a second or third TWOC still succeeds in over a quarter.
- Recurrence after success is common, mostly within **6 months**; then offer surgery.

After TWOC

- **Success**: continue α-blocker; add a **5-ARI (5α-reductase inhibitor)** for large glands, because 5-ARIs, not α-blockers, prevent further AUR.
- **Failure**: TURP (transurethral resection of prostate) or other surgery; CISC or long-term catheter if unfit.
- **Also know:** immediate surgery has higher morbidity; TWOC also follows prostate surgery.

Diagram to draw: algorithm: AUR → catheter + α-blocker → TWOC at 2–3 days → success / failure.

> **Key points to remember:**
> - TWOC is first choice after AUR; success about 60%.
> - ALFAUR: alfuzosin 10 mg raises success to 62% vs 48%.
> - Fails more with age, >1 L drained, spontaneous AUR; 5-ARI prevents recurrence.

Also, from the condensed book (not in the paper answer):

- CombAT: AUR fell 68% versus tamsulosin but only 18% (NS) versus dutasteride; α-blockers do not prevent AUR (ALTESS, MTOPS).
- About two-thirds of AUR is spontaneous, with a worse TWOC outcome than precipitated AUR.

ALGORITHM

1. Catheterise (†suprapubic if urethral fails).
2. Start an **α1-blocker**.
3. Do a **TWOC at ~3–5 days**.
4. If he voids, continue (add a 5-ARI for a large gland).
5. If he fails, try up to **3 TWOCs**.
6. If AUR recurs or TWOC fails: PFS and surgery (see Ch 151).

- TWOC success is ≈ **60%**, and recurrence is mostly within **6 months**.
- Failure predictors: age, severe LUTS, drained volume **>1 L**, spontaneous AUR and high-grade IPP. 5-ARIs prevent AUR (PLESS, MTOPS).

> **KEY POINT**
>
> **The α-blocker rescues the TWOC** (dynamic component); **the 5-ARI prevents the next episode** (static component).

19 · Landmark trials

Trial	Design	Result	Take-home
VA transurethral resection (TURP) vs WW	Randomised trial, moderate LUTS	TURP better; most WW did not fail; crossover by bother	WW safe

Trial	Design	Result	Take-home
PLESS	Finasteride vs placebo, 4 yr, PV ~55 mL	AUR 7% to 3%, surgery 10% to 5%; combined fall 51%	5-ARI prevents AUR/surgery
Dutasteride phase III	2 yr, PV ≥30 mL	DHT fell 90%, PV 26%, AUR 57%, surgery 48%	≈ finasteride
MTOPS	Doxazosin/finasteride/combination/placebo, ≥4 yr	Progression fell 39/34/66%; AUR/surgery only with 5-ARI	Combination for high risk
CombAT	Dutasteride + tamsulosin, PV ≥30, PSA ≥1.5, 4 yr	AUR/surgery fell 66% vs tamsulosin	Combination for large gland
SMART	α-blocker withdrawal	No deterioration after 6 months	Stop α-blocker at 6 months
PCPT / REDUCE	5-ARI prevention	Fewer cancers, more high grade	Not licensed
ALFAUR	Alfuzosin vs placebo before TWOC	Success 62% vs 48%	α-blocker before TWOC

151 Minimally Invasive and Endoscopic Management of Benign Prostatic Hyperplasia

1 · Scope and principles

- **American Urological Association (AUA) 2023 operations**: transurethral incision (TUIP), resection (TURP) and vaporisation (TUVP) of the prostate; photoselective vaporisation (PVP); prostatic urethral lift (PUL; UroLift); water vapour thermal therapy (WVTT; Rezūm); temporary implanted prostatic device (TIPD; iTind); robotic waterjet treatment (RWT; Aquablation); holmium or thulium laser enucleation (HoLEP/ThuLEP); simple prostatectomy; prostate artery embolisation (PAE).
- Transurethral microwave therapy (TUMT) was removed in 2023. Because the sham effect is large, new devices need sham-controlled trials.

> **KEY POINT**
>
> Volume predicts progression (acute urinary retention (AUR), surgery) but correlates poorly with symptoms. **Symptoms decide whether to operate; size and shape decide which operation.** A gland >30 cc triples AUR risk (Olmsted County).

2 · Work-up and indications for surgery

- History, examination, IPSS and urinalysis at baseline (Statement (St) 1). International Prostate Symptom Score (IPSS) bands: **0–7 mild, 8–19 moderate, 20–35 severe**.
- Before surgery, post-void residual (PVR) is **required** (St 6). **Consider** size and shape by transrectal ultrasound (TRUS), abdominal ultrasound, cystoscopy, MRI or CT (St 5), uroflowmetry (St 7), and pressure–flow study if the diagnosis is uncertain (St 8).
- A raised prostate-specific antigen (PSA) needs biopsy **before** BPH surgery. Counsel on failure and retreatment (St 9).

AUA Statement 26 (surgery recommended)	† European Association of Urology (EAU): absolute vs relative
Renal insufficiency from BPH; refractory retention; recurrent urinary tract infections (UTIs); recurrent bladder stones or gross haematuria from BPH; lower urinary tract symptoms (LUTS) refractory to, or unwilling to take, medical therapy	**Absolute** (complicated BPH): refractory retention, upper-tract dilatation or renal impairment, recurrent UTI, bladder stones, recurrent haematuria refractory to 5α-reductase inhibitor (5-ARI). **Relative**: moderate–severe bothersome LUTS failing or declining medical therapy

- **Antibiotics (AUA 2019)**: one dose suffices for most transurethral cases. Cover gram-negatives and enterococci (1st/2nd-generation cephalosporin or co-trimoxazole), guided by antibiogram and urine culture. **Defer surgery if UTI is active and symptomatic**.

3 · Choosing the operation

Volume / situation	AUA 2023 options	Restrictions
<30 cc	TUIP, TURP, TUVP, PVP, HoLEP, ThuLEP, TIPD	TUIP only ≤30 cc; TIPD 25–75 cc, no obstructing middle lobe
30–80 cc	TURP, TUVP, PVP, HoLEP, ThuLEP, PUL, WVTT, RWT, TIPD	PUL: no obstructing median lobe; WVTT and RWT treat median lobes

Volume / situation	AUA 2023 options	Restrictions
>80–150 cc / >150 cc	HoLEP, ThuLEP, simple prostatectomy	TURP, PVP, TUVP stop at 80 cc
Higher bleeding risk	HoLEP, ThuLEP, PVP (St 43)	Otherwise hold anticoagulant/antiplatelet after liaising with prescriber
Preserve erection and ejaculation	PUL or WVTT (St 37)	Within 30–80 cc

> **KEY POINT**
>
> Only enucleation spans every size band, joined above 80 cc only by simple prostatectomy. Minimally invasive surgical therapies (MISTs) and RWT live in 30–80 cc, so volume and median lobe assessment come first.

4 · Monopolar TURP (m-TURP): the benchmark

Every newer operation is judged against TURP.

SAQ TURP instruments, techniques and monopolar vs bipolar

Paper answer · P1 · P2 · merged from 2

Asked as: P1 SAQ "TURP instruments and techniques" (TURP instruments; TURP techniques) · P2 SAQ "Monopolar v/s Bipolar TURP."

Introduction

- **TURP (transurethral resection of the prostate):** an electrified (wire) loop removes transition-zone adenoma **from bladder neck to verumontanum, down to the surgical capsule**. The benchmark BPH (benign prostatic hyperplasia) operation; gold standard for glands up to 80 mL (AUA [American Urological Association] 2023); larger glands need enucleation or simple prostatectomy.

Resectoscope

Part	Description
Outer sheath	24–28 Fr; **insulated beak** prevents arcing to the urethra
Inner sheath	Creates **continuous flow**: inflow through the inner, outflow between sheaths (passive or suction). Lower bladder pressure, clear vision, no repeated emptying
Obturator	Blunt or visual; atraumatic passage
Working element	Moves the loop. **Passive:** spring holds loop inside sheath; cutting stroke by spring recoil. **Active:** surgeon's thumb pulls the extended loop in
Telescope	**Hopkins rod-lens**, usually **30°** (12° alternative)

Electrodes

- **Cutting loop:** tungsten (monopolar); **platinum–iridium (bipolar plasma)**. The plasma loop cuts at lower temperature and seals vessels.
- Roller ball or barrel (coagulation), vaporisation button, **Collins knife** (TUIP [transurethral incision of the prostate]).

Energy: generator and waveforms

- **Generator:** about 500 kHz, too fast to stimulate nerves.

- **Cut:** continuous waveform. **Coagulation:** interrupted, higher voltage. **Blend:** better haemostasis, cuts less well.

Monopolar versus bipolar: principles

- **Monopolar (m-TURP):** current runs from the loop through the patient to a grounding (dispersive) pad. Saline would disperse the current, so a **non-conducting, hypo-osmolar irrigant** (glycine 1.5%, sorbitol–mannitol, water) is needed.
- **Bipolar (b-TURP):** active and return poles are on the same instrument (the loop), so current stays local and **normal saline** can be used. The Plasma system turns saline into a plasma layer (orange glow) that cuts at lower temperature and seals vessels.
- **AUA:** no preference between monopolar and bipolar; choose by expertise.

Monopolar versus bipolar: comparison

Feature	Monopolar	Bipolar
Current path	Loop → patient → **dispersive pad**	Active and return poles on the loop; stays within the loop
Irrigant	**Non-conducting, hypo-osmolar:** glycine 1.5%, sorbitol, water	**Normal saline**
TUR (transurethral resection) syndrome	Yes; keep resection <90 min	Practically (virtually) eliminated
Resection time	Limited (speed over haemostasis)	No time pressure; large glands
Loop	Tungsten	Platinum–iridium (Plasma)
Bleeding, transfusion, clot retention	Higher	Lower
Catheter time, hospital stay	Longer	Shorter
Obturator reflex	Risk	Reduced
Efficacy: IPSS (International Prostate Symptom Score), QoL, PVR	Equivalent (Qmax marginally better with bipolar)	Equivalent
Stricture, BNC (bladder neck contracture), sexual function, retreatment	Similar	Similar

Imaging and irrigation

- **Light source:** xenon or LED (light-emitting diode) with fibreoptic cable; camera and monitor.
- **Irrigant** at body temperature; bag about **60 cm** high (each extra 10 cm more than doubles absorption); lowest height giving vision.

Accessories

- **Ellik evacuator or Toomey syringe** for chips.
- Van Buren sounds or Otis urethrotome: dilate meatus 2 Fr above sheath size (prevents stricture).
- **Three-way Foley catheter** (22–24 Fr) with catheter guide; continuous irrigation set.

Preparation

- Sterile urine; spinal or general anaesthesia; lithotomy with peroneal padding.
- **Inspect bladder first** (tumour, diverticula); identify ureteric orifices, verumontanum, external sphincter.

Resection techniques

- **Median lobe first**, then lateral lobes.
- **Modified Nesbit "encirclement":** start at 11–9 and 1–3 o'clock; expose but do not resect bladder neck fibres.
- **Floor-first:** channel at 5 or 7 o'clock to the capsule (sets depth early); widen laterally; lateral lobes "fall" into the fossa.
- **Anterior start:** 12 o'clock bladder neck, then quadrants down to the apex.
- **Chips:** long "canoe-like" chips with synchronised rocking.
- **Anterior tissue and apex last** in a bloodless field; leaving a little tissue is better than incontinence.
- Haemostasis: arterial bleeders must be fulgurated; venous ooze settles with catheter traction.
- Evacuate chips (Ellik evacuator); three-way catheter.
- **Also know:** TUVP (transurethral vaporisation, button electrode; ≤80 mL); TUIP (transurethral incision, <30 mL, no median lobe); bipolar enucleation; high bleeding risk: AUA prefers HoLEP (holmium laser enucleation) or PVP (photoselective vaporisation).

Troubleshooting a loop that will not cut

`Scope–generator connection → foot pedal → irrigant matches generator (glycine for monopolar, saline for bipolar) → patient grounding (pad contact)`

TUR syndrome (monopolar)

- Irrigant absorbed through open sinuses (venous pressure about 10 mmHg) → **dilutional hyponatraemia**: confusion, visual disturbance, nausea, bradycardia, hypertension, seizures; glycine also yields ammonia.
- **Risk factors:** gland >45 mL, resection >90 minutes, high irrigation bag.
- **Treatment:** stop surgery, check sodium; 3% saline 100 mL boluses for neurological symptoms; avoid over-rapid correction (central pontine myelinolysis).

Outcomes and evidence

- About 80% symptom improvement; low retreatment; retrograde ejaculation about two-thirds.
- Meta-analyses: monopolar and bipolar have equal efficacy up to 5 years; b-TURP safer (NNT about 50 to prevent one TUR syndrome, about 20 to prevent one clot retention).

Recent advances

- Bipolar plasma resection; chip-on-tip digital optics; laser enucleation scopes; single-use resectoscopes.
- **Also know:** TURis (sheath as return) caused strictures; bipolar enucleation and vaporisation (PlasmaButton).

Diagrams to draw

- Assembled resectoscope labelling sheaths, working element, loop and flow channels.
- Resection field with clock positions, bladder neck, verumontanum and sphincter.
- Monopolar circuit (loop → patient → pad) versus bipolar (loop to return electrode on the same resectoscope).

> **Key points to remember:**
>
> - Continuous-flow 24–28 Fr sheath, 30° telescope.
> - Tungsten loop (monopolar) vs platinum–iridium (bipolar).
> - Glycine 1.5% for monopolar, saline for bipolar; bag 60 cm.
> - Bladder neck to verumontanum, depth to capsule.
> - Median lobe first; apex last.
> - Bipolar allows saline; TUR syndrome practically eliminated.
> - Equal efficacy; bipolar has less bleeding and shorter catheter time.
> - TUR syndrome risk: >45 mL gland, >90 minutes.

SAQ Measures to decrease complications of TURP

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 9 Paper 2, Q3b (TUR syndrome also answered in 2017-Winter Paper 3, Q3f)

Introduction

- TURP (transurethral resection of the prostate) is the benchmark; overall morbidity approaches one-fifth, mostly preventable.

Preoperative measures

- **Selection:** AUA (American Urological Association) 2023: TURP for glands up to 80 mL; larger glands need HoLEP (holmium laser enucleation) or simple prostatectomy.
- **Sterile urine:** treat UTI (urinary tract infection) first; single-dose prophylaxis covering gram-negatives and enterococci.
- **Anticoagulants:** stop after advice from the prescriber; if they cannot be stopped, choose HoLEP, ThuLEP or PVP (photoselective vaporisation).
- **Finasteride** for recurrent BPH (benign prostatic hyperplasia) bleeding.
- Spinal anaesthesia shows early TUR syndrome; pad the peroneal nerve.

Intraoperative measures

Complication	Prevention
TUR syndrome (hypo-osmolar fluid absorbed, dilutional hyponatraemia)	**Bipolar TURP with saline**; else irrigant ≤60 cm high, resection <90 min, avoid capsular perforation, check sodium in long resections
Bleeding	Systematic resection to capsule; control arterial bleeders as you go; bipolar "cut and seal"; catheter traction for venous ooze
Ureteric orifice injury	Identify orifices first; take a large median lobe down first
Perforation	Respect the capsule; visible fat means perforation

Complication	Prevention
Incontinence	**Stop at the verumontanum**; resect apex last in a bloodless field
Stricture	Calibrate or dilate meatus, lubricate generously, correct sheath size
Bladder neck contracture	Avoid over-resection and excess fulguration; TUIP (transurethral incision of the prostate) for small glands

Postoperative measures

- Large-bore three-way catheter; careful **continuous bladder irrigation** (blocked outflow distends the bladder and worsens bleeding).
- Even routine cases absorb 800–1000 mL of irrigant; check sodium if confused.
- TUR syndrome treatment: **3% saline 100 mL** boluses; raise sodium by no more than 12 mEq/L in 24 h (pontine myelinolysis).
- Stool softeners; no cycling for 4–6 weeks; counsel about **retrograde ejaculation** (about two-thirds).

Diagram to draw: resection field from bladder neck to verumontanum with the external sphincter marked.

> **Key points to remember:**
> - Bipolar saline TURP removes TUR syndrome risk; fewer transfusions.
> - Monopolar: irrigant ≤60 cm, resection <90 min, gland <45 mL safer.
> - Verumontanum is the distal limit.

Also, from the condensed book (not in the paper answer):

- AUA: TURP for <80 cc (St 28), monopolar or bipolar by expertise (St 29). Monopolar irrigant is non-ionic (water, **1.5% glycine**†, sorbitol) and **hypo-osmolar**, which causes TUR syndrome.

ALGORITHM

1. Inspect the bladder. If tumour, resect it and defer TURP.
2. Identify orifices, neck, trigone, verumontanum (veru), sphincter; if orifices hidden, give indigo carmine or methylene blue.
3. Resect median lobe, then lateral lobes, then anterior tissue and apex last.
4. Evacuate chips, secure haemostasis, insert a large three-way catheter.

- **Anterior and apex last**, because the anterior fossa perforates easily and the sphincter lies more proximally anteriorly.
- **Arterial bleeding** is bright and must be fulgurated. **Venous bleeding** is darker and vanishes with a full bladder; once arteries are secured, use balloon traction.
- Even routine TURP absorbs **800–1000 mL** irrigant. The catheter comes out at 24–48 h.
- Warn of dysuria for weeks and eschar bleeding at **1–4 weeks**. Results last about 10 years.

5 · TURP complications

Missed TUR syndrome can kill.

SAQ TUR syndrome

Paper answer · P1 · P3 · merged from 2

Asked as: P1 SAQ "TUR Syndrome" · P3 SAQ "TUR Syndrome."

Definition

- **TUR (transurethral resection) syndrome:** neurological and cardiovascular disturbance from **absorption of hypo-osmolar, non-conducting irrigant** during monopolar TURP (transurethral resection of the prostate). Incidence low and falling; about 1–2% in large series.

Pathophysiology

- Prostatic venous pressure is about **10 mmHg**; irrigant at higher pressure enters open sinuses, or the periprostatic space after perforation.
- Even routine resection absorbs 800–1000 mL.
- Monopolar needs ion-free fluid: **glycine 1.5% (~200 mOsm/kg)**, sorbitol, mannitol or water (all hypo-osmolar; plasma ~290).
- **Fluid overload** → hypertension, pulmonary oedema.
- **Dilutional hyponatraemia and hypo-osmolality** → water moves into brain → **cerebral oedema**.
- **Glycine toxicity:** ammonia → encephalopathy; inhibitory retinal neurotransmitter → **transient blindness**.
- Sorbitol → hyperglycaemia; mannitol → osmotic diuresis.

Risk factors

- Gland **>45 mL**, resection **>90 minutes** (Veterans Affairs Cooperative Study), high irrigation pressure, capsular perforation, sterile water.

Clinical features

- Awake (spinal) patient: restlessness, confusion, **nausea, vomiting, visual disturbance**.
- Cardiovascular: **hypertension with bradycardia**, tachypnoea, later hypotension, ECG (electrocardiogram) changes.
- Severe: seizures, coma, herniation, death. **Na <120 mEq/L** is dangerous.

Prevention

- **Bipolar TURP or laser with saline** (eliminates it; NNT [number needed to treat] about 50).
- Bag **~60 cm** above patient (each extra 10 cm more than doubles absorption); continuous-flow resectoscope; low-pressure irrigation.
- Limit resection time; stage huge glands; check sodium in long cases.
- Spinal anaesthesia for early detection; **1% ethanol-tagged irrigant** (breath test) monitors absorption.

Treatment

1. Stop surgery, secure haemostasis, insert catheter.
2. ABC (airway, breathing, circulation), oxygen, monitoring; send **sodium, osmolality**, glucose.
3. Fluid overload: **furosemide**; restrict fluids.

4. Severe neurological signs or Na <120: **3% saline 100 mL bolus** (raises Na 2–3 mEq/L), repeated at 10 minutes, maximum 300 mL. **Target rise 4–6 mEq/L.**
5. Correction limit: **≤12 mEq/L in 24 h** for acute post-TURP hyponatraemia (8–10 mEq/L if symptoms start >48 h after surgery), avoiding **osmotic demyelination (central pontine myelinolysis)**.
6. Seizures: benzodiazepines; ICU (intensive care unit) care.

Other settings

- **Also know:** occurs in TURBT (transurethral resection of bladder tumour), PCNL (percutaneous nephrolithotomy) and hysteroscopy.

Diagram to draw

- Flowchart: absorption → overload + hyponatraemia → cerebral oedema.

> **Key points to remember:**
> - Venous pressure 10 mmHg; risk gland >45 mL, time >90 min.
> - Glycine: ammonia and transient blindness.
> - 3% saline 100 mL boluses; ≤12 mEq/L in 24 h.
> - Bipolar saline TURP prevents it.

Also, from the condensed book (not in the paper answer):

- Risk factors are a gland **>45 cc**, resection **>90 min**, perforation and irrigant height. Keep the bag at about **60 cm**, because prostatic venous pressure is only about **10 mm Hg**. Na^+ **<120 mEq/L** is severe.

ALGORITHM

1. Stop or expedite surgery, secure haemostasis, stop irrigation; send Na^+.
2. If overloaded, give a †loop diuretic.
3. If severe central nervous system signs: **3% saline 100 mL** (+2–3 mEq/L), repeat ×2 at 10 min (max 300 mL).
4. Target +4–6 mEq/L; ≤12 mEq/L in 24 h (≤8–10 if presentation >48 h). Monitor in high dependency.

> **KEY POINT**
> Acute post-TURP hyponatraemia is safer to correct than chronic: the brain has not yet shed its organic osmolytes, so osmotic demyelination is less likely. Bipolar saline TURP eliminates TUR syndrome.

- **Perforation** shows glistening fat. A limited extraperitoneal tear needs only a longer catheter, an extensive one needs drainage, and an intraperitoneal tear needs laparotomy. A persistent intra-operative erection responds to intracorporeal phenylephrine.

Late complication	Cause / features	Management
Bladder neck contracture (BNC)	Over-resection or over-fulguration (small glands); onset ~6 months, flow declines	Dilate, then incise over a wire into perivesical fat. Recurrent: add mitomycin C (10 mL of 0.4 mg/mL). Intractable: Y-V plasty

Late complication	Cause / features	Management
Stricture	Meatus or fossa navicularis	Dilate meatus 2 Fr above sheath; plenty of gel
Retrograde ejaculation	About **two-thirds**	Counsel; erection, orgasm, desire unaffected
Incontinence	Internal sphincter lost; apical over-resection causes total stress incontinence (SUI). Rare, usually transient	Veru is the distal limit; †pelvic floor exercises, then †sling or artificial sphincter
Bleeding	Early: incomplete haemostasis. Delayed (1–4 weeks): eschar	Early major: theatre. Delayed mild: fluids; †clot evacuation if retention
Erectile dysfunction (ED)	No conclusive causal evidence	—
Retreatment	Residual apical/anterior tissue, >80 cc, regrowth, BNC, stricture	Rises with size and pre-op retention

6 · Bipolar TURP (b-TURP)

*See the paper answer **TURP instruments, techniques and monopolar vs bipolar** (Section 16, Ch 151).*

Also, from the condensed book (not in the paper answer):

- It is equally effective and safer, so it may replace m-TURP as gold standard.

7 · TUIP

- TUIP releases capsule and bladder neck without removing adenoma. It is for **<30 cc only** (St 31) without a median lobe (relative contraindication; EAU the same). It suits a young man with a small gland who wants ejaculation.

ALGORITHM

1. Use a Collins knife, loop, or holmium/thulium laser.
2. Incise at 5 and/or 7 o'clock from distal to the orifice, through the bladder neck, to proximal to the veru.
3. Deepen to capsule (with or without fat); cauterise.
4. Catheter out and home on day 0–1.

- Compared with TURP it is shorter, bleeds less and relieves symptoms similarly, but gains less Qmax and needs **more reoperation**; retrograde ejaculation is rarer and BNC does not occur.

8 · Bipolar TUVP

- A broad electrode (PlasmaButton) vaporises and coagulates together, so haemostasis is excellent, but there is **no specimen**. Only **bipolar** is recommended (St 32: <30 and 30–80 cc); monopolar was abandoned for irritation, SUI and poor durability.
- Compared with TURP it needs a shorter catheter and fewer transfusions, but causes more dysuria, AUR and retreatment; retrograde ejaculation is as common.

> **KEY POINT**
>
> Vaporisation trades haemostasis for a coagulated, specimen-free bed. The coagulum causes dysuria and retention; with no anatomical plane, durability is weaker. PVP makes the same trade.

9 · Laser physics

SAQ Lasers in BPH, including green light

Paper answer · P2 · P3 · P4 · merged from 3

Asked as: P3 SAQ "Lasers in BPH, including HoLEP" (Various lasers in treatment of BPH; HOLEP) · P2 SAQ "LASERS in BPH." · P4 SAQ "Green light lasers"

HOLEP read as: holmium laser enucleation of the prostate for BPH (benign prostatic hyperplasia).

Principle

- LASER (light amplification by stimulated emission of radiation) turns light into local heat. The tissue effect depends on wavelength and **chromophore** (water or haemoglobin), which set **penetration depth**.
- About 60 °C coagulates tissue (delayed sloughing); ≥100 °C vaporises it (immediate removal). Coagulation lasers are obsolete.

Lasers used in BPH

Laser	Wavelength	Chromophore	Penetration	Procedure
Ho:YAG (holmium:yttrium-aluminium-garnet), pulsed	~2120 nm	Water	0.4 mm	**HoLEP** (holmium laser enucleation of the prostate)
Thulium:YAG (continuous) / **TFL** (thulium fibre laser)	2013 / 1940 nm	Water (TFL at absorption peak, about 4× holmium)	0.2 mm	ThuLEP (enucleation), ThuVEP (vapoenucleation), ThuVARP (vaporesection), vaporisation
KTP (potassium titanyl phosphate) / LBO (lithium triborate)	532 nm	Haemoglobin	0.8 mm	**PVP** (photoselective vaporisation), 120–180 W
Diode	940–1470 nm (various)	Water and haemoglobin (mixed)	0.5–5 mm	Vaporisation, enucleation
Nd:YAG (neodymium)	1064 nm	Weak	About 1 cm	Coagulation; abandoned (sloughing)

Enucleation versus vaporisation

Feature	Enucleation (HoLEP, ThuLEP)	Vaporisation (PVP)
Tissue	Whole transition zone to the capsule, then morcellated	Removed layer by layer
Histology	Yes	No
Gland size	**Any (size-independent)**	Up to 80 cc

Feature	Enucleation (HoLEP, ThuLEP)	Vaporisation (PVP)
Retreatment	Rare	Higher, especially >80 cc

HoLEP

- **HoLEP** removes the whole adenoma along the **surgical capsule**: the endoscopic equivalent of open simple prostatectomy.
- **Ho:YAG, 2120 nm**: water absorbs it; pulsed; **0.4 mm penetration**; coagulates vessels. Saline irrigation, so no TUR (transurethral resection) syndrome.
- **Indications: size-independent** (<30, 30–80, >80 mL), with or without median lobe; **anticoagulated or high bleeding risk** (AUA [American Urological Association]); alternative to open prostatectomy for large glands.
- **Steps:** 60–150 W laser, 550 µm end-firing fibre, 24–28 Fr scope; **about 2.0 J × 40–50 Hz** for enucleation. 1. Bladder-neck incisions at 5 and 7 o'clock (median lobe) or one at 6 o'clock. 2. Perimontanal incision; **early apical release** protects the sphincter. 3. Develop the capsular plane to 12 o'clock; push lobes into the bladder (**en bloc, two- or three-lobe**). 4. Haemostasis with a defocused beam. 5. **Morcellation** with a full bladder; three-way catheter.
- **Limitation: steep learning curve** (about 50 cases).

Green light (KTP/LBO) lasers: physics

- A 1064 nm Nd:YAG (neodymium:yttrium-aluminium-garnet) beam is passed through a **KTP or LBO crystal**. This doubles its frequency and halves the wavelength to **532 nm** (green).
- Green light is absorbed by **haemoglobin**, not water, so vascular prostate absorbs it avidly and vaporisation is bloodless.
- Optical penetration **0.8 mm**, tissue effect 1–3 mm, thin (0.2 mm) coagulation rim. The result is bloodless **photoselective vaporisation**.

Green light platforms

System	Power	Note
KTP	80 W (later 100 W)	Popularised PVP (photoselective vaporisation of the prostate)
LBO HPS (High Performance System)	120 W	Faster vaporisation
LBO XPS	**180 W**	750 µm MoXy fibre: metal tip, fluid cooling

PVP technique

- 23–24 Fr cystoscope; side-firing fibre emitting at **70°**; saline irrigation.
- Open the bladder neck first, away from the orifices; sweep 15–30° in 1–2 seconds at ≤2 mm.
- Large bubbles mean efficient vaporisation. Pulsatile bleeding: coagulation mode (about 30 W).
- More than 5 kJ/cc predicts complete vaporisation.

Indications by guideline (AUA)

- HoLEP or ThuLEP for any size.
- PVP for prostates <30 cc and 30–80 cc (under 80 cc); not >80 cc (high retreatment).

- HoLEP, ThuLEP or PVP for men at **higher bleeding risk** (anticoagulated men, who can continue anticoagulants with PVP).

Evidence and outcomes

- **HoLEP: size-independent**; the only endoscopic procedure **superior to TURP** (transurethral resection of the prostate) for IPSS (International Prostate Symptom Score) and Qmax (maximum flow rate), durable. Less bleeding, shorter catheter time and stay; longer operating time. Retreatment rare (whole transition zone removed).
- **ThuLEP:** outcomes equal to HoLEP.
- **GOLIATH RCT** (randomised controlled trial; 29 European centres, about 270 men, prostate ≤100 cc; 180 W PVP vs TURP): PVP **non-inferior** for IPSS, Qmax and complications at 2 years; shorter catheter time and stay; TURP faster. Retrograde ejaculation (about two-thirds) and reoperation equal. Retreatment high above 80 cc.

Complications

- Dysuria and storage symptoms (dysuria worst after PVP, the highest of transurethral techniques); UTI (urinary tract infection) and epididymitis from coagulum after PVP.
- Transient storage symptoms and early / transient stress incontinence (SUI, stress urinary incontinence).
- Retrograde ejaculation in most (**about 75%**, three-quarters, after HoLEP).
- Bladder neck contracture (BNC) 0–3.2% (P2: 0–3%); stricture; capsular perforation (0.2–1% with PVP); ureteric orifice injury; **morcellator bladder injury**.
- No TUR syndrome (saline); transfusion very rare; erectile function unchanged; PVP leaves **no tissue for histology**.
- **Safety:** wavelength-specific eyewear for all; laser safety officer.

Recent advances

- **Moses pulse modulation**; ThuLEP (thulium enucleation); median-lobe-only HoLEP (antegrade ejaculation about 90%); same-day discharge.
- GreenLight enucleation; office and same-day catheter-free pathways.

Diagrams to draw

- HoLEP: sagittal prostate with enucleation plane along the capsule; adenoma enucleated to the capsule, pushed into the bladder, then morcellated.
- PVP: side-firing fibre at 70° vaporising a lateral lobe.

> **Key points to remember:**
>
> - Chromophore sets penetration: thulium 0.2, holmium 0.4, KTP 0.8 mm, Nd:YAG 1 cm.
> - Ho:YAG 2120 nm, water chromophore, 0.4 mm.
> - HoLEP is size-independent and the only endoscopic method superior to TURP; suits anticoagulated men.
> - Retrograde ejaculation about 75% after HoLEP.
> - Green light: 532 nm, frequency-doubled Nd:YAG; haemoglobin chromophore; 0.8 mm penetration; 180 W XPS with MoXy fibre.
> - GOLIATH: PVP non-inferior to TURP; best for anticoagulated men; avoid PVP above 80 cc.

Also, from the condensed book (not in the paper answer):

- Deeper penetration leaves more coagulum, hence Nd:YAG's sloughing and retention.

10 · PVP (GreenLight)

PVP suits anticoagulated men with glands under 80 cc.

*See the paper answer **Lasers in BPH, including green light** (Section 16, Ch 151).*

Also, from the condensed book (not in the paper answer):

- AUA St 33 recommends 120 W or 180 W PVP for <30 and 30–80 cc; anticoagulated men improve equally without more transfusion.

11 · HoLEP

Size-independent, and the only endoscopic operation beating TURP.

P4 P1 *HoLEP — size-independent (St 38/43), only endoscopic method superior to TURP; settings, steps, learning curve 20/50, BNC if <40 cc*

- It is durable to 18 years because the whole transition zone (TZ) goes, whereas TURP leaves a peripheral rim.

ALGORITHM

1. Incise the bladder neck at 5 and 7 o'clock (at 6 o'clock if no median lobe); carry to the veru and enucleate the median lobe.
2. Make a perimontanal inverted-U incision; open the apical plane.
3. Sweep anteriorly to 11–1 o'clock; incise the apical mucosal strip (no sphincter traction).
4. Develop the anterior plane to the neck ("Baleen" sign = bladder); release lobes into the bladder.
5. Haemostasis with a defocused beam; morcellate via nephroscope; 22 Fr three-way catheter, ~60 mL balloon.

- The smooth reflective capsule is the depth reference. If the morcellator engages mucosa, stop cutting and suction.
- **Transient SUI** is common early, in proportion to volume removed; early apical release reduces it.
- Retrograde ejaculation occurs in about **three-quarters**, but median-lobe-only HoLEP preserves it in ~90%. Erections are unchanged or improved and retreatment is rare.

12 · Thulium (ThuLEP) and variants

- Variants include ThuVARP (vaporesection), ThuVEP (vapo-incision) and ThuLEP (blunt dissection); steps match HoLEP.
- Function about equals HoLEP, with less bleeding than TURP (St 38 and 43). Retrograde ejaculation is ~four-fifths; erections are preserved.

	HoLEP	ThuLEP / ThuVEP
Learning curve	Steep (plateau ≥50)	Claimed shorter (~30)
Evidence	Largest; 18-year follow-up	Shorter follow-up

13 · MISTs

The ideal MIST is office-based, single session, needs no general anaesthesia, spares sexual function and ends medication. **Trade-off**: retreatment within 5 years exceeds surgery, and durability beyond 5 years is uncertain.

SAQ UroLift (prostatic urethral lift)

Paper answer · P4

Asked as:

- Urolift. — *Winter 2024 Paper IV Q3d (SAQ)*
- Urolift. — *Practice Set 6 Paper IV Q3b (earlier version) (SAQ)*

Type: SAQ (10 marks)
(also answered in 2024-Winter Paper 4, Q3d)

Read as PUL (prostatic urethral lift, UroLift) for BPH (benign prostatic hyperplasia).

Introduction

- A **non-ablative MIST (minimally invasive surgical therapy)**. Permanent implants pull (retract) the lateral lobes apart and open the prostatic urethra.
- **FDA (Food and Drug Administration) approval 2013**; extended to ≤100 mL in 2020.
- **Main advantage:** preserves erectile and **ejaculatory** function.

Implant and mechanism

- Spring-loaded implant with T-shaped ends:
- **Capsular tab:** nitinol.
- **Urethral end:** stainless steel (resists encrustation).
- **Connector:** polyethylene terephthalate suture under tension.
- Delivered through a **19-gauge needle** from a cystoscopic device.
- The gland is more compliant than the capsule, so tension compresses the lobe outward; later local atrophy and fibrosis. Implants are MRI (magnetic resonance imaging)-conditional (≤3 T).

Indications

- **AUA (American Urological Association): prostate 30–80 mL, no obstructing median lobe** (confirm by cystoscopy or ultrasound). EAU (European Association of Urology): <70 mL, no middle lobe, men wanting ejaculation preserved.
- Men wishing to stop medication.
- **Avoid:** obstructing median lobe (MedLift data non-randomised), large gland, active infection.

Technique

1. Local anaesthesia; rigid cystoscope.
2. Compress lateral lobe, fire needle, deploy implant; repeat on the other side.
3. **Usually 4–6 implants**, anterolateral, from bladder neck to apex, avoiding the neurovascular bundles.
4. **Keep implants >1.5 cm distal to the bladder neck** (encrustation).
5. Most go home without a catheter.

Evidence

	L.I.F.T. (2013)	BPH6 (2015)
Design	PUL vs sham, 2:1, 19 centres	PUL vs TURP (transurethral resection of prostate), 10 European centres
Result	IPSS (International Prostate Symptom Score) about halved, twice the sham effect; Qmax (maximum flow) +4 mL/s; no new sexual dysfunction	Composite end point met 52.3% vs 20.0%; antegrade ejaculation 100% vs 60.6%
Durability / retreatment	13.6% surgical retreatment at 5 years	13.6% vs 5.7% at 2 years; TURP better IPSS, Qmax, PVR (post-void residual)

- TURP wins on IPSS, Qmax and PVR (post-void residual).

Complications

- Transient dysuria, haematuria, urgency (<2 weeks); encrustation; rare pelvic haematoma; device malfunction (misfire).
- Lower major adverse events but **higher retreatment than TURP**.

Diagram to draw: axial prostate with implants pulling lateral lobes to the capsule.

> **Key points to remember:**
> - 30–80 mL, no median lobe; local anaesthesia.
> - Ejaculation preserved; L.I.F.T. retreatment 13.6% at 5 years.
> - TURP wins (better) on IPSS, flow and durability.

Also, from the condensed book (not in the paper answer):

- AUA keeps 30–80 cc and has not adopted the FDA ≤100 cc label.
- PUL typically uses **4–6 implants**, placed anterolaterally between dorsal veins and neurovascular bundles.
- **WVTT (Rezūm)** is **convective**: radiofrequency steam is injected into the TZ, and condensation releases latent heat, causing 1.5–2 cm of necrosis confined by the capsule.
- Injections are given every 1 cm from neck to veru, with extra ones for a **median lobe**, under sedation or prostatic block.
- **Rezūm II** showed ~50% IPSS gain to 5 years, retreatment 4.4% and **no de novo ED**; reduced ejaculate settles by 12 months. (St 36/37).

> **KEY POINT**
>
> PUL moves tissue; WVTT ablates it. So PUL cannot treat a median lobe but needs no catheter and causes little dysuria, whereas WVTT treats median lobes at the cost of a catheter and weeks of irritation. Both preserve the bladder neck, and so antegrade ejaculation.

14 · Robotic waterjet (RWT, Aquablation)

- A non-thermal saline waterjet ablates tissue mapped on real-time biplanar TRUS. It **needs general or spinal anaesthesia, so it is not a MIST**. It suits 30–80 cc (St 39), treats

median lobes and gives no specimen.

- **Haemostasis is the weakness**: traction alone led to bleeding and transfusion, so **focal bladder-neck cautery** via resectoscope is now used.
- **WATER** showed non-inferiority to TURP to 5 years, with retrograde ejaculation 10% vs 36%.

15 · Temporary devices

- **iTind (TIPD)** is a nitinol device left **5–7 days** to cause ischaemic remodelling, then removed. St 41: 25–75 cc, no obstructing median lobe. In its sham trial the responder end point was met, but the absolute IPSS difference was not significant.
- **Optilume** (paclitaxel balloon) has single-arm data only and is not on the AUA list.

16 · Prostate artery embolisation (PAE)

No surgery, but less effective.

ALGORITHM

1. Femoral access under local anaesthesia; pelvic angiography with the catheter balloon as landmark.
2. Pass a 1.9–2.4 Fr microcatheter into the prostatic arteries (mostly from internal pudendal; often duplicated).
3. Inject microspheres to stasis; bilateral is better.

- **Contraindications**: underactive, acontractile or neurogenic bladder; stricture or BNC; large diverticula or stones; contrast allergy; chronic kidney disease.
- AUA St 40: by trained clinicians; no size limit; a role in refractory BPH haematuria. EAU: for men accepting results inferior to TURP.
- Compared with TURP, IPSS, Qmax and volume reduction are inferior but the stay is shorter; most men reintervene by 10 years.
- Complications include non-target embolisation, post-embolisation syndrome and AUR.

17 · Overall comparison

This is the counselling sheet.

Procedure	Volume (AUA)	Anaesthesia / catheter	Efficacy vs TURP	Antegrade ejaculation	Key hazard	Retreatment
m-/b-TURP	<80 cc	Spinal/general (GA); 1–2 d	Benchmark	Lost ~⅔	Bleeding, TUR syndrome (mono)	Low
TUIP	<30 cc	Day 0–1	Slightly less	Mostly kept	—	Higher
b-TUVP	<80 cc	Short catheter	Similar early, less durable	Lost	Dysuria, AUR	Higher
PVP	<80 cc; anticoagulated	Day 0–1	Non-inferior (GOLIATH)	Lost ~⅔	Dysuria, UTI	Higher than TURP

Procedure	Volume (AUA)	Anaesthesia / catheter	Efficacy vs TURP	Antegrade ejaculation	Key hazard	Retreatment
HoLEP/ThuLEP	Any; anticoagulated	GA/spinal; day 0–1	**Superior** (HoLEP)	Lost ~¾–⅘	Transient SUI, morcellator	Lowest
RWT	30–80 cc	GA/spinal	Non-inferior	Mostly kept	Bleeding	Low (short data)
PUL	30–80 cc, no median lobe	Local; often none	Less	**Preserved**	Encrustation	~14% at 5 y
WVTT	30–80 cc	Sedation; few days	Less	**Preserved**	Irritation	~4–8% at 5 y
iTind	25–75 cc, no median lobe	Local; device 5–7 d	Uncertain	**Preserved**	—	Unknown
PAE	Any	Local; day case	Inferior	Largely kept	Non-target embolisation	High

KEY POINT

The "gold standard" for a given man is the operation that meets his goals safely: a shared decision on symptoms, size and shape, bleeding risk, ejaculation wishes and surgeon expertise.

18 · Landmark trials

Trial	Design	Result	Take-home
L.I.F.T.	PUL vs sham, 2:1, 30–80 cc	IPSS ~halved; 13.6% surgical retreatment at 5 y	PUL works, preserves sexual function
BPH6	PUL vs TURP, ≤60 cc	Composite favoured PUL; TURP better IPSS/Qmax/PVR	"Superior" refers only to composite
Rezūm II	WVTT vs sham, 30–80 cc incl. median lobe	IPSS −11.2 vs −4.3; durable 5 y; no de novo ED	Convective MIST incl. median lobe
WATER	RWT vs TURP, double-blind	Non-inferior; better in 50–80 cc; less retrograde ejaculation	Aquablation for 30–80 cc
GOLIATH	180 W PVP vs TURP, ≤100 cc	Non-inferior to 2 y; shorter catheter and stay	PVP alternative to TURP
UK-ROPE	PAE vs TURP, matched registry	PAE inferior for IPSS/QoL; shorter stay	Lesser-efficacy option

152 Simple Prostatectomy: Open, Laparoscopic and Robotic

1 · Definition and surgical anatomy

- It enucleates only the benign prostatic hyperplasia (BPH) adenoma and is the gold standard for outlet obstruction with glands **>80 g**.
- The plane is the **surgical capsule (pseudocapsule)**: peripheral zone compressed by the enlarging transition zone (see Ch 148).
- **Urethral arteries enter posterolaterally just distal to the bladder neck**, so a figure-of-eight suture goes at **5 and 7 o'clock**.
- Eponyms MCQ : **Freyer** is suprapubic transvesical and **Millin (1945)** is retropubic.

2 · Indications

- Gland **>80–100 g** with bothersome lower urinary tract symptoms (LUTS) refractory to medical or endoscopic therapy.
- Retention, recurrent urinary tract infection (UTI), **bladder stones**, haematuria or obstructive nephropathy; a diverticulum needing surgery (relative); or inability to lie in lithotomy for transurethral resection (TURP).

3 · Pre-operative evaluation

- Check bleeding history, anticoagulants, prior pelvic surgery (which dictates the route), **urine culture** (treat first) and coagulation.

> **KEY POINT**
>
> Work-up asks three questions. Is the gland big enough (transrectal ultrasound)? Is there cancer (digital rectal examination, prostate-specific antigen, MRI)? Will the bladder empty (urodynamics)?

4 · Open techniques

Freyer opens the bladder; Millin opens the capsule.

SAQ Millin's prostatectomy

Paper answer · P1 · P2 · merged from 2

Asked as: P1 SAQ "Millin's prostatectomy" · P2 SAQ "Millin's prostatectomy"

Definition

- **Retropubic transcapsular (extravesical) simple prostatectomy** (Millin, 1945): adenoma enucleated through a transverse anterior capsulotomy **without opening the bladder**. Plane: the surgical capsule (compressed peripheral zone around the transition-zone adenoma).

Indications

- BPH (benign prostatic hyperplasia), gland **>80–100 g**, with refractory LUTS (lower urinary tract symptoms), retention, recurrent UTI (urinary tract infection), haematuria or upper-tract damage (obstructive complications), or when lithotomy is impossible.

- Prefer suprapubic (Freyer) for a large median lobe, diverticulum or bladder stones.
- EAU (European Association of Urology) and AUA (American Urological Association): simple prostatectomy or HoLEP (holmium laser enucleation) for large glands.

Work-up

- Exclude cancer: PSA (prostate-specific antigen), MRI (magnetic resonance imaging) ± biopsy; TRUS (transrectal ultrasound) volume; cystoscopy; urodynamics if bladder function is doubtful.

Steps

1. Lower midline or Pfannenstiel; extraperitoneal approach to the **space of Retzius**.
2. Control the dorsal venous complex: optional prophylactic ligation of DVC (dorsal venous complex) and pedicles (Walsh–Oesterling).
3. **Transverse capsulotomy ~2 cm distal to (below) the bladder neck** (a longitudinal cut may extend into the sphincter).
4. Find the adenoma–capsule plane with scissors; **finger enucleation**.
5. **Divide the apical urethra under vision** proximal to the sphincter, sparing it.
6. Trim the bladder neck lip; advance bladder neck mucosa into the fossa.
7. **Figure-of-eight sutures at 5 and 7 o'clock**, where prostatic arteries enter.
8. 22 Fr three-way catheter, **watertight capsule closure**, drain.

Millin versus Freyer

Aspect	Millin (retropubic)	Freyer (suprapubic)
Bladder	Not opened	Opened
Apex	Divided under vision	Pinched by finger
Fossa haemostasis	Directly visible	Less visible
Stones, diverticulum, median lobe	Difficult	Ideal
Specific risk	(Dorsal) venous bleeding	Cystotomy leak

Complications

- Bleeding (transfusion <5%): control by catheter traction, Malament purse-string, O'Conor capsular plication.
- Urine leak, UTI (infection), ureteric injury, bladder neck contracture 2–6%, stricture, DVT (deep vein thrombosis) and thromboembolism.
- **Retrograde ejaculation 80–90%**; ED (erectile dysfunction) 3–5%; incontinence very rare.

Versus TURP (transurethral resection of prostate)

- Complete removal, low (lower) retreatment, no TUR syndrome; but incision, longer stay.
- **Also know:** vesicocapsular variant.

Recent advances and modern alternatives

- HoLEP; laparoscopic, robotic multiport and single-port transvesical simple prostatectomy: less bleeding, shorter stay.

Diagram to draw

- Sagittal view of transverse capsulotomy and enucleation plane.

Key points to remember:

- Millin 1945: transverse capsulotomy ~2 cm below bladder neck; bladder not opened.
- Apex divided under vision; 5 and 7 o'clock sutures.
- Retrograde ejaculation 80–90%; transfusion <5%; bladder neck contracture 2–6%.

Also, from the condensed book (not in the paper answer):

- **Freyer** uses a transverse **cystotomy 1–2 cm proximal to the bladder neck**. It treats **median lobe, diverticula and stones**, but the apex is blind.
- **Millin**'s drawback is **dorsal venous complex (DVC) bleeding**.

KEY POINT

Every open variant: enter, find the pseudocapsular plane, divide the urethra at the apex **proximal to the verumontanum** without pulling on the sphincter, then 5/7 o'clock haemostasis with bladder neck mucosal advancement.

5 · Haemostasis

Technique	When / how
Pack fossa 5 min	First
5 and 7 o'clock figure-of-eight (2-0 chromic/polyglactin)	Every case; deep through proximal capsule
Bladder neck mucosal advancement ("trigonisation")	Haemostasis; prevents contracture
11 and 1 o'clock sutures	Anterior branches, occasionally
Walsh prophylactic control	Before Millin: ligate DVC and lateral pedicles, sparing neurovascular bundles (NVBs)
O'Conor plication	Diffuse bleeding: plicate the posterior capsule
Malament purse-string	Persistent bleeding: purse-string round the neck, tied suprapubically, partitions fossa from bladder

ALGORITHM

1. Catheter traction (50 mL balloon), with or without continuous bladder irrigation (CBI); exclude coagulopathy.
2. If bleeding continues, evacuate clot endoscopically and fulgurate.
3. If severe, reopen the fossa.

6 · Postoperative care (open)

- On day 2, if urine is clear, remove the Foley for a trial of void. If drain output is high, check **drain creatinine**.

7 · Laparoscopic and robotic

SAQ Laparoscopic Trans vesical Prostatectomy

Paper answer · P3

Type: SAQ (10 marks)
Source: 2025-Winter Paper 3, Q3d

Assumed meaning: laparoscopic transvesical (suprapubic) simple prostatectomy for large BPH (benign prostatic hyperplasia).

Introduction

- Laparoscopic version of the Freyer suprapubic operation: the adenoma is enucleated through a **cystotomy** in the **surgical capsule plane** (compressed peripheral zone). First LSP (laparoscopic simple prostatectomy): Mariano, 2002.

Indications

- Gland **>80–100 g** with refractory LUTS (lower urinary tract symptoms), retention, recurrent UTI (urinary tract infection), haematuria or obstructive nephropathy.
- **Transvesical route preferred** for a large intravesical median lobe, bladder stones or a diverticulum needing surgery.
- Work-up: PSA (prostate-specific antigen) ± MRI (magnetic resonance imaging) to exclude cancer; TRUS (transrectal ultrasound) volume; cystoscopy.

Steps

1. General anaesthesia, supine, Trendelenburg. **Ports:** 10-mm umbilical camera, two 10-mm working ports, 5-mm and 12-mm assistant ports. Transperitoneal or extraperitoneal (Retzius).
2. **Cystotomy** with stay sutures; identify ureteric orifices.
3. **Circumferential bladder-neck incision** (start 3 to 9 o'clock posteriorly), away from orifices.
4. Develop the **posterior plane first** to the apex, then the lateral lobes.
5. Divide anterior commissure; **transect urethra just proximal to the verumontanum** (protects sphincter).
6. **Figure-of-eight sutures at 5 and 7 o'clock** (prostatic arteries enter posterolaterally).
7. **Bladder-neck mucosal advancement** to urethral mucosa covers the fossa and reduces bleeding.
8. 22 Fr three-way catheter; two-layer cystotomy closure.

Outcomes versus open

- Less blood loss, shorter catheter time and stay, less analgesia; **same functional results** (IPSS (International Prostate Symptom Score), flow rate, residual).
- Limitations: steep learning curve, poor ergonomics; **robotic simple prostatectomy** has largely replaced it. **Single-port transvesical robotic enucleation** (2.7 cm suprapubic incision, pneumovesicum) allows same-day discharge.

Complications

- Bleeding (transfusion rare), urine leak, ureteric orifice injury, bladder neck contracture (<1%; open 2–6%).
- **Retrograde ejaculation 80–90%**; ED (erectile dysfunction) 3–5%; incontinence rare.

Diagram to draw: sagittal view of cystotomy, bladder-neck incision and enucleation plane.

Key points to remember:

- Indication: gland >80 g, especially with median lobe, stones or diverticulum.
- Posterior plane first; divide urethra proximal to verumontanum.
- 5 and 7 o'clock sutures plus mucosal advancement.
- Same function as open, less morbidity.

Also, from the condensed book (not in the paper answer):

- Mucosal advancement usually means **no CBI, drain or suprapubic tube**, and the catheter comes out on day 5–7.
- Avoid posterolateral cautery, because thermal NVB injury causes erectile dysfunction (ED).

ALGORITHM

1. Vertical cystotomy; identify orifices.
2. Incise the bladder neck 3→9 o'clock under the median lobe; develop the **posterior plane** to the apex, then the lobes.
3. Divide the anterior commissure, see the veru, and transect the urethra **just proximal to the veru**.
4. Place 5/7 barbed sutures; do **vesicourethral mucosal advancement**; close in two layers.

- **Single-port transvesical surgery** with **pneumovesicum 8 mm Hg** allows same-day discharge.

8 · Outcomes and comparisons

The route changes recovery, not function.

Comparison	Bottom line
Robotic vs open	Less bleeding, shorter catheter and stay, fewer complications; longer operation. Function **equal**
Single-port vs multiport	Shorter stay, less CBI and opioid; same complications
vs TURP	Lower retreatment, **no TUR syndrome**, treats stones and diverticula; but more bleeding, longer stay
†vs holmium laser enucleation (HoLEP)	†Equivalent outcomes; HoLEP bleeds less, but has a long learning curve and cannot treat diverticula

KEY POINT

Minimally invasive routes improve only perioperative morbidity; function is identical because enucleation is identical. Open transfusion is now **≤5%**.

9 · Complications

Complication	Cause / rate	Management
Extravasation	Incomplete closure; capsular tear	If drain creatinine equals urine, keep drain and Foley

Complication	Cause / rate	Management
Ureteric orifice injury	Neck incision or sutures	Stent
Bladder neck contracture	Open **2–6%** ; minimally invasive <1%	Dilation or Collings knife incision
Storage LUTS	Weeks–months	Anticholinergic or β3-agonist
Stress incontinence	Rare	Bulking (mild); artificial sphincter (severe)
Erectile dysfunction	**3–5%**	—
Retrograde ejaculation	**80–90%**	Counsel pre-op
UTI, stricture, venous thromboembolism (VTE)	VTE <1%	Prophylaxis

KEY POINT

The adenoma lies inside the pseudocapsule and proximal to the veru. Removal loses the bladder neck (retrograde ejaculation) but spares sphincter and NVBs, so incontinence and ED are rare.

★ Putting it together: the BPH pathway

A · Man with LUTS

ALGORITHM

1. **1. Assess:** International Prostate Symptom Score (IPSS: 0–7 mild, 8–19 moderate, 20–35 severe), rectal examination, urinalysis, PSA if it changes management; uroflow and post-void residual (PVR) before intervention. **2. If complications, operate:** refractory retention, recurrent UTI, stones, prostatic haematuria, renal impairment. **3. If mild**, watchful waiting. **4. If bothersome**, drugs (B); if they fail, a procedure (C).

B · Medical therapy

Drug	Use	Notes
α1-blocker	First line; works in days	Ejaculatory dysfunction; floppy iris
5α-reductase inhibitor (5-ARI)	Prostate >30–40 mL	Less retention and surgery; halves PSA
α-blocker and 5-ARI	Large gland, progression risk	Best (MTOPS, CombAT)
Add antimuscarinic or β3-agonist	Persistent storage LUTS	Check PVR first
Tadalafil 5 mg daily	LUTS with ED	No nitrates
Desmopressin	Nocturnal polyuria	Check sodium if ≥65

C · Choosing a procedure

Situation	Options
<30 mL	TUIP or TURP
30–80 mL	Bipolar TURP, PVP, HoLEP/ThuLEP, Aquablation; MISTs (urethral lift, water vapour)
>80 mL	HoLEP/ThuLEP; simple prostatectomy (stones/diverticulum)
Anticoagulation continued	HoLEP, ThuLEP, PVP
Preserve ejaculation	Urethral lift (no median lobe) or water vapour
Unfit	Long-term catheter or †prostatic artery embolisation

D · TURP complications

KEY POINT

TUR syndrome causes confusion and hyponatraemia. Stop, check sodium, give a diuretic if overloaded and 3% saline if severe; bipolar prevents it. Late problems are retrograde ejaculation (~two-thirds), stricture, bladder neck contracture and incontinence (see Ch 151).

Other paper questions in this section

Asked in the papers but not tied to a specific place in the book above.

LAQ HoLEP

Paper answer · P1 · P4 · merged from 2

Asked as: P4 LAQ "HoLEP (holmium laser enucleation of the prostate)" (Holep; also SAQ: HOLEP.) · P1 SAQ "HoLEP in BPH"

Read as HoLEP (holmium laser enucleation of the prostate) for BPH (benign prostatic hyperplasia).

Introduction and principle

- **HoLEP** removes the whole transition-zone adenoma endoscopically **along the surgical capsule**. It is the endoscopic equivalent of open simple prostatectomy.
- **Evolution:** HoLAP (holmium laser ablation, 1994, slow) → HoLRP (resection, strips removed with a grasper) → **HoLEP (1998)**, made practical by the **tissue morcellator**.
- "Enucleation is enucleation": any energy giving a true **AEEP (anatomic endoscopic enucleation of the prostate)** gives durable results.

Laser physics

- **LASER:** light amplification by stimulated emission of radiation. Ho:YAG (holmium:yttrium–aluminium–garnet) = holmium atoms in a YAG (yttrium–aluminium–garnet) crystal.
- **Wavelength about 2120 nm**; chromophore **water** (absorbed by water); **pulsed**; optical penetration **0.4 mm**. It cuts by vaporisation with a thin coagulation zone; heat spreads 2–3 mm and coagulates small and medium vessels.
- Flexible quartz fibres; **normal saline** irrigation, so no TUR (transurethral resection) syndrome.

Laser	Wavelength	Chromophore	Penetration
Holmium:YAG	2120 nm	Water	0.4 mm
Thulium fibre	1940 nm	Water	0.2 mm
KTP (potassium-titanyl-phosphate, green light)	532 nm	Haemoglobin	0.8 mm
Nd:YAG	1064 nm	–	About 1 cm (sloughing)

MOSES (pulse modulation)

- **Moses effect:** the first part of each pulse boils a **vapour channel** in the water. Vapour absorbs the light far less, so the rest of the pulse passes through to the tissue ("parting the water").
- **Moses technology** splits each pulse to exploit this: more energy reaches tissue, and it **may shorten operating time**.

Indications

- **Size-independent** (<30, 30–80, >80 mL), with or without a median lobe (AUA [American Urological Association] Statement 38). Only enucleation spans every size band.
- **High bleeding risk or anticoagulated men** (AUA recommendation, Statement 43).
- Large glands otherwise needing open prostatectomy (alternative to open prostatectomy for large glands); retention; failed medical therapy.

Technique (steps)

- **Equipment:** 60–150 W Ho:YAG laser; **550 µm end-firing fibre**; 24–28 Fr continuous-flow laser resectoscope with 30° lens; saline.
- **Settings:** about 2.0 J × 40–50 Hz to enucleate; 1.0–2.0 J × 20 Hz to coagulate.
- **Three-lobe (bottom-up, early apical release):** 1. Median lobe: **bladder-neck incisions at 5 and 7 o'clock** to the capsule, joined proximal to the verumontanum; median lobe enucleated into the bladder. No median lobe: one 6 o'clock incision. 2. **Perimontanal inverted-U incision**; scope pushes laterally to open the lobe–capsule plane at the apex; **early apical release** to protect (spare) the sphincter. 3. Capsular plane developed anteriorly to 11–1 o'clock (round to 12 o'clock); **apical mucosal strip incised early** to avoid sphincter traction. 4. Anterior commissure divided; lateral lobes enucleated (detached) into the bladder (**en bloc, 2-lobe or 3-lobe**). 5. **Haemostasis** with a defocused beam.
- **Two-lobe:** 6 and 12 o'clock grooves; each half enucleated with its share of median lobe.
- **En bloc:** whole adenoma in one piece after circumferential apical release; usually paired with early apical release.
- **Morcellation:** nephroscope, two open inflow lines, **full bladder** and good haemostasis. Morcellators: Piranha (rotating), Versacut (oscillating). Fragments sent for histology.
- **Catheter:** 22 Fr three-way; same-day or next-day removal.

Outcomes and trials

- **Versus TURP (transurethral resection of the prostate):** RCT (randomised controlled trial) in glands <100 mL: better symptom score and PVR (post-void residual), less haemoglobin drop (less bleeding), shorter catheter time and stay, longer operation.
- **Meta-analyses:** better **IPSS (International Prostate Symptom Score) and Qmax (maximum flow rate)**, durable at **3–8 years**. **HoLEP is the only endoscopic procedure with efficacy superior to TURP.**
- **Versus open prostatectomy** for large glands (Naspro 2006; Kuntz 2008, 5 years): equal function, less morbidity, less blood loss, shorter catheter time and stay.
- **Durable; retreatment rare**: whole transition zone removed.
- **Anticoagulated men:** feasible; slightly longer stay; warfarin worse than DOACs (direct oral anticoagulants).
- **Guidelines:** EAU (European Association of Urology) and AUA list HoLEP (enucleation) as a first-line alternative to TURP and to open surgery for large (>80 mL) glands.

Complications

- **Intraoperative:** capsular perforation; ureteric orifice injury (glands >80 mL); **morcellator bladder injury** (stop and drain; intraperitoneal → repair); retained adenoma.
- **Early:** transient storage symptoms and dysuria; **transient (early) SUI (stress urinary incontinence)** from adenoma volume removed, falling over 6 months; persistent SUI rare.
- **Late: retrograde ejaculation about 75%**; BNC (bladder neck contracture) **0–3.2%** (small glands <40 mL, consider bladder-neck incision); meatal stricture; urethral stricture.
- Erectile function unchanged. Transfusion rare; conversion to TURP almost never.

Limitation: learning curve

- The main barrier is the **steep learning curve**: from 20 supervised cases; **efficiency plateaus after about 50 cases**. Mentoring and simulators help.

Recent advances

- High-power **Moses pulse modulation**; **ThuLEP** (thulium laser enucleation) and bipolar enucleation.
- **Ejaculation-preserving (ejaculation-sparing) and median-lobe-only HoLEP** (antegrade ejaculation about 90%).
- **Same-day discharge** and catheter removal.

Diagram to draw

- Sagittal prostate showing the enucleation plane along the surgical capsule, with 5 and 7 o'clock incisions.

> **Key points to remember:**
> - Ho:YAG 2120 nm, water chromophore, 0.4 mm; saline, no TUR syndrome.
> - Size-independent; best (preferred) in anticoagulated men.
> - Only endoscopic method superior to TURP; better IPSS and Qmax; durable; retreatment rare.
> - Retrograde ejaculation about 75%; learning curve about 50 cases.

SAQ Prostatic stents

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 13 Paper 2, Q3f (urethral stents also answered in Practice Set 4 Paper 4, Q3f and 2015-Summer Paper 4, Q3d)

Definition

- A device placed in the **prostatic urethra**, between bladder neck and verumontanum, to relieve BOO (bladder outlet obstruction) without resection. Mainly for men **unfit for surgery**.

Classification

Type	Examples	Features
Temporary, first generation	Urospiral, Prostakath (spirals)	Steel or gold-plated coil; not epithelialised
Temporary, second generation	**Memokath** (thermo-expandable nitinol); Allium (covered nitinol)	Expands with warm water, softens with cold water for removal
Temporary polyurethane	Intraurethral catheter, Spanner	Bulbar anchor; sphincter left free
Biodegradable	Polylactic acid stents	Dissolve; trial use
Permanent, epithelialising	**UroLume** (woven mesh)	Urothelium covers the mesh; withdrawn

Indications

- Retention or severe BOO from BPH (benign prostatic hyperplasia) in men unfit for surgery: avoids a long-term catheter.
- Drainage after thermal treatments; trial of de-obstruction.
- **Also know:** DSD (detrusor–sphincter dyssynergia), now abandoned; bulbar stricture.

Contraindications

- Poor detrusor contractility (do urodynamics if in doubt); untreated UTI (urinary tract infection); large median lobe; bladder stones; stricture.

Technique

- Measure prostatic urethral length cystoscopically. Place under local anaesthesia with endoscopic or fluoroscopic guidance. The stent must **end above the external sphincter**, or incontinence follows.

Complications

- **Migration** (commonest with temporary stents); **encrustation**; UTI; irritative symptoms, perineal pain, haematuria.
- **Hyperplastic ingrowth** and restenosis with permanent stents (difficult removal); incontinence if the sphincter is crossed.

Current status

- Most unfit men void after placement, but many stents are later removed.
- **EAU (European Association of Urology):** alternative to catheterisation in men unfit for surgery (weak). Not in the **AUA (American Urological Association) 2023** BPH list.
- **Recent: iTind**: temporary nitinol device left 5–7 days; it remodels the urethra by ischaemic pressure necrosis. AUA 2023 option for 25–75 mL without median lobe. **Optilume BPH** (paclitaxel balloon).

Diagram to draw: sagittal prostate with a Memokath from bladder neck to verumontanum, above the external sphincter.

> **Key points to remember:**
> - Temporary (Memokath) vs permanent (UroLume, withdrawn).
> - Need detrusor contractility; never cross the sphincter.
> - Migration, encrustation, ingrowth; iTind is the modern temporary device.

Rapid Revision Sheet · Section 16

Prostate anatomy, neurovascular bundles and surgical concepts in localised CA prostate (LAQ)

- PZ gives 70% of cancers; TZ gives BPH; CZ around ejaculatory ducts; no true capsule.
- NVB between prostatic and levator fascia, 5 and 7 o'clock; apex is the commonest margin and nerve-injury site.
- Interfascial plane is the standard nerve-sparing plane; avoid heat and traction; clips, cold scissors.
- Ligate DVC; excise Denonvilliers fascia; spare accessory pudendals.
- Veil and high release for low-risk sides (Tsuzuki).
- NeuroSAFE frozen section: more sparing, safe margins.
- Retzius-sparing: earlier continence, more positive margins.
- Extended PLND captures about 63% of drainage.

IPSS score (SAQ)

- 7 items × 0–5 = 0–35; bands 0–7, 8–19, 20–35.
- QoL 0–6 is separate; IPSS = AUA-SI + QoL.
- 3-point change is noticeable; IPSS does not diagnose BOO.

Methods to estimate prostate size (SAQ)

- Ellipsoid volume = 0.52 × T × AP × L; 1 mL ≈ 1 g.
- DRE underestimates above 30 mL; PSA >1.6/2.0/2.3 by decade predicts >40 mL.
- Planimetry most precise; MRI most reproducible. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Role of UDS before prostate surgery (SAQ)

- PFS is the only test that separates BOO from DUA.
- BOOI >40 obstructed; BCI <100 weak.
- EAU indications: failed surgery, <150 mL voided, PVR >300 mL, age <50 or >80.
- UPSTREAM: routine UDS did not change outcomes.

BPH: medical management and surgical modalities (LAQ)

- α-blockers: rapid relief, no prevention.
- 5-ARIs: prevent AUR and surgery in glands >30–40 mL.
- MTOPS: combination cut progression 66% (↓66%); only 5-ARI-containing arms cut AUR and surgery.
- AUA sizes: TUIP ≤30; TURP/PVP/MISTs (minimally invasive surgical therapies) 30–80; HoLEP any size; simple prostatectomy >80 mL.
- TUR syndrome: gland >45 mL, resection >90 min; bipolar TURP avoids it.

Tamsulosin (SAQ)

- 0.4 mg once daily; α1a/α1d selective; no titration.
- IFIS highest with tamsulosin; persists after stopping.
- MET benefit in distal stones 5–10 mm.

Silodosin (SAQ)

- Most α1A-selective α-blocker; 8 mg with a meal; no titration.
- 4 mg if CrCl 30–50; avoid if <30.
- Equal efficacy to tamsulosin 0.4 mg; greatest BOOI fall; better for MET.
- Anejaculation is the signature effect; least hypotension.

Dutasteride. (SAQ)

- Dual type 1 + 2 inhibitor; 0.5 mg; DHT down 90%; half-life about 5 weeks.
- Volume down 25%; AUR and surgery down about half; needs 6 months.
- Double the PSA; CombAT and REDUCE.

Catheter free trial (SAQ)

- TWOC is first choice after AUR; success about 60%.
- ALFAUR: alfuzosin 10 mg raises success to 62% vs 48%.
- Fails more with age, >1 L drained, spontaneous AUR; 5-ARI prevents recurrence.

TURP instruments, techniques and monopolar vs bipolar (SAQ)

- Continuous-flow 24–28 Fr sheath, 30° telescope.
- Tungsten loop (monopolar) vs platinum–iridium (bipolar).
- Glycine 1.5% for monopolar, saline for bipolar; bag 60 cm.
- Bladder neck to verumontanum, depth to capsule.
- Median lobe first; apex last.
- Bipolar allows saline; TUR syndrome practically eliminated.
- Equal efficacy; bipolar has less bleeding and shorter catheter time.
- TUR syndrome risk: >45 mL gland, >90 minutes.

Measures to decrease complications of TURP (SAQ)

- Bipolar saline TURP removes TUR syndrome risk; fewer transfusions.
- Monopolar: irrigant ≤60 cm, resection <90 min, gland <45 mL safer.
- Verumontanum is the distal limit.

TUR syndrome (SAQ)

- Venous pressure 10 mmHg; risk gland >45 mL, time >90 min.
- Glycine: ammonia and transient blindness.
- 3% saline 100 mL boluses; ≤12 mEq/L in 24 h.
- Bipolar saline TURP prevents it.

Lasers in BPH, including green light (SAQ)

- Chromophore sets penetration: thulium 0.2, holmium 0.4, KTP 0.8 mm, Nd:YAG 1 cm.
- Ho:YAG 2120 nm, water chromophore, 0.4 mm.
- HoLEP is size-independent and superior to TURP (only endoscopic method superior to TURP); suits anticoagulated men.
- Retrograde ejaculation about 75%.
- 532 nm, frequency-doubled Nd:YAG; haemoglobin chromophore; 0.8 mm penetration.
- 180 W XPS with MoXy fibre.
- GOLIATH: PVP non-inferior to TURP; best for anticoagulated men; avoid PVP above 80 cc.

UroLift (prostatic urethral lift) (SAQ)

- 30–80 mL, no median lobe; local anaesthesia.
- Ejaculation preserved; L.I.F.T. retreatment 13.6% at 5 years.
- TURP wins (better) on IPSS, flow and durability.

Millin's prostatectomy (SAQ)

- Millin 1945: transverse capsulotomy ~2 cm below bladder neck; bladder not opened.
- Apex divided under vision; 5 and 7 o'clock sutures.
- Retrograde ejaculation 80–90%; transfusion <5%; bladder neck contracture 2–6%.

Laparoscopic Trans vesical Prostatectomy (SAQ)

- Indication: gland >80 g, especially with median lobe, stones or diverticulum.
- Posterior plane first; divide urethra proximal to verumontanum.
- 5 and 7 o'clock sutures plus mucosal advancement.
- Same function as open, less morbidity.

HoLEP (LAQ)

- Ho:YAG 2120 nm, water chromophore, 0.4 mm; saline, no TUR syndrome.
- Size-independent; best (preferred) in anticoagulated men.
- Only endoscopic method superior to TURP; better IPSS and Qmax; durable; retreatment rare.
- Retrograde ejaculation about 75%; learning curve about 50 cases.

Prostatic stents (SAQ)

- Temporary (Memokath) vs permanent (UroLume, withdrawn).
- Need detrusor contractility; never cross the sphincter.
- Migration, encrustation, ingrowth; iTind is the modern temporary device.

SECTION 17

Prostate Cancer

153 Epidemiology, Aetiology and Prevention of Prostate Cancer

1 · Epidemiology — the headline

- It is the **second commonest** non-skin cancer and the **fifth** cause of cancer death in men; 2020 saw ≈ **1.4 million cases, 375,000 deaths**.
- Incidence varies ~13-fold (South Central Asia lowest, Northern Europe highest). US lifetime risk is ≈ **1 in 8**. 5-year survival is 97%, yet it is the second cause of cancer death in US men.
- **Incidence and mortality are decoupled.** Incidence follows screening and wealth, while mortality peaks where African ancestry meets poor access: the **Caribbean has the highest mortality in the world** MCQ.
- **Latent (autopsy) cancer** is ~5% under 30 y, 29% at 60–69 and **59% >79 y**, and higher in Black men at every age.

> **KEY POINT**
>
> In a prostate-specific antigen (PSA)-screened population, risk-factor studies may find determinants of being tested, so the preferred end point is **aggressive disease**. **Healthy-user bias**: health-conscious men seek PSA tests, so healthy habits falsely seem to raise incidence while lowering mortality.

2 · Established risk factors

Only three are unequivocal: **age, race (ancestry) and family history**.

- **Age**: mean age at diagnosis is 66 and it is rare under 40. Diagnosis at ≥70 carries a higher risk of cancer death.
- **Race**: Black men have the highest incidence (172.6/100,000 in the US) and **twice** the cancer-specific mortality of White men, presenting younger and more metastatic. Drivers are unequal access, comorbidity and **8q24** (and 17q21) alleles; with **equal access, outcome is equal** in non-metastatic disease.
- **Family history** is present in ~20%. Twin-study heritability is up to **58%**, so genes outweigh shared lifestyle.

Family history	Relative risk
Any first-degree relative (father or brother)	2–3×
Two first-degree relatives	~3.5×
Both father and brother	**9×**
Relative diagnosed <60 vs ≥60	~2.5× vs ~1.6×
Father/brother died of prostate cancer	2× risk of prostate cancer death

3 · Germline genetics

The key inherited genes are **DNA-repair genes**, which predict **aggressive** cancer.

Gene	Pathway	Association	Implication
BRCA2, BRCA1	Homologous recombination (HR)	BRCA2 familial, aggressive	Early screening; poly(ADP-ribose) polymerase (PARP) inhibitor; caution in active surveillance (AS)
ATM	HR	Metastatic disease	As for BRCA
PALB2, CHEK2	HR / DNA repair	Familial; PALB2 lethal	CHEK2: targeted screening
MLH1, MSH2, MSH6, PMS2 (**Lynch**)	Mismatch repair (MMR)	2–5% of cases	Checkpoint inhibitor; very sensitive to androgen deprivation therapy (ADT)
HOXB13	Prostate-specific homeobox	Hereditary prostate cancer	Targeted screening
8q24, 17q21	Genome-wide association study (GWAS) loci	Commoner in African ancestry	Explain part of the racial excess

- **Polygenic risk scores** predict any cancer, not aggressive cancer, so they are not yet practice-changing.

SAQ Indications of genomic testing in carcinoma prostate

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 4 Paper 4, Q3e (PARP [poly-ADP-ribose polymerase] inhibitors also answered in 2024-Winter Paper 4, Q3f)

Types of test

Type	Sample	What it finds	Use
Germline	Blood or saliva	Inherited variants: BRCA2, BRCA1, ATM, CHEK2, PALB2, HOXB13, MMR (mismatch repair: MLH1, MSH2, MSH6, PMS2)	Therapy, family testing, screening
Somatic	Tumour tissue (usually primary) or ctDNA (circulating tumour DNA)	HRR (homologous recombination repair) genes, CDK12, MSI-H (microsatellite instability-high)/dMMR (MMR deficiency)	Targeted therapy
Gene-expression classifiers	Biopsy or prostatectomy tissue	Decipher (22 RNA markers); Oncotype DX GPS (Genomic Prostate Score, 17 genes); Prolaris (cell-cycle genes)	Prognosis

Indications: germline testing

- **All metastatic** disease (AUA [American Urological Association], NCCN [National Comprehensive Cancer Network]): over 10% carry actionable DNA-repair mutations.
- **High- or very-high-risk** localised disease.
- **Intermediate risk with intraductal or cribriform histology** (consider).

- **Family history:** early-onset or metastatic prostate cancer (relative <60 years); BRCA-related (breast, ovarian, pancreatic) or Lynch-related (colorectal, endometrial, upper tract) cancers; known familial mutation.
- **Ashkenazi Jewish ancestry.**
- **Why:** PARP inhibitor eligibility; caution with active surveillance in BRCA2; carrier screening; cascade testing of relatives with counselling.

Indications: somatic testing

- **Metastatic disease, especially mCRPC (metastatic castration-resistant prostate cancer):** HRR alterations in up to 20–25%.
- **HRR (BRCA2 most)** → PARP inhibitors (olaparib, rucaparib, niraparib, talazoparib).
- **MSI-H/dMMR** (2–5%) → pembrolizumab (tumour-agnostic FDA [Food and Drug Administration] approval, 2017); then germline Lynch testing.
- Tissue preferred; ctDNA alternative, but **CHIP (clonal haematopoiesis of indeterminate potential)** gives false calls.

Indications: tissue classifiers

- **Low and favourable-intermediate risk:** active surveillance versus treatment; surveillance intensity. Never the sole basis.
- **Intermediate and high risk:** metastasis risk guides treatment intensity.
- **After prostatectomy:** high Decipher score may favour adjuvant or early salvage radiotherapy (retrospective evidence).
- **Limits:** tumour heterogeneity, retrospective data, cost.
- **Also know:** ConfirmMDx methylation before repeat biopsy; polygenic risk scores.

Diagram to draw: flowchart: metastatic → germline + somatic; localised high-risk or family history → germline; localised low or intermediate → tissue classifier.

> **Key points to remember:**
> - Germline: all metastatic, high-risk, family history, Ashkenazi.
> - Somatic: HRR → PARP inhibitors; MSI-H → pembrolizumab.
> - BRCA2: aggressive; avoid casual active surveillance.
> - Decipher, Oncotype GPS, Prolaris: prognosis only.

Also, from the condensed book (not in the paper answer):

- GWAS has found >200 single-nucleotide polymorphisms, each of small effect.

4 · Molecular aetiology and inflammation

- **Chronic inflammation** is in nearly every biopsy and may be causal; **proliferative inflammatory atrophy** is the proposed precursor, via high-grade prostatic intraepithelial neoplasia (HGPIN).
- **TMPRSS2-ERG** fusion puts the ERG oncogene under an androgen-regulated promoter. It is in ~half of Western tumours but uncommon in East Asians. See Ch 154

5 · Possible risk factors — body, metabolism, lifestyle

Most lifestyle factors act on **progression** rather than initiation; evidence is observational.

Factor	Total incidence	Aggressive / fatal
Height	~5% higher per 10 cm	More high-grade cancer and death
Obesity	Mixed; less low-grade	**Consistently raised**
Smoking	Not raised (smokers screen less)	**Mortality raised ~40%**; no excess after quitting ≥10 y
Metabolic syndrome, hypertension, cardiovascular disease (CVD)	Slightly raised	Metabolic syndrome: worse pathology
Diabetes	Lower (low testosterone, IGF-1 and PSA; metformin)	—
Vigorous activity	No clear effect	Less advanced disease
Alcohol	Unclear	Mortality raised at ≥45 g/day; binge drinking raises it

- Adiposity raises insulin, IGF-1, leptin, inflammation and oestrogen, which drives proliferation and blocks apoptosis.
- **Adult serum testosterone does not predict risk** MCQ.
- A **composite healthy lifestyle** means **30–50%** less lethal cancer; in the highest polygenic-risk quartile, lifetime lethal risk falls from 5.3% to 1.6%.
- **Diet**: World Cancer Research Fund 2018 found **no dietary factor with strong evidence**; dairy, calcium and lycopene were downgraded.
- **Vasectomy** is not a risk factor (detection bias), and the AUA says it need not be discussed.
- Transgender women on oestrogen have a median PSA ≈ **0.02 ng/mL** (50-fold lower), so cut-points must be lowered.

6 · Supplements

SAQ Antioxidants in urological malignancy

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 1 Paper 4, Q3a (prevention trials also in 2017-Summer Paper 4, Q3a)

Rationale

- ROS (reactive oxygen species) cause **oxidative DNA damage**; polyunsaturated fat and excess iron increase it. Antioxidants scavenge ROS, so diet studies suggested protection.
- Randomised trials of single high-dose supplements have been **null or harmful**.

Prostate cancer

Agent	Evidence	Result
SELECT (Selenium and Vitamin E Cancer Prevention Trial)	35,533 men; selenium 200 µg, vitamin E 400 IU, both, or placebo	Stopped early (median 5.46 years) for futility. No benefit. Vitamin E alone trended to more cancer (P = 0.06); extended follow-up showed a significant **increase (about 17%)**
ATBC (Alpha-Tocopherol, Beta-Carotene)	Low-dose vitamin E in heavy smokers	Secondary analysis: prostate cancer ↓32%, deaths ↓41%; beta-carotene ↑ lung cancer

Agent	Evidence	Result
Selenised yeast	Skin-cancer trial	Secondary fall in prostate cancer prompted SELECT
Lycopene (cooked tomato)	Meta-analysis: modestly lower risk, mainly advanced cancer	WCRF (World Cancer Research Fund) downgraded "probable" (2007) to "no conclusion" (2018); supplement trials mostly null
Green tea catechins	HGPIN (high-grade prostatic intraepithelial neoplasia)/ASAP trials	Large fall in progression to cancer (meta-analysis); dose impractical
Vitamin C, high-dose antioxidants	High doses act as pro-oxidants	Supplements above median → more advanced cancer

- **Also know:** vitamin D (VITAL, no effect); coffee (modestly lower advanced risk); pomegranate, soy and curcumin unproven.

Bladder cancer

- Supplemental beta-carotene, vitamins C and E and folate do not lower risk; SELECT showed no effect.
- Fruit and vegetable benefit disappears after adjusting for smoking. Smoking cessation is the real prevention.

Kidney cancer

- No antioxidant trial has shown benefit.

Why the trials failed

- Single nutrient at high dose; wrong chemical form; non-smoking populations; PSA-era indolent cancers; U-shaped dose effects.

Current position

- Antioxidant supplements are **not recommended** for cancer prevention. Advise a balanced diet, no smoking and normal weight.

Diagram to draw: ROS → DNA damage → cancer, with antioxidants blocking the first arrow.

> **Key points to remember:**
> - SELECT: no benefit; vitamin E alone increased prostate cancer.
> - ATBC benefit limited to smokers on low dose.
> - Lycopene downgraded to "no conclusion" (2018).

SAQ Prostate cancer prevention trials

Paper answer · P4

Type: SAQ (10 marks)
Source: 2017-Summer Paper 4, Q3a

Rationale

- Prostate cancer is common, slow and androgen-dependent. Men with 5α-reductase deficiency have rudimentary prostates. This supported **5ARI (5α-reductase inhibitor)** trials.

Landmark trials

	PCPT (2003)	REDUCE (2010)	SELECT (2009)
Agent	Finasteride 5 mg	Dutasteride 0.5 mg	Selenium 200 µg ± vitamin E 400 IU
Men	18,882; PSA (prostate-specific antigen) ≤3	PSA 2.5–10, prior negative biopsy	35,533
Design	7 years; end-of-study biopsy	Biopsy at 2 and 4 years	Stopped early for futility
All cancers	↓24.8%	↓22.8%	No benefit
High grade	Gleason 7–10 ↑ (6.4% vs 5.1%)	Gleason 8–10 ↑ (0.9% vs 0.6%)	Vitamin E alone ↑ risk

The high-grade controversy

- Proposed artefacts: better PSA performance and a smaller gland make cancer easier to find.
- The FDA (Food and Drug Administration) did **not accept** these. In 2011 it refused a prevention licence and added a **high-grade warning** to 5ARI labels.
- PCPT at 18.4 years: non-significant 25% fewer prostate cancer deaths. Real-world data show no mortality effect.
- **Conclusion:** 5ARIs reduce detection of low-grade cancer, not death.

Vitamins and other agents

- **ATBC** trial: low-dose vitamin E cut prostate cancer 32% in smokers (secondary end point). SELECT then failed.
- **VITAL** (vitamin D): no prostate effect. High-dose zinc, calcium and antioxidants may harm.
- **Aspirin/NSAIDs** (non-steroidal anti-inflammatory drugs): small effect. **Statins:** observational benefit only. **Metformin:** none.
- Diet: whole foods; avoid obesity.

Current position

- No agent is approved for prevention. 5ARIs may be discussed with men taking them for BPH (benign prostatic hyperplasia).

Diagram to draw: bar chart: all cancers ↓ vs high-grade ↑ in PCPT and REDUCE.

> **Key points to remember:**
> - PCPT ↓24.8%; REDUCE ↓22.8%; both ↑ high grade.
> - FDA warning; no approved preventive agent.
> - SELECT negative; vitamin E alone harmful.

Also, from the condensed book (not in the paper answer):

- Calcium supplements raise fatal cancer (relevant on ADT), and zinc ≥75 mg/day raises lethal cancer.
- No supplement benefits healthy men on a balanced diet.

KEY POINT

Vitamin E worked in the ATBC trial but failed in SELECT: ATBC used a tenfold lower dose in **heavy smokers** with little PSA screening (aggressive cancers); SELECT used 400 IU in non-smokers with widespread screening (indolent cancers). Different disease, different answer.

7 · Chemoprevention

*See the paper answer **Prostate cancer prevention trials** (Section 17, Ch 153).*

Also, from the condensed book (not in the paper answer):

- No preventive agent is FDA-approved. 5α-reductase inhibitors shrink the gland about 25%, so on a 5ARI you **double the PSA**. See Ch 154
- **Secondary prevention**: in the Prostate Cancer Lifestyle Trial (men on AS), plant-based diet, exercise and stress management meant **5% vs 27%** treated at 2 y.
- Activity (≥150 min/week moderate or ≥75 min vigorous) lowers cancer-specific mortality ~31%, while a Western diet and continued smoking raise cancer death.

KEY POINT

Cardiovascular disease kills more men with non-metastatic prostate cancer than the cancer does, so lifestyle advice benefits both.

8 · PSA screening

LAQ PSA screening for prostate cancer: evidence, problems and solutions

Paper answer · P2

Asked as:

- What are the various problems in PSA screening — *Practice Set 8 Paper II Q1 (LAQ)*
- Screening of CA prostate. — *Practice Set 13 Paper II Q3d (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

Introduction (concept)

- Screening tests healthy, symptom-free (asymptomatic) men; its aim is lower prostate cancer mortality through earlier treatment (cure). The price is **overdiagnosis** (cancers that would never cause harm) and **overtreatment**.
- PSA (prostate-specific antigen) is a kallikrein (hK3) made by prostatic epithelium. It reaches blood when glandular architecture is disrupted. It is **organ-specific, not cancer-specific**.
- Since PSA testing began, (US) mortality has fallen by >40% (over 40%; SEER data), but at the price of overdiagnosis.

Tools

- **PSA:** biopsy threshold 4 ng/mL (many use 2.5–3; ERSPC used 3). Repeat a raised PSA before biopsy.
- **DRE (digital rectal examination):** little value alone; abnormal DRE raises risk.
- **Refinements:** PSA density ≥0.15; free PSA ≤25%; PHI (Prostate Health Index), 4Kscore, Stockholm3; urine tests (PCA3 [prostate cancer antigen 3], ExoDx, SelectMDx).

- **MRI (magnetic resonance imaging) before biopsy:** fewer biopsies and less low-grade overdiagnosis.

Evidence from screening trials

Trial	Design	Result
ERSPC (European Randomised Study of Screening for Prostate Cancer)	>182,000 men; PSA every 2–4 years vs none	**20% relative fall** in (prostate) cancer death at 16 years; 570 invited and 18 extra diagnoses per death prevented
PLCO (Prostate, Lung, Colorectal and Ovarian), USA	Annual PSA vs usual care	**No difference** at 17 years: control arm heavily PSA-tested (contamination), lower power
Göteborg-1 (Sweden)	Biennial PSA, men 50–64	44% fall in cancer death at 14 years (larger mortality reduction with biennial screening)
CAP (Cluster randomised trial of PSA testing for prostate cancer), UK	Single PSA invitation	No mortality difference at 10 years; only a small fall at 15 years
Göteborg-2 (2022)	MRI-targeted biopsy only	Halved overdiagnosis
- Corrected for non-attendance and contamination, ERSPC benefit is 27–31%; it appears only after about 10 years.		

Problems in PSA screening

1. Poor specificity (false positives)

- BPH (benign prostatic hyperplasia), prostatitis, retention, catheterisation, biopsy and TURP (transurethral resection of prostate) all raise PSA.
- At PSA 4–10 ng/mL with a normal DRE (digital rectal examination), only 17–32% of biopsies find cancer. Most men biopsied therefore have no cancer.

2. Poor sensitivity: no safe cut-off

- PCPT (Prostate Cancer Prevention Trial): risk is continuous; about 1 in 11 men biopsied at PSA 0–1 ng/mL has cancer.
- A high threshold misses curable cancer; a low threshold adds biopsies and overdiagnosis. Thresholds used: 4, 3 or 2.5 ng/mL.

3. Biological variability of PSA

- Rises with age, prostate volume (about 4% per mL) and Black race; physiological fluctuation; assay differences.
- Lowered by 5α-reductase inhibitors (about 50% at 12 months, so double the value), obesity, hypogonadism and statins.

4. Overdiagnosis (the central problem)

- Detection of cancers that would never have caused symptoms or death. Autopsy prevalence is high: 29% at 60–69 years and 59% after 79.
- Mostly Grade Group 1 cancers.

5. Overtreatment

- Overdiagnosed men treated radically suffer ED (erectile dysfunction), incontinence and bowel toxicity for no gain, especially above 65 years.

6. Harms of biopsy

- Pain, haematuria, haematospermia, retention and sepsis (transrectal route); anxiety from repeat biopsies.

7. Modest and delayed benefit

- Screening prevents only 20–30% of cancer deaths; no overall survival benefit has been shown. Men with life expectancy <10 years cannot benefit.

8. Biases in interpreting screening data

- **Lead-time bias**: earlier diagnosis lengthens apparent survival without delaying death.
- **Length-time bias**: screening preferentially finds slow-growing tumours.
- **Healthy-user bias**: health-conscious men seek testing, confounding risk-factor studies.

9. Unsettled policy

- Start age, stop age, interval and threshold are all debated; further large randomised trials are unlikely.
- Opportunistic screening is common and inequitable.

Guideline positions (guidelines)

Body	Recommendation
USPSTF (US Preventive Services Task Force) 2012	Grade D: against screening
USPSTF 2018 (current)	**Grade C, 55–69 years: individual shared decision; Grade D ≥70** (not ≥70; was Grade D in 2012)
AUA (American Urological Association)	Shared decision 55–69; every 2 years (2-yearly interval)
NCCN (National Comprehensive Cancer Network)	Baseline PSA and discussion at 45
EAU (European Association of Urology)	Risk-adapted early detection: informed men from 50; earlier start (45) with family history or African descent, 40 for BRCA2 carriers
- **No body recommends (advises) unselected population screening.**	
- **High risk:** Black men, father or brother diagnosed <65, BRCA2 and other DNA-repair mutations: start earlier.	
- Stop when life expectancy is under 10–15 years.	

Solutions: risk-adapted, MRI-first pathway (reducing harm)

```
Informed man, life expectancy >10–15 y → PSA (repeat if raised) → risk stratify: PSA density, free/total PSA, risk calculator, PHI / 4Kscore / Stockholm3 → MRI (magnetic resonance imaging) PI-RADS 1–2, low density → no biopsy, keep in follow-up PI-RADS ≥3 → targeted ± systematic biopsy (transperineal) → Grade Group 1 → active surveillance
```

- **Baseline PSA** predicts future risk (Malmö): low PSA at 40–60 allows longer intervals (intervals by baseline PSA).
- **PROMIS (2017):** MRI sensitivity 93% vs 48% for TRUS (transrectal ultrasound) biopsy; 27% could avoid biopsy.
- **PRECISION (2018):** MRI-targeted pathway found more significant cancer (38% vs 26%) and less insignificant cancer; 28% avoided biopsy.
- **Göteborg-2 (2022):** MRI with targeted biopsy in screening roughly halved overdiagnosis.
- **Active surveillance** of low-risk disease (Grade Group 1) uncouples overdiagnosis from overtreatment; MRI and biomarkers before biopsy.
- **Also know:** germline testing (BRCA2, ATM, HOXB13); polygenic risk scores; transperineal biopsy lowers sepsis.

Diagram to draw: iceberg of latent vs clinically significant cancer, plus the MRI-first flowchart (pathway: PSA → repeat → risk calculator or MRI → targeted biopsy → active surveillance or treatment).

> **Key points to remember:**
> - ERSPC: (about) 20% fall in prostate cancer death; PLCO null from contamination.
> - Core problems: low specificity, no safe cut-off, overdiagnosis, overtreatment.
> - USPSTF 2018: Grade C at 55–69, Grade D ≥70.
> - Fix: risk-adapted PSA, MRI before biopsy, active surveillance for GG1; MRI-first and active surveillance reduce overdiagnosis and overtreatment.

Also, from the condensed book (not in the paper answer):

- Screening lowers cancer death ~**20–30%**, and the benefit builds with time.

9 · Landmark trials

Trial	Design	Result	Take-home
ERSPC (European screening study)	>182,000; PSA every 2–4 y vs none	Cancer death fell 16% (9 y), then ~20% (16 y); excess incidence fell 85% to 36%	Benefit is time-dependent
PLCO (US)	Annual PSA vs usual care	Control arm heavily screened	Contamination dilutes the effect
Göteborg-1	Swedish screening	Significantly fewer cancer deaths	Supports ERSPC
CAP (UK)	Single PSA invitation	No benefit at 10 y; tiny fall in death at 15 y	One-off screening is weak
PCPT	18,882, PSA ≤3; finasteride 5 mg 7 y	Cancer fell **24.8%** (18.4 vs 24.4%); Gleason 7–10 6.4 vs 5.1%; 18-y death fell 25% (not significant)	Fewer cancers, more high-grade
REDUCE	PSA 2.5–10, prior negative biopsy; dutasteride 0.5 mg 4 y	Cancer fell **22.8%**; Gleason 8–10 0.9 vs 0.6%	FDA warning
SELECT	35,533; selenium 200 µg with or without vitamin E 400 IU	Stopped for futility; vitamin E raised risk (P = 0.06)	No supplements

Trial	Design	Result	Take-home
VITAL	>25,000; vitamin D3 2000 IU 5 y	No prostate effect	Not preventive

154 Prostate Cancer Biomarkers

1 · Why biomarkers beyond PSA

- Prostate-specific antigen (PSA) is **organ-specific, not cancer-specific**, which causes negative biopsies and overdiagnosis of Grade Group (GG) 1. Guidelines advise **secondary testing** (imaging or biomarker) before biopsy; the target is **clinically significant cancer (GG ≥2)**. See Ch 153
- Settings are before initial or repeat biopsy, favourable-risk cancer (active surveillance (AS) vs treatment), and after radical prostatectomy (RP).

2 · PSA biology

- PSA is **hK3** (gene KLK3 on **19q13**), an androgen-regulated serine protease that **liquefies semen**. Half-life is **2–3 days**.
- It is made as **proPSA**, and **hK2 cleaves the leader** to give active PSA.
- Serum PSA rises when it leaks through a lost basal layer; per cell, cancer makes less PSA than benign tissue.

Serum form	% of PSA	Measured?
PSA–α1-antichymotrypsin (ACT)	60–90	Yes (predominant, inactive)
PSA–α2-macroglobulin	10–20	**No**: epitopes masked
PSA–α1-protease inhibitor	1–5	Yes
Free PSA	5–45	Yes (proPSA, benign PSA (BPSA), intact PSA)

Confounders

- After biopsy the rise lasts ≥**4 weeks**; ejaculation raises it for 24 h, so repeat after abstinence.
- **5α-reductase inhibitors (5ARIs)** lower PSA ~**50%** by 12 months, so **double** it.
- Age-specific upper limits (ng/mL): 40–49, 2.5; 50–59, 3.5; 60–69, 4.5; 70–79, 6.5.

3 · PSA derivatives

Derivative	Definition / values		Use
PSA density	PSA ÷ volume; cut-off (0.08–0.15)	**0.15**	After negative/equivocal magnetic resonance imaging (MRI); ≤0.15 = Johns Hopkins AS entry; predicts AS reclassification
PSA velocity	Change per year		Minimal value; **never an isolated biopsy trigger**

Derivative	Definition / values	Use
% free PSA	Cancer PSA binds ACT, so **low %fPSA suggests cancer**; thresholds 15–25%	FDA: PSA 4–10, benign digital rectal examination (DRE); mainly **after negative biopsy**; unaffected by 5ARI

> **KEY POINT**
>
> %fPSA was validated against any cancer, not GG ≥2, so it overestimates significant risk.

- **[–2]proPSA** rises with Gleason and underpins the Prostate Health Index (PHI); **BPSA** tracks benign hyperplasia and has no prognostic value.

4 · Blood and urine tests (NCCN 2023 pre-biopsy)

Six tests estimate GG ≥2 risk; **thresholds are setting-specific** (higher for repeat biopsy).

Test	Components	Threshold / key fact
PHI	Blood: **(p2PSA ÷ fPSA) × √tPSA**	24–28.6 proposed; ~30% biopsies avoided
4Kscore	Blood: tPSA, fPSA, iPSA and **hK2**, with age, DRE, prior biopsy	**7.5%**: sensitivity 94%, negative predictive value (NPV) 95%; ≤7.5% = low long-term metastasis risk
IsoPSA	Blood: PSA isoform structure	Unaffected by 5ARI or α-blockers
SelectMDx	Post-DRE urine DLX1, HOXC6 mRNA with clinical data	NPV ~95%; needs prostate volume
ExoDx (EPI)	Urine exosomes: PCA3, ERG, SPDEF	**No DRE**; 15.6 initial, 20 repeat
MyProstateScore	Post-DRE urine PCA3, TMPRSS2:ERG and PSA	10 initial (sensitivity 97%), 15 repeat; validated in Black men

- **PCA3** is a long non-coding RNA. Score = (PCA3 ÷ KLK3 mRNA) × 1000 after an **attentive DRE**; thresholds 25/35. It is **independent of prostate size** and FDA-approved after a **previous negative biopsy**.
- **TMPRSS2:ERG** is **~100% specific** but in only ~half of tumours, so it is combined with PCA3.

> **KEY POINT**
>
> Urine tests follow a DRE because cancer RNA comes only from shed cells massaged into the urethra; ExoDx avoids this, as exosomes are shed spontaneously. Urinary markers are highly specific, but their coverage is limited, hence panels.

5 · Tissue-based biomarkers

SAQ Molecular staging of CA prostate

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 14 Paper 2, Q3e (related: prognosis in 2019-Summer Paper 4, Q3e)

Definition and rationale

- Clinical staging (DRE (digital rectal examination), PSA (prostate-specific antigen), grade, imaging) misses occult spread and cannot fully predict behaviour.
- **Molecular staging** uses DNA, RNA or protein markers in tissue, blood or nodes to find hidden disease and predict outcome.

1. Detection of occult spread

- **RT-PCR (reverse transcription polymerase chain reaction)** for PSA or PSMA (prostate-specific membrane antigen) mRNA in blood, marrow or "negative" nodes: detects micrometastases, but inconsistent, so not used clinically.

- **CTCs (circulating tumour cells):** counts are prognostic in metastatic disease; **AR-V7** (splice variant lacking the ligand-binding domain) predicts resistance to ARPIs (androgen receptor pathway inhibitors).
- **ctDNA (circulating tumour DNA)** liquid biopsy: DNA-repair mutations; clonal haematopoiesis gives false calls.

2. Tissue genomic classifiers

Test	Basis	Predicts
Decipher	22-RNA classifier	Metastasis, cancer death; post-prostatectomy radiotherapy decision
Oncotype DX GPS (Genomic Prostate Score)	17 genes (12 cancer, 5 reference) by RT-PCR	Adverse pathology; AS (active surveillance) choice
Prolaris	31 cell-cycle progression genes	Progression, cancer death

- **Also know:** PORTOS (24-gene radiotherapy response score); ConfirmMDx (methylation, after negative biopsy).

3. Genetic markers

- **Germline:** BRCA2 (aggressive disease), BRCA1, ATM, CHEK2, Lynch genes, HOXB13 (risk, not aggressiveness); test in metastatic or high-risk disease.
- **Somatic:** PTEN loss, TMPRSS2:ERG fusion, homologous recombination repair defects (guide PARP (poly ADP-ribose polymerase) inhibitors).

4. Molecular imaging

- **PSMA PET (positron emission tomography):** more sensitive and specific than CT and bone scan for nodes and metastases (proPSMA), but cannot exclude micrometastases.

Clinical use

- Refines **AS vs treatment** and post-operative radiotherapy decisions; guides targeted therapy.

Limitations

- Mostly retrospective; **multifocality and heterogeneity** (the sampled focus may not be the worst); cost; added value still being tested (G-MAJOR trial).

Diagram to draw: clinical stage vs molecular stage (tissue, blood, nodes, imaging).

> **Key points to remember:**
> - Molecular staging finds occult disease and predicts behaviour beyond TNM.
> - Decipher 22 genes; Oncotype 17 genes; Prolaris cell-cycle genes.
> - AR-V7 on CTCs predicts ARPI resistance; BRCA2 means aggressive disease.
> - Limitation: heterogeneity and mostly retrospective data.

Also, from the condensed book (not in the paper answer):

- **ConfirmMDx** measures GSTP1, APC and RASSF1 methylation in negative cores (NPV ~88–90%), using the **field ("halo") effect** around a missed tumour.
- **Decipher ≥0.4 post-RP** suggests benefit from radiotherapy.

6 · Germline testing

Up to 40–50% of risk is familial, mostly via **DNA-damage-repair** genes.

Gene	Risk	Phenotype
BRCA2	4.5× <65 y; ~8.6× overall	Younger, higher grade, shorter survival
BRCA1	1.8× <65 y	—
ATM, CHEK2	CHEK2 odds ratio ~2–3.4	ATM aggressive
Lynch (MSH2 etc.)	MSH2 ~10×	Also colorectal, upper tract
HOXB13 G84E	4.5×	**Not** more aggressive

Testing is for BRCA1/2 alone or a panel (BRCA1/2, ATM, CHEK2, MLH1, MSH2, TP53, EPCAM). Indications: see Ch 153.

7 · Clinical algorithm

ALGORITHM

1. **Raised PSA**: repeat it, excluding infection, instrumentation, ejaculation and 5ARI. Then do MRI and/or a biomarker (PHI, 4Kscore, IsoPSA, SelectMDx, EPI, MPS; PSA density). If risk is high, biopsy.
2. **Negative biopsy, persistent suspicion**: MRI, %fPSA, PCA3/EPI/MPS (repeat thresholds), ConfirmMDx. If positive, targeted re-biopsy.
3. **Favourable-risk**: PSA density and Prolaris/Oncotype/Decipher decide AS vs treatment.
4. **After RP**: PSA should be undetectable. If ≥0.2 ng/mL and confirmed, this is biochemical recurrence; Decipher/PORTOS then guides adjuvant or salvage radiotherapy.

155 Prostate Biopsy: Techniques and Imaging

1 · Indications and contraindications

On transrectal ultrasound (TRUS) the peripheral (PZ) and central zones are homogeneous and echogenic; the transition zone (TZ) is heterogeneous. **39%** of cancers are isoechoic, so a normal grey scale never precludes biopsy.

SAQ TRUS and prostate biopsy

Paper answer · P1 · P2 · merged from 3

Asked as: P2 SAQ "Transrectal ultrasonography (TRUS)" (Transrectal ultrasonography; TRUS) · P2 SAQ "TRUS / prostate biopsy" (TRUS biopsy.; Prostate biopsy) · P1 SAQ "Transrectal ultrasound study of prostate"

Read as: transrectal ultrasonography (TRUS) of the prostate, and TRUS-guided prostate biopsy.

Physics, principle and technique of TRUS

- Piezoelectric crystals; higher frequency gives better resolution but less penetration (depth).
- The prostate lies directly in front of the rectum. Probes: **6–10 MHz** end-fire, side-fire or biplane.
- Left lateral position (or lithotomy for transperineal work), after DRE (digital rectal examination); transverse and sagittal planes, base to apex.

Normal sonographic anatomy

Zone (McNeal)	Glandular share	TRUS appearance	Cancer share
Peripheral	~70%	Homogeneous, echogenic (posterior)	~70%
Transition	5–10% (BPH [benign prostatic hyperplasia])	Heterogeneous, anterior	~20–25%
Central	~25%	Homogeneous, echogenic, at base	~5–10%
Anterior fibromuscular stroma	Non-glandular	Hypoechoic	–

- Zones are not seen as such; TRUS separates echogenic posterior PZ (peripheral zone) from heterogeneous TZ (transition zone).
- **Corpora amylacea** mark the surgical capsule.
- Seminal vesicles about 4.5–5.5 × 2 cm; absent seminal vesicle → ipsilateral renal agenesis in 79%.

Measurements

- **Volume (ellipsoid) = 0.52 (π/6) × transverse × AP (anteroposterior) × length**; 1 mL ≈ 1 g; normal 20–25 g. **Planimetry** (serial sections) for brachytherapy.
- **PSAD (PSA [prostate-specific antigen] density)** = PSA/volume; **>0.15 ng/mL/cc** is suspicious (suggests cancer).

Findings

- **Cancer:** classically a hypoechoic peripheral-zone lesion, but **39% are isoechoic**; false positives are BPH nodules, prostatitis and infarcts. Capsular bulge suggests extracapsular (extraprostatic) extension.
- **Cysts:** Müllerian (midline); ejaculatory duct cyst (just off midline); seminal vesicle cyst (lateral; renal agenesis).
- Abscess, calculi.
- After ADT (androgen deprivation therapy) or radiotherapy the gland shrinks and becomes diffusely hypoechoic.

Indications for TRUS

- **Guided biopsy:** 12-core systematic ± MRI (magnetic resonance imaging)-targeted fusion.
- Volume before BPH surgery, brachytherapy or cryotherapy; fiducial markers.
- **Azoospermia:** ejaculatory duct obstruction (seminal vesicles >1.5 cm wide, dilated ducts, midline cyst).
- Drainage of prostatic abscess or cysts; haematospermia.

Prostate biopsy: introduction

- TRUS-guided systematic 12-core biopsy has been the standard for diagnosing prostate cancer; about 30% are positive at first biopsy.

Prostate biopsy: indications and contraindications

- Raised PSA with risk factors; suspicious DRE; MRI PIRADS (Prostate Imaging Reporting and Data System) ≥3; suspected metastasis.
- **Repeat:** rising PSA; ASAP (atypical small acinar proliferation; 50% cancer risk); positive biomarkers (PHI [Prostate Health Index], 4Kscore, PCA3); active surveillance. Saturation biopsy only after negative biopsies.
- **Contraindications:** coagulopathy, acute UTI (urinary tract infection) or prostatitis, severe immunosuppression; no anus (use transperineal route).

MRI-first pathway

- **PROMIS (2017):** mpMRI (multiparametric MRI) sensitivity 93% vs 48% for TRUS biopsy.
- **PRECISION (2018):** MRI-targeted biopsy found more significant cancer (38% vs 26%) and avoided 28% of biopsies.
- Targeting: cognitive, MRI/US fusion or in-bore; 2–3 cores per target ± systematic cores.

Prostate biopsy: preparation

- Continue low-dose aspirin; stop warfarin and clopidogrel 5–7 days, DOACs (direct oral anticoagulants) 2–5 days before.
- **Transrectal: antibiotics for all** (contaminated route), ≤24 hours: fluoroquinolone, or cephalosporin ± aminoglycoside. Augmented or swab-directed if risk factors (antibiotics in last 6 months, diabetes, previous infection). Povidone-iodine rectal cleansing.
- Enema optional. **Periprostatic block:** 1–2% lidocaine, 5 mL each side at the junction of prostate base and seminal vesicle.

Prostate biopsy: technique

	Transrectal	Transperineal
Position	Left lateral	Lithotomy
Needle route	Rectal wall (contaminated)	Sterile perineal skin
Antibiotics	Mandatory	Skin-flora only or none
Anterior/apex	Poor	Good
Anaesthesia	Periprostatic block	Perineal and periapical triangle block

- Left lateral position; DRE; 6–10 MHz probe; measure volume (PSA density).
- 18-gauge spring-loaded needle; indent the capsule, withdraw slightly, then fire, so the core samples peripheral zone.
- **12 cores:** base, mid-gland and apex on each side, medial and lateral (sextant alone is inadequate; it missed lateral cancers). Add MRI-targeted cores (cognitive or fusion). TZ (transition zone) cores if gland >50 mL.
- **EAU (European Association of Urology) 2023 prefers transperineal biopsy.**

Complications (transrectal)

Complication	Rate
Haematuria	23–63%
Haematospermia	10–50%
Rectal bleeding	2–22%
Infection	0.1–7%; sepsis 0.3–3%
Retention	0.2–0.4%
Vasovagal reaction	1–5%

- Bleeding is commonest; sepsis is most serious.
- **Sepsis** from fluoroquinolone-resistant *E. coli*: carbapenem until cultures return; prevent with povidone-iodine rectal cleansing plus targeted or augmented prophylaxis.
- Rectal bleeding: probe pressure, balloon tamponade, endoscopy, embolisation.

Recent advances

- **Transperineal biopsy** under local anaesthesia: fewer infections, better apex and anterior sampling; preferred by EAU 2023.
- **MRI-first pathway** (PROMIS, PRECISION, above).
- Colour/power Doppler, contrast-enhanced TRUS, **elastography**, multiparametric ultrasound.
- **Micro-ultrasound (29 MHz; 70 µm resolution)** with PRI-MUS scoring, detection comparable to mpMRI.
- **Also know:** PSMA (prostate-specific membrane antigen) PET.

Diagrams to draw

- 12-core biopsy template on a transverse prostate outline.
- Transverse and sagittal TRUS views labelling PZ, TZ, seminal vesicles and urethra.

Key points to remember:

- Probe 6–10 MHz; volume = 0.52 × T × AP × L; PSAD >0.15 suspicious.
- 39% of cancers isoechoic; hypoechoic PZ lesion is classic.
- Seminal vesicle >1.5 cm wide: ejaculatory duct obstruction.
- MRI before biopsy: PROMIS and PRECISION established the MRI-first pathway.
- 12 cores; antibiotics ≤24 hours for all (transrectal); sepsis 0.3–3%.
- Bleeding commonest; sepsis most serious.
- Transperineal now preferred.

Also, from the condensed book (not in the paper answer):

- **Repeat** biopsy follows rising prostate-specific antigen (PSA), atypical small acinar proliferation (ASAP; **≈50%** cancer risk), positive PCA3 or ConfirmMDx, suspicious MRI, or active surveillance (AS).
- National Comprehensive Cancer Network (NCCN): biopsy for abnormal digital rectal examination (DRE) with raised PSA, or repeat PSA >3 ng/mL with PI-RADS ≥3; PSA velocity alone, or abnormal DRE with normal PSA, is not enough. See Ch 154
- PSA density **>0.15 ng/mL/cc** suggests cancer. Volume = π/6 × transverse × anteroposterior × length.

2 · Pre-biopsy mpMRI

SAQ Multiparametric MRI, PI-RADS and fusion biopsy

Paper answer · P1 · P4 · merged from 2

Asked as: P1 SAQ "Prostate multiparametric MRI and endorectal coil MRI" (Multiparametric MRI : its components and interpretation for prostate cancer.; Endorectal coil MRI) · P4 SAQ "PI-RADS and MRI–ultrasound fusion biopsy"

Introduction

- **mpMRI (multiparametric magnetic resonance imaging):** anatomical plus functional sequences, usually at 3 T without endorectal coil. Reported by **PI-RADS v2.1 (Prostate Imaging Reporting and Data System, 2019)**.
- PI-RADS scores the likelihood of **clinically significant cancer** (Gleason ≥7, volume >0.5 cm^3 or EPE [extraprostatic extension]).

Components

Sequence	Principle	Cancer appearance
T2W (T2-weighted), triplanar	Anatomy; normal PZ (peripheral zone) bright	Low-signal lesion; "erased charcoal" in TZ (transition zone); shows EPE and SVI (seminal vesicle invasion)
DWI (diffusion-weighted imaging) with **ADC (apparent diffusion coefficient)** map; high b-value (≥1400)	Cellularity: cellular tumour restricts water movement	Bright on high-b DWI, **dark on ADC**; ADC falls as Gleason rises

Sequence	Principle	Cancer appearance
DCE (dynamic contrast-enhanced) T1 with gadolinium	Vascularity (angiogenesis): early enhancement and washout	Early focal enhancement (type 3 curve)

- **Biparametric MRI (T2W + DWI):** no contrast, cheaper, similar detection; poorer staging.
- **PI-QUAL (Prostate Imaging Quality):** image quality score (1–5) before reporting.

Interpretation: PI-RADS v2.1

Score	Clinically significant cancer
1	Highly unlikely
2	Unlikely
3	Equivocal
4	Likely; lesion <1.5 cm
5	Highly likely; ≥1.5 cm or definite EPE

Zone	Dominant sequence	Tie-breaker at score 3
Peripheral	**DWI**	DWI 3 + DCE positive → 4
Transition	**T2W**	T2W 3 + DWI 5 → 4; (v2.1) T2W 2 + DWI ≥4 → 3

- **DCE only upgrades PZ 3 to 4**; it never changes 4–5 or TZ scores.
- Cancer in about one-third of PI-RADS 3, over half of 4, 90% of 5.
- **PRECISE** score (1–5) tracks change during active surveillance.

Clinical use and evidence for MRI

- **PROMIS (2017):** MRI sensitivity 93% vs 48% for TRUS (transrectal ultrasound) biopsy; 27% could avoid biopsy.
- **PRECISION (2018):** MRI pathway with targeted biopsy of PI-RADS ≥3 detected more significant cancer (38% vs 26%); 28% avoided biopsy.
- EAU (European Association of Urology) and AUA (American Urological Association): MRI **before biopsy**; also staging, active surveillance, focal therapy planning.
- Limitations: low specificity (Cochrane 0.37); reader-dependent.

MRI–ultrasound fusion biopsy: principle

- Lesion contoured on MRI → software **co-registers it with live TRUS** in three dimensions; **elastic deformation** corrects probe distortion → needle aimed at the displayed target.
- Freehand electromagnetic or mechanical-arm tracking; transrectal or **transperineal** (fewer infections).

	Cognitive	Software fusion	In-bore MRI
Method	Aims from memory of MRI	Software overlay	Needle under MRI
Drawback	Operator-dependent; small lesion in large gland	Cost; registration error	Time, cost

- Meta-analysis (2022): no significant difference among the three; one RCT (randomised controlled trial) favoured fusion over cognitive.

Fusion biopsy: indications and evidence

- PI-RADS ≥3 lesion with raised PSA (prostate-specific antigen), first or repeat biopsy; active surveillance; focal therapy.
- **AUA/SUO (American Urological Association/Society of Urologic Oncology) 2023:** 2–3 cores per target ± systematic cores.
- Targeting reduces overdiagnosis; MRI-invisible cancers keep systematic cores relevant.

Endorectal coil MRI: principle

- **ERC (endorectal coil):** a receive-only surface coil inside an inflatable balloon placed in the rectum against the prostate, used with an external pelvic phased-array coil.
- **"The currency of MRI is signal."** A coil close to the prostate gives higher **SNR (signal-to-noise ratio)**, allowing **higher spatial resolution**.

Endorectal coil MRI: technique

- Antispasmodic to reduce rectal motion artefact; balloon filled with air or perfluorocarbon (fluid reduces susceptibility artefact).
- Sequences as for mpMRI: **T2W** triplanar for anatomy, **DWI with ADC** map, **DCE**; ERC was traditionally needed for **MRSI (MR spectroscopic imaging)**.

Endorectal coil MRI: uses

- **Local staging:** extraprostatic extension and seminal vesicle invasion (both low T2 signal) before radical prostatectomy.
- **Local recurrence** after prostatectomy: recurrence isointense on T1, slightly hyperintense on T2; DCE raises accuracy at low PSA.
- **Also know:** focal therapy planning (ERC recommended at 1.5 T); seminal vesicles; female urethral diverticulum.

ERC versus no ERC

	With ERC	**Without ERC (3 T, pelvic coil)**
Signal and resolution	Higher	Adequate at 3 T
csPCa (clinically significant cancer) detection	GG (Grade Group) ≥3 equal; GG2 slightly better	GG ≥3 equal
Comfort, cost, time	Worse	Better
Gland	Deformed (fusion and radiotherapy planning harder)	Natural shape

ERC: disadvantages and contraindications

- Artefacts: **near-field signal flare** next to the coil (needs intensity correction), air susceptibility at the rectal wall, malposition.
- **Contraindications:** absent rectum (abdominoperineal resection), recent rectal surgery, anorectal fistula or inflammatory bowel disease, severe haemorrhoids or stricture, latex allergy.

ERC: current position

- At **3 T the ERC is largely historical**; PI-RADS v2.1 does not require it. **Consider it at 1.5 T** or with older scanners, or when body-coil quality is poor.

Diagrams to draw

- Prostate zones with a PZ lesion dark on ADC.
- MRI target on live TRUS with needle path.
- Sagittal pelvis with inflated endorectal coil against the posterior prostate.

Key points to remember:

- T2W anatomy, DWI/ADC cellularity, DCE vascularity.
- PZ ruled by DWI, TZ by T2W; DCE only lifts PZ 3 to 4.
- PRECISION and PROMIS: MRI before biopsy.
- Fusion: co-registration with elastic deformation; cognitive, fusion, in-bore: similar yield.
- Endorectal coil close to prostate: higher SNR and resolution.
- Useful at 1.5 T; largely unnecessary at 3 T (GG ≥3 detection equal).
- Drawbacks: discomfort, gland deformation, near-field flare.

Also, from the condensed book (not in the paper answer):

- Multiparametric MRI is a **triage test**: with no lesion biopsy may be avoided. A lesion **>1.5 cm** or definite extraprostatic extension scores PI-RADS 5.

Feature	Peripheral zone	Transition zone
Dominant sequence	**DWI**	**T2WI**
Dominant score 3	Positive DCE upgrades to 4	DWI 5 upgrades to 4
v2.1 (2019)	—	T2WI 2 with DWI ≥4 becomes 3

KEY POINT

DCE never creates or downgrades PI-RADS 4–5 and never upgrades a TZ lesion. mpMRI misses a minority of significant cancers (mostly 3+4), so systematic cores remain relevant.

- Targeting is cognitive, software fusion or in-bore. Fusion-targeted cores are ≈3× likelier than systematic ones to find intermediate/high-risk disease.
- Take **2–4 cores per target** (AUA/SUO 2023: 2–3, with or without systematic cores) whenever biopsy is indicated and a PI-RADS ≥3 lesion exists.

3 · Systematic templates and technique

- **Extended 12-core** biopsy is the AUA standard, replacing sextant. Going from 6 to 12 cores raises detection, but 18–21 do not: **geometry, not core number**.
- Add TZ/anterior cores for repeat biopsy or gland >50 mL, and seminal vesicle cores for a palpable abnormality.
- Use an 18-gauge gun, and never reposition with the needle on the rectal wall.

4 · Transrectal vs transperineal (TP)

- TP passes clean perineal skin, causes no rectal bleeding and reaches apex and anterior gland (anterior-only cancers 6–20%).
- European Association of Urology 2023: TP is **preferred** (strong), with non-inferior detection. Give skin-flora cover or none; omitting antibiotics did not change the very low sepsis rate.

- Access is by brachytherapy grid, freehand guide (PrecisionPoint) or two 14-gauge cannulas.

5 · Antibiotic prophylaxis

- Every transrectal patient gets prophylaxis. AUA 2019: fluoroquinolone; 1st/2nd-generation cephalosporin with or without aminoglycoside; or 3rd-generation cephalosporin. Give ≥1 h before, for **≤24 h**.
- Single doses: levofloxacin 500 mg, gentamicin 5 mg/kg, amikacin 15 mg/kg, ceftriaxone 1–2 g. Aztreonam replaces aminoglycoside in renal impairment.
- The FDA 2016 fluoroquinolone warning covers tendon rupture, neuropathy, CNS effects and myasthenia.
- Infection risk factors are antibiotics within **6 months** (most consistent), diabetes, immunosuppression, prior UTI, travel and rectal resistant E. coli. These men need augmented or swab-targeted prophylaxis, or TP.
- Endocarditis risk needs IV ampicillin (vancomycin if allergic) and gentamicin. An enema has no proven infection benefit.

6 · Anticoagulation and anaesthesia

Contraindications are coagulopathy, severe immunosuppression, and acute urinary tract infection (UTI) or prostatitis.

Agent	Action
Low-dose aspirin	Continue
Warfarin, clopidogrel	Stop 5–7 days; biopsy when INR <1.5
Direct oral anticoagulants (DOACs)	Stop 2–5 days
Mechanical valve	Bridge with heparin

- Transrectal: periprostatic block, 1–2% lidocaine 5 mL each side at the seminal vesicle–prostate junction (beats intrarectal gel).
- Transperineal: local to the skin, then 5 mL into the **periapical triangle** and 3 mL along the capsule.

7 · Complications and management

Complication	Rate	Management
Haematuria	23–63%	Clot retention (0.7%): catheter, washout
Haematospermia	10–50%	Lasts ≈1 month
Rectal bleeding	2–22%	Ladder (below)
Infection / UTI	0.1–7% / 2–6%	Oral or IV antibiotics
Sepsis	**0.3–3.1%**	Carbapenem until cultures
Vasovagal	1.4–5.3%	Trendelenburg, fluids
Retention	0.2–0.4%	Temporary catheter

ALGORITHM

1. **Post-biopsy fever**: assess for sepsis and take cultures.
2. Give **no empirical fluoroquinolone**, as E. coli is resistant. Start a carbapenem (or amikacin/3rd-generation cephalosporin by sensitivity) and resuscitate.

ALGORITHM

1. **Persistent rectal bleeding**: apply pressure.
2. If it continues, tamponade (sheath or Foley balloon). If still bleeding: endoscopy with adrenaline/sclerosant, then angioembolisation, then transrectal suture.

KEY POINT

Bleeding is the commonest complication overall; infection leads serious ones and admissions (1.0% rising to 4.1% in the resistance era). This explains universal prophylaxis and the move to the perineum.

8 · Repeat and saturation biopsy

SAQ Saturation biopsy of prostate

Paper answer · P2 · P4 · merged from 2

Asked as: P2 SAQ "Saturation biopsy in CA prostate" · P4 SAQ "Saturation biopsies of prostate"

Definition

- Systematic prostate biopsy with **more cores than the extended 12-core scheme**, usually **18–21 or more** (often about 20–30), sampling the peripheral zone, transition zone, anterior gland and apex.
- Route: transrectal, or **transperineal template mapping biopsy** through a brachytherapy grid with a core every **5 mm**.

Rationale

- Sextant biopsy (Hodge, 1989) under-sampled the lateral peripheral zone; the 12-core scheme (adds apical and far-lateral cores) is today's standard.
- Saturation reduces sampling error, mainly for **anterior and apical cancers** missed transrectally (anterior-only cancer 6–20%).

Indications

- **Repeat biopsy** after a negative biopsy with persistent suspicion: rising PSA (prostate-specific antigen), ASAP (atypical small acinar proliferation), negative or unavailable MRI (magnetic resonance imaging).
- **Active surveillance** confirmation.
- **Focal therapy** mapping (shows multifocality; TRUS [transrectal ultrasound] biopsy misses almost half of disease relevant to hemiablation).

Templates and technique

Template	Division	Cores
Original Barzell	Octants (anterior/posterior × apex/base, each side)	4–8 per octant
Modified Barzell	Apex and base of each side × 5 sectors	20

Template	Division	Cores
Ginsburg	Anterior, mid, posterior sectors per side	4 per sector

- Lithotomy, stepper and grid; general or spinal anaesthesia, or periprostatic block (up to 22 mL of 1% lidocaine transrectally).

Results

- **Initial biopsy:** 12 to 18–21 cores did not raise detection.
- **Repeat setting:** early series gained; a later large prospective study did not. Extra cancers are often insignificant (overdiagnosis).
- **PROMIS (2017)** used 5-mm template mapping as the reference: mpMRI (multiparametric MRI) sensitivity 93% vs TRUS biopsy 48%.

Complications

- More haematuria, haematospermia, pain and **urinary retention** than standard biopsy; anaesthesia and theatre time. Transperineal route has less infection.

Current position

- Largely replaced by **mpMRI with targeted biopsy**: **PRECISION (2018)** detected more significant cancer (38% vs 26%) and 28% avoided biopsy.
- **EAU (European Association of Urology) 2023** prefers the transperineal route.
- Still used for MRI-negative high suspicion and focal therapy planning (mapping).

Diagram to draw

- Transperineal grid over the prostate with sampling sectors.

> **Key points to remember:**
> - Saturation = >12 cores (18–21 or more); 5-mm template mapping.
> - No gain at initial biopsy; used after negatives and for focal therapy.
> - PROMIS: template mapping as reference; mpMRI sensitivity 93%.

Also, from the condensed book (not in the paper answer):

- Micro-ultrasound (29 MHz) approaches mpMRI without contrast (emerging).

ALGORITHM

1. Cancer yield on successive biopsies in ERSPC: **22% → 10% → 5% → 4%**.

- Take a third or fourth biopsy only for high suspicion, using mpMRI with targeting, ConfirmMDx, anterior/TZ cores and the TP route.

9 · Landmark trials

Trial	Design	Result	Take-home
PROMIS (2017)	576: mpMRI, TRUS, 5-mm template mapping (reference)	Sensitivity 93% vs 48%; negative predictive value 89% vs 74%; 27% avoid biopsy	mpMRI better triage

Trial	Design	Result	Take-home
PRECISION (2018)	500 biopsy-naive: MRI with targeted biopsy vs 12-core TRUS	Significant cancer 38% vs 26%; 28% avoided biopsy	MRI pathway superior
MRI-FIRST (2019)	Paired systematic and targeted	Each alone missed cancers	Combine for maximal detection
PREVENT (2024)	TP without antibiotics vs transrectal targeted prophylaxis	Infection not significantly different; detection comparable	Both acceptable

10 · MCQ one-liners

MCQ ONE-LINERS

1. Absent seminal vesicle: ipsilateral renal agenesis in 79%.
2. Commonest false-positive hypoechoic lesion: BPH foci and prostatitis.
3. Müllerian cysts are midline; wolffian cysts are lateral.
4. Significant cancer in PROMIS: Grade Group ≥3 or core length ≥6 mm.

156 Pathology of Prostatic Neoplasia

1 · Precursor and intraductal lesions

P2 *PIN — definition, histology, precursor evidence, LGPIN not reported, isolated HGPIN needs no repeat biopsy, cribriform PIN reclassified*

- ~**20%** of high-grade prostatic intraepithelial neoplasia (HGPIN) carries the **TMPRSS2:ERG** fusion.
- **Intraductal carcinoma of the prostate (IDC-P)** is dense cribriform atypia with **basal cells retained**, usually **retrograde spread of high-grade invasive cancer**. It independently predicts early biochemical recurrence (BCR), and Grade Group (GG) 1 with IDC-P is upgraded at radical prostatectomy (RP) in **>80%**.

Lesion	Defining feature	Action on isolated finding
Low-grade PIN (LGPIN)	Mild atypia; irreproducible	Not reported
HGPIN	Benign architecture, atypical cells; non-cribriform	No repeat biopsy
PINATYP	HGPIN with adjacent small atypical glands	Manage as atypical
Atypical intraductal proliferation (AIP)	Atypia beyond HGPIN, below IDC-P	Consider re-biopsy (if benign or GG1 elsewhere)
IDC-P	Dense cribriform, basal cells retained	Definitive therapy (or re-biopsy); **excludes active surveillance (AS)**

> **KEY POINT**
>
> HGPIN is a weak risk marker; IDC-P usually marks high-grade cancer already present. Traps: re-biopsy for HGPIN; surveillance for GG1 with IDC-P.

2 · Adenocarcinoma: location, spread, staging

- **Zones (McNeal)**: the peripheral zone (PZ) holds ~70% of cancers, including **85%** of T1c; they are posterolateral and palpable.
- The transition zone (TZ) holds ~20% of cancers, which are anterior, impalpable, under-sampled and lower grade. The central zone holds ~5%, rare but aggressive.
- Tumours are **multifocal in >85%**. With no true capsule, say "**extraprostatic extension**" (EPE); tumour exits via **perineural spaces**, posterolaterally.
- **Seminal vesicle invasion (SVI)** means tumour in the SV muscular wall, usually reached from the base through periseminal tissue.
- EPE is uncommon below **0.5 cm^3**; nodes and SVI are uncommon below **4 cm^3**.

Stage	Definition
T1a / T1b	Incidental at TUR of prostate (TURP)/enucleation; ≤5% / >5% of tissue
T1c	Non-palpable, found on needle biopsy
pT2	Organ-confined at RP
pT2+ (pT2x)	Positive margin at an intraprostatic incision or ill-defined boundary (apex)
pT3a	EPE; **includes microscopic bladder neck invasion**

Stage	Definition
pT3b	SVI
pT4	Other adjacent structures (rectum, levators, external sphincter, pelvic wall)

Trap: microscopic bladder neck invasion is pT3a; T1a/b depend on the 5% cut-off, not grade.

3 · Grading: Gleason and grade groups

SAQ Gleason's score and its significance

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 3 Paper 2, Q3f

Introduction

- **Gleason grading** (Donald Gleason, late 1960s) grades prostate adenocarcinoma by **glandular architecture at low magnification**, not cytology. Patterns run from 1 (most differentiated) to 5. Modified by ISUP (International Society of Urological Pathology) in **2005** and **2014**.

Patterns (modern)

Pattern	Architecture
1–2	Not used on biopsy now
3	Discrete, well-formed glands of any size
4	**Cribriform** (including glomeruloid), poorly formed or fused glands
5	Sheets, cords, single cells, solid nests, **comedonecrosis**

Scoring rules

- **Radical prostatectomy:** primary (commonest) + secondary pattern.
- **Biopsy (2005):** commonest + **highest** pattern, however small. Single pattern is doubled (3+3).
- A tertiary pattern 5 under 5% at prostatectomy is reported as a comment; it adds risk but does not change the grade group.
- Report **% pattern 4** in score 7; report cribriform and intraductal carcinoma (both adverse).

ISUP Grade Groups (2014; WHO [World Health Organization] 2016)

Grade group	Gleason score	5-year BCR (biochemical recurrence)-free after prostatectomy
1	3+3=6	96%
2	3+4=7	88%
3	4+3=7	63%
4	8 (4+4, 3+5, 5+3)	48%
5	9–10	26%

- Grade groups separate 3+4 from 4+3 and 8 from 9–10. Starting at "1" avoids patients misreading the lowest score, 6, as moderate cancer.

Significance

- **Most powerful biopsy predictor** of pathological stage, EPE (extraprostatic extension), nodal spread (0% in GG1, 4.3% in GG3), recurrence and death.
- **Risk groups** (D'Amico, NCCN [National Comprehensive Cancer Network]) combine grade, PSA (prostate-specific antigen) and stage.
- **Treatment choice:** GG1 → active surveillance; selected GG2 with little pattern 4 → surveillance; GG ≥4 → high risk, longer ADT (androgen deprivation therapy) with radiotherapy; also selects for brachytherapy and focal therapy.
- **Limitations:** biopsy undergrades about one-third of GG1 (sampling error); inter-observer variation; after ADT or radiotherapy, grading is unreliable.
- **Also know:** pure GG1 does not metastasise, so "is it cancer?" is debated.

Diagram to draw: Gleason's pattern diagram 3 → 4 → 5.

> **Key points to remember:**
> - Biopsy score = commonest + highest pattern.
> - 3+4 (GG2) and 4+3 (GG3) differ: BCR-free 88% vs 63%.
> - GG4 = score 8; GG5 = 9–10.

Also, from the condensed book (not in the paper answer):

- **Biopsy score = commonest plus highest pattern**. The lowest score in practice is **6**, which has **no nodal metastatic capacity**.

Pattern	Morphology			
3	Discrete well-formed glands			
4	**Cribriform (incl. glomeruloid)**, poorly formed or fused glands			
5	Sheets, cords, single cells, solid nests, **comedonecrosis**			
	GG (International Society of Urological Pathology (ISUP); World Health Organization 2016)	Gleason	Nodes at RP	5-yr BCR-free
1		3+3	0%	**96%**
2		3+4	0.6%	**88%**
3		4+3	4.3%	**63%**
4		4+4, 3+5, 5+3	—	**48%**
5		9–10	—	**26%**

> **KEY POINT**
>
> The steep drop is between GG2 and GG3: never lump them as "7". 3+5 and 5+3 are GG4; GG5 starts at 9. Pure GG1 cannot metastasise, so calling it "cancer" is debated.

- Report **% pattern 4** for GG2–3 (ISUP 2014). GG2 with limited pattern 4 may suit AS.
- **Tertiary pattern 5** (<5% at RP) raises BCR risk but **stays in its GG**, because nodal risk is not raised. On biopsy any higher pattern counts as secondary, however small.

Issue	ISUP 2019	Genitourinary Pathology Society (GUPS) 2019
% pattern 4 in GG2–3; cribriform	Report	Report
IDC-P with invasive cancer	**Incorporated into grade**	**Not graded**; comment
Targeted cores of one lesion	**Aggregate grade** (plus per-core)	Each core; aggregate optional

KEY POINT

Cribriform pattern and IDC-P mark aggressive disease and argue against AS even in GG2, so they must be reported. Aggregate grading of a targeted lesion avoids under- or over-grading.

Upgrading setting	Risk
Biopsy GG1, then RP	~⅓ upgraded
Biopsy GG2, then RP	~20% become GG3
GG1 with negative mpMRI (PI-RADS 1–2)	Still ~20% at 6 yr

Reclassification within 3 yr usually means missed disease rather than progression.

4 · Needle biopsy assessment

- Use **separate jars per site**. Pitfalls: **at the base, SV/central zone mimic HGPIN; at the apex, Cowper gland mimics cancer**.
- Report the **number of positive cores plus one other extent measure**, and report **perineural invasion** on biopsy (it raises EPE risk).
- Basal cell markers, **high-molecular-weight cytokeratin (HMWCK) and p63**, stain benign glands, so their **absence supports cancer**. **AMACR** is an adjunct only.
- **Adenosis**, in the TZ, is the chief mimic of low-grade cancer and carries **no increased cancer risk**.
- **Atypical small acinar proliferation (ASAP)**: cancer follows in ~**40%**. Get **expert review before re-biopsy**, then re-biopsy the atypical site.

KEY POINT

Sampling error is one-directional: a bad biopsy is believed, a good one is provisional, and an atypical one is re-examined before it is repeated.

5 · TUR and RP specimens

- At TUR, **grade each major nodule**. Nodes are uncommon with Gleason <8.
- **EPE**: focal vs nonfocal extent predicts progression. **SVI** carries **65%** 5-yr progression.
- **Margins** are positive in ~15% (robotic), but only ~50% progress. Report **extent (focal vs nonfocal, 3 mm) and grade at the margin**.
- Independent predictors of BCR are grade, EPE extent, margins, SVI and nodes. **Tumour volume is not independent**.

6 · Treatment effect

Treatment	Histological effect	Gleason score?
Hormone therapy	Grade looks artefactually higher	**No** where effect is present
Finasteride	No consistent change	Yes
Radiotherapy	Radiation atypia (up to 72 months)	Only if no treatment effect
Cryotherapy / high-intensity focused ultrasound (HIFU)	Infarction; residual cancer unaltered	**Yes**

After radiotherapy, cancer with treatment effect has a prognosis close to no cancer.

7 · Subtypes and other tumours

Subtype	Grading	Behaviour
Ductal (pure 0.4–0.8%; mixed ~5%)	**Pattern 4** (PIN-like 3; comedonecrosis 5)	Exophytic at the verumontanum; PSA may be normal; behaves like Gleason 8; **visceral recurrence at low PSA**
Mucinous	As usual	As acinar
Neuroendocrine differentiation (~half of acinar)	As usual	Not prognostic
Small cell (~50% mixed)	**Not graded**	Survival <1 yr
Squamous	—	**Osteolytic** metastases; hormone-unresponsive
Sarcomatoid	—	Dismal

- **Urothelial carcinoma in the prostate**: suburethral invasion is pT1 and stromal pT2, but **transmural spread from the bladder is pT4**. Urothelial cancer stains **GATA3**; prostate stains PSA and NKX3.1.
- Sarcomas are **rhabdomyosarcoma in children and leiomyosarcoma in adults**.

157 Diagnosis and Staging of Prostate Cancer

1 · Clinical presentation

- Early cancer is **asymptomatic**; symptoms imply advanced disease.
- **Locally advanced** disease causes voiding or ureteric obstruction and haematuria; **metastatic** disease causes bone pain, fracture, leg oedema and cord compression.

2 · Screening in brief

*See the paper answer **PSA screening for prostate cancer: evidence, problems and solutions** (Section 17, Ch 153).*

3 · DRE and PSA

- **Digital rectal examination (DRE)** has low screening value at PSA <3, but an abnormal DRE raises the risk of Grade Group (GG) 4–5 at every PSA level.
- **Prostate-specific antigen (PSA)** is organ-specific, not cancer-specific, and rises ≈4% per mL of volume.
- PSA is lowered by hypogonadism, obesity, statins and 5α-reductase inhibitors (5-ARIs). On a 5-ARI, **double PSA** after ≥12 months, or treat the nadir as the new baseline.
- Risk is **continuous**: the Prostate Cancer Prevention Trial (PCPT) found cancer in 15% of men with PSA ≤4.
- Thresholds are 4 or 2.5–3 (ERSPC used 3). Repeat a raised PSA before biopsy, and give no empirical antibiotics without urinary symptoms.

PSA (ng/mL)	Yield, normal DRE	Yield, suspicious DRE	High-grade (normal / suspicious)
0–1	8.8%	—	0.9%
4–10	17–32%	45–51%	4.1% / 11.7%
>10	43–65%	70–90%	19.4% / 50.5%

- Baseline PSA in the 40s predicts metastasis and death decades later (Malmö).
- Derivatives: PSA density ≥0.15; velocity >0.75 ng/mL/yr; free PSA ≤25% (at PSA 4–10). See Ch 154, 155

4 · Staging: TNM (AJCC 8th, 2017)

Pathological stage is more accurate and prognostic than clinical stage.

Category	Definition
T1a / T1b	Incidental at transurethral resection (TUR): ≤5% / >5% of tissue
T1c	Impalpable, found on needle biopsy
T2a / T2b / T2c	Palpable, confined: ≤½ of one side / >½ of one side / both sides
T3a / T3b	Extraprostatic extension (EPE) / seminal vesicle invasion (SVI)
T4	Fixed, or invading sphincter, rectum, bladder, levators or pelvic wall
N1	Regional (pelvic) nodes
M1a / M1b / M1c	Non-regional nodes / bone / other sites with or without bone

AJCC prognostic groups add PSA and GG.

KEY POINT

T1 is defined by how the cancer was **found**, T2 by what is **felt**. Traps: SVI is T3b; non-regional nodes are M1a; a biopsy-found impalpable cancer is T1c even if bilateral; clinical T is set by DRE, not imaging.

5 · Predicting extent and risk

- **PSA** cannot stage an individual. Organ-confined disease is ≈80% at PSA <4 and <50% above 10; nodes are positive in **75% at >50**.
- **Biopsy grade** is the most important biopsy finding, and tertiary pattern 5 on biopsy is recorded as the secondary pattern. See Ch 156

D'Amico	Stage	PSA	Gleason	10-yr disease-free after RP
Low (all criteria)	T1–T2a	≤10	≤6	83%
Intermediate (any one)	T2b	>10–20	7	46%
High (any one)	T2c	>20	8–10	29%

National Comprehensive Cancer Network (NCCN) group	Criteria
Very low	T1c, GG1, PSA <10, PSA density <0.15, <3 positive cores, ≤50% of any core
Low	T1–T2a, GG1, PSA <10
Favourable intermediate	One intermediate factor (T2b–c, GG2–3, PSA 10–20), GG1–2, <50% cores positive
Unfavourable intermediate	≥2 intermediate factors, or GG3, or ≥50% cores positive
High	T3a, or GG4–5, or PSA >20
Very high	T3b–T4, primary pattern 5, >4 cores GG4–5, or ≥2 high-risk features

- CAPRA and **nomograms** predict better than groups for individuals; nodal nomograms decide pelvic lymph node dissection (PLND). Tissue genomic tests refine risk but await prospective validation.

6 · Staging imaging and nodes

Risk	Imaging
Low / intermediate, asymptomatic	No routine computed tomography (CT) or bone scan
High	Bone scan with CT or mpMRI; consider PSMA PET

- **Multiparametric MRI** is used for local staging (EPE, SVI), treatment planning and surveillance selection.
- **PSMA PET** (68Ga-PSMA-11, piflufolastat F-18) is FDA-approved for high-risk staging and is more sensitive and specific for nodes and metastases.
- **PLND** has no consistent oncological benefit; its value is **staging**. Do it when the nomogram nodal risk exceeds the guideline threshold.

KEY POINT

Imaging yield follows pre-test probability: in low-risk men most positive scans are false positives, leading to more tests, anxiety and wrongly denied cure.

7 · Landmark trials

Trial	Design	Result	Take-home
ERSPC	European screening randomised trial, from 1993	**20% relative reduction** in cancer death at 16 y	Saves lives, at the cost of overdiagnosis
PLCO	US screening randomised trial, from 1993	No mortality difference	Contaminated control arm
PCPT	Placebo arm biopsied regardless of PSA	15% cancer at PSA ≤4	No safe PSA level
proPSMA (2020)	PSMA PET-CT vs CT and bone scan, high-risk	Accuracy ≈92% vs 65%; less radiation	PSMA PET is the better first-line staging test

8 · MCQ one-liners

MCQ ONE-LINERS

1. Finasteride 1 mg lowers PSA as much as 5 mg.

158 Active Management Strategies for Localised Prostate Cancer

1 · The modern question: timing as well as modality

- In 15-year randomised data on favourable risk, surveillance gave **more metastasis but no difference in cancer mortality**.
- **Spectrum**: a man likely to die of competing risks gets **watchful waiting (WW)**; good longevity with unclear benefit means **active surveillance (AS)**; a man who needs therapy gets radical prostatectomy (RP), external-beam radiation therapy (EBRT), brachytherapy, or whole/partial-gland ablation.

> **KEY POINT**
>
> Cancer mortality is always weighed against competing mortality. A competing risk argues against treatment only when it will act before the cancer does.

2 · Host evaluation

- NCCN estimates life expectancy from Social Security Administration or WHO tables.
- In the Prostate Cancer Outcomes Study (PCOS), 10-year prostate cancer mortality was **3–18%** by risk, but other-cause mortality was **≥33%** with any comorbidity.
- The blunt **"10-year life expectancy" rule** undertreats older and overtreats younger men, so individualise.
- The **Charlson index** (1-year inpatient mortality) is reweighted as **PCCI** for 10-year mortality; NNT rises steeply with Charlson score.
- The automated **Care Assessment Needs (CAN) score** (Veterans Health Administration) best predicts 5-year survival MCQ.

Baseline function (IPSS, UCLA Prostate Cancer Index)

Baseline state	Favours	Reason
Obstructive lower urinary tract symptoms (LUTS), large gland	Surgery	RP relieves obstruction; brachytherapy increases retention
Severe irritative LUTS	Surgery	Worst after radiation
Gland >60 cm^3	Surgery (or neoadjuvant androgen-deprivation therapy, ADT)	Pubic arch interference; if >80 g, ADT (shrinks 25–40%) before radiotherapy (RT)
Chronic retention / neurogenic bladder	Surgery	RT causes bladder fibrosis; if RT is chosen, switch to intermittent self-catheterisation
Prior transurethral resection (TUR) defect	Surgery	Brachytherapy dosimetry is hard and incontinence increases. Do TURP before RT and delay EBRT **6–8 weeks**
Inflammatory bowel disease (IBD), bowel dysfunction, prior rectal surgery	Surgery	RT bowel toxicity

Baseline state	Favours	Reason
Poor sexual function	—	Recovery depends on age, baseline erection, PDE5 inhibitor use and vascular risk; poor baseline function means non-nerve-sparing surgery is used more.

Modality	Absolute contraindication	Relative contraindication
Radiotherapy	Previous radical pelvic irradiation	Bowel in field (prior surgery, hernia); uncontrolled diabetes; IBD; bowel dysfunction
Surgery	—	Previous pelvic RT; chronic steroids; severe cardiac/pulmonary/haematological comorbidity

3 · Risk stratification

NCCN group	Criteria	Staging imaging
Very low	**All**: T1c; grade group (GG) 1; prostate-specific antigen (PSA) <10; <3 positive cores, ≤50% each; PSA density <**0.15**	Not indicated
Low	**All**: T1–T2a; GG1; PSA <10	Not indicated
Favourable intermediate	Intermediate-risk factors (IRFs): T2b–c, GG2, PSA 10–20. **All**: 1 IRF; GG1–2; <50% cores positive	
Unfavourable intermediate	**Any**: 2–3 IRFs; GG3; ≥50% cores positive	Bone scan and CT/MRI, plus chest imaging; prostate-specific membrane antigen (PSMA) PET may replace these (NCCN 2023)
High	**Any**: T3a; GG4–5; PSA >20	
Very high	**Any**: T3b–T4; primary pattern 5; >4 cores GG4–5	

> **KEY POINT**
>
> Favourable groups need ALL criteria; adverse groups ANY one. PSA density appears only in very low risk; GG3 alone makes intermediate unfavourable. AUA/ASTRO/SUO use very low/low/intermediate/high without substratification.

- **Epstein criteria** (insignificant cancer): PSA density <0.15, Gleason ≤6, <3 positive cores, ≤50% of any core.
- Nomograms: **Partin** predicts pathological stage (accurate 72.4%); **Kattan** predicts relapse after RP; with **UCSF-CAPRA**, risk jumps at score ≥6.
- **proPSMA** (302 high-risk men): PSMA PET had accuracy **92% vs 65%** and sensitivity 85% vs 38% over conventional imaging, but a negative scan cannot exclude small nodes.
- Genomic tests are not routine (AUA 2022); use them only where they change an AS decision.

4 · Observation vs active surveillance

SAQ Expectant treatment of CA prostate

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 14 Paper 2, Q3d

Rationale

- Overdiagnosis: autopsy prevalence of microfocal cancer roughly equals age as a percentage; lifetime risk of death is about 3%.
- **Grade group 1 (Gleason 6) has virtually no metastatic potential.** Deaths on observation come from higher grade missed at biopsy (about 30% upgraded at prostatectomy).

Two forms of expectant treatment

	AS (active surveillance)	**WW (watchful waiting)**
Patient	Fit for radical therapy, life expectancy ≥10 years	Limited life expectancy (age, comorbidity)
Intent	**Curative, deferred**	**Palliative**
Monitoring	PSA (prostate-specific antigen), DRE (digital rectal examination), MRI, biopsy	Infrequent PSA and review; no biopsy
Action	Radical treatment on reclassification	ADT (androgen deprivation therapy) or palliation for symptoms

Evidence

Trial	**Result**
SPCG-4 (pre-PSA)	Prostatectomy cut cancer death (19.6% vs 31.3%), mainly <65 years and higher grade
PIVOT	No significant mortality benefit; low-risk cancer death 2.7%
ProtecT (15 years)	Cancer death <3% in all arms; metastasis doubled on active monitoring (9.4% vs about 5%)

Selection for AS

- **Very low risk (Epstein):** grade group 1, ≤2 positive cores, no core >50%, PSA density <0.15.
- **Low risk:** grade group 1, PSA <10 ng/mL; all guidelines prefer AS.
- **Favourable intermediate risk** (small-volume grade group 2): selected cases.
- Caution: **BRCA2** carriers, high-volume disease.

AS protocol

- MRI and **confirmatory biopsy** within 1–3 years; PSA 6-monthly; DRE every 1–2 years; MRI every 18–24 months; biopsy every 3–5 years.
- **Triggers:** grade group ≥3 → treat; volume or PSA rise → MRI or biopsy first.

Outcomes

- Cancer mortality **0.5–5% at 15 years** by entry criteria (Johns Hopkins vs Toronto); about **25% treated by 5 years**.

Watchful waiting

- Switch AS to WW when life expectancy falls below about 5–10 years.

- **Also know:** genomic tests refine selection; 5ARI (5α-reductase inhibitor) has no proven role.

Diagram to draw: AS flowchart: diagnosis → MRI + confirmatory biopsy → surveillance → trigger → treatment.

Key points to remember:

- AS is deferred cure; WW is palliation.
- Grade group 1 hardly ever metastasises; risk lies in missed higher grade.
- ProtecT: no mortality difference at 15 years; more metastases on monitoring.
- About 25% leave AS by 5 years.

Also, from the condensed book (not in the paper answer):

- AS is **preferred for very low and most low risk** (AUA, ASTRO, SUO, NCCN); selected low-volume GG2 is acceptable.
- Limitations are **25–45%** occult high grade on systematic biopsy (mostly GG2), declining adherence and no agreed triggers. PSA density >0.15 predicts reclassification.

5 · Comparative oncological evidence

- Population studies favour RP (Swedish subdistribution HR ≈1.8; PCOS disease-specific HR 0.35, curves apart by 3 years), but this is largely **selection bias**.

6 · Landmark trials

Trial	Design	Result	Take-home
SPCG-4	RP vs WW; 695; pre-PSA; ~90% palpable	23.2 yr: cancer deaths 63 vs 99, relative risk (RR) **0.56**; absolute 11%; NNT **8**; most benefit <65 (RR 0.45), intermediate risk	Benefit in younger men with intermediate risk
PIVOT	RP vs observation; 731 (2000 planned); PSA era; older, comorbid	10 yr: no difference; benefit if PSA >10. 19.5 yr: all-cause ARR 5.5% and cancer 4.0% (both not significant); significant only in **intermediate risk** (ARR 14.5%). Bone metastases 4.7% vs 10.6%	No benefit in low risk; underpowered
ProtecT	1643 screen-detected, 50–69; monitoring vs RP vs RT with 3–6 mo ADT; 77% low risk	15-yr cancer mortality **2.7% vs 2.2% vs 2.9%** (P = 0.53); more metastasis with monitoring; 60% of monitored treated by 15 yr	AS safe in favourable risk; RP equals RT

KEY POINT

Competing mortality and low-risk enrolment dilute trial benefit, which appears only in younger intermediate-risk men with long follow-up.

7 · Functional outcomes compared

Cancer control is similar, so side effects usually decide.

Domain (vs AS; Chen, JAMA 2017)	RP	EBRT	Brachytherapy
Sexual	**Worst**; persists at 24 mo	Mild early decline; near AS by 24 mo	Mild early decline
Incontinence	**Worst**; persists at 24 mo	No change	No change
Obstructive/irritative	Worse at 3 mo, then **better than AS**	Worse at 3 mo; resolves by 12	**Worst acute**; resolves by 24 mo
Bowel	No change	**Worse**; persists at 24 mo	No change

- **PCOS** found differences at 2 and 5 years but **none at 15 years**. **ProtecT** found rising bloody stools after RT, with no difference in anxiety, depression or quality of life.
- In **CEASAR**, the erectile disadvantage of RP was confined to the top quartile of baseline function. In open vs robotic RP, the surgeon matters more than the approach.

> **KEY POINT**
>
> Surgery front-loads urinary and sexual harm that partly recovers; radiation spares early function but adds bowel toxicity and slow decline. Curves converge by 15 years.

8 · Shared decision-making

- No treatment is clearly superior, so **patient preference carries unusual weight** (AUA, ASTRO, SUO).
- The **urologist's recommendation is the most influential factor**, so beware your own bias. **Longevity assessment may be the most important part of counselling**.

9 · Stepwise decision algorithm

ALGORITHM

1. **Treat or not?** Weigh disease and host factors. If longevity is short, then WW; if very low/low or selected favourable intermediate, then AS.
2. **Partial or whole gland?** Weigh focality, size, grade, potency, LUTS. If limited-focality intermediate risk, then focal ablation, ideally in a trial.
3. **Surgery, radiation or whole-gland ablation?** Compare outcomes, toxicities, baseline function and risk of secondary treatment.
4. **Delivery:** RP approach and node dissection, or EBRT vs brachytherapy, fractionation and ADT.

Risk / longevity	Usual recommendation
Limited longevity, low lethal potential	WW; ADT at symptoms
Very low	AS (best option)
Low	AS preferred for most; all therapies are options
Favourable intermediate	RP or RT; AS for select (low-volume GG2); focal therapy in trials
Unfavourable intermediate	RP, or RT with ADT; AS strongly discouraged

Risk / longevity	Usual recommendation
High / very high	RP, or RT with ADT; consider even with competing risks if the cancer threatens within 10 yr; no focal therapy

- **Surgery:** early incontinence and erectile dysfunction (ED) are the norm, so quote your own outcomes; LUTS may improve.
- **Radiation:** it spares short-term function but causes yearly decline; secondary rectal/bladder cancer at 15–20 years is controversial.
- **Whole-gland ablation:** long-term data are scant and fibrosis hampers salvage. Cryoablation almost always causes ED; HIFU preserves function better, but retreatment is **29–38%**.
- EBRT, brachytherapy, SBRT and protons give comparable 10-year cancer control.

159 Active Surveillance of Prostate Cancer

1 · Rationale: overdiagnosis

- **Latent reservoir**: autopsy cancer prevalence roughly equals age as a percentage. PSA testing and biopsy detect cancer in 15–25%, yet lifetime risk of cancer death is only **3%**; overdiagnosis is 87–94%.
- Active surveillance (AS) rests on **two premises**: progression takes a long time, and cure is still possible at the first sign of higher risk.

2 · AS versus watchful waiting

*See the paper answer **Expectant treatment of CA prostate** (Section 17, Ch 158).*

Also, from the condensed book (not in the paper answer):

- The switch to watchful waiting (WW) is typically near age 80.
- **Pitfall**: elderly men with unrecognised hypogonadism may progress (neuroendocrine dedifferentiation) without a PSA rise, so measure testosterone and get interval imaging.

> **KEY POINT**
>
> AS defers curative intent; WW abandons it. Life expectancy, not grade, chooses between them.

3 · Biology of grade group 1

- **Pattern 3 is molecularly near-normal.** Patterns 4–5 show EGFR, HER2 and BCL2 changes and **PTEN deletion in 70–90%**; **MYC amplification with PTEN deletion** drives metastasis.
- **Metastatic potential of pure grade group (GG) 1 is effectively zero**, though occasional extraprostatic extension occurs.
- True grade progression is ~**1–2% per year**, mostly in high-volume Gleason 6, so follow-up must be long term.
- Deaths in low-risk series come from **occult higher grade** missed at biopsy, present in ~**30%** after systematic biopsy (85% of it 3+4).

Hazard	Misattribution of grade	True progression
Frequency	**Commonest (25–30%)**	1–2.3%/year
Usual result	Mostly GG2	Many 3+4 stay indolent
Remedy	MRI, targeted biopsy, predictors	Long-term surveillance

> **KEY POINT**
>
> The main hazard of AS is **sampling error at entry**, not biology over time. First job: get the entry grade right.

4 · Patient selection

Most GG1 should be on AS (AUA, EAU, ASCO, NICE and Cancer Care Ontario agree).

	Very low risk (Epstein)	Low risk
Definition	GG1; **≤2 positive cores; no core >50%; PSA density <0.15**	Any GG1, PSA <10
Progression to treatment	Lower	Higher
Metastasis	Exceptionally low	Exceptionally low

High-volume GG1 is still a candidate, but check it with multiparametric MRI (mpMRI) and targeted biopsy, re-biopsy or a genomic assay.

- **Toronto** is inclusive (all Gleason 6 and selected intermediate), with 15-yr cancer mortality of **5%** actuarial; **Johns Hopkins** is restrictive (Epstein very low), with **0.5%**. So **inclusion criteria set mortality, while intervention stays constant (~25% at 5 yr)**.
- In Toronto, baseline Gleason 7 had ≥20% metastasis at 15 yr (HR 3.75). In PRIAS (n=5302), cancer mortality was <1% at 10 yr, and cardiovascular disease was the commonest cause of death.
- **Favourable intermediate**: offer AS selectively for small-volume GG2. Only NICE lists AS and radical treatment as equal for intermediate risk.
- Exclude GG ≥3; intraductal or cribriform histology raises concern.
- **Black men** are AS candidates, perhaps with closer surveillance. **Family history** rarely justifies treatment, except metastatic disease <60 yr in a first-degree relative, which calls for germline testing.
- **BRCA2** carries the highest upgrading risk; consider definitive therapy even for low grade, or AS only with very close follow-up.
- Risk factors for higher grade are **PSA density >0.15**, Black race and biopsy volume; restricted diffusion on MRI is a powerful predictor.

5 · Monitoring protocol

ALGORITHM

1. At diagnosis, obtain mpMRI.
2. Do a **confirmation biopsy** (systematic + targeted) within 1–3 yr.
3. Check PSA 6-monthly, DRE every 1–2 yr and MRI ≤ every 18–24 months.
4. Repeat biopsy every 3–5 yr while stable.
5. If a trigger appears, then give definitive therapy; if life expectancy falls, then switch to WW.

- Confirmation biopsy may be omitted if the first biopsy was MRI-targeted. **Compliance is paramount**: compliers had better cancer mortality and metastasis-free survival.

Guideline	Intermediate risk	Schedule / MRI
Cancer Care Ontario	Treat; AS if selected	PSA 3–6-monthly; DRE yearly; biopsy within 6–12 mo then 3–5-yearly; MRI if discordant
AUA/ASTRO 2022	AS if selected	PSA 6–12-monthly; DRE 1–2-yearly; biopsy 1–4-yearly; MRI before initial or confirmation biopsy
NICE 2019	Radical treatment or AS	PSA 3–4-monthly in year 1, then 6-monthly; **MRI at enrolment**

Prompts definitive therapy	Prompts evaluation only
GG ≥3 (unless life expectancy significantly shortened)	More Gleason 6 volume: get MRI or a biomarker
Traditionally any Gleason ≥7 (now relaxed)	Small-volume GG2: may continue AS
	PSA rise alone: repeat MRI or biopsy

> **KEY POINT**
>
> Grade triggers treatment; volume and PSA trigger investigation. 10–15% leave AS by choice (anxiety) without a trigger.

6 · MRI in AS

- **At entry the evidence is strong**: mpMRI before confirmation biopsy, with **systematic and targeted cores**, upgrades about ⅓. NICE recommends it; NCCN lists it as an option.
- **Serial MRI alone is weak** (NPV ~0.8–0.9 for GG ≥2), so it must never be the only way to exclude progression. MRI-invisible cancers behave better.

7 · Biomarkers and genomics

- Decipher predicts upgrading in GG1, not GG2; Oncotype DX evidence is mixed and Prolaris has none specific to AS.
- **Bayesian problem**: with a 3% prior metastasis risk, a 90%-specific test labels 10% "significant", so test only higher-risk candidates (high volume, high PSA density, GG2).
- Use the tests to set surveillance intensity, **never as the sole basis for AS**; a favourable score does not override adverse features.

8 · Lifestyle, drugs and psychology

- **Stop smoking**, exercise vigorously (ERASE: lowered PSA), keep a healthy weight and eat a vegetable-based diet.
- Only vitamin D 1000–2000 IU/day is reasonable. **Selenium and vitamin E are contraindicated** (SELECT).
- 5α-reductase inhibitors are not used to prevent progression (REDEEM); metformin, apalutamide and enzalutamide are investigational.

9 · Landmark trials

Trial	Design	Result	Take-home
SPCG-4	695, pre-PSA, palpable; RP vs WW; 23 yr	Cancer death 19.6% vs 31.3% (relative risk 0.55)	Benefit at age <65 and GG ≥3; none in GG1–2
PIVOT	731 US veterans; RP vs observation; 18.6 yr	All-cause death 68% vs 73% (not significant); low-risk cancer death 2.7%	Modest benefit in intermediate risk
ProtecT	1643 PSA-detected; monitoring vs RP vs radiotherapy; 15 yr	Cancer mortality <3%, no difference; **metastases 9.4% vs 4.5% vs 5.0%**; 61% of monitoring arm treated	Active monitoring is not AS (no scheduled biopsy, >⅓ intermediate/high)

Trial	Design	Result	Take-home
ASIST	GG1 on AS; MRI-targeted + systematic vs systematic confirmation biopsy	Upgrading 33% vs 27% (not significant); ~50% less AS failure at 2 yr	MRI benefit shows over follow-up
REDEEM	Dutasteride vs placebo on AS	38% less "progression" (less GG1 volume), no upgrading difference	Cytoreduction, not disease modification

160 Open Radical Prostatectomy

1 · Principles

- Radical prostatectomy (RP) alone has been shown in a randomised controlled trial (RCT) to reduce metastasis and cancer death (**SPCG-4**). Most RPs are now robotic, without proven benefit.
- **Goals, in order**: cancer control, then avoiding technical complications, then continence, then potency.

> **KEY POINT**
>
> The order of goals settles every intraoperative trade-off: when nerve sparing and a clear margin conflict, the margin wins.

2 · Surgical anatomy

P1 P4 *Anatomy of prostate and neurovascular bundles; Prostate anatomy, newer surgical concepts and NVB preservation — fascial layers table, NVB of Walsh, inferior vesical urethral/capsular branches, accessory pudendals, Santorini plexus, nerve-sparing planes*

- Inferior vesical urethral branches enter at the bladder neck (**5 and 7 o'clock** pedicles); capsular branches mark the cavernous nerves.
- The pelvic plexus combines parasympathetic **S2–S4** and hypogastric sympathetic fibres. It lies beside the rectum, **5–11 cm** from the anal verge, centred on the seminal vesicle tip.
- The **striated sphincter** is a vertical tubular sheath, not a flat diaphragm, so apical dissection decides continence; it and the levator ani are both mainly **pudendal**.

> **KEY POINT**
>
> Nerves and vessels run between levator and prostatic fascia. Leaving prostatic fascia on the gland and Denonvilliers fascia on its back lets nerve sparing and clear margins coexist.

3 · Selection and preoperative care

- **Harder, not contraindications**: prior pelvic surgery or radiotherapy, transurethral resection of the prostate (TURP), hernia mesh, large gland.
- **Defer surgery 6 weeks after biopsy and 12 weeks after TURP**, so haematoma settles.
- Inguinal hernia affects ~15% before and up to 20% after RP, so examine standing and repair it preperitoneally at RP.
- Give cephalosporin prophylaxis; the **fibreoptic headlight** is the most useful instrument.

4 · Retropubic RP — operative steps

ALGORITHM

1. Lower midline extraperitoneal incision; pelvic lymph node dissection (PLND).
2. Open endopelvic fascia; divide puboprostatic ligaments.
3. Control and divide the DVC, then the urethra and posterior sphincter.
4. Release the NVBs; divide pedicles, bladder neck and seminal vesicles.
5. Reconstruct the bladder neck and make the anastomosis.

- **PLND** is for intermediate/high risk, omitted for Grade Group 1–2 low-volume disease.
- **Endopelvic fascia**: open at the sidewall reflection, because too medial causes Santorini bleeding. Divide the **puboprostatic ligaments** but spare the **pubourethral** part, which anchors the sphincter.

DVC control

Feature	Stapler (preferred)	Suture ligation
Method	Vascular load; fire only if the catheter moves freely	3-0 figure-of-eight to pubic perichondrium, plus a back-bleeding stitch
Advantage	~200–400 mL less blood loss; less traction	Recreates puboprostatic support

- The **anterior apex is the commonest positive margin site**, so get absolute venous control before touching it.
- **Urethra**: transect close to the apex and place six 3-0 sutures through **mucosa and submucosa only**, because muscle bites delay continence.

Nerve sparing

- The **interfascial** plane (levator vs prostatic fascia) is correct; Walsh never uses intrafascial because of margin risk. Wide (extrafascial) excision is for induration, a fixed bundle or doubtful margin.
- Both NVBs are usually kept, because extracapsular extension rarely exceeds 1–2 mm. Decide in theatre: **if induration or a fixed bundle, excise that side**.
- **Technique**: release the prostate from the bundle, not the bundle from the prostate, starting **at the apex**. Use small clips and **never thermal energy**.
- **High anterior release** (before urethral division) means less traction and faster recovery.
- Avoid it if ≥2 **Tsuzuki criteria** are present: PSA >10, Gleason >6, >20% core involvement, >33% cores positive, perineural invasion, abnormal DRE.
- For **wide excision**, clamp **medial to lateral** on the rectum, since the reverse risks rectal injury.

Pedicles, bladder neck, reconstruction

- **Complete seminal vesicle excision is essential**; the pelvic plexus lies on their lateral surface.
- Close the bladder neck as a **tennis racquet** with mucosal eversion, and make a tension-free anastomosis; a peritoneal window lets haematoma decompress.

> **KEY POINT**
>
> Every step does one of three jobs: absolute venous control before the apex; the right fascia on the specimen; no traction or heat on the NVB.

5 · Perineal RP (brief)

- Perineal RP reflects the fasciae off the prostate and spares the DVC, so it bleeds less, but PLND needs a separate approach.

6 · Intraoperative complications

Complication	Setting	Management
Haemorrhage (commonest, venous)	Endopelvic fascia, DVC	**Divide DVC completely and oversew** (best measure)
Obturator nerve	PLND	Reanastomose with fine non-absorbable suture
Rectal injury	Small prostate; apical/posterior plane	Two-layer repair and omentum; colostomy if prior radiotherapy
Ureteric injury	Dissecting in trigone	Reimplantation

- Typical loss is 300–1000 mL; partial DVC division bleeds more.
- **Rectal repair**: finish the prostatectomy, dilate the anus, close in 2 layers and place omentum between rectum and anastomosis to prevent a rectourethral fistula.

7 · Postoperative care

- Discharge on day 1–2. Give ketorolac only if creatinine <1.2 mg/dL and there is no haematuria.
- Keep the **catheter ~1 week**, because day-4 removal causes frequent retention; never remove it with significant extravasation.

8 · Postoperative complications

- Mortality is **0.2%**. **Delayed haemorrhage** needs **early exploration and evacuation**, which shortens stay.
- **Thromboembolism** is the major cause of death. It is highest with PLND and peaks at **14–28 days**. Use compression and early ambulation; treat with low-molecular-weight heparin (LMWH).
- **Bladder neck contracture** appears at ~3 months. Treat with dilatation, then cold-knife incision at 3, 6 and 9 o'clock with self-catheterisation and triamcinolone; reconstruction with an artificial sphincter is rarely needed.

*See the paper answer **Post-prostatectomy incontinence** (Section 8, Ch 123).*

Also, from the condensed book (not in the paper answer):

- Continence means **0–1 pad/24 h**. Prevent incontinence with a preserved apical sphincter, a precise tension-free mucosal anastomosis and a small supple neck.
- **Erectile dysfunction (ED)**: recovery is predicted by **age <65, preoperative potency, no diabetes and bilateral nerve sparing**, and continues up to 4 years. PDE5 inhibitor rehabilitation helps; nightly is not better than on-demand.

> **KEY POINT**
>
> Most late complications trace to the pelvic haematoma: it distracts the anastomosis (contracture, incontinence) and inflames the NVBs. Hence early re-exploration, clips near the bundles and a peritoneal window.

9 · Outcomes

Time	Continent	Potent (Walsh)
3 months	80%	38%
12 months	93%	73%
18–24 months	97%	86%

- These are single-expert figures; elsewhere only a third regained baseline potency by 2 years. RP improves lower urinary tract symptoms for up to 15 years.

10 · Landmark trials

Trial	Design	Result	Take-home
SPCG-4	RCT, RP vs observation	Less metastasis and cancer death	RP cuts cancer death
Padma-Nathan	RCT, nightly sildenafil vs placebo	Better erections at 1 year	Rehabilitation works
REINVENT	RCT, vardenafil on-demand vs nightly	On-demand better	Daily dosing unproven

161 Laparoscopic and Robotic-Assisted Laparoscopic Radical Prostatectomy and Pelvic Lymphadenectomy

1 · Background

SAQ Robotic radical prostatectomy

Paper answer · P1

Type: SAQ (10 marks)
Source: 2023-Winter Paper 1, Q3b

Introduction

- RARP (robot-assisted radical prostatectomy): Walsh's anatomical nerve-sparing principles done with **3D, 10× magnified vision and wristed instruments**. The robot removed the laparoscopic suturing barrier; da Vinci from 2000 (S, Si, Xi, SP).

Indications and contraindications

- Localised cancer (cT1–T2, selected cT3); life expectancy over 10 years.
- **Absolute:** uncorrectable bleeding diathesis; unfit for general anaesthesia.
- **Relative:** morbid obesity, prostate >80 g, median lobe, prior TURP (transurethral resection of the prostate), salvage surgery.

Set-up

- Low lithotomy, arms tucked, **steep Trendelenburg (22–25°)**, padding; pressure 12–15 mmHg; camera port 15–17 cm above pubis, three robotic and two assistant ports.

Steps (transperitoneal anterior approach)

1. Drop bladder; enter space of Retzius.
2. Incise endopelvic fascia; ligate DVC (dorsal venous complex).
3. Divide bladder neck (fat line, balloon traction); check for median lobe.
4. Seminal vesicles and vasa; incise Denonvilliers fascia.
5. Clip pedicles athermally; **interfascial nerve sparing** (extrafascial if high risk).
6. Apical dissection preserving urethral length.
7. Extended PLND (pelvic lymph node dissection) if indicated.
8. Posterior (Rocco) reconstruction; **Van Velthoven running anastomosis**.

Postoperative care

- ERAS (enhanced recovery after surgery); discharge within 24 h; catheter about 1 week (not before day 5: retention); pelvic floor training; penile rehabilitation with PDE5 (phosphodiesterase type 5) inhibitors.

Outcomes

- **Yaxley 2016** randomised trial: equal function and early margins versus open; less blood loss (300 versus 1400 mL) and shorter stay.
- Margins about 15% (9% if pT2); 1-year continence mid-90s% in expert hands.

Complications

- Transfusion 1–2%; bladder neck stricture <2% (open up to 16%); rectal injury; ureteric injury; lymphocele; positioning neuropathy, corneal abrasion; robot failure <0.5%.

Recent advances

- **Retzius-sparing** (earlier continence, more positive margins); Hood technique; single-port robot; same-day discharge.

Diagram to draw: port layout and fascial planes (intra-, inter-, extrafascial).

> **Key points to remember:**
> - Surgeon, not platform, decides outcome (Yaxley 2016).
> - Interfascial plane is the standard nerve-sparing plane.
> - Bladder neck stricture <2%; transfusion 1–2%.

Also, from the condensed book (not in the paper answer):

- Avoid relative contraindications (including neoadjuvant hormones) early in the learning curve.

> **KEY POINT**
>
> The robot did not change the operation; it removed the suturing barrier. Surgical anatomy (dorsal venous complex [DVC], neurovascular bundle [NVB], fasciae, sphincter) is as in open surgery.

2 · Patient selection

- The only absolute contraindications are uncorrectable bleeding diathesis and unfitness for general anaesthesia.
- After extraperitoneal mesh repair, use the **transperitoneal** route.
- **Obesity**: place longer trocars more cephalad and lower pneumoperitoneum to 10 mm Hg. A **large gland or median lobe** takes longer and bleeds more, but continence is comparable; **prior TURP** brings the ureteric orifices close to the bladder neck.
- Multiparametric MRI (mpMRI) for extracapsular extension (ECE) has specificity **0.88** and sensitivity **0.61**; it guides nerve sparing.

3 · Preparation and anaesthesia

- Consent for **conversion to open**; **general anaesthesia is mandatory**.
- **Positioning**: tuck and pad arms; use **no fixed shoulder rests** (brachial plexus injury) and avoid exaggerated hip flexion (femoral neuropraxia).
- CO2 pneumoperitoneum causes vagal response, oliguria and hypercarbia, so raise ventilation and lower pressure. Restrict fluids to 1.5–2 L, mostly after the anastomosis.
- **Steep Trendelenburg** causes facial, laryngeal and corneal oedema and raises intraocular pressure, so protect the eyes and consider delayed extubation if the face is very swollen.

4 · Access

- **Extraperitoneal** access gives a smaller space, protects bowel and needs less tilt (suits obesity and lung disease), but absorbs more CO2. **Transperitoneal** gives a larger space. Neither has a consistent advantage.
- Transperitoneal **anterior** (commonest; Retzius first) differs from **retrovesical** (seminal vesicles first, useful with a median lobe).
- Place the camera port **15–17 cm** from the symphysis; insufflate to 12–15 mm Hg.

5 · Operative steps (transperitoneal anterior RALRP)

ALGORITHM

1. Drop the bladder (Retzius); open the endopelvic fascia; ligate the DVC.
2. Divide the bladder neck; free the seminal vesicles and vasa.
3. Develop the posterior (Denonvilliers) plane; control pedicles and release the NVBs.
4. At the apex, divide the DVC and urethra; inspect the specimen.
5. Do pelvic lymph node dissection (PLND); entrap the specimen.
6. Posterior reconstruction, anastomosis, extraction.

- Open the endopelvic fascia with **no cautery**. Pneumoperitoneum tamponades DVC bleeding, so it can be sutured, stapled, or divided late and oversewn.

Bladder neck — no tactile feedback

- Divide anteriorly in the midline, lift the catheter, inspect for median lobe and orifices, then divide the posterior neck angled **45°** downward.
- After TURP the balloon falls into the defect, so division drifts too distal; a balloon off the midline suggests a median lobe. When in doubt, **divide more proximally**.

Posterior plane, pedicles, nerve sparing

P1 P4 *Anatomy of prostate and NVB; NVB preservation in RP — nerve-sparing planes, Denonvilliers excision for posterolateral ECE, injury mechanisms, apex = commonest margin*

- **Athermal (veil) release**: incise levator fascia anteromedially and sweep the NVB posterolaterally off the prostatic fascia, then Hem-o-lok the pedicle and divide sharply. The **intrafascial** plane is accepted in selected men.
- Pedicle clips near the anastomosis can migrate into the bladder, causing haematuria, infection and stones.

Apex and reconstruction

- Preserve urethral length; a small urethral rim on the apex lowers margins without harming continence.
- Reconstruct the bladder neck (tennis racquet) only for a large gland, median lobe or prior TURP, and protect the orifices.
- The **Rocco stitch** joins the Denonvilliers remnant and posterior bladder to the posterior rhabdosphincter. Its continence benefit is inconsistent; its agreed benefit is a **tension-free anastomosis**.
- The **Van Velthoven anastomosis** runs two arms anteriorly from 6 o'clock.

6 · Continence-preserving variants

*See the paper answer **Saturation biopsy of prostate** (Section 17, Ch 155).*

Also, from the condensed book (not in the paper answer):

- Retzius-sparing leaves the detrusor apron, puboprostatic ligaments and DVC untouched, on the idea that preserving anterior support beats reconstructing it.

7 · Pelvic lymph node dissection

P4 P1 *PLND templates; Lymphocele — templates (extended ~63%), staging not survival; lymphocele drain, then fenestration*

- **Indications**: may omit in low risk (nodes <5%). AUA 2018: unfavourable intermediate or high risk. EAU 2021: nomogram risk **>5%**. AUA/ASTRO 2022: risk-stratified by nomogram (Briganti and MSKCC nomograms predate PSMA PET).
- The standard template misses most drainage, so **if PLND is done, use the extended template**: proximal 2 cm above the common iliac bifurcation (ureter crossing), distal node of Cloquet, lateral **genitofemoral nerve**, medial bladder wall.
- **Clipping lymphatics** means fewer lymphoceles; node count is a poor quality surrogate.

PLND complications (~1 in 7)

- For the obturator nerve, remove a clip or repair a transection primarily. Minimise heat proximally, where the ureter is at risk.
- **Lymphocele is commonest** (**2–5%** significant). **Transperitoneal access is not protective**, because the bladder walls off the fossa; a **peritoneal interposition flap** reduces it. It classically presents with ipsilateral deep vein thrombosis (DVT).

8 · Postoperative care

- Use enhanced recovery, with same-day discharge for selected men; a drain is not routine.

Scenario	Catheter	Cystogram
Routine	~1 week, not before day 5	No
Extensive neck reconstruction / doubtful	≥10 days	Yes
Salvage	2–3 weeks	Strongly consider

- Removal at day 3–4 causes more retention. Extravasation delays but does not reduce long-term continence.
- **Penile rehabilitation**: PDE5 inhibitors help on-demand erections, but early rehabilitative use does not improve later response (3 RCTs). Use injections for non-responders.

9 · MIS-specific complications

Complication	Rate	Management
Transfusion	1–2%	Low, because pneumoperitoneum compresses veins
Bladder neck stricture	<2% (open up to 16%)	Dilatation or incision; rarely urethroplasty
Thromboembolism	~0.5%	Pneumatic compression; drugs only with multiple risk factors

Complication	Rate	Management
Robotic failure	<0.5%	Cancel or convert
Recognised rectal injury	Rare; higher in salvage	Multilayer repair and flap, no colostomy
Unrecognised rectal injury	Rare	Presents as rectovesical fistula; colostomy, later repair
Urine leak	Uncommon	Drain, longer catheter, cystogram

- **Ureteric injury** occurs behind the vas (posterior approach), at the posterior bladder neck after TURP or with a median lobe, and during proximal PLND.
- With DVT, image the pelvis for lymphocele or haematoma; with a significant lymphocele, do a duplex for occult DVT.

> **KEY POINT**
>
> Robotic injuries are fewer but less visible: bleeding and strictures fall with magnified vision, while bowel injury and positioning neuropathy happen outside the camera's field and must be anticipated.

10 · Outcomes

- **Oncologic**: positive surgical margin (PSM) rates are equal open and robotic (~15%; ~9% organ-confined). PSM depends on tumour (Gleason ≥7, perineural invasion, stage) and surgeon volume, **not** on obesity, gland size or prior surgery.
- **Continence**: no-pad rates are in the mid-90s% at 1 year. Persistent mild leakage needs a sling or ProACT; severe (<2%) an artificial urinary sphincter.
- **Potency**: up to 90% at 1 year in expert hands; recovery continues to 5 years.
- **Salvage RALRP**: exclude metastases with PSMA PET and use full bowel prep and an air-bubble test. Continence is only 33–80%, and most men are impotent.

11 · Landmark trials

Trial	Design	Result	Take-home
Yaxley 2016	RCT, one open vs one robotic surgeon	Equal continence, potency, early oncology at 3 months; blood loss ~1400 vs 300 mL; shorter stay	Robot wins on recovery, not cancer or function
Retzius-sparing RCTs (Cochrane)	Single-centre RCTs	Earlier continence; equal at 12 months; more PSM	Use selectively

162 Radiation Therapy for Prostate Cancer

LAQ Radiotherapy and brachytherapy for CA prostate

Paper answer · P2 · P4 · merged from 2

Asked as: P4 LAQ "Role and impact of Radiotherapy methods for management of focal and advanced carcinoma prostate?" · P2 SAQ "Radiotherapy and brachytherapy for CA prostate" (Radiotherapy for CA prostate.; Brachytherapy for CA prostate)

"Focal" is taken to mean localised (organ-confined and locally advanced) disease.

Radiobiology

- RT (radiotherapy) causes DNA double-strand breaks. Prostate cancer has a **low α/β ratio (~1.5 Gy)**, lower than the rectum (~3 Gy). So larger fractions kill more tumour for the same late toxicity: the basis of **hypofractionation**. The low α/β ratio also favours high dose per fraction (HDR brachytherapy).

EBRT (external beam radiotherapy) techniques

Step	Advance
3D-CRT (three-dimensional conformal RT)	CT (computed tomography)-planned beams shaped to the prostate
IMRT (intensity-modulated RT)/VMAT (volumetric modulated arc therapy)	Inverse planning, steep dose gradients; less rectal toxicity; now standard; allows dose escalation
IGRT (image-guided RT)	Daily fiducials or cone-beam CT; margins 3–7 mm
MRI fusion, PSMA (prostate-specific membrane antigen) PET (positron emission tomography)	Better targeting; PSMA PET changes plans in up to half of high-risk men

EBRT fractionation schedules

Schedule	Dose per fraction	Example	Evidence
Conventional	1.8–2 Gy	78 Gy in 39	Dose escalation (RTOG 0126): better PSA control, no survival gain
Moderate hypofractionation (standard)	2.25–4 Gy	**60 Gy in 20**	**CHHiP** (non-inferior to 74 Gy in 37), **PROFIT**: non-inferior
SBRT (stereotactic body RT)	≥5 Gy	36.25 Gy in 5	**HYPO-RT-PC** non-inferior; **PACE-B** equal toxicity

Brachytherapy: definition and principle

- Brachytherapy (BT; Greek *brakhus*, short): radioactive sources placed **within the prostate**.
- **Steep dose fall-off** gives an intraprostatic dose far above what EBRT can safely give.

Brachytherapy: types

Feature	LDR (low dose rate), permanent seeds	HDR (high dose rate), temporary
Source	^{125}I (half-life 59.6 days), ^{103}Pd (17 days), ^{131}Cs (9.7 days)	**^{192}Ir** via remote afterloader
Monotherapy dose	^{125}I 145 Gy (P2: 140–160 Gy); ^{103}Pd 110–125 Gy	27 Gy in 2, 31.5 Gy in 3, 34–38 Gy in 4 fractions (single 19 Gy fraction fails)

Feature	LDR (low dose rate), permanent seeds	HDR (high dose rate), temporary
Monotherapy use	Low and favourable intermediate risk	
Dosimetry / quality	Post-implant CT at 14–30 days: **D90** (dose covering 90% of gland) **>90%** of prescription	Real-time planning; none afterwards

Brachytherapy: technique

- **TRUS (transrectal ultrasound)-guided transperineal** implant under anaesthesia: high lithotomy, stepper, template grid, needles, seeds deployed on withdrawal.
- **Peripheral loading** spares the urethra. Intraoperative planning gives best coverage.
- Quality: **D90 >90%** of prescription and **V100 >85%** of gland; urethral and rectal (V100 <1 cm^3) limits. Low D90 predicts failure.

Brachytherapy: indications

- **Monotherapy:** low and favourable intermediate risk (**RTOG 0232**: adding EBRT to implant gave no benefit in intermediate risk, more toxicity).
- **Boost** (EBRT 45 Gy + implant ± ADT [androgen deprivation therapy]) for unfavourable intermediate and high risk: **ASCENDE-RT:** LDR boost halved biochemical failure vs EBRT boost, with more grade 3 urinary toxicity (18.4% vs 5.2%).
- Also salvage and focal BT.

Brachytherapy: contraindications (relative)

- **IPSS (International Prostate Symptom Score) >20, Qmax <10 mL/s**, large median lobe; recent or large TURP (transurethral resection of the prostate) defect; gland >60 cm^3 (cytoreduce with ADT); pubic arch interference; prior pelvic RT; inflammatory bowel disease; cannot lie in lithotomy.

Brachytherapy: complications and follow-up

- Acute obstructive and irritative symptoms; retention (29% if IPSS >20, 2% if <10); α-blockers help.
- Chronic grade ≥2 urinary toxicity about 20%; urethral stricture; rectal complications 1–2%; rare rectourethral fistula; ED (erectile dysfunction).
- Nadir is deeper and later (about 2.5 years). **Benign PSA bounce** in about one-third: watch for 30 months before biopsy, do not salvage.

ADT with RT

- **Why:** ADT radiosensitises (less DNA repair) and causes apoptosis; RT up-regulates androgen-receptor signalling, which ADT blocks.

Risk group	ADT with EBRT
Low, favourable intermediate	None
Unfavourable intermediate	**4–6 months**
High / locally advanced	**18–36 months** (P2: 24–36 months) (EORTC 22863, PCS IV); add **abiraterone** if very high risk (STAMPEDE)

- **RT added to ADT** improves survival in locally advanced disease: **SPCG-7, NCIC PR.3/MRC PR07** (about 8–10% absolute).

After radical prostatectomy

- **Adjuvant RT** (pT3/positive margin): EORTC 22911, ARO 96-02 (biochemical control), **SWOG 8794** (metastasis-free and overall survival).
- **RADICALS-RT, RAVES, GETUG-AFU 17** (ARTISTIC meta-analysis): **early salvage at PSA (prostate-specific antigen) 0.1–0.2 ng/mL** equals adjuvant RT and spares over 60% of men RT.
- Salvage dose ≥64–66 Gy. **RTOG 9601:** 2 years bicalutamide improved overall survival. **SPPORT:** pelvic nodes + short ADT improve freedom from progression.

Advanced and metastatic disease

Setting	RT role	Evidence
Low-volume de novo mHSPC (metastatic hormone-sensitive)	**RT to prostate** + ADT	**STAMPEDE arm H**: overall survival HR (hazard ratio) 0.68 (3-year 73% → 81%); none in high volume
Oligometastases (≤3–5)	SBRT to all lesions (MDT, metastasis-directed therapy)	STOMP, **ORIOLE** (less progression), SABR-COMET
Painful bone metastases	Palliative EBRT, e.g. **8 Gy single fraction**; cord compression	Equal pain relief to multiple fractions
mCRPC (metastatic castration-resistant), bone-only symptomatic	**Radium-223** (α-emitter, calcium mimic, range <100 µm) 50 kBq/kg 4-weekly × 6	**ALSYMPCA**: overall survival 14.9 vs 11.3 months; fewer skeletal events
mCRPC, PSMA-positive, after taxane and ARPI (androgen-receptor pathway inhibitor)	**^{177}Lu-PSMA-617** (β-emitter radioligand)	**VISION**: survival 15.3 vs 11.3 months (HR 0.62); **TheraP**: better PSA response than cabazitaxel

- **Never combine radium-223 with abiraterone** (more deaths and fractures).
- **Also know:** protons (no proven gain over IMRT); strontium-89, samarium-153 (pain palliation); elective pelvic nodal RT (RTOG 94-13, GETUG-01: no survival gain).

Outcomes

- **ProtecT** (2016; 15-year 2023): equally low prostate-cancer mortality with RT, surgery or active monitoring; RT more bowel effects, surgery more incontinence and ED.

Response assessment

- **Phoenix failure:** PSA nadir + 2 ng/mL.
- Biopsy-proven local failure: salvage prostatectomy, brachytherapy, HIFU (high-intensity focused ultrasound) or cryotherapy.

Toxicity of RT

Type	Features and management
Acute (weeks)	Frequency, dysuria, diarrhoea; α-blockers
Late GI (gastrointestinal)	**Radiation proctitis** with bleeding at 6–18 months: sucralfate/steroid enemas, argon plasma coagulation, hyperbaric oxygen

Type	Features and management
Late GU (genitourinary)	Cystitis, haematuria, stricture
Sexual	ED, vascular; worse with ADT; PDE5 (phosphodiesterase type 5) inhibitors work in most
Rare	Second cancers, fistula

Recent advances

- **Hydrogel rectal spacer** (late rectal toxicity 2% vs 7%); MR-linac adaptive SBRT; **FLAME** focal boost to the MRI lesion; protons; PSMA radioligands moving earlier (PSMAfore).

Diagram to draw

- TRUS-guided perineal template brachytherapy (transperineal implant set-up).

Key points to remember:

- α/β ~1.5 Gy: basis of hypofractionation, 60 Gy/20 (CHHiP) and SBRT 36.25 Gy/5.
- ADT 4–6 months unfavourable intermediate; 18–36 months (P2: 24–36 months) high risk.
- Early salvage at PSA 0.1–0.2 = adjuvant (RADICALS, RAVES, GETUG-AFU 17).
- STAMPEDE H: prostate RT for low-volume M1; ALSYMPCA, VISION for mCRPC.
- Phoenix: nadir + 2 ng/mL.
- LDR ^{125}I 145 Gy; HDR ^{192}Ir; D90 >90%, V100 >85%.
- IPSS >20 or Qmax <10 = relative contraindication.
- ASCENDE-RT: boost halves biochemical failure.

Also, from the condensed book (not in the paper answer):

- A single 19 Gy high-dose-rate fraction fails (§5).
- RADAR and RTOG 99-10 on ADT duration (§6).
- Salvage prostatectomy incontinence ~63% (§11).

1 · Radiobiology and fractionation

- Conventional fractionation gives **1.8–2 gray (Gy)** daily.
- In the **linear–quadratic model**, the **α/β ratio** sets a tissue's response to fraction size: low (2–4 Gy) in late-responding normal tissue, high (≥8 Gy) in most tumours.
- Prostate cancer has few cycling cells, so its α/β is **≈1.5 Gy**, **lower than the rectum (~3 Gy)**. Fractions >2 Gy therefore favour tumour kill over late toxicity, which is the rationale for hypofractionation.

Category	Dose/fraction	Typical course
Conventional	1.8–2 Gy	~8 wk (78 Gy in 39 fractions)
Moderate hypofractionation	2.25–4 Gy	~4 wk (**60 Gy/20**)
Ultrahypofractionation = stereotactic body RT (SBRT)	≥5 Gy	≤5–7 fractions (**36.25 Gy/5**)

2 · EBRT techniques

- MRI is fused with planning CT; PSMA PET changes management in up to half of higher-risk men (see Ch 152).

- **IMRT or VMAT** lowers gastrointestinal (GI) but not genitourinary (GU) toxicity versus 3D conformal RT, as the urethra lies inside the target.

Volume	Prostate application
Gross tumour volume (GTV)	Dominant intraprostatic lesion (DIL) on MRI
Clinical target volume (CTV)	Prostate, with or without proximal seminal vesicles and pelvic nodes
Planning target volume (PTV)	CTV plus a margin for motion and set-up: **3–7 mm** with image guidance

- **Image-guided RT (IGRT)** with daily gold fiducials or cone-beam CT is the NCCN standard; its smaller margins make large fractions safe.
- A **rectal spacer** (hydrogel or hyaluronic acid) is injected transperineally into Denonvilliers' fascia. Late rectal toxicity falls to **2% vs 7%** (hydrogel, 79.2 Gy).
- The MR-linac allows 2 mm margins, and protons have shown no randomised benefit over IMRT.

3 · Dose escalation

- Six randomised trials, including RTOG 0126, raised **biochemical control in all risk groups** with **no overall survival (OS) gain**, because salvage works and metastasis fell only slightly. Still, ≥~78 Gy is standard.
- **FLAME**: a focal boost to the MRI DIL improved biochemical progression-free survival (HR 0.45) without more toxicity.

4 · Hypofractionation

- **Moderate**: CHHiP 57 Gy/19 was not shown noninferior; PROFIT 60/20 was noninferior; RTOG 0415 70 Gy/28 was noninferior but more toxic late; HYPRO 64.6 Gy/19 was **not superior**.
- After prostatectomy, HYPORT 62.5 Gy/25 was noninferior to 66.6 Gy/37.
- **SBRT**: HYPO-RT-PC gave ~84% 5-yr failure-free survival, with more acute toxicity.
- **Dose ceiling**: there is no gain above **40 Gy/5**. Give it on alternate days with 3–5 mm margins.

5 · Brachytherapy (BT)

Urinary function decides eligibility.

	Low-dose-rate (LDR)	High-dose-rate (HDR)
Source	Permanent seeds (^{125}I, ^{103}Pd, ^{131}Cs)	Temporary ^{192}Ir (380 keV) in a remote afterloader
Technique	TRUS-guided transperineal template; **peripheral loading** spares urethra	Dose shaped by dwell times and positions; no post-implant dosimetry
Uses	Monotherapy for low/intermediate risk; boost	Boost (e.g. 15 Gy × 1 after 46 Gy); monotherapy 27 Gy/2, 34–38 Gy/4
Pitfall	Oedema, seed migration	**19 Gy × 1 fails**: 5-yr local failure 29% vs 3% (27 Gy/2)

Isotope	Half-life	Monotherapy	Boost (with 45 Gy EBRT)
^{125}I	**59.6 d**	140–160 Gy	108–110 Gy
^{103}Pd	17 d	110–125 Gy	90–100 Gy

Isotope	Half-life	Monotherapy	Boost (with 45 Gy EBRT)
^{131}Cs	9.7 d	100–115 Gy	80–85 Gy

- **Post-implant dosimetry** (CT at 14–30 days): **D90** >90% of prescription; **V100** >85%; urethral V150 <0.05 cm³; rectal V100 <1 cm³. A low D90 predicts biochemical failure. **MCQ** D90 is a dose, V100 a volume.
- **Relative contraindications**: **IPSS >20**; **Qmax <10 mL/s**; large median lobe; large TURP defect or TURP within 3–6 mo; prior pelvic RT; inflammatory bowel or connective tissue disease; **pubic arch interference**; inability to adopt lithotomy.
- A large gland shrinks ~30% with ADT. Retention risk is ≈29% if IPSS >20 vs ≈2% if <10, so start α-blockers before the implant.
- **With EBRT**: 45 Gy plus an implant at 60–70% of monotherapy dose. **RTOG 0232**: BT alone equalled EBRT plus BT in intermediate risk with less toxicity, so BT alone is standard.
- **ASCENDE-RT**: no OS difference; LDR boost grade 3 GU toxicity **18.4% vs 5.2%**.

> **KEY POINT**
>
> Why BT works: the steep fall-off lets intraprostatic dose far exceed EBRT. Hence outcome depends on implant accuracy (D90), eligibility on whether the urethra tolerates swelling, and EBRT is added only where disease may lie beyond the implant.

6 · ADT with RT

- **Synergy**: ADT causes apoptosis, shrinks the target and lowers **Ku70**, impairing non-homologous end-joining repair (see Ch 164).

Risk group (NCCN)	With EBRT	With EBRT + BT	Key evidence
Low / favourable intermediate	None	—	RTOG 94-08: no benefit in low risk
Unfavourable intermediate	**4–6 mo**	—	RTOG 94-08, DFCI 95-096, TROG 96.01 (6 mo better than 3); RTOG 99-10: 9 mo no better than 4
High / very high	**24–36 mo** (≥18)	12–36 mo	EORTC 22863, RTOG 92-02, EORTC 22961 (36 mo better than 6); PCS IV: 18 mo = 36 mo

- Needed duration falls as RT dose rises: in **RADAR**, 18 mo beat 6 mo only without an HDR boost.
- **STAMPEDE**: adding 2 yr abiraterone in high-risk N0/N1 M0 improved metastasis-free survival and OS.

7 · Pelvic nodal RT

- RTOG 94-13 and GETUG-01 showed no durable benefit, mainly because a field border at L5/S1 misses the **common iliac nodes**; modern fields extend to **L4/L5**.
- **POP-RT** (node risk >20%, long ADT): whole-pelvis RT raised 5-yr biochemical failure-free survival (~95% vs ~81%), with no OS difference and more acute GI toxicity.

8 · Post-prostatectomy RT

- RADICALS-RT, GETUG-AFU 17 and RAVES, with their meta-analysis, make **early salvage preferred** over adjuvant RT, which may still suit very high risk (Gleason 8–10, pT3b) (see Ch 164).
- Dose is ≥64 Gy, and PSMA PET localises recurrence.

9 · Follow-up and biochemical failure

Do not mistake a benign bounce for failure.

- Benign glands persist, so PSA stays detectable. The nadir is ≤0.1 ng/mL after EBRT plus BT and higher after EBRT alone; it is reached at ~1.5 yr after EBRT and ~2.5 yr after BT.

	ASTRO 1996	**Phoenix 2005 (current)**
Failure	3 consecutive rises	PSA **≥ nadir + 2 ng/mL**
Date	Backdated to midway between nadir and first rise	When reached (or when salvage started)
Notes	Delays diagnosis ≥18 mo; invalid with ADT or BT	Best correlates with clinical relapse; valid with short ADT; fewer false calls from bounces

- Poor prognosis is signalled by failure <18 mo and especially **PSA doubling time <3 mo**, which predicts cancer death.
- A **bounce** (rise >0.2 ng/mL, then durable fall) occurs in ~1/3 after LDR, mostly in the first 3 yr, and predicts a better outcome.
- **Biopsy** is best at **30–36 mo**, since irradiated cells die slowly. Marked radiation effect is not graded and behaves as negative.

10 · Toxicity

Toxicity	**Features**	**Management / prevention**
Acute GU	Peaks ~2 mo, settles by 12 mo	α-blockers; catheter for retention
Late GU	Grade ≥2 ~20%; stricture; incontinence ~3%, more after TURP	IGRT halves late GU toxicity; peripheral loading
Late GI (proctitis)	Bleeding at 6–18 mo; IMRT 81 Gy grade ≥2 ~5%; BT 1–2%; colostomy ~0.3%	Steroid or sucralfate enemas, then **argon plasma coagulation** or hyperbaric oxygen, then colostomy
Erectile dysfunction	Vascular, over 24 mo; often due to ADT; best with BT alone (~76% potent at 5 yr)	PDE5 inhibitors work in ~¾
Second cancers	Small excess of bladder and rectal cancers	Investigate haematuria or rectal bleeding fully

11 · Salvage after local failure

- Salvage needs **biopsy-proven, local-only** recurrence. Options are salvage RP (incontinence ~63%), BT (preferred), SBRT re-irradiation, cryotherapy and HIFU.

12 · Landmark trials

Trial	Design	Result	Take-home
RTOG 0126	70.2 vs 79.2 Gy, low–intermediate	8-yr biochemical failure-free 65% vs 80%; OS same	Dose escalation improves control, not OS
CHHiP	74/37 vs 60/20 vs 57/19	60/20 noninferior	60 Gy/20 standard
HYPO-RT-PC	42.7/7 vs 78/39	Noninferior; equal late toxicity	Ultrahypofractionation valid
PACE-B	36.25/5 vs conventional/moderate	Similar 2-yr toxicity	SBRT safe
ASCENDE-RT	EBRT vs LDR boost (plus 12 mo ADT)	Biochemical failure HR 2.04 for EBRT boost; more GU toxicity with LDR	BT boost for high risk
TROG 96.01	RT with no, 3 or 6 mo ADT	6 mo improved OS; 3 mo did not	≥6 mo short-course
RTOG 94-08	66.6 Gy with or without 4 mo ADT, T1b–T2b	10-yr OS 57% vs 62%; intermediate risk only	ADT for intermediate, not low
EORTC 22863	RT with or without 3 yr goserelin, T3–T4	10-yr OS 39.8% vs 58.1%	Long ADT in high risk
EORTC 22961	6 vs 36 mo ADT	Short course not noninferior	Long ADT for high risk
PCS IV	18 vs 36 mo	No survival difference	18 mo acceptable
SPCG-7; NCIC PR.3/MRC PR07	Lifelong ADT with or without RT	RT lowered cancer death and raised OS	Never ADT alone if fit
RTOG 94-13; GETUG-01	Pelvic vs prostate-only RT	No durable benefit	Field height matters
SPPORT	Salvage bed RT, with or without ADT and nodes	Pelvic RT plus ADT best	Treat nodes in salvage

163 Focal Therapy of Prostate Cancer

1 · Definition and rationale

- **Focal therapy (FT)** is guided ablation of an image-defined, biopsy-confirmed lesion with a **safety margin**. Sparing the neurovascular bundles (NVBs), sphincter and rectum preserves function.
- FT rests on mpMRI, targeted biopsy and the fact that **Gleason pattern 4/5** is lethal.

Index lesion hypothesis

- Cancer is multifocal in **50–91%** of radical prostatectomy (RP) specimens. The **index lesion (IL)**, the largest or highest-grade focus, is held to drive natural history and metastasis.
- **For**: the IL holds **80–90%** of tumour volume, and satellites are mostly <**0.5 mL** (Stamey "insignificant").
- **Against**: a satellite is the aggressive focus in 12–38%, and IL Gleason matches overall grade in only ~⅔.

> **KEY POINT**
>
> "Multifocal, therefore unsuitable for FT" is a trap: what matters is one significant lesion plus insignificant satellites, not the count of foci. The hypothesis's exceptions are why FT is always paired with surveillance.

- **FT and active surveillance (AS)** form one strategy: ablate grade 4 disease, watch the low-grade remainder, and retreat new foci.

2 · Patient selection

- The patient must **accept close surveillance**.
- **TRUS biopsy** misses almost half the disease relevant to hemiablation compared with **transperineal mapping biopsy (TMB, 5-mm grid)**.
- **mpMRI** detects high-grade and anterior tumours. In **PROMIS**, its sensitivity for clinically significant cancer (csPCa) was **93%** and specificity 41%, so MRI lesions need biopsy and MRI-negative areas still need systematic biopsy.
- mpMRI **underestimates volume**, so a margin is needed: **5 mm** ablates >95% of csPCa. About **90%** of outfield csPCa lies within **1 cm** of the target.

> **KEY POINT**
>
> Imaging finds the lesion, only biopsy grades it, and only systematic biopsy vouches for the rest of the gland. Every consensus has kept all three.

Consensus criteria (2016–2019)

Variable	Consensus
Work-up	mpMRI plus targeted and systematic biopsy
Risk group	**Intermediate**; Gleason **3+4 = "sweet spot"**; 4+3 maximum; no cribriform or ductal histology
Size	≤1.5 mL (≤3 mL if not crossing midline); 20–25% of gland if crossing

Variable	Consensus
Residual untreated disease	Gleason 3+3 only (≤1 mm core length in 2016, tightened from 5 mm)
Life expectancy	Per guidelines; no fixed limit
Erectile dysfunction	**Not** a contraindication
Not suitable	High risk (no long-term data); low risk usually goes to AS

- The goal is now to treat and retreat so as to **delay or avoid radical therapy**. FT remains investigational, ideally within a trial.

3 · Ablation patterns and energies

- FT means **lesion, quadrant or hemiablation**; "hockey-stick" (subtotal) ablation is partial-gland ablation, not FT.

SAQ HIFU

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 5 Paper 2, Q3d

Definition and principle

- HIFU (high-intensity focused ultrasound) focuses ultrasound from a bowl-shaped transducer or lens onto a point. Energy is absorbed only where the beams converge, so tissue there is heated while the path is spared.
- **Intensity 1500–10,000 W/cm²** (diagnostic ultrasound causes negligible heating).
- **Mechanisms: thermal coagulative necrosis** (≥55 °C; prostate exceeds 100 °C within seconds) plus **cavitation** (bubbles collapse and damage cells).
- Each lesion is a **rice grain**; the target is "painted" with overlapping lesions. Ablated and viable tissue look alike on ultrasound, so gaps occur.

Devices and routes

- **Transrectal** probe that both monitors and treats (posterior zone).
- **TULSA (MRI [magnetic resonance imaging]-guided transurethral ultrasound ablation): MR thermometry** confirms lethal temperature; reaches the anterior gland.

Indications

Setting	Use
Focal therapy, prostate cancer	MRI-visible, unilateral, grade group 2–3 index lesion; hemiablation; based on the index-lesion hypothesis
Whole-gland ablation	Localised cancer; retreatment 29–38%
Salvage after EBRT (external beam radiotherapy)	Biopsy-proven local recurrence, **≥2 years after EBRT**
BPH (benign prostatic hyperplasia)	Approved
Kidney	Investigational (incomplete ablation)

Limitations

- **Anterior lesions** (energy lost, oedema displaces target), large gland, gland height and **calcification** (some advise TURP [transurethral resection of the prostate] first).

Outcomes

- **Focal HIFU:** continence 90–95%, potency 50–78%; infield positive biopsy 16.7–54% (csPCa [clinically significant prostate cancer] 0–21.8%).
- **HEAT registry** (1379 men): 7-year failure-free survival 69%.
- **Salvage HIFU:** biochemical progression-free survival about 48% at 3 years; continence 64–87.5%.

Complications

- Retention (about 8%), UTI (urinary tract infection) 6–18%, stricture (4%), ED (erectile dysfunction); rectourethral fistula rare (about 2% after salvage); osteitis pubis.

Follow-up

- PSA (prostate-specific antigen) falls only 50–80%; **mpMRI (multiparametric MRI) + targeted biopsy at 3–6 months**, then at 1–2 and 5 years. EAU (European Association of Urology): focal therapy within trials or registries.

Also know: histotripsy (>10,000 W/cm², mechanical liquefaction, visible bubble cloud).

Diagram to draw: transrectal transducer focusing beams on a rice-grain lesion in the posterior prostate.

> **Key points to remember:**
> - Focused ultrasound → ≥55 °C coagulative necrosis plus cavitation.
> - Transrectal HIFU suits posterior lesions; TULSA for anterior.
> - Focal HIFU: continence 90–95%; HEAT 7-year failure-free 69%.

SAQ Cryotherapy for CA prostate

Paper answer · P2 · P4 · merged from 2

Asked as: P4 SAQ "Cryotherapy for carcinoma prostate" · P2 SAQ "Biological effects of cryosurgery"

Principle

- Tissue destruction by extreme cold. **Argon** expanding in the probe tip cools by the **Joule–Thomson effect**; **helium** expanding warms (active thaw).
- Used for small renal masses and prostate cancer.
- Prostate cryotherapy is **transperineal, image-guided** freezing; suits **anterior** tumours. HIFU (high-intensity focused ultrasound) is the main alternative.
- Thermocouples guard sphincter and neurovascular bundles; a **urethral warmer** prevents sloughing.
- Generations: liquid nitrogen without imaging; TRUS (transrectal ultrasound) plus warmer; argon/helium 17-gauge probes; focal with MRI (magnetic resonance imaging) fusion.

Biological effects: mechanisms of cell death

Tissue dies during **both freezing and thawing**, by four mechanisms:

Mechanism	What happens
1. Direct cell injury	*Freeze:* extracellular water freezes first and becomes hyperosmolar; cells dehydrate and membranes distort. Rapid cooling forms **intracellular ice**, which ruptures organelles. *Thaw:* water rushes back, cells swell and burst; ice recrystallises into larger crystals
2. Vascular injury	Endothelial damage, platelet activation and **thrombosis** cause **vascular stasis** and ischaemic necrosis; the main delayed killing effect
3. Apoptosis	Sublethal cold at the periphery triggers programmed cell death
4. Immunomodulation	Necrotic cells release antigens and DAMPs (damage-associated molecular patterns); dendritic cells activate T cells against tumour

- **End result:** coagulative necrosis, then fibrous scar.

Factors deciding lethality

- **Temperature:** normal renal tissue dies at about −19.4 °C; tumour may need −40 to −50 °C. **Target ≤−40 °C.**
- **Fast freeze, slow thaw** maximise injury.
- **Double freeze–thaw cycle** enlarges the necrotic zone (standard of care); each freeze about 8–10 minutes.
- **Heat sink:** nearby large vessels warm tissue; more probes or longer freezing needed.

Zones of the ice ball

- Centre: complete necrosis. Edge: 0 °C and sublethal; below −20 °C only about 3 mm inside the edge. The lethal −40 °C isotherm lies well inside the visible ice ball.
- So the ice ball must extend **5–10 mm beyond the tumour**.

Indications in prostate cancer

1. **Salvage after radiotherapy** (main role): biopsy-proven local recurrence, no metastases, **PSA (prostate-specific antigen) <10 (ideally <4)**, doubling time ≥16 months.
2. **Primary whole-gland:** men unfit for surgery or radiotherapy.
3. **Focal or hemi-gland:** MRI-visible grade group 2–3 lesion; rests on the index-lesion hypothesis.

Technique

- TRUS-guided transperineal template: place cryoprobes, thermocouples and urethral warmer; real-time imaging shows the ice ball; freeze anterior to posterior until −40 °C is reached; helium thaw; repeat the cycle.
- **Also know:** passive versus active thaw (both used); insulated probe shafts reduce skin injury.

Outcomes

- Salvage: biochemical control **55–71% at 3–5 years** (Phoenix: nadir + 2).
- **Donnelly, Cancer 2010** (RCT, randomised controlled trial): **non-inferior** to external beam radiotherapy; worse sexual function.

Complications

	Primary / focal	Salvage
ED (erectile dysfunction)	**Almost always** (whole gland)	Very common

	Primary / focal	Salvage
Incontinence	Rare	Single digit
Rectourethral fistula	≤0.8% (focal)	1–3.4%
Retention	Commonest after focal	~4%

- **Also know:** urethral sloughing, stricture; focal ablation preserves erections better (COLD [Cryo On-Line Data] registry).

Comparison with other ablative energies

	Cryotherapy	HIFU	IRE (irreversible electroporation)
Energy	Freezing	Focused ultrasound heat (≥55 °C)	Electric pulses, non-thermal
Route	Transperineal	Transrectal	Transperineal
Best for	Anterior lesions	Posterior lesions	Lesions near nerves, vessels

Guidelines and advances

- **EAU (European Association of Urology):** primary or focal only in trials or registries. PSMA (prostate-specific membrane antigen) PET selects salvage patients.

Diagram to draw

- Prostate (or tumour) with probe and ice ball: −40 °C isotherm inside the 0 °C edge, and tumour margin.

> **Key points to remember:**
> - Argon freezes, helium thaws (Joule–Thomson); ≤−40 °C, two freeze–thaw cycles, margin 5–10 mm, urethral warmer.
> - Four mechanisms: direct injury, vascular thrombosis, apoptosis, immune effect.
> - Main role is salvage: PSA <10, doubling time ≥16 months.
> - Whole-gland ED near universal; fistula ≤0.8% focal, up to 3.4% salvage.

Also, from the condensed book (not in the paper answer):

- Focal cryotherapy 5-yr biochemical progression-free survival (BPFS) ~62–70%; COLD registry: partial equals whole-gland control with ~2× better erections.

Modality	Mechanism / route	Pros	Cons
TULSA (MRI-guided transurethral ultrasound ablation)	Transurethral transducer with MR thermometry	Anterior gland; control within 1.3 mm	In-bore MRI; csPCa on 12-month biopsy ~⅓
Focal laser	Interstitial fibre ≥60 °C; MR thermometry	Most predictable zone	Narrow margin, so **highest infield failure** (~40% at 6 yr)
Irreversible electroporation (IRE)	Non-thermal pulses >1600 V/cm open membrane pores	Spares connective tissue; good function	Needs deep anaesthesia; NVB and urethral injury on histology

Modality	Mechanism / route	Pros	Cons
Vascular-targeted photodynamic therapy (VTP)	IV padeliporfin (TOOKAD) 4 mg/kg activated by 753-nm light occludes vessels	**Only randomised evidence** (PCM301)	Gleason 6 only; perineal pain ~15%; light precautions
Radiofrequency	~100 °C	Fat limits conduction	Infield csPCa 20–30%
Focal brachytherapy	Low-dose-rate seeds or high-dose-rate (HDR) catheters	Mild early toxicity	HDR: stricture ≤8.2%, ED in half; biopsy only at 24 mo

KEY POINT

Every energy fails at the edge of the ablation zone, where temperature or field strength becomes sublethal. Margins, thermometry and adjuvants all aim to make that edge lethal, so laser's precise but narrow zone gives the most infield recurrence.

4 · Outcomes

Modality	Infield positive biopsy	Outfield	Continence	Erections
Cryotherapy	7–39% overall (~¼ csPCa)		95–100%	47–100%
HIFU	17–54% (csPCa 0–22%)	7–43% (csPCa 2–8%)	90–95%	50–78%
Laser	11–50%	17–60%	95–100%	Minimal change
IRE	csPCa 1–10%	csPCa 5–16%	~All	Some decline
VTP	19–37%	17–19%	Preserved	Nil to mild fall

- HEAT (HIFU): 7-yr failure-free survival was **69%** (68% intermediate, 65% high risk).

5 · Complications

- The commonest is **urinary retention** (0–9%); others are infection, sloughing and stricture (~4–5% after HIFU or IRE).
- **Rectourethral fistula** occurs in ≤2% and is the most feared; some heal with a catheter for 4–6 weeks.

6 · Follow-up and failure

Follow-up relies on MRI and biopsy, not PSA.

- **PSA** falls only **50–80%**, and the Phoenix/ASTRO definitions are not validated after FT. A rise to nadir + **1.0** ng/mL at 12 months, or + 1.5 at 24–36 months, predicts csPCa.

Success (ICUD–SIU 2015)	Infield	Outfield
Short term	Eradication of treated focus	No csPCa — any outfield cancer = **selection failure**
Intermediate/long term	No csPCa	No csPCa (new cancer may be de novo)

ALGORITHM

1. At **3–6 months**: mpMRI and targeted biopsy (4–6 cores) of the treated zone.
2. At 1–2 and 5 years: mpMRI; if a new lesion appears, then targeted biopsy.
3. At 1–2 and 5 years: systematic biopsy of the untreated gland.

- **PI-FAB**: read contrast-enhanced (**DCE) images first**, as scar obscures T2 and diffusion. Score 1 (both negative) is monitored, score 2 (DCE positive only) is biopsied if PSA rises, and score 3 (both positive) is biopsied.

7 · Salvage

- After FT failure, options are **focal retreatment** or whole-gland RP or radiotherapy.

Focal salvage for radiorecurrent disease

	Cryotherapy	HIFU	IRE
Cancer control	5-yr BPFS ~50%	3-yr BPFS 48%	Progression-free survival 60%, metastasis-free survival 91% (48 months)
Continence	87–100%	64–87.5%	93%
Signature complication	Rectourethral fistula ~1.4%	Bladder-neck stricture, fistula ~2%; grade ≥3 ~¼	Urethral sludge ~9%

- Focal salvage controls cancer as well as whole-gland salvage, with better function.

8 · Landmark trials

Trial	Design	Result	Take-home
PCM301 (Azzouzi 2017)	415 low-risk men; padeliporfin VTP vs AS	Progression 28% vs 58%; conversion to radical therapy at 4 yr **24% vs 53%**; continence preserved	First level 1 evidence for FT — but Gleason 6 only, surrogate end points
PROMIS	mpMRI vs 5-mm TMB, 11 centres	csPCa sensitivity 93%, specificity 41%	MRI lesions need biopsy; systematic biopsy still needed
HEAT registry	Prospective focal HIFU, 1379 men	7-yr failure-free survival 69%	Largest FT dataset; includes high risk

KEY POINT

Open problems are leaving satellites, the cost of retreatment and adjuvants. Lumpectomy works because the breast gets adjuvant radiotherapy, whereas prostate FT relies on surveillance and retreatment.

164 Treatment of Locally Advanced Prostate Cancer

LAQ Staging and management of locally advanced (non-organ-confined) CA prostate

Paper answer · P1 · P2 · merged from 2

Asked as: P2 LAQ "Staging of prostate cancer and management of locally advanced (non-organ-confined) prostate cancer; complications of bilateral orchidectomy" (Describe staging of CA prostate and management of non-organ confirmed CA prostate; Desc. mgt optns for locally adv. CaP. Discuss complic. of b/l orchidectomy.; Describe management of LAPC) · P1 LAQ "Describe staging of CA prostate and management of nonorgan confined CA prostate"

Read as: staging of prostate cancer, and management of non-organ-confined (locally advanced and node-positive, non-metastatic) disease; describe the management options for locally advanced carcinoma prostate, and discuss the complications of bilateral orchidectomy. LAPC is read as locally advanced prostate cancer.

Introduction and definition

- **Locally advanced CaP (carcinoma prostate) / LAPC: cT3–4, N0 or N1 (N+), M0:** extracapsular extension (cT3a), seminal vesicle invasion (cT3b), fixed or invading adjacent organs (cT4), ± pelvic nodes.
- At least 10% of new prostate cancers are locally advanced; these men contribute a disproportionate share of cancer deaths.
- Non-organ-confined disease has a high chance of micrometastases. Any single treatment carries a high recurrence risk (single-modality treatment usually fails), so **multimodal (combined) therapy**, local therapy plus systemic ADT (androgen deprivation therapy), is the rule.
- In the PSA (prostate-specific antigen) era, most "high-risk" disease is defined by grade and PSA rather than palpation.
- Untreated, 15-year disease-specific survival is 57% for T3–T4 vs 81% for T1–T2 (Swedish cohort). Untreated, Gleason 8–10 disease carries high cancer mortality, so fit men need treatment.

Staging: TNM (tumour, node, metastasis) 2017 (AJCC [American Joint Committee on Cancer] 8th edition)

Stage	Definition
T1a / T1b	Not palpable; incidental at TURP (transurethral resection of prostate): ≤5% / >5% of resected tissue
T1c	Not palpable; found on needle biopsy (for raised PSA)
T2a / T2b / T2c	Palpable, confined: ≤ half of one lobe / > half of one lobe / both lobes
T3a	Extraprostatic extension (unilateral or bilateral), including microscopic bladder neck invasion
T3b	**Seminal vesicle invasion** (not T4)
T4	Fixed or invading external sphincter, rectum, bladder, levators, pelvic wall
N0 / N1	No nodes / regional (pelvic) nodes

Stage	Definition
M1a / M1b / M1c	Non-regional nodes / bone / other sites ± bone

- **Traps:** seminal vesicle invasion is T3b, not T4; non-regional nodes are M1a.
- **Prognostic stage groups** combine T, N, M, PSA and grade group: stage III = T3–4, PSA ≥20 ng/mL or grade group 5; IVA = N1; IVB = M1. pT2 is not subdivided.
- **Grade group (ISUP, International Society of Urological Pathology):** GG1 ≤6; GG2 3+4; GG3 4+3; GG4 8; GG5 9–10.

Risk groups

System	High risk	Very high / locally advanced
D'Amico	T2c, PSA >20 or Gleason 8–10 (any one)	–
NCCN (National Comprehensive Cancer Network)	cT3a, GG4–5 or PSA >20	cT3b–T4, primary Gleason pattern 5, ≥2 high-risk features, or >4 cores GG4–5
EAU (European Association of Urology)	PSA >20 ng/mL, ISUP 4–5 or cT2c	**Locally advanced: cT3–4 or cN+ (cN1), any PSA, any grade**

- Nomograms (CAPRA [Cancer of the Prostate Risk Assessment] score, Partin tables, Kattan nomograms) and genomic classifiers (Decipher) refine risk.

Staging work-up (evaluation)

- **DRE (digital rectal examination)** sets clinical T stage but under- and over-stages. Biopsy ISUP grade group.
- PSA predicts extent: organ-confined about 80% if PSA <4, under 50% if >10; nodes in about 20% if PSA >20 ng/mL and about 75% if >50 ng/mL.
- **mpMRI (multiparametric magnetic resonance imaging):** local extent; separates T2 from T3 (extraprostatic extension and seminal vesicle invasion; PI-RADS v2 reporting); best specificity in higher-risk men.
- **Bone scan + CT (computed tomography) or MRI** for high risk, PSA >20 or GG4–5.
- **PSMA (prostate-specific membrane antigen) PET-CT (positron emission tomography)** is more accurate; **proPSMA:** accuracy 92% vs 65% for conventional imaging.
- Assess life expectancy: a fit man in his 70s with high-grade disease will live long enough to die of it.

Management of locally advanced N0 (cT3–4 N0 M0)

A. Radiotherapy (EBRT, external-beam RT) + long-term ADT: the standard

- Dose-escalated IMRT (intensity-modulated RT) with image guidance, ≥78–81 Gy (P1: about 76–78 Gy), with or without pelvic node RT, with or without brachytherapy boost (**ASCENDE-RT:** LDR [low-dose-rate] boost halved biochemical failure, improving biochemical control; overall survival (OS) equal).
- **Why add ADT:** it shrinks the target and is synergistic with radiation: neoadjuvant, concurrent and adjuvant, **18–36 months** for high risk (NCCN; EAU 2–3 years).

Trial	Finding
EORTC 22863	3 years of goserelin with RT improved 5-year OS, 78% vs 62% (other sources: 79% vs 62%)
RTOG 85-31 / 92-02	Long adjuvant ADT better; RTOG 92-02: 28 vs 4 months ADT better on all endpoints except overall OS; OS gain in Gleason 8–10
TROG 96.01	6 months better than 3 months
SPCG-7; MRC PR07/NCIC PR.3	**RT is still needed with ADT:** adding RT to ADT reduced cancer death and gave an overall survival gain: SPCG-7 cancer death RR (relative risk) 0.44; MRC PR07 OS HR (hazard ratio) 0.77 (P1: HR 0.46). ADT alone is inferior
STAMPEDE (high-risk M0)	Adding 2 years of abiraterone: metastasis-free survival HR 0.53
RTOG 0521; GETUG 12	Docetaxel added modest benefit

- **Duration:** NCCN: ADT 18–36 months with EBRT (12–36 with brachytherapy boost); 18 vs 36 months gave similar OS.
- Pelvic nodal RT: RTOG 94-13 inconclusive; POP-RT (Tata Memorial) improved biochemical failure-free survival in high-risk N0.

B. Systemic intensification

- **STAMPEDE (abiraterone, M0):** for N1, or two of cT3–4, Gleason ≥8 or PSA ≥40 ng/mL, **abiraterone + prednisolone for 2 years** with ADT + RT gave metastasis-free survival HR 0.53 and OS HR 0.60. NCCN now lists it for very high risk.
- Docetaxel with RT and ADT (RTOG 0521, OS HR 0.69): not routine.

C. Radical prostatectomy (RP) + extended PLND (pelvic lymph node dissection, ePLND), within multimodal therapy: selected men

- For fit men, life expectancy >10 years, with a completely resectable tumour (mainly cT3a; selected cT3b, limited nodes). Clinical overstaging of cT3 occurs in 7–27%, so cT3 alone is no bar.
- **Advantages:** accurate pathological staging; removes the primary, the source of local morbidity (an untreated prostate causes retention, haematuria and ureteric obstruction that ADT does not prevent); RT kept for salvage.
- RP alone gives disease-free survival in at least half at 8–10 years; disease-specific survival 85–92% at 5 years and 79–82% at 10 years (10-year disease-specific survival about 80–90%), but biochemical recurrence is common, so multimodal care is planned.
- **Neoadjuvant ADT before RP:** clinical downstaging and fewer positive margins, but **no survival or PSA benefit**.
- **Adjuvant vs early salvage RT after RP:** EORTC 22911, SWOG 8794 (metastasis-free survival and OS improved) and ARO 96-02 favour adjuvant RT over observation. RADICALS-RT, RAVES and GETUG-AFU 17 favour **observation with early salvage RT** (at PSA rise) for most men with pT3 or a positive margin. Adjuvant RT is still considered for very high-risk pathology (pT3b, GG4–5 / Gleason 8–10, positive margins). Dose 64–66 Gy; adding short-term ADT to salvage RT improved progression-free survival (GETUG-AFU 16).

D. ADT alone

- Only for men unfit for or declining local therapy. Consider immediate ADT if symptomatic, PSA doubling time is short (<12 months) or PSA is very high (>50 ng/mL); otherwise

deferred ADT is acceptable. Early ADT delays progression; survival gain mainly in aggressive disease. In fit men it is inferior to ADT + RT.

- Options: bilateral orchidectomy, LHRH (luteinising hormone-releasing hormone) agonist or antagonist; bicalutamide 150 mg monotherapy spares sexual interest (harmful in localised disease); intermittent ADT.

E. Others

- Watchful waiting with deferred ADT in elderly comorbid men.
- Cryotherapy and HIFU (high-intensity focused ultrasound): not standard, investigational in high risk.
- **Also know:** TURP or stents for local obstruction; trial enrolment.

Node-positive disease (N1 M0)

Setting	Management
cN1 M0	RT to prostate and pelvic nodes + long-term ADT 2–3 years + abiraterone 2 years (STAMPEDE). RP + ePLND in selected men
pN1 (pN+) after RP	**Immediate ADT** (ECOG 7887: improved overall survival) **or ADT + adjuvant pelvic RT** (beats ADT alone); observation with undetectable PSA and ≤2 nodes

Metastatic (M1), in brief

- ADT plus an **ARPI (androgen receptor pathway inhibitor)** (abiraterone, enzalutamide, apalutamide) or a triplet with docetaxel; prostate RT for low-volume disease (see 2025-Winter Paper 3, Q3b).

Algorithm

```
Non-organ-confined, cT3–4 / cN1, M0 (PSMA PET or CT + bone scan) ├─ Fit, cT3a, resectable → RP + ePLND → early salvage RT ± ADT ├─ Most (cT3–4 or cN1) → EBRT + ADT 18–36 months │ very high risk / N1 → + abiraterone 2 years └─ Unfit / refuses → ADT alone (immediate if symptomatic, or deferred)
```

Complications and local sequelae

- ADT: hot flushes, osteoporosis, metabolic syndrome, cardiovascular risk, loss of libido; give bone protection.
- RT: proctitis, cystitis. RP: incontinence, erectile dysfunction.
- A prostate left in place can cause bladder outlet obstruction (channel TURP), ureteric obstruction (stent or nephrostomy) and haematuria (palliative RT).

Bilateral orchidectomy

- Removes about 95% of testosterone: **castrate level (<20 ng/dL) within 12 hours**. Cheap, no compliance issues, no flare, so useful in impending cord compression. Simple or subcapsular (all Leydig tissue must be removed).
- Versus LHRH agonists: fewer cardiac events, fractures and peripheral arterial disease reported.

Complications of bilateral orchidectomy

A. Surgical complications

- Scrotal haematoma (commonest), wound infection, chronic scrotal or groin pain.

- Psychological: empty scrotum and altered self-image. Prevent with subcapsular or epididymis-sparing technique, or a testicular prosthesis.
- Irreversible: intermittent ADT is impossible.

B. Systemic effects of castration

Effect	Details and management
Hot flushes	Commonest (50–80%); loss of hypothalamic oestrogen signalling; venlafaxine, gabapentin, megestrol or cyproterone if bothersome
Sexual dysfunction	Loss of libido, ED (erectile dysfunction), penile shortening; ED aids help selected men
Osteoporosis, fractures	Fracture risk up 20–45%; baseline DEXA (dual-energy X-ray absorptiometry), FRAX; calcium, vitamin D, exercise; **denosumab 60 mg 6-monthly** or zoledronic acid
Metabolic	Weight gain, fat gain, **sarcopenia**, insulin resistance, diabetes (HR about 1.34), metabolic syndrome
Cardiovascular	Coronary events, sudden cardiac death, thromboembolism
Anaemia	Normocytic; reversible after stopping
Others	Gynaecomastia (≤5%), fatigue, depression, cognitive change

- **Also know:** risk of castration resistance over time; check testosterone with rising PSA.

Follow-up

- PSA every 3–6 months; bone health, metabolic and cardiovascular monitoring on ADT.

Recent advances

- PSMA PET staging; trials adding apalutamide or enzalutamide (ATLAS, ENZARAD); SPCG-15 (RP vs RT + ADT).

Diagrams to draw

- TNM T-stage diagram / sagittal pelvis: T2 confined, T3a extracapsular, T3b seminal vesicle, T4 adjacent organs.
- Subcapsular orchidectomy.

> **Key points to remember:**
> - Seminal vesicle invasion is T3b; non-regional nodes are M1a.
> - Locally advanced = T3–4, N0/N+, M0; treat with combined therapy.
> - RT + long-term ADT (18–36 months; EAU 2–3 years) is the standard; ADT alone is inferior (SPCG-7).
> - EORTC 22863 overall survival 79% vs 62%.
> - STAMPEDE: abiraterone 2 years for N1 or very high-risk M0 (metastasis-free survival HR 0.53).
> - Neoadjuvant ADT before RP: no survival gain.
> - Early salvage RT preferred over routine adjuvant RT for most men after RP.
> - Orchidectomy: castrate in 12 hours; flushes, osteoporosis (denosumab), metabolic, sexual effects.

Also, from the condensed book (not in the paper answer):

- pN1: hormones plus RT beat hormones alone (OS 74% vs 55%) (§3).
- Bicalutamide 150 mg raised death in localised disease (HR 1.23) (§5).

1 · Definition and natural history

- The working definition is **T3–4, N0 or N1, M0**, and it calls for **multimodal therapy**.
- NCCN high risk is cT3a, PSA >20 or grade group (GG) 4–5. Very high risk is cT3b–T4, primary pattern 5, ≥2 high-risk features, or >4 cores of GG 4–5.
- Because of stage migration, high risk now mostly means **high grade**, not a palpable T3.
- **Grade drives death**: untreated Gleason 8–10 had 15-yr cancer mortality of 60–87% before the PSA era.

2 · Staging and risk assessment

- Clinical staging and imaging both **understage**.
- NCCN orders a bone scan and CT or MRI for cT3–T4, PSA >20, GG 4–5 or primary pattern 5; PSMA PET may substitute.
- In **proPSMA**, PSMA PET accuracy was **92% vs 65%**.

3 · Radical prostatectomy in high-risk disease

- **Clinical cT3 is overstaged in 7–27%**, and >⅓ of high-risk specimens are confined, so do not exclude men on clinical stage alone.
- NCCN advises RP with extended pelvic node dissection (**ePLND**) only if the tumour is **completely excisable**; otherwise give androgen deprivation (ADT, "AD" in trial reports) with radiotherapy (RT).
- In cT3, DSS after RP is **~80–90% at 10 yr**.

> **KEY POINT**
>
> Operative principle: a **positive surgical margin** carries the highest relative risk of progression (4.37), higher than extracapsular extension (ECE), seminal vesicle invasion (SVI) or nodes, so remove all prostatic tissue regardless of stage.

Pathology after RP	5-yr PSA-free survival	Relative risk of progression
Organ-confined	95%	Reference
ECE	76%	2.2–2.7
SVI	37%	2.6
Node-positive	18%	3.3
Positive margin	—	**4.37** (highest)

Neoadjuvant AD before RP

- **Clinical downstaging occurs in 32–90%, but pathological in only 8–31%**. It has **no** effect on PSA recurrence, DSS or overall survival (OS).
- **CALGB 90203** (AD plus docetaxel before RP) missed its primary endpoint but improved MFS and OS.

Adjuvant and salvage RT after RP

Trial	Population / comparison	Result
EORTC 22911	pT3 and/or positive margin; immediate RT vs observation	Biochemical disease-free survival 74% vs 53%; **benefit only with positive margins**
SWOG 8794	Same	**MFS (HR 0.71) and OS (HR 0.72)** — the only OS-level evidence
ARO 96-02	pT3N0, undetectable PSA	Biochemical HR 0.53
RADICALS-RT, GETUG-AFU 17, RAVES	Adjuvant vs early salvage (64–66 Gy)	**Early salvage preferred**; 5-yr event-free 87–94% in both arms

- **Trap**: very high-risk men (Gleason 8–10, pT3b) were only ~20% of the timing trials, so adjuvant RT may still help them, as may extensive margins.
- Dose is **≥64 Gy**. **GETUG-AFU 16**: **3 months** of AD with salvage RT gave 10-yr progression-free survival 64% vs 49% (HR 0.54) (see Ch 162).

Adjuvant AD and pN1

- In **pT3N0**, adjuvant AD reduces progression but gives **no DSS or OS gain**.
- **ECOG 7887** (nodes found at RP): immediate AD improved OS, DSS and PFS.
- **AD plus RT beats AD alone** in pN1: 10-yr DSS 86% vs 70%, OS 74% vs 55%.

ALGORITHM

1. **pN1 after RP**
2. If favourable (few nodes, Gleason ≤8), then observe with PSA monitoring.
3. If adverse (>3 nodes, Gleason >8), then give AD plus RT (best OS).
4. Avoid AD alone in a fit man.

4 · RT + ADT

- The standard is **dose-escalated RT (>72, ideally ~78–81 Gy) with ADT**, often with pelvic RT (see Ch 162).
- ADT shrinks the target and is synergistic with RT.

Trial	Question	Verdict
RTOG 86-10	4 mo neoadjuvant/concurrent AD, bulky T2–T4	Disease-specific mortality fell from 36% to 23%; **OS unchanged**
TROG 96.01	RT with no, 3 or 6 mo NAD	6 mo lowered all-cause mortality (HR 0.63); 3 mo did not
RTOG 85-31	Indefinite adjuvant goserelin	Better control and MFS; OS gain only in Gleason 8–10
PCS IV	18 vs 36 mo	OS comparable; 18 mo better quality of life
RTOG 94-13	Pelvic vs prostate RT, and neoadjuvant vs adjuvant AD	Early PFS gain lost; whole pelvis with adjuvant AD had worst OS

- **Duration**: intermediate risk 4–6 mo; high risk (NCCN) **18–36 mo** with external beam RT (EBRT), 12–36 mo with a brachytherapy boost.
- **ASCENDE-RT**: a low-dose-rate boost halved biochemical recurrence vs an EBRT boost (HR 2.04), with equal OS.

- **Trap**: AD plus RT benefits often exclude OS. The OS gain is seen with long AD in high risk (EORTC 22863; RTOG 92-02 Gleason 8–10 OS 81% vs 71%).

Adding RT to ADT

- **SPCG-7**: cancer death relative risk **0.44**; overall mortality RR 0.68. **PR.3/PR07**: DSS HR 0.54, and 7-yr OS rose from 66% to 74%. So RT adds **8–10% OS**.

Systemic intensification

Trial	Added agent	Verdict
RTOG 0521	Docetaxel to long-term AD and RT	OS HR 0.69
GETUG 12	Docetaxel and estramustine to AD	Relapse-free 62% vs 50%
RTOG 9902	Paclitaxel, estramustine and etoposide	No benefit; thromboembolism
STAMPEDE (high-risk M0/N1)	**Abiraterone and prednisolone for 2 yr**, added to 3 yr AD and RT	**MFS HR 0.53, OS HR 0.60**

- STAMPEDE eligibility: **N1, or two of Gleason ≥8, cT3–4 and PSA ≥40**. **NCCN 2023 advises RT, ADT and 2 yr abiraterone for very high risk.**

5 · ADT alone and other options

- **MRC**: immediate rather than deferred AD gave fewer cancer deaths, mainly in M0.
- **Bicalutamide 150 mg** improved PFS in locally advanced disease but **raised death in localised disease** (HR 1.23), so never use it at low risk.
- AD alone does not stop local progression, and over half need a TURP.
- **RP vs RT with ADT**: there is no completed randomised trial; **SPCG-15** (cT3N0M0) is ongoing.

ALGORITHM

1. **Locally advanced / high-risk M0**
2. Stage with PSMA PET, or CT and bone scan.
3. If fit and resectable, then RP with ePLND. If pathology is adverse, then early salvage RT (adjuvant if pT3b, Gleason 8–10 or positive margins). If pN1, then observe or give AD plus RT.
4. Alternative: dose-escalated RT (with or without brachytherapy boost and pelvic nodes) plus 18–36 mo ADT.
5. If very high risk or N1, then add 2 yr abiraterone.
6. If unfit for local therapy, then ADT alone.

KEY POINT

Viva: "Why not ADT alone for T3 disease?" SPCG-7 and PR.3 show RT added to ADT halves cancer death and improves OS; AD alone neither cures nor controls the primary, and it carries its own toxicity.

6 · Landmark trials

Trial	Design	Result	Take-home
SPCG-7	ADT with or without RT, mostly T3	Cancer death RR 0.44	Add RT to ADT
NCIC PR.3/MRC PR07	Lifelong ADT with or without RT	OS HR 0.77	Add RT to ADT
EORTC 22863	RT with or without 3 yr goserelin	5-yr OS 62% vs 79%	Long ADT in high risk
RTOG 92-02	4 vs 28 mo ADT	OS gain in Gleason 8–10	Prolong ADT for high grade
STAMPEDE (abiraterone)	ADT and RT, with or without 2 yr abiraterone	MFS HR 0.53, OS HR 0.60	Standard for very high risk/N1
SWOG 8794	Adjuvant RT vs observation	MFS and OS improved	Postoperative RT works
RADICALS-RT / RAVES / GETUG-AFU 17	Adjuvant vs early salvage	No difference	Early salvage for most
ECOG 7887	pN+ immediate vs delayed AD	OS improved	Treat pN+ early
proPSMA	PSMA PET vs conventional imaging	Accuracy 92% vs 65%	PSMA PET for staging

165 Management of Recurrent and Newly Metastatic Prostate Cancer

1 · Biochemical recurrence: concept and definitions

Biochemical recurrence (BCR) is a rise in prostate-specific antigen (PSA) after curative therapy, before clinical recurrence. Clinical recurrence is almost always preceded by it.

- After radical prostatectomy (RP), **¼–⅖** develop BCR within 10 years; only ≈half of higher-risk men with BCR fail clinically.
- At MSK, PSA **≥0.4 ng/mL followed by a further rise** best predicted metastasis.
- **Ultrasensitive PSA** is prognostic, not diagnostic; about half of early values fall on retesting, so repeat it.

Setting	Definition	Logic
Post-RP — AUA	PSA **≥0.2 ng/mL**	No gland remains, so any PSA is suspect
Post-RP — NCCN	Never undetectable, two rises, or PSA ≥0.1	Any of three triggers
Post-RP — EAU	No threshold; risk groups (below)	Kinetics plus grade
After external beam RT (EBRT) ± androgen deprivation therapy (ADT)	**Phoenix**: **nadir + 2 ng/mL**	Validated against metastasis and death
After low-dose-rate (LDR) brachytherapy	≥0.2 may suit better; **<0.2 at 4 years ≈ cure**	The very low nadir means Phoenix may flatter brachytherapy

EAU BCR risk group	After RP	After radiotherapy (RT)
Low risk (needs BOTH)	PSA doubling time (PSADT) **>1 year** AND grade group (GG) <4	Interval to failure **>18 months** AND GG <4
High risk (EITHER)	PSADT ≤1 year OR GG 4–5	Interval ≤18 months OR GG 4–5

> **KEY POINT**
>
> **PSA bounce** after RT is a transient rise of 0.1–0.5 ng/mL at a mean of ≈ **9 months**, followed by a fall. It is **favourable**, so monitor; never salvage on a bounce.

2 · Risk prediction and natural history

- A **positive margin** is the only modifiable BCR predictor.
- **CAPRA** is the pre-operative score and **CAPRA-S** the post-operative one MCQ. Only the MSK nomograms predict 15-year cancer-specific survival.
- **Pound**: untreated, a median **8 years** passes from BCR to metastasis, then **5 years** to death (P2-144).
- After Phoenix failure, metastasis follows at ≈5.4 years. Follow-up is lifelong, because ¼ recur after 5 years.

3 · Imaging at recurrence

P2 *65 yr male underwent RP... PSA has risen — BCR definitions, PSMA PET/mpMRI work-up, salvage RT dose/timing, RTOG 9601, GETUG-AFU 16, SPPORT, STOMP/ORIOLE, EMBARK, Pound prognosis*

- Bone scan and CT are rarely positive below PSA **10 ng/mL**.
- **Prostate-specific membrane antigen (PSMA) PET** is positive in ≈½ at PSA **<0.5**, and finds disease outside the salvage field in ≈1 in 5. Fluciclovine-guided salvage improved control (**EMPIRE-1**).
- Multiparametric MRI (mpMRI) is best for local recurrence. **Biopsy after RT** is mandatory before local salvage.

> **KEY POINT**
>
> Salvage RT works best at the lowest PSA, but imaging improves only as PSA rises, so risk features (grade, margins, kinetics) guide early action.

4 · Salvage after radical prostatectomy

- **Salvage RT (SRT)** is the only curative option (P2-144). Give **≥64–65 Gy** to the fossa.
- Delay costs control: ≈10% relapse-free survival is lost per 0.1 ng/mL in men with ≥2 adverse features.
- No visible lesion on mpMRI predicts SRT failure, because the disease is probably occult metastasis.

5 · Salvage after radiotherapy

Prerequisites are biopsy-proven local recurrence, no metastasis on imaging, excellent performance status, life expectancy ≥5–10 years and a shared decision. Every modality gives ≈**50–70%** control, so selection (local-only disease, low PSA, GG <4) matters more than modality.

Option	Selection / features	Signature morbidity
Salvage RP (open/robotic)	EAU: life expectancy ≥10 years, PSA **<10**, biopsy GG ≤3, N0M0 on imaging, initial cT1–T2.	Incontinence, bladder neck contracture, rectal injury, near-universal impotence
Salvage brachytherapy (LDR ≈ high-dose-rate)	Only after EBRT	Grade ≥3 urinary toxicity ≈1 in 6
Salvage stereotactic body RT (SBRT)	After EBRT	Low toxicity, modest control
Salvage cryotherapy	PSA <10 (ideally <4), PSADT **≥16 months**; a post-cryo nadir <0.2 predicts success	**Urethrorectal fistula** (1–3%), erectile dysfunction; incontinence single-digit with **urethral warming**
Salvage high-intensity focused ultrasound (HIFU)	≥2 years after EBRT	Fistula, osteitis pubis, stricture, incontinence

Recurrence after **focal therapy** is worked up like RT failure.

6 · Non-curative approach to BCR: observation vs ADT

- AUA: observe (or enrol in a trial) after maximal local therapy; give ADT only for higher metastatic risk (PSADT **<12 months** or high grade), preferably intermittent.
- EAU advises observing low-risk BCR, limited life expectancy, or men declining local therapy. Primary ADT for localised disease is **not** recommended.
- **Messing (ECOG)**: immediate ADT for node-positive (pN+) disease after RP improved overall survival (OS), but the trial was small and pre-dated modern imaging.
- Low-volume pN+ with undetectable PSA: **20–40%** are cured by surgery alone.
- Immediate ADT: TOAD OS hazard ratio (HR) 0.55; EORTC 30891 benefit only if PSA >50 or PSADT <12 months.
- **EMBARK** entry: PSADT **≤9 months**, with PSA ≥1 after RP or ≥2 above nadir after RT.

Intermittent vs continuous ADT

- **NCIC PR.7** (rising PSA after RT): intermittent was **non-inferior** (OS 8.8 vs 9.1 years).
- **SWOG 9346** (new metastatic): OS 5.1 vs 5.8 years, HR 1.10 (90% CI 0.99–1.23). This was **inconclusive**; intermittent may be worse here.

7 · Androgen deprivation therapy: biology and agents

Castration removes the ligand, not the receptor, so castration-resistant prostate cancer (CRPC) keeps androgen receptor (AR) signalling alive through AR amplification, ligand-binding-domain mutations, splice variants (**AR-V7**, poor prognosis) and tumour steroid synthesis. **Castrate testosterone: <50 ng/dL**.

SAQ GnRH (LHRH) analogues: agonists and antagonists

Paper answer · P2 · P4 · merged from 2

Asked as: P4 SAQ "GnRH (LHRH) analogues: agonists and antagonists" (GNRH Analogue; LHRH agonist) · P2 SAQ "LHRH analogue"

Read as: GnRH (gonadotrophin-releasing hormone, also LHRH, luteinising hormone-releasing hormone) agonists and antagonists, chiefly in prostate cancer.

Physiology

- **LHRH (GnRH)** is a hypothalamic decapeptide released in **pulses**. Pulsatile GnRH drives pituitary LH (luteinising hormone) and FSH (follicle-stimulating hormone). LH → Leydig cells → testosterone (about 95% testicular; about 95% of total).
- **Continuous** exposure **desensitises and down-regulates** the pituitary receptor. This is the basis of agonist therapy.

GnRH (LHRH) agonists: drugs

Drug	Formulation
Leuprolide	Depot 1, 3, 6 monthly
Goserelin	SC (subcutaneous) implant 1 or 3 monthly
Triptorelin	IM (intramuscular) depot
Histrelin	Implant, 1 year
Buserelin	Nasal or SC

- Substitution at **position 6** of the decapeptide raises potency and half-life; depots last 1 month to 1 year.

Agonists: mechanism

`Agonist → initial LH surge (flare, testosterone up) → receptor down-regulation (1–2 weeks) → LH falls → castrate testosterone in 2–4 weeks`

- Survival **equals orchidectomy**.

Flare phenomenon

- LH rises up to **10-fold for 10–20 days** → bone pain, **spinal cord compression**, urinary retention.
- Dangerous with spinal or weight-bearing metastases.
- **Prevent with an antiandrogen** (e.g. bicalutamide, flutamide) started with or before the agonist, for **21–28 days**.
- Impending cord compression → **antagonist** (degarelix) or orchidectomy instead.

GnRH antagonists

- Immediate competitive receptor block: LH falls 84% in 24 hours, **no flare**, FSH falls.
- **Degarelix** (SC monthly): castrate by day 3 in >95%; injection-site reactions.
- **Relugolix** (oral daily, FDA 2020). **HERO** vs leuprolide: sustained castration 97% vs 89%; faster recovery after stopping; **fewer MACE (major adverse cardiovascular events)** (about 3% vs 6%).
- Preferred for **impending cord compression** or severe bone pain when orchidectomy is unsuitable.
- **Also know:** abarelix restricted (severe allergy); PRONOUNCE (degarelix vs leuprolide, stopped early) showed no MACE difference.

Agonist vs antagonist

Agonist	Antagonist
Flare; needs antiandrogen	No flare; castrate by day 3 (degarelix)
FSH rises	FSH falls
Depot to yearly	Monthly SC or oral (relugolix; HERO)

Castrate level

- **<50 ng/dL** (traditional); **<20 ng/dL** is achievable (orchidectomy median about 15) and may be better (aim **<20 ng/dL**). Check testosterone when PSA (prostate-specific antigen) rises.

Indications in urology

- Metastatic hormone-sensitive prostate cancer (with ARPI [androgen receptor pathway inhibitor] or docetaxel).
- With radiotherapy for intermediate/high-risk and locally advanced disease.
- Recurrent prostate cancer: BCR (biochemical recurrence) with short doubling time; continued in CRPC (castration-resistant prostate cancer).
- **Also know:** intermittent ADT (androgen deprivation therapy) acceptable (PR.7); BPH (benign prostatic hyperplasia) volume reduction (not routine); central precocious puberty;

cryptorchidism (GnRH or hCG [human chorionic gonadotrophin]: low success, not first-line); endometriosis.

Adverse effects

- Hot flushes (50–80%), loss of libido, ED (erectile dysfunction).
- **Osteoporosis:** DEXA (dual-energy X-ray absorptiometry); calcium, vitamin D, denosumab or zoledronic acid.
- Diabetes, cardiovascular events, sarcopenia, anaemia, gynaecomastia (≤5%), cognitive change.
- Slow testosterone recovery after stopping.

Diagram to draw

- Hypothalamic–pituitary–testis axis showing agonist and antagonist sites, and the flare.

> **Key points to remember:**
> - Agonist: surge, then continuous stimulation → down-regulation → castration in 2–4 weeks.
> - Flare: LH 10-fold for 10–20 days; cover with antiandrogen 21–28 days.
> - Antagonist: no flare; degarelix castrate by day 3; relugolix oral (HERO, fewer MACE); preferred in cord compression.
> - Castrate <50 ng/dL; aim <20 ng/dL.

SAQ Maximum / combined androgen blockade

Paper answer · P2 · P4 · merged from 2

Asked as: P2 SAQ "Maximum (combined) androgen blockade and bicalutamide" (Maximum androgen blockade; Bicalutamide) · P4 SAQ "Combined Androgen Blockade."

Definition and rationale

- **MAB (maximum androgen blockade)**, also **CAB (combined androgen blockade)** or total androgen blockade: castration (orchidectomy or LHRH [luteinising hormone-releasing hormone] agonist) **plus a non-steroidal antiandrogen**.
- **Rationale (Labrie):** castration removes about 90–95% of testosterone, but **adrenal androgens** (DHEA, dehydroepiandrosterone; androstenedione) are converted to DHT (dihydrotestosterone) inside the prostate. The antiandrogen blocks these at the AR (androgen receptor).

Antiandrogens

Drug	Dose	Signature toxicity
Flutamide	250 mg three times daily	Diarrhoea, hepatotoxicity
Nilutamide	300 mg, then 150 mg daily	Delayed dark adaptation, pneumonitis
Bicalutamide	**50 mg daily** (with LHRH agent)	Best tolerated; gynaecomastia
Cyproterone (steroidal)	100 mg two to three times daily	Cardiovascular, hepatic; no CAB benefit, not used for CAB now

- Class effects: gynaecomastia, hepatotoxicity, pneumonitis.

Evidence for MAB

- **Crawford (1989):** leuprolide + flutamide improved survival (partly flare control); **Eisenberger (1998):** orchidectomy + flutamide, **no** benefit.
- **PCTCG (Prostate Cancer Trialists' Collaborative Group) meta-analysis, 2000** (27 trials, about 8,000 men): with non-steroidal antiandrogens, 5-year survival improved by only **about 2–3%** (borderline significance); cyproterone gave no benefit (slightly worse).
- Small gain, more cost and toxicity: **AUA (American Urological Association) advises against long-term CAB**.

Current uses of MAB

1. **Flare prevention:** an LHRH agonist raises LH (luteinising hormone) up to 10-fold, and testosterone, for 10–20 days, risking cord compression, pain or obstruction with bone metastases. Start antiandrogen **with or before** the agonist for **21–28 days**. Antagonists (degarelix, relugolix) or orchidectomy cause no flare.
2. **Antiandrogen withdrawal:** on progression, stopping bicalutamide gives a PSA (prostate-specific antigen) fall >50% in **15–30%** for 3.5–5 months (mutant AR turns the drug into an agonist).

Bicalutamide: introduction and mechanism

- Non-steroidal **AR antagonist**, called first- or second-generation. It competes with testosterone and DHT at the ligand-binding domain.
- Given alone, it blocks central feedback, so **LH and testosterone rise** (about 1.5 times); aromatisation to oestradiol causes gynaecomastia; libido is partly preserved.
- **Half-life about 6 days** (once daily); unaffected by age or renal impairment. Most potent and best tolerated of its class.

Uses of bicalutamide

Use	Dose	Evidence
CAB with LHRH agonist, metastatic	50 mg daily	Small survival gain (2–3%, PCTCG)
Flare prevention	50 mg, with or before the agonist, 2–4 weeks	Main use today
Monotherapy, locally advanced	150 mg daily	**EPC (Early Prostate Cancer) programme:** better progression-free survival; **harm in localised disease** (death HR [hazard ratio] 1.23)
With salvage radiotherapy	150 mg for 2 years	**RTOG 9601:** 12-year overall survival 76.3% vs 71.3%
CRPC (castration-resistant prostate cancer)	50 mg	Second-line step; largely replaced by newer agents

- 150 mg monotherapy roughly equals castration in locally advanced disease, with better sexual interest; not FDA (US Food and Drug Administration) approved alone, but used in some countries.
- **Also know:** stuttering priapism; feminising hormone therapy; studied in BPH (benign prostatic hyperplasia).

Adverse effects of bicalutamide

- **Gynaecomastia and breast pain** (two-thirds at 150 mg): prevent with breast radiotherapy (10–15 Gy single fraction) or tamoxifen 10 mg daily.
- **Hepatotoxicity** (monitor liver tests); rare **interstitial pneumonitis**; erectile dysfunction; fewer hot flushes than castration.

Modern intensification (replaces CAB in mHSPC, metastatic hormone-sensitive prostate cancer)

Trial	Addition to ADT (androgen deprivation therapy)	Result
LATITUDE, STAMPEDE (arm G)	Abiraterone + prednisone	OS (overall survival) HR 0.62 / 0.63
ENZAMET	Enzalutamide (vs ADT + older antiandrogen, a CAB control arm)	OS HR 0.67; enzalutamide beat CAB
ARCHES	Enzalutamide	Radiographic PFS (progression-free survival) HR 0.39
TITAN	Apalutamide	OS HR 0.67
PEACE-1, ARASENS	**Triplet**: + docetaxel + abiraterone or darolutamide	OS benefit (ARASENS HR 0.68)

- Docetaxel is also an option with ADT.
- **Comparison:** enzalutamide beat bicalutamide in CRPC (TERRAIN, STRIVE) and in **ENZAMET**.

Diagrams to draw

- Testicular and adrenal androgen sources with LHRH agonist and antiandrogen sites of action.
- Hypothalamus–pituitary–testis axis with bicalutamide blocking the AR.

Key points to remember:

- MAB/CAB = castration + non-steroidal antiandrogen (blocks adrenal androgens).
- PCTCG 2000: only 2–3% 5-year survival gain.
- Main use now: flare cover for 21–28 days.
- ARPIs (androgen receptor pathway inhibitors) or triplets now standard in mHSPC; newer AR pathway inhibitors beat CAB (ENZAMET).
- Bicalutamide half-life about 6 days; 50 mg with LHRH agonist, 150 mg monotherapy.
- Monotherapy raises testosterone; gynaecomastia two-thirds.
- EPC: harm in localised disease; RTOG 9601: survival gain with salvage radiotherapy.
- Withdrawal response 15–30%.

Method	Key facts
Bilateral orchiectomy	Castrate within hours. Fewer cardiac events and fractures than agonists with equal survival, so offer it for lifelong ADT
LHRH agonists / antagonists	See P4-103. HERO: castrate by day 4 in 56% vs 0%
Cyproterone acetate	Inferior to goserelin; cardiovascular events up to 10%; used 50–100 mg/day for hot flushes

Method	Key facts
AR-targeting agents (ARTAs): enzalutamide 160 mg, apalutamide 240 mg, darolutamide 600 mg twice daily	Seizures with enzalutamide; rash and hypothyroidism with apalutamide; darolutamide has the fewest effects
Abiraterone (selective CYP17), ketoconazole (non-selective)	Give abiraterone with prednisone to prevent ACTH-driven mineralocorticoid excess. Ketoconazole castrates within 4 h: an emergency option

KEY POINT

Flare matters with spinal or weight-bearing bone metastases: give an antiandrogen with the agonist for **21–28 days**. For **impending cord compression**, use an **antagonist or orchiectomy** (or ketoconazole). **Antiandrogen withdrawal**: 15–30% respond with bicalutamide and 0–10% with ARTAs, with no OS gain.

8 · ADT side effects and their management

Effect	Key facts	Management
Osteoporosis / fracture	Loss of peripheral oestrogen; fracture risk rises 20–45%	Baseline DEXA and FRAX. For all: calcium 1000–1200 mg and vitamin D 400–1000 IU/day, exercise, stop smoking. Give a drug if T-score is −1 to −2.5 with 10-year hip fracture risk **>3%** or major fracture **>20%** (or osteoporosis): denosumab 60 mg SC 6-monthly, zoledronic acid 4–5 mg IV yearly or alendronate 70 mg weekly. Repeat DEXA at 1 year
Hot flushes (commonest, 50–80%)	Hypothalamic oestrogen loss	Oestrogens best; megestrol, cyproterone, venlafaxine, gabapentin. Clonidine is ineffective
Metabolic	Fat rises with sarcopenia; diabetes (agonist HR 1.44)	Exercise; monitor glucose and lipids
Cardiovascular	Agonists raise coronary events, sudden death and thromboembolism	Risk-factor control; orchiectomy or antagonist favoured
Anaemia	Normocytic; pre-ADT Hb <12 g/dL predicts shorter survival	Reversible ≈1 year after stopping
Gynaecomastia	≈½ with bicalutamide 150 mg or enzalutamide; ≤5% with castration	Prophylactic breast RT 10–15 Gy or tamoxifen

Follow-up on ADT

Check PSA and bloods 3–6-monthly. **If PSA rises, check testosterone first.** A rising PSA with castrate testosterone means **CRPC** (see Ch 166), so restage. The germline HSD3B1 (1245A>C) variant predicts ADT resistance MCQ.

9 · Metastatic hormone-sensitive prostate cancer (mHSPC)

LAQ Metastatic prostate cancer, including newly diagnosed and bone metastases

Paper answer · P2 · P3 · merged from 2

Asked as: P2 LAQ "Metastatic prostate cancer with bone metastases (retention and bone pain; bone mets; zoledronic acid)" (Treatment of metastatic CA prostate in 60 yr male with retention of urine and bone pain; Management of bone mets in prostatic CA; also SAQ: Zolendronic acid) · P3 SAQ "Management of newly Metastatic prostate cancer"

Zolendronic acid: read as zoledronic acid.

Introduction

- Bone is the commonest metastatic site; visceral metastasis occurs in <20%.
- Aims: control the cancer, relieve pain, **prevent SREs (skeletal-related events)** and treat emergencies (cord compression, fracture).
- **Retention + bone pain case:** this is newly diagnosed (synchronous) **mHSPC (metastatic hormone-sensitive prostate cancer)** with two urgent problems: an obstructed outlet and painful bone metastases. Treat the emergencies first, start castration safely (no flare), then intensify systemic therapy.

Definitions

- **mHSPC:** no prior hormones. **Synchronous** (at diagnosis) or **metachronous** (after local therapy).
- **CHAARTED high volume:** ≥4 bone metastases with ≥1 outside the axial skeleton, or any visceral metastasis.
- **LATITUDE high risk:** ≥2 of Gleason ≥8, ≥3 bone lesions, visceral metastasis.

Pathophysiology of bone metastases

- Tumour cells lodge in axial marrow, adhering through integrins and collagen receptors.
- Metastases are mainly **osteoblastic (sclerotic)**, but resorption also rises. Tumour cytokines induce **RANKL (receptor activator of nuclear factor-κB ligand)**, which activates osteoclasts; resorbed bone releases growth factors that feed the tumour (**vicious cycle**). ADT (androgen deprivation therapy) adds osteoporosis through loss of oestrogen.
- **SREs:** pathological fracture, spinal cord compression, need for radiotherapy or surgery to bone. Marrow replacement causes anaemia. Hypercalcaemia is rare and suggests neuroendocrine disease.

Evaluation (immediate assessment and work-up)

- **Clinical:** pain, neurological signs, performance status. **Ask and examine for cord compression:** back pain, leg weakness, sensory level, bowel symptoms. DRE (digital rectal examination).
- **Bloods:** PSA (prostate-specific antigen), ALP (alkaline phosphatase), calcium, creatinine, haemoglobin, testosterone.
- **Imaging:** ultrasound for hydronephrosis and residual; **bone scan and CT (computed tomography)** define volume; PSMA (prostate-specific membrane antigen) PET (positron emission tomography) finds more and is more sensitive, but trials used conventional imaging. **Plain X-ray or CT of painful weight-bearing sites** (lytic lesion, impending

fracture). **Whole-spine MRI (magnetic resonance imaging)** if any back pain or neurological sign.
- **Histology:** prostate biopsy (or TURP [transurethral resection of the prostate] chips).
- **Germline (and somatic) genetic testing** for all men with metastatic disease (>10% have DNA-repair mutations).
- **Volume (CHAARTED):** see definitions above.

Step 1: Relieve the retention

- **Urethral catheter** (suprapubic if urethral fails). Watch for post-obstructive diuresis.
- Persisting renal impairment with hydronephrosis means ureteric obstruction by tumour at the trigone: **nephrostomy or stent**.
- ADT shrinks the prostate, so try **TWOC (trial without catheter)** after a few months of ADT.
- If TWOC fails or symptoms persist: **channel (palliative) TURP**. ADT alone does not prevent local progression (older series: about a third needed TURP within 60 days of orchidectomy).

Step 2: Systemic treatment, hormone-sensitive (mHSPC): start ADT at once, without flare

- **ADT is the backbone, never alone** (orchidectomy, LHRH [luteinising hormone-releasing hormone] agonist or antagonist):

Option	Point
Bilateral orchidectomy	Castrate within hours; best for impending cord compression
GnRH (gonadotropin-releasing hormone) antagonist: degarelix 240 mg subcutaneously, then 80 mg monthly; oral relugolix	**No flare**; castrate by day 3 in >95% (degarelix); for impending cord compression
LHRH agonist + antiandrogen	Agonist raises LH (luteinising hormone) up to 10-fold, raising testosterone, for 10–20 days (**flare**). Give bicalutamide 50 mg daily with or before the agonist, for 21–28 days

- **Why avoid flare:** the testosterone surge can worsen bone pain, retention and cord compression in spinal metastases. Cover it with an antiandrogen for 21–28 days, or use an antagonist (degarelix) or orchidectomy.
- Target castrate testosterone **<50 ng/dL**. Antiandrogen alone is not recommended.

Intensified systemic therapy in mHSPC (doublets and triplets)

- ADT alone is no longer standard. Classify volume (CHAARTED) and risk (LATITUDE).

Add to ADT	Trial	Overall survival HR (hazard ratio)
Docetaxel 75 mg/m², 6 cycles	**CHAARTED** (benefit in high volume only); STAMPEDE	0.61; 0.78
Abiraterone 1000 mg + prednisone 5 mg	**LATITUDE**; STAMPEDE arm G	0.62; 0.63
Enzalutamide (ARPI, androgen receptor pathway inhibitor)	ARCHES (radiographic PFS [progression-free survival]); ENZAMET	0.67 (ENZAMET)

Add to ADT	Trial	Overall survival HR (hazard ratio)
Apalutamide (ARPI)	**TITAN**	0.67
Triplet: docetaxel 75 mg/m² × 6 + abiraterone	**PEACE-1** (synchronous)	0.75 (with docetaxel)
Triplet: docetaxel + darolutamide	**ARASENS**	0.68

- Hormonal agents work **regardless of volume**; docetaxel helps only high volume (GETUG-AFU 15 negative), so it is now given within a triplet. Triplets suit chemotherapy-fit men, especially high volume.
- *Guideline note:* the chapter lists ADT plus docetaxel as an option; current EAU (European Association of Urology) guidance gives docetaxel only within a triplet.

Local and metastasis-directed therapy in mHSPC

- **RT (radiotherapy) to the prostate in low-volume synchronous disease only:** STAMPEDE arm H, survival HR 0.68 (3-year 81% vs 73%); it also helps local control. HORRAD negative overall.
- Cytoreductive prostatectomy: trials only.
- **MDT (metastasis-directed therapy)** / **SBRT (stereotactic body RT)** for oligometastases (STOMP, ORIOLE) delays ADT; SBRT also for spinal re-irradiation.

Castration-resistant disease (mCRPC)

- ARPIs, docetaxel then cabazitaxel, PARP (poly-ADP-ribose polymerase) inhibitors for HRR (homologous recombination repair) mutations.
- **Radium-223** (alpha emitter, path <100 µm, spares marrow): for **symptomatic bone-only** disease (licensed only in castration-resistant disease); 50 kBq/kg every 4 weeks × 6. **ALSYMPCA:** overall survival 14.9 vs 11.3 months; delayed SREs. **Never with abiraterone** (more fractures and deaths).
- **^{177}Lu-PSMA-617** (beta emitter, PSMA-avid disease at any site): after ARPI and taxane; **VISION:** survival 15.3 vs 11.3 months.

Step 3: Bone pain

- **Analgesia** by the WHO (World Health Organization) analgesic ladder; neuropathic pain needs gabapentinoids or tricyclics; steroids for diffuse pain.
- **Focal pain: EBRT (external beam radiotherapy) single 8 Gy fraction.** Pain relief equals multiple fractions (30 Gy in 10 fractions, RTOG 9714); retreatment is commoner.
- X-ray or CT of painful weight-bearing sites to exclude lytic lesions and impending fracture; **orthopaedic fixation** then RT.
- **Diffuse pain:** steroids, wide-field RT; radiopharmaceuticals (strontium-89, samarium-153 relieve pain only; radium-223 adds survival, licensed only in castration-resistant disease).

Bone-targeted agents

Feature	Zoledronic acid	Denosumab
Mechanism / target	Bisphosphonate: inhibits osteoclast enzyme	Antibody against RANKL
SRE dose (mCRPC) / route	4 mg IV every 4 weeks, adjusted for creatinine clearance	120 mg SC (subcutaneous) every 4 weeks

Feature	Zoledronic acid	Denosumab
Renal adjustment	Yes	No
Evidence	Fewer SREs than placebo	Time to first SRE 20.7 vs 17.1 months vs zoledronate
Toxicity	Fever, myalgia, renal impairment, hypocalcaemia, ONJ	Hypocalcaemia, hypophosphataemia, ONJ 2–4%

- **Neither improves survival**; they protect the skeleton only.
- **In mHSPC, SRE doses are not indicated** (no benefit: CALGB 90202, STAMPEDE). SRE doses (zoledronic acid 4 mg or denosumab 120 mg 4-weekly) are for CRPC (castration-resistant prostate cancer). Use osteoporosis doses for ADT bone loss: denosumab 60 mg 6-monthly or zoledronic acid yearly, by DEXA (dual-energy X-ray absorptiometry) and FRAX.
- **ONJ (osteonecrosis of the jaw) prevention:** dental check before starting; calcium 1000 mg and vitamin D 800 IU daily.
- Give a bone agent with radium-223 or any ARPI.

Zoledronic acid (in detail)

- **Introduction:** a potent IV (intravenous) **nitrogen-containing bisphosphonate**.
- **Mechanism:**
- Binds hydroxyapatite at sites of active bone turnover and is taken up by **osteoclasts**.
- Inhibits **farnesyl pyrophosphate synthase** (mevalonate pathway), causing osteoclast dysfunction and apoptosis, so resorption falls.
- Breaks the "vicious cycle".
- Excreted unchanged by the kidneys.

Indication	Dose
SRE prevention, mCRPC (metastatic castration-resistant prostate cancer) and other bone metastases (RCC [renal cell carcinoma], bladder)	**4 mg IV over ≥15 min every 4 weeks**, adjusted for creatinine clearance
ADT bone loss (by DEXA and FRAX)	4–5 mg IV yearly
Hypercalcaemia of malignancy	4 mg IV with saline hydration

- 12-weekly dosing after the first doses is non-inferior to 4-weekly (CALGB 70604).
- **Not indicated at SRE doses in mHSPC:** no benefit (CALGB 90202, STAMPEDE).
- **Evidence: Zometa 039 trial (mCRPC):** fewer SREs than placebo (about 33% vs 44%). **Denosumab vs zoledronate (1904 men, mCRPC):** time to first SRE 20.7 vs 17.1 months. **Neither improves survival.**
- **Precautions and monitoring:** creatinine before every dose; avoid if creatinine clearance <30 mL/min. **Dental review before starting**; avoid invasive dental work during therapy. Calcium and vitamin D supplements.
- **Adverse effects: acute-phase reaction** (fever, myalgia) after the first infusion; nephrotoxicity, hypocalcaemia, hypophosphataemia; **ONJ**, atypical femoral fracture; ocular inflammation (rare).

Emergencies

Spinal cord compression: watch constantly (every focal back pain is cord compression until proven otherwise): `Back pain ± weakness, sensory level, sphincter change →`

dexamethasone 10 mg IV load, then 4–10 mg 6–12-hourly → urgent whole-spine MRI → decompressive surgery + RT (unstable spine, progression on RT, prior RT) or RT alone → orchidectomy or antagonist if hormone-naive (never agonist alone)

- Surgery plus RT beat RT alone for walking ability (**Patchell, 2005**).

Pathological or impending fracture: orthopaedic fixation then RT. **Mirels score** ≥9 (site, pain, lytic type, size) favours prophylactic fixation.

Algorithms

Retention + bone pain + metastases ├─ Catheter ± nephrostomy; cord signs? → dexamethasone → MRI → RT ± surgery ├─ Orchidectomy / antagonist / agonist + antiandrogen ├─ Analgesia + 8 Gy single-fraction RT; bone protection ├─ Add ARPI or triplet; prostate RT if low volume └─ TWOC after ADT → channel TURP if it fails

- Newly diagnosed mHSPC: split by volume and fitness to doublet or triplet ± prostate RT.

Follow-up

- PSA, ALP, liver tests, calcium, renal function and blood count every 3–6 months; testosterone if PSA rises; new scan if symptoms or PSA rise. Rising PSA with castrate testosterone = CRPC.
- PSA nadir ≤0.2 ng/mL at 7 months predicts long survival (SWOG 9346).
- PCWG3 (Prostate Cancer Working Group 3): bone scan progression needs ≥2 new lesions, confirmed later (early "flare" misleads).
- Manage ADT effects: bone loss (DEXA), diabetes, cardiovascular risk, hot flushes.
- **Also know:** stent or nephrostomy for obstruction; marrow-failure anaemia; hospice care.

Diagrams to draw

- The retention + bone pain flowchart above.
- Vicious cycle: tumour → RANKL → osteoclast → growth factors → tumour, with the sites blocked by denosumab and zoledronate.
- Algorithm splitting newly metastatic disease by volume and fitness to doublet or triplet ± prostate RT.

Key points to remember:

- ADT alone is no longer standard: ADT plus abiraterone or an ARPI, or a triplet (PEACE-1, ARASENS), for every fit man.
- CHAARTED volume vs LATITUDE risk definitions.
- Prostate RT only for low-volume synchronous disease (STAMPEDE arm H).
- Never start an LHRH agonist alone with spinal metastases: antiandrogen cover, antagonist or orchidectomy.
- Single 8 Gy fraction gives the same pain relief for focal pain; focal back pain = cord compression until MRI proves otherwise; dexamethasone and MRI.
- Denosumab 120 mg or zoledronate 4 mg IV 4-weekly prevent SREs in mCRPC, not in mHSPC; adjust zoledronate for creatinine clearance; no survival benefit.
- Dental check first (ONJ); calcium and vitamin D.
- Denosumab delays SREs longer (20.7 vs 17.1 months).
- Radium-223 (ALSYMPCA) for symptomatic bone-only mCRPC; never with abiraterone.

Also, from the condensed book (not in the paper answer):

- **SWOG 9346 PSA nadir at 7 months**: ≤0.2 gives median OS **75** months; 0.2–4 gives 44; >4 gives 13.
- Metachronous low volume does best (median OS ≈92 vs ≈43 months for synchronous high volume).
- Genes to test: germline BRCA1/2, ATM, PALB2, CHEK2, HOXB13, mismatch repair; somatic CDK12, MSI-H.

Definition	Criteria	Use
CHAARTED high volume	**≥4 bone metastases with ≥1 outside the axial skeleton**, OR any visceral metastasis (no grade criterion)	Docetaxel benefit
LATITUDE high risk	**≥2 of**: Gleason ≥8, ≥3 bone lesions, visceral	Abiraterone entry

- **ADT is the floor, never the ceiling**; long-term combined blockade is not recommended.
- Docetaxel is **75 mg/m² 3-weekly × 6** (meta-analysis HR 0.77).
- Observe only if asymptomatic with life expectancy <5 years. Cytoreductive prostatectomy is for trials only.
- **Metastasis-directed therapy (MDT)**: oligometastases are **≤3 sites** on PET, treated by SBRT or salvage node dissection.

10 · Landmark trials

Trial	Design	Result	Take-home
RADICALS, RAVES, GETUG-AFU 17 (ARTISTIC)	Adjuvant vs early salvage RT (PSA 0.1–0.2)	Equivalent; ≥60% spared RT	Early salvage preferred
RTOG 0534 (SPPORT)	Bed vs bed plus 4–6 months ADT vs pelvis plus ADT	5-year freedom from progression 70.9 / 81.3 / **87.4%**	Add ADT and pelvic nodes (not OS)

Trial	Design	Result	Take-home
RTOG 9601	SRT ± 2 years bicalutamide 150 mg	12-year OS 76.3 vs 71.3%	OS benefit
GETUG-AFU 16	SRT ± 6 months goserelin	10-year BCR-free 64 vs 49% (HR 0.54); MFS HR 0.73; OS unchanged	Short ADT: no OS gain
EMBARK	High-risk BCR; leuprolide ± enzalutamide vs enzalutamide	MFS 87.3% (combination) / 80.0% (enzalutamide) / 71.4% (leuprolide); HR 0.42 and 0.63	Enzalutamide for high-risk non-metastatic HSPC
CHAARTED (ECOG 3805)	ADT ± docetaxel	OS 57.6 vs 44.0 months (HR 0.61); high volume **49.2 vs 32.2** months	High volume only
GETUG-AFU 15	ADT ± docetaxel	OS HR 0.88, not significant	Negative docetaxel trial
STAMPEDE (docetaxel; arm G abiraterone)	Multi-arm; metastatic and high-risk M0	Docetaxel OS HR 0.78; abiraterone OS HR 0.63, failure-free survival HR 0.29	Both work
LATITUDE	High-risk mHSPC; ADT ± abiraterone	OS HR 0.62; rPFS 33.0 vs 14.8 months	Abiraterone in high risk
ARCHES	ADT ± enzalutamide	rPFS HR 0.39	Volume-independent
ENZAMET	ADT plus enzalutamide vs ADT plus older antiandrogen	OS HR 0.67	Control was not placebo
TITAN	ADT ± apalutamide	rPFS HR 0.48; OS HR 0.67	Volume-independent
PEACE-1	Synchronous; 2×2, ± abiraterone and prostate RT	All: rPFS HR 0.54, OS HR 0.82; with docetaxel: rPFS 0.50, OS 0.75	Triplet with abiraterone
ARASENS	ADT and docetaxel, ± darolutamide	OS HR 0.68	Darolutamide in mHSPC only with docetaxel
STAMPEDE arm H	ADT ± prostate RT	Overall HR 0.92; low volume HR 0.68 (3-year OS 81 vs 73%)	RT to prostate in low volume

166 Management of Castration-Resistant Prostate Cancer

1 · Definition and biology

CRPC means progression (biochemical, radiographic or clinical) despite castrate testosterone **<50 ng/dL**. On androgen deprivation therapy (ADT) alone, metastatic overall survival (OS) was historically 24–36 months.

LAQ Castration-resistant prostate cancer (CRPC): definition and treatment

Paper answer · P2

Asked as:

- CRPC in Ca prostate. — *Summer 2020 Paper II Q3a (SAQ)*
- Discuss treatment of CRPC — *Practice Set 6 Paper II Q1 (LAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(also answered in 2020-Summer Paper 2, Q3a)

Definition (EAU [European Association of Urology])

- CRPC (castration-resistant prostate cancer): castrate testosterone **<50 ng/dL (1.7 nmol/L)** plus either:
- **Biochemical progression:** three consecutive PSA (prostate-specific antigen) rises 1 week apart, two of them 50% above nadir, with PSA >2 ng/mL; or
- **Radiological progression:** ≥2 new bone lesions, or soft-tissue progression by RECIST (Response Evaluation Criteria in Solid Tumours).

Why it happens: the AR still drives growth

- Castration-resistant is **not** androgen-independent, so treatment first targets the AR (androgen receptor) more deeply.
- Mechanisms: AR overexpression and amplification; AR mutation (activated by other steroids); **splice variants lacking the ligand-binding domain**; adrenal and intratumoural androgen synthesis (CYP17 overexpressed); bypass pathways (PI3K/AKT, PTEN loss in ~40%).
- **Lineage plasticity:** neuroendocrine change with low PSA, visceral and lytic disease (visceral disease).

Evaluation

- Testosterone (exclude non-compliance), PSA, **PSADT (PSA doubling time)**; CT and bone scan; PSMA (prostate-specific membrane antigen) PET increasingly.
- Prognosis: performance status, pain, haemoglobin, ALP (alkaline phosphatase), LDH (lactate dehydrogenase), visceral disease.
- **Germline testing** in all metastatic disease; somatic sequencing of tumour tissue (germline and tumour testing) for HRR (homologous recombination repair) genes such as BRCA2, and MSI (microsatellite instability).
- Biopsy a metastasis if disease progresses fast with a low PSA (neuroendocrine change).

- PCWG3 (Prostate Cancer Clinical Trials Working Group 3): early bone-scan lesions may be **flare**; re-baseline at 12–16 weeks, then progression = ≥2 new lesions.

Clinical states model (PCWG3): `Localised → rising PSA → mCSPC (castration-sensitive) ↓ (on ADT) nmCRPC → mCRPC 1st line → 2nd line → 3rd line`

General principles of management

1. **Continue ADT (androgen deprivation therapy) lifelong**, including during chemotherapy; try antiandrogen withdrawal.
2. **Antiandrogen withdrawal** (e.g. bicalutamide): watch PSA 4–8 weeks; brief responses in a minority.
3. Choose by prior therapy, symptoms, visceral disease, genomics and comorbidity.
4. **Sequencing: switch mechanism**; a second ARPI rarely works (cross-resistance). Cross-resistance means a second ARPI (AR-pathway inhibitor) helps only 15–30%.

nmCRPC (non-metastatic)

- **PSADT ≤10 months:** add an ARPI. Each improved MFS (metastasis-free survival) and overall survival (improve metastasis-free and overall survival):
- Apalutamide 240 mg (**SPARTAN**: MFS HR 0.28); enzalutamide 160 mg (**PROSPER**); darolutamide 300 mg twice daily (**ARAMIS**: MFS 40.4 vs 18.4 months).
- Slower PSADT: continue ADT and monitor.

mCRPC: life-prolonging options

Class	Agent and dose	Key trial (overall survival)
ARPI (AR-pathway inhibitor)	**Abiraterone** 1000 mg fasting + prednisone	COU-AA-301: 14.8 vs 10.9 months; COU-AA-302 (pre-chemotherapy)
	Enzalutamide 160 mg	AFFIRM: 18.4 vs 13.6 months; PREVAIL
Taxane	**Docetaxel** 75 mg/m² 3-weekly + prednisone	TAX 327: 18.9 vs 16.4 months (mitoxantrone)
	Cabazitaxel 20–25 mg/m² after docetaxel	TROPIC: 15.1 vs 12.7; CARD: beats a second ARPI
Radioligand (mCRPC)	**Radium-223** 50 kBq/kg 4-weekly ×6 (symptomatic bone-only, no visceral)	ALSYMPCA: 14.9 vs 11.3 months
	^{177}Lu-PSMA-617 (PSMA-positive, after ARPI + taxane)	VISION: 15.3 vs 11.3 months
PARP (poly-ADP-ribose polymerase) inhibitor	Olaparib (HRR genes); rucaparib (BRCA)	PROfound; TRITON3
	Olaparib + abiraterone; niraparib + abiraterone; talazoparib + enzalutamide (first line)	PROpel; MAGNITUDE; TALAPRO-2
Immunotherapy	**Sipuleucel-T** (asymptomatic, no visceral disease)	IMPACT: 25.8 vs 21.7 months
	Pembrolizumab for MSI-high/dMMR (mismatch repair-deficient), 2–5%	Tumour-agnostic approval

- **Abiraterone** blocks CYP17, which also lowers cortisol. ACTH (adrenocorticotropic hormone) then rises and causes mineralocorticoid excess (hypokalaemia, hypertension, oedema), so prednisone is co-prescribed. Never combine with radium-223 (more deaths and fractures).
- **Enzalutamide:** seizures (~1%), fatigue, falls. Apalutamide: rash, hypothyroidism.
- **PARP inhibitors** cause synthetic lethality in HRR-deficient cells; benefit is greatest with BRCA2 and weak with ATM.

Sequencing by prior mCSPC therapy

Prior therapy	Next option
Docetaxel only	ARPI
ARPI only	Docetaxel, radium-223, sipuleucel-T
Both	Cabazitaxel, ^{177}Lu-PSMA, PARP inhibitor if HRR mutation

Neuroendocrine/aggressive variant

- Neuroendocrine variant: low PSA, liver or brain metastases, lytic bone lesions with hypercalcaemia. Treat with **platinum + etoposide** (or carboplatin + taxane) ± radiotherapy; survival <12 months.

Supportive and palliative care

- **Bone health:** denosumab 120 mg SC or zoledronic acid 4 mg IV (renal dose adjustment) every 4 weeks against skeletal events, with calcium and vitamin D. Denosumab delayed the first SRE (skeletal-related event) longer (20.7 vs 17.1 months); neither prolongs life. Risk: osteonecrosis of the jaw.
- Focal bone pain: external-beam radiotherapy (radiotherapy for bone pain).
- **Cord compression** needs high-dose dexamethasone → urgent MRI → radiotherapy ± decompression (Patchell trial: surgery + radiotherapy better).
- Ureteric obstruction: stent or nephrostomy. Anaemia; hospice care.

Recent advances

- PSMA-ligand therapy earlier (PSMAfore); AR degraders (ARV-110); AKT inhibition in PTEN loss; bipolar androgen therapy; antibody–drug conjugates.

Diagram to draw: clinical states: rising PSA → nmCRPC → mCRPC first, second, third line.

Diagram to draw: clinical states model with drugs placed at each state.

> **Key points to remember:**
> - Testosterone <50 ng/dL with PSA or radiological progression; keep ADT going.
> - AR still drives CRPC.
> - nmCRPC with PSADT ≤10 months: apalutamide, enzalutamide or darolutamide.
> - mCRPC: ARPI, taxanes (docetaxel, cabazitaxel), radium-223, ^{177}Lu-PSMA, PARP inhibitors, sipuleucel-T.
> - Switch mechanism; never radium-223 with abiraterone.

Also, from the condensed book (not in the paper answer):

- A prostate-specific antigen (PSA) doubling time (PSADT) **<3 months** predicts a rapid course, so consider aggressive therapy.
- Bone pain or visceral-predominant disease calls for **upfront chemotherapy**.

- **Castration resistance is not androgen independence.** The androgen receptor (AR) still drives growth through amplification, mutations, the splice variant **AR-V7** and tumour steroid synthesis, so AR-pathway inhibitors (ARPIs) work.
- Once AR dependence is lost, use taxanes, radium-223, PSMA therapy, PARP inhibitors (PARPIs) or immunotherapy.

2 · Clinical states and progression criteria

The **Prostate Cancer Clinical Trials Working Group 3 (PCWG3)** clinical states model classifies men by prior therapy, from rising PSA through non-metastatic (nmCRPC) to metastatic CRPC (mCRPC).

- PCWG3 PSA progression is a rise of ≥25% and ≥2 ng/mL above nadir, confirmed ≥3 weeks later.

Modality	Pitfall	PCWG3 rule
PSA	Early rise with chemotherapy; flare with radium-223 and ^{177}Lu-PSMA	Confirm any rise
Bone scan	Early new lesions may be **flare** of healing disease	Rescan at **12–16 weeks** as the new baseline; progression needs **≥2 new lesions** confirmed by ≥2 more ("2 + 2")
CT	Soft tissue is little of the burden	RECIST; PFS preferred over response rate
PSMA PET	Apparent progression at 4 weeks resolves by 3 months in ≈13%	Criteria not defined

> **KEY POINT**
>
> Every marker (PSA, bone scan, PSMA PET) can flare early, so call progression only on confirmed, combined evidence.

- **Continue ADT indefinitely** during every later therapy.

3 · Germline and somatic testing

- **Germline testing** suits all metastatic disease, relevant family history (BRCA, Lynch, HOXB13) and Ashkenazi ancestry.
- **Somatic** sequencing usually uses the archival primary. In liquid biopsy, **clonal haematopoiesis (CHIP)** can give false DNA-repair calls.
- Somatic homologous recombination repair (HRR) mutations occur in **20–25%** of mCRPC. Mismatch repair deficiency (dMMR/MSI-H) occurs in **2–5%** and should prompt Lynch counselling.

4 · Non-metastatic CRPC (nmCRPC)

nmCRPC is a rising PSA on castration with no metastases on **conventional** imaging. PSA kinetics are the strongest predictor. **Add an ARPI (apalutamide, enzalutamide or**

darolutamide) to ADT when PSADT is ≤10 months. All three improve metastasis-free survival (MFS) and OS. Abiraterone is active but **not approved**.

5 · mCRPC: androgen-signalling inhibitors

Agent	Mechanism	Dose	Toxicity / notes
Abiraterone	Selective CYP17 inhibitor (17α-hydroxylase and 17,20-lyase)	1000 mg **fasting** with prednisone (10 mg/day in mCRPC)	Mineralocorticoid excess, hepatotoxicity; correct potassium first. Food raises absorption up to 10× (alternative: 250 mg with a low-fat meal)
Enzalutamide	Pure AR antagonist; blocks translocation and DNA binding	160 mg, no steroid	**Seizures (≈1%)**, fatigue, falls; CYP3A4/2C8 interactions
Apalutamide	Does not cross the blood–brain barrier (BBB)	240 mg	**Rash, hypothyroidism**, fractures
Darolutamide	Structurally distinct; does not cross the BBB	600 mg twice daily	Least CNS toxicity

KEY POINT

Prednisone with abiraterone: blocking CYP17 lowers cortisol, so ACTH rises and causes mineralocorticoid excess; prednisone suppresses ACTH. Enzalutamide needs no steroid. All ARPIs cause falls and fractures, so take care over 75.

6 · Cytotoxic chemotherapy

Cabazitaxel has poor P-glycoprotein affinity, so it works after docetaxel.

Agent	Dose / line	Signature toxicity
Docetaxel	**75 mg/m² 3-weekly** with prednisone 5 mg twice daily; first line	Neutropenia, **neuropathy**, oedema, hyperlacrimation, nail dystrophy
Cabazitaxel	**20 mg/m²** (25 if low neutropenic risk) 3-weekly with prednisone and G-CSF; after docetaxel	**Neutropenia** (grade 3–4 82%), **diarrhoea**
Mitoxantrone	12 mg/m² with prednisone; palliation	No survival benefit
Platinum (with etoposide or taxane)	Neuroendocrine variant; DNA-repair-deficient	—

7 · Radiopharmaceuticals

Feature	Radium-223	^{177}Lu-PSMA-617
Emitter	**Alpha** (path **<100 µm**, so marrow is spared); calcium mimetic	Beta; needs PSMA PET-positive disease
Eligible	**Symptomatic bone metastases, no visceral or bulky nodal disease**	PSMA-avid, any site, after ARPI and taxane
Schedule	50 kBq/kg IV 4-weekly × 6	6-weekly, up to 6 cycles
Toxicity	Diarrhoea, anaemia	Fatigue, **dry mouth**, anaemia

Feature	Radium-223	^{177}Lu-PSMA-617
Caution	**Never with abiraterone/prednisone**; bone agent with any ARPI	—

Strontium-89 and samarium-153 only palliate pain.

8 · PARP inhibitors

SAQ PARP inhibitors in Prostate cancer.

Paper answer · P4

Type: SAQ (10 marks)
Source: 2024-Winter Paper 4, Q3f

Introduction

- **PARP (poly-ADP-ribose polymerase)** repairs single-strand DNA breaks. **PARPIs (PARP inhibitors):** olaparib, rucaparib, niraparib, talazoparib.

Mechanism: synthetic lethality

- **Catalytic inhibition** and **PARP trapping** on DNA leave single-strand breaks unrepaired.
- At replication these become **double-strand breaks**, normally repaired by **HRR (homologous recombination repair)**.
- Tumours lacking HRR (e.g. **BRCA2** loss) die; normal cells survive.
- **AR (androgen receptor)–PARP synergy:** AR maintains HRR genes, so AR blockade adds repair deficiency: basis for PARPI + ARPI (androgen receptor pathway inhibitor).

Who benefits: testing

- **HRR alterations in mCRPC (metastatic castration-resistant prostate cancer): up to 20–25%.** Genes: **BRCA2 (largest benefit)**, BRCA1, ATM (weak), PALB2, CHEK2, CDK12.
- **Germline testing** in all metastatic disease, strong family history or Ashkenazi ancestry; **somatic testing** on tissue or ctDNA (circulating tumour DNA), where clonal haematopoiesis can mislead.

Key trials

Trial	Setting	Result
PROfound	Olaparib vs enzalutamide/abiraterone after ARPI	BRCA/ATM cohort: imaging PFS (progression-free survival) **7.4 vs 3.6 months (HR [hazard ratio] 0.34)**; OS (overall survival) 19.1 vs 14.7 months in the final analysis (chapter quotes the interim 18.5 vs 15.1)
TRITON3	Rucaparib vs physician's choice	BRCA: PFS 11.2 vs 6.4 months (HR 0.50); no ATM benefit
PROpel	Olaparib + abiraterone, first-line mCRPC	OS 42.1 vs 34.7 months (P = 0.054, not significant); greatest in BRCA
MAGNITUDE	Niraparib + abiraterone, HRR-altered	BRCA rPFS (radiographic PFS) 19.5 vs 10.9 months (HR 0.55); no benefit without HRR alteration
TALAPRO-2	Talazoparib + enzalutamide, first line	All-comers rPFS not reached vs 21.9 months (HR 0.63); largest in BRCA

- **Also know:** TOPARP-A/B, TRITON2, GALAHAD (phase II).

Approvals (FDA [Food and Drug Administration])

- **Olaparib monotherapy:** HRR-mutated mCRPC after ARPI (2020).
- **Rucaparib:** BRCA-mutated mCRPC.
- **Olaparib or niraparib + abiraterone:** BRCA-mutated; **talazoparib + enzalutamide:** HRR-mutated (both 2023).

Toxicity

- **Anaemia** (commonest grade 3–4; monitor counts), fatigue, nausea; rare **MDS/AML (myelodysplastic syndrome/acute myeloid leukaemia)**, pulmonary embolism.

Recent advances

- **mCSPC (metastatic castration-sensitive prostate cancer)** trials: AMPLITUDE, TALAPRO-3; ATR inhibitors for ATM loss.

Diagram to draw: synthetic lethality: PARP block + BRCA loss → cell death.

> **Key points to remember:**
> - Synthetic lethality; BRCA2 gains most, ATM least.
> - PROfound: olaparib, PFS 7.4 vs 3.6 months.
> - Combinations: PROpel, MAGNITUDE, TALAPRO-2.

Also, from the condensed book (not in the paper answer):

- PARPIs are ineffective in unselected men; haemoglobin can fall suddenly, so watch blood counts.

9 · Immunotherapy

- **Sipuleucel-T**, an autologous dendritic-cell vaccine, was the first approved therapeutic cancer vaccine.
- It suits **asymptomatic/minimally symptomatic mCRPC with no visceral metastases or opioid pain**. It prolongs OS **without** PSA or PFS change.
- **Pembrolizumab** 200 mg IV 3-weekly is approved for **MSI-H/dMMR** or tumour mutational burden **≥10 mutations/megabase** after ≥1 line.

10 · Neuroendocrine/aggressive-variant CRPC

Feature	Usual mCRPC	Neuroendocrine variant
AR / PSA	Retained; PSA tracks disease	Lost; **PSA low despite progression**
Pattern	Osteoblastic bone ± nodes; visceral <20%	Bulky masses, liver, **osteolytic** lesions with hypercalcaemia, brain
Molecular	AR alterations	≥2 of PTEN loss, TP53 mutation, RB1 loss; low PSMA
Treatment	ARPI, taxane, radioligand, PARPI	**Platinum + etoposide** (or taxane + carboplatin) ± radiotherapy (RT)
Survival	Years	**<12 months**

Biopsy a metastasis if progression is rapid with a low PSA.

11 · Sequencing and cross-resistance

A second ARPI helps only **15–30%** because of cross-resistance, so **switch mechanism** (CARD). See the P2-140 sequencing table; add pembrolizumab for MSI-H/dMMR and a PARPI if HRR-mutated.

- Enzalutamide induces CYP3A4/2C9/2C19, so check anticoagulants and antiepileptics.
- AR-V7 in circulating tumour cells predicts resistance to abiraterone and enzalutamide, not to taxanes.

12 · Complications and palliative care

*See the paper answer **Metastatic prostate cancer, including newly diagnosed and bone metastases** (Section 17, Ch 165).*

Also, from the condensed book (not in the paper answer):

- Obstructive uropathy must be **relieved before chemotherapy**, but not every unilateral case needs drainage.

Every focal back pain in a man with bone metastases is cord compression until proven otherwise.

ALGORITHM

1. Back pain with or without deficit or sphincter change: suspect compression.
2. Give **high-dose IV dexamethasone** (10 mg IV, then 4–10 mg 6–12-hourly; 16–100 mg/day).
3. Arrange **urgent whole-spine MRI**.
4. Give RT with or without decompressive surgery.
5. Taper steroids over 2–3 weeks.

- Operate for progression during RT, unstable fracture, or recurrence after RT.
- Treat neuropathic pain with tricyclics or gabapentinoids, and start hospice care in time. Erythropoiesis-stimulating agents are not recommended for anaemia.

13 · Bone-targeted therapy

Two uses: ADT osteoporosis, and **prevention of skeletal-related events (SREs)** in CRPC bone metastases. There is **no survival effect**.

Feature	Zoledronic acid	Denosumab
Dose	**4 mg IV 4-weekly**, renal adjustment	**120 mg subcutaneous (SC) 4-weekly**; no renal adjustment
SRE	Better than placebo	**Better than zoledronic acid** (20.7 vs 17.1 months)
Toxicity	Flu-like, creatinine rise, hypocalcaemia, osteonecrosis of the jaw (ONJ)	Hypocalcaemia (grade ≥3 5%), ONJ **2–4%**

KEY POINT

ONJ follows dental extraction or poor teeth, so check the teeth first. Give calcium 1000 mg and vitamin D 800 IU daily with both agents. Bone agents are strongly recommended with radium-223.

14 · Landmark trials

Trial	Design	Result	Take-home
SPARTAN	nmCRPC, PSADT ≤10 mo; apalutamide	MFS 40.5 vs 16.2 mo (HR 0.28); OS HR 0.78	First nmCRPC drug
PROSPER	nmCRPC; enzalutamide	MFS improved; death reduced 27%	Enzalutamide for nmCRPC
ARAMIS	nmCRPC; darolutamide	MFS 40.4 vs 18.4 mo; OS HR 0.69	Least toxic
TAX 327	Docetaxel 3-weekly vs weekly vs mitoxantrone	OS 18.9 vs 17.3 vs 16.4 mo (HR 0.76)	First chemo to prolong survival
TROPIC	Post-docetaxel; cabazitaxel vs mitoxantrone	OS 15.1 vs 12.7 mo (HR 0.70)	Second-line chemo
PROSELICA / FIRSTANA	Cabazitaxel 20 vs 25 / vs docetaxel first line	20 non-inferior / not superior	20 mg/m²; docetaxel first
CARD	After docetaxel and 1 ARPI: cabazitaxel vs 2nd ARPI	OS 13.6 vs 11.0 mo (HR 0.64)	Switch mechanism
COU-AA-301	Post-docetaxel; abiraterone	OS 14.8 vs 10.9 mo (HR 0.65)	Post-chemo
COU-AA-302	Chemo-naive, asymptomatic	rPFS 16.5 vs 8.3 mo	Pre-chemo
AFFIRM	Post-docetaxel; enzalutamide	OS 18.4 vs 13.6 mo (HR 0.63)	Post-chemo
PREVAIL	Chemo-naive; enzalutamide	Death reduced 29%; radiographic progression reduced 81%	Pre-chemo
ALSYMPCA	Symptomatic bone-only; radium-223	OS 14.9 vs 11.3 mo (HR 0.70); SRE delayed	Alpha emitter prolongs life
VISION	Post-ARPI and taxane, PSMA-positive; ^{177}Lu-PSMA	OS 15.3 vs 11.3 mo (HR 0.62); rPFS HR 0.40	Approved 2022
TheraP	Phase II; ^{177}Lu-PSMA vs cabazitaxel	PSA response 65 vs 37%; fewer grade 3–4; OS equal	Lutetium ≥ cabazitaxel
IMPACT	Sipuleucel-T vs placebo	OS 25.8 vs 21.7 mo (HR 0.78); no PSA/PFS change	Vaccine approved 2010
PROfound	Post-ARPI; olaparib vs 2nd ARPI	BRCA/ATM rPFS 7.4 vs 3.6 mo (HR 0.34)	Olaparib, HRR panel
TRITON2/3	Rucaparib after ARPI	BRCA rPFS HR 0.50; no ATM benefit	BRCA only
PROpel / MAGNITUDE / TALAPRO-2	First-line PARPI with abiraterone / with enzalutamide	rPFS gains, largest in BRCA; PROpel OS 42.1 vs 34.7 mo (P = 0.054)	HRR/BRCA

Trial	Design	Result	Take-home
Denosumab vs zoledronic acid	Bone-metastatic CRPC	First SRE 20.7 vs 17.1 mo; OS equal	Denosumab for SRE

★ Putting it together

A · PSA to diagnosis

ALGORITHM

1. **Raised prostate-specific antigen (PSA) or abnormal DRE**: repeat it; exclude infection and instrumentation; double the PSA on 5α-reductase inhibitors (→ Ch 154).
2. **Multiparametric MRI (mpMRI) before biopsy**: PI-RADS (→ Ch 155).
3. **Biopsy** PI-RADS 3–5 targets plus systematic cores, preferably transperineal.
4. Assign **ISUP Grade Group (GG)** (→ Ch 156) and risk group (→ Ch 157). If unfavourable intermediate or worse, stage with PSMA PET (or CT and bone scan).

B · Localised disease: matching risk to treatment

NCCN favourable groups need ALL criteria, adverse groups ANY one.

Risk group	Defined by	Management
Very low	T1c, GG1, PSA <10, density <0.15, <3 positive cores (≤50% each)	Active surveillance (AS)
Low	T1–T2a, GG1, PSA <10	AS preferred
Favourable intermediate	One factor (T2b–c, GG2, PSA 10–20); <50% cores	Radical prostatectomy (RP) or radiotherapy (RT); AS if low-volume GG2
Unfavourable intermediate	2–3 factors, or GG3, or ≥50% cores	RP, or RT with 4–6 months ADT
High / very high	T3a, GG4–5 or PSA >20 / T3b–T4, primary pattern 5, ≥2 high-risk features	RP with extended node dissection, or RT with ADT 18–36 months; add abiraterone 2 years if N1 or ≥2 of Gleason ≥8, cT3–4, PSA ≥40 (STAMPEDE)

KEY POINT

SPCG-4, PIVOT and ProtecT show that radical treatment helps only younger men with ≥intermediate-risk disease. ProtecT: RP = RT for 15-year cancer survival.

C · Surgery vs radiotherapy vs focal therapy

	RP (Ch 160–161)	RT (Ch 162)	Focal (Ch 163)
Main toxicity	Incontinence, ED	Rectal/urinary irritation	Least; retreatment common
Advantages	Pathological staging	Suits comorbid men	Preserves function
Failure	PSA ≥0.2 ng/mL (confirmed)	Phoenix: nadir + 2 ng/mL	Undefined; biopsy-based
Salvage	Bed RT early at PSA 0.1–0.2 ± ADT	RP, cryotherapy, HIFU, brachytherapy	Repeat focal, RP, RT

D · Advanced disease

ALGORITHM

1. **Biochemical recurrence** (Ch 165): PSMA PET. After RP, salvage RT; after RT, local salvage if biopsy-proven. If high risk (PSADT ≤9 months), give ADT and enzalutamide (EMBARK).
2. **mHSPC**: never ADT alone; add an ARPI, or a docetaxel triplet if fit and high volume. If low volume, add prostate RT (STAMPEDE arm H). Test genes in all.
3. **nmCRPC** with PSADT ≤10 months: ARPI (SPARTAN, PROSPER, ARAMIS).
4. **mCRPC** (Ch 166): switch mechanism. Docetaxel → cabazitaxel (CARD); ^{177}Lu-PSMA (VISION); radium-223 for symptomatic bone-only disease (ALSYMPCA); PARP inhibitor if BRCA/HRR mutation (PROfound); pembrolizumab if MSI-H; protect bone.

Other paper questions in this section

Asked in the papers but not tied to a specific place in the book above.

LAQ 65 yr male underwent RP for stage T1c Gleason's score (4+3). At 6 months post surgery, his S. PSA has risen to 2 ng/ml above his preop PSA value. Discuss treatment and prognosis

Paper answer · P2

Type: LAQ (25 marks)
Source: Practice Set 12 Paper 2, Q1 (biochemical recurrence also discussed in 2019-Summer Paper 4, Q2)

Assumed meaning: 6 months after RP (radical prostatectomy) the PSA (prostate-specific antigen) never became undetectable and is now ≥2 ng/mL, i.e. PSA persistence with early rise, suggesting residual or systemic disease.

Introduction

- After RP, PSA (half-life about 3 days) should become undetectable (<0.1 ng/mL) within about 6 weeks.
- **PSA persistence:** failure to become undetectable after RP. **BCR (biochemical recurrence):** a rise after an undetectable nadir.
- **Gleason 4+3 = grade group 3** (unfavourable intermediate risk).

Definitions of BCR after RP

Body	Definition
AUA (American Urological Association)	PSA ≥0.2 ng/mL
NCCN (National Comprehensive Cancer Network)	Failure to reach undetectable, two consecutive rises, or PSA ≥0.1 ng/mL
EAU (European Association of Urology) risk groups	**Low risk:** PSADT (PSA doubling time) >1 year **and** grade group <4. **High risk:** PSADT ≤1 year **or** grade group 4–5

Why is the PSA high? (sources)

1. **Residual local disease:** positive margin (the only modifiable factor), residual benign tissue (low, stable PSA).
2. **Occult nodal metastases** (limited or no lymph node dissection).
3. **Distant micrometastases:** likeliest with persistent PSA and a fast rise.

Evaluation

- **Repeat PSA** to confirm; calculate PSADT.
- **Review pathology:** pT stage, margin (length >1 mm, Gleason 4 at margin), seminal vesicles, nodes. **Also know:** CAPRA-S (Cancer of the Prostate Risk Assessment–Surgery) nomogram; Decipher genomic classifier.
- **PSMA (prostate-specific membrane antigen) PET/CT (positron emission tomography/computed tomography):** best test; detection about half below 0.5 ng/mL and nearly 100% above 2 ng/mL. Choline or fluciclovine if PSMA is unavailable.
- **mpMRI (multiparametric MRI) pelvis:** best for local recurrence in the prostatic fossa.

- CT and bone scan are rarely positive below PSA 10 ng/mL.

Treatment

```
Confirm PSA, PSADT, pathology → PSMA PET ± mpMRI ├ Negative or fossa only → salvage RT to bed + ADT ├ Pelvic nodes only → bed + pelvic node RT + ADT ├ Oligometastatic (≤3) → ADT ± metastasis-directed therapy └ Polymetastatic → treat as mHSPC: ADT + ARPI ± docetaxel
```

1. Salvage RT (radiotherapy): the only curative option

- Dose **≥64–66 Gy** to the bed (SAKK 09/10: 70 Gy gave no gain over 64 Gy); 40–45 Gy to pelvic nodes.
- Best at the lowest PSA: each 0.1 ng/mL costs about 2.6% recurrence-free survival.
- **RADICALS-RT, RAVES, GETUG-AFU 17 (ARTISTIC meta-analysis):** early salvage at PSA 0.1–0.2 equals adjuvant RT and spares ≥60% from RT. These apply to men who become undetectable, not to this patient.

2. Add ADT (androgen deprivation therapy) to salvage RT: essential here

Trial	Arms	Result
RTOG 9601 (Radiation Therapy Oncology Group)	RT ± 2 years bicalutamide 150 mg; PSA 0.2–4.0	12-year OS (overall survival) 76.3% vs 71.3%
GETUG-AFU 16	RT ± 6 months goserelin; PSA 0.2–2.0	10-year BCR-free 64% vs 49%; MFS (metastasis-free survival) better; OS unchanged
SPPORT (RTOG 0534)	Bed / bed + 4–6 months ADT / pelvis + ADT	5-year freedom from progression 70.9% / 81.3% / 87.4%
RADICALS-HD	None vs 6 vs 24 months ADT	24 months better MFS than 6 months
- At PSA ≥2 ng/mL with grade group 3, give whole-pelvis RT with **long-course (about 2 years) ADT**.		

3. Systemic therapy

- **PSMA-positive metastases:** ADT + ARPI (androgen receptor pathway inhibitor: abiraterone, enzalutamide, apalutamide) ± docetaxel (LATITUDE, ARCHES, PEACE-1, ARASENS).
- **Oligometastases (≤3):** MDT (metastasis-directed therapy) by SBRT (stereotactic body RT) delays ADT (**STOMP:** ADT-free survival 21 vs 13 months; **ORIOLE:** progression 19% vs 61%).
- **High-risk BCR not fit for salvage: EMBARK** (PSADT ≤9 months, PSA ≥1 ng/mL): enzalutamide + leuprolide, MFS HR (hazard ratio) 0.42.
- **Also know:** castrate testosterone <50 ng/dL; intermittent ADT acceptable (PR.7); observation only for EAU low-risk BCR or short life expectancy; salvage lymph node dissection is investigational.

Prognosis

- **Untreated BCR (Pound):** metastasis in about one-third, median 8 years; death a median 5 years later. Predictors: **BCR <2 years, PSADT <10 months, Gleason 8–10.**

- **This patient is poor-risk:** persistent PSA, high level within 6 months, likely short PSADT. Persistent PSA carries a higher risk of metastasis than late BCR.
- **Salvage RT:** one-third to one-half BCR-free at 4–5 years; poorer above PSA 0.5–1 ng/mL. Cancer-specific survival gain mainly with PSADT <6 months treated within 2 years.
- Lifelong PSA follow-up.

Diagram to draw: the treatment flowchart above, with PSA kinetics curve after RP.

> **Key points to remember:**
> - EAU high-risk BCR: PSADT ≤1 year or grade group 4–5.
> - PSMA PET first: almost 100% detection above PSA 2 ng/mL.
> - Salvage RT ≥64 Gy + ADT: RTOG 9601 (OS), GETUG-AFU 16, SPPORT (add pelvis).
> - Pound: BCR <2 years, PSADT <10 months, Gleason 8–10 predict death.

LAQ 48 yr male presented with mixed voiding symptoms with PSA 18. Discuss management option in brief.

Paper answer · P2

Type: LAQ (25 marks)
Source: Practice Set 14 Paper 2, Q1 (related: prognosis in 2019-Summer Paper 4, Q3e; staging of non-organ-confined disease in Practice Set 10 Paper 2, Q1)

"Mixed voiding symptoms" is read as mixed storage and voiding LUTS (lower urinary tract symptoms).

Introduction

- In a 48-year-old, a PSA (prostate-specific antigen) of 18 ng/mL is far above what BPH (benign prostatic hyperplasia) usually produces. **Prostate cancer must be excluded first**; the LUTS are then treated on their own merits.
- PSA is organ-specific, not cancer-specific: BPH, prostatitis, UTI (urinary tract infection), retention and instrumentation also raise it. At PSA >10 ng/mL, biopsy yield is 43–65% with a normal DRE (digital rectal examination) and 70–90% with a suspicious one.

History

- **LUTS:** IPSS (International Prostate Symptom Score) with bother score; storage versus voiding; retention, haematuria, recurrent UTI.
- **Red flags:** bone pain, weight loss, leg oedema, cord compression.
- Family history (prostate cancer <65 years; BRCA (breast cancer gene)-related cancers), recent catheter, fever or perineal pain (prostatitis), 5ARI (5α-reductase inhibitor) use, neurological disease, erectile function.

Examination

- **DRE:** size, nodule, induration, extension, fixity (clinical T stage). Palpable bladder, neurological and spinal examination.

Investigations

Test	Purpose
Urinalysis, culture	Exclude UTI or prostatitis; treat and repeat PSA later (no empirical antibiotics without symptoms)

Test	Purpose
Repeat PSA, free PSA	Confirm the rise; low free PSA favours cancer
Creatinine, ultrasound KUB (kidney, ureter, bladder)	Upper tracts, prostate volume (PSA density)
Uroflowmetry, PVR (post-void residual)	Baseline LUTS assessment
mpMRI (multiparametric MRI) before biopsy	PI-RADS (Prostate Imaging Reporting and Data System) score; guides targeted cores; local staging
Biopsy	Targeted (PI-RADS ≥3) **plus systematic** 12 cores; **transperineal** preferred (lower sepsis)

- At PSA 18, biopsy is needed **even if MRI is negative**. Blood or urine biomarkers (PHI (Prostate Health Index), 4Kscore, PCA3) matter mainly in the 2–10 ng/mL grey zone.

Staging if cancer is found

- Grade group, cores involved, PSA, T stage → risk group. PSA 10–20 makes him at least **intermediate risk** (D'Amico, NCCN (National Comprehensive Cancer Network)).
- **High risk:** bone scan with CT (computed tomography), or **PSMA PET (prostate-specific membrane antigen positron emission tomography)**. Organ-confined disease is <50% at PSA >10; nodes about 20% at PSA >20.
- **Germline testing** (BRCA2, mismatch repair genes) for high-risk or metastatic disease or strong family history.

Management of the cancer (life expectancy >25 years)

Situation	Management
Low or favourable intermediate risk	AS (active surveillance) is possible, but uncommon at PSA 18
Localised intermediate or high risk	**RP (radical prostatectomy)**, usually robotic, with extended PLND (pelvic lymph node dissection) when node risk exceeds the nomogram threshold; or **EBRT (external beam radiotherapy) + ADT (androgen deprivation therapy)**: 4–6 months intermediate, 18–36 months high risk
Locally advanced (T3–T4, N1)	Multimodal: RP ± adjuvant or early salvage RT, or EBRT + long-term ADT; abiraterone added for high-risk node-positive or high-risk N0 disease (STAMPEDE)
Metastatic	ADT + ARPI (androgen receptor pathway inhibitor) ± docetaxel; RT to the prostate for low volume

- **Why surgery suits him:** young, gives pathological staging, and **removes the obstructing gland**, curing the LUTS. SPCG-4: surgical benefit is concentrated in men <65.
- Brachytherapy is relatively contraindicated with severe LUTS, a large gland or prior TURP (transurethral resection of the prostate) (retention, incontinence).

If biopsy is negative

- PSA 18 still needs explaining (prostatitis, large gland, missed anterior tumour): repeat MRI-targeted transperineal biopsy; %free PSA, PCA3, ConfirmMDx; close PSA follow-up.

Management of LUTS

- **Behavioural measures** (fluid timing, less caffeine) and an **α-blocker** (e.g. tamsulosin) at once; it does not alter PSA.

- **Avoid starting a 5ARI until cancer is excluded:** it halves PSA and masks cancer; if used later, record a baseline PSA and double readings after 12 months.
- Persistent storage symptoms with low PVR: add an antimuscarinic or β3-agonist (mirabegron).
- **Surgery** (TURP, HoLEP (holmium laser enucleation of the prostate)) for retention, refractory symptoms or complications, only once cancer is excluded. In locally advanced cancer with retention, a channel TURP relieves obstruction.
- **Also know:** pressure-flow study before surgery at age <50 with mainly voiding LUTS (EAU (European Association of Urology)).

```
PSA 18 + LUTS → history, DRE, urine culture → repeat PSA → mpMRI → targeted + systematic biopsy (transperineal) Cancer → risk group + staging → RP / RT + ADT / systemic Benign → explain PSA, re-biopsy if needed → BPH pathway LUTS throughout: α-blocker; no 5ARI until cancer excluded
```

Diagram to draw: the flowchart above.

> **Key points to remember:**
> - PSA 18 at 48: exclude cancer first; MRI before biopsy, transperineal targeted + systematic cores.
> - PSA >10: organ-confined <50%; PSA >20: nodes about 20%.
> - Young fit man: RP treats both cancer and obstruction.
> - α-blocker for LUTS; no 5ARI until cancer excluded (halves PSA).

LAQ Discuss management of organ confined CA prostate

Paper answer · P2

Type: LAQ (25 marks)
Source: Practice Set 15 Paper 2, Q2

Introduction

- Organ-confined prostate cancer is **cT1–T2 N0 M0**. Treatment must balance cancer mortality against **competing mortality**: life expectancy, comorbidity, baseline urinary, sexual and bowel function, and patient preference (shared decision-making).

Risk stratification (NCCN, National Comprehensive Cancer Network)

Risk group	Criteria
Very low	All of: T1c, GG (grade group) 1, PSA (prostate-specific antigen) <10 ng/mL, <3 positive cores with ≤50% cancer in each, PSA density <0.15
Low	All of: T1–T2a, GG1, PSA <10
Favourable intermediate	1 intermediate factor (T2b–c, GG2–3, PSA 10–20), GG1–2, <50% cores positive
Unfavourable intermediate	2–3 intermediate factors, GG3, or ≥50% cores positive
High	Any of: T3a, GG4–5, PSA >20

- **Grade groups:** GG1 = 3+3; GG2 = 3+4; GG3 = 4+3; GG4 = 8; GG5 = 9–10.

Evaluation

- PSA, DRE (digital rectal examination), mpMRI (multiparametric MRI) with targeted and systematic biopsy.
- **Staging imaging** from unfavourable intermediate risk: bone scan + CT/MRI, or PSMA (prostate-specific membrane antigen) PET (positron emission tomography). **proPSMA:** accuracy 92% vs 65% for conventional imaging.
- Life expectancy (life tables, Charlson index); IPSS (International Prostate Symptom Score) and erectile function.
- **Also know:** Partin tables, Kattan nomogram, CAPRA (Cancer of the Prostate Risk Assessment) score; genomic tests (Decipher, Prolaris, Oncotype DX) for borderline surveillance decisions.

Options by risk

Risk	Preferred options
Very low / low	**Active surveillance (AS)**; watchful waiting if life expectancy short
Favourable intermediate	AS in selected men; RP (radical prostatectomy); EBRT (external-beam radiotherapy) or brachytherapy alone
Unfavourable intermediate	RP + PLND (pelvic lymph node dissection); EBRT + **4–6 months ADT** (androgen deprivation therapy)
High (organ-confined, e.g. GG4–5 or PSA >20)	RP + extended PLND (multimodal); EBRT + **18–36 months ADT**

1. Active surveillance vs watchful waiting

- **AS:** deferred treatment with **curative intent**. Monitor by PSA every 3–6 months, DRE, **confirmatory biopsy or MRI within 6–12 months**, then periodic MRI and biopsy. Trigger for treatment: grade progression (GG ≥2–3), increasing volume or patient choice.
- Entry criteria: Johns Hopkins (strict, Epstein), PRIAS (Prostate Cancer Research International Active Surveillance), Toronto (inclusive). About half are treated by 10–15 years.
- **Watchful waiting:** no curative intent; ADT for symptomatic progression. For limited life expectancy.

2. Radical prostatectomy

- Open retropubic or **robot-assisted** (similar outcomes in the randomised trial). Nerve sparing when there is no same-side extracapsular risk.
- **PLND** when nodal risk is significant on nomograms (extended template).
- **Complications:** incontinence, ED (erectile dysfunction), anastomotic stricture, rectal injury.
- **Advantages:** pathological staging; relieves obstruction; RT kept for salvage.

3. Radiotherapy

- **EBRT:** IMRT (intensity-modulated RT) with image guidance. Fractionation: conventional 78 Gy in 39; **moderate hypofractionation 60 Gy in 20** (CHHiP: non-inferior); **SBRT (stereotactic body RT) 36.25 Gy in 5** (PACE-B: similar toxicity).
- **Why add ADT:** synergistic cell kill and target shrinkage. **RTOG 94-08:** 4 months ADT improved OS in intermediate risk; **DFCI 95-096:** 6 months ADT improved 8-year OS (61% → 74%).

- **Brachytherapy:** LDR (low-dose-rate) seeds for low and favourable intermediate risk. Relative contraindications: gland >60 mL, high IPSS, recent TURP. HDR (high-dose-rate) or LDR **boost** with EBRT for higher risk (ASCENDE-RT: half the biochemical failure).
- **Toxicity:** proctitis, irritative urinary symptoms, late ED, second cancers (controversial).

4. Focal and whole-gland ablation

- HIFU (high-intensity focused ultrasound), cryotherapy, focal laser, IRE (irreversible electroporation), VTP (vascular-targeted photodynamic therapy).
- Based on the **index lesion hypothesis**. Best for MRI-visible, biopsy-confirmed, unifocal **intermediate-risk (GG2)** disease. Investigational; ideally in trials. VTP is the only modality with a randomised trial (versus AS in low risk).

Landmark trials

Trial	Comparison	Result
SPCG-4 (pre-PSA era)	RP vs watchful waiting	Cancer death RR (relative risk) 0.56 at 23 years; NNT 8; benefit <65 years
PIVOT	RP vs observation (PSA era)	No mortality difference; benefit only in intermediate risk
ProtecT	Monitoring vs RP vs RT + ADT	15-year cancer mortality 2.7%, 2.2%, 2.9% (no difference); more metastases with monitoring

- **ProtecT function:** RP worst for continence and sexual function; RT transient bowel effects.

Algorithm

```
Organ-confined CaP → risk group + life expectancy ├─ short life expectancy → watchful waiting ├─ very low / low → active surveillance ├─ favourable intermediate → AS (selected), RP, EBRT, brachytherapy └─ unfavourable intermediate / high → RP + PLND or EBRT + ADT (4–6 or 18–36 months)
```

Follow-up

- PSA every 6–12 months. **Biochemical recurrence:** after RP, PSA ≥0.2 ng/mL confirmed; after RT, **nadir + 2 ng/mL (Phoenix)**.

Recent advances

- PSMA PET staging; MRI-led surveillance; SBRT; genomic classifiers.

Diagram to draw: stepwise decision tree (treat or not → partial or whole gland → surgery or radiation).

> **Key points to remember:**
> - Organ-confined = cT1–T2 N0 M0; NCCN risk groups drive choice.
> - AS is preferred for low risk; ProtecT: no mortality difference at 15 years.
> - EBRT + ADT: 4–6 months (intermediate), 18–36 months (high).
> - Focal therapy rests on the index lesion hypothesis; still investigational.

LAQ Discuss advances in management of CA prostate

Paper answer · P4

Type: LAQ (25 marks)
Source: Practice Set 5 Paper 4, Q2 (related: 2025-Winter Paper 3, Q3b; 2024-Winter Paper 4, Q3f)

Introduction

- Three aims drive recent change: **find significant cancer and avoid diagnosing insignificant cancer**; **match local treatment to risk** with less morbidity; give **earlier, intensified and targeted** systemic therapy.

1. Screening and diagnosis

- **ERSPC (European Randomized Study of Screening for Prostate Cancer)** and Göteborg-1: PSA (prostate-specific antigen) screening lowers cancer-specific mortality, but PSA alone causes negative biopsies and overdiagnosis of GG (Grade Group) 1. Hence risk-stratified pathways.
- **Biomarkers** (reflex tests before biopsy):

Test	Sample	Content
PHI (Prostate Health Index)	Blood	(p2PSA ÷ free PSA) × √total PSA
4Kscore	Blood	Four kallikreins + clinical data; % risk of GG ≥2
STHLM3	Blood	4Kscore kallikreins + 232 risk SNPs (single-nucleotide polymorphisms) + clinical variables
PCA3 (prostate cancer antigen 3), SelectMDx	Urine after DRE (digital rectal examination)	RNA ratio (PCA3); DLX1 and HOXC6 (SelectMDx)
ExoDx IntelliScore	Urine, no DRE	Exosomal RNA of PCA3, ERG, SPDEF

- Tissue genomic classifiers (Decipher, Prolaris, Oncotype DX) refine risk for surveillance and adjuvant decisions.
- **MRI-first pathway:** mpMRI (multiparametric magnetic resonance imaging), scored by **PI-RADS (Prostate Imaging–Reporting and Data System)** 1–5, before biopsy.
- **PROMIS (2017):** mpMRI sensitivity 93% vs 48% for TRUS (transrectal ultrasound) biopsy; NPV (negative predictive value) 89%; 27% could avoid biopsy.
- **PRECISION (2018):** MRI with targeted biopsy of PI-RADS ≥3 found more significant cancer (38% vs 26%); 28% avoided biopsy.
- So PI-RADS ≥3 lesions are biopsied, while PI-RADS 1–2 with low PSA density may safely avoid biopsy.
- **Transperineal biopsy** under local anaesthesia: near-zero sepsis; MRI–ultrasound fusion targeting. **Micro-ultrasound** (PRI-MUS scoring) matches mpMRI in registry data.
- **PSMA (prostate-specific membrane antigen) PET (positron emission tomography): proPSMA:** accuracy 92% vs 65% for CT (computed tomography) plus bone scan (sensitivity 85%, specificity 98%). Used for high-risk staging, biochemical recurrence and selecting radioligand therapy.

2. Localised disease

- **AS (active surveillance):** standard for GG1; extended to low-volume favourable intermediate risk (GG2), with MRI and PSA density. **ProtecT (15 years):** prostate cancer mortality about 2–3% in monitoring, surgery and radiotherapy arms; more metastasis on monitoring, no survival difference.

- **Robotic RP (radical prostatectomy): Retzius-sparing** (Bocciardi) gives earlier continence (1 week to 3 months; equal by 12 months); single-port robot; frozen-section-guided nerve sparing (NeuroSAFE).
- **Radiotherapy:**
- Moderate hypofractionation **60 Gy in 20** (CHHiP) is standard.
- **SBRT (stereotactic body radiotherapy) 36.25 Gy in 5:** PACE-B showed similar late toxicity at 2 years (chapter); its 5-year report (2024) showed non-inferior biochemical or clinical control. HYPO-RT-PC (42.7 Gy in 7) was also non-inferior.
- **FLAME:** MRI-lesion boost improved biochemical PFS (progression-free survival) (HR [hazard ratio] 0.45). **Hydrogel rectal spacer** reduces rectal toxicity.
- High-risk M0: **abiraterone added to ADT (androgen deprivation therapy) and RT (STAMPEDE):** MFS (metastasis-free survival) HR 0.53.
- **Focal therapy:** based on the **index lesion hypothesis** (the largest or highest-grade focus drives progression). Energies: HIFU (high-intensity focused ultrasound), cryotherapy, IRE (irreversible electroporation), focal laser, VTP (vascular-targeted photodynamic therapy, padeliporfin). **HEAT registry (1379 men):** failure-free survival 68% (intermediate risk); continence 90–95%. Still within trials or registries.

3. Recurrent and advanced disease

Setting	Advance	Evidence
High-risk BCR (biochemical recurrence)	PSMA-guided salvage; enzalutamide + leuprolide	**EMBARK:** MFS HR 0.42
mHSPC (metastatic hormone-sensitive)	ADT + ARPI (androgen receptor pathway inhibitor)	LATITUDE, TITAN, ENZAMET, ARCHES
	Triplet (ADT + docetaxel + ARPI)	**PEACE-1, ARASENS**
	Prostate RT, low volume	STAMPEDE arm H
	MDT (metastasis-directed therapy)	STOMP, ORIOLE
nmCRPC (non-metastatic castration-resistant)	Apalutamide, enzalutamide, darolutamide	SPARTAN, PROSPER, ARAMIS
HRR (homologous recombination repair)-mutated mCRPC	**PARP (poly-ADP-ribose polymerase) inhibitors**	**PROfound:** olaparib imaging PFS 7.4 vs 3.6 months; PROpel, MAGNITUDE, TALAPRO-2
PSMA-positive mCRPC	**^{177}Lu-PSMA-617**	**VISION:** OS (overall survival) 15.3 vs 11.3 months (HR 0.62); TheraP: PSA response 65% vs 37% vs cabazitaxel

- **ADT alone is no longer standard** for mHSPC; every fit man gets intensification.
- **Germline and somatic testing** for all metastatic disease; **pembrolizumab** for MSI-high (microsatellite instability-high) tumours.
- **Relugolix:** oral GnRH (gonadotropin-releasing hormone) antagonist; HERO showed better sustained castration than leuprolide.

- **Also know:** radium-223 (ALSYMPCA, OS HR 0.70); PSMAfore (lutetium before chemotherapy); AI in MRI reading and pathology.

Diagram to draw: MRI-first pathway: raised PSA → mpMRI → PI-RADS ≥3 targeted transperineal biopsy; PI-RADS 1–2 with low PSA density → surveillance.

> **Key points to remember:**
> - PROMIS and PRECISION: MRI before biopsy finds more significant cancer and avoids biopsies.
> - proPSMA: PSMA PET accuracy 92% vs 65%.
> - ProtecT: no mortality difference at 15 years; AS is standard for GG1.
> - VISION (Lu-PSMA, HR 0.62); PEACE-1 and ARASENS triplets; EMBARK.

SAQ Docetaxel

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 8 Paper 1, Q3b

Introduction

- Semisynthetic **taxane** cytotoxic. Since **TAX 327 (2004)** it is the standard first-line chemotherapy for mCRPC (metastatic castration-resistant prostate cancer).

Mechanism of action

- Binds **β-tubulin** and **stabilises microtubules** (GTP [guanosine triphosphate]-independent polymerisation). Cells arrest in **G2/M**.
- Induces apoptosis independent of TP53, through **BCL2 phosphorylation** and p27 induction.
- Prostate-specific: blocks microtubule transport of ligand-bound **AR (androgen receptor)** into the nucleus.

Dose and administration

- **75 mg/m² IV (intravenous) every 3 weeks with prednisone 5 mg twice daily**; 6 cycles in mHSPC (metastatic hormone-sensitive prostate cancer). Steroid premedication limits fluid retention and hypersensitivity. Hepatic metabolism: caution in liver dysfunction.

Uses in urology

Setting	Evidence
mCRPC	**TAX 327:** overall survival (OS) 18.9 vs 16.4 months versus mitoxantrone (HR [hazard ratio] 0.76); better pain and PSA (prostate-specific antigen) response. SWOG 9916 (with estramustine) also positive
mHSPC	**CHAARTED:** OS 57.6 vs 44.0 months (HR 0.61), benefit in **high volume**. STAMPEDE: OS HR 0.78. GETUG-AFU 15 not significant
Triplet	ADT (androgen deprivation therapy) + docetaxel + abiraterone (**PEACE-1**) or darolutamide (**ARASENS**) improves OS
High-risk localised	RTOG 0521: added to RT (radiotherapy) + long-term ADT, OS HR 0.69

Setting	Evidence
Bladder	Second-line single agent (response about 13%); **intravesical gemcitabine–docetaxel** for BCG (bacillus Calmette–Guérin)-unresponsive or BCG-naive high-grade NMIBC (non-muscle-invasive bladder cancer)
Penis	**TPF** (docetaxel, cisplatin, 5-fluorouracil)

CHAARTED high volume: ≥4 bone metastases with ≥1 outside the axial skeleton, or visceral metastasis.

Toxicity

- **Neutropenia** and febrile neutropenia (dose-limiting), fatigue, **fluid retention** (peripheral oedema), **peripheral neuropathy**, hyperlacrimation, **nail dystrophy**, alopecia, liver dysfunction, hypersensitivity.
- 3-weekly: more neutropenia; weekly: more fatigue, diarrhoea, tearing.

Resistance and sequencing

- P-glycoprotein efflux pump mediates resistance. **Cabazitaxel** has poor P-glycoprotein affinity, so it works after docetaxel (**TROPIC**; **CARD** beat a second ARPI [androgen receptor pathway inhibitor]).
- **Also know:** failed combinations (bevacizumab, atrasentan).

Diagram to draw: microtubule stabilised by docetaxel causing G2/M arrest and blocking AR entry into the nucleus.

> **Key points to remember:**
> - 75 mg/m² 3-weekly + prednisone; TAX 327 HR 0.76.
> - CHAARTED: benefit in high-volume mHSPC (HR 0.61).
> - Neutropenia, oedema, neuropathy, nail change; cabazitaxel after failure.

SAQ PIN

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 1 Paper 2, Q3f

Assumed meaning: prostatic intraepithelial neoplasia.

Definition

- PIN (prostatic intraepithelial neoplasia): **architecturally benign** acini or ducts lined by **cytologically atypical** cells, with an intact or patchy **basal cell layer**.
- Graded **LGPIN (low-grade PIN)** and **HGPIN (high-grade PIN)**. LGPIN is not reported: it is irreproducible and carries no excess risk.

Histology

- HGPIN: nuclear enlargement, hyperchromasia, **prominent nucleoli**.
- Patterns: tufting, micropapillary, flat. **Cribriform lesions are now classed as IDC-P (intraductal carcinoma of the prostate) or AIP (atypical intraductal proliferation).**
- Immunohistochemistry: basal cells stain with **HMWCK (high-molecular-weight cytokeratin) and p63**; AMACR (α-methylacyl-CoA racemase) positive. Mainly in the peripheral zone.

Evidence that HGPIN is a precursor

- More and larger HGPIN foci in cancerous prostates; carcinoma seen budding from HGPIN.
- About 20% carry **TMPRSS2:ERG fusion** (about 50% of cancers). Extra genomic changes are needed for invasion.

Incidence

- About **4–5%** of biopsies (range 0–25%, due to observer variation).

Related lesions

Lesion	Meaning	Action on isolated finding
HGPIN	Weak risk marker	**No repeat biopsy**
PINATYP (HGPIN with adjacent atypical glands)	As "atypical"	Manage as ASAP (atypical small acinar proliferation)
ASAP	Cancer on re-biopsy about 40%	Repeat biopsy ± MRI (magnetic resonance imaging)
AIP	May herald IDC-P	Consider re-biopsy
IDC-P	Usually high-grade cancer present	Definitive treatment; excludes active surveillance

Management

- Isolated HGPIN, unifocal or multifocal: **no immediate repeat biopsy**. Decide further testing on PSA (prostate-specific antigen), DRE (digital rectal examination) and **mpMRI (multiparametric MRI)**.
- HGPIN on TURP (transurethral resection of the prostate) chips: no further sampling on current data.
- **Chemoprevention:** none proven; green tea showed reduced progression in small trials.
- *Guideline note:* EAU (European Association of Urology) guidelines still list extensive HGPIN (≥3 biopsy sites) as a reason for repeat biopsy; the chapter does not.

Diagram to draw: benign gland, HGPIN gland (atypical cells, intact basal layer) and cancer (no basal layer).

> **Key points to remember:**
> - Atypical cells, benign architecture, basal layer retained.
> - Isolated HGPIN needs no repeat biopsy; ASAP does.
> - Cribriform "PIN" is now IDC-P or AIP.

SAQ Imaging measures in CA prostate

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 13 Paper 2, Q3e (mpMRI detail in 2024-Winter Paper 1, Q3d; PSMA PET in 2019-Summer Paper 4, Q3a)

1. Ultrasound

- **TRUS (transrectal ultrasound):** guides biopsy, measures volume (PSA [prostate-specific antigen] density). Cancer is a hypoechoic PZ (peripheral zone) lesion, but **39% are isoechoic**.

- **Also know:** Doppler, contrast and elastography (adjuncts); **micro-ultrasound** (29 MHz; PRI-MUS [Prostate Risk Identification using Micro-Ultrasound] score; close to MRI accuracy).

2. MRI: detection and local staging

- **mpMRI (multiparametric magnetic resonance imaging):**

Sequence	Shows	Cancer
T2-weighted	Anatomy; EPE (extraprostatic extension), SVI (seminal vesicle invasion)	Low signal
DWI (diffusion-weighted imaging) with ADC (apparent diffusion coefficient)	Cellularity	Bright on DWI, dark on ADC
DCE (dynamic contrast-enhanced)	Vascularity	Early enhancement, washout

- **PI-RADS v2.1 (Prostate Imaging Reporting and Data System)** 1–5: DWI dominant in PZ, T2 in TZ (transition zone); DCE only upgrades PZ 3 to 4.
- **PROMIS** (sensitivity 93% vs 48% for TRUS biopsy) and **PRECISION** (more significant, less insignificant cancer): **MRI before biopsy**. Cochrane: sensitivity 0.91, specificity 0.37.
- **Also know:** biparametric MRI; **PI-QUAL** (Prostate Imaging Quality); **PRECISE** (scores change on surveillance); fusion biopsy.

3. Staging by risk group

- **Low or intermediate risk:** no routine CT (computed tomography) or bone scan.
- **High risk:** CT plus **^{99m}Tc-MDP (methylene diphosphonate) bone scan**, or **PSMA (prostate-specific membrane antigen) PET (positron emission tomography)**.
- **proPSMA:** accuracy 92% vs 65% for conventional imaging. Nodal sensitivity only 30–40%, specificity >90%: a negative scan does not exclude micrometastases.

4. Biochemical recurrence

- **PSMA PET** works at low PSA. **Calais:** 56% vs 26% for fluciclovine. **CONDOR:** management changed in 63.9%.
- Choline and fluciclovine PET are older; FDG (fluorodeoxyglucose) PET is poor except in neuroendocrine change.

5. Advanced disease

- PSMA PET selects men for **^{177}Lu-PSMA-617** (VISION); whole-body MRI. No proof yet that PSMA-guided treatment improves survival.

Diagram to draw: prostate zones with a PZ lesion dark on ADC; PSMA PET nodal uptake.

> **Key points to remember:**
> - MRI before biopsy (PROMIS, PRECISION); PI-RADS v2.1.
> - High risk: bone scan + CT or PSMA PET (proPSMA 92% vs 65%).
> - PSMA nodal sensitivity 30–40%.

SAQ Methods to predict prognosis of patient of Cancer prostate

Paper answer · P4

Type: SAQ (10 marks)
Source: 2019-Summer Paper 4, Q3e

1. Clinical and pathological factors

- **PSA (prostate-specific antigen):** organ-confined about 80% at <4 but <50% at >10 ng/mL; density and velocity add information.
- **Clinical stage (DRE (digital rectal examination), TNM).**
- **Grade group (ISUP (International Society of Urological Pathology)) – strongest single factor:**

Grade group	Gleason	5-year BCR (biochemical recurrence)-free
1	≤6	96%
2	3+4	88%
3	4+3	63%
4	8	48%
5	9–10	26%

- Biopsy: % positive cores, **cribriform and intraductal** patterns, perineural invasion.
- Prostatectomy: margin (extent, grade), EPE (extraprostatic extension), seminal vesicle and node status.

2. Risk groups

D'Amico	Low	Intermediate	High
PSA (ng/mL)	≤10	10–20	>20
Gleason	≤6	7	8–10
Stage	T1–T2a	T2b	T2c

- **NCCN (National Comprehensive Cancer Network):** very low, low, favourable/unfavourable intermediate, high, very high.
- **CAPRA (Cancer of the Prostate Risk Assessment)** score 0–10 (age, PSA, Gleason, stage, % cores); **CAPRA-S** after surgery.

3. Nomograms

- **Partin tables:** organ-confined disease, EPE, seminal vesicle, node involvement.
- **MSKCC (Memorial Sloan Kettering) nomograms:** pre- and post-treatment BCR and 15-year cancer-specific survival.
- **Briganti nomogram:** risk of nodal invasion → decides PLND (pelvic lymph node dissection).

4. Imaging

- mpMRI (multiparametric MRI): PI-RADS (Prostate Imaging Reporting and Data System) score, tumour contact length, EPE.
- PSMA PET (prostate-specific membrane antigen PET) for metastatic burden.

5. Tissue biomarkers

Test	Basis	Use
Decipher	22-RNA classifier	Metastasis risk; post-op radiotherapy
Oncotype DX GPS	17 genes	Adverse pathology; active surveillance

Test	Basis	Use
Prolaris	Cell-cycle progression genes	Progression, cancer death

- Germline BRCA2 (BReast CAncer gene 2): aggressive disease.

6. After treatment

- **PSA doubling time (PSADT):** <3 months highest risk; Pound: time to BCR <2 years, PSADT <10 months, Gleason 8–10 predict metastasis.
- **EAU (European Association of Urology) BCR risk groups:** low risk = PSADT >1 year and grade group <4.
- PSA nadir on ADT (androgen deprivation therapy); **Phoenix** failure (nadir + 2) after radiotherapy.

Diagram to draw: flowchart from risk group to nomogram, biomarker and treatment decision.

> **Key points to remember:**
> - Grade group is the strongest predictor.
> - D'Amico, NCCN and CAPRA combine PSA, grade and stage.
> - Decipher, Oncotype, Prolaris add genomic risk.

Rapid Revision Sheet · Section 17

Indications of genomic testing in carcinoma prostate (SAQ)

- Germline: all metastatic, high-risk, family history, Ashkenazi.
- Somatic: HRR → PARP inhibitors; MSI-H → pembrolizumab.
- BRCA2: aggressive; avoid casual active surveillance.
- Decipher, Oncotype GPS, Prolaris: prognosis only.

Antioxidants in urological malignancy (SAQ)

- SELECT: no benefit; vitamin E alone increased prostate cancer.
- ATBC benefit limited to smokers on low dose.
- Lycopene downgraded to "no conclusion" (2018).

Prostate cancer prevention trials (SAQ)

- PCPT ↓24.8%; REDUCE ↓22.8%; both ↑ high grade.
- FDA warning; no approved preventive agent.
- SELECT negative; vitamin E alone harmful.

PSA screening for prostate cancer: evidence, problems and solutions (LAQ)

- ERSPC: (about) 20% fall in prostate cancer death; PLCO null from contamination.
- Core problems: low specificity, no safe cut-off, overdiagnosis, overtreatment.
- USPSTF 2018: Grade C at 55–69, Grade D ≥70.
- Fix: risk-adapted PSA, MRI before biopsy, active surveillance for GG1; MRI-first and active surveillance reduce overdiagnosis and overtreatment.

Molecular staging of CA prostate (SAQ)

- Molecular staging finds occult disease and predicts behaviour beyond TNM.
- Decipher 22 genes; Oncotype 17 genes; Prolaris cell-cycle genes.
- AR-V7 on CTCs predicts ARPI resistance; BRCA2 means aggressive disease.
- Limitation: heterogeneity and mostly retrospective data. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

TRUS and prostate biopsy (SAQ)

- Probe 6–10 MHz; volume = 0.52 × T × AP × L; PSAD >0.15 suspicious.
- 39% of cancers isoechoic; hypoechoic PZ lesion is classic.
- Seminal vesicle >1.5 cm wide: ejaculatory duct obstruction.
- MRI before biopsy: PROMIS and PRECISION established the MRI-first pathway.
- 12 cores; antibiotics ≤24 hours for all (transrectal); sepsis 0.3–3%.
- Bleeding commonest; sepsis most serious.
- Transperineal now preferred.

Multiparametric MRI, PI-RADS and fusion biopsy (SAQ)

- T2W anatomy, DWI/ADC cellularity, DCE vascularity.
- PZ ruled by DWI, TZ by T2W; DCE only lifts PZ 3 to 4.
- PRECISION and PROMIS: MRI before biopsy.
- Fusion: co-registration with elastic deformation.
- Cognitive, fusion, in-bore: similar yield.

- Endorectal coil close to prostate: higher SNR and resolution.
- Useful at 1.5 T; largely unnecessary at 3 T (GG ≥3 detection equal).
- Drawbacks: discomfort, gland deformation, near-field flare.

Saturation biopsy of prostate (SAQ)

- Saturation = >12 cores (18–21 or more); 5-mm template mapping.
- No gain at initial biopsy; used after negatives and for focal therapy.
- PROMIS: template mapping as reference; mpMRI sensitivity 93%.

Gleason's score and its significance (SAQ)

- Biopsy score = commonest + highest pattern.
- 3+4 (GG2) and 4+3 (GG3) differ: BCR-free 88% vs 63%.
- GG4 = score 8; GG5 = 9–10.

Expectant treatment of CA prostate (SAQ)

- AS is deferred cure; WW is palliation.
- Grade group 1 hardly ever metastasises; risk lies in missed higher grade.
- ProtecT: no mortality difference at 15 years; more metastases on monitoring.
- About 25% leave AS by 5 years.

Robotic radical prostatectomy (SAQ)

- Surgeon, not platform, decides outcome (Yaxley 2016).
- Interfascial plane is the standard nerve-sparing plane.
- Bladder neck stricture <2%; transfusion 1–2%.

Radiotherapy and brachytherapy for CA prostate (LAQ)

- α/β ~1.5 Gy: basis of hypofractionation, 60 Gy/20 (CHHiP) and SBRT 36.25 Gy/5.
- ADT 4–6 months unfavourable intermediate; 18–36 months (P2: 24–36 months) high risk.
- Early salvage at PSA 0.1–0.2 = adjuvant (RADICALS, RAVES, GETUG-AFU 17).
- STAMPEDE H: prostate RT for low-volume M1; ALSYMPCA, VISION for mCRPC.
- Phoenix: nadir + 2 ng/mL.
- LDR ^{125}I 145 Gy; HDR ^{192}Ir.
- D90 >90%, V100 >85%.
- IPSS >20 or Qmax <10 = relative contraindication.
- ASCENDE-RT: boost halves biochemical failure.

HIFU (SAQ)

- Focused ultrasound → ≥55 °C coagulative necrosis plus cavitation.
- Transrectal HIFU suits posterior lesions; TULSA for anterior.
- Focal HIFU: continence 90–95%; HEAT 7-year failure-free 69%.

Cryotherapy for CA prostate (SAQ)

- Argon freezes, helium thaws (Joule–Thomson); −40 °C, two cycles, urethral warmer.
- Four mechanisms: direct injury, vascular thrombosis, apoptosis, immune effect.
- ≤−40 °C, two freeze–thaw cycles, margin 5–10 mm.
- Main role is salvage: PSA <10, doubling time ≥16 months.
- Whole-gland ED near universal; fistula ≤0.8% focal, up to 3.4% salvage.

Staging and management of locally advanced (non-organ-confined) CA prostate (LAQ)

- Seminal vesicle invasion is T3b; non-regional nodes are M1a.
- Locally advanced = T3–4, N0/N+, M0; treat with combined therapy.
- RT + long-term ADT (18–36 months; P1: 2–3 years) is the standard; ADT alone is inferior (SPCG-7).
- RT + ADT: EORTC 22863 overall survival 79% vs 62%.
- STAMPEDE: abiraterone 2 years for N1 or very high-risk M0 (metastasis-free survival HR 0.53).
- Neoadjuvant ADT before RP: no survival gain.
- After RP: early salvage RT preferred over routine adjuvant RT for most men.
- Orchidectomy: castrate in 12 hours; flushes, osteoporosis (denosumab), metabolic, sexual effects.

GnRH (LHRH) analogues: agonists and antagonists (SAQ)

- Agonist: surge, then continuous stimulation → down-regulation → castration in 2–4 weeks.
- Flare: LH 10-fold for 10–20 days; cover with antiandrogen 21–28 days.
- Antagonist: no flare; degarelix castrate by day 3; relugolix oral (HERO, fewer MACE); preferred in cord compression.
- Castrate <50 ng/dL; aim <20 ng/dL.

Maximum / combined androgen blockade (SAQ)

- MAB/CAB = castration + non-steroidal antiandrogen (blocks adrenal androgens).
- PCTCG 2000: only 2–3% 5-year survival gain.
- Main use now: flare cover for 21–28 days.
- ARPIs (androgen receptor pathway inhibitors) or triplets now standard in mHSPC; newer AR pathway inhibitors beat CAB (ENZAMET).
- Bicalutamide half-life about 6 days; 50 mg with LHRH agonist, 150 mg monotherapy.
- Monotherapy raises testosterone; gynaecomastia two-thirds.
- EPC: harm in localised disease; RTOG 9601: survival gain with salvage radiotherapy.
- Withdrawal response 15–30%.

Metastatic prostate cancer, including newly diagnosed and bone metastases (LAQ)

- ADT alone is no longer standard: ADT plus abiraterone or an ARPI, or a triplet (PEACE-1, ARASENS), for every fit man.
- CHAARTED volume vs LATITUDE risk definitions.
- Prostate RT only for low-volume synchronous disease (STAMPEDE arm H).
- Never start an LHRH agonist alone with spinal metastases: cover the flare with antiandrogen, or use an antagonist or orchidectomy.
- Single 8 Gy fraction gives the same pain relief as longer courses for focal pain; dexamethasone and MRI for cord compression.
- Focal back pain = cord compression until MRI proves otherwise.
- Denosumab 120 mg or zoledronate 4 mg IV 4-weekly prevent SREs in mCRPC, not in mHSPC; adjust zoledronate for creatinine clearance.
- No survival benefit from bone agents.
- Dental check first (ONJ); calcium and vitamin D.

- Denosumab delays SREs longer (20.7 vs 17.1 months).
- Radium-223 (ALSYMPCA) for symptomatic bone-only mCRPC; never with abiraterone.

Castration-resistant prostate cancer (CRPC): definition and treatment (LAQ)

- Testosterone <50 ng/dL with PSA or radiological progression; keep ADT going.
- AR still drives CRPC.
- nmCRPC with PSADT ≤10 months: apalutamide, enzalutamide or darolutamide.
- mCRPC: ARPI, taxanes (docetaxel, cabazitaxel), radium-223, ^{177}Lu-PSMA, PARP inhibitors, sipuleucel-T.
- Switch mechanism; never radium-223 with abiraterone.

PARP inhibitors in Prostate cancer. (SAQ)

- Synthetic lethality; BRCA2 gains most, ATM least.
- PROfound: olaparib, PFS 7.4 vs 3.6 months.
- Combinations: PROpel, MAGNITUDE, TALAPRO-2.

65 yr male underwent RP for stage T1c Gleason's score (4+3). At 6 months post surgery, his S. PSA has risen to 2 ng/ml above his preop PSA value. Discuss treatment and prognosis (LAQ)

- EAU high-risk BCR: PSADT ≤1 year or grade group 4–5.
- PSMA PET first: almost 100% detection above PSA 2 ng/mL.
- Salvage RT ≥64 Gy + ADT: RTOG 9601 (OS), GETUG-AFU 16, SPPORT (add pelvis).
- Pound: BCR <2 years, PSADT <10 months, Gleason 8–10 predict death.

48 yr male presented with mixed voiding symptoms with PSA 18. Discuss management option in brief. (LAQ)

- PSA 18 at 48: exclude cancer first; MRI before biopsy, transperineal targeted + systematic cores.
- PSA >10: organ-confined <50%; PSA >20: nodes about 20%.
- Young fit man: RP treats both cancer and obstruction.
- α-blocker for LUTS; no 5ARI until cancer excluded (halves PSA).

Discuss management of organ confined CA prostate (LAQ)

- Organ-confined = cT1–T2 N0 M0; NCCN risk groups drive choice.
- AS is preferred for low risk; ProtecT: no mortality difference at 15 years.
- EBRT + ADT: 4–6 months (intermediate), 18–36 months (high).
- Focal therapy rests on the index lesion hypothesis; still investigational.

Discuss advances in management of CA prostate (LAQ)

- PROMIS and PRECISION: MRI before biopsy finds more significant cancer and avoids biopsies.
- proPSMA: PSMA PET accuracy 92% vs 65%.
- ProtecT: no mortality difference at 15 years; AS is standard for GG1.
- VISION (Lu-PSMA, HR 0.62); PEACE-1 and ARASENS triplets; EMBARK.

Docetaxel (SAQ)

- 75 mg/m² 3-weekly + prednisone; TAX 327 HR 0.76.
- CHAARTED: benefit in high-volume mHSPC (HR 0.61).

- Neutropenia, oedema, neuropathy, nail change; cabazitaxel after failure. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

PIN (SAQ)

- Atypical cells, benign architecture, basal layer retained.
- Isolated HGPIN needs no repeat biopsy; ASAP does.
- Cribriform "PIN" is now IDC-P or AIP.

Imaging measures in CA prostate (SAQ)

- MRI before biopsy (PROMIS, PRECISION); PI-RADS v2.1.
- High risk: bone scan + CT or PSMA PET (proPSMA 92% vs 65%).
- PSMA nodal sensitivity 30–40%.

Methods to predict prognosis of patient of Cancer prostate (SAQ)

- Grade group is the strongest predictor.
- D'Amico, NCCN and CAPRA combine PSA, grade and stage.
- Decipher, Oncotype, Prolaris add genomic risk.